
**January 2026
IEEE Thesaurus**

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**Created by
The Institute of
Electrical and
Electronics
Engineers
(IEEE)**



January 2026 IEEE Thesaurus

The IEEE Thesaurus is a controlled vocabulary of almost 12,760 descriptive engineering, technical and scientific terms, as well as IEEE-specific society terms [referred to as “descriptors” or “preferred terms”]. * Each descriptor included in the thesaurus represents a single concept or unit of thought. The descriptors are considered the preferred terms for use in describing IEEE content. The scope of descriptors is based on the material presented in IEEE journals, conference papers, standards, and/or IEEE organizational material. A controlled vocabulary is a specific terminology used in a consistent and controlled fashion that results in better information searching and retrieval.

Thesaurus construction is based on the ANSI/NISO Z39.19-2005(2010) standard, *Guidelines for the Construction, Format, and Management of Monolingual Controlled Vocabulary*. The Thesaurus vocabulary uses American-based spellings with cross references to British variant spellings. The scope and structure of the IEEE Thesaurus reflects the engineering and scientific disciplines that comprise the Societies, Councils, and Communities of the IEEE in addition to the technologies IEEE serves.

IEEE has developed its controlled vocabulary through consultation with subject matter experts (specialists in a particular field) and specialists in information analysis. The IEEE Thesaurus thus provides a controlled vocabulary of subject headings to help people categorize or search for engineering and computing concepts, especially IEEE published content.

The IEEE Thesaurus also provides a conceptual map through the use of semantic relationships such as broader terms (BT), narrower terms (NT), 'used for' relationships (USE/UF), and related terms (RT). These semantic relationships identify theoretical connections between terms.
Italic text denotes Non-preferred terms.

Bold text is used for preferred headings.

Abbreviations used in the Thesaurus:

BT - Broader term
NT - Narrower term
RT - Related term
USE- Use preferred term
UF - Used for

*Refer to ANSI/NISO NISO Z39.19-2005 (R2010) Sections 5 through 8 for detailed information on controlled vocabularies, display formats, usage and spelling, and selection criteria for descriptors
(<http://www.niso.org/kst/reports/standards>).



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1/f noise

UF: 1f
Pink noise
BT: Noise

1f

USE: 1/f noise

2 DOF

USE: 2-DOF

2-D displays

USE: Two-dimensional displays

2-d hole gas

USE: Two dimensional hole gas

2-D photonic crystals

USE: Photonic crystals

2-DOF

UF: 2 DOF
two degrees of freedom
BT: Robot motion

21CN

USE: Next generation networking

21st century networks

USE: Next generation networking

2D displays

USE: Two-dimensional displays

2d hole gas

USE: Two dimensional hole gas

2D photonic crystals

USE: Photonic crystals

3 degrees of freedom

USE: 3-DOF

3 DOF

USE: 3-DOF

3-D displays

USE: Three-dimensional displays

3-D modeling

USE: Three-dimensional displays

3-D modelling

USE: Three-dimensional displays

3-D reconstruction

USE: Three-dimensional displays

3-DOF

UF: 3 DOF
3 degrees of freedom
BT: Robot motion
RT: 5-DOF
6-DOF
Motion analysis

3D accelerators

USE: Hardware acceleration

3D audio

BT: Audio systems
Augmented reality
RT: Ambisonics
Virtual reality

3D displays

USE: Three-dimensional displays

3D integrated circuits

USE: Three-dimensional integrated circuits

3D integration

USE: Three-dimensional integrated circuits

3D interconnect

USE: Three-dimensional integrated circuits

3D modeling

USE: Three-dimensional displays

3D modelling

USE: Three-dimensional displays

3D printing

USE: Three-dimensional printing

3D reconstruction

USE: Three-dimensional displays



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3D/

USE: Three-dimensional
integrated circuits

3G

USE: 3G mobile communication

3G mobile communication

UF: 3G
3rd generation mobile

communication

IMT-2000
Third generation mobile

communication

UMTS
Universal mobile

telecommunication service

BT: Cellular technology
Mobile communication

RT: 4G mobile communication
Ambient networks
Cellular radio
MIMO
Multiaccess communication
Next generation networking
OFDM
Radio access networks
Spread spectrum

communication

Telecommunication

computing

Time division synchronous

code division multiple access

3G partnership project

USE: 3GPP

3GPP

UF: 3G partnership project
3rd generation partnership

project

BT: Standards organizations
RT: New Radio

3GPP Standards

BT: Standards publications
NT: Long Term Evolution

3rd generation mobile communication

USE: 3G mobile communication

3rd generation partnership project

USE: 3GPP

mobile communication

UF: 4th generation mobile
communication

BT: Cellular technology
Mobile communication

RT: 3G mobile communication
5G mobile communication
Cellular radio
Enhanced mobile

broadband

Long Term Evolution
Next generation networking
Radio access networks
Spread spectrum

communication

Telecommunication

computing

Time division synchronous

code division multiple access

4K UHD

USE: UHDTV

4th generation mobile communication

USE: 4G mobile communication

5 degrees of freedom

USE: 5-DOF

5 DOF

USE: 5-DOF

5-day biochemical oxygen demand

USE: BOD5

5-day BOD

USE: BOD5

5-DOF

UF: 5 DOF
5 degrees of freedom

BT: Robot motion

RT: 3-DOF
6-DOF
Motion analysis

5G

USE: 5G mobile communication

5G mobile communication

UF: 5G
5th generation mobile

systems

5th generation systems



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systems	5th generation wireless	<i>802.11ax</i>	USE:	IEEE 802.11ax Standard
BT:	Cellular technology			
	Mobile communication	<i>802.11e</i>	USE:	IEEE 802.11e Standard
RT:	4G mobile communication			
	6G mobile communication	<i>802.11g</i>	USE:	IEEE 802.11g Standard
	Aerial computing			
	Cellular radio	<i>802.11n</i>	USE:	IEEE 802.11n Standard
	Edge computing			
	Land mobile radio			
	Next generation networking			
	Tactile Internet	<i>802.15</i>	USE:	IEEE 802.15 Standard
NT:	Ultra-dense networks			
broadband	Enhanced mobile	<i>802.16</i>	USE:	IEEE 802.16 Standard
	New Radio			
	Sidelink			
<i>5IR</i>		<i>802.3</i>	USE:	IEEE 802.3 Standard
USE:	Fifth Industrial Revolution			
<i>5th generation mobile systems</i>		<i>8K UHD</i>	USE:	UHDTV
USE:	5G mobile communication			
<i>5th generation systems</i>		<i>9/11</i>	USE:	Terrorism
USE:	5G mobile communication			
<i>5th generation wireless systems</i>		<i>9/11 attack</i>	USE:	Terrorism
USE:	5G mobile communication			
<i>6 degrees of freedom</i>		<i>911 attack</i>	USE:	Terrorism
USE:	6-DOF			
<i>6 DOF</i>		<i>A-bomb</i>	USE:	Nuclear weapons
USE:	6-DOF			
6-DOF		<i>A/D</i>	USE:	Analog-digital conversion
UF:	6 DOF			
	6 degrees of freedom	<i>A/D conversion</i>	USE:	Analog-digital conversion
BT:	Robot motion			
RT:	3-DOF	<i>A/D converter</i>	USE:	Analog-digital conversion
	5-DOF			
	Motion analysis			
6G mobile communication		<i>A2G</i>	USE:	Air to ground
BT:	Mobile communication			
RT:	5G mobile communication	communication		
	Aerial computing			
	Cellular radio	<i>AAL</i>	USE:	Ambient assisted living
	Space-air-ground			
integrated networks		<i>AAV</i>	USE:	Autonomous aerial vehicles
<i>802.11</i>				
USE:	IEEE 802.11 Standard			



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ABC algorithms

USE: Artificial bee colony
algorithm

Abdomen

BT: Body regions

Abrasive water jet cutting

USE: Water jet cutting

Abrasives

BT: Production materials

Absorption

BT: Materials science and

technology

RT: Semiconductor detectors

Abstract algebra

BT: Algebra

NT: Galois fields

Modules (abstract algebra)

Abstracts

BT: Writing

NT: Text summarization

AC generators

UF: Alternating current

generators

BT: Generators

RT: Pulse width modulation

NT: Induction generators

Synchronous generators

AC light emitting diode lamps

USE: LED lamps

AC machines

UF: Alternating current

machines

BT: Electric machines

RT: AC-AC converters

Pulse width modulation

Sensorless control

Windings

NT: AC motors

Induction machines

Synchronous machines

AC motors

UF: Alternating current motors

BT: AC machines

Motors

RT: Pulse width modulation

inverters

modulation

NT:

Pulse width modulation

Space vector pulse width

Hysteresis motors

Induction motors

AC power

USE: AC power transmission

AC power transmission

UF: AC power

Alternating current power

transmission

BT: Power transmission

NT: Flexible AC transmission

systems

AC-AC converters

UF: AC-AC convertors

AC-AC power conversion

BT: Converters

Power conversion

RT: AC machines

AC-AC convertors

USE: AC-AC converters

AC-AC power conversion

USE: AC-AC converters

AC-DC power converters

UF: AC-DC power convertors

AC/DC power converters

Analog-to-digital converter

Analog-to-digital convertor

BT: Power conversion

RT: Machine vector control

Pulse width modulation

inverters

Voltage multipliers

Voltage-source converters

NT:

Rectifying circuits

AC-DC power convertors

USE: AC-DC power converters

AC-LED lamps

USE: LED lamps

AC/DC power converters

USE: AC-DC power converters



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Accelerated aging

BT: Aging
Materials testing

Accelerated computing

USE: Hardware acceleration

Accelerated testing

USE: Life estimation

Acceleration measurement

USE: Accelerometers

Accelerator architectures

BT: Computer architecture

Accelerator beams

USE: Particle beams

Accelerator magnets

BT: Magnetic devices
Particle accelerators

Accelerometers

UF: Acceleration measurement
BT: Measurement
RT: Fall detection

Access charges

BT: Economics
Multiaccess communication

Access control

BT: Security
RT: Biometric authentication
Biometrics
Building services
Capability-based security
Communication system

security

Computer security
Identification of persons
Smart cards
Trust management
Zero Trust
NT: Accesslists
Authorization
Blocklists
Multi-factor authentication
Non-repudiation
Passwords

Access lists

USE: Accesslists

Access point base station

USE: Femtocell networks

Access protocols

BT: Protocols
RT: CAPTCHAs
NT: Media Access Control
NOMA

Access rights

USE: Permission

Accesslists

UF: Access lists
Allow lists
BT: Access control
Information filters

Accident and emergency medicine

USE: Emergency medicine

Accident prevention

BT: Industry applications
RT: Explosion protection
Preventive maintenance
Risk analysis
Safety devices
NT: Accidents

Accidents

BT: Accident prevention
RT: Domestic safety
Electric shock
Emergency services
Explosions
Fires
Hazardous areas
Occupational health
Occupational safety
Oil pollution
Product safety
Risk analysis
NT: Aerospace accidents
Electrical accidents
Industrial accidents
Marine accidents
Railway accidents
Road accidents

Accreditation

BT: Educational programs
RT: Conformance testing
Training



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Accuracy

BT: Mathematical analysis

Achilles tendon

BT: Tendons

Acidification

BT: Chemical processes
Pollution

NT: Ocean acidification

Acidosis

BT: Medical conditions

Acoustic applications

UF: Ultrasonic applications

BT: Acoustics

RT: Acoustic measurements

Biomedical acoustics

NT: Acoustic communication

(telecommunication)

Acoustic imaging

Acoustic testing

Acoustic arrays

BT: Acoustic transducers

RT: Acoustic signal processing

Array signal processing

Sonar

Acoustic beams

BT: Beams

Acoustic communication

(telecommunication)

BT: Acoustic applications

Telecommunication

services

RT: Mobile communication

OFDM

Wireless networks

Acoustic devices

UF: Ultrasonic devices

BT: Acoustics

RT: Piezoelectric devices

NT: Acoustic waveguides

Acousto-optical devices

Acoustoelectric devices

Bulk acoustic wave devices

Film bulk acoustic

resonators

Surface acoustic wave

devices

Acoustic diffraction

BT: Acoustic propagation

Acoustic distortion

BT: Distortion

RT: Acoustic noise

Acoustic signal processing

Loudspeakers

Nonlinear acoustics

Acoustic distortion measurement

USE: Distortion measurement

Acoustic emission

BT: Acoustics

RT: Acoustic noise

Acoustic testing

Nondestructive testing

Acoustic field

BT: Acoustics

Acoustic imaging

BT: Acoustic applications

RT: Acoustic testing

Oceanographic techniques

Acoustic materials

UF: Acoustic metamaterials

BT: Materials

RT: Piezoelectric materials

Acoustic measurements

BT: Measurement

RT: Acoustic applications

Acoustic testing

Anechoic chambers

Biomedical acoustics

Frequency measurement

Phase measurement

Seismic measurements

Wavelength measurement

Acoustic metamaterials

USE: Acoustic materials AND
Metamaterials

Acoustic noise

UF: Audible noise

Audio restoration

BT: Acoustics

RT: Acoustic distortion

Acoustic emission

Acoustic signal detection

Environmental factors



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NT: Mechanical factors
Vibrations
Background noise
Noise cancellation
Noise level
Noise reduction
Working environment noise

Acoustic phonetics

BT: Acoustics
Phonetics

Acoustic propagation

BT: Acoustics
RT: Acoustic pulses
Waves
NT: Acoustic diffraction

Acoustic pulses

BT: Acoustics
RT: Acoustic propagation

Acoustic reflection

BT: Reflection
RT: Acoustic scattering

Acoustic refraction

BT: Acoustic waves

Acoustic scattering

BT: Scattering
RT: Acoustic reflection
Waves

Acoustic sensors

BT: Sensors

Acoustic signal detection

BT: Signal detection
RT: Acoustic noise
NT: Sonar detection

Acoustic signal processing

UF: Acoustical signal processing
BT: Audio enhancement
Signal processing
RT: Acoustic arrays
Acoustic distortion
Acoustic transducers
NT: Active noise reduction
Speech processing

Acoustic surface waves

USE: Surface acoustic waves

Acoustic testing

BT: Acoustic applications
Materials testing
RT: Acoustic emission
Acoustic imaging
Acoustic measurements
Photoacoustic effects

Acoustic transducers

BT: Transducers
RT: Acoustic signal processing
Array signal processing
NT: Acoustic arrays

Acoustic wave attenuation

USE: Acoustic waves AND
Attenuation

Acoustic waveguides

BT: Acoustic devices

Acoustic waves

UF: Acoustic wave attenuation
BT: Acoustics
RT: Seismic waves
NT: Acoustic refraction
Acoustoelectric effects
Surface acoustic waves

Acoustical engineering

BT: Engineering - general

Acoustical signal processing

USE: Acoustic signal processing

Acoustics

UF: Ultrasonics
BT: Music
Physics
RT: Acoustoelectric effects
Fourier transforms
Magnetoacoustic effects
Phonons
Resonators
Vibrations
NT: Acoustic applications
Acoustic devices
Acoustic emission
Acoustic field
Acoustic noise
Acoustic phonetics
Acoustic propagation
Acoustic pulses
Acoustic waves
Acoustooptic effects



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Biomedical acoustics
Cepstral analysis
Nonlinear acoustics
Psychoacoustics
Reverberation
Spectral shape
Underwater acoustics

Acousto-optic devices

USE: Acousto-optical devices

Acousto-optical devices

UF: Acousto-optic devices
Acoustooptic devices
BT: Acoustic devices
RT: Acoustooptic effects

Acoustoelectric devices

UF: Electroacoustic devices
BT: Acoustic devices
RT: Acoustoelectric effects
Piezoelectric devices
Pulsed electroacoustic

methods

Surface acoustic wave

devices

Acoustoelectric effects

UF: Electroacoustic effects
BT: Acoustic waves
Electric fields
RT: Acoustics
Acoustoelectric devices
Semiconductor materials
NT: Pulsed electroacoustic

methods

Acoustomagnetic effects

USE: Magnetoacoustic effects

Acoustooptic devices

USE: Acousto-optical devices

Acoustooptic effects

BT: Acoustics
RT: Acousto-optical devices
NT: Piezooptic effects

Acquired immune deficiency syndrome

UF: AIDS
Acquired immunodeficiency

syndrome

BT: Diseases
RT: Human immunodeficiency

virus

Acquired immunodeficiency syndrome

USE: Acquired immune deficiency syndrome

Action potentials

UF: Bioelectric potentials
BT: Physiology
RT: Axons
Membrane potentials
Neurons
White matter

Action units

BT: Facial muscles
RT: Emotion recognition
Facial expressions

Activated sludge process

USE: Sludge treatment

Activation analysis

BT: Chemical analysis

Active appearance model

BT: Computer vision

Active circuits

BT: Circuits
NT: Active inductors
Gyrators
Operational amplifiers

Active contours

BT: Motion analysis

Active distribution networks

BT: Power distribution networks

Active disturbance rejection control

USE: Robust control

Active filters

BT: Filters
NT: Band-pass filters

Active inductors

BT: Active circuits
Inductors
RT: Gyrators
Integrated circuits
MOSFET circuits

Active learning

BT: Learning systems



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Active matrix addressing

BT: Active matrix technology

Active matrix liquid crystal displays

UF: AMLCDs
Active-matrix liquid-crystal

displays

BT: Active matrix technology
Liquid crystal displays

Active matrix organic LEDs

USE: Active matrix organic light emitting diodes

Active matrix organic light emitting diodes

UF: AMOLEDs
Active matrix organic LEDs
Active matrix organic light-

emitting diodes

BT: Active matrix technology
Diodes
Organic light emitting

diodes

Active matrix organic light-emitting diodes

USE: Active matrix organic light emitting diodes

Active matrix technology

UF: Active-matrix
BT: Displays
NT: Active matrix addressing
Active matrix liquid crystal

displays

Active matrix organic light emitting diodes

Thin film transistors

Active networking

BT: Network architecture

Active noise reduction

BT: Acoustic signal processing
Noise reduction
NT: Echo cancellers

Active perception

BT: Psychology
RT: Cognition
Control systems
Sensor fusion

Active pixel sensors

BT: Image sensors

Active RFID tags

BT: RFID tags

Active shape model

BT: Image processing
Pattern recognition

Active-matrix

USE: Active matrix technology

Active-matrix liquid-crystal displays

USE: Active matrix liquid crystal displays

Activity recognition

BT: Cognition
Pattern recognition
Sensor systems
RT: Computer vision
NT: Human activity recognition

Actuators

UF: Dielectric electroactive polymer actuators

Electroactive polymer

actuators

Electrostrictive polymer

actuators

Ionomeric polymer-metal

composite actuators

BT: Nanoactuators
Control equipment
RT: Control systems
Servomechanisms
Servosystems
Shape memory alloys
NT: Dielectric elastomer

actuators

Electrostatic actuators

Electrothermal actuators

Hydraulic actuators

Intelligent actuators

Microactuators

Piezoelectric actuators

Pneumatic actuators

Acupuncture

BT: Medical treatment

Acute respiratory distress syndrome

UF: ARDS
BT: Medical conditions
RT: Lungs



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Ad hoc networks

BT: Computer networks
 RT: Cross layer design
 Data communication
 Land mobile radio
 Mobile computing
 Multicast communication
 Protocols
 Wireless LAN
 Wireless sensor networks
 NT: AODV
 Mesh networks
 Mobile ad hoc networks
 Vehicular ad hoc networks

Ad hoc On Demand Distance Vector

USE: AODV

Ad hoc vehicle networks

USE: Vehicular ad hoc networks

Ad hoc vehicular networks

USE: Vehicular ad hoc networks

Adaptation models

BT: Adaptive algorithms

Adaptive algorithms

BT: Algorithms
 NT: Adaptation models

Adaptive antenna arrays

USE: Adaptive arrays

Adaptive arrays

UF: Adaptive antenna arrays
 BT: Antenna arrays
 RT: Adaptive signal detection
 Array signal processing
 Radar countermeasures
 Radio communication

countermeasures

Adaptive codes

USE: Adaptive coding

Adaptive coding

UF: Adaptive codes
 BT: Data compression

Adaptive control

UF: Self-tuning regulators
 BT: Adaptive systems
 RT: Cognitive systems
 Control systems

Disturbance observers
 Iterative learning control

Adaptive equalisers

USE: Adaptive equalizers

Adaptive equalizers

UF: Adaptive equalisers
 BT: Equalizers

Adaptive estimation

BT: Statistics

Adaptive filters

BT: Adaptive signal processing

Adaptive learning

BT: Education
 RT: Distance learning
 Human computer

interaction

User interfaces

Adaptive mesh refinement

BT: Numerical analysis

Adaptive optics

BT: Optics

Adaptive scheduling

BT: Scheduling
 RT: Adaptive systems
 Production control

Adaptive signal detection

BT: Adaptive signal processing
 RT: Adaptive arrays
 Blind source separation
 Source separation

Adaptive signal processing

BT: Signal processing
 NT: Adaptive filters
 Adaptive signal detection

Adaptive systems

BT: Cybernetics
 Systems engineering and

theory

RT: Adaptive scheduling
 Extensibility
 Learning systems
 Neural networks
 NT: Adaptive control
 Cognitive radar



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Line enhancers
Multi-agent systems
Variable structure systems

ADAS
USE: Advanced driver assistance systems

Add-drop multiplexers
BT: Multiplexing equipment
NT: Optical add-drop multiplexers

Added delay
BT: Delay systems

Adders
BT: Circuits
RT: Digital integrated circuits
Logic circuits

Addiction
UF: Substance abuse
BT: Medical conditions

Additive manufacturing
USE: Three-dimensional printing

Additive metric
USE: Maximum likelihood detection

Additive noise
BT: Noise
NT: AWGN
Additive white noise

Additive white noise
BT: Additive noise
RT: Gaussian noise

Additives
UF: Fuel additives
BT: Materials
RT: Production materials

Adenoviridae
USE: Adenoviruses

Adenoviruses
UF: Adenoviridae
BT: Microorganisms

Adhesive bonding
USE: Adhesives

Adhesive strength
BT: Materials testing

Adhesives
UF: Adhesive bonding
BT: Bonding
NT: Conductive adhesives
Nonconductive adhesives

Adiabatic processes
BT: Thermodynamics

Adjacent channel interference
USE: Interchannel interference

Admission control
BT: Quality of service
RT: Bandwidth

Admittance
UF: Electric admittance
BT: Electric variables
RT: Admittance measurement
Impedance
NT: Admittance control

Admittance control
BT: Admittance
Control systems
RT: Human-robot interaction
Position control
Torque control

Admittance measurement
BT: Electric variables
measurement
RT: Admittance
Impedance measurement

Adrenal glands
BT: Glands

Adsorption
BT: Surface morphology
RT: Interface phenomena
Molecular sieves
Surfactants

Advanced driver assistance systems
UF: ADAS
BT: Traffic control
Vehicle safety
RT: Automotive electronics
Collision avoidance
Computer vision



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systems
Driver behavior
Intelligent transportation
Intelligent vehicles
Mobile robots
Object detection
Steering systems
Vehicle-to-everything

Advanced renewable tariff
USE: Feed in tariff

Advanced Research Projects Agency Network
USE: ARPANET

Advanced TV
USE: HDTV

Advanced video coding
USE: Video coding

Adversarial learning
USE: Adversarial machine
learning

Adversarial machine learning
UF: Adversarial learning
Adversarial metric learning
BT: Machine learning

Adversarial metric learning
USE: Adversarial machine
learning

Advertising
BT: Marketing management

Aerial computing
BT: Computational modeling
RT: 5G mobile communication
6G mobile communication
Autonomous aerial vehicles
Data analysis
Edge computing
Internet of Things
Radio access networks

Aerial photography
USE: Satellite images

Aerial robots
USE: Autonomous aerial vehicles

Aero-engines
USE: Aircraft propulsion

Aerodynamics
BT: Dynamics
Mechanical factors
RT: Aerospace control
Flight dynamics
Shock waves
Wind tunnels

Aeroengines
USE: Aircraft propulsion

Aeroponics
BT: Agriculture

Aerosols
BT: Electrostatic processes
RT: Liquids
Particle production
Spraying

Aerospace accidents
BT: Accidents
RT: Aerospace safety
Space vehicles
NT: Air accidents

Aerospace and electronic systems
RT: Auditory displays
Digital signal processing
Programming
Systems engineering and

theory
NT: Aerospace control
Aerospace engineering
Aerospace materials
Aircraft manufacture
Aircraft navigation
Aircraft propulsion
Command and control

systems
Electronic warfare
Military equipment
Sensor systems
Sonar
Telemetry

Aerospace biophysics
BT: Aerospace engineering
Biophysics
RT: Human factors

Aerospace components
BT: Aerospace materials



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Aerospace control

UF: Aircraft control
Flight control
BT: Aerospace and electronic
systems
RT: Aerodynamics
Aerospace simulation
Aircraft
Hardware-in-the-loop
simulation
Hypersonic vehicles
Military systems
Missiles
Motion control
Space vehicles
Velocity control
NT: Air traffic control
Attitude control
Ground support

Aerospace electronics

UF: Aerospace instrumentation
Aeronautical engineering
Aircraft electronics
Aircraft instrumentation
Avionics
Space vehicle electronics
Space vehicle
instrumentation
BT: Aerospace engineering
RT: Aircraft
Gamma-ray telescopes
Space vehicles
Total ionizing dose
X-ray telescopes

Aerospace engineering

BT: Aerospace and electronic
systems
RT: Aerospace industry
Aerospace materials
Lightweight structures
NT: Aerospace biophysics
Aerospace electronics
Aerospace safety
Aerospace simulation
Aerospace testing
Artificial satellites
Flight dynamics
Space technology

Aerospace ground equipment

USE: Ground support

Aerospace ground services

USE: Ground support

Aerospace industry

BT: Manufacturing industries
RT: Aerospace engineering
Aerospace materials
Aerospace safety
Aircraft manufacture
Lightweight structures

Aerospace instrumentation

USE: Aerospace electronics

Aerospace materials

UF: Aircraft materials
Spacecraft materials
BT: Aerospace and electronic
systems
Production materials
RT: Aerospace engineering
Aerospace industry
Aircraft manufacture
Lightweight structures
Space vehicles
NT: Aerospace components

Aerospace navigation

USE: Aircraft navigation

Aerospace propulsion

BT: Propulsion

Aerospace safety

BT: Aerospace engineering
Safety
RT: Aerospace accidents
Aerospace industry
Ion beam effects
Space shuttles
Space vehicles
NT: Air safety

Aerospace simulation

UF: Flight simulation
BT: Aerospace engineering
RT: Aerospace control
Aerospace testing
Wind tunnels

Aerospace testing

BT: Aerospace engineering
Testing
RT: Aerospace simulation
NT: Wind tunnels



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Aeurnautical engineering

USE: Aerospace electronics

Affective computing

BT: Artificial intelligence
Human computer

interaction

RT: Behavioral sciences
Cognitive systems
Emotion recognition
Human factors
Psychology
User experience

Affinity propagation

BT: Clustering algorithms

Affordances

BT: Object recognition
Optimization methods
RT: Interactive systems
User interfaces
Virtual reality

Afforestation

BT: Forestry

Africa

BT: Continents

Afterburners

USE: Incineration

Ag

USE: Silver

Age factors

USE: Aging

Age of information

USE: Information age

Aged

USE: Aging

Ageing

USE: Aging

Agent-based modeling

BT: Computational modeling
Software agents
RT: Multi-agent systems

Agentic AI

UF: Agentic artificial intelligence
BT: Artificial intelligence
RT: Autonomous systems

Agentic artificial intelligence

USE: Agentic AI

Aggregate concrete

UF: Concrete aggregate
BT: Concrete

Aggregate signature schemes

BT: Communication system
security

Aggregates

BT: Materials
RT: Building materials

AGI

USE: Artificial general intelligence

Agile computing

USE: Agile software development

Agile manufacturing

BT: Manufacturing systems
RT: Computer integrated
manufacturing
Flexible manufacturing
systems

Agile project management

BT: Project management
Software development
management
RT: Scrum (Software
development)

Agile software development

UF: Agile computing
BT: Software development
management
RT: Hackathon
NT: Scrum (Software
development)

Aging

UF: Age factors
Aged
Ageing
BT: Materials science and
technology
RT: Alzheimer's disease



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		Ambient assisted living		Farming
		Assisted living		Food products
		Cataracts		Food security
		Electric breakdown		Irrigation
		Energy storage		Pesticides
		Gerontology	NT:	Cotton
		Insulation life		Crops
		Life estimation		Dairy products
		Older adults		Sugar
		Reliability		Wool
	NT:	Accelerated aging		
<i>Agri-IoT</i>			Agricultural robots	
	USE:	Agriculture Internet of	BT:	Agricultural machinery
Things			RT:	Robots
				Agricultural engineering
				Agricultural products
				Mobile robots
Agribusiness				
	UF:	Agrobusiness		
	BT:	Agriculture	Agriculture	
		Business	BT:	Industries
	RT:	Agricultural products	RT:	Agricultural engineering
		Crop yield		Agricultural machinery
		Forestry		Animals
		Smart agriculture		Crop yield
				Dairy products
<i>Agrichemicals</i>				Farming
	USE:	Agrochemicals		Food waste
				Genetic engineering
Agricultural engineering				Livestock
	BT:	Engineering - general		Pest control
	RT:	Agricultural machinery		Seeds (agriculture)
		Agricultural robots		Soil pollution
		Agriculture		Vegetation mapping
		Farming	NT:	Aeroponics
	NT:	Deforestation		Agribusiness
				Agricultural pollution
				Agricultural products
				Agriculture Internet of
Agricultural machinery				
	UF:	Combine harvesters	Things	Agrochemicals
		Tractors		Aquaculture
	BT:	Machinery		Greenhouses
	RT:	Agricultural engineering		Irrigation
		Agriculture		Precision agriculture
		Applicators		Regenerative agriculture
		Blades		Smart agriculture
	NT:	Agricultural robots		Urban agriculture
Agricultural pollution				
	BT:	Agriculture		
		Pollution	Agriculture Internet of Things	
	RT:	Agrochemicals	UF:	Agri-IoT
			BT:	Agriculture
Agricultural products				Internet of Things
	BT:	Agriculture		
	RT:	Agribusiness	<i>Agrobusiness</i>	
		Agricultural robots	USE:	Agribusiness



January 2026 IEEE Thesaurus

Agrochemicals

UF: Agrichemicals
BT: Agriculture
Chemical products
RT: Agricultural pollution
NT: Fertilizers
Pesticides

AI

USE: Artificial intelligence

AI accelerators

UF: AI chips
Artificial intelligence chips
BT: Artificial intelligence
Microprocessor chips
RT: Application specific

integrated circuits

Coprocessors
Field programmable gate

arrays

Graphics processing units
Learning (artificial

intelligence)

Multiprocessing systems
Neural network hardware
Neural networks
Neuromorphic engineering
System-on-chip

AI chips

USE: AI accelerators

AI generated content

USE: Generative AI

AI technologies

USE: Artificial intelligence

AI-generated content

USE: Generative AI

AIDS

USE: Acquired immune
deficiency syndrome

Aids for the handicapped

USE: Assistive technologies

AIEE Standards

BT: IEEE Standards

Air accidents

BT: Aerospace accidents
RT: Air safety

Air traffic control

Air bags

USE: Automotive components

Air cleaners

UF: Air filters
Air purifiers
BT: Machine components
RT: Air pollution
Cleaning
Purification

Air conditioning

BT: Cooling
RT: Building services
Buildings
Compressors
Ducts
Fans
HVAC
Ventilation
Vents
NT: Central air conditioning

Air filters

USE: Air cleaners

Air gaps

UF: Air-gap
Airgaps
BT: Electromagnetic analysis
RT: Electrodes
Spark gaps

Air interface

USE: Communication channels

Air pollutants

USE: Air pollution

Air pollution

UF: Air pollutants
BT: Air quality
Pollution
RT: Air cleaners
Ash
Atmospheric

measurements

Carbon footprint
Carbon sequestration
Exhaust gases
Flue gases
Fossil fuels
Global warming



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	Incineration		Airline industry
	Industrial pollution		Global Positioning System
	Meteorology	NT:	Aircraft
	Natural gas		Airports
	Thermal pollution		
NT:	Air pollution control	<i>Air-gap</i>	
		USE:	Air gaps
Air pollution control		<i>Air-ground communication</i>	
BT:	Air pollution	USE:	Air to ground
	Atmospheric	communication	
measurements		<i>Air-to-ground communication</i>	
RT:	Carbon emissions	USE:	Air to ground
<i>Air purifiers</i>		communication	
USE:	Air cleaners		
Air quality		Airborne radar	
BT:	Atmosphere	BT:	Radar
RT:	Atmospheric	RT:	Synthetic aperture radar
measurements			
	Environmental factors	Aircraft	
NT:	Air pollution	BT:	Air transportation
	Indoor air quality	RT:	Aerospace control
			Aerospace electronics
Air safety			Aircraft manufacture
BT:	Aerospace safety		Aircraft navigation
RT:	Air accidents		Aircraft propulsion
			Airline industry
Air to ground communication			Ground support
UF:	A2G		Military aircraft
	Air-ground communication		Propellers
	Air-to-ground	NT:	Airplanes
communication			Vertical takeoff and landing
BT:	Communication systems	aircraft	
RT:	Autonomous aerial vehicles	<i>Aircraft control</i>	
	Global Positioning System	USE:	Aerospace control
	Military communication		
	Surveillance	<i>Aircraft electronics</i>	
Air traffic control		USE:	Aerospace electronics
UF:	Air traffic management		
BT:	Aerospace control	<i>Aircraft engines</i>	
RT:	Air accidents	USE:	Aircraft propulsion
	Air transportation		
	Control systems	<i>Aircraft instrumentation</i>	
	Radio navigation	USE:	Aerospace electronics
	Urban air mobility		
	Vertiports	Aircraft manufacture	
<i>Air traffic management</i>		BT:	Aerospace and electronic
USE:	Air traffic control	systems	
Air transportation		RT:	Aerospace industry
BT:	Transportation		Aerospace materials
RT:	Air traffic control		Aircraft



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Aircraft materials

USE: Aerospace materials

Aircraft navigation

UF: Aerospace navigation
Entry, descent and landing

BT: Aerospace and electronic

systems

RT: Navigation

RT: Aircraft

Course correction

Vertiports

NT: Instrument landing systems

Urban air mobility

Aircraft propulsion

UF: Aero-engines

Aeroengines

Aircraft engines

BT: Aerospace and electronic

systems

RT: Propulsion

RT: Aircraft

Engines

Jet engines

Turbines

NT: Propellers

Airfields

USE: Airports

Airfoils

USE: Automotive components

Airgaps

USE: Air gaps

Airline industry

BT: Industries

RT: Air transportation

Aircraft

Airports

Airplanes

BT: Aircraft

Airports

UF: Airfields

BT: Air transportation

RT: Airline industry

Freeports

NT: Vertiports

Al

USE: Aluminum

Al₂O₃

USE: Aluminum oxide

Alarm systems

UF: Warning systems

BT: Security

RT: Fall detection

Monitoring

Motion detection

Safety

Safety devices

NT: Smoke detectors

Alcoholic beverages

BT: Ethanol

RT: Fermentation

Alcoholism

BT: Diseases

Alexa

USE: Virtual assistants

Algae

BT: Organisms

RT: Eutrophication

NT: Seaweed

Algebra

BT: Mathematics

RT: Lie groups

Nonlinear equations

NT: Abstract algebra

Boolean algebra

Linear algebra

Set theory

Algorithm design and analysis

BT: Algorithms

RT: Fish schools

NT: Algorithmic efficiency

Computational statistics

Generative adversarial

networks

Model compression

Algorithm design and theory

BT: Algorithms

NT: Backtracking

Consensus algorithm

Flower pollination algorithm



January 2026 IEEE Thesaurus

Algorithmic efficiency

BT: Algorithm design and
analysis
RT: Computational complexity
Software performance
Software quality
NT: Metaheuristics

algorithms

detection

Machine learning

Matching pursuit algorithms
Maximum likelihood

Multicast algorithms
Parallel algorithms
Particle swarm optimization
Partitioning algorithms
Prediction algorithms
Procedural generation
Projection algorithms
Pursuit algorithms
Signal processing

Algorithms

UF: Subroutines
BT: Mathematics
RT: Biometrics
Ciphers
Cyclic redundancy check
Huffman coding
Linear programming
Maximum likelihood

algorithms

Software algorithms
Viterbi algorithm
Whale optimization

decoding

Model checking
Numerical stability
Random processes
Software
Software libraries
Stability analysis
NT: Adaptive algorithms
Algorithm design and

algorithms

All optical networks

USE: All-optical networks

All-optical networks

UF: All optical networks
BT: Optical fiber networks

analysis

Algorithm design and

Allergies

BT: Diseases
Immune system
RT: Medical conditions

theory

Approximation algorithms
Artificial bee colony

algorithm

Backpropagation algorithms
Basis algorithms
Bees algorithm
Change detection

Allocation

USE: Resource management

algorithms

Classification algorithms
Clustering algorithms
Compression algorithms
Density estimation robust

Allow lists

USE: Accesslists

algorithm

Detection algorithms
Distributed algorithms
Dynamic programming
Filtering algorithms
Genetic algorithms
Glowworm swarm

Alloying

BT: Metals
RT: Aluminum alloys
Aluminum compounds
Barium compounds
Bismuth compounds
Calcium
Cobalt
Cobalt alloys
Copper alloys
Gallium alloys
Gallium compounds
Germanium alloys
Gold alloys
Hafnium compounds
Indium compounds
Iron alloys

optimization

Hash functions
Heuristic algorithms
Inference algorithms
Krill herd algorithm
MLFMA



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		Lithium		<i>Aluminium alloys</i>	
		Lithium compounds		USE:	Aluminum alloys
		Neodymium alloys		<i>Aluminium compounds</i>	
		Nickel alloys		USE:	Aluminum compounds
		Niobium alloys		<i>Aluminium industry</i>	
		Platinum alloys		USE:	Metals industry
		Silicon alloys		<i>Aluminium oxide</i>	
		Strontium compounds		USE:	Aluminum oxide
		Tin alloys		Aluminum	
		Titanium alloys		UF:	Al
		Yttrium compounds		BT:	Aluminium
NT:		Intermetallic		BT:	Chemical elements
		Shape memory alloys		NT:	Metals
					Aluminum alloys
<i>Alloys</i>					Aluminum compounds
	USE:	Metals		Aluminum alloys	
Alpha particles				UF:	Aluminium alloys
BT:		Nuclear physics		BT:	Aluminum
RT:		Ions		RT:	Alloying
<i>Alphavoltaic power sources</i>				Aluminum compounds	
USE:		Radioactive materials		UF:	Aluminium compounds
<i>Alternating current generators</i>				BT:	Aluminum
USE:		AC generators		RT:	Alloying
<i>Alternating current machines</i>				NT:	Aluminum gallium nitride
USE:		AC machines			Aluminum nitride
<i>Alternating current motors</i>					Aluminum oxide
USE:		AC motors		Aluminum gallium nitride	
<i>Alternating current power transmission</i>				BT:	Aluminum compounds
USE:		AC power transmission			Gallium compounds
<i>Alternative energy resources</i>				RT:	III-V semiconductor
USE:		Renewable energy sources		materials	Transistors
<i>Alternative energy sources</i>				<i>Aluminum industry</i>	
USE:		Renewable energy sources		USE:	Metals industry
Alternators				Aluminum nitride	
BT:		Electric machines		BT:	Aluminum compounds
RT:		Synchronous generators		Aluminum oxide	
Altimetry				UF:	Al ₂ O ₃
BT:		Pressure measurement			Aluminium oxide
RT:		Atmospheric		BT:	Aluminum compounds
measurements				RT:	Ceramics
<i>Aluminium</i>				Alzheimer's disease	
USE:		Aluminum		BT:	Dementia
					Diseases



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RT:	Aging Gerontology Hippocampus Older adults	Amino acids	BT:	Biochemistry
Ambient assisted living		<i>AML</i>	USE:	Automated machine learning
UF:	AAL	<i>AMLCDs</i>	USE:	Active matrix liquid crystal displays
BT:	Assisted living Information and communication technology	Ammeters	BT:	Electric variables measurement
RT:	Aging Assistive devices	RT:	Current measurement	
Ambient intelligence		Ammonia	BT:	Nitrogen compounds
BT:	Consumer electronics Telecommunications	Amniocentesis	BT:	Medical tests
RT:	Intelligent systems Internet of Things Ubiquitous computing User interfaces	RT:	Genetics Ultrasonic imaging	
Ambient networks		NT:	Amniotic fluid Birth disorders	
UF:	AN Project	Amniotic fluid	BT:	Amniocentesis Fluids and secretions
BT:	Mobile communication	<i>AMOLEDs</i>	USE:	Active matrix organic light emitting diodes
RT:	3G mobile communication	Amorphous magnetic materials	BT:	Magnetic materials
Ambisonics		Amorphous materials	BT:	Materials
BT:	Audio systems	NT:	Diamond-like carbon Glass	
RT:	3D audio Immersive audio Spatial audio	Amorphous semiconductors	BT:	Semiconductor materials
<i>Amblyopia</i>		RT:	Silicon Thin film devices	
USE:	Vision defects	Amorphous silicon	BT:	Silicon
Ambulatory surgery		Amperometric sensors	BT:	Electrochemical devices Gas detectors
BT:	Surgery			
<i>American Express</i>				
USE:	Credit cards			
<i>American National Institute of Standards</i>				
USE:	ANSI			
<i>American Standard Code for Information Interchange</i>				
USE:	ASCII			
American Standards Association				
UF:	ASA			
BT:	Standards organizations			
RT:	ANSI			
Americium				
BT:	Chemical elements			



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Amphibious robots

BT: Marine vehicles
Robots
RT: Autonomous aerial vehicles
Mobile robots

Amplifiers

BT: Signal processing
RT: Frequency response
Klystrons
Optical fiber amplifiers
Rail to rail amplifiers
Rail to rail operation
NT: Broadband amplifiers
Differential amplifiers
Distributed amplifiers
Low-noise amplifiers
Operational amplifiers
Power amplifiers
Preamplifiers
Pulse amplifiers
Radiofrequency amplifiers
Resonators

Amplify-and-forward cooperative communication

USE: Cooperative communication

Amplitude estimation

BT: Parameter estimation
RT: Reflection coefficient

Amplitude modulation

BT: Modulation
RT: Demodulation
Intensity modulation
NT: Amplitude shift keying
Quadrature amplitude

modulation

Amplitude shift keying

UF: ASK
BT: Amplitude modulation

Amputation

BT: Injuries
Medical treatment

Amygdala

UF: Amygdalae
Corpus amygdaloideum
BT: Brain

Amygdalae

USE: Amygdala

AN Project

USE: Ambient networks

Anaemia

USE: Anemia

Anaerobic digestion

BT: Biological processes
Microorganisms
RT: Biodegradation
Biogas
Green energy
Methane
Renewable energy sources
Smart agriculture
Soil remediation
Sustainable development
Waste management

Anaesthesia

USE: Anesthesia

Analog circuits

BT: Circuits
RT: Microwave circuits
Millimeter wave circuits
Neuromorphics
Submillimeter wave circuits
Switched capacitor

networks

UHF circuits
VHF circuits
NT: Analog integrated circuits
Analog processing circuits

Analog CMOS integrated circuits

USE: CMOS analog integrated circuits

Analog computers

UF: Analogue computers
BT: Computers
RT: Summing circuits

Analog digital integrated circuits

USE: Analog-digital integrated circuits

Analog integrated circuits

UF: Analogue integrated circuits
Linear integrated circuits
BT: Analog circuits
Integrated circuits
RT: Analog processing circuits
MMICs



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circuits	Microwave integrated	Digital transformation
		Digitization
circuits	Millimeter wave integrated	
	Neural network hardware	Analog-digital integrated circuits
	Submillimeter wave	UF: Analog digital integrated
integrated circuits		circuits
	UHF integrated circuits	Analogue digital integrated
NT: CMOS analog integrated		circuits
circuits		Analogue-digital integrated
	Field programmable analog	circuits
arrays		Minimum analog-digital
		integrated circuits
Analog memory		BT: Integrated circuits
BT: Memory		NT: Mixed analog-digital
RT: Analog processing circuits		integrated circuits
		<i>Analog-to-digital conversion</i>
Analog processing circuits		USE: Analog-digital conversion
BT: Analog circuits		<i>Analog-to-digital converter</i>
RT: Analog integrated circuits		USE: AC-DC power converters
	Analog memory	AND
	Application specific	Analog-digital conversion
integrated circuits		
	Mixed analog-digital	<i>Analog-to-digital convertor</i>
integrated circuits		USE: AC-DC power converters
	Signal processing	
		<i>Analogue CMOS integrated circuits</i>
<i>Analog to digital conversion</i>		USE: CMOS analog integrated
USE: Analog-digital conversion		circuits
<i>Analog to digital converter</i>		
USE: Analog-digital conversion		<i>Analogue computers</i>
		USE: Analog computers
Analog TV		
UF: Analogue TV		<i>Analogue digital integrated circuits</i>
BT: TV		USE: Analog-digital integrated
		circuits
<i>Analog-digital</i>		
USE: Analog-digital conversion		<i>Analogue integrated circuits</i>
		USE: Analog integrated circuits
Analog-digital conversion		
UF: A/D		<i>Analogue TV</i>
	A/D conversion	USE: Analog TV
	A/D converter	
	Analog to digital conversion	<i>Analogue-digital conversion</i>
	Analog to digital converter	USE: Analog-digital conversion
	Analog-digital	
	Analog-to-digital conversion	<i>Analogue-digital converters</i>
	Analog-to-digital converter	USE: Analog-digital conversion
	Analogue-digital conversion	
	Analogue-digital converters	<i>Analogue-digital integrated circuits</i>
BT: Data conversion		USE: Analog-digital integrated
RT: Data acquisition		circuits
	Quantization (signal)	
NT: Delta modulation		



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Analysis of variance

UF: ANOVA
BT: Statistical analysis

Analytic hierarchy process

UF: Analytical hierarchy process
BT: Decision making
RT: Management
Strategic planning

Analytical hierarchy process

USE: Analytic hierarchy process

Analytical models

BT: Modeling
RT: Neuroinformatics
NT: Common Information Model
(computing)

Anatomical structure

BT: Medical diagnostic imaging

Anatomy

BT: Biological systems
NT: Auditory system
Biological tissues
Body regions
Cardiovascular system
Circulatory system
Digestive system
Embryonic structures
Endocrine system
Fluids and secretions
Human anatomy
Immune system
Integumentary system
Lymphatic system
Musculoskeletal system
Nervous system
Neuroanatomy
Respiratory system
Sense organs
Stomatognathic system
Urogenital system

Android (operating system)

USE: Operating systems

Androids

BT: Robots
RT: Human factors
Human-machine systems

Anechoic chambers

BT: Test facilities
RT: Acoustic measurements
Antenna measurements
Electromagnetic measurements
Immunity testing
TEM cells

Anemia

UF: Anaemia
BT: Medical conditions
RT: Blood flow
Red blood cells

Anemometers

USE: Fluid flow measurement

Anesthesia

UF: Anaesthesia
BT: Medical treatment
NT: Anesthetic drugs

Anesthesiology

BT: Medical specialties

Anesthetic drugs

BT: Anesthesia

Aneurism

USE: Aneurysm

Aneurysm

UF: Aneurism
BT: Medical conditions

Angiocardiology

BT: Biomedical imaging
RT: Biomedical applications of radiation

Angiography

UF: Arteriography
BT: Biomedical imaging

Angioplasty

BT: Medical treatment

Angiosperms

USE: Flowering plants

Angular velocity

BT: Mechanical variables
measurement
RT: Velocity control



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Velocity measurement

Angular velocity control

BT: Velocity control

Animal behavior

BT: Behavioral sciences

Animal structures

BT: Animals

NT: Beak

Feathers

Tail

Animals

BT: Organisms

Zoology

RT: Agriculture

Biological systems

Life sciences

NT: Animal structures

Birds

Bovine

Cats

Dinosaurs

Dogs

Endangered species

Horses

Insects

Livestock

Marine animals

Mice

Rabbits

Rats

Rodents

Wildlife

Animation

UF: Computer animation

BT: Graphics

RT: Computer graphics

Motion capture

Text to video

Visual effects

Visualization

NT: Facial animation

Animatronics

BT: Robotics and automation

Anisotropic

BT: Filters

Anisotropic conductive films

BT: Conductive films

Anisotropic diffusion

USE: Anisotropic
magnetoresistance

Anisotropic effects

USE: Anisotropic
magnetoresistance

Anisotropic magnetoresistance

UF: Anisotropic diffusion

Anisotropic effects

Anisotropic

magnetoresistance sensors

Anisotropic material

Anisotropic processing

Anisotropically

Anisotropy

BT: Magnetoresistance

Anisotropic magnetoresistance sensors

USE: Anisotropic
magnetoresistance

Anisotropic material

USE: Anisotropic
magnetoresistance

Anisotropic processing

USE: Anisotropic
magnetoresistance

Anisotropically

USE: Anisotropic
magnetoresistance

Anisotropy

USE: Anisotropic
magnetoresistance

Ankle

BT: Joints

RT: Foot

Legs

Annealing

UF: Annealing temperature

BT: Heat treatment

Materials processing

RT: Quantum annealing

Simulated annealing

Softening

Thermal factors

NT: Rapid thermal annealing



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Annealing temperature

USE: Annealing

Annotations

BT: Metadata
RT: Text analysis

Anodes

BT: Electrodes
RT: Electron tubes

Anomaly detection

UF: Novelty detection
Outlier detection
BT: Data mining

Anorexia

USE: Eating disorders

Anorexia nervosa

USE: Eating disorders

ANOVA

USE: Analysis of variance

ANPR

USE: License plate recognition

ANSI

UF: American National Institute
of Standards
BT: Standards organizations
RT: American Standards
Association

ANSI Standards

BT: Standards publications
RT: ASA Standards
IEEE Standards
ISO Standards
NT: National Electric Code

Answer set programming

BT: Programming
RT: Knowledge representation

Ant colony optimization

BT: Probability
Swarm intelligence
RT: Graph theory
Metaheuristics

Antarctic Ocean

UF: Southern Ocean
BT: Oceans

Antarctica

BT: Geoscience
NT: South Pole

Antenna accessories

UF: Antenna components
BT: Antennas
NT: Radomes

Antenna arrays

UF: Distributed antennas
BT: Antennas
RT: Broadband antennas
SIMO
SISO
NT: Adaptive arrays
Butler matrices
Linear antenna arrays
Log periodic antennas
Microstrip antenna arrays
Microwave antenna arrays
Phased arrays
Planar arrays

Antenna components

USE: Antenna accessories

Antenna diversity

USE: Spatial diversity

Antenna feeds

BT: Feeds
RT: Antennas
Aperture coupled antennas

Antenna measurements

BT: Measurement
RT: Anechoic chambers
Antennas
Electromagnetic

measurements

Antenna phased arrays

USE: Phased arrays

Antenna radiation patterns

UF: Radiation pattern
BT: Antennas
RT: Antenna theory
NT: Near-field radiation pattern

Antenna theory

BT: Antennas
RT: Antenna radiation patterns
Current distribution



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surfaces	NT:	Mode matching methods Frequency selective	Slot antennas Steerable antennas Textile antennas Transmission line antennas Transmitting antennas UHF antennas Wearable antennas Yagi-Uda antennas
	BT:	Antennas Electronics packaging System-in-package	
Antenna-in-package			
Antennas	BT:	Antennas and propagation	
	RT:	Antenna feeds Antenna measurements Beam steering Butler matrices Fractals IEEE 802.11n Standard Microstrip antenna arrays Radio communication	Antennas and propagation RT: Communication systems Communications technology NT: Signal processing Antennas Electromagnetic propagation Radio astronomy
equipment	NT:	Spatial diversity Waveguide theory Antenna accessories Antenna arrays Antenna radiation patterns Antenna theory Antenna-in-package Apertures Broadband antennas Dielectric resonator	Anthropology BT: Social sciences RT: Behavioral sciences Cultural differences Humanities Linguistics NT: Ethnicity
			Anthropometry BT: Measurement RT: Biomechanics Biomedical measurement Ergonomics Human factors
antennas		Dipole antennas Directional antennas Directive antennas Feeds Fractal antennas Helical antennas Horn antennas Leaky wave antennas Loaded antennas Log-periodic dipole	Anthropomorphism BT: Human factors
		Loop antennas Microstrip antennas Microwave antennas Mobile antennas Multifrequency antennas Omnidirectional antennas Optical antennas Patch antennas Radar antennas Receiving antennas Rectennas Reflector antennas Satellite antennas	<i>Anti freeze</i> USE: Anti-freeze <i>Anti virus software</i> USE: Anti-virus software <i>Anti-bacterial</i> USE: Antibacterial activity <i>Anti-biotics</i> USE: Antibiotics Anti-freeze UF: Anti freeze Antifreeze BT: Chemical compounds RT: Methanol



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Anti-fungal

UF: Antifungal
BT: Antibiotics

Anti-parasitical

UF: Antiparasitical
BT: Antibiotics

Anti-reflective coatings

USE: Antireflection coatings

Anti-virus software

UF: Anti virus software
BT: Software
RT: Computer viruses
Malware
Security

Antibacterial

USE: Antibacterial activity

Antibacterial activity

UF: Anti-bacterial
Antibacterial
BT: Antibiotics
RT: Bacteria

Antibiotics

UF: Anti-biotics
BT: Drugs
NT: Anti-fungal
Anti-parasitical
Antibacterial activity

Antibodies

UF: Immunoglobulin
BT: Immune system
RT: Microorganisms
NT: Monoclonal antibodies

Anticoagulants

BT: Drugs
RT: Blood

Antidepressants

BT: Drugs

Antiderivatives

USE: Integral equations

Antiferroelectric materials

USE: Dielectric materials

Antiferromagnetic materials

BT: Magnetic materials

RT: Antiferromagnetic
resonance

Antiferromagnetic resonance

BT: Magnetic resonance
RT: Antiferromagnetic materials

Antifreeze

USE: Anti-freeze

Antifreeze materials

USE: Coolants

Antifungal

USE: Anti-fungal

Antigens

BT: Immunology
RT: Immune response
Immunofluorescence

Antimicrobial resistance

BT: Drug resistance
RT: Antimicrobials

Antimicrobials

BT: Microbiology
RT: Antimicrobial resistance

Antimony

BT: Chemical elements

Antioxidants

BT: Chemical products
RT: Oxidation

Antiparasitical

USE: Anti-parasitical

Antireflection coatings

UF: Anti-reflective coatings
Antireflective coatings
BT: Coatings
RT: Optical reflection

Antireflective coatings

USE: Antireflection coatings

Anxiety disorders

BT: Mental disorders
RT: Emotion recognition
Medical conditions
Mental health
Sentiment analysis



January 2026 IEEE Thesaurus

AODV

UF: Ad hoc On Demand
Distance Vector
BT: Ad hoc networks
Wireless networks

AOI

USE: Information age

Aortic semilunar valves

USE: Heart valves

Apache hadoop

USE: Cluster computing

Apache spark

USE: Cluster computing

APCVD

USE: Atmospheric pressure
chemical vapor deposition

APDs

USE: Avalanche photodiodes

Aperture antennas

BT: Apertures
RT: Aperture coupled antennas
Reflector antennas

Aperture coupled antennas

BT: Apertures
RT: Antenna feeds
Aperture antennas
Microstrip antenna arrays
Microstrip antennas

Apertures

BT: Antennas
RT: Couplers
NT: Aperture antennas
Aperture coupled antennas

Aphasia

BT: Medical conditions
RT: Brain injuries
Patient rehabilitation
Speech
Stroke (medical condition)

API

USE: Application programming
interfaces

Apicology

UF: Melittology
BT: Entomology
RT: Insects

Appearance matching

USE: Image matching

Apple watch

USE: Wearable Health Monitoring
Systems

Appliances

USE: Home appliances

Application programming interfaces

UF: API
Mobile application
development
BT: Computer interfaces
RT: Software defined
networking
NT: Restful API
WebRTC

Application security

BT: Computer security

Application software

BT: Software
NT: Decentralized applications

Application specific integrated circuits

UF: ASIC
Custom integrated circuits
Semicustom integrated
circuits
BT: Circuits
Integrated circuits
RT: AI accelerators
Analog processing circuits
CMOS logic circuits
Field programmable analog
arrays
NT: System-on-chip

Application specific processors

BT: Program processors

Application virtualization

UF: Cross platform virtualization
Cross-platform virtualization
BT: Computer applications
RT: Emulation



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virtualization	Network function	BT:	Agriculture
		RT:	Marine animals
	Simulation	NT:	Coral reef
NT:	Edge computing		Eutrophication
			Fisheries
Applicators			
BT:	Production equipment	Aquatic ecosystems	
RT:	Agricultural machinery	UF:	Hydrosphere
	Labeling	BT:	Ecosystems
		RT:	Lakes
Appraisal			Ocean dynamics
BT:	Human resource		Ocean salinity
management			Oceans
RT:	Incentive schemes		Rivers
	Personnel		Sea floor
			Wetlands
Appropriate technology		NT:	Marine ecosystems
BT:	Technology		
RT:	Microhydro power	Aquatic robots	
	Picohydro power	UF:	Swimming robots
		BT:	Robots
Approximate computing			
BT:	Computers and information	<i>Aquatic vehicles</i>	
processing		USE:	Underwater vehicles
Approximation algorithms			
BT:	Algorithms	Aquifers	
		BT:	Hydrology
Approximation error			Water resources
BT:	Approximation methods	RT:	Rocks
			Water storage
Approximation methods		<i>Ar</i>	
UF:	Approximation theory	USE:	Argon
BT:	Numerical analysis		
RT:	Least squares	Arc discharges	
approximations		UF:	Arc flash
	Minimization methods		Arc-flash
	Signal representation	BT:	Dielectric breakdown
NT:	Approximation error	RT:	Electrostatic discharges
	Chebyshev approximation		High intensity discharge
	Curve fitting	lamps	
	Extrapolation		Light sources
	Function approximation		Plasmas
	Interpolation		
	Linear approximation	<i>Arc flash</i>	
	Mean square error methods	USE:	Arc discharges
	Perturbation methods		
<i>Approximation theory</i>		<i>Arc lamps</i>	
USE:	Approximation methods	USE:	Lighting
<i>APS</i>		<i>Arc-flash</i>	
USE:	Automated parking	USE:	Arc discharges
Aquaculture		Archaea	
		BT:	Organisms



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Archaeology

USE: Archeology

Archeology

UF: Archaeology

BT: Humanities

Architectural description languages

USE: Architecture description languages

Architecture

BT: Industries

RT: Building information management

Buildings
Structural engineering

Architecture (computer)

USE: Computer architecture

Architecture description languages

UF: Architectural description languages

BT: Computer languages

Arctic

BT: Geoscience

NT: North Pole

Arctic Ocean

BT: Oceans

ARDS

USE: Acute respiratory distress syndrome

Area measurement

BT: Measurement

RT: Size measurement

Argon

UF: Ar

BT: Gases

Arithmetic

BT: Mathematics

NT: Digital arithmetic
Fixed-point arithmetic
Floating-point arithmetic

Arm

BT: Limbs

NT: Elbow
Wrist

Armature

BT: Electromechanical devices

Armpit

USE: Axilla

ARPANET

UF: Advanced Research

Projects Agency Network

DARPANET

BT: Communication systems

RT: Internet

Packet switching

ARQ

USE: Automatic repeat request

Array

USE: Arrays

Array processing

USE: Arrays AND
Parallel processing

Array signal processing

UF: Beamforming

BT: Signal processing

RT: Acoustic arrays
Acoustic transducers

Adaptive arrays

Blind source separation

Direction-of-arrival

estimation

Semidefinite programming

Signal resolution

Source separation

Time of arrival estimation

Arrayed waveguide gratings

UF: AWG device

BT: Optical waveguides

RT: Demultiplexing

Integrated optics

Multiplexing

Arrays

UF: Array

Array processing

BT: Data structures

NT: Sensor arrays



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Arresters

BT: Surge protection
RT: Power system protection
Power system transients
Varistors

Arrhythmia

BT: Medical conditions
RT: Cardiology
Heart beat

Arsenic

BT: Chemical elements
NT: Arsenic compounds

Arsenic compounds

UF: Arsenite
Arsine
BT: Arsenic

Arsenite

USE: Arsenic compounds

Arsine

USE: Arsenic compounds

Art

BT: Humanities
RT: Computer graphics
Graphics
Layout
Museums
Photorealism
NT: Digital art
Fractal art

Arterial blood circulation

BT: Arteries

Arterial blood pressure

BT: Arteries

Arterial occlusion

BT: Arteries

Arterial pressure

USE: Blood pressure

Arterial wall structures

USE: Arteries

Arterial walls

USE: Arteries

Arteries

UF: Arterial wall structures
Arterial walls
Artery
BT: Blood vessels
NT: Arterial blood circulation
Arterial blood pressure
Arterial occlusion
Carotid arteries

Arteriography

USE: Angiography

Arteriosclerosis

BT: Diseases
NT: Atherosclerosis
Coronary arteriosclerosis

Artery

USE: Arteries

Arthritis

BT: Diseases

Artificial bee colony algorithm

UF: ABC algorithms
BT: Algorithms
RT: Cooperative systems
Optimization
Particle swarm optimization
Search problems

Artificial biological organs

UF: Artificial organs
BT: Prosthetics
RT: Biological systems
Bioprinting
NT: Artificial heart
Artificial limbs
Artificial muscles

Artificial fibers

USE: Synthetic fibers

Artificial fibres

USE: Synthetic fibers

Artificial general intelligence

UF: AGI
BT: Artificial intelligence
RT: Intelligent systems

Artificial heart

BT: Artificial biological organs



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Artificial immune systems

BT: Immune system

Artificial intelligence

UF: AI
AI technologies
Open AI

BT: Computational and artificial

intelligence

RT: Autonomous vehicles
Computational intelligence
Data mining
Digital economy
Digital humans
Digital twins
Feedforward neural

networks

Few shot learning
Generative adversarial

networks

analysis

Independent component

Machine ethics
Minimax techniques
Model compression
Natural languages
Neural networks
Neurocontrollers
Pervasive computing
Posthuman
Prediction theory
Radial basis function

networks

Reinforcement learning
Robot learning
Semantic Web
Semisupervised learning
Software agents
Support vector machines
Synapses

NT: Text summarization
AI accelerators
Affective computing
Agentic AI
Artificial general intelligence
Autonomous robots
Bio-inspired computing
Cognitive systems
Commonsense reasoning
Context awareness
Context management
Conversational artificial

intelligence

Cooperative systems
Edge AI

Explainable AI
Foundation models
Generative AI
Intelligent systems
Knowledge based systems
Knowledge engineering
Large language models
Learning (artificial

intelligence)

Learning systems
Machine learning
Machine listening
Prediction methods
Retrieval augmented

generation

Swarm intelligence
Text to image
Text to video
Virtual artifact

Artificial intelligence chips

USE: AI accelerators

Artificial light

UF: Light pollution
BT: Lighting

Artificial limbs

BT: Artificial biological organs
Prosthetics
RT: Limbs

Artificial muscles

UF: Muscle-like actuators
BT: Artificial biological organs

Artificial neural networks

UF: Deep neural networks
BT: Neural networks
RT: Fish schools
Large language models
Mathematical models
Neural machine translation
Neuromorphic engineering
Synapses
NT: Autoencoders
Convolutional neural

networks

Hebbian theory
Long short term memory
Neural architecture search
Residual neural networks
Self-organizing feature

maps

Spiking neural networks



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Artificial organs

USE: Artificial biological organs

Artificial satellites

BT: Aerospace engineering
RT: Satellite communications
Satellites
Space technology
Space vehicles
NT: Earth Observing System
Low earth orbit satellites
Military satellites
Space stations

ASA

USE: American Standards Association

ASA Standards

BT: Standards publications
RT: ANSI Standards

ASCII

UF: American Standard Code for Information Interchange
BT: Encoding

Ash

BT: Industrial waste
RT: Air pollution
Exhaust gases
Incineration
Volcanic ash
NT: Fly ash

Asia

BT: Continents

ASIC

USE: Application specific integrated circuits

ASK

USE: Amplitude shift keying

Asphalt

UF: Bitumen
BT: Building materials

Aspheric lenses

USE: Aspherical optics

Aspheric mirrors

USE: Aspherical optics

Aspheric optics

USE: Aspherical optics

Aspherical lenses

USE: Aspherical optics

Aspherical mirrors

USE: Aspherical optics

Aspherical optics

UF: Aspheric lenses
Aspheric mirrors
Aspheric optics
Aspherical lenses
Aspherical mirrors
Optics
BT: Optics

Asphyxia

BT: Death

Aspirin

BT: Drugs

ASR

USE: Automatic speech recognition

Assemblers (program)

USE: Program processors

Assembly

BT: Manufacturing
RT: Assembly systems
Manipulators
Manufacturing automation
NT: Fitting
Microassembly
Preforms
Retrofit
Soldering

Assembly robots

USE: Robotic assembly

Assembly systems

BT: Industrial electronics
Manufacturing
Production systems
RT: Assembly
Fitting
Industrial control
Manipulators
Manufacturing automation
Mobile robots
Robots



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NT: Flexible electronics
Robotic assembly

Asset management

UF: Asset-management
BT: Management
NT: Public infrastructure

Asset-management

USE: Asset management

Assisted living

BT: Medical services
RT: Aging
Assistive robots
Fall detection
Geriatrics
Older adults
NT: Ambient assisted living

Assistive devices

BT: Assistive technologies
RT: Ambient assisted living

Assistive robots

UF: Rehabilitation robotics
Rehabilitation robots
BT: Assistive technologies
Medical robotics
Service robots
RT: Assisted living
Exoskeletons
Field robots
Geriatrics
Older adults
Patient monitoring
Patient rehabilitation
Social robots
Soft robotics

Assistive technologies

UF: Aids for the handicapped
Handicapped aids
BT: Biomedical equipment
RT: Braille
Communication aids
Gaze tracking
Gerontechnology
Medical control systems
Orthotics
People with disabilities
Prosthetics
Sensory aids
Sign language
Social robots

NT: Wearable robots
Assistive devices
Assistive robots
Closed captioning
Video description
Wheelchairs

Association rule learning

UF: Association rules
BT: Data mining
Machine learning

Association rules

USE: Association rule learning

Associative memory

UF: Content addressable
memory
BT: Memory
RT: Neural networks

Associative processing

BT: Data processing
RT: Computers and information
processing

Asteroids

BT: Planets
NT: Meteoroids

Asteroseismology

BT: Astrophysics
NT: Helioseismology

Asthma

BT: Pulmonary diseases
RT: Chronic obstructive
pulmonary disease
Emphysema
Pneumonia

Astrobiology

BT: Biology

Astrochemistry

UF: Planetary chemistry
BT: Chemistry

Astronomy

BT: Science - general
RT: Extraterrestrial
measurements
Gamma-ray detectors
Telescopes
NT: Astrophysics



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	Extrasolar planets Gravitational waves Observatories Radio astronomy Solar system Stars X-ray astronomy		
Astrophysics	BT: Astronomy Physics RT: Gravity measurement NT: Asteroseismology Dark matter Orbits Stellar dynamics	Atmosphere	BT: Geoscience RT: Atmospheric measurements Atmospheric science Meteorology NT: Air quality Atmospheric modeling Atmospheric waves
Asymptotic stability	BT: System analysis and design RT: Discrete-time systems Eigenvalues and Stability	Atmospheric measurements	BT: Measurement RT: Air pollution Air quality Altimetry Atmosphere Atmospheric science Barometers Geophysical measurements Global warming Meteorology Pressure gauges Remote sensing Terrestrial atmosphere NT: Air pollution control
Asynchronous circuits	BT: Circuits	Atmospheric modeling	BT: Atmosphere Modeling RT: Sentinel-3 Sentinel-5P
Asynchronous communication	BT: Data communication RT: Web services	Atmospheric pressure chemical vapor deposition	UF: APCVD BT: Chemical vapor deposition
<i>Asynchronous machines</i>	USE: Induction motors	Atmospheric science	BT: Science - general RT: Atmosphere Atmospheric measurements NT: Climatology
Asynchronous transfer mode	UF: ATM BT: Data communication Protocols RT: B-ISDN Broadband communication ISDN Multiprotocol label SONET	<i>Atmospheric sintering</i>	USE: Materials preparation
Atherosclerosis	BT: Arteriosclerosis	Atmospheric waves	BT: Atmosphere Waves
Atlantic Ocean	BT: Oceans	Atmospheric-pressure plasmas	BT: Plasmas
<i>ATM</i>	USE: Asynchronous transfer		
mode AND	Automated teller machine		



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<i>Atom bomb</i>			Radiation detectors
USE:	Nuclear weapons		Spectroscopy
Atom lasers		<i>Atomic optics</i>	
UF:	Single atom lasers	USE:	Atom optics
BT:	Lasers		
RT:	Atom optics	<i>Atomic warfare</i>	
	Atomic beams	USE:	Nuclear weapons
	Gas lasers		
Atom optics		<i>Atomic weapons</i>	
UF:	Atomic optics	USE:	Nuclear weapons
BT:	Particle beam optics		
RT:	Atom lasers	Atoms	
	Atomic beams	BT:	Materials, elements, and
		compounds	
		NT:	Ultracold atoms
Atomic batteries		<i>Atopic dermatitis</i>	
BT:	Energy conversion	USE:	Eczema
	Nuclear power generation		
Atomic beams		<i>ATPG</i>	
BT:	Particle beams	USE:	Automatic test pattern
RT:	Atom lasers	generation	
	Atom optics		
Atomic clocks		Atrial fibrillation	
UF:	Atomic frequency standards	BT:	Fibrillation
BT:	Clocks		
RT:	Frequency measurement	<i>Atrioventricular valves</i>	
	International Atomic Time	USE:	Heart valves
	Masers		
<i>Atomic energy</i>		Atrophy	
USE:	Nuclear power generation	BT:	Medical conditions
Atomic force microscopy		Attention mechanisms	
BT:	Microscopy	BT:	Machine learning
RT:	Casimir effect	RT:	Deep learning
	Magnetic force microscopy		
	Nanotechnology	Attenuation	
	Scanning microwave	UF:	Acoustic wave attenuation
microscopy			Electromagnetic wave
		attenuation	
			Light attenuation
		BT:	Propagation
<i>Atomic frequency standards</i>		RT:	Attenuation measurement
USE:	Atomic clocks		Attenuators
			Diagnostic radiography
<i>Atomic lasers</i>			Insertion loss
USE:	Gas lasers	Attenuation measurement	
Atomic layer deposition		BT:	Electric variables
BT:	Chemical vapor deposition	measurement	
		RT:	Attenuation
Atomic measurements			Loss measurement
BT:	Measurement		
RT:	Nuclear measurements		



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Attenuators

BT: Signal processing
RT: Attenuation
NT: Optical attenuators

Attitude control

BT: Aerospace control
RT: Position control

Attitude determination

USE: Position measurement

ATV

USE: HDTV

Au

USE: Gold

Audible noise

USE: Acoustic noise

Audio coding

BT: Encoding
Information theory
RT: MPEG 7 Standard
Rate distortion theory
Speech coding

Audio compression

BT: Data compression
RT: HDMI

Audio databases

BT: Database systems
RT: File systems
Multimedia databases

Audio enhancement

USE: Acoustic signal processing

AND

Noise reduction

Audio recording

BT: Recording

Audio restoration

USE: Acoustic noise

Audio signal processing

USE: Machine listening

Audio streaming

BT: Streaming media

Audio systems

UF: Phonographs
Sound systems
Stereophonic systems
BT: Consumer electronics
RT: Digital audio broadcasting
Music
NT: 3D audio
Ambisonics
Audio tapes
Audio-visual systems
Auditory displays
Headphones
Immersive audio
Loudspeakers
Microphones
Pitch control (audio)
Portable media players
Sonification
Spatial audio
Video description

Audio tapes

BT: Audio systems

Audio user interfaces

UF: Auditory icons
BT: User interfaces
RT: Multimedia computing

Audio video

USE: Audio-visual systems

Audio visual systems

USE: Audio-visual systems

Audio watermarking

USE: Watermarking

Audio-visual instructional aids

USE: Educational technology

Audio-visual systems

UF: Audio video
Audio visual systems
Audiovisual systems
BT: Audio systems
RT: Educational technology
Text to speech
NT: HDMI

Audiovisual systems

USE: Audio-visual systems



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Auditory displays

BT: Audio systems
Communication equipment
RT: Aerospace and electronic
systems
Communication aids

Auditory icons

USE: Audio user interfaces

Auditory implants

UF: Auditory midbrain implants
BT: Implants

Auditory midbrain implants

USE: Auditory implants

Auditory system

UF: Hearing
BT: Anatomy
RT: Biomedical acoustics
Chatbots
Head
Hearing aids
Psychoacoustics
NT: Psychoacoustic models

Augmented reality

BT: Programming
Virtual reality
RT: Digital representation
Digital transformation
Digital twins
Extended reality
Metaverse
Mixed reality
Network slicing
Spatial computing
Virtual museums
NT: 3D audio
Human augmentation
Immersive audio
Immersive experience
Spatial augmented reality

Augmented virtuality

BT: Virtual reality

Austenite

UF: Gamma phase iron
BT: Iron alloys
RT: Materials science and
technology
Smart materials

Australia

BT: Continents

Authentication

BT: Computer security
RT: Blockchains
CAPTCHAs
Federated identity
Image processing
Interactive systems
Passwords
Video signal processing
Zero Trust
NT: Multi-factor authentication
Nonfungible tokens

Authoring systems

UF: Authoring tools
BT: Software tools
RT: Computer aided instruction
Courseware
Multimedia systems
Web design

Authoring tools

USE: Authoring systems

Authorisation

USE: Authorization

Authorization

UF: Authorisation
BT: Access control
RT: Privacy

Autism

BT: Medical conditions

Auto correlation

USE: Autocorrelation

Auto-pilot

USE: Autopilot

Autobiographies

UF: Memoirs
BT: Biographies

Autocorrelation

UF: Auto correlation
BT: Correlation
RT: Signal analysis
Time series analysis



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Autoencoders

BT: Artificial neural networks
RT: Decoding
Encoding
Unsupervised learning

Autoimmune diseases

UF: Autoimmune disorders
BT: Diseases

Autoimmune disorders

USE: Autoimmune diseases

Automata

UF: Finite state machines
BT: Robots
RT: Cognitive systems
Cybernetics
Intelligent systems
NT: Turing machines

Automated automobile parking

USE: Automated parking

Automated car parking

USE: Automated parking

Automated guided vehicles

USE: Remotely guided vehicles

Automated highways

BT: Automation
Intelligent transportation

systems

RT: Road safety
Smart transportation

Automated indexing

USE: Machine assisted indexing

Automated machine learning

UF: AML
AutoML
Automated ML
BT: Automation
Machine learning
RT: Neural architecture search

Automated meter reading

USE: Automatic meter reading

Automated ML

USE: Automated machine
learning

Automated parking

UF: APS
Automated automobile
parking
Automated car parking
Smart automobile parking
Smart car parking
Smart parking
BT: Automation
Car parking
Mechanical systems
RT: Smart transportation

Automated storage and retrieval systems

USE: Storage automation

Automated teller machine

UF: ATM
BT: Banking
Financial services
RT: Transaction databases

Automatic control

BT: Control systems
NT: Power generation control

Automatic frequency control

BT: Frequency control

Automatic gain control

USE: Gain control

Automatic generation control

BT: Automation
Control systems
Power generation

Automatic indexing

USE: Machine assisted indexing

Automatic logic units

BT: Microprocessors

Automatic meter reading

UF: Automated meter reading
BT: Meter reading
RT: Flowmeters
Smart meters

Automatic number plate recognition

USE: License plate recognition

Automatic optical inspection

BT: Inspection
RT: Machine vision



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	Manufacturing automation		Voltage measurement
	Pattern recognition		
<i>Automatic pilot</i>		<i>Automatic-number plate recognition</i>	
USE:	Autopilot	USE:	License plate recognition
Automatic programming		Automation	
UF:	Program generators	BT:	Robotics and automation
BT:	Programming	RT:	Bagging
			Biometrics
			Flash memories
<i>Automatic protection switching</i>			Home automation
USE:	Protection switching		Information technology
			Substation automation
<i>Automatic question generation</i>			Zigbee
USE:	Question generation	NT:	Automated highways
			Automated machine
Automatic repeat request		learning	
UF:	ARQ		Automated parking
BT:	Feedback communications		Automatic generation
		control	
Automatic speech recognition			Automatic testing
UF:	ASR		Autonomous networks
BT:	Speech recognition		Building automation
RT:	Speech to text		Fifth Industrial Revolution
			Fourth Industrial Revolution
<i>Automatic summarization</i>			Intelligent automation
USE:	Text summarization		Manufacturing automation
			Office automation
Automatic test equipment			Storage automation
BT:	Test equipment		Vehicular automation
RT:	Automatic testing		
<i>Automatic Test Markup Language</i>		<i>AutoML</i>	
USE:	XML	USE:	Automated machine
		learning	
Automatic test pattern generation		<i>Automobile engineering</i>	
UF:	ATPG	USE:	Automotive engineering
BT:	Automatic testing		
RT:	Design automation	Automobile manufacture	
NT:	Test pattern generators	BT:	Manufacturing systems
		RT:	Automobiles
Automatic testing			Automotive components
UF:	Self testing		Automotive engineering
BT:	Automation		Automotive materials
	Testing		Die casting
RT:	Automatic test equipment		Engines
	Maintenance engineering		Tires
NT:	Automatic test pattern		Wheels
generation		<i>Automobile materials</i>	
	Ring generators	USE:	Automotive materials
Automatic voltage control		<i>Automobile parts</i>	
UF:	AVC	USE:	Automotive components
BT:	Voltage control		
RT:	Voltage		



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Automobile safety

USE: Vehicle safety

RT: Automobiles
Automotive components
Automotive electronics
Intelligent transportation

Automobiles

UF: Cars
Motor vehicles
BT: Road vehicles
RT: Automobile manufacture
Automotive components
Automotive control
Automotive engineering
Automotive materials
NT: Car parking

systems
Road traffic control
Road vehicles

Automotive electronics

BT: Automotive engineering
RT: Advanced driver assistance

systems
Automotive control

Automotive

USE: Automotive engineering

Automotive applications

BT: Automotive engineering

Automotive engineering

UF: Automobile engineering
Automotive

BT: Vehicular and wireless
technologies

RT: Automobile manufacture
Automobiles
Automotive components
Diesel engines
Road safety
Wheels

NT: Automotive applications
Automotive control
Automotive electronics
Power steering
Vehicle crash testing
Vehicle detection
Vehicle driving
Vehicle dynamics
Vehicle safety

Automotive components

UF: Air bags
Airfoils
Automobile parts
Radiators (automotive)
Starter motors (automotive)
Windscreen wipers
Windscreens
Windshield wipers
Windshields
BT: Mechanical products
RT: Automobile manufacture
Automobiles
Automotive control
Automotive engineering
Axles
Belts
Brakes
Camshafts
Gears
Hoses
Internal combustion

engines

Shock absorbers
Steering systems
Suspensions (mechanical

systems)

Tires
Torque converters
Water pumps
Wheels

Automotive control

BT: Automotive engineering
Control systems

Automotive materials

UF: Automobile materials
BT: Production materials
RT: Automobile manufacture
Automobiles

Automotive safety

USE: Vehicle safety

Autonomic computing

USE: Distributed computing

Autonomic nervous system

BT: Nervous system
NT: Parasympathetic nervous

system

Sympathetic nervous

system



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Autonomic systems

BT: Network operating systems

Autonomous aerial vehicles

UF: AAV
Aerial robots
Micro air vehicles
UAV
Uncrewed aerial systems
Uncrewed aircraft
Unmanned aerial vehicles
Unmanned air vehicles
Unmanned airborne

vehicles

Unpiloted aerial vehicles
Unpiloted air vehicles
BT: Autonomous vehicles
RT: Aerial computing
Air to ground

communication

Amphibious robots
Drones
Military robotics
Military systems
Quadrotors
Urban air mobility

Autonomous agents

BT: Software agents

Autonomous automobiles

UF: Autonomous cars
Driver free automobiles
Driver free cars
Driver-free car
Driverless automobiles
Driverless cars
Robot automobiles
Robot cars
Self-driving automobiles
Self-driving car
Unmanned automobiles
Unmanned autonomous

cars

BT: Autonomous vehicles

Autonomous cars

USE: Autonomous automobiles

Autonomous driving

BT: Autonomous vehicles
Vehicle driving
RT: Scenario generation

Autonomous machines

USE: Autonomous systems

Autonomous mental development

BT: Computational and artificial intelligence

Autonomous navigation

USE: Autonomous robots

Autonomous networks

UF: Zero touch networks
Zero-touch networks
BT: Automation
Autonomous systems
Network systems

Autonomous robots

UF: Autonomous navigation
BT: Artificial intelligence
Autonomous systems
Robots
RT: Cognitive robotics
Field robots
Intelligent robots

Autonomous systems

UF: Autonomous machines
Self managing systems
Self-managing systems
Unmanned systems
BT: Intelligent systems
Robotics and automation
RT: Agentic AI
Border Gateway Protocol
NT: Autonomous networks
Autonomous robots
Autonomous vehicles

Autonomous trucks

USE: Autonomous vehicles

Autonomous underwater vehicles

UF: Underwater autonomous vehicles
Underwater drones
Underwater exploration
robots
Underwater robots
Unmanned underwater vehicles
BT: Autonomous vehicles
Underwater vehicles
RT: Marine robots
Military robotics



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Autonomous vehicles

UF: Autonomous trucks
Uncrewed
Unmanned autonomous
vehicles
Unmanned surface vehicles
Unmanned vehicles
BT: Autonomous systems
Intelligent vehicles
RT: Artificial intelligence
Mechatronics
Multi-agent systems
Vehicular automation
NT: Autonomous aerial vehicles
Autonomous automobiles
Autonomous driving
Autonomous underwater

vehicles

Autopilot

UF: Auto-pilot
Automatic pilot
BT: Control systems

Autopsy

BT: Medical diagnosis
RT: Pathology

Autoregressive moving average models

USE: Autoregressive processes

Autoregressive processes

UF: Autoregressive moving
average models
Box Jenkins models
BT: Statistics
RT: Noise
Time series analysis

Auxetic materials

UF: Auxetics
BT: Materials

Auxetics

USE: Auxetic materials

Auxiliary transmitters

BT: Transmitters

Availability

BT: Reliability
RT: Maintenance engineering

Avalanche breakdown

BT: Electric breakdown

Avalanche photodiodes

UF: APDs
BT: Photodiodes
RT: Optical fiber communication
Photomultipliers
NT: Single-photon avalanche

diodes

Avatars

BT: Graphical user interfaces
Virtual reality
RT: Metaverse
NT: Digital humans

AVC

USE: Automatic voltage control

Avionics

USE: Aerospace electronics

Awards

BT: IEEE indexing
RT: IEEE Awards activities

AWG device

USE: Arrayed waveguide gratings

AWGN

BT: Additive noise
Gaussian noise
White noise

AWGN channels

BT: Gaussian channels
RT: Intersymbol interference
White noise

Axial force

BT: Force

Axilla

UF: Armpit
Underarm
BT: Shoulder

Axles

BT: Mechanical products
RT: Automotive components
Wheels

Axons

BT: Nerve fibers
RT: Action potentials
Myelin
White matter



January 2026 IEEE Thesaurus

Azimuth

BT: Mathematics
NT: Azimuthal angle
Azimuthal component
Azimuthal current
Azimuthal harmonics
Azimuthal plane

Azimuthal angle

BT: Azimuth

Azimuthal component

BT: Azimuth

Azimuthal current

BT: Azimuth

Azimuthal harmonics

BT: Azimuth

Azimuthal plane

BT: Azimuth

Azobenzene

BT: Polymers
RT: Smart materials

B-cells

USE: Lymphocytes

B-ISDN

UF: Broadband ISDN
BT: Broadband communication
ISDN
RT: Asynchronous transfer

mode

Data communication
Frame relay
Image communication
Multimedia communication

B-Spline

USE: Splines (mathematics)

Ba

USE: Barium

Babies

USE: Pediatrics

Baby

USE: Pediatrics

Back

BT: Body regions

Back propagation

USE: Backpropagation

Back propagation through time

USE: Backpropagation through time

Background noise

BT: Acoustic noise

Backhaul networks

BT: Telecommunication network topology

Backpacks

BT: Consumer products

Backplanes

BT: Data buses

Backpropagation

UF: Back propagation
Backward propagation
Backwards propagation of errors
BT: Learning systems
RT: Backpropagation algorithms
Neural networks

Backpropagation algorithms

BT: Algorithms
RT: Backpropagation
NT: Backpropagation through time

Backpropagation through time

UF: BPTT
Back propagation through time
BT: Backpropagation algorithms
RT: Recurrent neural networks

Backscatter

BT: Reflection
RT: Meteorological radar
NT: Electron backscatter diffraction

Backscattering

USE: Scattering

Backsheets

BT: Solar panels



January 2026 IEEE Thesaurus

Backstepping

BT: Control nonlinearities

Backtracking

BT: Algorithm design and theory

Backward propagation

USE: Backpropagation

Backwards propagation of errors

USE: Backpropagation

Bacteria

BT: Microorganisms
RT: Antibacterial activity
Bacterial infections
Bacteriophages
NT: Cyanobacteria
Escherichia coli
Salmonella

Bacterial infections

BT: Diseases
RT: Bacteria
Escherichia coli

Bacteriophages

UF: Phages
BT: Viruses (medical)
RT: Bacteria

Bag of words model

UF: Bag-of-words model
BT: Natural language processing

Bag-of-words model

USE: Bag of words model

Bagging

BT: Packaging
RT: Automation
Packaging machines
Plastic packaging

Baidu

USE: Web and internet services

Ball bearings

BT: Machinery
RT: Mechanical bearings
Metal products
Rolling bearings

Ball grid arrays

BT: Electronics packaging
Surface mount technology

Ball milling

BT: Production
RT: Milling machines

Ball screws

USE: Mechanical products

Ballasts

USE: Electronic ballasts

Ballistic transport

BT: Electron emission
NT: Electronic ballasts

Baluns

BT: Electromagnetic devices
Impedance matching
Microwave technology
Transformers
RT: Transmission lines

Bamboo

BT: Flowering plants

Band gap

USE: Photonic band gap

Band pass filters

USE: Band-pass filters

Band structures

BT: Energy states

Band-gap

USE: Photonic band gap

Band-pass filters

UF: BPF
Band pass filters
Bandpass filters
BT: Active filters
RT: Frequency
Signal processing
NT: Filter banks

Band-stop filters

USE: Notch filters

Bandgap

USE: Photonic band gap



January 2026 IEEE Thesaurus

Bandpass filters

USE: Band-pass filters

Bandwidth

BT: Frequency
RT: Admission control
Coherence time
Computer network

management

Direct sequence spread

spectrum communication

Information theory
Radio communication
Signal processing
Spectral efficiency
Spectroscopy
NT: Narrowband
Wideband

Bandwidth allocation

USE: Channel allocation

Bandwidth efficiency

USE: Spectral efficiency

Bang bang control

USE: Bang-bang control

Bang-bang control

UF: Bang bang control
BT: Optimal control
RT: Time factors

Banking

BT: Financial industry
RT: Finance
NT: Automated teller machine
Online banking
Open banking

Bankruptcy

BT: Finance
RT: Business
Commercial law
Economics

Bar codes

BT: Optical detectors
Product codes
RT: Internet of Things
Inventory management
NT: QR codes

Barges

USE: Boats

Barium

UF: Ba
BT: Metals
NT: Barium compounds

Barium compounds

BT: Barium
RT: Alloying
Yttrium barium copper

oxide

Barometers

BT: Instruments
RT: Atmospheric
measurements
Meteorology

Baroreceptor reflex

USE: Baroreflex

Baroreflex

UF: Baroreceptor reflex
BT: Cardiovascular system

Bars

BT: Structural shapes
NT: Billets

Basal cell carcinoma

BT: Skin cancer

Basal ganglia

BT: Brain

Base stations

BT: Radio communication
equipment
RT: Device-to-device
communication
Femtocell networks
NT: Femtocells

Baseball

USE: Sports

Baseband

BT: Digital communication
Radio communication
RT: Passband

Basis algorithms

BT: Algorithms

Batch manufacturing

USE: Batch production systems



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Batch normalization		NT:	Charging stations Electric vehicle charging State of charge
BT:	Deep learning		
<i>Batch processing</i>			
USE:	Batch production systems		
Batch production systems			
UF:	Batch manufacturing Batch processing		
BT:	Manufacturing systems		
Bathymetry			
BT:	Measurement Sea floor		
RT:	Hydrography Side-scan sonar		
Batteries			
UF:	Flow batteries Secondary cells Storage batteries Storage battery		
BT:	Electrochemical devices Energy conversion Energy storage		
RT:	Battery charge		
measurement	Battery chargers Emergency power supplies Lithium Lithium compounds Power generation Uninterruptible power		
systems	NT:		
	Lead acid batteries Lithium batteries Lithium-ion batteries Lithium-sulfur batteries Nickel cadmium batteries Solid state batteries		
Battery charge measurement			
BT:	Charge measurement		
RT:	Batteries Battery chargers Battery powered vehicles		
Battery chargers			
UF:	Charging devices Device chargers		
BT:	Power supplies		
RT:	Batteries Battery charge		
measurement			
Battery energy storage system			
BT:	Energy storage		
Battery management systems			
BT:	Electrochemical devices		
Battery powered vehicles			
BT:	Electric vehicles		
RT:	Battery charge		
measurement	Charging stations Energy storage Hybrid electric vehicles Solar powered vehicles Traction motors Vehicle-to-grid		
Battery storage plants			
BT:	Electricity supply industry Energy storage		
Bayes methods			
UF:	Bayesian approach Bayesian belief networks Bayesian estimation Bayesian inference Bayesian learning Bayesian methods Bayesian networks		
BT:	Probability		
RT:	Belief propagation Posterior probability Reinforcement learning Relevance vector machines Semisupervised learning		
NT:	Naive Bayes methods Recursive estimation		
<i>Bayesian approach</i>			
USE:	Bayes methods		
<i>Bayesian belief networks</i>			
USE:	Bayes methods		
<i>Bayesian estimation</i>			
USE:	Bayes methods		
<i>Bayesian inference</i>			
USE:	Bayes methods		



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Bayesian learning

USE: Bayes methods

Bayesian methods

USE: Bayes methods

Bayesian networks

USE: Bayes methods

BCI

USE: Brain-computer interfaces

Beak

BT: Animal structures

RT: Birds

Beam bunches

USE: Particle beam bunching

Beam steering

BT: Microwave technology

RT: Antennas

NT: Steerable antennas

Beamforming

USE: Array signal processing

Beams

UF: Electromagnetic beams

BT: Physics

NT: Acoustic beams

Laser beams

Molecular beams

Optical beams

Particle beams

Bean model

UF: Pry and Bean model

BT: Superconductivity

Bearing estimation

USE: Direction-of-arrival estimation

Bees algorithm

BT: Algorithms
Optimization

Behavior recognition

BT: Behavioral sciences

RT: Deep learning

Machine learning

Pattern recognition

Behavioral economics

UF: Behavioural economics

BT: Behavioral sciences

RT: Psychology

Behavioral sciences

BT: Social sciences
Systems, man, and

cybernetics

RT: Affective computing
Anthropology

Bio-inspired computing

Cultural competence

Cyberbullying

Cyberethics

Digital intelligence

Emotion recognition

Ergonomics

Fish schools

Human factors

Intent recognition

Medical services

Mental health

Persuasive systems

Social Internet of Things

Social computing

System dynamics

NT: Animal behavior

Behavior recognition

Behavioral economics

Cognition

Consumer behavior

Psychiatry

Psychology

Social intelligence

User preference

Behavioural economics

USE: Behavioral economics

Belief functions

USE: Evidence theory

Belief propagation

UF: Sum product message passing

BT: Inference mechanisms

RT: Bayes methods

Evidence theory

Graph theory

Iterative methods

Markov processes

Message passing

Probability



January 2026 IEEE Thesaurus

Bellows

BT: Mechanical products
RT: Pistons
Pneumatic systems
Pumps
Vacuum systems

Belts

UF: Cambelts
Seat belts
BT: Machine components
Machinery
RT: Automotive components
Camshafts
Fasteners

Bench to bedside

USE: Translational research

Benchmark problems

USE: Benchmark testing

Benchmark tasks

USE: Benchmark testing

Benchmark testing

UF: Benchmark problems
Benchmark tasks
Benchmarking
BT: Testing
RT: NSL-KDD
Performance evaluation

Benchmarking

USE: Benchmark testing

Bending

BT: Mechanical factors

Benign masses

USE: Benign tumors

Benign tumors

UF: Benign masses
BT: Tumors

BER

USE: Bit error rate

BER analysis

USE: Bit error rate

BER performance

USE: Bit error rate

Berry phase

BT: Waves

BERT

USE: Bit error rate

Beryllium

BT: Chemical elements

Bespoke production

USE: Job production systems

Best practices

BT: Management
Quality assurance
RT: Business communication
Enterprise architecture

management

Beta rays

BT: Nuclear physics
RT: Electrons

Betavoltaic power sources

USE: Radioactive materials

Bevel gears

USE: Gears

Beverage industry

BT: Industries
RT: Bottling
Food industry

Beyond CMOS

BT: Integrated circuit
technology

BGP

USE: Border Gateway Protocol

BI

USE: Business intelligence

Bi

USE: Bismuth

Bi-directional long short term memory

USE: Bidirectional long short term
memory

Bi-directional LSTM

USE: Bidirectional long short term
memory



January 2026 IEEE Thesaurus

Bi-directional power flow

USE: Bidirectional power flow

Bi-stable circuits

USE: Bistable circuits

Bibliographies

BT: Writing
RT: Publishing

Bibliometrics

BT: Publishing
NT: Citation analysis

BiCMOS integrated circuits

UF: BiMOS integrated circuits
BT: Bipolar transistor circuits

Bicuspid valves

USE: Heart valves

Bicycle sharing

USE: Shared transport

Bicycles

BT: Land vehicles
Vehicles
RT: Micromobility
Sports equipment

Bidirectional communication

USE: Bidirectional control

Bidirectional control

UF: Bidirectional
communication
Bidirectional reflectance
BT: Control systems
RT: Bidirectional long short term
memory

Bidirectional long short term memory

UF: Bi-directional LSTM
Bi-directional long short
term memory
BiLSTM
Bidirectional LSTM
BT: Long short term memory
Recurrent neural networks
RT: Bidirectional control
Natural language
processing

Bidirectional LSTM

USE: Bidirectional long short term
memory

Bidirectional power flow

UF: Bi-directional power flow
BT: Power system control

Bidirectional reflectance

USE: Bidirectional control

Bifurcation

BT: Nonlinear equations
RT: Chaos

Big Data

BT: Data collection
RT: Buffer storage
Data handling
Data lakes
Data mining
Information management
Information processing
Information retrieval
Linked data
Neuroinformatics
NoSQL databases
NT: Big Data applications

Big Data applications

BT: Big Data
Computer applications
RT: Cloud computing
Data analysis
Data systems
Information analysis
Information systems

Bike sharing

USE: Shared transport

Bilinear systems

USE: Nonlinear systems

Billets

BT: Bars

Bills of materials

BT: Inventory management
Materials requirements
planning

BiLSTM

USE: Bidirectional long short term
memory



January 2026 IEEE Thesaurus

BIM

USE: Building information management

BiMOS integrated circuits

USE: BiCMOS integrated circuits

Binary codes

BT: Codes
NT: Reflective binary codes
Webassembly

Binary decision diagrams

BT: Data structures

Binary phase shift keying

UF: BPSK
Binary phase-shift keying
BT: Phase shift keying

Binary phase-shift keying

USE: Binary phase shift keying

Binary search trees

BT: Binary trees

Binary sequences

UF: Bit streams
Bitstreams
BT: Sequences

Binary trees

BT: Tree data structures
NT: Binary search trees

Binge eating

USE: Eating disorders

Bio inspired materials

USE: Bio-inspired materials

Bio-agents

USE: Biological weapons

Bio-computing

USE: Bio-inspired computing

Bio-inspired computing

UF: Bio-computing
Biocomputing
Bioinspired computing
Biologically inspired computing
BT: Artificial intelligence
Bio-inspired engineering

RT: Behavioral sciences
Bio-inspired materials
Biology
Machine learning
Mathematics
Neural networks
Social factors

Bio-inspired control

UF: Biologically inspired control
BT: Bio-inspired engineering

Bio-inspired engineering

UF: Biologically inspired engineering
BT: Engineering - general
RT: Biology
Biomimetics
Complex systems
NT: Bio-inspired computing
Bio-inspired control
Bio-inspired robotics

Bio-inspired materials

UF: Bio inspired materials
Bioinspired materials
BT: Materials
RT: Bio-inspired computing
Biomimetics
Nanotechnology
Organic inorganic hybrid materials
NT: Self-assembly

Bio-inspired robotics

UF: Biologically inspired robots
Biologically-inspired robots
bioinspired robotics
BT: Bio-inspired engineering
Robots

Bio-MEMs

USE: Biomedical microelectromechanical systems

Bio-nanotechnology

USE: Bionanotechnology

Bio-sensors

USE: Biosensors

Bio-weapons

USE: Biological weapons



January 2026 IEEE Thesaurus

Bioacoustics

USE: Biomedical acoustics

Bioagents

USE: Biological weapons

Bioceramics

BT: Biological materials
Biomedical materials
Ceramics
RT: Ceramics industry
Prosthetics

Biochemical analysis

BT: Biochemistry
RT: Biochips

Biochemical oxygen demand

UF: BOD
BT: Biochemistry
Microorganisms
RT: Biodegradation
Composting
NT: BOD5

Biochemistry

UF: Hormones
Metabolic networks
BT: Biology
Chemistry
RT: Biogeochemistry
Biological cells
Bioreactors
Cell motility
Cell signaling
Chemical reaction network

theory

Computational biochemistry
Drugs
Entomology
Hydrolysis
Molecular biophysics
Pharmaceutical technology
Pharmaceuticals
Pharmacodynamics
Pharmacokinetics
Pharmacology

NT: Amino acids
Biochemical analysis
Biochemical oxygen

demand

Metabolism
Peptides
Proteins
Receptor (biochemistry)

Biochips

BT: Molecular biology
RT: Biochemical analysis
Microfluidics
NT: Digital microfluidic biochips

Biocompatibility

BT: Material properties
RT: Medical devices

Biocomputing

USE: Bio-inspired computing

Biocontrol

USE: Biological control systems

Biocybernetics

USE: Cybernetics

Biodegradable

BT: Biodegradation

Biodegradable materials

BT: Biodegradation
RT: Cellulose

Biodegradation

BT: Environmental
management
RT: Anaerobic digestion
Biochemical oxygen
demand
Waste management
Weathering
NT: Biodegradable
Biodegradable materials
Composting

Biodiesel

BT: Biofuels

Biodiversity

BT: Biology
RT: Circular economy
NT: Biogeography
Endangered species

Bioelectric phenomena

UF: Bioelectronics
Electrobiology
BT: Biology
RT: Brain
Electrical accidents
Electroencephalography
Electromyography



January 2026 IEEE Thesaurus

	Electrooculography	Bioimpedance	
	Nervous system	BT:	Biomedical engineering
NT:	Electric shock	RT:	Current
			Blood flow
<i>Bioelectric potentials</i>		Bioinformatics	
USE:	Action potentials	UF:	Biomedical informatics
			Health informatics
<i>Bioelectronics</i>		BT:	Biomedical computing
USE:	Bioelectric phenomena		Informatics
		RT:	Biology
<i>Bioengineering</i>			Computational biochemistry
USE:	Biomedical engineering		Computational biology
			Computational biophysics
<i>Biofeedback</i>		NT:	Neuroinformatics
USE:	Biological control systems		
Biofilms		<i>Bioinspired computing</i>	
BT:	Biological materials	USE:	Bio-inspired computing
	Microorganisms		
Biofuels		<i>Bioinspired materials</i>	
BT:	Fuels	USE:	Bio-inspired materials
RT:	Food waste		
NT:	Biodiesel	<i>bioinspired robotics</i>	
	Biogas	USE:	Bio-inspired robotics
Biogas		<i>Biological agent</i>	
BT:	Biofuels	USE:	Biological weapons
RT:	Anaerobic digestion		
Biogeochemistry		Biological cells	
BT:	Chemistry	UF:	Cell biology
	Geology		Chromosomes
RT:	Biochemistry	BT:	Biology
		RT:	Biochemistry
Biogeography			Biological materials
BT:	Biodiversity		Biomembranes
NT:	Biomes		DNA
			Microorganisms
Biographies		NT:	Self-assembly
BT:	Writing		Cell signaling
RT:	Engineering profession		Cells (biology)
NT:	Autobiographies		Chromosome mapping
			Endothelial cells
Biohazards			Fibroblasts
UF:	Germ warfare		RNA
BT:	Hazards		Stem cells
RT:	Chemical hazards	<i>Biological clocks</i>	
	Green products	USE:	Chronobiology
	Medical treatment		
	Terrorism	Biological control systems	
NT:	Biological weapons	UF:	Biocontrol
			Biofeedback
<i>Bioimaging</i>		BT:	Systems, man, and
USE:	Biomedical imaging	cybernetics	
		RT:	Immune system



January 2026 IEEE Thesaurus

	Legged locomotion	Neurophysiology
	Prosthetics	
NT:	Biomarkers	
<i>Biological effects of protons</i>		<i>Biological organs</i>
USE:	Proton effects	USE: Biological systems
Biological effects of radiation		Biological processes
UF:	Biological radiation effects	BT: Biology
BT:	Radiation effects	RT: Phenology
RT:	Biomedical applications of radiation	Photosynthesis
	Neutron capture therapy	NT: Anaerobic digestion
	Occupational health	Biological interactions
	Proton therapy	Biosynthesis
	Radiation protection	Cell motility
		Chronobiology
		Circadian rhythm
		Coagulation
		Molecular biology
		Symbiosis
		Synaptic communication
<i>Biological EPR resonance</i>		
USE:	Electron paramagnetic resonance	
<i>Biological imaging</i>		<i>Biological radiation effects</i>
USE:	Biomedical imaging	USE: Biological effects of radiation
Biological information theory		<i>Biological sensors</i>
BT:	Biology	USE: Biosensors
	Information theory	
RT:	DNA	Biological system modeling
	Genetic communication	BT: Biology
		RT: Mechanobiology
Biological interactions		Neuromorphics
BT:	Biological processes	Synthetic biology
<i>Biological macromolecules</i>		
USE:	Molecular biophysics	<i>Biological systematics</i>
<i>Biological markers</i>		USE: Systematics
USE:	Biomarkers	Biological systems
Biological materials		UF: Biological organs
UF:	Biomaterials	Organs (biological)
BT:	Materials	BT: Biology
RT:	Biological cells	RT: Animals
	Biomedical materials	Artificial biological organs
	Fats	Biomedical engineering
	Tissue engineering	Organoids
NT:	Bioceramics	NT: Anatomy
	Biofilms	Molecular communication
		Organisms
<i>Biological membranes</i>		Biological techniques
USE:	Biomembranes	BT: Biomedical engineering
		RT: Biomedical equipment
Biological neural networks		<i>Biological therapy</i>
UF:	Neuronal networks	USE: Immunotherapy
BT:	Neural networks	



January 2026 IEEE Thesaurus

Biological threat agents

USE: Biological weapons

Biological tissue

USE: Biological tissues

Biological tissues

UF: Biological tissue
Tissues
BT: Anatomy
RT: Electrosurgery
NT: Bone tissue
Breast tissue
Cardiac tissue
Connective tissue
Glands
Neoplasms

Biological warfare

USE: Biological weapons

Biological warfare agents

USE: Biological weapons

Biological weapons

UF: Bio-agents
Bio-weapons
Bioagents
Biological agent
Biological threat agents
Biological warfare
Biological warfare agents
Bioweapons
Germ warfare
BT: Biohazards
Weapons
RT: Chemical weapons
National security
Nuclear weapons
US Department of

Homeland Security

destruction Weapons of mass

Biologically inspired computing

USE: Bio-inspired computing

Biologically inspired control

USE: Bio-inspired control

Biologically inspired engineering

USE: Bio-inspired engineering

Biologically inspired robots

USE: Bio-inspired robotics

Biologically-inspired robots

USE: Bio-inspired robotics

Biology

BT: Engineering in medicine
and biology

RT: Science - general
Bio-inspired computing
Bio-inspired engineering
Bioinformatics
Computational biology
Environmental science
Immune system
Life sciences
Paleontology
Pharmacology
NT: Astrobiology
Biochemistry
Biodiversity
Bioelectric phenomena
Biological cells
Biological information

theory

Biological processes
Biological system modeling
Biological systems
Biology computing
Biophotonics
Biophysics
Biostatistics
Botany
Cryobiology
Embryology
Evolution (biology)
Genetics
Homeostasis
Mechanobiology
Microbiology
Microinjection
Nanobioscience
Phenology
Physiology
Predator prey systems
Structural biology
Synthetic biology
Systematics
Systems biology
Vegetation
Zoology

Biology computing

BT: Biology
RT: Biomedical computing
Computers and information

processing



January 2026 IEEE Thesaurus

Bioluminescence

BT: Chemiluminescence

Biomagnetics

UF: Biomagnetism
BT: Biophysics
Magnetism
RT: Biomedical engineering
Magnetic fields
Magnetic materials
Magnetic particles
NT: Magnetoencephalography

Biomagnetism

USE: Biomagnetics

Biomarkers

UF: Biological markers
Human disease markers
BT: Biological control systems
Biomedical measurement
NT: Molecular biomarkers

Biomass

BT: Renewable energy sources
RT: Renewable Portfolio

Standard

Biomaterials

USE: Biological materials

Biomechanics

BT: Mechanical factors
RT: Anthropometry
Cell motility
Cell signaling
Entomology
Human activity recognition
Mechanobiology
Motion capture
Wearable robots
NT: Fall detection

Biomechatronics

BT: Mechatronics

Biomedical acoustics

UF: Bioacoustics
Biomedical ultrasonics
BT: Acoustics
RT: Acoustic applications
Acoustic measurements
Auditory system

Biomedical applications of electromagnetic radiation

USE: Biomedical applications of radiation

Biomedical applications of radiation

UF: Biomedical applications of electromagnetic radiation

BT: Biomedical engineering
Nuclear and plasma

sciences

RT: Angiocardiology
Biological effects of

radiation

Biomedical imaging
Cancer

Collimators
Computed tomography
Gamma-ray detectors
Medical treatment
Positron emission

tomography

Radiation effects
Radiography
Synchrotron radiation
NT: Radiation therapy

Biomedical communication

BT: Communication systems
Engineering in medicine

and biology

RT: Fall detection
Medical devices
Nanocommunication
Picture archiving and
communication systems

NT: Point of care
Biomedical telemetry
Telemedicine

Biomedical computing

UF: Medical computing
BT: Engineering in medicine

and biology

RT: Biology computing
Biomedical signal

processing

Computer applications
Picture archiving and
communication systems

NT: Signal processing
Bioinformatics
Medical expert systems
Medical information

systems



January 2026 IEEE Thesaurus

Biomedical electrodes

BT: Biomedical equipment
RT: Biomedical engineering
Biomedical measurement
Dry electrodes
Electrophysiology
Wet electrodes

microelectromechanical systems

Biomedical electronics
Biomedical measurement
Biomedical
Collimators
Electrocardiography
Electroencephalography
Endomicroscopy

Biomedical electronics

BT: Biomedical engineering
RT: Biomedical equipment

Insulin pumps
Medical control systems
Medical robotics
Molecular biophysics
Nanosensors

Biomedical engineering

UF: Bioengineering
BT: Engineering in medicine

and biology

RT: Biological systems
Biomagnetics
Biomedical electrodes
Biomedical engineering

Needles
Orthotics
Prosthetics
Sensory aids
Speech synthesis
Surgery
Zigbee

education

NT:

Biomedical monitoring
Biomedical optical imaging
Colloidal lithography
Genetic communication
Genetic engineering
Hospitals
Microfluidics
Orthotics
Bioimpedance
Biological techniques
Biomedical applications of

Assistive technologies
Biomedical electrodes
Biomedical telemetry
Biomedical transducers
Catheters
Endoscopes
Gerontechnology
Hypodermic needles
Implants
Intracranial pressure

radiation

sensors

Biomedical electronics
Biomedical signal

Lithotriptors
Medical devices
Medical instruments
Pacemakers
Petri dishes
Pulse oximeter
Stethoscope
Surgical instruments
Ventilators

processing

Bioprinting
Biotechnology
Cloning
Drug delivery
Neural engineering
Protein engineering
Tissue engineering
Translational research

Biomedical image processing

UF: Medical image processing

BT: Biomedical signal

processing

RT: Biomedical imaging
Biomedical optical imaging
Functional magnetic

resonance imaging

Biomedical engineering education

BT: Engineering education
RT: Biomedical engineering

Magnetoencephalography
Medical robotics
Subtraction techniques
Imaging phantoms
Motion artifacts
Neuroimaging

Biomedical equipment

UF: Clinical equipment
Medical equipment
BT: Engineering in medicine

and biology

RT: Biological techniques



January 2026 IEEE Thesaurus

enhancement
Radiographic image
Radiology
Radiomics
Ultrasonography
Whole body imaging

Biomedical imaging

UF: Bioimaging
Biological imaging
Biomedical X-ray imaging
Medical imaging
Tomosynthesis
BT: Engineering in medicine
and biology
RT: Imaging
radiation Biomedical applications of
processing Biomedical image
Data visualization
Isosurfaces
Medical diagnosis
Molecular biophysics
Nanobiophotonics
Picture archiving and
communication systems
Radiation imaging
Tomography
Ultrasonic imaging
NT: Angiocardiology
Angiography
Biomedical optical imaging
Cardiology
Contrast agents
DICOM
Elastography
Encephalography
Mammography
Medical diagnostic imaging
Molecular imaging
Phantoms
Photoacoustic imaging

Biomedical informatics

USE: Bioinformatics

Biomedical infrared imaging

USE: Biomedical optical imaging

Biomedical instruments

USE: Biomedical measurement

Biomedical materials

BT: Materials
RT: Biological materials
Diamond-like carbon
Molecular biophysics
NT: Bioceramics
Biomembranes

Biomedical measurement

UF: Biomedical instruments
Biomedical measurements
BT: Measurement
RT: Anthropometry
Biomedical electrodes
Biomedical equipment
Biosensors
Pulse oximetry
NT: Biomarkers
Biomedical monitoring
Body mass index
Electroencephalography
Electromyography
Electrooculography
Electrophysiology
Photoplethysmography
Plethysmography
Pulse oximeter
Sensitivity and specificity
Spirometry

Biomedical measurements

USE: Biomedical measurement

Biomedical microelectromechanical systems

UF: Bio-MEMs
BT: Micromechanical devices
RT: Biomedical equipment

Biomedical monitoring

BT: Biomedical measurement
RT: Biomedical engineering
Epidemiology
Internet of Medical Things
Phonocardiography
NT: Continuous glucose
monitoring
Nanomedicine

Biomedical MRI

USE: Magnetic resonance
imaging

Biomedical optical imaging

UF: Biomedical infrared imaging
BT: Biomedical imaging



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RT: Biomedical engineering
Biomedical image processing

Endomicroscopy
Endoscopes
Infrared imaging
Optical communication
equipment
Optical devices

Biomedical signal processing

BT: Biomedical engineering
RT: Biomedical computing
Fall detection
Neurophysiology
Time-frequency analysis
NT: Biomedical image processing

Biomedical telemetry

UF: Biotelemetry
BT: Biomedical communication
Biomedical equipment
Telemetry

Biomedical transducers

BT: Biomedical equipment
Transducers

Biomedical ultrasonics

USE: Biomedical acoustics

Biomedical X-ray imaging

USE: Biomedical imaging

Biomembranes

UF: Biological membranes
Membranes
BT: Biomedical materials
RT: Biological cells

Biomes

BT: Biogeography

Biometric authentication

BT: Biometrics
RT: Access control
Security

Biometric identification

BT: Biometrics
Identification of persons

Biometric systems

USE: Biometrics

Biometrics

UF: Biometric systems
BT: Identification of persons
RT: Access control
Algorithms
Automation
Handwriting recognition
Information technology
Security
Speaker recognition
NT: Biometric authentication
Biometric identification
Face recognition
Fingerprint recognition
Gait recognition
Iris recognition
Keystroke dynamics
Palmprint recognition

Biometry

USE: Biostatistics

Biomimetic

USE: Biomimetics

Biomimetic materials

BT: Biomimetics
Smart materials

Biomimetic microelectronics

USE: Biomimetics

Biomimetics

UF: Biomimetic
Biomimetic microelectronics
Biomimicry
Bionics
BT: Microprocessors
RT: Bio-inspired engineering
Bio-inspired materials
Cradle to cradle
Intelligent materials
Soft robotics
Whale optimization

algorithms

NT: Biomimetic materials

Biomimicry

USE: Biomimetics

Biomolecular electronics

USE: Molecular electronics

Biomolecules

USE: Molecular biophysics



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Bionanotechnology

UF: Bio-nanotechnology
BT: Engineering in medicine
and biology
Nanotechnology

Bionics

USE: Biomimetics

Biophotonics

BT: Biology
Photonics

Biophysics

BT: Biology
Physics
RT: Computational biophysics
NT: Aerospace biophysics
Biomagnetics
Cellular biophysics
Molecular biophysics

Biopolymers

BT: Polymers

Bioprinting

BT: Biomedical engineering
Three-dimensional printing
RT: Artificial biological organs
Tissue engineering

Biopsies

USE: Biopsy

Biopsy

UF: Biopsies
BT: Medical tests

Bioreactors

BT: Chemical reactions
RT: Biochemistry

Biorthogonal modulation

BT: Wavelet transforms

Biosensors

UF: Bio-sensors
Biological sensors
BT: Chemical and biological
sensors
RT: Biomedical measurement
Nanobiophotonics
Wearable sensors

Biosphere

BT: Environmental factors
Geoscience
RT: Carbon cycle

Biostatistics

UF: Biometry
BT: Biology
Statistics

Biosynthesis

BT: Biological processes

Biotechnology

BT: Biomedical engineering
RT: Genetic engineering
Hydrolysis

Biotelemetry

USE: Biomedical telemetry

Bioterrorism

BT: Engineering in medicine
and biology
Terrorism

Bioweapons

USE: Biological weapons

Bipartite graph

BT: Graph theory

Biped locomotion

USE: Legged locomotion

Bipolar integrated circuits

BT: Bipolar transistor circuits
RT: Bipolar transistors

Bipolar transistor circuits

BT: Circuits
RT: Parameter extraction
NT: BiCMOS integrated circuits
Bipolar integrated circuits

Bipolar transistors

BT: Power semiconductor
switches
RT: Bipolar integrated circuits
Proton radiation effects
Semiconductor epitaxial
layers
Transistors
NT: Insulated gate bipolar
transistors



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		Kirk field collapse effect		<i>Bit error rate test</i>	
				USE:	Bit error rate
Birds					
	BT:	Animals		<i>Bit interleaved coded</i>	
	RT:	Beak		USE:	Interleaved codes
		Ornithology			
Birefringence			Bit rate		
	BT:	Optics		UF:	Bit allocation
	RT:	Photorefractive effect			Bitrate
		Photorefractive materials		BT:	Timing
		Refractive index		RT:	Communication system
		Thermooptic effects		signaling	Computer networks
					Signal processing
Birth disorders				<i>Bit streams</i>	
	BT:	Amniocentesis		USE:	Binary sequences
Bismuth				<i>Bit-interleaved coded</i>	
	UF:	Bi		USE:	Interleaved codes
	BT:	Metals			
	RT:	Bismuth compounds		Bitcoin	
Bismuth compounds				BT:	Cryptocurrency
	UF:	BSCCO		RT:	Blockchains
	BT:	Compounds			Cryptography
	RT:	Alloying			Finance
		Bismuth			Nonfungible tokens
					Online banking
<i>BIST</i>				<i>Bitrate</i>	
	USE:	Built-in self-test		USE:	Bit rate
<i>Bistability (optical)</i>				<i>Bitstreams</i>	
	USE:	Optical bistability		USE:	Binary sequences
Bistable circuits				<i>Bitumen</i>	
	UF:	Bi-stable circuits		USE:	Asphalt
	BT:	Circuits			
	NT:	Latches		<i>Bixby</i>	
<i>Bistable multivibrator</i>				USE:	Virtual assistants
	USE:	Pulse circuits		<i>Black box</i>	
Bistatic radar				USE:	Flight recording
	BT:	Radar		Black carbon	
<i>Bit allocation</i>				BT:	Carbon emissions
	USE:	Bit rate		<i>Blackberry</i>	
Bit error rate				USE:	Personal digital devices
	UF:	BER		<i>Blackbox</i>	
		BER analysis		USE:	Closed box
		BER performance			
		BERT		Blackwater	
		Bit error rate test		BT:	Wastewater treatment
	BT:	Error analysis			



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Bladder	RT:	Greywater	BT:	Mixed source separation
				RT:
Blades	BT:	Urogenital system	analysis	Adaptive signal detection
				RT:
	UF:	Vanes		Independent component
	BT:	Mechanical products		Signal analysis
	RT:	Agricultural machinery	Blindness	Signal detection
		Cutting tools		
		Fans	BT:	Medical conditions
		Impellers		RT:
		Propellers		Visual impairment
		Turbomachinery		
Blanking			Blob detection	
	BT:	Manufacturing systems		BT:
	RT:	Metal products		Image processing
		Metals		RT:
		Sheet metal processing	Block chain	
				USE:
Blast furnaces	BT:	Furnaces	Block codes	
	RT:	Smelting		UF:
Bleaching			BT:	Channel coding
	BT:	Materials processing		RT:
	RT:	Manufacturing systems	NT:	Linear codes
		Paper making		
		Process control	Block coding	
		Textile technology		USE:
Bleeding			Block listing	
	USE:	Hemorrhaging		USE:
Blended learning			Block lists	
	USE:	Hybrid learning		USE:
Blind channel estimation			Block signaling	
	USE:	Blind equalizers		USE:
Blind equalisers			Block signalling	
	USE:	Blind equalizers		UF:
Blind equalizers			BT:	Signaling block systems
	UF:	Blind channel estimation		
		Blind equalisers	RT:	Control systems
	BT:	Source signal equalizers		
		Equalizers		Collision avoidance
Blind signal separation			Blockchains	
	USE:	Blind source separation		UF:
Blind source separation				Chain codes
	UF:	Blind signal separation		

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organization	BT: RT:	Permissioned blockchains	<i>Blood brain barrier</i>	USE:	Blood-brain barrier
		Private blockchains			
		Public blockchains	<i>Blood clots</i>	USE:	Coagulation
		Cryptography			
		Digital economy	Blood flow	BT:	Blood pressure
		Distributed databases		RT:	Anemia
		Authentication			Bioimpedance
		Bitcoin			Hematology
		Computer security			Hemorrhaging
		Content management		NT:	Hemodynamics
	NT:	Cryptocurrency	Blood platelets	BT:	Blood
		Data collection		RT:	Coagulation
		Decentralized autonomous			Hematology
		Decentralized identity	Blood pressure	UF:	Arterial pressure
		Directed acyclic graph		BT:	Blood vessels
		Distributed ledger		NT:	Blood flow
		Federated identity			Blood pressure
		Metaverse	measurement		Blood pressure variability
		Micropayments			Hypertension
		Proof of Work			Hypotension
Blocklists	UF:	Trustless services	Blood pressure measurement	BT:	Blood pressure
		Consensus protocol			
		Nonfungible tokens	Blood pressure variability	BT:	Blood pressure
		Proof of stake			
		Block listing	Blood vessels	BT:	Cardiovascular system
		Block lists		RT:	Endothelial cells
		Deny lists			Hematology
		Access control		NT:	Arteries
		Information filters			Blood
					Blood pressure
<i>Blogging</i>	USE:				Veins
		Blogs	Blood-brain barrier	UF:	Blood brain barrier
				BT:	Brain
			Blue carbon	BT:	Carbon capture and storage
			Blue hydrogen	BT:	Carbon capture and storage
Blogs	UF:	Blogging			
		Twitter			
		Weibo			
		Information retrieval			
		Electronic mail			
		Internet			
		Social networking (online)			
Blood	BT: RT:	Blood vessels			
		Anticoagulants			
		Leukemia			
		Blood platelets			
		Coagulation			
		Red blood cells			
		White blood cells			



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Bluetooth

BT: Personal area networks
Radio communication
RT: Cellular radio
Communication equipment
Digital communication
IEEE 802.11 Standard
IEEE 802.11g Standard
IEEE 802.11n Standard
IEEE 802.15 Standard
Land mobile radio
Protocols
Spread spectrum

communication

Wireless LAN
Wireless communication
Zigbee
NT: Bluetooth Low Energy

Bluetooth Low Energy

BT: Bluetooth

BMI

USE: Body mass index

BNCT

USE: Neutron capture therapy

BNSC

UF: British National Space
Centre
BT: Organizations

Boat building industry

USE: Shipbuilding industry

Boats

UF: Barges
Yachts
BT: Marine vehicles
RT: Marine robots

BOD

USE: Biochemical oxygen

demand

BOD5

UF: 5-day BOD
5-day biochemical oxygen

demand

BT: Biochemical oxygen

demand

Body area networking

USE: Body area networks

Body area networks

UF: Body area networking
BT: Personal area networks
RT: Human activity recognition
Wearable antennas

Body borne computers

USE: Wearable devices

Body mass index

UF: BMI
BT: Biomedical measurement

Body regions

BT: Anatomy
NT: Abdomen
Back
Breast
Extremities
Head
Neck
Pelvis
Perineum
Thorax
Torso
Viscera

Body sensor networks

BT: Personal area networks
Wireless sensor networks
RT: Fall detection
Human activity recognition
Wearable sensors

Boilers

BT: Heating systems
RT: Heat recovery
Steam engines
Turbines
Waste heat

Bolometers

BT: Radiation detectors
RT: Infrared detectors
Temperature measurement

Bolts

USE: Fasteners

Boltzmann distribution

BT: Statistics
NT: Lattice Boltzmann methods



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Boltzmann equation

UF: Boltzmann transport equation
BT: Equations

Boltzmann transport equation

USE: Boltzmann equation

Bomb

USE: Weapons

Bonding

BT: Bonding processes
RT: Manufacturing
Materials processing
NT: Adhesives

Bonding forces

BT: Materials testing

Bonding processes

BT: Fabrication
Joining processes
RT: Soldering
Welding
NT: Bonding
Diffusion bonding
Wafer bonding

Bone density

UF: Bone mineral density
BT: Bones
RT: Density measurement

Bone diseases

BT: Diseases
NT: Osteoarthritis
Osteoporosis

Bone mineral density

USE: Bone density

Bone tissue

BT: Biological tissues
RT: Bones
NT: Cancellous bone
Cortical bone

Bones

BT: Skeleton
RT: Bone tissue
Skull
NT: Bone density
Pelvic bones

Bonuses

USE: Incentive schemes

Book reviews

BT: IEEE indexing

Boolean algebra

BT: Algebra
RT: Logic
Logic gates
Set theory
NT: Boolean functions

Boolean functions

BT: Boolean algebra
RT: Fault trees
NT: Logic functions

Boosting

BT: Machine learning
Supervised learning

Booting

BT: Operating systems

Border Gateway Protocol

UF: BGP
BT: Protocols
RT: Autonomous systems
Information exchange
Peer-to-peer computing
Routing protocols
Telecommunication

network management

Telecommunication traffic

Boring

BT: Machining
RT: Drilling
Milling
Turning

Boron

BT: Chemical elements
Metals
RT: Fertilizers
NT: Boron alloys

Boron alloys

BT: Boron
RT: Magnetic materials

Boron neutron capture therapy

USE: Neutron capture therapy



January 2026 IEEE Thesaurus

Bot

USE: Chatbots

Bot (Internet)

UF: WWW robot
Web robot
BT: Computer applications
Internet
World Wide Web
RT: Crawlers

Botany

BT: Biology
RT: Seeds (agriculture)

Botnet

BT: Interconnected systems
Internet
Software agents
RT: Computer crime
Distributed denial-of-service

attack

Robots

Bottling

BT: Packaging
RT: Beverage industry
Glass products
Packaging machines
Plastic products

Boundary conditions

BT: Boundary value problems
NT: Lower bound
Upper bound

Boundary element methods

USE: Boundary-element methods

Boundary value problems

BT: Mathematics
RT: Partial differential equations
NT: Boundary conditions

Boundary-element methods

UF: Boundary element methods
BT: Partial differential equations
RT: Integral equations
Method of moments

Bovine

BT: Animals
NT: Cows

Box Jenkins models

USE: Autoregressive processes

BPEL

USE: Business Process
Execution Language

BPF

USE: Band-pass filters

BPR

USE: Business process re-engineering

BPSK

USE: Binary phase shift keying

BPTT

USE: Backpropagation through
time

Brachial

USE: Brachytherapy

Brachytherapy

UF: Brachial
BT: Medical treatment
RT: Radiation effects

Bradycardia

BT: Heart rate
RT: Tachycardia

Bragg gratings

UF: Fiber Bragg gratings
Fiber-Bragg gratings
BT: Filters
Optical devices
RT: Diffraction
Diffraction gratings
Laser beams
Optical beams
Optical transmitters
Temperature sensors
Wavelength division

multiplexing

NT: Fiber gratings

Braille

BT: Writing
RT: Assistive technologies
Blindness
Haptic interfaces
Tactile sensors



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Brain

BT: Nervous system
 RT: Bioelectric phenomena
 Cerebrospinal fluid
 Cognition
 Cognitive informatics
 Cognitive science
 Diffusion tensor imaging
 Electroencephalography
 Encephalography
 Head
 Intracranial pressure

sensors

Magnetoencephalography
 Soma
 Synapses
 White matter
 NT: Amygdala
 Basal ganglia
 Blood-brain barrier
 Brain cells
 Brain injuries
 Brain modeling
 Brain tumors
 Brain ventricles
 Brainstem
 Cerebellum
 Cerebral cortex
 Cerebrum
 Corpus callosum
 Forebrain
 Frontal lobe
 Hindbrain
 Hydrocephalus
 Hypothalamus
 Limbic system
 Midbrain
 Neural activity
 Neural implants
 Neurodynamics
 Neurophysiology
 Neuroprostheses
 Neuropsychology
 Neurotechnology
 Occipital Lobe
 Parietal lobe
 Primary motor cortex
 Sleep
 Temporal lobe
 Thalamus
 Ventricle system
 Virtual artifact

Brain cancer

BT: Brain tumors

Cancer

Brain cells

BT: Brain

Brain computer interfaces

USE: Brain-computer interfaces

Brain imaging

USE: Neuroimaging

Brain implants

USE: Neural implants

Brain injuries

BT: Brain
 Injuries
 RT: Aphasia
 Cerebral palsy
 Spasticity

Brain interfaces

USE: Brain-computer interfaces

Brain machine interfaces

USE: Brain-computer interfaces

Brain mapping

BT: Nervous system
 NT: Neuroimaging

Brain modeling

UF: Brain modelling
 BT: Brain
 Modeling

Brain modelling

USE: Brain modeling

Brain plasticity

USE: Neuroplasticity

Brain stem

USE: Brainstem

Brain stem implants

USE: Brainstem implants

Brain stimulation

BT: Medical treatment
 NT: Deep brain stimulation

Brain tumors

BT: Brain
 Tumors



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NT:	Brain cancer	USE:	Brand management
Brain ventricles		Brazing	
BT:	Brain	BT:	Soldering
		RT:	Welding
<i>Brain-computer interaction</i>		Breadboard	
USE:	Brain-computer interfaces	UF:	Plugboard
Brain-computer interfaces			Solderless breadboard
UF:	BCI	BT:	Electronic circuits
	Brain computer interfaces		Prototypes
	Brain interfaces		
	Brain machine interfaces	<i>Breakdown</i>	
	Brain-computer interaction	USE:	Electric breakdown
	Brain-computer-interfaces		
	Brain-machine interfaces	Breakdown voltage	
	Mind-machine interfaces	BT:	Voltage
BT:	User interfaces	RT:	Current
RT:	Fifth Industrial Revolution		Diodes
	Neural engineering		Insulators
	Neuroprostheses		
<i>Brain-computer-interfaces</i>		Breast	
USE:	Brain-computer interfaces	BT:	Body regions
<i>Brain-machine interfaces</i>		NT:	Breast biopsy
USE:	Brain-computer interfaces		Breast cancer
<i>BrainLobe</i>			Breast tissue
USE:	Frontal lobe		Breast tumors
Brainstem		Breast biopsy	
UF:	Brain stem	BT:	Breast
BT:	Brain		
Brainstem implants		Breast cancer	
UF:	Brain stem implants	UF:	Breast-cancer
BT:	Implants	BT:	Breast
			Cancer
Brakes		RT:	Mastectomy
BT:	Control systems		
	Mechanical products	Breast neoplasms	
RT:	Automotive components	UF:	Mammary neoplasms
		BT:	Neoplasms
Branching factor		Breast tissue	
BT:	Game theory	UF:	Breast tissues
	Tree data structures	BT:	Biological tissues
			Breast
Brand management		<i>Breast tissues</i>	
UF:	Branding	USE:	Breast tissue
BT:	Marketing management	<i>Breast tumor</i>	
RT:	Market research	USE:	Breast tumors
	Product development	Breast tumors	
<i>Branding</i>		UF:	Breast tumor
			Breast tumour



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BT:	Breast tumours	Broadband antennas	
	Breast Tumors	UF:	Wideband antennas
<i>Breast tumour</i>	USE: Breast tumors	BT:	Antennas
		RT:	Antenna arrays
<i>Breast tumours</i>	USE: Breast tumors		Microstrip
			Microwave propagation
			UHF propagation
		NT:	Ultra wideband antennas
			Vivaldi antennas
Breast-cancer		Broadband communication	
USE:	Breast cancer	UF:	Broadband networks
Bremsstrahlung	BT: Electromagnetic radiation	BT:	Communication systems
		RT:	Asynchronous transfer
Bridge circuits	BT: Circuits	mode	Cable TV
			Frequency division
	RT: Rectifiers	multiaccess	IEEE 802.16 Standard
<i>Bridge converters</i>	USE: Switched reluctance motors		IPTV
			Multimedia communication
			Optical fiber communication
			Ultra wideband
Bridges	BT: Structural shapes	communication	Video on demand
		NT:	B-ISDN
	RT: Civil engineering		Broadband amplifiers
	Structural engineering		
	Transportation		
Brightness		<i>Broadband ISDN</i>	
BT:	Optics	USE:	B-ISDN
NT:	Brightness temperature	<i>Broadband networks</i>	
Brightness temperature		USE:	Broadband communication
BT:	Brightness	Broadcast technology	
Brillouin scattering		NT:	Broadcasting
BT:	Scattering	Broadcasting	
Bring your own device		UF:	Broadcasts
UF:	BYOD	BT:	Broadcast technology
BT:	Information technology	RT:	Entertainment industry
RT:	Mobile computing		Journalism
	Office automation	NT:	Digital audio broadcasting
	Personnel		Digital multimedia
	Security	broadcasting	Digital video broadcasting
	Smart phones		Motion pictures
			NVIS
			Radio broadcasting
			Satellite broadcasting
			Web TV
<i>British National Space Centre</i>		<i>Broadcasts</i>	
USE:	BNSC	USE:	Broadcasting
Broadband amplifiers			
UF:	Wideband amplifiers		
BT:	Amplifiers		
	Broadband communication		



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Bromine

BT: Chemical elements
NT: Bromine compounds

Bromine compounds

UF: Organobromine compounds
BT: Bromine
Chemical compounds
RT: Flame retardants

Bronchi

USE: Respiratory system

Bronchoscopy

BT: Medical diagnosis

Brownian motion

UF: Fractional brownian motion
BT: Random processes
RT: Diffusion processes

Browsers

UF: Google Chrome
Web browsers
BT: Computer interfaces
RT: User interfaces
Webassembly

Brushes

BT: Contacts
RT: Rotating machines

Brushless DC motors

UF: Brushless direct current
motors
BT: Brushless motors
DC motors

Brushless direct current motors

USE: Brushless DC motors

Brushless machines

BT: Electric machines

Brushless motors

BT: Motors
RT: Switched reluctance motors
NT: Brushless DC motors

Brute force attacks

UF: Brute-force attacks
BT: Cryptography

Brute-force attacks

USE: Brute force attacks

BSCCO

USE: Bismuth compounds

Buck converters

BT: DC-DC power converters

Buckeyballs

USE: Fullerenes

Buckminsterfullerene

USE: Fullerenes

Buckyballs

USE: Fullerenes

Buckytubes

USE: Fullerenes

Buffer layers

BT: Thin films
RT: Diffusion processes
Semiconductor films
Semiconductor growth

Buffer overflows

BT: Computer crashes

Buffer storage

UF: Data storage systems
BT: Data systems
RT: Big Data
In-memory computing
Random access memory
Storage area networks
NT: Triples (Data structure)

Bugs

USE: Computer bugs

Building automation

BT: Automation
Building services
RT: Construction industry

Building information management

UF: BIM
Building information

modelling

BT: Buildings
Modeling
RT: Architecture
Building management

systems

Building services
Design engineering



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	Facilities management		Facilities management
	Project management		
	Structural engineering		
<i>Building information modelling</i>		<i>Building-integrated photovoltaics</i>	
USE:	Building information management	USE:	Building integrated photovoltaics
Building integrated photovoltaics		Buildings	
UF:	Building-integrated photovoltaics	UF:	Space habitats
	Roof mounted photovoltaics	BT:	Construction
	Roof mounted solar cell arrays	RT:	Air conditioning
BT:	Photovoltaic systems		Architecture
RT:	Building services		Civil engineering
	Solar power generation		Construction industry
			Elevators
			Escalators
			Industrial power systems
			Lighting
			Modular construction
			Prefabricated construction
			Smart cities
			Space cooling
			Vents
Building management systems		NT:	Building information management
BT:	Buildings Management		
RT:	Building information management		
Building materials			Building management systems
BT:	Buildings Materials		Building materials
RT:	Aggregates		Building services
	Construction		Flexible structures
	Construction industry		Intelligent structures
	Prefabricated construction		Smart buildings
	Solar panels		Smart homes
	Structural beams		
NT:	Asphalt	Built-in self-test	
	Concrete	UF:	BIST
	Floors		Self-testing
	Mortar	BT:	Testing
	Tiles	RT:	Circuit testing
	Windows		Design for testability
Building services		<i>Bulimia</i>	
BT:	Buildings	USE:	Eating disorders
RT:	Access control		
	Air conditioning	<i>Bulimia nervosa</i>	
	Building information management	USE:	Eating disorders
	Building integrated photovoltaics		
	Furnaces	Bulk acoustic wave devices	
	Lighting	BT:	Acoustic devices
	Solar panels	RT:	Film bulk acoustic resonators
	Space heating		
	Wiring	Bulk storage	
NT:	Building automation	BT:	Material storage
	Elevators	RT:	Containers



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Bundle adjustment

BT: Three-dimensional displays

Buoyancy

BT: Fluid dynamics

RT: Fluids
Physics

NT:

Buried object detection

UF: Buried objects
Underground object

detection

BT: Underground objects
Object detection
RT: Geophysical measurements
Ground penetrating radar
NT: Landmine detection

responsibility

Industries
International trade
Leadership
Manufacturing
Productivity
Service computing
Agribusiness
Business data processing
Business intelligence
Commerce and trade
Corporate social
Disruptive innovation
Entrepreneurship
Franchising
Industrial relations
Management
Operations research
Organizations

Buried objects

USE: Buried object detection

burn injuries

USE: Skin burns

burn wounds

USE: Skin burns

Burnishing

BT: Surface finishing
RT: Machining

Burst switching

BT: Packet switching
NT: Optical burst switching

Bushings

USE: Insulators

Business

BT: Engineering management
RT: Bankruptcy
Business process

integration

management

Business process

Commercial law
Consortia
Contracts
Digital economy
Digital transformation
Employment
Enterprise resource

planning

Finance
Industrial communication

Business communication

BT: Organizational aspects
RT: Best practices

Business continuity

BT: Management
RT: Security
System recovery
Venture capital

Business data processing

BT: Business
Data processing
RT: Data governance
Information processing

Business intelligence

UF: BI
BT: Business
Data analysis
RT: Competitive intelligence
Data mining
Strategic planning

Business organisation

USE: Organizational aspects

Business organization

USE: Organizational aspects

Business Process Execution Language

UF: BPEL
WP-BPEL
Web services business

process execution language



January 2026 IEEE Thesaurus

BT: Computer languages
Web services
RT: Business process re-
engineering
Information processing

Business process integration

BT: Enterprise resource
planning
Process planning
RT: Business
Resource management
Supply chain management
Systems engineering and
theory

Business process management

BT: Management
Process planning
RT: Business
Resource management
Supply chain management
Systems engineering and
theory
NT: Task analysis

Business process re-engineering

UF: BPR
BT: Management
RT: Business Process
Execution Language
Corporate acquisitions
Organizational aspects
Total quality management

Business social impact

USE: Corporate social
responsibility

Business writing

USE: Writing

Butler matrices

UF: Butler matrix
BT: Antenna arrays
RT: Antennas
IEEE 802.11 Standard
Phase shifters
Wireless LAN

Butler matrix

USE: Butler matrices

Butter

USE: Dairy products

Buttocks

BT: Extremities

By-product

USE: Byproduct

BYOD

USE: Bring your own device

Byproduct

UF: By-product
BT: Recycling

C band

USE: C-band

C languages

BT: Computer languages
RT: Object oriented
programming
NT: C# languages
C++ languages

C sharp languages

USE: C# languages

C# languages

UF: C sharp languages
BT: C languages
RT: Object oriented
programming

C++ languages

BT: C languages
RT: Webassembly

C-band

UF: C band
BT: Microwave bands

C2C

USE: Cradle to cradle

Ca

USE: Calcium

Cable insulation

BT: Insulation
RT: Cables
Oil filled cables
NT: Power cable insulation

Cable shielding

BT: Electromagnetic shielding
RT: Cables



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Cable splicing

USE: Splicing

Cable TV

UF: Community antenna
television
BT: TV
RT: Broadband communication
Image communication
NT: Must-carry regulations

Cables

BT: Transmission lines
RT: Cable insulation
Cable shielding
Conductors
Fault location
Winches
Wiring
NT: Coaxial cables
Communication cables
Mechanical cables
Optical fiber cables
Power cables
Underwater cables

Cables (mechanical)

USE: Mechanical cables

Cache memory

BT: Memory
RT: In-memory computing
NT: Cache storage

Cache storage

BT: Cache memory

CAD

USE: Design automation

Cadaver

UF: Corpse
BT: Pathological processes

CADCAM

BT: Computer aided
manufacturing
RT: Design automation
Computer integrated
manufacturing
Integrated manufacturing
systems
Rapid prototyping
Virtual manufacturing

Cadmium

BT: Metals
NT: Cadmium compounds

Cadmium compounds

BT: Cadmium

CAE

USE: Computer aided
engineering

Caesium

USE: Cesium

CAI

USE: Computer aided instruction

Calamities

USE: Disasters

Calcination

BT: Heat treatment
RT: Kilns

Calcium

UF: Ca
BT: Metals
RT: Alloying
NT: Calcium compounds

Calcium carbonate

USE: Calcium compounds

Calcium compounds

UF: Calcium carbonate
Calcium phosphate
BT: Calcium

Calcium phosphate

USE: Calcium compounds

Calculators

BT: Computers
RT: Digital arithmetic
NT: Difference engines

Calculus

BT: Mathematics
NT: Differential equations
Integral equations
Level set



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Calibration

UF: Intercalibration
BT: Measurement techniques
RT: Odometry

Californium

BT: Chemical elements

Call admission control

BT: Telecommunication
congestion control
RT: Internet telephony

Call conference

BT: Collaborative tools

Callosal commissure

USE: Corpus callosum

Calorimetry

BT: Measurement
RT: Energy measurement
Thermal variables

measurement

CAM

USE: Computer aided
manufacturing

CAMAC

UF: Computer automated
measurement and control
BT: Control systems
RT: Data buses
Data communication
Data processing
Nuclear measurements

Cambelts

USE: Belts

Camcorders

USE: Video equipment

Cameras

BT: Imaging
RT: Digital photography
Image capture
Image sensors
Motion pictures
Photography
Photorealism
NT: Digital cameras
Smart cameras
Webcams

Cams

BT: Machine components
RT: Engines
Mechanical power

transmission

NT: Camshafts

Camshafts

BT: Cams
Shafts
RT: Automotive components
Belts
Engines

CAN

USE: Controller area networks

Canadian Standards Association

USE: CSA Group

Cancellous bone

BT: Bone tissue
RT: Osteoporosis

Cancer

UF: Malignancy
Malignant
BT: Diseases
RT: Biomedical applications of

radiation

Chemotherapy
Medical diagnostic imaging
Oncological surgery
Oncology
Single photon emission
computed tomography

Tumor microenvironment
Tumors
NT: Brain cancer
Breast cancer
Cervical cancer
Colorectal cancer
Leukemia
Liver cancer
Lung cancer
Metastasis
Ovarian cancer
Pancreatic cancer
Prostate cancer
Skin cancer
Thyroid cancer

Cancer detection

BT: Medical tests



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Cancer drugs

BT: Drugs

Capacitors
Dielectric measurement
Supercapacitors

Cancer treatment

BT: Medical treatment
RT: Monoclonal antibodies
Radiation therapy
NT: Cell therapy
Microwave ablation

Capacitance-voltage characteristics

BT: Electric variables
RT: Capacitance
Voltage

Canning

BT: Packaging
RT: Containers
Material storage
Materials handling
Materials processing

Capacitive sensors

UF: Strain based sensors
Strain sensors
BT: Mechanical sensors

Cantilever beams

USE: Structural beams

Capacitive transducers

BT: Transducers
RT: Capacitance
Position control
Sensors

Cap and trade

USE: Emissions trading

Capacitor testing

USE: Capacitors

Cap-and-trade

USE: Emissions trading

Capacitors

UF: Capacitor testing
Electric condensers
BT: Dielectric devices
Electronic components
Voltage multipliers
RT: Capacitance
Capacitance measurement
Dielectric constant
Electrets
MOS capacitors
Q-factor
Switched capacitor

Capability engineering

BT: Systems engineering and theory

Capability maturity model

BT: Software engineering
RT: Software performance
Software reusability

Capability-based security

BT: Security
RT: Access control

networks

NT: Ceramic capacitors
Power capacitors
Varactors

Capacitance

BT: Electric variables
RT: Capacitance measurement
Capacitance-voltage

characteristics

Capacitive transducers
Capacitors
Supercapacitors
Transmission line theory
NT: Parasitic capacitance
Quantum capacitance

Capacity factor

BT: Power system
measurements
RT: Power generation reliability

Capacity planning

BT: Production planning
RT: Supply chain management
NT: Storage management

Capacitance measurement

BT: Electric variables
measurement
RT: Capacitance

Capital cost reduction

USE: Costing



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CAPTCHAs

BT: Symbols
RT: Access protocols
Authentication

Car parking

BT: Automobiles
NT: Automated parking

Car plate recognition

USE: License plate recognition

Car pools

USE: Shared transport

Car sharing

USE: Shared transport

Car-sharing

USE: Shared transport

Carbinol

USE: Methanol

CarboFullerene

USE: Fullerenes

Carbon

BT: Chemical elements
RT: Carbon compounds
Low carbon economy
Organic compounds
NT: Carbon cycle
Carbon nanotubes
Diamond
Fullerenes
Graphene
Graphite

Carbon accounting

BT: Environmental economics

Carbon budget

BT: Environmental economics

Carbon calculator

BT: Environmental economics
RT: Carbon footprint
Ecological footprint

Carbon capture and storage

BT: Carbon dioxide
NT: Blue carbon
Blue hydrogen
Carbon sequestration

Carbon compounds

BT: Organic compounds
RT: Carbon
NT: Carbon dioxide
Carbon emissions
Carbon monoxide

Carbon cycle

BT: Carbon
RT: Biosphere

Carbon dioxide

BT: Carbon compounds
RT: Carbon footprint
Carbon tax
GOSAT
Greenhouse gases
Net zero

NT: Carbon capture and storage

Carbon dioxide equivalent
Carbon neutral
Tons of CO₂

Carbon dioxide equivalent

UF: CO₂e
BT: Carbon dioxide
RT: Greenhouse gases

Carbon disclosure

BT: Environmental economics

Carbon emissions

BT: Carbon compounds
Gases
RT: Air pollution control
Carbon emissions scope
Climate change
Emissions trading
Global warming
Green communications
Green hydrogen
Greenhouse effect
Methane
Pollution control
NT: Black carbon
Carbon footprint
Carbon neutral

Carbon emissions scope

BT: Environmental economics
RT: Carbon emissions



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Carbon footprint

BT: Carbon emissions
RT: Air pollution
Carbon calculator
Carbon dioxide
Carbon neutral
Climate change
Climate positive
Ecological footprint
Environmental factors
Greenhouse effect

Carbon markets

BT: Environmental economics

Carbon monoxide

BT: Carbon compounds

Carbon nanotube

USE: Carbon nanotubes

Carbon nanotube FETs

USE: CNTFETs

Carbon nanotube field effect transistors

USE: CNTFETs

Carbon nanotube field-effect transistors

USE: CNTFETs

Carbon nanotubes

UF: Carbon nanotube
Carbon-nanotube
Single-wall carbon

nanotubes

BT: Carbon
Nanotubes
RT: CNTFETs

Carbon neutral

UF: Carbon neutralization
BT: Carbon dioxide
Carbon emissions
RT: Carbon footprint
Climate change
Electrification

Carbon neutralization

USE: Carbon neutral

Carbon offsets

BT: Environmental economics

Carbon sequestration

BT: Carbon capture and
storage

RT: Air pollution
Environmental factors
Global warming
Greenhouse effect

Carbon sink

BT: Environmental economics

Carbon tax

BT: Environmental economics
RT: Carbon dioxide

Carbon trading

USE: Emissions trading

Carbon-nanotube

USE: Carbon nanotubes

Carbon-nanotube FETs

USE: CNTFETs

Carbon-nanotube field effect transistors

USE: CNTFETs

Carbon-nanotube field-effect transistors

USE: CNTFETs

Cardiac arrest

UF: Cardiopulmonary arrest
Heart arrest
BT: Cardiovascular diseases
RT: Cardiopulmonary

resuscitation

Cardiac cath

USE: Cardiac catheterization

Cardiac catheterization

UF: Cardiac cath
Heart cath
BT: Medical diagnosis
RT: Heart

Cardiac disease

BT: Cardiovascular diseases

Cardiac function

BT: Heart

Cardiac tissue

BT: Biological tissues
Cardiology



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Cardiac transplants

USE: Heart transplantation

Cardiography

BT: Biomedical imaging
RT: Cardiology
Sputter etching
NT: Echocardiography
Electrocardiography
Phonocardiography
Seismocardiography

Cardiology

BT: Medical specialties
RT: Arrhythmia
Cardiography
Defibrillation
Heart
Pacemakers
Phonocardiography
NT: Cardiac tissue
Cardiopulmonary

resuscitation

Cardiopulmonary arrest

USE: Cardiac arrest

Cardiopulmonary resuscitation

UF: CPR
BT: Cardiology
RT: Cardiac arrest

Cardiovascular diseases

UF: Heart failure
BT: Diseases
NT: Cardiac arrest
Cardiac disease
Myocardial infarction

Cardiovascular system

BT: Anatomy
NT: Baroreflex
Blood vessels
Heart

Career development

BT: Education
NT: Continuing education
Jobs listings
Mentoring

Careers

USE: Engineering profession

Cargo handling

USE: Freight handling

Carotid arteries

UF: Carotoid arteries
BT: Arteries

Carotoid arteries

USE: Carotid arteries

Carpal tunnel syndrome

BT: Medical conditions
RT: Wrist

Carpools

USE: Shared transport

Carrier aggregation

BT: Mobile communication

Carrier confinement

BT: Charge carrier processes

Carrier density

USE: Charge carrier density

Carrier lifetime

USE: Charge carrier lifetime

Carrier processes

USE: Charge carrier processes

Carrier sense multiaccess

USE: Multiaccess communication

Cars

USE: Automobiles

Cartilage

BT: Musculoskeletal system

Cartography

BT: Geography
RT: Geographic information

systems

Cascade lasers

USE: Quantum cascade lasers

Cascading style sheets

BT: Style sheet languages
RT: Markup languages



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Casimir effect

UF: Casimir energy
Casimir force
BT: Electric fields
Nanotechnology
RT: Atomic force microscopy
Elementary particle vacuum
Vacuum systems

Casimir energy

USE: Casimir effect

Casimir force

USE: Casimir effect

Cast iron

BT: Iron
RT: Casting
Production materials

Castellations

USE: Flip chip solder joints

Casting

BT: Materials processing
RT: Cast iron
Foundries
NT: Die casting
Tape casting

Catalysis

BT: Chemical reactions
RT: Hydrolysis
NT: Electrocatalysis
Photocatalysis

Catalysts

BT: Materials
NT: Electrocatalysts
Photocatalysts

Catalytic converters

USE: Exhaust systems

Catalytic convertors

USE: Exhaust systems

Cataracts

BT: Eyes
Medical conditions
RT: Aging
Visual impairment

Catheterization

BT: Medical services
RT: Catheters

Catheters

BT: Biomedical equipment
RT: Catheterization
Surgery

Cathode ray tubes

UF: CRT
BT: Displays
Electron devices
RT: Flyback transformers

Cathode-ray oscilloscopes

USE: Oscilloscopes

Cathodes

BT: Electrodes
RT: Electron emission
Electron tubes
NT: Photocathodes

Cats

UF: Felines
BT: Animals

Cattle

USE: Cows

Causality

USE: Cause effect analysis

Cause effect analysis

UF: Causality
Fishbone diagrams
Ishikawa diagrams
BT: Process planning
RT: Expert systems
Failure analysis
Fault diagnosis
Pareto analysis
Testing

Cavity perturbation methods

BT: Perturbation methods
RT: Cavity resonators

Cavity resonators

BT: Resonators
RT: Cavity perturbation

methods

Klystrons
Microcavities



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	Resonance		Mechanobiology
NT:	Laser cavity resonators		
<i>CCD</i>		Cell therapy	
USE:	Charge coupled devices	UF:	Cell transplantation Cellular therapy Cytotherapy
CCD image sensors		BT:	Cancer treatment
BT:	Image sensors	RT:	Oncology
RT:	Digital photography		
CD recording		<i>Cell transplantation</i>	
UF:	Compact disk	USE:	Cell therapy
BT:	Optical recording		
RT:	Laser applications	Cells (biology)	
NT:	CD-ROMs	BT:	Biological cells
		NT:	Cell membrane Cytometry Cytoplasm Cytoskeleton Cytotoxicity Extracellular Ganglia Glial cells Membrane potentials Microtubules Mitochondria Organelles Organoids Progenitor cells Vesicles
<i>CD-ROM</i>			
USE:	CD-ROMs		
CD-ROMs			
UF:	CD-ROM		
BT:	CD recording		
RT:	Electronic publishing Information systems		
<i>CDMA</i>			
USE:	Multiaccess communication		
<i>Cell biology</i>			
USE:	Biological cells		
<i>Cell biophysics</i>		Cellular biophysics	
USE:	Cellular biophysics	UF:	Cell biophysics
		BT:	Biophysics
<i>Cell clones</i>		RT:	Molecular biophysics Nanomedicine
USE:	Cloning	NT:	Tumor microenvironment
Cell membrane			
UF:	Plasma membrane	<i>Cellular land mobile radio</i>	
BT:	Cells (biology)	USE:	Cellular radio
Cell motility			
UF:	Cellular motility	Cellular manufacturing	
BT:	Biological processes	BT:	Manufacturing systems
RT:	Biochemistry Biomechanics	RT:	Flexible manufacturing Production control
<i>Cell phones</i>		systems	
USE:	Cellular phones	<i>Cellular motility</i>	
		USE:	Cell motility
Cell signaling		Cellular networks	
UF:	Cell signaling	BT:	Cellular radio
BT:	Biological cells	RT:	Cellular technology Cloud radio access
RT:	Biochemistry Biomechanics	networks	



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communication	Device-to-device	4G mobile communication
	Handover	5G mobile communication
	Sidelink	GSM
NT:	Femtocell networks	Land mobile radio networks
	Macrocell networks	Multiaccess communication
	Microcell networks	
	Ultra-dense networks	
Cellular neural networks		<i>Cellular therapy</i>
BT:	Neural networks	USE: Cell therapy
Cellular phones		Cellulose
UF:	Cell phones	BT: Polymers
BT:	Telephone equipment	RT: Biodegradable materials
		Paper pulp
		Polymer fibers
Cellular radio		Cement industry
UF:	Cellular land mobile radio	BT: Manufacturing industries
	Land mobile radio cellular	
systems	Land-mobile radio cellular	Censorship
systems		BT: Law
BT:	Land mobile radio	RT: Consumer protection
RT:	3G mobile communication	Government policies
	4G mobile communication	Law enforcement
	5G mobile communication	Legal factors
	6G mobile communication	
	Bluetooth	Central air conditioning
	Channel estimation	UF: Central air-conditioning
	Code division multiplexing	BT: Air conditioning
	Cross layer design	
	Digital multimedia	<i>Central air-conditioning</i>
broadcasting	Downlink	USE: Central air conditioning
	Film bulk acoustic	
resonators	Intercell interference	<i>Central heating</i>
	Location awareness	USE: District heating
	Multiuser detection	
	NOMA	Central nervous system
	Network resource	BT: Nervous system
management		RT: Hypothalamus
networks	Personal communication	NT: Grey matter
	Software radio	Midbrain
	Time division synchronous	White matter
code division multiple access		Central office
NT:	Cellular networks	BT: Communication networks
	Paging systems	
Cellular technology		Central Processing Unit
BT:	Mobile communication	UF: CPU
	Radio communication	BT: Electronic circuits
RT:	Cellular networks	RT: Hardware acceleration
NT:	3G mobile communication	NT: VLIW
		Centralized control
		UF: Integrated control
		BT: Control systems



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Cepstral analysis

BT: Acoustics
RT: Music information retrieval
Speech analysis
Speech recognition
NT: Cepstrum
Mel frequency cepstral

coefficient

Cepstrum

BT: Cepstral analysis
RT: Fourier transforms

Ceramic capacitors

BT: Capacitors
Ceramic products

Ceramic glazes

BT: Glazes

Ceramic products

BT: Manufactured products
RT: Ceramics
Glass products
Insulators
Porcelain
Tiles
NT: Ceramic capacitors

Ceramics

UF: Glass ceramics
BT: Insulation
Materials
RT: Aluminum oxide
Ceramic products
Ceramics industry
Cermets
Dielectric materials
Diffusion bonding
Electrets
Firing
Glass
Glass products
Glazes
High-temperature

superconductors

Magnesium oxide
Powders
Tape casting
Tiles
NT: Bioceramics
Porcelain

Ceramics industry

BT: Manufacturing industries
RT: Bioceramics
Ceramics
Porcelain

Cerebellum

BT: Brain

Cerebral cortex

BT: Brain

Cerebral palsy

BT: Medical conditions
RT: Brain injuries
Spasticity

Cerebrospinal fluid

BT: Fluids and secretions
Spinal cord
RT: Brain

Cerebrum

BT: Brain

Cerenkov lasers

USE: Free electron lasers

Cerium

BT: Chemical elements

Cermet

BT: Composite materials
RT: Ceramics
Metallic materials

CERN

UF: European Organization for
Nuclear Research
BT: Organizations
RT: High energy physics
Laboratories
Nuclear physics
Particle accelerators
Research and development

Certification

BT: Training
RT: Conformance testing

Cervical cancer

BT: Cancer
RT: Cervix



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Cervix	BT: Uterus RT: Cervical cancer	BT: Communication channels RT: Cellular radio Equalizers Land mobile radio Multipath channels Signal detection Spread spectrum
Cesium	UF: Caesium BT: Chemical elements	communication
<i>CFC</i>	USE: Chlorofluorocarbons	Channel hot electron injection
<i>CFD</i>	USE: Computational fluid dynamics	UF: Channel hot-electron injection BT: Hot carrier injection
<i>CGM</i>	USE: Computer generated music	<i>Channel hot-electron injection</i> USE: Channel hot electron injection
<i>Chain codes</i>	USE: Blockchains	Channel impulse response UF: CIR BT: Channel state information RT: Frequency response Impulse testing
Chalcogenides	BT: Chemical compounds	Channel models BT: Communication channels
Change detection algorithms	BT: Algorithms	Channel rate control BT: Rate distortion theory
Channel allocation	UF: Bandwidth allocation BT: Communication channels NT: Spectral efficiency	Channel spacing BT: Communication channels RT: Optical fiber applications Rate distortion theory
Channel bank filters	BT: Filters	<i>Channel state estimation</i> USE: Channel estimation
Channel capacity	BT: Communication channels RT: Quantum communication	Channel state information BT: Communication channels NT: Channel impulse response
Channel coding	BT: Encoding Information theory RT: Communication channels Convolutional codes Polar codes Rate distortion theory Space-time codes NT: Block codes Combined source-channel coding Turbo codes	<i>Channel-state estimation</i> USE: Channel estimation
Channel estimation	UF: Channel state estimation Channel-state estimation	Chaos BT: Nonlinear systems RT: Bifurcation Econophysics Fractals Nonlinear circuits Nonlinear dynamical systems Pattern formation Predator prey systems



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NT: Random media
Chaotic communication
Complexity theory
Spatiotemporal phenomena

Electron mobility
Electron traps
Excitons
Space charge

Chaotic communication

BT: Chaos
RT: Cryptography
Synchronization
Time series analysis

Chaotic mapping

BT: Cryptography
RT: Encryption
Nonlinear equations
Random sequences

Character generation

BT: Graphics
RT: Computer graphics
Displays
Printing

Character recognition

UF: Print readers
BT: Pattern recognition
RT: License plate recognition
Text recognition

Characteristic mode analysis

BT: Electromagnetic analysis

Charge carrier density

UF: Carrier density
BT: Charge carriers

Charge carrier lifetime

UF: Carrier lifetime
BT: Charge carriers

Charge carrier mobility

BT: Charge carriers

Charge carrier processes

UF: Carrier processes
Charge carrier trapping
Electron carriers
Hole carriers
Semiconductor charge

carriers

BT: Charge carriers
RT: Diffusion processes
Semiconductor impurities
NT: Carrier confinement
Charge transfer

Charge carrier trapping

USE: Charge carrier processes

Charge carriers

BT: Elementary particles
RT: Conductivity
Impact ionization
Semiconductivity
Semiconductor materials
NT: Charge carrier density
Charge carrier lifetime
Charge carrier mobility
Charge carrier processes
Hot carriers

Charge coupled devices

UF: CCD
Charge injection devices
Charge transfer devices
Charge-injection devices
Charge-transfer devices
BT: MIS devices

Charge injection devices

USE: Charge coupled devices

Charge measurement

BT: Electrostatic measurements
RT: Pulsed electroacoustic
methods
NT: Battery charge
measurement

Charge pumps

UF: Charge-pumping
BT: Circuits
RT: Voltage multipliers

Charge transfer

BT: Charge carrier processes

Charge transfer devices

USE: Charge coupled devices

Charge-coupled image sensors

BT: Image sensors
Optoelectronic devices

Charge-injection devices

USE: Charge coupled devices



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Charge-pumping

USE: Charge pumps

Charge-transfer devices

USE: Charge coupled devices

Charged device model

USE: Electrostatic discharges

Charging devices

USE: Battery chargers

Charging stations

BT: Power supplies
RT: Battery chargers
Battery powered vehicles
Electric vehicles
Hybrid electric vehicles
Plug-in hybrid electric

vehicles

NT: Docking stations
Fast charging

Chatbots

UF: Bot
ChatGPT
BT: Conversational artificial

intelligence

Human computer

interaction

Natural language

processing

RT: Speech synthesis
Auditory system
Digital humans
Foundation models
Generative Pre-Trained

Transformer

Human factors
Prompt engineering
Question answering

(information retrieval)

Question generation

ChatGPT

USE: Chatbots

CHCP

USE: Trigeration

Chebyshev approximation

BT: Approximation methods
RT: Discrete cosine transforms

Checkpointing

BT: System recovery

Cheese

USE: Dairy products

Chemical analysis

BT: Chemistry
Materials science and

technology

RT: Chemical technology
Drugs
Fractionation
Activation analysis
Chemical processes
Chemicals
Electronic noses
pH measurement

Chemical and biological sensors

BT: Sensors
NT: Biosensors
Gas detectors

Chemical compounds

BT: Chemistry
NT: Anti-freeze
Bromine compounds
Chalcogenides
Chlorophyll
Ethanol
Methanol
Radiotracer

Chemical elements

BT: Materials, elements, and
compounds

NT: Aluminum
Americium
Antimony
Arsenic
Beryllium
Boron
Bromine
Californium
Carbon
Cerium
Cesium
Chlorine
Dysprosium
Europium
Fluorine
Gadolinium
Hafnium
Helium



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Holmium
Hydrogen
Iodine
Iridium
Isotopes
Krypton
Lutetium
Mercury (metals)
Molybdenum
Neon
Neptunium
Nitrogen
Osmium
Oxygen
Phosphorus
Plutonium
Polonium
Potassium
Praseodymium
Promethium
Protactinium
Radium
Radon
Rhenium
Rhodium
Roentgenium
Rubidium
Ruthenium
Scandium
Selenium
Sodium
Sulfur
Tantalum
Technetium
Tellurium
Terbium
Thallium
Thorium
Thulium
Titanium
Uranium
Vanadium
Ytterbium
Yttrium
Zirconium

RT: Biohazards
Contamination
Explosions
Hazardous materials
Toxicology
NT: Chemical weapons
Toxic chemicals

Chemical industry

BT: Industries
RT: Chemical engineering
Chemical reactions
Chemical technology
Electrochemical processes
Pesticides
Petrochemicals
Petroleum industry
Pipelines
Plastic products
Plastics industry
Rubber industry

Chemical lasers

BT: Lasers
RT: Gas lasers

Chemical mechanical planarisation

USE: Planarization

Chemical mechanical planarization

USE: Planarization

Chemical processes

BT: Chemical analysis
NT: Acidification
Fermentation
Hydrophobicity
Leaching
Molecular sieves
Osmosis
Oxidation
Reverse osmosis
Solvents
Thermolysis
Water splitting

Chemical engineering

BT: Engineering - general
RT: Chemical industry
Chemical products
Chemical technology
Process design

Chemical products

BT: Manufactured products
RT: Chemical engineering
Chemistry
Glass products
Pesticides
Plastic products
Production materials
NT: Agrochemicals

Chemical hazards

BT: Hazards



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Antioxidants

Fats

Inhibitors

Lacquers

Mortar

Paints

Petrochemicals

Petroleum

Pharmaceuticals

Phthalates

Plastics

Propellants

Chemical reaction network theory

BT: Mathematics

RT: Biochemistry

Chemical reactions

Chemical reactions

UF: CSTR

BT: Chemical technology

RT: Chemical industry

Chemical reaction network

theory

Crystallizers

Process control

Water splitting

NT: Bioreactors

Catalysis

Chemical reduction

Chemiluminescence

Continuous-stirred tank

reactor

Hydrolysis

Ignition

Solvation

Chemical reduction

BT: Chemical reactions

NT: Redox

Chemical sensors

BT: Chemical technology

RT: Detectors

Chemical technology

BT: Industry applications

RT: Chemical analysis

Chemical engineering

Chemical industry

Chemistry

Decontamination

Refining

NT: Chemical reactions

Chemical sensors

Crystallizers

Distillation equipment

Fluidization

Pharmaceutical technology

Vitrification

Chemical transducers

BT: Transducers

RT: Gas detectors

Chemical vapor deposition

UF: CVD

Chemical vapour deposition

BT: Plasma materials

processing

RT: Coatings

Epitaxial layers

Films

NT: Atmospheric pressure

chemical vapor deposition

Atomic layer deposition

MOCVD

Pulsed laser deposition

Chemical vapour deposition

USE: Chemical vapor deposition

Chemical warfare

USE: Chemical weapons

Chemical weapons

UF: Chemical warfare

BT: Chemical hazards

Weapons

RT: Biological weapons

National security

Nuclear weapons

Terrorism

US Department of

Homeland Security

Weapons of mass

destruction

Chemicals

BT: Chemical analysis

RT: Plastics

Chemiluminescence

UF: Chemoluminescence

BT: Chemical reactions

Luminescence

RT: Electroluminescence

Photoluminescence

Thermoluminescence

NT: Bioluminescence



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Chemistry

BT: Science - general
 RT: Chemical products
 Chemical technology
 Drugs
 Petrochemicals
 Pharmaceutical technology
 Pharmaceuticals
 Pickling
 Plastic products
 NT: Astrochemistry
 Biochemistry
 Biogeochemistry
 Chemical analysis
 Chemical compounds
 Computational chemistry
 Electrochemistry
 Geochemistry
 Inorganic chemicals
 Interstellar chemistry
 Organic chemicals
 Photochemistry
 Physical chemistry
 Quantum chemistry

Chemoluminescence

USE: Chemiluminescence

Chemotherapy

BT: Medical treatment
 RT: Cancer
 Drugs
 Medical services
 Oncology
 Patient monitoring

Chest medicine

USE: Pulmonology

Child

USE: Pediatrics

Children

USE: Pediatrics

Chip design

USE: Chip scale packaging

Chip development

USE: Chip scale packaging

Chip fabrication

USE: Chip scale packaging

Chip scale packaging

UF: CSP
 Chip design
 Chip development
 Chip fabrication
 Chip-making process
 BT: Electronics packaging
 RT: Integrated circuit packaging
 NT: System-in-package

Chip-making process

USE: Chip scale packaging

Chipless RFID

BT: RFID tags

Chipleths

BT: Integrated circuits

Chirp

BT: Signal processing

Chirp modulation

UF: Linear frequency
 modulation

BT: Modulation
 RT: Sonar
 Spread spectrum

communication

Spread spectrum radar

Chlorine

BT: Chemical elements
 NT: Chlorine compounds

Chlorine compounds

UF: Chlorine dioxide
 Chloroform
 Hydrogen chloride
 Sodium chloride
 BT: Chlorine
 NT: Salt

Chlorine dioxide

USE: Chlorine compounds

Chlorofluorocarbons

UF: CFC
 BT: Fluorine compounds
 RT: Greenhouse gases

Chloroform

USE: Chlorine compounds



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Chlorophyll

BT: Chemical compounds
Plants (biology)
RT: Photosynthesis

Chokes

USE: Inductors

Cholera

BT: Diseases

Choppers (circuits)

BT: Switching circuits
RT: Power conversion

CHP

USE: Cogeneration

Chromatic dispersion

BT: Dispersion

Chrome plating

BT: Plating
RT: Coatings

Chromium

UF: Cr
BT: Metals
NT: Chromium alloys

Chromium alloys

BT: Chromium

Chromosome mapping

BT: Biological cells

Chromosomes

USE: Biological cells

Chronic kidney disease

UF: CKD
BT: Kidney
Medical conditions

Chronic obstructive pulmonary disease

UF: COPD
BT: Pulmonary diseases
RT: Asthma
Emphysema
Pneumonia

Chronobiology

UF: Biological clocks
BT: Biological processes
RT: Phenology

Churn

BT: Customer satisfaction

CIM

USE: Common Information Model
(computing) AND
Common Information Model
(electricity) AND
Computer integrated
manufacturing AND
In-memory computing

Cinema

USE: Motion pictures

Cinematography

BT: Photography
RT: Motion pictures
Object tracking

Ciphers

UF: Cryptograph
Cyphers
BT: Cryptography
RT: Algorithms
Codes
Encryption
S Box

CIR

USE: Channel impulse response

Circadian rhythm

UF: Circadian rhythms
BT: Biological processes

Circadian rhythms

USE: Circadian rhythm

Circuit analysis

BT: Circuits
RT: Frequency-domain analysis
SPICE
Sensitivity
Tolerance analysis
Yield estimation
NT: Circuit analysis computing
Coupled mode analysis
Nonlinear network analysis

Circuit analysis computing

BT: Circuit analysis

Circuit boards

USE: Printed circuits



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Circuit breakers

BT: Switchgear
RT: Interrupters
Power system protection
Protection
Switching circuits
NT: Molded case circuit

breakers

Circuit CAD

USE: Design automation

Circuit complexity

USE: Complexity theory

Circuit design

USE: Circuit synthesis

Circuit design (CAD)

USE: Design automation

Circuit design (logic)

USE: Logic design

Circuit faults

BT: Circuits
NT: Electrical fault detection

Circuit feedback

USE: Feedback circuits

Circuit layout CAD

USE: Design automation

Circuit noise

BT: Circuits
RT: Transmission lines
NT: Thermal noise

Circuit optimisation

USE: Circuit optimization

Circuit optimization

UF: Circuit optimisation
Circuit performance
Circuit tuning
BT: Optimization methods
RT: Tolerance analysis

Circuit performance

USE: Circuit optimization

Circuit reliability

BT: Circuits

Circuit simulation

BT: Circuits
RT: Design automation
Semiconductor process

modeling

Circuit stability

BT: Stability
RT: Grounding
Jitter

Circuit subsystems

BT: Solid state circuits
RT: Circuits and systems

Circuit synthesis

UF: Circuit design
BT: Circuits
RT: Control system synthesis
Logic design
Solid state circuit design
NT: High level synthesis
Integrated circuit synthesis
Register transfer level

Circuit testing

BT: Testing
RT: Built-in self-test
NT: Integrated circuit

measurements

Circuit theory

BT: Solid state circuits
NT: Inductive coupling

Circuit tolerance analysis

USE: Tolerance analysis

Circuit topology

BT: Digital circuits
RT: Graph theory
Tree graphs
NT: Current mirrors

Circuit tuning

USE: Circuit optimization

Circuits

BT: Circuits and systems
RT: Flow graphs
Impedance matching
Oscillators
Phase transformers
Poles and zeros
Scattering parameters



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NT:	Active circuits		Pulse circuits
	Adders		Quantum circuit
	Analog circuits		RLC circuits
integrated circuits	Application specific		Radiation detector circuits
			Rail to rail operation
	Asynchronous circuits		Rectifiers
	Bipolar transistor circuits		Sampled data circuits
	Bistable circuits		Sequential circuits
	Bridge circuits		Silicon-on-insulator
	Charge pumps		Submillimeter wave circuits
	Circuit analysis		Summing circuits
	Circuit faults		Switched circuits
	Circuit noise		Switching circuits
	Circuit reliability		Thick film circuits
	Circuit simulation		Thin film circuits
	Circuit synthesis		Thyristor circuits
	Coprocessors		Time varying circuits
	Counting circuits		Trigger circuits
	Coupling circuits		UHF circuits
	Digital circuits		UHF integrated circuits
	Digital signal processors		Ultra large scale integration
	Distributed parameter		VHF circuits
circuits			Very large scale integration
	Driver circuits		Voltage multipliers
	Electronic circuits		Wafer scale integration
	Equivalent circuits		
	Feedback	Circuits and systems	
	Hybrid integrated circuits	RT:	Circuit subsystems
	Integrated circuits		Formal verification
	Isolators		Solid state circuits
	Large scale integration	NT:	Circuits
	Linear circuits		Contacts
	Logic arrays		Filtering
	Logic circuits		Integrated circuit
	MOSFET circuits	technology	
	Magnetic circuits		Logic devices
	Microprocessors		Oscillators
	Microwave circuits		Single electron devices
	Millimeter wave circuits		Tunable circuits and
	Millimeter wave integrated	devices	
circuits			
	Monolithic integrated	Circular economy	
circuits		BT:	Environmental economics
	Multiplying circuits		Resource management
	Neural circuits	RT:	Biodiversity
	Nonlinear circuits		Cradle to cradle
	Passive circuits		Energy management
	Phase shifters		Environmental
	Power dissipation	management	
	Power integrated circuits		Recycling
	Printed circuits		Regeneration engineering
	Programmable circuits		Renewable energy sources
	Programmable logic arrays		Sustainable development
	Programmable logic		Waste management
devices			



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Circular polarisation

USE: Polarization

Circular polarization

USE: Polarization

Circular waveguides

BT: Electromagnetic waveguides

Circulators

BT: Ferrite devices
Microwave technology
RT: Electromagnetic coupling
Waveguide components

Circulatory system

UF: Vascular system
BT: Anatomy

Citation analysis

UF: Citation identification
Citation studies
BT: Bibliometrics

Citation identification

USE: Citation analysis

Citation studies

USE: Citation analysis

Cities and towns

USE: Urban areas

City

USE: Urban areas

City planning

USE: Urban planning

Civil engineering

BT: Engineering - general
RT: Bridges
Buildings
Construction
Energy resources
Environmental factors
Power systems
Road transportation
Roads
Transmission lines
NT: Geotechnical engineering
Geotechnical structures
Railway engineering
Structural engineering

CKD

USE: Chronic kidney disease

Cladding techniques

BT: Coating techniques
RT: Claddings

Claddings

BT: Coatings
Optical fiber cables
RT: Cladding techniques
NT: Laser cladding

Cladistics

USE: Phylogeny

Clamping

USE: Clamps

Clamps

UF: Clamping
BT: Production equipment
RT: Machine tools
Machining

Classification algorithms

BT: Algorithms
NT: Multi label classification
Relevance vector machines

Classification tree analysis

BT: Decision trees
RT: Formal concept analysis

Clean energy

USE: Green energy AND
Renewable energy sources

Cleaning

BT: Materials handling
RT: Air cleaners
Refining
NT: Purification
Surface cleaning

Client server model

USE: Client-server systems

Client server systems

USE: Client-server systems

Client-server model

USE: Client-server systems



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Client-server systems

UF: Client server model
Client server systems
Client-server model
Clientserver systems
BT: Distributed computing
Software architecture
RT: Dew computing
Unified modeling language
NT: Middleware
Servers

Clientserver systems

USE: Client-server systems

Climate

BT: Meteorology
NT: Climate variability

Climate change

UF: Climate fluctuations
Climate mitigation
Climate warming
Climatic change
BT: Environmental factors
RT: Carbon emissions
Carbon footprint
Carbon neutral
Climate variability
Emissions trading
Long-term climate

strategies

Loss and damage
Paris Agreement
Phenology
Science based targets
Tipping point
Transparency (climate

reporting)

NT: Climate change adaptation
Climate change mitigation
Climate crisis
Climate justice
Climate overshoot
Climate positive
Climate resilience
Climate security
Conference of the Parties
Eco-conscious
Eco-friendly
Ecodistricts
Global warming
Intergovernmental Panel on

Climate Change

Nationally Determined

Contributions

United Nations Framework

Convention on Climate Change

Climate change adaptation

BT: Climate change

Climate change mitigation

BT: Climate change

Climate crisis

BT: Climate change

Climate finance

BT: Environmental economics

Climate fluctuations

USE: Climate change

Climate justice

BT: Climate change

Climate mitigation

USE: Climate change

Climate overshoot

BT: Climate change

Climate positive

BT: Climate change
RT: Carbon footprint

Climate resilience

BT: Climate change

Climate science

USE: Climatology

Climate security

BT: Climate change

Climate variability

UF: Climate variations
BT: Climate
RT: Climate change
Weather modification
NT: El Nino
El Nino-Southern

Oscillation

Climate variations

USE: Climate variability



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Climate warming

USE: Climate change

Climatic change

USE: Climate change

Climatology

UF: Climate science
BT: Atmospheric science
RT: Meteorology

Climbing robots

BT: Mobile robots

Clinical analysis

USE: Clinical diagnosis AND
Clinical trials

Clinical diagnosis

UF: Clinical analysis
Clinical engineering
Clinical information
BT: Medical services
RT: Point of care
NT: Clinical neuroscience

Clinical engineering

USE: Clinical diagnosis

Clinical equipment

USE: Biomedical equipment

Clinical information

USE: Clinical diagnosis

Clinical neuroscience

BT: Clinical diagnosis
Neuroscience

Clinical trials

UF: Clinical analysis
BT: Medical treatment

Clock synchronization

USE: Synchronization

Clocks

BT: Time measurement
RT: Timing
NT: Atomic clocks
Watches

Clone

USE: Cloning

Clones

USE: Cloning

Cloning

UF: Cell clones
Clone
Clones
Human cloning
Molecular clones
Reproductive cloning
BT: Biomedical engineering
RT: DNA
Stem cells

Closed box

UF: Blackbox
Closed-box
BT: Software
System analysis and design
RT: Software testing
System testing

Closed captioning

UF: Vision captioning
Visual captioning
BT: Assistive technologies
Communication aids
RT: Media
TV

Closed form solution

USE: Closed-form solutions

Closed loop systems

UF: Closed-loop systems
BT: Control systems
RT: H infinity control

Closed-box

USE: Closed box

Closed-form expression

USE: Closed-form solutions

Closed-form solutions

UF: Closed form solution
Closed-form expression
BT: Mathematics

Closed-loop systems

USE: Closed loop systems

Cloth face masks

USE: Face masks



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Clothing

UF: Garments
BT: Consumer products
RT: Clothing industry
Fabrics
Wool
NT: Footwear
Protective clothing

Clothing industry

UF: Fashion industry
Garment industry
BT: Manufacturing industries
RT: Clothing
Footwear
Footwear industry
Protective clothing
Textile industry

Clotting

USE: Coagulation

Cloud computing

BT: Internet
RT: Big Data applications
Dew computing
Distributed computing
Edge computing
Grid computing
Internet of Things
Network function

virtualization

Service computing
Software as a service
Software defined

networking

Virtual power plants
Web services
NT: Cloud computing security
Cloud gaming
Cloud robotics
Elastic computing
Platform as a service
Serverless computing

Cloud computing security

UF: Cloud security
BT: Cloud computing
Computer security

Cloud gaming

UF: Gaming on demand
BT: Cloud computing
Games
RT: Online services

Cloud radio access networks

BT: Network function
virtualization
Radio access networks
RT: Cellular networks

Cloud robotics

UF: Cloud robots
BT: Cloud computing
Robotics and automation

Cloud robots

USE: Cloud robotics

Cloud security

USE: Cloud computing security

Cloud-dew architecture

USE: Dew computing

Cloud-dew computing

USE: Dew computing

Clouds

BT: Terrestrial atmosphere

Cluster computing

UF: Apache hadoop
Apache spark
Network of workstations
Workstation clusters
BT: Distributed computing
RT: Distributed processing
Message systems
Parallel processing
Peer-to-peer computing
Resource management
Workstations

Clustered regularly interspaced short palindromic repeats

USE: CRISPR

Clustering algorithms

BT: Algorithms
NT: Affinity propagation

Clustering methods

BT: Pattern recognition
RT: Extreme learning machines
Fish schools
NT: Pattern clustering

Clutter

BT: Interference



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	RT:	Echo interference	<i>CMOS memory integrated circuits</i>	
<i>CMM</i>			USE:	CMOS memory circuits
	USE:	Coordinate measuring machines	CMOS process	
			BT:	CMOS technology
CMOS analog integrated circuits			CMOS technology	
UF:	Analog CMOS integrated circuits		BT:	Integrated circuit technology
	Analogue CMOS integrated circuits		RT:	MOSFET Microcontrollers Microprocessors Transistors
	CMOS analogue integrated circuits		NT:	CMOS process Silicon on sapphire
BT:	Analog integrated circuits CMOS integrated circuits			
<i>CMOS analogue integrated circuits</i>			CMOSFET circuits	
USE:	CMOS analog integrated circuits		BT:	MOSFET circuits
			RT:	Rail to rail operation
CMOS digital integrated circuits			CMOSFET logic devices	
BT:	CMOS integrated circuits		BT:	CMOSFETs
NT:	CMOS logic circuits		RT:	MISFETs MOSFET P-i-n diodes
CMOS image sensors			CMOSFETs	
BT:	Image sensors		BT:	MOSFET
CMOS integrated circuits			RT:	CMOS integrated circuits Semiconductor-insulator interfaces
BT:	Integrated circuits		NT:	CMOSFET logic devices
RT:	CMOSFETs Neuromorphic engineering		<i>CNC</i>	
NT:	CMOS analog integrated circuits		USE:	Computer numerical control
	CMOS digital integrated circuits		<i>CNFETs</i>	
	CMOS logic circuits CMOS memory circuits Transconductors		USE:	CNTFETs
CMOS logic circuits			<i>CNN</i>	
BT:	CMOS digital integrated circuits		USE:	Convolutional neural networks
	CMOS integrated circuits		CNTFETs	
RT:	Application specific integrated circuits		UF:	CNFETs Carbon nanotube FETs Carbon nanotube field-effect transistors
	Power dissipation			Carbon nanotube field-effect transistors
CMOS memory circuits				Carbon-nanotube FETs Carbon-nanotube field-effect transistors
UF:	CMOS memory integrated circuits			Carbon-nanotube field-effect transistors
BT:	CMOS integrated circuits			
RT:	Memory SRAM chips			



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BT:	Field effect transistors	Coating techniques	BT:	Coatings
RT:	Carbon nanotubes		NT:	Cladding techniques
	Quantum capacitance			
<i>Co-channel interference</i>		Coatings	BT:	Materials processing
USE:	Interchannel interference		RT:	Chemical vapor deposition
<i>CO₂e</i>				Chrome plating
USE:	Carbon dioxide equivalent			Corrosion
<i>Coagulate</i>				Films
USE:	Coagulation			Magnetic multilayers
Coagulation				Painting
UF:	Blood clots			Spraying
	Clotting			Sputtering
	Coagulate	NT:		Vapor deposition
BT:	Biological processes			Antireflection coatings
	Blood			Claddings
RT:	Blood platelets			Coating techniques
	Hematology			Dip coating
Coal				Epitaxial layers
BT:	Fuels			Glazes
RT:	Coal gas			Lacquers
	Coal mining			Paints
<i>Coal ash</i>		Coaxial cables		Powders
USE:	Fly ash	BT:		Cables
		RT:		Electromagnetic
Coal gas		waveguides		
UF:	Coal gasification			Transmission lines
	Illumination gas	NT:		Coaxial components
	Town gas			Hybrid fiber coaxial cables
BT:	Gases	Coaxial components		
RT:	Coal	BT:		Coaxial cables
	Fuels	Cobalt		
<i>Coal gasification</i>		BT:		Metals
USE:	Coal gas	RT:		Alloying
				Magnets
Coal industry		NT:		Cobalt alloys
BT:	Industries	Cobalt alloys		
RT:	Coal mining	BT:		Cobalt
		RT:		Alloying
Coal mining		<i>Cobots</i>		
BT:	Mining industry	USE:		Collaborative robots
RT:	Coal			
	Coal industry	<i>Cochannel interference</i>		
<i>Coal tar</i>		USE:		Interchannel interference
USE:	Fuel processing industries	Cochlear implants		
Coastlines		BT:		Implants
BT:	Water	RT:		Ear



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<i>Code division multiaccess</i>			Encoding
USE:	Multiaccess communication		Error correction
<i>Code division multiple access</i>			Redundancy
USE:	Multiaccess communication		Sequences
<i>Code division multiplexed</i>		NT:	Vector quantization
USE:	Code division multiplexing		Binary codes
			Convolutional codes
			Cyclic redundancy check
Code division multiplexing		codes	Error correction codes
UF:	Code division multiplexed		Parity check codes
	OCDM		Product codes
	Optical code division		Space-time codes
multiplexing			Zero correlation zone
BT:	Communication switching		
	Multiplexing	<i>Coding</i>	
RT:	Cellular radio	USE:	Encoding
	Codes		
	Multicarrier code division	<i>Coding theory</i>	
multiple access		USE:	Information theory
	Optical fiber applications		
	Software radio	Codons	
	Spread spectrum	BT:	Genomics
communication		RT:	DNA
			RNA
Code refracting		Coercive force	
UF:	Refracting	UF:	Coercivity
BT:	Encoding	BT:	Magnetic forces
RT:	Information theory		
	Software engineering	<i>Coercivity</i>	
<i>Code-division multiple access</i>		USE:	Coercive force
USE:	Multiaccess communication		
<i>Code-division multiple-access</i>		<i>Cogen</i>	
USE:	Multiaccess communication	USE:	Cogeneration
Codecs		Cogeneration	
UF:	Coder-decoders	UF:	CHP
BT:	Communication equipment		Cogen
RT:	Decoding		Combined heat and power
	Encoding	BT:	Heating systems
	Speech codecs		Power generation
NT:	Video codecs	RT:	Industrial power systems
<i>Coder-decoders</i>			Trigeneration
USE:	Codecs		Waste heat
Codes		<i>Cogging</i>	
UF:	Parity check	USE:	Forging
BT:	Information theory		
RT:	Ciphers	Cognition	
	Code division multiplexing	UF:	Reasoning
	Cryptography	BT:	Behavioral sciences
	Decoding	RT:	Active perception
			Brain
			Cognitive systems



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	Cultural competence		Computational and artificial
	Digital intelligence	intelligence	
	Psychology		Human factors
NT:	Activity recognition		Inference mechanisms
	Cognitive load		Logic
	Cognitive neuroscience		Psychology
	Cognitive processes		Uncertainty
	Commonsense reasoning	NT:	Human intelligence
	Metacognition		Problem-solving
	Self-aware		
<i>Cognitive computing</i>		Cognitive systems	
USE:	Cognitive systems	UF:	Cognitive computing
			Reasoning
Cognitive informatics		BT:	Artificial intelligence
BT:	Cybernetics		Learning systems
	Informatics	RT:	Adaptive control
RT:	Brain		Affective computing
			Automata
Cognitive load			Cognition
BT:	Cognition		Cybernetics
RT:	Cognitive processes		Fuzzy cognitive maps
	Memory		Machine learning
		Coherence	
Cognitive neuroscience		UF:	Coherent detection
BT:	Cognition	BT:	Electromagnetic scattering
	Neuroscience	RT:	Interference
			Quantum decoherence
Cognitive processes		NT:	Coherence time
BT:	Cognition		
RT:	Cognitive load	Coherence time	
		BT:	Coherence
Cognitive radar			Quantum mechanics
BT:	Adaptive systems	RT:	Bandwidth
	Radar		Electromagnetic scattering
			Quantum computing
Cognitive radio			Qubit
UF:	Cognitive radio network		
BT:	Wireless communication	<i>Coherent detection</i>	
<i>Cognitive radio network</i>		USE:	Coherence
USE:	Cognitive radio		
Cognitive robotics		Coilguns	
UF:	Cognitive robots	BT:	Electromagnetic launching
BT:	Robots		
RT:	Autonomous robots	Coils	
<i>Cognitive robots</i>		UF:	Electric coils
USE:	Cognitive robotics	BT:	Electronic components
		RT:	Electromagnets
Cognitive science			Generators
UF:	Mental models		Inductance
BT:	Cybernetics		Inductors
RT:	Brain		Magnetic circuits
			Motors
			Rotating machines
			Transformers



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NT: Windings
Superconducting coils

Cold chain

UF: Cold chain supply
Cold chain transportation
BT: Supply chains

Cold chain supply

USE: Cold chain

Cold chain transportation

USE: Cold chain

Cold plates

BT: Cooling

Collaboration

BT: Professional
communication
RT: Collective intelligence
Cyber-physical systems
Information sharing
Interoperability
Social Internet of Things
NT: Collaborative robots
Collaborative tools
Discussion forums
Hackathon
Teamwork
Virtual groups

Collaborative filtering

BT: Filtering theory
RT: Recommender systems

Collaborative intelligence

BT: Federated learning
Multi-agent systems
RT: Distributed management
Intelligent systems

Collaborative learning

USE: Federated learning

Collaborative networking

USE: Federated learning

Collaborative problem solving

USE: Federated learning

Collaborative robots

UF: Cobots
BT: Collaboration
Robots

RT: Field robots

Collaborative software

UF: Groupware
BT: Collaborative tools
RT: Communication system
software
Federated learning

Collaborative tools

BT: Collaboration
NT: Call conference
Collaborative software
Videoconferences

Collaborative work

USE: Federated learning

Collective bargaining

USE: Industrial relations

Collective intelligence

BT: Decision making
Intelligent systems
RT: Collaboration
Crowdsensing
Crowdsourcing
Sociology
NT: Fish schools

Collective learning

BT: Learning systems

Colleges

USE: Educational institutions

Colliding beam accelerators

BT: Colliding beam devices
Particle accelerators
RT: Klystrons
Particle beams
Synchrotrons

Colliding beam devices

BT: Nuclear and plasma
sciences
RT: Particle accelerators
NT: Colliding beam accelerators
Muon colliders

Collimators

UF: Multileaf collimators
BT: Optical devices
RT: Biomedical applications of
radiation



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	Biomedical equipment	<i>Colon cancer</i>	
	Dosimetry	USE:	Colorectal cancer
	Gamma-rays		
	Linear accelerators	Colonic polyps	
computed tomography	Single photon emission	BT:	Colon
	X-ray applications		Tumors
	X-rays	RT:	Colorectal cancer
Collision avoidance		Colonography	
UF:	Collision detection	BT:	Medical diagnosis
	Obstacle avoidance		
	Sense and avoid	Colonoscopy	
BT:	Motion control	BT:	Medical tests
RT:	Advanced driver assistance	NT:	Virtual colonoscopy
systems		Color	
	Block signalling	BT:	Optics
	Lane departure warning	RT:	Electrochromism
systems			Imaging
	Lane detection		Photochromism
	Vehicle crash testing	NT:	Color gamut
<i>Collision detection</i>			Color temperature
USE:	Collision avoidance		Pigmentation
Collision mitigation		<i>Color blindness</i>	
BT:	Motion control	USE:	Vision defects
<i>Collision theory</i>		<i>Color center lasers</i>	
USE:	Kinetic theory	USE:	Solid lasers
Colloidal crystals		Color gamut	
BT:	Crystals	UF:	Colour gamut
RT:	Crystallizers	BT:	Color
NT:	Ice clouds	RT:	Image color analysis
			Light sources
			Printing
Colloidal lithography		Color temperature	
BT:	Lithography	UF:	Colour temperature
	Nanopatterning	BT:	Color
RT:	Biomedical engineering		Temperature
	Nanobioscience		
	Nanotopography	Color TV	
	Polymers	BT:	TV
	Surface treatment		
	Tissue engineering	Colorectal cancer	
Colloidal nanocrystals		UF:	Colon cancer
BT:	Nanocrystals	BT:	Cancer
	Optical materials		Colon
		RT:	Colonic polyps
Colon		Colored noise	
BT:	Digestive system	UF:	Coloured noise
NT:	Colonic polyps	BT:	Noise
	Colorectal cancer		



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Colossal magnetoresistance

BT: Magnetoresistance

Colour gamut

USE: Color gamut

Colour temperature

USE: Color temperature

Coloured noise

USE: Colored noise

Comb filters

BT: Filters

Combinational circuits

UF: Combinational logic circuits

BT: Logic circuits

Combinational logic circuits

USE: Combinational circuits

Combinatorial mathematics

BT: Mathematics

NT: Graph theory

Steiner trees

Combinatorial software testing

USE: Combinatorial testing

Combinatorial testing

UF: Combinatorial software

testing

BT: Software testing

RT: Design for testability

Combine harvesters

USE: Agricultural machinery

Combined heat and power

USE: Cogeneration

Combined heat, cooling and power

USE: Trigeneration

Combined heat, cooling, and power

USE: Trigeneration

Combined source channel coding

USE: Combined source-channel

coding

Combined source-channel coding

UF: Combined source channel

coding

BT: Channel coding

Combustion

BT: Oxidation

RT: Exhaust gases

NT: Plasma-assisted

combustion

Comets

BT: Planets

RT: Meteors

NT: Meteoroids

Command and control systems

UF: Military command and

control

BT: Aerospace and electronic

systems

RT: Military communication

Military systems

Command languages

BT: Computer languages

Commerce and trade

BT: Business

NT: Free economic zones

Product delivery

Commercial law

BT: Law

RT: Bankruptcy

Business

Consumer products

Consumer protection

Economics

Commercial power systems

USE: Industrial power systems

Commercialization

BT: Engineering management

Common Information Model (computing)

UF: CIM

BT: Analytical models

Information management

RT: Information exchange

Common Information Model (electricity)

UF: CIM

BT: Information management

Power transmission

RT: IEC Standards

Information exchange



January 2026 IEEE Thesaurus

	Interoperability	RT:	Channel coding
	Open systems		Communication systems
	Unified modeling language		IEEE 802.11e Standard
			IEEE 802.11n Standard
<i>common sense knowledge</i>			Multicarrier code division
USE:	Commonsense reasoning	multiple access	
<i>common sense reasoning</i>			OFDM
USE:	Commonsense reasoning		Synapses
<i>common-sense knowledge</i>		NT:	Channel allocation
USE:	Commonsense reasoning		Channel capacity
<i>common-sense reasoning</i>			Channel estimation
USE:	Commonsense reasoning		Channel models
			Channel spacing
			Channel state information
			Gaussian channels
			Multipath channels
			Multiuser channels
			Partial response channels
			Quantum channels
			Throughput
			Time-varying channels
Commonsense reasoning		<i>Communication complexity</i>	
UF:	common sense knowledge	USE:	Complexity theory
	common sense reasoning		
	common-sense knowledge	Communication effectiveness	
	common-sense reasoning	BT:	Professional
	commonsense knowledge	communication	
	reasoning about programs	RT:	Cooperative communication
BT:	Artificial intelligence		Federated learning
	Cognition		
<i>Communicable disease</i>		Communication engineering education	
USE:	Infectious diseases	BT:	Engineering education
Communication aids		Communication equipment	
BT:	Professional	BT:	Communications
communication		technology	
RT:	Assistive technologies	RT:	Bluetooth
	Auditory displays		Communication systems
NT:	Closed captioning		Multiplexing equipment
Communication cables			Satellite ground stations
UF:	Underground	NT:	Auditory displays
communication cables			Codecs
BT:	Cables		Modems
RT:	Fault location		On board unit
	Wire		Optical communication
<i>Communication cables (optical)</i>		equipment	
USE:	Optical fiber cables	equipment	
Communication channels			Radio communication
UF:	Air interface		Receivers
	Telecom channels		Repeaters
	Telecommunication		Speech codecs
channels			TV equipment
BT:	Information theory		Telephone equipment
			Transceivers



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	Transmitters		
	Transponders		
	Video codecs		
	Video equipment		
	Vocoders		
Communication industry		Communication switching	
BT:	Industries	BT:	Communications
RT:	Communication systems	technology	
		RT:	IEEE 802.3 Standard
			Switching systems
		NT:	Code division multiplexing
			Electronic switching
		systems	
			Frame relay
			Handover
			Multiprotocol label
<i>Communication network reliability</i>		switching	
USE:	Telecommunication		Packet switching
network reliability			
Communication networks		Communication symbols	
UF:	PSTN	BT:	Professional
	Public switched telephone	communication	
network			Symbols
BT:	Communication systems	RT:	Pragmatics
RT:	Network security		Syntactics
NT:	Central office	NT:	Semiotics
	Cyberspace		
	Industrial communication	Communication system control	
	Maritime communications	BT:	Communication systems
	Radio access technologies	RT:	Control systems
	Relay networks	NT:	Telecommunication control
	Telecommunication		
network performance		Communication system operations and management	
	Virtual links	BT:	Management
		RT:	Communication system
<i>Communication protocols</i>		signaling	
USE:	Protocols		Communication systems
<i>Communication satellites</i>		<i>Communication system privacy</i>	
USE:	Satellite communications	USE:	Communication system
		security	
Communication standards		Communication system security	
UF:	Telecommunication	UF:	Communication system
standards		privacy	
BT:	Standards categories		Telecommunication security
RT:	Communication systems		Wireless security
	FDDI	BT:	Communication systems
	IEC	RT:	Access control
	ISO		Cryptography
	ISO Standards		Data security
	Radio spectrum		Electronic warfare
management			Hardware security
NT:	Data over cable service		Privacy
interface specification			Aggregate signature
	Long Term Evolution	NT:	
	Near field communication	schemes	
	SONET		Denial-of-service attack
	Synchronous digital		Impersonation attacks
hierarchy			Quantum key distribution
	Universal Serial Bus		



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countermeasures	Radio communication	security	Communication system
Communication system signaling		signaling	Communication system
UF:	Signaling systems	software	Communication system
	Telecom signaling	traffic	Communication system
	Telecom system signaling	traffic control	Communication system
	Telecommunication		Computer networks
signalling	BT: Communication systems		Cross layer design
	RT: Bit rate		Data buses
	Communication system		Data communication
operations and management	Handover		Device-to-device
NT: Received signal strength			Digital communication
indicator		communication	Duplex communication
Communication system software			FDDI
BT: Communication systems		systems	Facsimile
RT: Collaborative software			IP networks
NT: Streaming media			ISDN
Communication system traffic			Indoor communication
UF: Mice flows			Integrated sensing and
BT: Communication systems		communication	Internet
RT: Telecommunication traffic			Local area networks
Communication system traffic control			Low latency communication
BT: Communication systems			Machine-to-machine
Communication systems		communications	Magnetic communication
BT: Communications			Metropolitan area networks
technology			Microwave communication
RT: Antennas and propagation			Military communication
	Communication channels		Mobile communication
	Communication equipment		Molecular communication
	Communication industry		Multiaccess communication
	Communication standards		Multicast communication
	Communication system		Multimedia communication
operations and management			NOMA
	Digital systems		Nanocommunication
	Huffman coding		Narrowband
	Information theory		Optical fiber communication
	Office automation		Personal communication
	Telecontrol equipment		Protocols
	Traffic control	networks	Quality of experience
NT: ARPANET			Quality of service
Air to ground			Quantum communication
communication			Radio communication
	Biomedical communication		Regional area networks
	Broadband communication		Routing
	Communication networks		SIMO
	Communication system		Satellite communications
control			



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	Satellite ground stations	UHF technology
	Semantic communication	Ultra wideband technology
	Spatial diversity	VHF devices
communication	Submillimeter wave	
		<i>Community antenna television</i>
	Subscriber loops	USE: Cable TV
	Switching systems	
hierarchy	Synchronous digital	Community networks
		BT: Social networking (online)
	Telecommunications	RT: Social computing
	Teleconferencing	
	Telegraphy	Commutation
	Telephony	BT: Motors
	Teleprinting	
	Teletext	Commutators
	Terahertz communications	BT: DC motors
	Terrestrial communications	
	Token networks	<i>Compact disk</i>
	UHF communication	USE: CD recording
	Underwater communication	
	Vehicle-to-everything	Compaction
	Video conferencing	BT: Waste reduction
	Videophone systems	RT: Materials handling
	Videotex	
	Visual communication	Companies
	Wide area networks	BT: Organizations
	Wideband	
	Wireless communication	Compass
	Wireless mesh networks	UF: Compasses
	Wireless sensor networks	BT: Instruments
		RT: Magnetic fields
		Navigation
<i>Communications computing</i>		
USE:	Telecommunication	
computing		<i>Compasses</i>
		USE: Compass
Communications technology		
RT:	Antennas and propagation	
NT:	Communication equipment	Competitive intelligence
	Communication switching	BT: Information management
	Communication systems	RT: Business intelligence
	Couplers	Decision support systems
	Green communications	Digital intelligence
	High-speed electronics	Knowledge management
	Image communication	Market research
	Information and	
communication technology		Competitive learning
	MIMO	BT: Unsupervised learning
	MISO	
	Message systems	<i>Compilers (program)</i>
	Modulation	USE: Program processors
	Multiplexing	
	Network topology	Complex instruction set computer
	Presence network agents	BT: Instruction sets
	SISO	RT: Reduced instruction set
	TV	computing



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Complex networks

BT: Network topology
RT: Social network theory
Social sciences
System dynamics
System of systems

manufacture

Integrated circuit

Integrated circuit packaging
Semiconductor device

packaging

Thermal management of

electronics

Complex systems

theory
BT: Systems engineering and
RT: Bio-inspired engineering
Configuration management
Large-scale systems

Composite materials

BT: Materials
RT: Intelligent materials
NT: Cermet

Complexity

USE: Complexity theory

Composite media

USE: Nonhomogeneous media

Complexity constrained detection

USE: Maximum likelihood
detection

Composite systems

USE: Interconnected systems

Complexity theory

UF: Circuit complexity
Communication complexity
Complexity
BT: Chaos
RT: Computation theory
Econophysics
NT: Computational complexity
NP-complete problem
NP-hard problem

Composting

BT: Biodegradation
Recycling
RT: Biochemical oxygen

demand

Fertilizers
Waste management

Compliance testing

USE: Conformance testing

Compliant mechanisms

USE: Manufacturing processes

Compounds

BT: Materials, elements, and
compounds
NT: Bismuth compounds
Gallium compounds
Indium compounds
Inorganic compounds
Lead compounds
Organic compounds
Silicon compounds

Component architectures

UF: Component-based systems
BT: Components, packaging,
and manufacturing technology

Compressed sensing

UF: Compressive sensing
BT: Sampling methods

Component-based systems

USE: Component architectures

Compression algorithms

BT: Algorithms

Components, packaging, and manufacturing technology

NT: Component architectures
Electronic components
Electronic equipment

Compression molding

UF: Compression moulding
BT: Production
RT: Injection molding

manufacture

Electronics packaging
Environmentally friendly

Compression moulding

USE: Compression molding

manufacturing techniques

Compressive sensing

USE: Compressed sensing



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Compressive stress

BT: Stress

Compressors

BT: Electric machines

RT: Air conditioning

Pumps

Turbines

Turbomachinery

Computation complexity

USE: Computational complexity

Computation theory

BT: Computational intelligence

RT: Complexity theory

NT: Computational chemistry

Computational complexity

Computational statistics

Concurrent computing

Greedy algorithms

Reservoir computing

Support vector machines

Turing completeness

Computational and artificial intelligence

RT: Cognitive science

Digital systems

NT: Artificial intelligence

Autonomous mental

development

Computational intelligence

Logic

Machine intelligence

Neural networks

Computational auditory scene analysis

USE: Machine listening

Computational biochemistry

BT: Computational biology

RT: Biochemistry

Bioinformatics

Computational biology

BT: Engineering in medicine

and biology

RT: Bioinformatics

Biology

Computational

neuroscience

Synthetic biology

NT: Computational biochemistry

Computational biophysics

biology

Computational systems

Computational biophysics

BT: Computational biology

RT: Bioinformatics

Biophysics

Computational chemistry

BT: Chemistry

Computation theory

Computational complexity

UF: Computation complexity

BT: Complexity theory

Computation theory

RT: Algorithmic efficiency

Graph drawing

Semidefinite programming

NT: Time complexity

Computational cultural dynamics

USE: Computational cultural

modeling

Computational cultural modeling

UF: Computational cultural

dynamics

Computational social and

behavioral modeling

BT: Computational modeling

Computational cybernetics

BT: Computational intelligence

Cybernetics

Computational efficiency

BT: Mathematics

Computational electrodynamics

USE: Computational

electromagnetics

Computational electromagnetics

UF: Computational

electrodynamics

BT: Electromagnetic analysis

RT: Computer applications

Electromagnetic field theory

Electromagnetic fields

Finite difference methods

Monte Carlo methods

Stochastic processes



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Computational ethics

USE: Machine ethics

NT: Time complexity
Aerial computing
Agent-based modeling
Computational cultural

Computational fluid dynamics

UF: CFD
BT: Fluid dynamics
RT: Isosurfaces

modeling
science
Computational materials

Computational geometry

BT: Geometry
RT: Computer graphics
Layered manufacturing
Surface fitting
NT: Fractals
Geometric modeling

Evolutionary dynamics
Multiphysics
Reversible computing

Computational morality

USE: Machine ethics

Computational neuroscience

UF: Theoretical neuroscience
BT: Computer science
Neuroscience
RT: Computational biology
Nervous system

Computational intelligence

BT: Computational and artificial intelligence
RT: Artificial intelligence
Synapses
NT: Computation theory
Computational cybernetics
Evolutionary computation
Fuzzy systems
Genetic algorithms

Computational science

USE: Scientific computing

Computational social and behavioral modeling

USE: Computational cultural

modeling

Computational life sciences

USE: Computational modeling
AND Life sciences

Computational statistics

UF: Statistical computing
BT: Algorithm design and

analysis

Computation theory
Statistics

Computational linguistics

BT: Systems, man, and cybernetics
RT: Context modeling
Natural language
generation
NT: Machine translation
Sentiment analysis

Computational systems biology

BT: Computational biology

Compute-in-memory

USE: In-memory computing

Computational materials science

BT: Computational modeling
Materials science and technology
RT: Informatics

Computed microtomography

USE: Computed tomography

Computed tomography

UF: CT scan
Computed
microtomography
tomography
Computerised axial
Computerised tomography
Computerized axial
tomography
Computerized tomography
BT: Tomography

Computational modeling

UF: Computational life sciences
Life sciences computing
ZINDO
BT: Modeling
RT: Fish schools
Neuroinformatics
Opinion dynamics



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RT: Biomedical applications of
radiation
NT: Single photon emission
computed tomography

Computer aided analysis

BT: Computer applications
RT: Digital simulation
Geophysics computing
Independent component
analysis
Simulation

Computer aided design

USE: Design automation

Computer aided diagnosis

UF: Computer assisted
diagnosis
Computer-aided diagnosis
Computer-assisted
diagnosis
BT: Medical diagnosis

Computer aided engineering

UF: CAE
BT: Computer applications

Computer aided instruction

UF: CAI
Computer aided learning
Computer-aided instruction
Computer-aided learning
Teaching machines
BT: Computer applications
Educational technology
RT: Authoring systems
Continuing education
Courseware
Educational courses
Electronic learning
MATLAB
Technology acceptance
model
NT: Hybrid learning
Learning management
systems

Computer aided learning

USE: Computer aided instruction

Computer aided manufacturing

UF: CAM
BT: Industrial electronics
Manufacturing automation

RT: Computer integrated
manufacturing
Integrated manufacturing
systems
NT: CAD/CAM
Silicon compiler

Computer aided software engineering

BT: Software engineering
RT: Programming environments
Software tools

Computer animation

USE: Animation

Computer applications

UF: Volunteer computing
BT: Computers and information
processing
RT: Biomedical computing
Computational
electromagnetics
Computerized monitoring
Edge computing
Electrical engineering

computing

Flexible manufacturing

systems

Humanities
Information technology
Learning management

systems

Middleware
Mobile agents
Software agents
Software packages
Application virtualization
Big Data applications
Bot (Internet)
Computer aided analysis
Computer aided

engineering

Computer aided instruction
Computer generated music
Computer integrated

manufacturing

Control engineering

computing

Engineering computing
Green computing
High energy physics

instrumentation computing

Knowledge management
Mathematics computing



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systems	Medical information	Computer crashes	BT: Computer errors
	Military computing	NT: Buffer overflows	Computer bugs
	Mobile applications		
	Physics computing		
	Power engineering	Computer crime	
computing	Power system analysis	UF: Cyber crime	Cyber-crime
computing			Cybercrime
	Publishing		Hacking
	Scientific computing		Piracy (software)
	Telecommunication		Software piracy
computing	Virtual assistants	BT: Computer security	
	Virtual enterprises	RT: Botnet	Computer viruses
	Virtual manufacturing		Computer worms
	Web sites		Control system security
	World Wide Web		Data security
			Digital rights management
			Distributed denial-of-service
Computer architecture		attack	
UF: Architecture (computer)			Hardware security
BT: Computers and information			Identity theft
processing			Privacy-invasive software
RT: Microprogramming			Threat modeling
NT: Accelerator architectures		NT: Counterfeiting	Unsolicited e-mail
	Data structures		Cyber terrorism
	Dynamic voltage scaling		Cyber threat intelligence
	Memory architecture		Cyberattack
	Memory management		SQL injection
	Multiprocessor		
interconnection	Parallel architectures		
	Reconfigurable	Computer displays	
architectures		BT: Displays	
	Redundant array of	RT: Computer graphics	Computer peripherals
independent disks			Workstations
<i>Computer arithmetic</i>		NT: Mesh generation	
USE: Digital arithmetic			Touch sensitive screens
<i>Computer assisted diagnosis</i>		<i>Computer documentation</i>	
USE: Computer aided diagnosis		USE: Documentation	
<i>Computer automated measurement and control</i>		<i>Computer engineering education</i>	
USE: CAMAC		USE: Computer science	education
Computer bugs		Computer errors	
UF: Bugs		BT: Computer performance	
BT: Computer crashes		NT: Computer crashes	
<i>Computer buses</i>		<i>Computer games</i>	
USE: Data buses		USE: Video games	
<i>Computer control</i>			
USE: Digital control			



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Computer generated music

UF: CGM
Computer music
BT: Computer applications
Music

RT: Manufacturing automation
Agile manufacturing
CAD/CAM
Computer aided
manufacturing
Virtual manufacturing

Computer graphics

BT: Graphics
RT: Animation
Art
Character generation
Computational geometry
Computer displays
Computer peripherals
Curve fitting
Fractals
Graphics processing units
Mesh generation
Modeling
Multimedia computing
Photorealistic images
Ray tracing
Simulation
Surface fitting
Visual effects
Workstations
NT: Data visualization
Emojis
Rendering (computer

graphics)

Shadow mapping
Sprites (computer)
Video sequences
Virtual reality
Visualization
X3D

Computer hacking

UF: Hacker
Hacks
BT: Computer security
NT: Hackathon

Computer hardware

USE: Hardware

Computer industry

UF: DP industry
BT: Industries
RT: Computers and information

processing

Computer integrated manufacturing

UF: CIM
BT: Computer applications

Computer interfaces

BT: Computers and information
processing
RT: Computer peripherals
Data buses
Interface management
User interfaces
Application programming
NT: Application programming
interfaces

Browsers
Field buses
Firewire
HDMI
Haptic interfaces
Hypertext systems
Input devices
Interface phenomena
Interface states
Musical instrument digital

interfaces

Ports (computers)
System buses

Computer languages

UF: Programming languages
BT: Formal languages
RT: Data structures
Natural languages
Software
NT: Architecture description

languages

Business Process
Execution Language

C languages
Command languages
Database languages
Hardware design

languages

High level languages
Markup languages
Python
R language
Specification languages
Style sheet languages
Systems Modeling

Language

Visual BASIC



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Computer mediated communication

UF: Computer-mediated communication

Mediated communication
Mediated discourse
Mediated interaction
Mediated technology
Technology mediated

communication

Technology-mediated

communication

BT: Social networking (online)

Computer music

USE: Computer generated music

AND

Music

interconnection

Computer network management

BT: Computer networks

RT: Bandwidth

Computer security
Data security
Network function

virtualization

Software defined

networking

NT:

NT: Traffic control
Computer network reliability
Disruption tolerant

networking

networks

management

networks

base

Management information

Middleboxes
Network address translation
Network synthesis
Source address validation

Computer network reliability

BT: Computer network management

NT: Network topology
Fault tolerant computer

networks

interconnection networks

virtualization

Computer networks

UF: Mice flows

BT: Communication systems
Computers and information

processing

RT: Bit rate
Cyber terrorism
Cyber warfare
Data communication
Delay estimation

networking

Distributed computing
Edge computing
File servers
Firewalls (computing)
Frame relay
Hypercubes
IEEE 802.11 Standard
IEEE 802.11g Standard
IEEE 802.11n Standard
IEEE 802.16 Standard
IEEE 802.3 Standard
IPTV
Internetworking
LAN interconnection
Middleware
Multiprocessing systems
Multiprocessor

Network security
Open systems
Personal area networks
Ports (computers)
Radial basis function

TCPIP
Web sites
Ad hoc networks
Computer network

Content distribution

Cyberspace
Diffserv networks
Domain Name System
Ethernet
Heterogeneous networks
IP networks
Internet
Intserv networks
Metropolitan area networks
Multiprocessor

Network function

Network servers
Next generation networking
Overlay networks
Peer-to-peer computing
Software defined

Storage area networks
Token networks
Unicast
Virtual private networks
Wide area networks



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Wireless access points

NT:
neuroscience

Computational

Computer numerical control

UF:	CNC
	NC machines
BT:	Manufacturing automation
RT:	Digital control
	Industrial control

- Formal languages
- Network theory (graphs)
- Programming

Computer operating systems

USE: Operating systems

Computer performance

BT:	Computers and information processing
NT:	Computer errors Hardware acceleration Performance loss

Computer peripherals

	UF:	Computer terminals Peripheral equipment
processing	BT:	Computers and information
	RT:	Computer displays Computer graphics Computer interfaces Device drivers Firewire Flash memories User interfaces
	NT:	Disk drives Keyboards Modems Printers

Computer science education

UF: Computer engineering
education

BT: Engineering education

Computer security

UF:	cyber security
	cybersecurity
BT:	Computers and information
processing	

RT: Security
Access control
Blockchains
Computer network

management

- Cryptography
- Cyber threat intelligence
- Data protection
- Data security
- Digital forensics
- Digital policy
- Eavesdropping
- Hardware security
- Operating systems
- Privacy
- Privacy enhancing

technologies

- Privacy-invasive software
- Session hijacking
- Threat modeling
- Trust management
- Application security
- Authentication
- Cloud computing security
- Computer crime
- Computer hacking
- Cross-site scripting
- Cyber espionage
- Cyber warfare
- Cyberattack
- Data integrity
- Denial-of-service attack
- Firewalls (computing)
- Honey pot (computing)
- Identity management

NT:

Computer pipeline processing

USE: Pipeline processing

Computer ports

USE: Ports (computers)

Computer programming

USE: Programming

Computer programming profession

USE: Programming profession

Computer science

BT:	Computers and information processing
RT:	Function approximation
	Logic
	Software

systems

- Internet security
- Mobile security
- Passwords



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	Penetration testing		Vision transformers
	Permission		Visual odometry
	Phishing		
	Proof of Work		
	Trusted computing		
Computer simulation		Computer worms	
BT:	Simulation	UF:	Worms (computer)
RT:	Digital twins	BT:	Malware
	EMTP	RT:	Computer crime
	Fish schools		Computer viruses
	Human in the loop		
	Multiphysics		
NT:	Synthetic data		
<i>Computer software</i>		<i>Computer-aided design</i>	
USE:	Software	USE:	Design automation
<i>Computer speech recognition</i>		<i>Computer-aided diagnosis</i>	
USE:	Speech to text	USE:	Computer aided diagnosis
<i>Computer terminals</i>		<i>Computer-aided instruction</i>	
USE:	Computer peripherals	USE:	Computer aided instruction
Computer viruses		<i>Computer-aided learning</i>	
UF:	Viruses (computer)	USE:	Computer aided instruction
BT:	Malware		
RT:	Anti-virus software	<i>Computer-assisted diagnosis</i>	
	Computer crime	USE:	Computer aided diagnosis
	Computer worms		
Computer vision		<i>Computer-mediated communication</i>	
BT:	Robots	USE:	Computer mediated
RT:	Activity recognition	communication	
	Advanced driver assistance	<i>Computerised axial tomography</i>	
systems		USE:	Computed tomography
	Conditional random fields	<i>Computerised instrumentation</i>	
recognition	Fine-grained image	USE:	Computerized
	Gait recognition	instrumentation	
	Gaze tracking	<i>Computerised monitoring</i>	
	Image capture	USE:	Computerized monitoring
	Indoor navigation	<i>Computerised tomography</i>	
	One shot learning	USE:	Computed tomography
	Pattern recognition	<i>Computerized axial tomography</i>	
	Pose estimation	USE:	Computed tomography
	Video restoration		
	Vision language model	Computerized instrumentation	
NT:	Active appearance model	UF:	Computerised
	Blob detection	instrumentation	
	Corner detection	BT:	Instrumentation and
	Face detection	measurement	
	Feature detection	Computerized monitoring	
	Interest point detection	UF:	Computerised monitoring
	Smart cameras	BT:	Monitoring
		RT:	Computer applications



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Computerized tomography

USE: Computed tomography

Computers

UF: Computing technology
BT: Computers and information

processing

RT: Cyberspace
NT: Analog computers
Calculators
Digital computers
Microcomputers
Parallel machines
Supercomputers
Tablet computers
Wearable devices

Computers and information processing

RT: Associative processing
Biology computing
Computer industry
Data processing
Electronic learning
Home computing
Information systems
Logic circuits
Multimedia computing
Multiprocessing systems
NT: Approximate computing
Computer applications
Computer architecture
Computer interfaces
Computer networks
Computer performance
Computer peripherals
Computer science
Computer security
Computers
Concurrency control
DNA computing
Data systems
Database machines
Digital systems
Distributed computing
File servers
Hardware
High performance

computing

Image processing
Memory
Mobile computing
Molecular computing
Multitasking
Open systems
Optical computing

Parallel processing
Pattern recognition
Pervasive computing
Petascale computing
Platform virtualization
Probabilistic computing
Probability computing
Quantum computing
Real-time systems
Software
Software engineering
System recovery
Time sharing computer

systems

Virtual machine monitors

Computing technology

USE: Computers

Concatenated codes

BT: Programming

Concave programming

BT: Optimization methods

Concentric tube robots

BT: Continuum robots
RT: Minimally invasive surgery

Concept drift

BT: Data science
Machine learning
Predictive analytics

Concrete

BT: Building materials
RT: Pressure vessels
NT: Aggregate concrete

Concrete aggregate

USE: Aggregate concrete

Concurrency

USE: Concurrent computing

Concurrency control

BT: Computers and information
processing
RT: Distributed computing
Distributed databases
Multiprocessing systems
Parallel processing
Protocols
Synchronization
NT: Processor scheduling



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Concurrent computing

UF: Concurrency
BT: Computation theory
RT: Granular computing
Model checking

Concurrent engineering

BT: Engineering - general
RT: Product design
Project management
Quality function deployment
Research and development

management

Time to market
Virtual manufacturing

Condensation trails

USE: Contrails

Condition monitoring

BT: Preventive maintenance

Conditional random fields

UF: CRF
BT: Graph theory
Random processes
Statistical analysis
RT: Computer vision
Feature extraction
Image segmentation
Learning (artificial

intelligence)

Natural language

processing

Text analysis

Conditions of employment

USE: Employee welfare

Conducting bodies

USE: Conductors

Conducting materials

BT: Materials
RT: Conductivity
Conductors
Semiconductor materials
NT: Electrolytes

Conductive adhesives

BT: Adhesives

Conductive films

BT: Films
NT: Anisotropic conductive films

Conductivity

UF: Electric conductivity
Electrical conductivity
Resistivity
BT: Electric variables
RT: Charge carriers
Conducting materials
Conductivity measurement
Grain boundaries
Impact ionization
Transmission line theory
NT: Photoconductivity
Semiconductivity
Transconductance

Conductivity measurement

UF: Resistivity measurement
BT: Electric variables
measurement
RT: Conductivity

Conductors

UF: Conducting bodies
BT: Electric machines
RT: Cables
Conducting materials
Core loss
Power cables
Power distribution lines
Power transmission lines
Proximity effects
Skin effect
Thermal noise
Three-phase electric power
Wire
Wireless power

transmission

Wiring

Conference management

BT: Management

Conference of the Parties

UF: COP
BT: Climate change

Conferences

UF: Meetings (technical)
Symposia
Workshops
BT: Meetings

Configuration management

BT: Systems engineering and
theory



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RT: Complex systems
Maintenance engineering
System analysis and design

Conformal mapping

BT: Mathematics
RT: Coplanar waveguides
Wave functions
Waveguide components
Waveguide theory

Conformance testing

UF: Compliance testing
Conformity assessment
Type testing
BT: Testing
RT: Accreditation
Certification
Quality of service
Standards
Surveillance

Conformity assessment

USE: Conformance testing

Confusion matrices

BT: Statistical learning
RT: Visualization

Congestive heart failure

BT: Medical conditions

Conifers

USE: Non-flowering plants

Connected cars

USE: Connected vehicles

Connected vehicles

UF: Connected cars
BT: Vehicles
RT: Controller area networks
Intelligent vehicles

Connecting

USE: Joining processes

Connective tissue

BT: Biological tissues

Connectors

BT: Electronic components
NT: Plugs
Sockets

Conscious capitalism

BT: Environmental economics

Consensus algorithm

BT: Algorithm design and theory

Consensus control

BT: Decentralized control
RT: Decentralized applications
Swarm robotics

Consensus mechanism

USE: Consensus protocol

Consensus protocol

UF: Consensus mechanism
BT: Blockchains
Protocols
RT: Ecosystems

Conservation areas

USE: Protected areas

Consortia

BT: Engineering management
RT: Business

Constellation diagram

UF: Signal constellation
BT: Digital modulation

Constraint handling

UF: Constraint programming
BT: Logic programming

Constraint optimization

BT: Design optimization
RT: Electronics packaging

Constraint programming

USE: Constraint handling

Constraint theory

UF: Theory of constraints
BT: Integer linear programming
RT: Stability plasticity

Construction

UF: Erection
BT: Industries
RT: Building materials
Civil engineering
Construction industry
Structural engineering



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NT:	Buildings Green buildings Modular construction Prefabricated construction Stairs	Consumer products BT: Manufactured products RT: Commercial law Consumer behavior Consumer electronics Domestic safety Electrical products Food industry Food manufacturing Food products Footwear industry Market research Plastic products Product liability Product safety Watches
Construction industry BT: Industries RT: Building automation Building materials Buildings Construction Excavation Floors Mortar Shipbuilding industry Smart cities NT: Prefabricated construction		NT: Backpacks Clothing Games Home appliances Microwave ovens
Consumer behavior UF: Consumer behaviour Customer behavior Customer behaviour BT: Behavioral sciences RT: Consumer products Customer profiles Customer relationship management Electronic commerce Food waste Market opportunities Technology acceptance model NT: Neuromarketing Self-service		Consumer protection BT: Law Product safety engineering RT: Censorship Commercial law Customer relationship management Quality assurance <i>Consumer satisfaction</i> USE: Customer satisfaction <i>Consumer-generated media</i> USE: User-generated content
<i>Consumer behaviour</i> USE: Consumer behavior		Consumption footprint BT: Environmental measurement RT: Energy consumption
Consumer electronics UF: Kindle RT: Consumer products Digital systems Firewire Flat panel displays Microcomputers Video equipment NT: Ambient intelligence Audio systems Home automation Home computing Low power electronics Microwave ovens Multimedia systems		Contact resistance BT: Contacts Contact tracing BT: Public healthcare RT: Epidemiology Infectious diseases Contactors BT: Switches Contacts BT: Circuits and systems



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RT: Semiconductor devices
NT: Brushes
Contact resistance
Ohmic contacts

Containers

BT: Material storage
Materials handling
equipment
RT: Bulk storage
Canning
Filling
Fuel storage
Loading
Measurement
Pallets
Production
Stacking
NT: Freight containers

Contamination

BT: Materials science and
technology
RT: Chemical hazards
Decontamination
Hazards
Impurities
Microfiltration
Pollution
Quality control
Radiation protection
Soil remediation
NT: Surface contamination

Content addressable memory

USE: Associative memory

Content addressable storage

BT: Memory
RT: Content-based retrieval

Content analysis

USE: Content management

Content based retrieval

USE: Content-based retrieval

Content delivery networks

USE: Content distribution
networks

Content distribution networks

UF: Content delivery networks
BT: Computer networks

Content management

UF: Content analysis
BT: Electronic publishing
Management
RT: Blockchains
Document handling
MPEG 7 Standard
Multimedia computing
Publish subscribe systems
Semantic Web
Web design
Web sites

Content-based retrieval

UF: Content based retrieval
BT: Information retrieval
RT: Content addressable
storage
Cross modal retrieval

Content-centric networking

USE: Information-centric
networking

Context

BT: Professional
communication
RT: Pragmatics

Context aware

USE: Context-aware services

Context awareness

BT: Artificial intelligence
RT: Context management
Intelligent control
Intelligent systems
Knowledge acquisition
Learning systems
Pervasive computing
Semantic search

Context management

BT: Artificial intelligence
RT: Context awareness

Context modeling

BT: Modeling
RT: Computational linguistics

Context-aware applications

USE: Context-aware services

Context-aware computing

USE: Context-aware services



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Context-aware services

UF: Context aware
Context-aware applications
Context-aware computing
BT: Ubiquitous computing

RT: Production management
Quality awards

Continental crust

BT: Geology
RT: Oceanic crust
Rocks
Tectonics

Continents

BT: Geoscience
NT: Africa
Asia
Australia
Europe
North America
South America

Contingency management

BT: Management
NT: Crisis management
Disaster management
Mission critical systems

Continual learning

USE: Continuing education

Continuing education

UF: Continual learning
Further education
BT: Career development
Educational programs
RT: Computer aided instruction
Engineering education
Management training
Training

Continuing professional development

UF: Life long learning
BT: Human resource
management
RT: Qualifications
Training

Continuous glucose monitoring

BT: Biomedical monitoring
Glucose
RT: Medical devices

Continuous improvement

UF: Kaizen
BT: Total quality management

Continuous integration

BT: Software development
management

Continuous phase modulation

BT: Phase modulation

Continuous production

BT: Flow production systems
Production control
RT: Process control
Production management

Continuous systems

USE: Continuous time systems

Continuous time models

USE: Continuous time systems

Continuous time systems

UF: Continuous systems
Continuous time models
BT: Time factors

Continuous wavelet transforms

BT: Wavelet transforms

Continuous-stirred tank reactor

BT: Chemical reactions

Continuously variable transmission

USE: Mechanical power
transmission

Continuum robots

BT: Robots
NT: Concentric tube robots

Contract law

BT: Law
RT: Contracts
Employment law

Contract management

BT: Contracts
Management
RT: Risk management

Contracts

BT: Management
RT: Business
Contract law



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	Procurement		
	Proposals		
NT:	Contract management	Control engineering education	BT: Engineering education
	Forward contracts		RT: Control engineering
	Licenses		Control engineering
	Service level agreements	computing	
	Smart contracts		
	Subcontracting	Control equipment	
Contrails		BT: Control systems	
UF:	Condensation trails	RT: Manipulators	
	Vapor trails		Mechatronics
	Vapour trails		Robots
BT:	Exhaust gases	NT: Actuators	
	Ice clouds		Fasteners
			Microcontrollers
			Regulators
Contrast agents			Remote control
UF:	Contrast media		Servosystems
BT:	Biomedical imaging		Switches
			Switchgear
<i>Contrast media</i>			Telecontrol equipment
USE:	Contrast agents		Thermostats
Contrast resolution		Control nonlinearities	
BT:	Image resolution	BT: Control theory	
RT:	Spatial resolution	RT: Nonlinear control systems	
		NT: Backstepping	
			Feedback linearization
Contrastive learning			
BT:	Self-supervised learning		
		Control system analysis	
Control charts		BT: System analysis and design	
UF:	Cusum charts	RT: Piecewise linear techniques	
	Shewhart charts	NT: State-space methods	
BT:	Production management		
RT:	Control systems	Control system security	
	Quality management	BT: Control engineering	
			Security
Control design		RT: Computer crime	
BT: Control systems			Control systems
RT: Feedback			Cyber-physical systems
	Lyapunov methods		Industrial control
			National security
Control engineering			Power system control
BT: Control systems			Safety
RT: Control engineering		NT: Physical unclonable	
education		function	
	Predictive control		
NT: Control system security		Control system synthesis	
		UF: Control systems synthesis	
Control engineering computing		BT: Control systems	
BT: Computer applications		RT: Circuit synthesis	
RT: Control engineering			Hardware-in-the-loop
education		simulation	
	Hardware-in-the-loop		Linearization techniques
simulation			Piecewise linear techniques



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Control systems

RT: Active perception
Actuators
Adaptive control
Air traffic control
Communication system

control

Control charts
Control system security
Cybernetics
Discrete-event systems
Discrete-time systems
Estimation
Flexible structures
Force control
Game theory
H infinity control
Interconnected systems
Inventory control
Legged locomotion
Linear systems
Linearization techniques
MIMO
Manipulators
Microcontrollers
Microsensors
Mobile robots
Neuromodulation
Nonlinear systems
Parameter estimation
Poles and zeros
Real-time systems
Robots
Robustness
Sensitivity
Stability
State estimation
Stochastic systems
Switched systems
Target tracking
Time-varying systems
Transfer functions
Uncertain systems
Underactuated surface

vessels

NT: Admittance control
Automatic control
Automatic generation

control

Automotive control
Autopilot
Bidirectional control
Block signalling
Brakes

CAMAC
Centralized control
Closed loop systems
Control design
Control engineering
Control equipment
Control system synthesis
Controllability
Controller area networks
Cruise control
Decentralized control
Delay systems
Digital control
Fault tolerant control
Feedback
Feedback linearization
Fluid flow control
Fluidics
Gaze tracking
Homeostasis
Linear feedback control

systems

Magnetic variables control
Mechanical variables

control

Medical control systems
Missile control
Moisture control
Motion compensation
Networked control systems
Nonlinear control systems
Open loop systems
Optical control
Optimal control
PD control
PI control
Pneumatic systems
Positive train control
Pressure control
Proportional control
Radio control
Robot control
SCADA systems
Sensorless control
Sliding mode control
Supervisory control
Thermal variables control
Traffic control

Control systems synthesis

USE: Control system synthesis

Control theory

BT: Cybernetics
RT: Dynamics



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NT:	Feedback circuits	converters	Power conversion
	Control nonlinearities		Pulse width modulation
Controllability	Iterative learning control	converters	Resonant converters
	Observability		Static power converters
BT:	Control systems		Voltage-source converters
			Wavelength converters
Controller area networks		<i>Convertors</i>	
UF:	CAN	USE:	Converters
BT:	Control systems		
RT:	Connected vehicles	Convex functions	
	Intelligent systems	UF:	Convex optimization
			Convex programming
Convection		BT:	Mathematics
UF:	Mixed convection	NT:	Convex hulls
	Rayleigh-Benard		Semidefinite programming
convection			
BT:	Heat transfer	Convex hulls	
		BT:	Convex functions
Convergence		<i>Convex optimization</i>	
BT:	Mathematics	USE:	Convex functions
Convergence of numerical methods		<i>Convex programming</i>	
BT:	Numerical analysis	USE:	Convex functions
<i>Conversational AI</i>			
USE:	Conversational artificial intelligence	Convolution	
		BT:	Signal processing
Conversational artificial intelligence		RT:	Deconvolution
UF:	Conversational AI		Numerical analysis
BT:	Artificial intelligence	NT:	Convolvers
RT:	Digital humans		
NT:	Chatbots	Convolutional codes	
	Virtual assistants	UF:	Trellis codes
		BT:	Codes
Converters		RT:	Channel coding
UF:	Convertors		Digital multimedia
	Switching convertors	broadcasting	
BT:	Power electronics		Error correction
RT:	Data conversion		Error correction codes
	Inrush current		Radio communication
	Pulse width modulation		Satellite communications
	Pulse width modulation		Telecommunications
inverters		<i>Convolutional networks</i>	
	Rectennas	USE:	Convolutional neural networks
	Space vector pulse width		
modulation		<i>Convolutional neural nets</i>	
NT:	AC-AC converters	USE:	Convolutional neural networks
	DC-AC power converters		
	Digital-to-frequency		
converters			
	Frequency conversion		
	Multilevel converters		



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Convolutional neural networks

UF: CNN
Convolutional networks
Convolutional neural nets
BT: Artificial neural networks
RT: Deep learning
Generative adversarial

networks

Graph neural networks
Image augmentation
Machine learning
Vision transformers
Graph convolutional

NT:
networks

Convolvers

BT: Convolution

Cookie hijacking

USE: Session hijacking

Cookie side-jacking

USE: Session hijacking

Cookie sidejacking

USE: Session hijacking

Coolants

UF: Antifreeze materials
BT: Cooling
RT: Space cooling
NT: Refrigerants

Cooling

BT: Temperature control
RT: Electronics cooling
Electronics packaging
HVAC
Heat pipes
Thermal engineering
Water pumps
NT: Air conditioning
Cold plates
Coolants
Evaporative cooling
Heat sinks
Immersion cooling
Laser cooling
Liquid cooling
Refrigeration
Solar cooling
Space cooling
Thermal quenching
Trigeneration
Ultracold atoms

Ventilation

Cooperative cache

USE: System performance

Cooperative caching

BT: System performance

Cooperative communication

UF: Amplify-and-forward
cooperative communication
BT: Federated learning
Wireless communication
RT: Communication
effectiveness
Professional
communication

Cooperative networks

USE: Cooperative systems

Cooperative operation

USE: Cooperative systems

Cooperative systems

UF: Cooperative networks
Cooperative operation
BT: Artificial intelligence
RT: Artificial bee colony
algorithm

Cooperative work

USE: Federated learning

Coordinate measuring machines

UF: CMM
BT: Measurement
RT: Inspection
Machine tools
Quality control

COP

USE: Conference of the Parties

COPD

USE: Chronic obstructive
pulmonary disease

Coplanar transmission lines

BT: Planar transmission lines
NT: Coplanar waveguides

Coplanar waveguides

UF: CPW
BT: Coplanar transmission lines



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RT: Conformal mapping
Electromagnetic
waveguides

Copper

UF: Cu
BT: Metals
NT: Copper alloys
Copper compounds

Copper alloys

BT: Copper
RT: Alloying

Copper compounds

BT: Copper

Coprocessors

BT: Circuits
Integrated circuits
Microprocessors
RT: AI accelerators
Digital arithmetic

Copyright protection

BT: Legal factors
RT: Digital rights management
Plagiarism
Public domain software
Publishing
Trademarks
Watermarking
NT: Intellectual property
Software protection

Coral reef

BT: Aquaculture

Core dumps

BT: System recovery

Core loss

UF: Core losses
BT: Energy loss
RT: Conductors
Transformers

Core losses

USE: Core loss

Core-shelf nanostructures

USE: Nanostructured materials

Cornea

BT: Eyes

RT: Ophthalmology

Corner cube prisms

USE: Retroreflectors

Corner detection

BT: Computer vision
Image processing
RT: Image edge detection
Motion detection

Corner-cube prisms

USE: Retroreflectors

Corona

BT: Electric breakdown
RT: Partial discharges

Corona virus

USE: Coronaviruses

Coronal mass ejections

BT: Sun
RT: Geomagnetic storms
Sunspots

Coronary arteriosclerosis

BT: Arteriosclerosis

Coronaviruses

UF: Corona virus
BT: Viruses (medical)
RT: Epidemics
Pandemics
NT: COVID-19

Corporate acquisitions

UF: Mergers
BT: Organizational aspects
RT: Business process re-engineering

Corporate ethical strategy

USE: Corporate social responsibility

Corporate ethics strategy

USE: Corporate social responsibility

Corporate social impact

USE: Corporate social responsibility



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Corporate social responsibility

UF: Business social impact
CSR
Corporate ethical strategy
Corporate ethics strategy
Corporate social impact

BT: Business
Environmental economics

RT: Environmental factors
Social factors
Socioeconomics

NT: ESG
Greenwashing
Long-term climate

strategies

investing

Socially responsible

Value chain emissions

Corpse

USE: Cadaver

Corpus amygdaloideum

USE: Amygdala

Corpus callosum

UF: Callosal commissure

BT: Brain

Correlation

BT: Statistics

RT: Correlation coefficient

NT: Autocorrelation

Correlation coefficient

BT: Statistics

RT: Correlation

Regression analysis

Correlators

BT: Electromagnetic radiation

RT: Signal detection
Signal processing

Corrosion

BT: Surfaces

RT: Coatings
Corrosion inhibitors
Galvanizing
Grain boundaries
Magnetic flux leakage
Passivation

Corrosion inhibitors

BT: Inhibitors

Materials

Corrosion

Galvanizing

Materials preparation

Materials processing

Corrugated surfaces

BT: Rough surfaces
Surfaces

Cortana

USE: Virtual assistants

Cortical bone

BT: Bone tissue

Cortical plasticity

USE: Neuroplasticity

Cosmic gamma ray bursts

USE: Gamma-ray bursts

Cosmic rays

BT: Extraterrestrial phenomena

RT: Electrons
Elementary particles
Mesons
Neutrons
Protons

Cost accounting

UF: Valuation

BT: Costing
Management accounting

RT: Costs
Economics
Profitability

Cost analysis

USE: Cost benefit analysis

Cost benefit analysis

UF: Cost analysis
Cost-benefit analysis

BT: Costs

RT: Functional point analysis

Cost function

BT: Optimization

Cost of living index

USE: Economic indicators

Cost-benefit analysis

USE: Cost benefit analysis



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Cost-of-living index

USE: Economic indicators

Costing

UF: Capital cost reduction
Operating cost reduction
BT: Financial management
NT: Cost accounting

Costs

BT: Economics
RT: Cost accounting
Econometrics
Exchange rates
Levelized cost of energy
NT: Cost benefit analysis

Cotton

BT: Agricultural products
Textiles
RT: Natural fibers
Textile fibers
Textile industry
Weaving

Counseling

USE: Employee welfare

Counselling

USE: Employee welfare

Counterfeit goods

USE: Counterfeiting AND
Manufactured products

Counterfeiting

UF: Counterfeit goods
BT: Computer crime

Counters

USE: Radiation detectors

Counting circuits

BT: Circuits
RT: Logic circuits
Radiation detector circuits

Coupled circuits

USE: Coupling circuits

Coupled mode analysis

BT: Circuit analysis
RT: Multiconductor transmission

lines

Couplers

BT: Communications
technology
RT: Apertures
Coupling circuits
Electromagnetic coupling
NT: Directional couplers

Coupling (process)

USE: Joining processes

Coupling circuits

UF: Coupled circuits
BT: Circuits
RT: Couplers

Couplings

UF: Linkages
BT: Mechanical products
RT: Fasteners
Joining processes
Machine components
Shafts

Course correction

UF: Course-correction
BT: Navigation
RT: Aircraft navigation
Path planning

Course-correction

USE: Course correction

Courseware

BT: Educational technology
RT: Authoring systems
Computer aided instruction
Hybrid learning
Software

Covariance matrices

UF: Covariance matrix
BT: Statistics

Covariance matrix

USE: Covariance matrices

COVID-19

BT: Coronaviruses
RT: Pandemics
Viruses (medical)



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Cows

UF: Cattle
BT: Bovine

CPR

USE: Cardiopulmonary resuscitation

CPU

USE: Central Processing Unit

CPW

USE: Coplanar waveguides

Cr

USE: Chromium

Cradle to cradle

UF: C2C
BT: Environmental economics
RT: Biomimetics
Circular economy

Cramer Rao bound

USE: Cramer-Rao bounds

Cramer Rao bounds

USE: Cramer-Rao bounds

Cramer-Rao bounds

UF: Cramer Rao bound
Cramer Rao bounds
Cramer-Rao inequality
Information inequality
BT: Estimation theory

Cramer-Rao inequality

USE: Cramer-Rao bounds

Cranes

BT: Lifting equipment

Cranial

BT: Nervous system
NT: Cranial pressure

Cranial pressure

BT: Cranial
NT: Intracranial system

Cranium

BT: Head

Crawlers

BT: Web search

RT: Bot (Internet)

CRC codes

USE: Cyclic redundancy check codes

Creativity

BT: Innovation management

Credit cards

UF: American Express
Mastercard
Visa gold
BT: Financial management

Creep

BT: Material properties

CRF

USE: Conditional random fields

Criminal law

BT: Law

Crimping

BT: Joining processes

Crisis management

BT: Contingency management

CRISPR

UF: Clustered regularly interspaced short palindromic repeats
BT: Genetic engineering
RT: Gene therapy
Genetic mutations

Critical current density

BT: Superconductivity
RT: Superconducting materials
Thermal factors

Critical infrastructure

UF: Critical national infrastructure
BT: Public infrastructure

Critical national infrastructure

USE: Critical infrastructure

Crohn's disease

BT: Inflammatory bowel disease
RT: Ulcerative colitis



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Crop yield

BT: Crops
Measurement units
RT: Agribusiness
Agriculture
Farming

Crops

BT: Agricultural products
Vegetation
RT: Fertilizers
Greenhouses
Irrigation
Water storage
Yield estimation
NT: Crop yield
Seeds (agriculture)

Cross cultural communication

USE: Cross-cultural communication

Cross layer design

UF: Cross-layer design
BT: Communication systems
RT: Ad hoc networks
Cellular radio
IEEE 802.16 Standard
Military communication
Radio communication

Cross lingual

UF: Cross-lingual
BT: Natural language processing
RT: Information retrieval
Knowledge transfer
Linguistics
Machine learning

Cross modal retrieval

UF: Cross-modal retrieval
BT: Information retrieval
RT: Content-based retrieval
Data retrieval
Image retrieval
Music information retrieval
Query processing
Search methods
Visual information retrieval

Cross phase modulation

USE: Cross-phase modulation

Cross platform virtualization

USE: Application virtualization

Cross-cultural communication

UF: Cross cultural communication
BT: Global communication
RT: Cultural differences

Cross-layer design

USE: Cross layer design

Cross-lingual

USE: Cross lingual

Cross-modal retrieval

USE: Cross modal retrieval

Cross-phase modulation

UF: Cross phase modulation
XPM
BT: Optical modulation
Phase modulation
RT: Kerr effect
Nonlinear optics
Refractive index

Cross-platform virtualization

USE: Application virtualization

Cross-site scripting

UF: XSS
BT: Computer security

Crosstalk

UF: Crosstalk noise
BT: Interference
RT: Electromagnetic interference
Interchannel interference
Transmission line theory

Crosstalk noise

USE: Crosstalk

Crowd dynamics

BT: Psychology
Sociology

Crowd funding

USE: Crowdfunding

Crowd funding

UF: Crowd funding
BT: Crowdsourcing



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	RT:	Finance Entrepreneurship Internet Investment Project management		Cryopreservation	UF:	Cryobanking
					BT:	Cryogenics
Crowdsensing	BT:	Crowdsourcing		Cryotherapy	BT:	Medical treatment Temperature measurement
	RT:	Collective intelligence Mobile computing		<i>Crypto currency</i>	USE:	Cryptocurrency
Crowdsourcing	BT:	Federated learning Internet		Cryptocurrency	UF:	Crypto currency
	RT:	Collective intelligence Distributed processing Mobile computing Outsourcing Social computing Social networking (online)			BT:	Cryptography Currencies Blockchains Digital economy Digital systems Distributed ledger Finance Fintech Micropayments Nonfungible tokens Online banking Proof of stake Bitcoin
	NT:	Crowdfunding Crowdsensing Product development			RT:	
<i>CRT</i>	USE:	Cathode ray tubes			NT:	
Cruise control	BT:	Control systems Electromechanical systems		<i>Cryptograph</i>	USE:	Ciphers
	RT:	Velocity control		<i>Cryptographic</i>	USE:	Cryptography
<i>Cryobanking</i>	USE:	Cryopreservation		<i>Cryptographic accelerators</i>	USE:	Hardware acceleration
Cryobiology	BT:	Biology Temperature measurement		Cryptographic hash function	BT:	Cryptography Hash functions
Cryogenic electronics	BT:	Industrial electronics		Cryptographic protocols	BT:	Protocols
	RT:	Cryogenics Superconducting devices Superconducting materials		Cryptography	UF:	Cryptographic
Cryogenics	UF:	Cryonics			BT:	Data security Security
	BT:	Industry applications Temperature measurement			RT:	Bitcoin Chaotic communication Codes Communication system
	RT:	Cryogenic electronics		security		
	NT:	Cryopreservation Liquid nitrogen				Computer security Data handling
<i>Cryonics</i>	USE:	Cryogenics				



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function	NT:	Encoding	Crystallizers	NT:	Crystal growth
		Hash functions		UF:	Crystallisers
		Message authentication		BT:	Chemical technology
		Physical unclonable		RT:	Chemical reactions
function	NT:	Privacy	Crystallography	NT:	Colloidal crystals
		Quantum key distribution			Crystallization
		Random sequences			Vacancy defects
		Trust management			
function	NT:	Blockchains	Crystals	BT:	Materials
		Brute force attacks			Crystallizers
		Chaotic mapping			Epitaxial growth
		Ciphers			Materials science and
generation	NT:	Cryptocurrency	technology	RT:	Molecular beam epitaxial
		Cryptographic hash			Phonons
		Encryption			Piezoelectric materials
		Multi-party computation			Semiconductor materials
generation	NT:	Nonce	growth	NT:	Solids
		Public key			Colloidal crystals
		Quantum cryptography			Crystal microstructure
		Random number			Crystallography
Crystal growth	BT:	S Box	growth	RT:	Grain boundaries
		Side-channel attacks			Grain size
		Steganography			Liquid crystals
		Zero knowledge proof			Quartz crystals
Crystal microstructure	BT:	Crystallization	CSA Group	UF:	Canadian Standards
		Epitaxial growth			Association
		Semiconductor growth			BT:
					Standards organizations
diffraction	RT:	Crystals	CSA Group Standards	BT:	Standards publications
		Electron backscatter			
		Microstructure			
Crystalline materials	NT:	Crystallization	CSMA	USE:	Multiaccess communication
		Martensite			
		Nanocrystals			
		Perovskites			
<i>Crystallisation</i>	USE:	Superlattices	CSP	USE:	Chip scale packaging
<i>Crystallisers</i>	USE:	Crystallization	CSR	USE:	Corporate social
		Crystallizers			responsibility
Crystallization	UF:	Crystallisation	responsibility	USE:	Corporate social
		Crystallography			



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<i>CSTR</i>			Current	
	USE:	Chemical reactions	UF:	Electric current
<i>CT scan</i>			BT:	Electric variables
	USE:	Computed tomography	RT:	Breakdown voltage
<i>CTI</i>				Current control
	USE:	Cyber threat intelligence		Current limiters
<i>Cu</i>				Current measurement
	USE:	Copper		Current supplies
CubeSat				Current transformers
	BT:	Small satellites	NT:	Electrosurgery
Cultural aspects				Bioimpedance
	BT:	Social implications of		Current slump
technology				Dark current
	RT:	Inclusive language		Fault currents
		Museums		Inrush current
	NT:	Cultural competence		Leakage currents
		Cultural differences		Persistent currents
Cultural competence				Short-circuit currents
	UF:	Intercultural competence		Threshold current
	BT:	Cultural aspects	Current control	
	RT:	Behavioral sciences		UF:
		Cognition		Current regulation
		Linguistics		BT:
Cultural differences				Electric variables control
	BT:	Cultural aspects		RT:
		Social implications of		Current
technology				Current measurement
	RT:	Anthropology		Regulators
		Cross-cultural		Switches
communication				Switchgear
		Developing countries	NT:	Electric current control
		Digital divide		Electrical ballasts
		Ethnicity	Current crowding	
		Memetics		USE:
		Museums		Proximity effects
		Social intelligence	Current density	
		Virtual museums		BT:
Curing				Current measurement
	BT:	Materials processing		RT:
	RT:	Heat treatment		Density measurement
		Kilns		Particle measurements
Currencies				Skin effect
	BT:	Finance	Current distribution	
	NT:	Cryptocurrency		BT:
				Current measurement
				RT:
				Antenna theory
			Current limiters	
				BT:
				Power electronics
				RT:
				Current
				Current measurement
			NT:	Fault current limiters
			Current measurement	
				UF:
				Electric current
				measurement
				BT:
				Electric variables
				measurement



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RT:	Ammeters		Least squares
	Current	approximations	
	Current control		Splines (mathematics)
	Current limiters		Surface fitting
	Inrush current		
NT:	Current density	<i>Custom integrated circuits</i>	
	Current distribution	USE:	Application specific
		integrated circuits	
<i>Current measurement (water)</i>		<i>Customer behavior</i>	
USE:	Sea measurements	USE:	Consumer behavior
Current mirrors		<i>Customer behaviour</i>	
BT:	Circuit topology	USE:	Consumer behavior
<i>Current mode circuits</i>			
USE:	Current-mode circuits	Customer profiles	
<i>Current regulation</i>		BT:	Customer relationship
USE:	Current control	management	
		RT:	Consumer behavior
Current slump			Market opportunities
BT:	Current	Customer relationship management	
Current supplies		BT:	Management
BT:	Power supplies	RT:	Consumer behavior
RT:	Current		Consumer protection
			Data-driven modeling
Current transformers			Management information
BT:	Transformers	systems	
RT:	Current		Public relations
<i>Current voltage characteristics</i>			Quality management
USE:	Current-voltage	NT:	Supply chain management
characteristics AND			Customer profiles
	Electric variables		Customer satisfaction
			Customer services
			Market research
			Stakeholders
Current-mode circuits		Customer satisfaction	
UF:	Current mode circuits	UF:	Consumer satisfaction
BT:	Integrated circuits	BT:	Customer relationship
Current-voltage characteristics		management	
UF:	Current voltage	RT:	Customer services
characteristics			Electronic commerce
BT:	Electric variables		Market research
			Product customization
Curriculum development			Quality management
BT:	Educational courses	NT:	Churn
RT:	Educational programs		Quality of experience
	STEM		Quality of service
Curve fitting		Customer services	
BT:	Approximation methods	BT:	Customer relationship
	Visualization	management	
RT:	Computer graphics	RT:	Customer satisfaction
	Interpolation	NT:	Self-service



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<i>Cusum charts</i>		<i>Cyber forensics</i>	
USE:	Control charts	USE:	Digital forensics
<i>Cut-off frequency</i>		<i>Cyber harrassment</i>	
USE:	Cutoff frequency	USE:	Cyberbullying
Cutoff frequency		<i>cyber security</i>	
UF:	Cut-off frequency	USE:	Computer security
BT:	Integrated circuit modeling	<i>Cyber sickness</i>	
<i>Cutting fluids</i>		USE:	Cybersickness
USE:	Lubricants	<i>Cyber spying</i>	
Cutting tools		USE:	Cyber espionage
BT:	Production equipment	Cyber terrorism	
RT:	Blades	UF:	Cyberterrorism
	Dies	BT:	Computer crime
	Machine tools		Terrorism
	Metalworking machines	RT:	Computer networks
	Milling machines	Cyber threat intelligence	
NT:	Metal cutting tools	UF:	CTI
	Water jet cutting	BT:	Computer crime
<i>CVD</i>		RT:	Computer security
USE:	Chemical vapor deposition		Cyberattack
Cyanobacteria			Information integrity
BT:	Bacteria		Information security
<i>Cyber attack</i>			Information sharing
USE:	Cyberattack		Threat modeling
<i>Cyber attacks</i>		Cyber warfare	
USE:	Cyberattack	UF:	Cyberwarfare
<i>Cyber bullying</i>		BT:	Computer security
USE:	Cyberbullying	RT:	Computer networks
<i>Cyber crime</i>			Cyberattack
USE:	Computer crime		National security
<i>Cyber eavesdropping</i>		<i>Cyber-crime</i>	
USE:	Eavesdropping	USE:	Computer crime
Cyber espionage		Cyber-physical systems	
UF:	Cyber spying	UF:	Cyberphysical
	Cyberespionage	BT:	System of systems
	Cyberspying	RT:	Collaboration
BT:	Computer security		Control system security
	Information security		Embedded systems
RT:	Malware		Human computer
	Trojan horses	interaction	
<i>Cyber ethics</i>			Internet of Things
USE:	Cyberethics		Operating systems
			Smart cities
			Smart grids
			Wireless sensor networks
		NT:	Digital twins



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<i>Cyber-space</i>			Privacy
USE:	Cyberspace	technology	Social implications of
Cyberattack		<i>Cyberforensics</i>	
UF:	Cyber attack	USE:	Digital forensics
	Cyber attacks		
BT:	Computer crime	<i>Cyberharrassment</i>	
	Computer security	USE:	Cyberbullying
RT:	Cyber threat intelligence		
	Cyber warfare	Cybernetics	
	Cyberethics	UF:	Biocybernetics
	Security information and	BT:	Systems, man, and
event management	Threat modeling	cybernetics	
	US Department of	RT:	Automata
Homeland Security			Cognitive systems
NT:	Session hijacking		Control systems
			Cyberspace
Cyberbullying			Econometrics
UF:	Cyber bullying		Ergonomics
	Cyber harrassment		Human-machine systems
	Cyberharrassment		Information theory
	Internet bullying		Learning systems
	Online bullying		Neural networks
BT:	Cyberspace	networks	Radial basis function
	Social computing		
	Social networking (online)	NT:	Robots
	User experience		Adaptive systems
RT:	Behavioral sciences		Cognitive informatics
	Cyberethics		Cognitive science
	Privacy		Computational cybernetics
	Social implications of		Control theory
technology	Unsolicited e-mail		Econophysics
			Emergent phenomena
			Intelligent control
			Linear feedback control
Cybercare		systems	
BT:	Medical services		
<i>Cybercrime</i>		<i>Cyberphysical</i>	
USE:	Computer crime	USE:	Cyber-physical systems
<i>Cybereavesdropping</i>		<i>cybersecurity</i>	
USE:	Eavesdropping	USE:	Computer security
<i>Cyberespionage</i>		Cybersickness	
USE:	Cyber espionage	UF:	Cyber sickness
		BT:	Medical conditions
			Virtual reality
Cyberethics			
UF:	Cyber ethics	Cyberspace	
BT:	Ethics	UF:	Cyber-space
RT:	Behavioral sciences	BT:	Communication networks
	Cyberattack		Computer networks
	Cyberbullying	RT:	Computers
	Intellectual property		Cybernetics



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	Electromagnetics	BT:	Particle accelerators
	Internet		
	Telematics	<i>Cyphers</i>	
	Virtual reality	USE:	Ciphers
	World Wide Web		
NT:	Cyberbullying	Cytometry	
<i>Cyberspying</i>		BT:	Cells (biology)
USE:	Cyber espionage		Measurement
<i>Cyberterrorism</i>		RT:	Medical tests
USE:	Cyber terrorism	NT:	Flow cytometry
<i>Cyberwarfare</i>		Cytoplasm	
USE:	Cyber warfare	BT:	Cells (biology)
<i>Cyborgs</i>		Cytoskeleton	
USE:	Human-machine systems	BT:	Cells (biology)
<i>Cycle sharing</i>		<i>Cytotherapy</i>	
USE:	Shared transport	USE:	Cell therapy
Cyclic redundancy check		Cytotoxicity	
BT:	Mathematics	BT:	Cells (biology)
RT:	Algorithms	D-HEMTs	
	Data communication	UF:	Depletion mode HEMTs
	Error analysis		Depletion-mode HEMTs
	Error correction	BT:	HEMTs
	Information theory	<i>D/A</i>	
	Noise	USE:	Digital-analog conversion
NT:	Cyclic redundancy check	<i>D/A conversion</i>	
codes		USE:	Digital-analog conversion
Cyclic redundancy check codes		<i>D/A converters</i>	
UF:	CRC codes	USE:	Digital-analog conversion
BT:	Codes	<i>D2D</i>	
	Cyclic redundancy check	USE:	Device-to-device
RT:	Decoding	communication	
	Error analysis	Dairy products	
	Error correction	UF:	Butter
Cyclones			Cheese
UF:	Polar cyclones		Milk
BT:	Geoscience	BT:	Agricultural products
NT:	Hurricanes		Food products
	Tropical cyclones	RT:	Agriculture
<i>Cyclonic storms</i>			Fermentation
USE:	Tropical cyclones	Damascene integration	
Cyclostationary process		BT:	Electronic equipment
BT:	Signal processing	manufacture	
	Statistics	RT:	Very large scale integration
Cyclotrons			



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Dampers

USE: Shock absorbers

BT: Darknet
Internet
Overlay networks
RT: Web services

Damping

BT: Mechanical factors
RT: Hysteresis
Impedance
Oscillators
Propagation
Shock absorbers
Stability
Transfer functions
Transient response
Vibration control
Vibrations

Darknet

USE: Dark Web

DARPANET

USE: ARPANET

Data acquisition

UF: DAQ
BT: Data systems
RT: Analog-digital conversion
Data handling
Data processing
High energy physics
instrumentation computing
Measurement
NT: Data transparency
Fastbus
User-generated content

Dams

BT: Geotechnical structures
Water storage
RT: Environmental
management
Hydroelectric power
generation
Reservoirs

Data aggregation

BT: Data collection
RT: Data assimilation
Data handling
Data integration
Database systems
Information management

Dance

USE: Humanities

DAO

USE: Decentralized autonomous organization

Data analysis

UF: Data analytics
BT: Data processing
RT: Aerial computing
Big Data applications
Data collection
Data mining
Digital economy
Digital twins
Formal concept analysis
R language
Text categorization
Tiny machine learning
Trend analysis
NT: Business intelligence
Data augmentation
Data science
Functional data analysis
Itemsets
Training data

dapp

USE: Decentralized applications

DAQ

USE: Data acquisition

Dark current

BT: Current

Dark energy

BT: Physics

Dark matter

BT: Astrophysics

Dark net

USE: Dark Web

Dark states

BT: Laser applications

Dark Web

UF: Dark net



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Data analytics

USE: Data analysis

Data assimilation

BT: Data handling
RT: Data aggregation
Meteorology

Data augmentation

BT: Data analysis
Data handling
RT: Deep learning
Image augmentation
Learning (artificial
intelligence)
Machine learning
Synthetic data

Data authentication

USE: Message authentication

Data breach

UF: Data spill
BT: Information security
RT: Data security
Privacy breach
NT: Network intrusion

Data buses

UF: Computer buses
BT: Communication systems
Data communication
RT: CAMAC
Computer interfaces
IEEE 1394 Standard
NT: Backplanes

Data center power

BT: Data centers
Power systems

Data centers

UF: Data centres
BT: Data systems
NT: Data center power

Data centres

USE: Data centers

Data collection

UF: Data generation
BT: Data processing
RT: Blockchains
Data analysis
Information processing

technologies

NT: Big Data
Data aggregation
Data lakes

Data communication

UF: Data transmission
BT: Communication systems
RT: Ad hoc networks
B-ISDN
CAMAC
Computer networks
Cyclic redundancy check
Data dissemination
Data security
Digital communication
Distributed computing
Extranets
File servers
Firewire
HTTP
IEEE 1394 Standard
ISDN
Modems
Multiprocessor

interconnection

Office automation
Packet loss
Personal area networks
TCPIP
Telecontrol equipment
Teletext
Videotex
Asynchronous

NT:
communication

mode

Asynchronous transfer
Data buses
Data transfer
LoRa
Telecommunication buffers
Telemetry
Teleprinting
Visible light communication

Data compression

BT: Data systems
RT: Encoding
Entropy coding
Fourier series
Quantization (signal)
Rate-distortion
Streaming media
Transcoding



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compression

NT: Video compression
Adaptive coding
Audio compression
Huffman coding
Neural network

Point cloud compression
Source coding
Test data compression
Transform coding

Data confidentiality

USE: Data privacy

Data conversion

BT: Data systems
RT: Converters
NT: Analog-digital conversion
Digital-analog conversion

Data dissemination

BT: Data handling
Information sharing
RT: Data communication
Data integration
Mobile computing

Data encapsulation

BT: Data handling

Data engineering

BT: Data systems

Data enhancement

BT: Data handling
Data integrity

Data envelopment analysis

BT: Linear programming

Data flow computing

BT: Multiprocessing systems

Data flow graphs

USE: Flow graphs

Data freshness

USE: Data integrity

Data fusion

USE: Data integration

Data generation

USE: Data collection

Data gloves

BT: Haptic interfaces

Data governance

BT: Data handling
Organizational aspects
RT: Business data processing
Data integrity
Data models
Data privacy
Data protection
Data security
Database systems
Quality management
NT: Government policies

Data handling

UF: Electronic data interchange
BT: Data systems
RT: Big Data
Cryptography
Data acquisition
Data aggregation
Data processing
Data security
Encoding
Enterprise resource

planning

Regulation

General Data Protection

NT: Data assimilation
Data augmentation
Data dissemination
Data encapsulation
Data enhancement
Data governance
Data ingestion
Data integrity
Data retrieval
Document handling
Merging
Open data
Sorting
Turing completeness

Data ingestion

BT: Data handling
RT: Intake systems

Data intake

USE: Intake systems

Data integration

UF: Data fusion
BT: Data processing



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RT:	Data aggregation Data dissemination Data mesh	Data models BT: Modeling RT: Data governance Database systems Semantic Web Semantic technology NT: Data-driven modeling High dimensional data Metadata NSL-KDD
Data integrity UF: Data freshness Data quality BT: Computer security Data handling RT: Data governance Decentralized identity Digital preservation Quality assurance Quality control NT: Data enhancement Non-repudiation		Data over cable service interface specification UF: DOCSIS BT: Communication standards
Data lakes UF: Data swamp BT: Data collection RT: Big Data Data mesh Data mining Data warehouses		Data preprocessing BT: Data processing
Data mesh BT: Database systems RT: Data integration Data lakes Data structures Decentralized applications Sociotechnical systems		Data privacy UF: Data confidentiality Privacy preserving data mining BT: Data mining RT: Data governance Decentralized identity Digital policy Privacy Privacy enhancing technologies Synthetic data Threat modeling NT: Data protection Differential privacy
Data mining BT: Pattern recognition RT: Artificial intelligence Big Data Business intelligence Data analysis Data lakes Data visualization Digital forensics Knowledge discovery NSL-KDD Naive Bayes methods Nearest neighbor methods Predictive analytics R language NT: Anomaly detection Association rule learning Data privacy Process mining Text analysis Text mining Web mining		Data processing BT: Data systems RT: CAMAC Computers and information processing Data acquisition Data handling Database systems Enterprise resource planning Signal processing Smart cards Technology management NT: Associative processing Business data processing Data analysis Data collection Data integration Data preprocessing Data transfer



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	Information exchange		Security information and
	Spreadsheet programs	event management	
	Text processing		Threat modeling
	Virtual enterprises		Virtual private networks
		NT:	Cryptography
Data protection			Message authentication
BT:	Data privacy		Tokenization
RT:	Computer security		
	Data governance	<i>Data sources</i>	
	Data security	USE:	Soft sensors
	Decentralized identity		
	Differential privacy	<i>Data spill</i>	
	Information security	USE:	Data breach
	Privacy		
NT:	General Data Protection	<i>Data storage</i>	
Regulation		USE:	Memory
<i>Data quality</i>		<i>Data storage systems</i>	
USE:	Data integrity	USE:	Buffer storage
<i>Data representation</i>		Data stream processing	
USE:	Data visualization	UF:	Stream processing
		BT:	Data systems
Data retrieval		Data structures	
BT:	Data handling	BT:	Computer architecture
RT:	Cross modal retrieval	RT:	Computer languages
	Information retrieval		Data mesh
	Query processing		Database systems
			File systems
Data science			NoSQL databases
BT:	Data analysis	NT:	Arrays
RT:	Knowledge discovery		Binary decision diagrams
	Neuroinformatics		Null value
NT:	Concept drift		Octrees
			Persistent identifiers
Data security			Table lookup
UF:	Security of data		Tree data structures
	System privacy		
management		<i>Data swamp</i>	
		USE:	Data lakes
BT:	Security	Data systems	
RT:	Communication system	BT:	Computers and information
security		processing	
			Information systems
	Computer crime		Big Data applications
	Computer network	RT:	Buffer storage
		NT:	Data acquisition
	Computer security		Data centers
	Data breach		Data compression
	Data communication		Data conversion
	Data governance		Data engineering
	Data handling		Data handling
	Data protection		Data processing
	Decentralized identity		
	Network intrusion		
	Privacy		
	Privacy breach		



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	Data stream processing	RT:	Database systems
	Data warehouses		Information systems
Data transfer		<i>Database management systems</i>	
BT:	Data communication	USE:	Database systems
	Data processing		
RT:	Packet switching	Database systems	
NT:	Handover	UF:	Database management
	Simultaneous wireless	systems	Technical data
information and power transfer		management	
<i>Data transmission</i>		BT:	Databases
USE:	Data communication		Information systems
Data transparency		RT:	Data aggregation
BT:	Data acquisition		Data governance
			Data models
<i>Data visualisation</i>			Data processing
USE:	Data visualization		Data structures
Data visualization			Database languages
UF:	Data representation		Database machines
	Data visualisation		File systems
BT:	Computer graphics		Hypertext systems
	User interfaces		Information architecture
RT:	Biomedical imaging	NT:	Linked data
	Data mining		Triples (Data structure)
	Graph neural networks		Audio databases
	Modeling		Data mesh
	R language		Deductive databases
	Vision language model		Image databases
NT:	Graph drawing		Indexes
	Heat maps		Multimedia databases
	Isosurfaces		NoSQL databases
			Object oriented databases
			Query processing
			Sharding
Data warehouses		Databases	
BT:	Data systems	BT:	Professional
RT:	Data lakes	communication	
	NoSQL databases	NT:	Database systems
Data-driven modeling			Deductive databases
BT:	Data models		Distributed databases
	Modeling		Image databases
RT:	Customer relationship		Multimedia databases
management			Object oriented databases
Database languages			Relational databases
UF:	Query languages		Spatial databases
BT:	Computer languages		Transaction databases
RT:	Database systems		Visual databases
NT:	Structured Query Language	Daylighting	
Database machines		BT:	Lighting
BT:	Computers and information	<i>DBR</i>	
processing		USE:	Distributed Bragg reflectors



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DBS

USE: Satellite broadcasting

DC distribution systems

BT: Network systems
Power distribution
RT: DC power transmission
HVDC transmission
Power transmission lines

DC fast charging

USE: Fast charging

DC generators

UF: Direct current generators
BT: DC machines
Generators
RT: Pulse width modulation
Rotating machines

DC machines

UF: Direct current machines
BT: Electric machines
RT: Pulse width modulation
Sensorless control
NT: DC generators
DC motors
Homopolar machines

DC motors

UF: Direct current motors
BT: DC machines
Motors
RT: Pulse width modulation
Pulse width modulation

inverters

Space vector pulse width

modulation

NT: Brushless DC motors
Commutators

DC power transmission

UF: Direct current power

transmission

BT: Power transmission
RT: DC distribution systems

DC-AC power converters

UF: DC-AC power converters
BT: Converters
Power conversion
RT: Microinverters

DC-AC power converters

USE: DC-AC power converters

DC-DC converters

USE: DC-DC power converters

DC-DC power conversion

USE: DC-DC power converters

DC-DC power converters

UF: DC-DC converters
DC-DC power conversion
DC-DC power converters
BT: Power conversion
RT: Machine vector control
Pulse width modulation

inverters

NT: Buck converters

DC-DC power converters

USE: DC-DC power converters

DCT

USE: Discrete cosine transforms

DDoS

USE: Distributed denial-of-service
attack

DDos attack

USE: Denial-of-service attack

DDSM

USE: Delta-sigma modulation

De broglie hypothesis

USE: Matter waves

De Broglie methods

USE: Matter waves

De-noising

USE: Noise reduction

Dead reckoning

BT: Navigation

Deadlocks (computers)

USE: System recovery

Deafness

BT: Medical conditions
RT: Sign language



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Death

BT: Pathological processes
NT: Asphyxia

Death rate

USE: Mortality

Deblurring

BT: Image restoration
RT: Image dehazing
Image quality
Superresolution
Video restoration

Debugging

BT: System recovery

Deburring

BT: Surface finishing
RT: Drilling
Machining
Polishing machines

Decarbonization

BT: Greenhouse gases
RT: Low carbon economy

Decentralised applications

USE: Decentralized applications

Decentralised apps

USE: Decentralized applications

Decentralised autonomous organization

USE: Decentralized autonomous organization

Decentralised control

USE: Decentralized control

Decentralised identity

USE: Decentralized identity

Decentralized applications

UF: Decentralised applications
Decentralised apps
Decentralized apps
dapp
BT: Application software
Distributed computing
RT: Consensus control
Data mesh
Peer-to-peer computing
Smart contracts

Decentralized apps

USE: Decentralized applications

Decentralized autonomous organization

UF: DAO
Decentralised autonomous

organization

BT: Organizations
RT: Blockchains

Decentralized control

UF: Decentralised control
Distributed control
Distributed generation
Distributed modeling
BT: Control systems
RT: Flexible structures
NT: Consensus control
Distributed parameter

systems

Decentralized identity

UF: Decentralised identity
BT: Digital control
RT: Blockchains
Data integrity
Data privacy
Data protection
Data security
Digital management
Identity management

systems

Information management
Information security
Privacy
Semantic Web

Decision analysis

BT: Decision making
Information analysis
RT: MCDM

Decision feedback equalizers

BT: Equalizers

Decision making

BT: Decision theory
RT: Deep reinforcement

learning

Expert systems
Fish schools
Fuzzy cognitive maps
Planning
Risk analysis
Signal detection



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	Stakeholders		Speech codecs
	Strategic planning		Video codecs
	TOPSIS	NT:	Maximum likelihood
	User preference	decoding	
NT:	Analytic hierarchy process		
	Collective intelligence	Decontamination	
	Decision analysis	BT:	Materials handling
	Decision support systems	RT:	Chemical technology
	Distributed decision making		Contamination
	Game theory		Environmental monitoring
	MCDM		Pollution control
	Pattern classification		Purification
	Persuasive systems	NT:	Disinfection
	Trust management		Sterilization
Decision support systems		Deconvolution	
BT:	Decision making	BT:	Inverse problems
	Information systems	RT:	Convolution
RT:	Competitive intelligence		Integral equations
	Knowledge based systems		Numerical analysis
	MCDM		Signal processing
			Signal restoration
Decision theory		Decorrelation	
BT:	Statistics	BT:	Signal processing
RT:	Probability		
	Statistical analysis	Dedicated short range communication	
	Statistical learning	UF:	DSRC
NT:	Decision making	BT:	Wireless communication
	Decision trees	RT:	Intelligent vehicles
	TOPSIS		On board unit
	Weighted sum model		Vehicular ad hoc networks
Decision trees		Deductive databases	
UF:	Tree searching	UF:	Intelligent databases
BT:	Decision theory	BT:	Database systems
RT:	Random forests		Databases
NT:	Classification tree analysis	RT:	Knowledge based systems
	Regression tree analysis		
<i>Decoder</i>		Deep architecture	
USE:	Decoding	UF:	Deep learning architecture
		BT:	Software architecture
Decoding			Systems architecture
UF:	Decoder	RT:	Machine learning
BT:	Information theory		Neural networks
RT:	Autoencoders	NT:	Deep learning
	Codecs	Deep brain stimulation	
	Codes	BT:	Brain stimulation
	Cyclic redundancy check		Neurosurgery
codes		RT:	Neural implants
	Demodulation		Neurostimulation
	Parity check codes	<i>Deep etching</i>	
	Product codes	USE:	Etching
	Signal processing		
	Space-time codes		



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Deep fake

USE: Deepfakes

Deep learning

UF: Deep structured learning
Hierarchical learning
BT: Deep architecture
Machine learning
RT: Attention mechanisms
Behavior recognition
Convolutional neural

networks

Data augmentation
Feature extraction
Fine-grained image

recognition

Generative Pre-Trained

Transformer

Graph neural networks
Image classification
Image segmentation
Large language models
Learning systems
Model compression
Neural networks
Recurrent neural networks
Reinforcement learning
Semantic segmentation
Supervised learning
Unsupervised learning
NT: Batch normalization
Deep reinforcement
learning
Few shot learning
Neural radiance field
Zero shot learning

Deep learning architecture

USE: Deep architecture

Deep level transient spectroscopy

BT: Semiconductor materials
Spectroscopy

Deep neural networks

USE: Artificial neural networks

Deep reinforcement learning

BT: Deep learning
Machine learning
Reinforcement learning
RT: Decision making

Deep space

USE: Deep-space
communications

Deep structured learning

USE: Deep learning

Deep-space communications

UF: Deep space
BT: Space communications
RT: Telemetry

Deepfakes

UF: Deep fake
BT: Fake news
Videos
RT: Information integrity
Photorealism

Defect detection

BT: Detection algorithms

Defence industry

USE: Defense industry

Defense industry

UF: Defence industry
BT: Industries
RT: Military equipment
Weapons

Defibrillation

BT: Medical treatment
RT: Cardiology
Fibrillation

Definitions

USE: Terminology

Deforestation

UF: Forest clearance
Forest clearing
BT: Agricultural engineering
Environmental degradation
Forestry
Land use planning
RT: Rainforests
Reforestation

Deformable models

BT: Modeling



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Deformation

BT: Mechanical factors
RT: Electron backscatter

diffraction

Degenerative disc disease

BT: Medical conditions
RT: Spine

Degenerative diseases

BT: Diseases

Degradation

BT: Materials science and
technology
NT: Desertification

Delamination

BT: Materials testing

Delay effects

UF: Time delay
BT: Electromagnetic analysis
RT: Delay lines
Delay systems
Distortion
Phase distortion
NT: Propagation delay

Delay estimation

BT: Delays
RT: Computer networks
Multiaccess communication
Speech processing

Delay lines

BT: Delay systems
RT: Delay effects

Delay lock loops

USE: Tracking loops

Delay systems

BT: Control systems
RT: Delay effects
Telerobotics
NT: Added delay
Delay lines

Delays

BT: Timing
RT: Quality of transmission
NT: Delay estimation

Delivery of goods

USE: Product delivery

Delta modulation

BT: Analog-digital conversion
Digital signal processing
NT: Delta-sigma modulation
Sigma-delta modulation

Delta sigma

USE: Sigma-delta modulation

Delta sigma modulators

USE: Delta-sigma modulation

Delta-sigma modulation

UF: DDSM
Delta sigma modulators
Delta-sigma modulators
BT: Delta modulation

Delta-sigma modulators

USE: Delta-sigma modulation

Demagnetisation

USE: Demagnetization

Demagnetization

UF: Demagnetisation
BT: Magnetism

Demand forecasting

BT: Forecasting
RT: Production planning

Demand response

BT: Power demand

Demand side management

BT: Energy management
RT: Power system planning
Vehicle-to-grid

Dementia

BT: Medical conditions
NT: Alzheimer's disease

Demodulation

UF: Demodulators
BT: Modulation
RT: Amplitude modulation
Decoding
Detectors
Frequency modulation
Mixers



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Modems
Phase modulation
Pulse modulation
Receivers
Signal detection

Demodulators

USE: Demodulation

Demography

BT: Social factors
NT: Population density

Dempster?Shafer theory

USE: Evidence theory

Demultiplexing

BT: Multiplexing
RT: Arrayed waveguide gratings

Dendrites (neurons)

UF: Dendrons
BT: Neurons

Dendrons

USE: Dendrites (neurons)

Dengue fever

BT: Diseases

Denial-of-service attack

UF: DDos attack
DoS attack
BT: Communication system

security

RT: Computer security
Proof of Work
NT: Distributed denial-of-service

attack

Denitrification

BT: Nitrogen

Denoising

USE: Noise reduction

Density estimation robust algorithm

UF: DER
BT: Algorithms

Density function

USE: Density functional theory

Density function theory

USE: Density functional theory

Density functional theory

UF: Density function
Density function theory
Density-function
BT: Quantum mechanics

Density measurement

BT: Measurement
RT: Bone density
Current density
Pressure gauges
NT: Hydrometers
Population density

Density-function

USE: Density functional theory

Dental

USE: Dentistry

Dentistry

UF: Dental
BT: Medical treatment

Deny lists

USE: Blocklists

Deoxyribonucleic acid

USE: DNA

Dependability management

BT: Management
RT: Reliability
Safety management

Depletion mode HEMTs

USE: D-HEMTs

Depletion-mode HEMTs

USE: D-HEMTs

Deployable structures

USE: Flexible structures

Depression

UF: Major depressive disorder
BT: Medical conditions
RT: Psychiatry

Depth estimation

USE: Depth measurement

Depth measurement

UF: Depth estimation
BT: Measurement



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	RT:	Distance measurement Geophysical measurements Semiconductor device			Design tools EMTDC Hardware design
measurement			languages		
<i>DER</i>					Laser sintering SPICE
	USE:	Density estimation robust	NT:		CAD/CAM Logic design PSCAD
algorithm					
<i>Derailments</i>					
	USE:	Railway accidents			
Dermatology			Design engineering		
	BT:	Medical specialties	BT:		Engineering - general
	RT:	Psoriasis	RT:		Building information management
Dermis					Design optimization Digital twins
	BT:	Skin	NT:		Design tools
Desalination			Design for disassembly		
	BT:	Water conservation Water resources	BT:		Design methodology
	RT:	Reverse osmosis	RT:		Pollution Process design Product design Waste reduction
Description logic			Design for experiments		
	BT:	Knowledge representation Ontologies	BT:		Design methodology
Descriptive statistics			<i>Design for manufacturability</i>		
	BT:	Statistical analysis	USE:		Design for manufacture
	RT:	Inferential statistics			
Desertification			Design for manufacture		
	BT:	Degradation	UF:		Design for manufacturability
	RT:	Deserts	BT:		Design methodology
			RT:		Design tools
Deserts			Design for quality		
	BT:	Geography	BT:		Design methodology
	RT:	Desertification	RT:		Process design Product design Quality assurance Quality control Quality management Total quality management
<i>Design</i>					
	USE:	Design methodology			
Design automation			Design for testability		
	UF:	CAD Circuit CAD Circuit design (CAD) Circuit layout CAD Computer aided design Computer-aided design	UF:		Design-for-test Design-for-testability Design-for-testing
	BT:	Electronic design	BT:		Design methodology
automation and methodology			RT:		Built-in self-test Combinatorial testing Logic design Logic testing
	RT:	Automatic test pattern			
generation					
		Circuit simulation			



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Design methodology

UF: Design
 BT: Electronic design
 automation and methodology
 RT: Ergonomics
 Industrial engineering
 Logic design
 Optical design techniques
 Rapid prototyping
 System analysis and design
 NT: Design for disassembly
 Design for experiments
 Design for manufacture
 Design for quality
 Design for testability
 Design standards
 Design tools
 Extensibility
 Graphics
 Green design
 Integrated design
 Inverse design
 Process design
 Product design
 Prototypes
 Technical drawing
 Time to market
 User centered design
 Virtual prototyping

Design optimization

BT: Optimization methods
 RT: Design engineering
 NT: Constraint optimization
 Space mapping

Design standards

BT: Design methodology

Design tools

BT: Design engineering
 Design methodology
 RT: Design automation
 Design for manufacture
 Instruments
 Media
 Product design
 Visualization

Design-for-test

USE: Design for testability

Design-for-testability

USE: Design for testability

Design-for-testing

USE: Design for testability

Desktop publishing

BT: Electronic publishing
 Publishing
 RT: Document handling
 Office automation
 Page description languages
 Text processing

Detection (signal)

USE: Signal detection

Detection algorithms

BT: Algorithms
 NT: Defect detection

Detectors

BT: Sensor systems and
 applications
 RT: Chemical sensors
 Demodulation
 Fall detection
 Nonlinear filters
 Readout electronics
 NT: Envelope detectors
 Semiconductor detectors

Deuterium

BT: Hydrogen

Developing countries

BT: Economics
 Social factors
 RT: Cultural differences
 Digital divide
 Globalization
 Government policies
 International trade
 Investment

Device chargers

USE: Battery chargers

Device drivers

BT: Input-output programs
 RT: Computer peripherals

Device under test

UF: Equipment under test
 Unit under test
 BT: Testing



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Device-to-device communication

UF: D2D
BT: Communication systems
RT: Base stations
Cellular networks
Sidelink

DevOps

BT: Information technology
Software development
management

Dew computing

UF: Cloud-dew architecture
Cloud-dew computing
Dew servers
BT: Distributed processing
Software architecture
RT: Client-server systems
Cloud computing
Edge computing

Dew servers

USE: Dew computing

DFA

USE: Doped fiber amplifiers

DFIG

USE: Doubly fed induction
generators

DFT

USE: Discrete Fourier transforms

DGPS

USE: Global Positioning System

DH-HEMTs

UF: Double heterojunction
HEMTs
BT: HEMTs

DHBTs

USE: Double heterojunction
bipolar transistors

Diabetes

UF: Diabetic
BT: Medical conditions
RT: Endocrinology
Insulin pumps
NT: Diabetic retinopathy
Gestational diabetes

Diabetic

USE: Diabetes

Diabetic retinopathy

BT: Diabetes
Retinopathy
RT: Retinal vessels

Diacs

USE: Thyristors

Diagnosis (medical)

USE: Medical diagnosis

Diagnostic expert systems

BT: Expert systems
RT: Failure analysis
Fault diagnosis

Diagnostic radiography

BT: Radiography
RT: Attenuation
Magnetic resonance

imaging

Medical diagnosis
X-ray detection

Diakoptics

BT: System analysis and design

Diamagnetic materials

BT: Magnetic materials

Diamond

BT: Carbon

Diamond carbon

USE: Diamond-like carbon

Diamond like carbon

USE: Diamond-like carbon

Diamond-like carbon

UF: Diamond carbon
Diamond like carbon
Hard amorphous carbon
BT: Amorphous materials
RT: Biomedical materials
Thin films
Tissue engineering

DICOM

UF: Digital Imaging and
Communications in Medicine
BT: Biomedical imaging



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	Digital communication	RT:	Dielectric materials
Dictionaries		antennas	Dielectric resonator
BT:	Information services		Electrets
	Terminology	NT:	Capacitors
	Writing		Ferroelectric devices
<i>Dictionary learning</i>			Piezoelectric devices
USE:	Machine learning		Pyroelectric devices
<i>Didi</i>		Dielectric elastomer actuators	
USE:	Ride hailing	BT:	Actuators
		NT:	Dielectric elastomers
<i>Die attach</i>		Dielectric elastomers	
USE:	Microassembly	UF:	Smart elastomers
<i>Die bonding</i>		BT:	Dielectric elastomer
USE:	Microassembly	actuators	
		RT:	Smart materials
Die casting		<i>Dielectric electroactive polymer actuators</i>	
BT:	Casting	USE:	Actuators
RT:	Automobile manufacture		
	Dies	Dielectric films	
	Materials handling	BT:	Dielectric materials
	Melt processing		Films
	Metals	RT:	Planarization
<i>Dielectric barrier discharges</i>			Thick films
USE:	Discharges (electric)		Thin films
		NT:	Dielectric thin films
Dielectric breakdown			Piezoelectric films
UF:	Dielectric strength	Dielectric liquids	
	Voltage breakdown	UF:	Liquid insulation
BT:	Electric breakdown	BT:	Dielectric materials
RT:	Dielectric measurement		
	Insulation	Dielectric loss measurement	
NT:	Lightning	BT:	Dielectric measurement
	Arc discharges	RT:	Dielectric losses
	Discharges (electric)	Dielectric losses	
	Electrostatic discharges	UF:	Dielectric loss
	Flashover	BT:	Dielectrics
	Glow discharges	RT:	Dielectric loss
	Partial discharges	measurement	
	Surface discharges		Insulation
	Vacuum breakdown		
Dielectric constant		Dielectric materials	
UF:	Solid permittivity	UF:	Antiferroelectric materials
BT:	Dielectrics		Paraelectric materials
RT:	Capacitors	BT:	Materials
	Permittivity	RT:	Ceramics
NT:	High-k gate dielectrics		Dielectric devices
Dielectric devices			Dielectric thin films
BT:	Dielectrics		Electrohydrodynamics
			Electrokinetics



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	Ferroelectric materials		BT:	Dielectrics and electrical
	Glass	insulation		
	Insulation		NT:	Dielectric constant
	Loaded waveguides			Dielectric devices
	Permittivity			Dielectric losses
	Plastic insulation			Dielectric substrates
NT:	Dielectric films			Dielectrophoresis
	Dielectric liquids			Electrohydrodynamics
	Electrets			Electrokinetics
	Epoxy resins			Electrostriction
	High-k dielectric materials			
	Piezoelectric materials			
Dielectric measurement		Dielectrics and electrical insulation		
UF:	Dielectric measurements	NT:	Dielectrics	
BT:	Electric variables		Electric breakdown	
measurement			Electromagnetic fields	
RT:	Capacitance measurement		Insulation	
	Dielectric breakdown	Dielectrophoresis		
	Electromagnetic	BT:	Dielectrics	
measurements		<i>Dielectric loss</i>		
NT:	Dielectric loss	USE:	Dielectric losses	
measurement				
	Permittivity measurement	Dies		
<i>Dielectric measurements</i>		UF:	Dies (machine tools)	
USE:	Dielectric measurement	BT:	Machine tools	
<i>Dielectric properties</i>		RT:	Cutting tools	
USE:	Dielectrics		Die casting	
			Presses	
Dielectric resonator antennas		<i>Dies (machine tools)</i>		
BT:	Antennas	USE:	Dies	
RT:	Dielectric devices	Diesel engines		
	Resonance	BT:	Internal combustion	
<i>Dielectric strength</i>		engines		
USE:	Dielectric breakdown	RT:	Automotive engineering	
<i>Dielectric substrate</i>		Difference engines		
USE:	Dielectric substrates	BT:	Calculators	
Dielectric substrates		Difference equations		
UF:	Dielectric substrate	BT:	Equations	
BT:	Dielectrics	RT:	Discrete-time systems	
Dielectric thin films			Numerical analysis	
BT:	Dielectric films		Piecewise linear techniques	
RT:	Dielectric materials	Differential algebraic equations		
	Polymer films	BT:	Differential equations	
	Semiconductor films	Differential amplifiers		
Dielectrics		BT:	Amplifiers	
UF:	Dielectric properties	Differential diagnosis		
	Electrical insulation	BT:	Medical diagnosis	



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Differential equations

BT: Calculus
 RT: Differential games
 Higher order statistics
 Integrodifferential equations
 Numerical analysis
 Predator prey systems
 Stability analysis
 Time invariant systems
 NT: Differential algebraic

equations

Differential operators
 Navier-Stokes equations
 Ordinary differential

equations

Partial differential equations
 Transfer functions

Differential games

BT: Game theory
 RT: Differential equations

Differential gears

USE: Gears

Differential GPS

USE: Global Positioning System

Differential operators

BT: Differential equations

Differential phase shift keying

UF: DPKS
 Differential phase-shift

keying

BT: Phase modulation

Differential phase-shift keying

USE: Differential phase shift

keying

Differential privacy

BT: Data privacy
 Statistics
 RT: Data protection
 Information security
 Privacy

Differential quadrature phase shift keying

BT: Quadrature phase shift

keying

Differentiated services networks

USE: Diffserv networks

Diffraction

UF: Wave diffraction
 BT: Interference
 RT: Bragg gratings
 Crystallography
 Fourier transforms
 X-ray detection
 NT: Electron backscatter

diffraction

Diffraction gratings

BT: Optical diffraction
 RT: Bragg gratings

DiffServ

USE: Diffserv networks

Diffserv networks

UF: DiffServ
 Differentiated services

networks

BT: Computer networks
 Distributed computing
 RT: Internet
 Multimedia communication

Diffuse reflectance spectra

USE: Diffuse reflectance

spectroscopy

Diffuse reflectance spectroscopy

UF: Diffuse reflectance spectra
 BT: Reflectivity
 Spectroscopy

Diffusion bonding

BT: Bonding processes
 RT: Ceramics

Diffusion models

UF: Diffusion probabilistic

models

BT: Machine learning

Diffusion probabilistic models

USE: Diffusion models

Diffusion processes

BT: Semiconductor device
 manufacture

RT: Brownian motion
 Buffer layers
 Charge carrier processes
 Image denoising
 Image processing



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NT:	Stochastic processes Electromigration	RT:	Digital communication Audio systems HDMI Portable media players
Diffusion tensor imaging		NT:	Digital Radio Mondiale Digital audio players
UF:	DT-MRI DTI Diffusion tensor magnetic resonance imaging		
BT:	Magnetic resonance	<i>Digital audio broadcasts</i>	
RT:	Brain	USE:	Digital audio broadcasting
<i>Diffusion tensor magnetic resonance imaging</i>		Digital audio players	
USE:	Diffusion tensor imaging	UF:	MP3
		BT:	Digital audio broadcasting
Digestive system		<i>Digital camera</i>	
BT:	Anatomy	USE:	Digital cameras
RT:	Endomicroscopy Inflammatory bowel disease Microbiome		
NT:	Colon Esophagus Gallbladder Gastrointestinal tract Intestines Liver Mouth Pancreas Pharynx Stomach Tongue	Digital cameras	
		UF:	Digital camera
		BT:	Cameras
		RT:	Digital photography
<i>Digital age</i>		<i>Digital certificates</i>	
USE:	Information age	USE:	Public key
<i>Digital agriculture</i>		Digital circuits	
USE:	Smart agriculture	BT:	Circuits
		RT:	Digital computers Logic circuits Pulse circuits Register transfer level Switching circuits
Digital alloys		NT:	Circuit topology Digital integrated circuits
BT:	Metals		
Digital arithmetic		Digital communication	
UF:	Computer arithmetic	UF:	Digital radio
BT:	Arithmetic	BT:	Communication systems
RT:	Calculators Coprocessors	RT:	Bluetooth Data communication Digital recording Musical instrument digital interfaces
			Synchronous digital
Digital art			TCPIP Teleprinting
BT:	Art	NT:	Baseband DICOM DSL Digital audio broadcasting Digital images Digital multimedia
RT:	Virtual museums		
Digital audio broadcasting		broadcasting	Digital video broadcasting
UF:	Digital audio broadcasts Podcast		
BT:	Broadcasting		



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	ISDN		Internet of Things
	Passband		Online banking
	Portable media players	NT:	Blockchains
	SONET		
	Spread spectrum		
communication		<i>Digital elevation modeling</i>	
		USE:	Digital elevation models
Digital computers		Digital elevation models	
BT:	Computers	UF:	Digital elevation modeling
RT:	Digital circuits		Digital terrain model
	Digital systems		Digital terrain modeling
	Digital-analog conversion		Digital terrain models
	Parallel processing	BT:	Modeling
	Programming		Terrain mapping
	Turing machines		
NT:	Mainframes	<i>Digital factories</i>	
		USE:	Virtual manufacturing
Digital control		Digital filters	
UF:	Computer control	BT:	Filters
BT:	Control systems	RT:	Frequency response
RT:	Computer numerical control		Line enhancers
NT:	Decentralized identity		Optical resonators
	Programmable control		Transversal filters
<i>Digital currency</i>		NT:	Finite impulse response
USE:	Online banking	filters	
<i>Digital data</i>		Digital forensics	
USE:	Digital information	UF:	Cyber forensics
			Cyberforensics
<i>Digital devices</i>		BT:	Forensics
USE:	Personal digital devices	RT:	Computer security
			Data mining
Digital divide			Law enforcement
UF:	Digital inclusion		Non-repudiation
BT:	Sociology		
RT:	Cultural differences	<i>Digital governance</i>	
	Developing countries	USE:	Digital policy
	Economics		
	Ethical aspects	<i>Digital health</i>	
	Gender issues	USE:	Electronic healthcare
	Social factors		
	Social implications of	<i>Digital healthcare</i>	
technology		USE:	Electronic healthcare
Digital economy		Digital humans	
BT:	Economics	BT:	Avatars
RT:	Artificial intelligence	RT:	Artificial intelligence
	Business		Chatbots
	Cryptocurrency		Conversational artificial
	Data analysis	intelligence	
	Digital systems		Human computer
	Electronic commerce	interaction	
	Information technology		Human factors
	Internet		



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processing	Natural language	Digital management	BT: Management
	Social factors	RT: Decentralized identity	
	Virtual assistants		
	Virtual reality	Digital microfluidic biochips	BT: Biochips
Digital images			
UF:	Digital imaging	<i>Digital micromirror devices</i>	USE: Micromirrors
BT:	Digital communication		
NT:	Pixel		
	Virtual museums	Digital modulation	BT: Modulation
<i>Digital imaging</i>		NT: Constellation diagram	Partial response signaling
USE:	Digital images		
<i>Digital Imaging and Communications in Medicine</i>		Digital multimedia broadcasting	UF: DMB
USE:	DICOM		Digital multimedia
<i>Digital inclusion</i>		broadcasts	
USE:	Digital divide	BT: Broadcasting	Digital communication
			Cellular radio
Digital information		RT: Convolutional codes	Digital TV
UF:	Digital data		MPEG 4 Standard
BT:	Information management		MPEG 7 Standard
RT:	Digital systems		MPEG standards
	Information systems		Multimedia communication
	Symbols		Radio broadcasting
			Video on demand
Digital integrated circuits		<i>Digital multimedia broadcasts</i>	USE: Digital multimedia
BT:	Digital circuits	broadcasting	
	Integrated circuits		
RT:	Adders	Digital photography	BT: Photography
	Logic circuits		RT: CCD image sensors
	Multiplying circuits		Cameras
NT:	Integrated memory circuits		Digital cameras
			Transform coding
			Virtual museums
Digital intelligence		Digital policy	UF: Digital governance
UF:	Digital literacy		Internet governance
BT:	Human intelligence	BT: Digital systems	Guidelines
	Human-machine systems	RT: Computer security	Data privacy
RT:	Behavioral sciences		Government policies
	Cognition		Information security
	Competitive intelligence		Privacy
	Digital systems		
	Ethics		
	Human factors		
	Psychology		
	Social intelligence		
<i>Digital literacy</i>			
USE:	Digital intelligence		
Digital magnetic recording			
BT:	Magnetic recording		



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Digital preservation

BT: Digital systems
Information management
RT: Data integrity
Virtual museums

Digital printing

BT: Printing
RT: Publishing

Digital publishing

USE: Electronic publishing

Digital radio

USE: Digital communication

Digital Radio Mondiale

BT: Digital audio broadcasting

Digital recording

BT: Recording
RT: Digital communication
Digital systems
Virtual museums

Digital relays

BT: Relays

Digital representation

BT: Encoding
Image representation
Information representation
RT: Augmented reality
Quantization (signal)
Virtual museums
Virtual reality

Digital rights management

BT: Intellectual property
RT: Computer crime
Copyright protection
Software protection

Digital sequences

USE: Sequences

Digital signal processing

UF: DSP
BT: Signal processing
RT: Aerospace and electronic

systems

Digital TV
Fast Fourier transforms
OFDM
NT: Delta modulation

Digital signal processing
chips

Digital signal processing chips

BT: Digital signal processing

Digital signal processors

BT: Circuits
RT: Signal processing

Digital signatures

BT: Security
RT: Message authentication
Message systems

Digital simulation

BT: Simulation
RT: Computer aided analysis
Modeling
Power system analysis

computing

NT: Digital twins
Discrete event simulation

Digital storage

BT: Digital systems
Storage management
NT: DNA data storage
Solid state drives

Digital subscriber lines

USE: DSL

Digital subscriber loops

USE: DSL

Digital systems

BT: Computers and information
processing
RT: Communication systems
Computational and artificial
intelligence

Consumer electronics
Cryptocurrency
Digital computers
Digital economy
Digital information
Digital intelligence
Digital recording
Digital transformation
Persistent identifiers
Personal communication

networks

NT: Digital policy
Digital preservation



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Digital storage
ISDN
Internet
Local area networks
Metropolitan area networks
Smart agriculture
Token networks
Virtual artifact

theory

Product design
Solid modeling
Systems engineering and
Virtual reality

Digital versatile discs

USE: DVD

Digital video broadcasting

UF: Digital video broadcasts
BT: Broadcasting
Digital communication

Digital video broadcasts

USE: Digital video broadcasting

Digital video discs

USE: DVD

Digital watermarking

USE: Watermarking

Digital-analog

USE: Digital-analog conversion

Digital-analog conversion

UF: D/A
D/A conversion
D/A converters
Digital to analog conversion
Digital to analog converters
Digital-analog
Digital-analog converters
Digital-analogue conversion
Digital-analogue converters
Digital-to-analog conversion
Digital-to-analog converters
BT: Data conversion
RT: Digital computers
Interpolation

Digital-analog converters

USE: Digital-analog conversion

Digital-analogue conversion

USE: Digital-analog conversion

Digital-analogue converters

USE: Digital-analog conversion

Digital-controlled oscillators

BT: Oscillators

Digital terrain model

USE: Digital elevation models

Digital terrain modeling

USE: Digital elevation models

Digital terrain models

USE: Digital elevation models

Digital to analog conversion

USE: Digital-analog conversion

Digital to analog converters

USE: Digital-analog conversion

Digital transformation

BT: Analog-digital conversion
RT: Augmented reality
Business
Digital systems
Digitization
Internet of Things
Virtual reality
NT: Fifth Industrial Revolution
Fourth Industrial Revolution

Digital TV

BT: TV
RT: Digital multimedia
broadcasting
Digital signal processing
HbbTV Standards
Text to speech
NT: HDTV
IPTV

Digital twins

UF: virtual twins
BT: Cyber-physical systems
Digital simulation
RT: Artificial intelligence
Augmented reality
Computer simulation
Data analysis
Design engineering
Internet of Things



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Digital-to-analog conversion

USE: Digital-analog conversion

Digital-to-analog converters

USE: Digital-analog conversion

Digital-to-frequency converters

BT: Converters

Digitization

BT: Analog-digital conversion

RT: Digital transformation

Dike

USE: Levee

DIL

USE: Electronics packaging

Dimension reduction

USE: Dimensionality reduction

Dimensionality reduction

UF: Dimension reduction

BT: Information retrieval

Machine learning

Statistics

NT: Manifold learning

Dinosaurs

BT: Animals

Diode lasers

UF: Laser diodes

BT: Diodes

Lasers

Diodes

BT: Electronic components

Voltage multipliers

RT: Breakdown voltage

Optical transmitters

Semiconductor diodes

NT: Active matrix organic light

emitting diodes

Diode lasers

Light emitting diodes

Organic light emitting

diodes

P-i-n diodes

Schottky diodes

Semiconductor lasers

Superluminescent diodes

DIP

USE: Electronics packaging

Dip coating

BT: Coatings

Dipole antennas

BT: Antennas

Direct broadcast satellites

USE: Satellite broadcasting

Direct current generators

USE: DC generators

Direct current machines

USE: DC machines

Direct current motors

USE: DC motors

Direct current power transmission

USE: DC power transmission

Direct sequence CDMA

USE: Direct-sequence code-division multiple access

Direct sequence code division multiple access

USE: Direct-sequence code-division multiple access

Direct sequence spread spectrum communication

BT: Radio spectrum management

RT: Bandwidth

Modulation

Direct-sequence CDMA

USE: Direct-sequence code-division multiple access

Direct-sequence code-division multiple access

UF: Direct sequence CDMA

Direct sequence code

division multiple access

Direct-sequence CDMA

BT: Multiaccess communication

Directed acyclic graph

BT: Graph theory

RT: Blockchains



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Directed graphs

BT: Graph theory
NT: Fuzzy cognitive maps

Direction of arrival

USE: Direction-of-arrival
estimation

Direction of arrival estimation

USE: Direction-of-arrival
estimation

Direction-finding

USE: Navigation

Direction-of-arrival estimation

UF: Bearing estimation
DOA estimation
Direction of arrival
Direction of arrival
estimation
Estimation of the direction
of arrival
BT: Parameter estimation
RT: Array signal processing
Position measurement
Spectral analysis
Time of arrival estimation

Directional antennas

BT: Antennas

Directional couplers

BT: Couplers
RT: Hybrid junctions

Directive antennas

BT: Antennas

Disabled people

USE: People with disabilities

Disabled persons

USE: People with disabilities

Disaster and recovery

USE: Disaster management

Disaster management

UF: Disaster and recovery
Disaster planning
Prevention and mitigation
BT: Contingency management
RT: Disasters
Stormwater

NT: Prevention and mitigation

Disaster planning

USE: Disaster management

Disasters

UF: Calamities
BT: Safety
RT: Disaster management
Hazards

Discharge lamps

BT: Lamps
NT: High intensity discharge
lamps

Discharges (electric)

UF: Dielectric barrier discharges
Gas discharges
Ozone generators
Ozonizers
BT: Dielectric breakdown
RT: Electrostatic processes
Gas discharge devices
Gases
Ionization
Plasmas

Discrete cosine transforms

UF: DCT
BT: Discrete transforms
RT: Chebyshev approximation

Discrete element method

USE: Finite element analysis

Discrete event simulation

BT: Digital simulation
NT: Time warp simulation

Discrete event systems

USE: Discrete-event systems

Discrete Fourier transforms

UF: DFT
BT: Fourier transforms
RT: Discrete Hartley transforms
Signal processing
Text detection

Discrete Hartley transforms

BT: Discrete transforms
RT: Discrete Fourier transforms



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Discrete time systems

USE: Discrete-time systems

Discrete transforms

BT: Transforms
NT: Discrete Hartley transforms
Discrete cosine transforms

Discrete wavelet transforms

BT: Wavelet transforms
RT: Finite impulse response

filters

Text detection

Discrete-event systems

UF: Discrete event systems
BT: Signal analysis
RT: Control systems
Manufacturing
Petri nets
Production systems

Discrete-time systems

UF: Discrete time systems
BT: Time factors
RT: Asymptotic stability
Control systems
Difference equations
Time invariant systems
NT: Sampled data systems

Discussion forums

BT: Collaboration

Diseases

BT: Medical conditions
RT: Epidemics
Medical diagnosis
Metastasis
Pandemics
Pathology
NT: Acquired immune
deficiency syndrome
Alcoholism
Allergies
Alzheimer's disease
Arteriosclerosis
Arthritis
Autoimmune diseases
Bacterial infections
Bone diseases
Cancer
Cardiovascular diseases
Cholera
Degenerative diseases

Dengue fever
Epilepsy
Eye diseases
Human immunodeficiency

virus

Infectious diseases
Influenza
Malaria
Multiple sclerosis
Muscular dystrophy
Neurological diseases
Parasitic diseases
Parkinson's disease
Pathogens
Plant diseases
Pulmonary diseases
Schizophrenia
Tuberculosis

Disentangled learning

USE: Disentangled
representation learning

Disentangled representation learning

UF: Disentangled learning
BT: Representation learning

Disinfection

BT: Decontamination

Disinformation

USE: Fake news

Disk drives

BT: Computer peripherals
RT: Disk recording
Perpendicular magnetic

recording

Disk recording

BT: Recording
RT: Disk drives

Disks (structures)

USE: Structural discs

Dismissal

USE: Termination of employment

Dispatching

BT: Object oriented
programming
RT: Materials handling



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Dispersed power generation

USE: Distributed power generation

Dispersion

UF: Dispersion effect
Dispersion measurement
Dispersion relations
Dispersive
BT: Signal processing
RT: Refractive index
NT: Chromatic dispersion
Optical fiber dispersion

Dispersion effect

USE: Dispersion

Dispersion measurement

USE: Dispersion

Dispersion relations

USE: Dispersion

Dispersive

USE: Dispersion

Displacement control

BT: Mechanical variables control

Displacement measurement

BT: Mechanical variables measurement

Display systems

BT: Displays

Displays

BT: Optical devices
RT: Character generation
Graphics
Thin film transistors
User interfaces
NT: Active matrix technology
Cathode ray tubes
Computer displays
Display systems
Flat panel displays
Head-mounted displays
Head-up displays
Liquid crystal devices
Microdisplays
Readout electronics
Three-dimensional displays
Two-dimensional displays

Disruption tolerant networking

BT: Computer network management

Disruptive innovation

BT: Business
RT: Disruptive technologies
Entrepreneurship
Market opportunities
Technological innovation

Disruptive technologies

BT: Technology
RT: Disruptive innovation
Market opportunities
Technological innovation

Dissolved air flotation

USE: Wastewater treatment

Dissolved gas analysis

BT: Fault diagnosis

Distance

USE: Distance measurement

Distance learning

UF: Remote learning
BT: Learning (artificial intelligence)
RT: Adaptive learning
Electronic learning
Hybrid learning
Mobile learning

Distance measurement

UF: Distance
Ranging
BT: Measurement
RT: Depth measurement
Micrometers
Position measurement
NT: Euclidean distance
Odometers

Distance relays

USE: Protective relaying

Distance work

USE: Remote working

Distillation columns

USE: Distillation equipment



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Distillation equipment

UF: Distillation columns
BT: Chemical technology

Distortion

UF: Distortion information
BT: Signal processing
RT: Delay effects
Distortion measurement
Image restoration
Interference
Noise
Rate distortion theory
Signal restoration
NT: Acoustic distortion
Four-wave mixing
Jitter
Nonlinear distortion
Phase distortion

Distortion information

USE: Distortion

Distortion measurement

UF: Acoustic distortion
measurement
Electric distortion
measurement
Optical distortion
measurement
BT: Measurement
RT: Distortion
Noise measurement
NT: Total harmonic distortion

Distributed algorithms

BT: Algorithms

Distributed amplifiers

BT: Amplifiers

Distributed antennas

USE: Antenna arrays

Distributed Bragg reflectors

UF: DBR
BT: Mirrors
RT: Integrated optics
Vertical cavity surface
emitting lasers

Distributed computing

UF: Autonomic computing
BT: Computers and information
processing

RT: Cloud computing
Computer networks
Concurrency control
Data communication
Distributed ledger
Local area networks
Metropolitan area networks
Mobile agents
Multiprocessing systems
Semantic Web
Software agents
Software architecture
NT: Client-server systems
Cluster computing
Decentralized applications
Diffserv networks
Distributed databases
Distributed information

systems

Internet
Metacomputing
Peer-to-peer computing

Distributed control

USE: Decentralized control

Distributed databases

BT: Databases
Distributed computing
RT: Concurrency control
Distributed ledger
NoSQL databases
NT: Blockchains

Distributed decision making

BT: Decision making

Distributed denial-of-service attack

UF: DDoS
BT: Denial-of-service attack
RT: Botnet
Computer crime

Distributed energy resources

USE: Distributed power
generation

Distributed feedback devices

UF: Distributed feedback lasers
BT: Laser applications
RT: Feedback circuits
Optical feedback



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Distributed feedback lasers

USE: Distributed feedback devices

Distributed generation

USE: Decentralized control AND Distributed power generation

Distributed information systems

BT: Distributed computing
Information systems
NT: Distributed management
Publish-subscribe

Distributed learning

USE: Hybrid learning

Distributed ledger

UF: DLT
Hyper ledger
Hyperledger
Shared ledger
BT: Online banking
RT: Blockchains
Cryptocurrency
Distributed computing
Distributed databases
Nonfungible tokens
Peer-to-peer computing

Distributed management

BT: Distributed information systems
Management
RT: Collaborative intelligence

Distributed management task force

USE: DMTF

Distributed modeling

USE: Decentralized control

Distributed parameter circuits

UF: Nonuniform transmission lines
Transmission line circuits
BT: Circuits
RT: Microwave circuits
Millimeter wave circuits
Transmission lines

Distributed parameter systems

BT: Decentralized control

Distributed power generation

UF: Dispersed power generation
Distributed energy resources
Distributed generation
Embedded power generation
BT: Power generation
RT: Hybrid power systems
Microgrids
Vehicle-to-grid
NT: Virtual power plants

Distributed processing

UF: Volunteer computing
BT: System analysis and design
RT: Cluster computing
Crowdsourcing
Software defined networking
NT: Dew computing
Edge computing
Message passing
Sharding

Distributed vision networks

BT: System analysis and design

Distribution functions

BT: Statistical distributions
RT: Probability
Probability density function

Distribution networks

BT: Supply chains

Distribution of electric power

USE: Power distribution

Distribution strategy

BT: Marketing management

District heating

UF: Central heating
District heating and cooling
Heat networks
Teleheating
BT: Heating systems
RT: Space heating
Water heating

District heating and cooling

USE: District heating



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Disturbance observers

BT: Observers
RT: Adaptive control
Robust control

Nucleic acids
Transcriptomics
DNA data storage
DNA sequencing
Genetic mutations

Diversity gain

USE: Diversity methods

Diversity methods

UF: Diversity gain
BT: Transmitters
RT: Fading channels
Multipath channels
Radio communication

Diversity reception

BT: Signal resolution
RT: SIMO
SISO
Telecommunications

Diversity schemes

BT: Telecommunication
network reliability
RT: Fading channels
Interference

Diving equipment

USE: Underwater equipment

DLT

USE: Distributed ledger

DMB

USE: Digital multimedia
broadcasting

DMTF

UF: Distributed management
task force
BT: Standards organizations

DNA

UF: Deoxyribonucleic acid
BT: Genetics
RT: Biological cells
Biological information

theory

Cloning
Codons
DNA computing
Electrophoresis
Epigenetics
Genetic communication
Molecular biophysics

DNA computing

BT: Computers and information
processing
RT: Nanobioscience
DNA
Molecular computing

DNA data storage

UF: DNA digital data storage
DNA storage
BT: DNA
Digital storage

DNA digital data storage

USE: DNA data storage

DNA sequencing

BT: DNA
RT: Nucleic acids
Sequential analysis

DNA storage

USE: DNA data storage

DOA estimation

USE: Direction-of-arrival
estimation

Docking stations

BT: Charging stations

DOCSIS

USE: Data over cable service
interface specification

Doctor

USE: Medical services

Document delivery

BT: Information services

Document handling

BT: Data handling
Information management
RT: Content management
Desktop publishing
Information retrieval
Office automation
Publishing



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NT: Semantic Web
Text processing
Document image
processing
Portable document format

Document image processing

BT: Document handling
RT: Portable document format

Documentation

UF: Computer documentation
Software documentation
BT: Writing
RT: Engineering drawings
Manuals
Software
NT: Point of care

DoD

USE: US Department of Defense

DoE

USE: US Department of Energy

Dogs

BT: Animals

Dolphins

BT: Marine animals

Domain Name System

BT: Computer networks

Domain specific languages

UF: Domain-specific languages
BT: Specification languages

Domain wall memory

USE: Racetrack memory

Domain-specific languages

USE: Domain specific languages

Domain-wall memory

USE: Racetrack memory

Domestic appliances

USE: Home appliances

Domestic induction appliances

USE: Home appliances

Domestic safety

UF: Safety in the home

BT: Safety
RT: Accidents
Consumer products
Electrical safety
Occupational health
Occupational safety
Smoke detectors
NT: Fall detection

Doped fiber amplifiers

UF: DFA
BT: Optical amplifiers

Doping

BT: Materials preparation
RT: Semiconductor device

doping

NT: Silicon devices
Doping profiles

Doping profiles

BT: Doping
RT: Optimization
Thin film devices

Doppler

USE: Doppler effect

Doppler effect

UF: Doppler
BT: Waves
RT: Doppler measurement
Doppler radar
NT: Doppler shift

Doppler measurement

BT: Measurement
RT: Doppler effect
Doppler radar
Frequency measurement
Motion measurement
Velocity measurement

Doppler radar

BT: Radar
RT: Doppler effect
Doppler measurement

Doppler shift

BT: Doppler effect

DoS attack

USE: Denial-of-service attack



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Dosimetry

UF: Radiation dosimetry
BT: Measurement
RT: Collimators
Neutron capture therapy
Phantoms
Radiation detectors
Radiation monitoring
Radiation protection

DOT

USE: US Department of Transportation

Double gate FETs

USE: Double-gate FETs

Double heterojunction bipolar transistors

UF: DHBTs
BT: Heterojunction bipolar transistors

Double heterojunction HEMTs

USE: DH-HEMTs

Double-gate FETs

UF: Double gate FETs
BT: Field effect transistors
RT: Silicon-on-insulator

Doubly fed induction generators

UF: DFIG
BT: Induction generators
RT: Wind turbines

Downlink

BT: Satellite communications
RT: Cellular radio

DP industry

USE: Computer industry

DPKS

USE: Differential phase shift keying

Drag

BT: Fluid dynamics
RT: Friction

DRAID

USE: Redundant array of independent disks

DRAM

USE: DRAM chips

DRAM chips

UF: DRAM
BT: Random access memory
RT: Solid state drives

Drama

USE: Humanities

Dredging

USE: Excavation

Drift velocity

USE: Electron mobility

Drilling

BT: Machining
RT: Boring
Deburring
Drilling machines
Geoengineering
Oil drilling

Drilling machines

BT: Machine tools
RT: Drilling

Drilling oil

USE: Oil drilling

Driver behavior

UF: Driver behaviour
Driving behavior
Driving behaviour
BT: Vehicle driving
RT: Advanced driver assistance

systems

Human computer

interaction

Human vehicle systems

Driver behaviour

USE: Driver behavior

Driver circuits

BT: Circuits
RT: Power transistors

Driver free automobiles

USE: Autonomous automobiles

Driver free cars

USE: Autonomous automobiles



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<i>Driver-free car</i>		RT:	Pharmacodynamics
USE:	Autonomous automobiles	NT:	Antimicrobial resistance
<i>Driverless automobiles</i>		Drugs	
USE:	Autonomous automobiles	BT:	Pharmaceuticals
		RT:	Biochemistry
<i>Driverless cars</i>			Chemical analysis
USE:	Autonomous automobiles		Chemistry
Drives			Chemotherapy
BT:	Machinery		Drug discovery
RT:	Mechanical power		Medicinal plants
transmission			Molecular biomarkers
	Sensorless control		Pharmaceutical industry
	Torque converters		Pharmacodynamics
NT:	Hydraulic drives		Pharmacokinetics
	Motor drives	NT:	Pharmacology
	Variable speed drives		Antibiotics
			Anticoagulants
<i>Driving behavior</i>			Antidepressants
USE:	Driver behavior		Aspirin
<i>Driving behaviour</i>			Cancer drugs
USE:	Driver behavior		Insulin
			Opioids
Drones		Dry electrodes	
UF:	Flying robots	BT:	Electrodes
BT:	Remotely guided vehicles	RT:	Biomedical electrodes
RT:	Autonomous aerial vehicles	Dry etching	
	Mobile robots	BT:	Etching
	Quadrotors	DSL	
Droughts		UF:	Digital subscriber lines
BT:	Meteorology		Digital subscriber loops
		BT:	Digital communication
Drug delivery		<i>DSP</i>	
UF:	Drug delivery systems	USE:	Digital signal processing
BT:	Biomedical engineering	<i>DSRC</i>	
RT:	Hydrogels	USE:	Dedicated short range
	Nanocarriers	communication	
NT:	Targeted drug delivery	<i>DT-MRI</i>	
<i>Drug delivery systems</i>		USE:	Diffusion tensor imaging
USE:	Drug delivery	<i>DTI</i>	
Drug discovery		USE:	Diffusion tensor imaging
BT:	Pharmacology	Dual band	
RT:	Drugs	UF:	Dual-band
	Pharmaceutical industry		Dualband
Drug interactions		BT:	Mobile communication
BT:	Pharmacodynamics	RT:	GSM
Drug resistance			Mobile handsets
BT:	Medical conditions		



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	Roaming	BT:	Algorithms
		RT:	Markov processes
<i>Dual inline packaging</i>			Neural networks
USE:	Electronics packaging		Temporal difference
		learning	
<i>Dual-band</i>			Viterbi algorithm
USE:	Dual band		
<i>Dualband</i>		Dynamic range	
USE:	Dual band	BT:	Measurement
		NT:	High dynamic range
Ducts		Dynamic response	
BT:	Structural shapes	BT:	Dynamics
RT:	Air conditioning		Mechanical engineering
	Vents		
Duplex communication systems		Dynamic scheduling	
BT:	Communication systems	BT:	Scheduling
NT:	Full-duplex system		
	Half-duplex system	<i>Dynamic service delivery</i>	
		USE:	Network resource
<i>Dust masks</i>		management	
USE:	Face masks		
<i>Dusty plasma</i>		Dynamic spectrum access	
USE:	Dusty plasmas	BT:	Radio transceivers
		RT:	Telecommunication
Dusty plasmas		network topology	
UF:	Dusty plasma		Wireless communication
BT:	Plasma properties	<i>Dynamic systems</i>	
		USE:	Dynamical systems
DVD		Dynamic voltage scaling	
UF:	DVD-ROM	UF:	Self-dynamic voltage
	Digital versatile discs	scaling	
	Digital video discs	BT:	Computer architecture
BT:	Video coding		Voltage
RT:	Video recording		
<i>DVD-ROM</i>		Dynamical systems	
USE:	DVD	UF:	Dynamic systems
		BT:	Mathematics
<i>Dynamic algorithms</i>		NT:	Nonlinear dynamical
USE:	Heuristic algorithms	systems	
Dynamic compiler		Dynamics	
BT:	Runtime	BT:	Mechanical factors
		RT:	Control theory
Dynamic equilibrium			Force
BT:	Measurement techniques		Friction
NT:	Steady-state		Vibrations
<i>Dynamic program analysis</i>		NT:	Aerodynamics
USE:	Performance analysis		Dynamic response
Dynamic programming			Elastodynamics
			Electrodynamics
			Hydrodynamics
			Magnetohydrodynamics



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Dynamo

USE: Generators

Dynamometers

UF: Dyno
BT: Force measurement
Meters
Power measurement
Torque measurement

Dyno

USE: Dynamometers

Dyslexia

BT: Medical conditions
Neurology
RT: Learning systems

Dysprosium

BT: Chemical elements

E health

USE: Electronic healthcare

E learning

USE: Electronic learning

e-agriculture

USE: Smart agriculture

E-banking

USE: Online banking

E-books

USE: Electronic publishing

E-coli

USE: Escherichia coli

E-commerce

USE: Electronic commerce

E-currency

USE: Online banking

E-government

USE: Electronic government

E-health records

USE: Electronic medical records

E-learning

USE: Electronic learning

E-mail

USE: Electronic mail

E-publishing

USE: Electronic publishing

E-reader

USE: Electronic publishing

E-skin

USE: Electronic skin

E-voting

USE: Electronic voting

E-wallets

USE: Online banking

E-waste

USE: Electronic waste

E. coli

USE: Escherichia coli

EAM

USE: Electro-absorption
modulators

Ear

BT: Head
Sense organs
RT: Cochlear implants

EAROM

USE: EPROM

Earphones

USE: Headphones

Earth

BT: Geoscience
Planets
RT: Geophysics
Remote sensing
Soil
Terrain factors
Terrain mapping

Earth atmosphere

USE: Terrestrial atmosphere

Earth observation images

USE: Satellite images



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Earth observation system

USE: Earth Observing System

Earth Observing System

UF: EOS
Earth observation system
BT: Artificial satellites
Observers
NT: GOSAT
Global Earth Observation

System of Systems

Landsat
Satellite images
Sentinel-1
Sentinel-2
Sentinel-3
Sentinel-5P

Earth science

USE: Geoscience

earthing

USE: Grounding

Earthquake engineering

UF: Seismic retrofitting
BT: Earthquakes
RT: Seismology

Earthquakes

BT: Geoscience
RT: Seismic waves
Seismology
NT: Earthquake engineering

Eating disorders

UF: Anorexia
Anorexia nervosa
Binge eating
Bulimia
Bulimia nervosa
BT: Mental disorders

Eavesdropping

UF: Cyber eavesdropping
Cybereavesdropping
BT: Privacy
RT: Computer security
Information leakage

EBSD

USE: Electron backscatter

diffraction

ECC

USE: Elliptic curve cryptography

AND

Error correction codes

ECCM

USE: Electronic countermeasures

ECG

USE: Electrocardiography

Echo cancellation

USE: Echo cancellers

Echo cancellers

UF: Echo cancellation
BT: Active noise reduction

Echo interference

BT: Interference
RT: Clutter
TV interference

Echo sounders

UF: Echosounders
BT: Sonar equipment
RT: Hydrophones
Sonar detection

Echo state networks

BT: Recurrent neural networks

Echocardiography

UF: ECHOEG
BT: Cardiology

ECHOEG

USE: Echocardiography

Echosounders

USE: Echo sounders

ECM

USE: Electronic countermeasures

Eco design

USE: Ecodesign

Eco-conscious

BT: Climate change

Eco-design

USE: Ecodesign



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Eco-friendly						Harmonised index of
BT:	Climate change			consumer prices		Harmonized index of
Ecodesign				consumer prices		Index of production
UF:	Eco design					Interest rates
	Eco-design					RPI
BT:	Green design					Retail price index
RT:	Energy conservation					Economics
	Environmental factors			BT:		Economic forecasting
				RT:		Exchange rates
Ecodistricts				NT:		Share prices
BT:	Climate change					
Ecological footprint				Economics		
BT:	Ecology			BT:		Engineering management
RT:	Carbon calculator			RT:		Bankruptcy
	Carbon footprint					Commercial law
Ecological restoration						Cost accounting
BT:	Ecology					Digital divide
Ecology						Econophysics
BT:	Environmental factors					Finance
RT:	Entomology					Food security
	Seeds (agriculture)					Freeports
NT:	Ecological footprint			NT:		Planning
	Ecological restoration					Access charges
	Habitats					Costs
Ecommerce						Developing countries
USE:	Electronic commerce					Digital economy
Econometrics						Econometrics
BT:	Economics					Economic indicators
RT:	Costs					Electronic commerce
	Cybernetics					Environmental economics
	Mathematics					Exchange rates
	Profitability					Free economic zones
	Regression analysis					Fuel economy
	Statistics					International trade
NT:	Economic forecasting					Macroeconomics
Economic forecasting						Mechanism design
BT:	Econometrics					Microeconomics
	Forecasting					Monopoly
RT:	Economic indicators					Oligopoly
Economic indicators						Power generation
UF:	Cost of living index					Profitability
	Cost-of-living index					Sharing economy
	GDP					Socioeconomics
	GNP					Stock markets
	Gross domestic product					Supply and demand
	Gross national product					Trade agreements
						Venture capital
						Virtual enterprises
				Economies of scale		
				BT:		Microeconomics
				RT:		Industrial economics



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Econophysics

BT: Cybernetics
RT: Chaos
Complexity theory
Economics
Fractals
Information theory
Knowledge acquisition
Nonlinear dynamical

systems

considerations

Philosophical

Science - general

Ecosystems

BT: Environmental factors
RT: Consensus protocol
Environmental degradation
Low carbon economy
NT: Aquatic ecosystems
Estuaries
Grasslands
Rainforests
Tundra
Wetlands

Eczema

UF: Atopic dermatitis
BT: Medical conditions

EDA

USE: Electronic design
automation and methodology

Eddy current losses

USE: Eddy currents

Eddy current testing

BT: Eddy currents
RT: Finite element analysis

Eddy currents

UF: Eddy current losses
BT: Electromagnetic induction
RT: Magnetic losses
NT: Eddy current testing

EDFA

USE: Erbium-doped fiber
amplifiers

Edge AI

UF: Edge artificial intelligence
BT: Artificial intelligence
Edge computing

Edge artificial intelligence

USE: Edge AI

Edge computing

UF: Fog computing
BT: Application virtualization
Distributed processing
RT: 5G mobile communication
Aerial computing
Cloud computing
Computer applications
Computer networks
Dew computing
Internet of Things
Mobile computing
Resource management
Wireless sensor networks
NT: Edge AI
Multi-access edge

computing

Edge detection

USE: Image edge detection

EDTV

USE: HDTV

Education

UF: Inverted classroom
Reverse teaching
Teaching
RT: Hybrid learning
Immersive learning
Personnel
NT: Adaptive learning
Career development
Educational courses
Educational institutions
Educational programs
Educational technology
Engineering education
Humanities
Training

Educational courses

BT: Education
RT: Computer aided instruction
Educational programs
Hybrid learning
STEM
NT: Curriculum development
Open Educational

Resources



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Educational institutions

UF: Colleges
Schools
Universities
BT: Education
RT: Hybrid learning
NT: Museums

Industrial waste
Waste disposal
Waste management
Water pollution

Ehealth

USE: Electronic healthcare

Educational programs

BT: Education
RT: Curriculum development
Educational courses
NT: Accreditation
Continuing education
Pre-college engineering
STEM
Scholarships
Self-study courses
Seminars
Tutorials

Eigenfunctions

USE: Eigenvalues and eigenfunctions

Eigenfunctions and eigenvalues

USE: Eigenvalues and eigenfunctions

Eigenplaces

USE: Eigenvalues and eigenfunctions

Eigenvalues

USE: Eigenvalues and eigenfunctions

Eigenvalues and eigenfunctions

UF: Eigenfunctions
Eigenfunctions and eigenvalues

Eigenplaces
Eigenvalues
BT: Mathematics
RT: Asymptotic stability
Functional analysis
Linear algebra
Vectors

Educational robots

BT: Robots
RT: Engineering education
Social robots

Educational technology

UF: Audio-visual instructional aids
Instructional aids
Programmed instruction
Smart education
Smart learning
BT: Education
RT: Audio-visual systems
Visualization
NT: Computer aided instruction
Courseware
Electronic learning

EKG

USE: Electrocardiography

EEG

USE: Electroencephalography

EEPROM

USE: EPROM

Effective mass

BT: Energy states

EFFF

USE: Field-flow fractionation

Effluents

BT: Waste materials
RT: Flue gases

El Nino

BT: Climate variability
RT: Sea surface temperature
NT: El Nino-Southern

Oscillation

El Nino-Southern Oscillation

UF: ENSO
El-Nino-Southern

Oscillation

BT: Climate variability
El Nino
RT: Sea surface temperature



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El-Nino-Southern Oscillation

USE: El Nino-Southern
Oscillation

Elastic computing

BT: Cloud computing
Resource management

Elastic optical networks

UF: EON
BT: Optical fiber networks

Elastic recovery

BT: Materials testing

Elasticity

BT: Material properties
RT: Strain

Elastodynamics

BT: Dynamics
RT: Seismic waves
Vibrations

Elastography

BT: Biomedical imaging

Elastomers

BT: Polymers

Elbow

BT: Arm

Elderly

USE: Older adults

Elearning

USE: Electronic learning

Electrets

BT: Dielectric materials
RT: Capacitors
Ceramics
Dielectric devices

Electric admittance

USE: Admittance

Electric ballast

USE: Electronic ballasts

Electric breakdown

UF: Breakdown
BT: Dielectrics and electrical

insulation

RT: Aging

Electrostatic discharge

protection

NT:

Fault currents
Avalanche breakdown
Corona
Dielectric breakdown
Sparks

Electric characteristics

USE: Electric variables

Electric coils

USE: Coils

Electric condensers

USE: Capacitors

Electric conductivity

USE: Conductivity

Electric current

USE: Current

Electric current control

BT: Current control
RT: Inrush current
Power control
Power transmission
Voltage control
NT: Power factor correction
Shunts (electrical)

Electric current measurement

USE: Current measurement

Electric distortion measurement

USE: Distortion measurement

Electric fences

BT: Electric machines

Electric field

USE: Electric fields

Electric fields

UF: Electric field
BT: Electromagnetic fields
RT: Electrohydrodynamics
Electrokinetics
Electrostatic analysis
Electrostatic discharge

protection

Electrostatic processes
Inrush current



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	Maxwell equations		
	Synchrotrons		
NT:	Acoustoelectric effects	Electric sensing devices	
	Casimir effect	BT:	Sensor systems and applications
	Nonuniform electric fields	Electric shock	
Electric generators		UF:	Electrocution
BT:	Generators		Shock
RT:	Nanogenerators	BT:	Bioelectric phenomena
		RT:	Accidents
<i>Electric heating</i>			Electrical accidents
USE:	Resistance heating		Grounding
			Occupational health
<i>Electric impedance</i>			Occupational safety
USE:	Impedance		Safety
Electric machines		<i>Electric stimulation therapy</i>	
BT:	Machinery	USE:	Electrical stimulation
RT:	Windings	<i>Electric utilities</i>	
NT:	AC machines	USE:	Electricity supply industry
	Alternators	AND	
	Brushless machines		Power industry
	Compressors	Electric variables	
	Conductors	UF:	Current voltage
	DC machines	characteristics	
	Electric fences		Electric characteristics
	Generators		Electrical characteristics
	Permanent magnet		Instrumentation and
machines		BT:	
	Rotating machines	measurement	
	Rotors	RT:	Electric variables control
	Stators		Electric variables
	Washing machines	measurement	
Electric motors			Frequency
BT:	Motors	NT:	Admittance
NT:	Planar motors		Capacitance
			Capacitance-voltage
Electric potential		characteristics	
BT:	Electric variables		Conductivity
RT:	Electrostatic discharge		Current
protection		characteristics	Current-voltage
<i>Electric power</i>			Electric potential
USE:	Power electronics AND		Gain
	Power systems		Impedance
			Impedance matching
<i>Electric power distribution</i>			Inductance
USE:	Electrical power distribution		Permittivity
			Piezoresistance
Electric resistance			Q-factor
UF:	Electrical resistivity		Resistance
BT:	Resistance		Voltage
			Wiring



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Electric variables control

BT: Power engineering and
energy
RT: Electric variables
Frequency control
Phase control
Regulators
NT: Current control
Gain control
Power control
Power system control
Reactive power control
Voltage control

Electric variables measurement

BT: Measurement
RT: Electric variables
Electromagnetic
measurements
Frequency measurement
Gain measurement
Integrated circuit
measurements
Noise measurement
Oscilloscopes
Phase measurement
Pulse measurements
Transducers
NT: Admittance measurement
Ammeters
Attenuation measurement
Capacitance measurement
Conductivity measurement
Current measurement
Dielectric measurement
Electrical resistance
measurement
Electrostatic measurements
Energy measurement
Impedance measurement
Inductance measurement
Partial discharge
measurement
Phasor measurement units
Power measurement
Q measurement
Rydberg atoms
Transmission line
measurements
Voltage measurement

Electric vehicle charging

UF: EV charging
BT: Battery chargers
Electric vehicles

NT: Smart charging

Electric vehicles

BT: Vehicles
RT: Charging stations
Electrification
Fast charging
Land vehicles
NT: Battery powered vehicles
Electric vehicle charging
Fuel cell vehicles
Hybrid electric vehicles
Plug-in electric vehicles
Solar powered vehicles
Vehicle-to-grid

Electrical accidents

BT: Accidents
RT: Bioelectric phenomena
Electric shock
Electrical safety

Electrical appliances

USE: Electrical products

Electrical ballast

USE: Electronic ballasts

Electrical ballasts

BT: Current control
Lighting
RT: High intensity discharge
lamps
Inductors
Resistors

Electrical brain stimulation

USE: Electrical stimulation

Electrical capacitance tomography

BT: Tomography

Electrical characteristics

USE: Electric variables

Electrical conductivity

USE: Conductivity

Electrical double layer capacitors

USE: Supercapacitors

Electrical engineering

BT: Engineering - general
RT: Engineering profession
Research and development



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NT: Electrical engineering
computing

Electrical engineering computing

BT: Electrical engineering
RT: Computer applications

Electrical engineering education

BT: Engineering education
NT: Electronics engineering
education

Electrical engineering industry

BT: Industries

Electrical equipment industry

BT: Power industry
RT: Electrical products industry
Electricity supply industry
Electronics industry

Electrical fault detection

BT: Circuit faults
RT: Electrostatic discharge
protection

Electrical impedance tomography

BT: Tomography

Electrical insulation

USE: Dielectrics

Electrical power distribution

UF: Electric power distribution
Electricity distribution
BT: Power distribution

Electrical products

UF: Electrical appliances
BT: Electrical products industry
Manufactured products
RT: Consumer products
NT: Washing machines

Electrical products industry

BT: Manufacturing industries
RT: Electrical equipment
industry
Electronics industry
NT: Electrical products

Electrical resistance measurement

UF: Ohmmeters
BT: Electric variables
measurement

RT: Resistance

Electrical resistivity

USE: Electric resistance

Electrical safety

BT: Power system protection
RT: Domestic safety
Electrical accidents
Islanding
Partial discharge

measurement

NT: Fault protection
Grounding

Electrical stimulation

UF: Electric stimulation therapy
Electrical brain stimulation
Microelectronic stimulation
Spinal cord stimulation
BT: Medical treatment

Electrically alterable read only memory

USE: EPROM

Electrically erasable programmable read only memory

USE: EPROM

Electricity

BT: Science - general
NT: Photoelectricity
Piezoelectricity
Pyroelectricity
Thermoelectricity
Triboelectricity
kWh

Electricity distribution

USE: Electrical power distribution

Electricity grids

USE: Power grids

Electricity market

USE: Electricity supply industry

Electricity markets

USE: Power markets

Electricity supply industry

UF: Electric utilities
Electricity market
Power supply industry
BT: Power industry



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industry	RT:	Electrical equipment			Power generation
					Renewable energy sources
		Feed in tariff			Sustainable development
		Power demand			
		Power distribution	<i>Electro hydraulics</i>		
		Power quality	USE:	Electrohydraulics	
		Power system faults			
		Power system planning	<i>Electro oculography</i>		
		Power system restoration	USE:	Electrooculography	
		Public infrastructure			
Standard		Renewable Portfolio	Electro-absorption modulators		
			UF:	EAM	
	NT:	Battery storage plants		Electroabsorption	
		Electricity supply industry	modulators		
	deregulation			Electroabsorptive	
			modulators		
				Franz-Keldysh effect	
				Optical modulators	
				Laser beams	
	UF:	Electricity supply industry	BT:		
liberalisation			RT:		
		Electricity supply industry			
liberalization			<i>Electro-chromic devices</i>		
		Electricity supply industry	USE:	Electrochromic devices	
privatisation					
		Electricity supply industry	<i>Electro-fluid dynamics</i>		
privatization			USE:	Electrohydrodynamics	
	BT:	Electricity supply industry			
		Power generation	<i>Electro-oculography</i>		
economics			USE:	Electrooculography	
	RT:	Power system economics			
	NT:	Power markets	Electro-optic deflectors		
			UF:	Electrooptic devices	
<i>Electricity supply industry liberalisation</i>				Electrooptical devices	
	USE:	Electricity supply industry	BT:	Electro-optic devices	
deregulation					
			Electro-optic devices		
<i>Electricity supply industry liberalization</i>			UF:	Electro-optical devices	
	USE:	Electricity supply industry		Electrooptic devices	
deregulation				Electrooptical devices	
			BT:	Lasers and electrooptics	
<i>Electricity supply industry privatisation</i>			RT:	Electro-optic effects	
	USE:	Electricity supply industry		Electroluminescent devices	
deregulation				Liquid crystal devices	
				Optical bistability	
<i>Electricity supply industry privatization</i>				Optoelectronic devices	
	USE:	Electricity supply industry	NT:	Electro-optic deflectors	
deregulation				Electrochromic devices	
<i>Electricity trading</i>			Electro-optic effects		
	USE:	Power markets	UF:	Electrooptic effects	
				Electrooptical effects	
Electrification			BT:	Lasers and electrooptics	
	BT:	Power engineering	RT:	Electro-optic devices	
	RT:	Carbon neutral		Electroluminescence	
		Electric vehicles		Nonlinear optics	
		Low carbon economy	NT:	Electrochromism	

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Kerr effect
Optical bistability
Stark effect

Electro-optic modulators

UF: Electro-optical modulators
Electrooptic modulators
Electrooptical modulators
Pockels readout optical
modulator
BT: Optical modulators
RT: Integrated optics
Intensity modulation
Laser beams
Microwave photonics
Optical waveguides
P-i-n diodes
Phase modulation
Quantum well devices

Electro-optical devices

USE: Electro-optic devices

Electro-optical modulators

USE: Electro-optic modulators

Electro-optical waveguides

UF: EO waveguides
Electrooptic waveguides
Electrooptical waveguides
BT: Optical waveguides

Electro-osmosis

BT: Osmosis

Electro-thermal actuators

USE: Electrothermal actuators

Electroabsorption modulators

USE: Electro-absorption
modulators

Electroabsorptive modulators

USE: Electro-absorption
modulators

Electroacoustic devices

USE: Acoustoelectric devices

Electroacoustic effects

USE: Acoustoelectric effects

Electroactive polymer actuators

USE: Actuators

Electroactive polymers

USE: Polymers

Electrobiography

USE: Bioelectric phenomena

Electrocardiography

UF: ECG
EKG
BT: Cardiography
RT: Biomedical equipment

Electrocatalysis

BT: Catalysis
Electrochemistry
RT: Electrocatalysts

Electrocatalysts

BT: Catalysts
RT: Electrocatalysis
Electrochemistry
Mesoporous materials

Electroceuticals

USE: Neurostimulation

Electrochemical deposition

UF: Electroplating
BT: Surface treatment
RT: Materials processing
Vapor deposition

Electrochemical devices

BT: Industry applications
RT: Electrochemical processes
Power engineering and

energy

Synapses
NT: Amperometric sensors
Batteries
Battery management

systems

Fuel cells
Supercapacitors

Electrochemical impedance spectroscopy

BT: Spectroscopy
RT: Electrochemical processes

Electrochemical machining

UF: Electrolytic machining
BT: Machining
RT: Micromachining



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Electrochemical processes

UF: Electrolysis
BT: Industry applications
RT: Chemical industry
Electrochemical devices
Electrochemical impedance

spectroscopy

Electrolytes

Electrochemistry

BT: Chemistry
RT: Electrocatalysts
NT: Electrocatalysis

Electrochromic devices

UF: Electro-chromic devices
BT: Electro-optic devices
RT: Electrochromism

Electrochromism

BT: Electro-optic effects
RT: Color
Electrochromic devices

Electrocution

USE: Electric shock

Electrodeless lamps

BT: Lamps

Electrodes

BT: Electronic components
RT: Air gaps
Electron emission
Electron tubes
Electrophysiology
Metal-insulator structures
Spark gaps
NT: Anodes
Cathodes
Dry electrodes
Microelectrodes
Textile electrodes
Wet electrodes

Electrodynamics

BT: Dynamics
Waves
RT: Electromagnetic fields
Electron beams
Electron optics
Electron tubes
Ion beams
Particle beam optics

NT: Electromagnetic wave
polarization

Electroencephalography

UF: EEG
BT: Biomedical measurement
RT: Bioelectric phenomena
Biomedical equipment
Brain
Electrooculography
Medical diagnosis

Electrofluid dynamics

USE: Electrohydrodynamics

Electrohydraulics

UF: Electro hydraulics
BT: Hydraulic systems
RT: Fluid flow
Liquids
Magnetohydrodynamics

Electrohydrodynamics

UF: Electro-fluid dynamics
Electrofluid dynamics
Electrostrictive

hydrodynamics

BT: Dielectrics
Hydrodynamics
RT: Dielectric materials
Electric fields
Electrokinetics

Electrokinetics

BT: Dielectrics
RT: Dielectric materials
Electric fields
Electrohydrodynamics

Electroluminescence

BT: Luminescence
RT: Chemiluminescence
Electro-optic effects
Organic light emitting

diodes

NT: Electroluminescent devices

Electroluminescent devices

BT: Electroluminescence
Light sources
Luminescent devices
RT: Electro-optic devices

Electrolysis

USE: Electrochemical processes



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Electrolytes

BT: Conducting materials
RT: Electrochemical processes
Supercapacitors

Electrolytic machining

USE: Electrochemical machining

Electromagnetic absorbers

BT: Electromagnetic wave
absorption

Electromagnetic analysis

BT: Electromagnetics
RT: Electrostatic analysis
Magnetic analysis
Mie scattering
NT: Air gaps
Characteristic mode
analysis
Computational
electromagnetics
Delay effects
Electromagnetic forces
Electromagnetic refraction
Permeability
Spark gaps
Time-domain analysis

Electromagnetic beams

USE: Beams

Electromagnetic compatibility

UF: EMC
BT: Electromagnetic
compatibility and interference
NT: Immunity testing
Reverberation chambers

Electromagnetic compatibility and interference

RT: Electromagnetic
interference
Open area test sites
TEM cells
NT: Electromagnetic
compatibility
Electromagnetics
Interference
Radar

Electromagnetic coupling

BT: Electromagnetics
RT: Circulators
Couplers

Electromagnetic induction
Electromagnetic shielding
NT: Mutual coupling
Optical coupling

Electromagnetic devices

BT: Electromagnetics
RT: Magnetic gears
NT: Baluns

Electromagnetic diffraction

BT: Electromagnetic
propagation
RT: Electromagnetic fields
NT: Optical diffraction
Physical theory of
diffraction
X-ray diffraction

Electromagnetic field theory

BT: Electromagnetic fields
RT: Computational
electromagnetics
Optical fiber theory

Electromagnetic fields

BT: Dielectrics and electrical
insulation
RT: Computational
electromagnetics
Electrodynamics
Electromagnetic diffraction
Electromagnetic
propagation
Electromagnetic radiation
Electromagnetic reflection
Electromagnetic refraction
Electromagnetic scattering
Magnetic fields
Mie scattering
NT: Electric fields
Electromagnetic field theory
Electromagnetic spectrum
Windings

Electromagnetic forces

BT: Electromagnetic analysis
RT: Electromagnetic launching
Magnetic forces
Mie scattering

Electromagnetic guns

USE: Electromagnetic launching



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Electromagnetic heating

UF: Microwave heating
BT: Heating systems
RT: Hyperthermia
Induction heating

NT: Reflectometry
Wavelength measurement
Electromagnetic modeling
Linearity
Microwave measurement
Millimeter wave

Electromagnetic induction

UF: Induction (electromagnetic)
BT: Electromagnetics
RT: Electromagnetic coupling
Geomagnetism
Magnetic communication
NT: Eddy currents
Inductive power

measurements

Parameter extraction
Polarimetry
Radiometry
Submillimeter wave

transmission

measurements

Electromagnetic interference

UF: EMI
Electromagnetic noise
RF interference
Radio interference
BT: Interference
RT: Crosstalk
Electromagnetic

Electromagnetic metamaterials

BT: Electromagnetics
Metamaterials
RT: Optical metamaterials
Photonics
Split ring resonators
NT: Microwave metamaterials
Terahertz metamaterials

compatibility and interference

Electromagnetic model

USE: Electromagnetic modeling

Environmental factors
Immunity testing
Noise
Open area test sites
TEM cells
NT: Radiofrequency

Electromagnetic modeling

UF: Electromagnetic model
Electromagnetic modelling
BT: Electromagnetic

interference

measurements

Electromagnetic modelling

USE: Electromagnetic modeling

Electromagnetic launching

UF: Electromagnetic guns
Electromagnetic propulsion
Launching

Electromagnetic noise

USE: Electromagnetic

(electromagnetic)

BT: Propulsion
RT: Electromagnetic forces
Electrothermal launching
NT: Coilguns
Railguns

interference

Electromagnetic propagation

UF: Electromagnetic wave
propagation
BT: Antennas and propagation
Propagation
RT: Electromagnetic fields
Electromagnetic transients
Electromagnetic

Electromagnetic measurements

BT: Measurement
RT: Anechoic chambers
Antenna measurements
Dielectric measurement
Electric variables

waveguides

Magnetostatic waves
Mie scattering
Waves

measurement

Frequency measurement
Mie scattering
Photogrammetry

NT: Electromagnetic diffraction
Electromagnetic

propagation in absorbing media

Electromagnetic reflection
Microwave propagation



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propagation
 Millimeter wave
 Optical propagation
 Propagation constant
 Propagation losses
 Radio propagation
 Radiowave propagation
 Submillimeter wave
 propagation
 UHF propagation

Electromagnetic propagation in absorbing media

BT: Electromagnetic
 propagation

Electromagnetic propulsion

USE: Electromagnetic launching

Electromagnetic pulse propagation

USE: Electromagnetic transients

Electromagnetic pulse scattering

USE: Electromagnetic transients

Electromagnetic pulses

BT: Electromagnetics
 RT: EMP radiation effects
 Geomagnetic storms
 Nuclear weapons
 Pulse generation
 Radiofrequency exposure

Electromagnetic radiation

BT: Electromagnetics
 RT: Electromagnetic fields
 Electromagnetic wave

polarization
 Radiofrequency exposure
 Radiofrequency safety
 Waves
 X-ray detection
 X-ray detectors
 X-rays
 NT: Bremsstrahlung
 Correlators
 Electromagnetic wave

absorption
 Frequency
 Gamma-rays
 Line-of-sight propagation
 Terahertz radiation

Electromagnetic radiative interference

BT: Interference

Electromagnetic reflection

UF: Electromagnetic wave
 reflection
 BT: Electromagnetic
 propagation
 Reflection
 RT: Electromagnetic fields
 Electromagnetic scattering
 Reflectometry
 NT: Optical reflection

Electromagnetic refraction

BT: Electromagnetic analysis
 RT: Electromagnetic fields

Electromagnetic scattering

UF: Electromagnetic wave
 scattering
 Electromagnetic waves
 BT: Scattering
 RT: Coherence time
 Electromagnetic fields
 Electromagnetic reflection
 Electromagnetic transients
 Waves
 NT: Coherence
 Microwave plasmas
 Mie scattering
 Optical scattering
 Polarization
 Radar scattering
 Raman scattering
 Rayleigh scattering

Electromagnetic shielding

BT: Electromagnetics
 RT: EMP radiation effects
 Electromagnetic coupling
 NT: Cable shielding
 Magnetic shielding

Electromagnetic spectrum

BT: Electromagnetic fields

Electromagnetic transient program

USE: EMTF

Electromagnetic transients

UF: Electromagnetic pulse
 propagation
 Electromagnetic pulse
 scattering
 BT: Electromagnetics



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RT: Electromagnetic
propagation
Electromagnetic scattering
Transient analysis
NT: EMP radiation effects
EMTDC
EMTP
Power system transients
Surges

Electromagnetic transients DC
USE: EMTDC

Electromagnetic transients including DC
USE: EMTDC

Electromagnetic wave absorption

BT: Electromagnetic radiation
NT: Electromagnetic absorbers
Terahertz wave absorption

Electromagnetic wave attenuation
USE: Attenuation

Electromagnetic wave polarisation
USE: Electromagnetic wave
polarization

Electromagnetic wave polarization

UF: Electromagnetic wave
polarisation
BT: Electrodynamics
RT: Electromagnetic radiation
Photonic band gap

Electromagnetic wave propagation
USE: Electromagnetic
propagation

Electromagnetic wave reflection
USE: Electromagnetic reflection

Electromagnetic wave scattering
USE: Electromagnetic scattering

Electromagnetic waveguides

BT: Transmission lines
RT: Coaxial cables
Coplanar waveguides
Electromagnetic
propagation
Helical antennas
Microwave devices
Microwave propagation
Optical fibers

Propagation
Waveguide discontinuities
NT: Circular waveguides
Gap waveguide
Hollow waveguides
Loaded waveguides
Planar waveguides
Rectangular waveguides
Waveguide components
Waveguide lasers
Waveguide theory

Electromagnetic waves
USE: Electromagnetic scattering

Electromagnetics

BT: Electromagnetic
compatibility and interference
RT: Cyberspace
Neuroradiology
NT: Electromagnetic analysis
Electromagnetic coupling
Electromagnetic devices
Electromagnetic induction
Electromagnetic

metamaterials

Electromagnetic pulses
Electromagnetic radiation
Electromagnetic shielding
Electromagnetic transients
Proximity effects

Electromagnets

BT: Magnets
RT: Coils
Magnetic confinement
Magnetic levitation
Magnetic levitation vehicles
NT: Superconducting magnets

Electromechanical devices

BT: Electromechanical systems
NT: Armature
SAW filters

Electromechanical sensors

BT: Sensors
NT: Microsensors

Electromechanical systems

BT: Industry applications
NT: Cruise control
Electromechanical devices



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Electromigration

BT: Diffusion processes

Electromyography

UF: EMG

BT: Biomedical measurement

RT: Bioelectric phenomena

Electron accelerators

BT: Particle accelerators

RT: Electron beams

Electron sources

Electrons

Electron backscatter diffraction

UF: EBSD

BT: Backscatter

Diffraction

RT: Crystal microstructure

Deformation

Grain boundaries

Grain size

Scanning electron

microscopy

Electron beam applications

BT: Electron beams

RT: Flyback transformers

Scanning electron

microscopy

Electron beam pumping

USE: Laser excitation

Electron beams

BT: Particle beams

RT: Electrodynamics

Electron accelerators

Electron emission

Electron sources

Electrons

Flyback transformers

Free electron lasers

Gyrotrons

Relativistic effects

Transmission electron

microscopy

NT: Electron beam applications

Electron carriers

USE: Charge carrier processes

Electron devices

RT: Threshold current

NT: Cathode ray tubes

Electron guns

Electron multipliers

Electron tubes

Mechatronics

Microelectromechanical

Microfluidics

Micromechanical devices

Photoelectricity

Photovoltaic cells

Quantum computing

Quantum well devices

Semiconductivity

Semiconductor devices

Single electron devices

Thick film devices

Thin film devices

Tunneling

Vacuum technology

Electron emission

UF: Field electron emission

Secondary electron

emission

BT: Nuclear and plasma

sciences

RT:

Cathodes

Electrodes

Electron beams

Electron guns

Electron multipliers

Electron sources

Electron tubes

Electrons

Photoelectricity

Thermionic emission

Vacuum arcs

Vacuum breakdown

NT:

Ballistic transport

Electron guns

BT: Electron devices

RT: Electron emission

Electron microscopy

BT: Microscopy

NT: Photoelectron microscopy

Scanning electron

microscopy

Transmission electron

microscopy

Electron mobility

UF: Drift velocity

BT: Charge carrier processes



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RT: Plasma properties

BT: Ballistic transport

Electron multipliers

BT: Electron devices
RT: Electron emission
Electron tubes
Photomultipliers

Electronic banking

USE: Online banking

Electronic books

USE: Electronic publishing

Electron optics

BT: Optics
Particle beam optics
RT: Electrodynamics

Electronic circuits

BT: Circuits
NT: Breadboard
Central Processing Unit
Multivibrators
Stripboard circuit

Electron paramagnetic resonance

UF: Biological EPR
Electron spin resonance
BT: Spectroscopy

Electronic commerce

UF: E-commerce
Ecommerce
Home shopping
Online shopping
BT: Economics
RT: Consumer behavior
Customer satisfaction
Digital economy
Financial management
Internet
Marketing management
Neuromarketing
Online banking
Product delivery
Supply chain management
Virtual enterprises
Web sites

Electron sources

BT: Electrons
RT: Electron accelerators
Electron beams
Electron emission

Electron spin resonance

USE: Electron paramagnetic resonance

Electron traps

BT: Charge carrier processes
RT: Leakage currents
Reliability

Electron tubes

UF: Thermionic valves
Tubes
Vacuum tubes
BT: Electron devices
RT: Anodes
Cathodes
Electrodes
Electrodynamics
Electron emission
Electron multipliers
Gettering
NT: Field emitter arrays
Klystrons
Magnetrons
Thyratrons
Traveling wave tubes

Electronic components

BT: Components, packaging, and manufacturing technology
NT: Capacitors
Coils
Connectors
Diodes
Electrodes
Fuses
Inductors
Resistors
Structural plates
Switches
Transducers

Electronic counter-countermeasures

USE: Electronic countermeasures

Electronic ballasts

UF: Ballasts
Electric ballast
Electrical ballast

Electronic countermeasures

UF: ECCM
ECM



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Electronic counter-
countermeasures
BT: Electronic warfare
RT: Jamming
Military communication
Radar countermeasures
Radio communication
Spread spectrum
Spread spectrum radar
Weapons

Electronic currency
USE: Online banking

Electronic data interchange
USE: Data handling

Electronic design automation and methodology
UF: EDA
RT: VHDL
NT: Design automation
Design methodology

Electronic education
USE: Electronic learning

Electronic equipment
BT: Electronics industry
RT: Electronic equipment
manufacture
Electronic equipment
testing
NT: Low power electronics
Electronic voting systems
Microelectronics
Organic electronics
Smart devices
Soft electronics

Electronic equipment manufacture
BT: Components, packaging,
and manufacturing technology
RT: Electronic equipment
Electronics industry
Optical device fabrication
NT: Damascene integration
Micromachining
Radiation hardening
(electronics)
Semiconductor device
manufacture

Electronic equipment testing
BT: Testing
RT: Electronic equipment
TEM cells
NT: Immunity testing

Electronic government
UF: E-government
BT: Government

Electronic health records
USE: Electronic medical records

Electronic healthcare
UF: Digital health
Digital healthcare
E health
Ehealth
BT: Information processing
Medical services
RT: Smart healthcare
Wearable Health Monitoring

Systems

Electronic learning
UF: E learning
E-learning
Elearning
Electronic education
Online learning
Virtual learning
BT: Educational technology
Learning systems
RT: Computer aided instruction
Computers and information
processing
Distance learning
Internet
Learning (artificial
intelligence)
Learning management
systems

Online services
TV
Training
Wide area networks
NT: Mobile learning

Electronic mail
UF: E-mail
Email
Mail (electronic)
BT: Message systems
RT: Blogs
Office automation



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	Postal services		Desktop publishing
	Social networking (online)		
	Voice mail		
NT:	Unified messaging	Electronic skin	
	Unsolicited e-mail	UF:	E-skin
		BT:	Flexible electronics
			Skin
Electronic medical prescriptions		RT:	Haptic interfaces
BT:	Medical treatment		Tactile sensors
RT:	Electronic medical records		
Electronic medical records		Electronic switching systems	
UF:	E-health records	BT:	Communication switching
	Electronic health records		Switching systems
BT:	Medical information	<i>Electronic textiles</i>	
systems		USE:	Smart textiles
RT:	Electronic medical	<i>Electronic visual prosthesis</i>	
prescriptions	Patient monitoring	USE:	Visual prosthesis
Electronic messaging		Electronic voting	
UF:	Text messaging	UF:	E-voting
BT:	Message systems		Online voting
RT:	Emojis	BT:	Voting
NT:	Instant messaging		
	Unified messaging	Electronic voting systems	
		BT:	Electronic equipment
Electronic music		<i>Electronic wallets</i>	
BT:	Music	USE:	Online banking
NT:	Synthesizers		
Electronic noses		Electronic warfare	
BT:	Chemical analysis	BT:	Aerospace and electronic
RT:	Intelligent sensors	systems	
		RT:	Communication system
Electronic packaging thermal management		security	
BT:	Thermal management of	countermeasures	Radio communication
electronics		communication	Spread spectrum
<i>Electronic portfolios</i>			Spread spectrum radar
USE:	Portfolios	NT:	Electronic countermeasures
			Jamming
Electronic publishing			Radar countermeasures
UF:	Digital publishing	Electronic waste	
	E-books	UF:	E-waste
	E-publishing		WEEE
	E-reader		Waste electrical and
	Electronic books	electronic equipment	
	Epublishing	BT:	Waste materials
	Kindle		
BT:	Publishing	Electronics cooling	
RT:	CD-ROMs	BT:	Thermal management of
	Journalism	electronics	
	Multimedia systems	RT:	Cooling
	Open data		
NT:	Content management		



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Electronics engineering education

BT: Electrical engineering
education

Electronics industry

UF: Integrated circuits industry
Semiconductor electronics

industry

BT: Semiconductor industry
Manufacturing industries
RT: Electrical equipment

industry

Electrical products industry
Electronic equipment

manufacture

NT: Toy manufacturing industry
Electronic equipment

Electronics packaging

UF: DIL
DIP
Dual inline packaging
PGA
Pin grid arrays
QFP
Quad flat packs
BT: Components, packaging,
and manufacturing technology
RT: Constraint optimization
Cooling
Plastic packaging
Printed circuits
NT: Antenna-in-package
Ball grid arrays
Chip scale packaging

Electrons

BT: Elementary particles
RT: Beta rays
Cosmic rays
Electron accelerators
Electron beams
Electron emission
Elementary particle
exchange interactions
Impact ionization
Phonons
Schrodinger equation
NT: Electron sources
Quantum wells
Trions

Electrooculography

UF: EOG
Electro oculography

BT: Electro-oculography
Biomedical measurement
Gaze tracking
RT: Bioelectric phenomena
Electroencephalography
Eyes

Electrooptic devices

USE: Electro-optic deflectors
AND Electro-optic devices

Electrooptic effects

USE: Electro-optic effects

Electrooptic modulators

USE: Electro-optic modulators

Electrooptic waveguides

USE: Electro-optical waveguides

Electrooptical devices

USE: Electro-optic deflectors
AND Electro-optic devices

Electrooptical effects

USE: Electro-optic effects

Electrooptical modulators

USE: Electro-optic modulators

Electrooptical waveguides

USE: Electro-optical waveguides

Electropermeabilization

USE: Electroporation

Electrophoresis

BT: Separation processes
RT: DNA
Medical diagnosis

Electrophotography

UF: Xerography
BT: Electrostatic processes
Photoconducting devices
RT: Gas discharge devices
Photography

Electrophotoluminescence

USE: Photoluminescence



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Electrophysiology

BT: Biomedical measurement
RT: Biomedical electrodes
Electrodes
Voltage

Electroplating

USE: Electrochemical deposition

Electroporation

UF: Electropermeabilization
BT: Medical treatment
Microbiology

Electrostatic actuation

USE: Electrostatic actuators

Electrostatic actuators

UF: Electrostatic actuation
BT: Actuators
RT: Electrostatic devices

Electrostatic analysis

BT: Electrostatic processes
RT: Electric fields
Electromagnetic analysis
Electrostatic measurements

Electrostatic charges

USE: Electrostatic discharges

AND

Electrostatics

Electrostatic devices

BT: Industry applications
RT: Electrostatic actuators
Electrostatic processes

Electrostatic discharge protection

UF: ESD protection
BT: Electrostatic discharges
Protection
RT: Electric breakdown
Electric fields
Electric potential
Electrical fault detection
Failure analysis
Integrated circuit reliability
Integrated circuit

technology

Semiconductor device

reliability

Triboelectricity

Electrostatic discharges

UF: Charged device model
ESD
Electrostatic charges
BT: Dielectric breakdown
RT: Arc discharges
Electrostatic interference
NT: Electrostatic discharge
protection

Electrostatic induction

UF: Induction (electrostatic)
BT: Electrostatic processes

Electrostatic interference

BT: Interference
RT: Electrostatic discharges
NT: Immunity testing

Electrostatic levitation

BT: Electrostatics
Levitation

Electrostatic measurements

BT: Electric variables
measurement
RT: Electrostatic analysis
NT: Charge measurement

Electrostatic precipitators

BT: Industry applications
RT: Pollution control

Electrostatic processes

BT: Industry applications
RT: Discharges (electric)
Electric fields
Electrostatic devices
Lightning
NT: Aerosols
Electrophotography
Electrostatic analysis
Electrostatic induction
Electrostatics
Particle charging
Particle production
Space charge
Surface charging
Triboelectricity

Electrostatic self assembly

USE: Electrostatic self-assembly



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Electrostatic self-assembly

UF: Electrostatic self assembly
BT: Self-assembly

Electrostatics

UF: Electrostatic charges
BT: Electrostatic processes
RT: Poisson equations
NT: Electrostatic levitation

Electrostriction

BT: Dielectrics
RT: Mechanical factors
Piezoelectricity

Electrostrictive hydrodynamics

USE: Electrohydrodynamics

Electrostrictive polymer actuators

USE: Actuators

Electrosurgery

BT: Medical treatment
Surgery
RT: Biological tissues
Current
Radio frequency

Electrothermal actuators

UF: Electro-thermal actuators
BT: Actuators
Thermal sensors
RT: Resistance heating
Thermal expansion

Electrothermal effects

BT: Thermoelectricity
RT: Electrothermal launching
NT: Proton effects

Electrothermal launching

UF: Launching (electrothermal)
BT: Propulsion
RT: Electromagnetic launching
Electrothermal effects

Elemental semiconductors

BT: Semiconductor materials
RT: Silicon

Elementary particle exchange interactions

BT: Elementary particles
RT: Electrons
Ions
Proton effects

Wave functions

Elementary particle vacuum

UF: Instanton vacuum
QCD vacuum
Quantum vacuum
String vacuum
Superstring vacuum
Vacuum energy
BT: Elementary particles
RT: Casimir effect

Elementary particles

UF: Particles (elementary)
BT: Nuclear and plasma
sciences
RT: Cosmic rays
High energy physics
instrumentation computing
Microwave photonics
Nuclear thermodynamics
Proton effects
NT: Charge carriers
Electrons
Elementary particle
exchange interactions
Elementary particle vacuum
Ions
Mesons
Neutrino sources
Neutrons
Particle beams
Particle collisions
Phonons
Positrons
Protons

Elementary school engineering

USE: Pre-college engineering

Elevators

BT: Building services
RT: Buildings
Stairs

Ellipsoids

BT: Elliptic design

Ellipsometry

BT: Optical variables
measurement
RT: Polarimetry



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Elliptic curve cryptography

UF: ECC
Elliptic curve cryptosystems
BT: Public key cryptography

Elliptic curve cryptosystems

USE: Elliptic curve cryptography

Elliptic curves

BT: Geometry

Elliptic design

UF: Elliptical design
BT: Geometry
NT: Ellipsoids

Elliptical design

USE: Elliptic design

Elongation

BT: Material properties
RT: Strain

Email

USE: Electronic mail

eMBB

USE: Enhanced mobile
broadband

Embedded computing

BT: Embedded systems

Embedded multicore processing

BT: Multicore processing

Embedded power generation

USE: Distributed power
generation

Embedded software

BT: Software

Embedded system

USE: Embedded systems

Embedded systems

UF: Embedded system
BT: Operating systems
RT: Cyber-physical systems
Hardware-in-the-loop
simulation
Microprocessors
NT: Embedded computing

Embolization

BT: Medical treatment
Noninvasive treatment

Embossing

BT: Manufacturing
Production
RT: Injection molding
Micromachining
Sheet metal processing
Watermarking

Embryo

BT: Embryonic structures
RT: Embryology

Embryology

BT: Biology
Medical specialties
RT: Embryo

Embryonic structures

BT: Anatomy
NT: Embryo
Fetus

EMC

USE: Electromagnetic
compatibility

Emergency lighting

BT: Lighting
RT: High intensity discharge
lamps

Emergency management

USE: Emergency services

Emergency medical services

USE: Medical services

Emergency medicine

UF: Accident and emergency
medicine
BT: Medical specialties

Emergency power generators

USE: Standby generators

Emergency power supplies

UF: Standby power supplies
BT: Power supplies
RT: Batteries
Standby generators



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systems	Uninterruptible power	<i>Emotion models</i>	USE: Emotion recognition
<i>Emergency response</i>	USE: Emergency services	Emotion recognition	UF: Emotion models Emotion theory Models of emotion
Emergency services	UF: Emergency management Emergency response	BT: User interfaces	RT: Action units Affective computing Anxiety disorders Behavioral sciences Emojis Emotional responses Facial expressions Image recognition Psychology Sentiment analysis Social robots Speech recognition
	BT: Safety		
	RT: Accidents Fires Food security Internet of Medical Things Medical services Rescue robots Stormwater		
Emergent phenomena	BT: Cybernetics		
	RT: System of systems		
<i>EMG</i>	USE: Electromyography	Emotion regulation	UF: Emotional regulation BT: Psychology RT: Emotional responses
<i>EMI</i>	USE: Electromagnetic	<i>Emotion theory</i>	USE: Emotion recognition
interference		<i>Emotional regulation</i>	USE: Emotion regulation
<i>Emirates Mars Mission</i>	USE: Interplanetary exploration	Emotional responses	BT: Psychology RT: Emotion recognition Emotion regulation
Emissions factor	BT: Environmental economics RT: Greenhouse gases	EMP radiation effects	BT: Electromagnetic transients Radiation effects RT: Electromagnetic pulses Electromagnetic shielding
Emissions trading	UF: Cap and trade Cap-and-trade Carbon trading	Emphysema	BT: Pulmonary diseases RT: Asthma Chronic obstructive
	BT: Environmental economics Pollution	pulmonary disease	Pneumonia
	RT: Carbon emissions Climate change Market opportunities Power markets Pricing	Empirical mode decomposition	UF: Hilbert?Huang transforms BT: Transforms RT: Signal processing
Emojis	BT: Computer graphics Symbols		
	RT: Electronic messaging Emotion recognition		



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Employee rights

BT: Employment

Employee welfare

UF: Conditions of employment
Counseling
Counselling
Maternity benefits
Sick pay
Working conditions
BT: Human resource

management

RT: Incentive schemes
Industrial psychology
Occupational health
Occupational safety
Occupational stress
Pensions
Psychology
Remuneration

Employment

UF: Work-place
Workplace
BT: Human resource

management

RT: Business
Employment law
Engineering profession
Jobs listings
Personnel
Programming profession
NT: Employee rights
Green jobs
Livelihood
Termination of employment

Employment law

BT: Law
RT: Contract law
Employment

EMTDC

UF: Electromagnetic transients
DC Electromagnetic transients

including DC

BT: Electromagnetic transients
Geophysics
Software packages
RT: Design automation
PSCAD

EMTP

UF: Electromagnetic transient
program
BT: Electromagnetic transients
RT: Computer simulation

Emulation

BT: Modeling
RT: Application virtualization
Simulation

Encapsulation

BT: Packaging
RT: Integrated circuit packaging
Plastic packaging

Encephalography

BT: Biomedical imaging
RT: Brain

Encoding

UF: Coding
BT: Information theory
RT: Autoencoders
Codecs
Codes
Cryptography
Data compression
Data handling
Hash functions
Modulation
Modulation coding
Quantization (signal)
Semantic technology
Signal processing
Vector quantization
NT: ASCII
Audio coding
Channel coding
Code refractoring
Digital representation
Entropy coding
Precoding
Source coding
Speech coding
Transcoding

Encryption

BT: Cryptography
RT: Chaotic mapping
Ciphers
Information leakage
Privacy enhancing

technologies

NT: Homomorphic encryption



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Encyclopedias

BT: Information services

End effectors

UF: End-effectors

BT: Manipulators

RT: Grippers

End-effectors

USE: End effectors

Endangered species

BT: Animals

Biodiversity

RT: Environmental

management

Endocrine glands

USE: Glands

Endocrine system

BT: Anatomy

Endocrinology

BT: Medical specialties

RT: Diabetes

Thyroid

Endomicroscopy

BT: Endoscopes

Microscopy

RT: Biomedical equipment

Biomedical optical imaging

Digestive system

Medical robotics

Optical devices

Real-time systems

Endoscopes

UF: Endoscopy

BT: Biomedical equipment

RT: Biomedical optical imaging

Image sensors

Laser applications

Surgery

NT: Endomicroscopy

Endoscopy

USE: Endoscopes

Endothelial cells

BT: Biological cells

RT: Blood vessels

Energy

BT: Power engineering and

energy

RT: Thermal energy

NT: Energy barrier

Energy capture

Energy consumption

Energy conversion

Energy dissipation

Energy exchange

Energy harvesting

Energy management

Energy resources

Energy states

Energy storage

Energy barrier

BT: Energy

Energy capture

BT: Energy

Energy conservation

UF: Energy saving

BT: Energy management

RT: Ecodesign

Energy resources

Feed in tariff

Power demand

Waste heat

NT: Green computing

Grid decarbonization

Potential energy

Renewable energy sources

Energy consumption

UF: Energy demand

BT: Energy

RT: Consumption footprint

Load monitoring

Energy conversion

BT: Energy

NT: Atomic batteries

Batteries

Fuel cells

Motors

Ocean thermal energy

conversion

Photovoltaic cells

Potential well

Solar heating

Thermoelectricity

Waste heat

Wave energy conversion



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	Wind energy conversion		Power distribution Power engineering
<i>Energy demand</i>		computing	
USE:	Energy consumption		Power generation Power grids Power system control Power system
Energy dissipation		interconnection	
BT:	Energy		Power system management Power transmission Smart grids
Energy efficiency			
UF:	Energy efficient		
BT:	Energy management		
RT:	Energy informatics		
NT:	Energy efficient computing Energy efficient ethernet	<i>Energy levels</i>	
		USE:	Energy states
<i>Energy efficient</i>		Energy loss	
USE:	Energy efficiency	BT:	Energy measurement
		NT:	Core loss
Energy efficient computing		Energy management	
UF:	Energy-efficient computing	BT:	Energy
BT:	Energy efficiency	RT:	Circular economy
		NT:	Demand side management Energy conservation Energy efficiency Energy informatics Energy management
Energy efficient ethernet			
UF:	EtherEEE		
BT:	Energy efficiency Ethernet	systems	
Energy exchange			Load management Transactive energy
UF:	Energy transfer		
BT:	Energy		
NT:	Inductive charging		
Energy harvesting		Energy management systems	
UF:	Energy scavenging Power harvesting	BT:	Energy management
BT:	Energy		
NT:	Nanogenerators	Energy measurement	
		BT:	Electric variables
Energy informatics		measurement	
BT:	Energy management Informatics	RT:	Calorimetry Enthalpy Watt-hour meters Energy loss
RT:	Energy efficiency Global warming Green design Information and communication technology Machine learning Smart cities Smart grids	NT:	
		Energy resolution	
Energy Internet		BT:	Nuclear imaging
UF:	Internet of Energy	RT:	Nuclear medicine Solid scintillation detectors
BT:	Power systems		
RT:	Internet Internet of Things	Energy resources	
		BT:	Energy
		RT:	Civil engineering Energy conservation Environmental economics Natural gas Power demand Fuels
		NT:	



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Geothermal energy
Nuclear energy
Solar energy
Wave power
Wind energy
Wind farms

Energy saving

USE: Energy conservation

Energy scavenging

USE: Energy harvesting

Energy states

UF: Energy levels
Levels, energy
BT: Energy
NT: Band structures
Effective mass
Fermi level
Orbital calculations
Polaritons
Quasi-Fermi level
Surface states

Energy storage

UF: Energy storage systems
Stored energy
BT: Energy
RT: Aging
Battery powered vehicles
Fuel cell vehicles
Fuel storage
Hybrid electric vehicles
Load management
Material storage
Pulsed power systems
Solar powered vehicles
NT: Batteries
Battery energy storage

system

Battery storage plants
Flywheels
Fuel cells
Hydrogen storage
Supercapacitors
Superconducting magnetic

energy storage

Energy storage systems

USE: Energy storage

Energy transfer

USE: Energy exchange

Energy-efficient computing

USE: Energy efficient computing

Engine cylinders

BT: Machine components
RT: Engines
Gaskets
Pistons
Structural rings

Engineering - general

RT: STEM
Technology
NT: Acoustical engineering
Agricultural engineering
Bio-inspired engineering
Chemical engineering
Civil engineering
Concurrent engineering
Design engineering
Electrical engineering
Engineering profession
Environmental engineering
Maintenance engineering
Marine engineering
Mechanical engineering
Optical engineering
Precision engineering
Production engineering
Research and development
Reverse engineering
Sanitary engineering
Standardization
Thermal engineering

Engineering computing

BT: Computer applications

Engineering drawings

BT: Graphics
RT: Documentation
Technical drawing
NT: Flowcharts

Engineering education

BT: Education
RT: Continuing education
Educational robots
Laboratories
Logic design
NT: Biomedical engineering

education

Communication

engineering education



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education	Computer science	Employment
education	Control engineering	Ethics
education	Electrical engineering	Programming profession
education	Engineering students	Research and development
education	Physics education	Research and development
education	Power engineering	management
education	Student experiments	NT: Professional aspects
education	Systems engineering	Engineering students
<i>Engineering geology</i>		UF: Student engineers
USE: Geoengineering		BT: Engineering education
Engineering in medicine and biology		<i>Engineering writing</i>
NT: Biology		USE: Writing
Biomedical communication		Engines
Biomedical computing		BT: Industry applications
Biomedical engineering		RT: Aircraft propulsion
Biomedical equipment		Automobile manufacture
Biomedical imaging		Cams
Bionanotechnology		Camshafts
Bioterrorism		Engine cylinders
Computational biology		Exhaust systems
Genetic engineering		Fuel pumps
Medical services		Gaskets
Medical specialties		Machine components
Nuclear medicine		Manifolds
Synthetic biology		Mechanical power
Engineering management		transmission
NT: Business		Oils
Commercialization		Pistons
Consortia		Propellers
Economics		Propulsion
Innovation management		Rockets
Legal factors		Torque converters
Market research		Turbomachinery
Planning		NT: Heat engines
Product development		Internal combustion
Project engineering		engines
Research and development		Jet engines
management	Research initiatives	Enhanced magnetoresistance
management	Software development	BT: Magnetoresistance
		RT: Nanocontacts
Engineering profession		Enhanced mobile broadband
UF: Careers		UF: eMBB
BT: Engineering - general		BT: 5G mobile communication
RT: Biographies		RT: 4G mobile communication
Electrical engineering		Enhanced vegetation index
		UF: EVI
		BT: Environmental
		measurement
		Indexes



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RT: Vegetation	Thermal management
<i>Ensemble approach</i>	Entomology
USE: Ensemble learning	BT: Zoology
Ensemble learning	RT: Biochemistry
UF: Ensemble approach	Biomechanics
Ensemble methods	Ecology
BT: Machine learning	Insects
Statistics	Morphology
RT: Predictive models	Physiology
NT: Fish schools	NT: Apicology
<i>Ensemble methods</i>	<i>Entrepreneurial</i>
USE: Ensemble learning	USE: Entrepreneurship
<i>ENSO</i>	Entrepreneurship
USE: El Nino-Southern	UF: Entrepreneurial
Oscillation	BT: Business
<i>Entangled states</i>	RT: Crowdfunding
USE: Quantum entanglement	Disruptive innovation
Enterprise architecture management	Innovation management
BT: Information architecture	Venture capital
Information management	
RT: Best practices	Entropy
Enterprise resource	BT: Physics
planning	RT: Heating systems
Enterprise resource planning	Nuclear thermodynamics
BT: Management	Entropy coding
RT: Business	BT: Encoding
Data handling	RT: Data compression
Data processing	NT: Huffman coding
Enterprise architecture	<i>Entry, descent and landing</i>
management	USE: Aircraft navigation
Software	Envelope detectors
System integration	BT: Detectors
Venture capital	Environmental degradation
NT: Business process	BT: Environmental factors
integration	RT: Ecosystems
Entertainment industry	Environmental impact
BT: Industries	Pollution
RT: Broadcasting	NT: Deforestation
Films	Habitat loss
Games	<i>Environmental design</i>
Motion pictures	USE: Green design
TV	Environmental economics
NT: Sports	BT: Economics
Enthalpy	Environmental factors
BT: Thermodynamics	RT: Energy resources
RT: Energy measurement	



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management	NT:	Environmental	Environmental science	
		Pollution	Epidemiology	
		Carbon accounting	Food security	
		Carbon budget	Green buildings	
		Carbon calculator	Green computing	
		Carbon disclosure	Green energy	
		Carbon emissions scope	Greenhouse effect	
		Carbon markets	Health and safety	
		Carbon offsets	International collaboration	
		Carbon sink	Meteorology	
		Carbon tax	Occupational health	
		Circular economy	Ozonation	
		Climate finance	Safety	
		Conscious capitalism	NT:	
		Corporate social	Biosphere	
responsibility			Climate change	
			Ecology	
			Ecosystems	
			Environmental degradation	
			Environmental economics	
			Environmental impact	
			Environmental	
			measurement	
			Environmental monitoring	
			Green products	
			Green transportation	
			Habitat loss	
			Life cycle assessment	
reporting)			Pollution	
Environmental engineering			Environmental impact	
	BT:	Engineering - general	BT:	Environmental factors
	RT:	Environmental factors	RT:	Environmental degradation
		Environmental		
Environmental management			Environmental management	
			BT:	Industry applications
			RT:	Circular economy
				Dams
				Endangered species
				Environmental economics
				Environmental engineering
				Environmental science
				Global warming
				International collaboration
				Low carbon economy
				Public infrastructure
				Sanitary engineering
			NT:	Biodegradation
				Land use planning
				National Adaptation Plans
				Nature-based solutions
				Net zero
				Pest control
				Pollution control
				Recycling
				Renewable energy



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	Sustainable development		<i>Enzymatic fuel cells</i>
	Waste management	USE:	Fuel cells
	Water conservation		
	Water resources		
Environmental measurement		Enzymes	
UF:	Environmental metrics	BT:	Proteins
BT:	Environmental factors	RT:	Hydrolysis
	Measurement		
NT:	Consumption footprint	<i>EO waveguides</i>	
	Enhanced vegetation index	USE:	Electro-optical waveguides
	Normalized difference		
vegetation index		<i>EOG</i>	
		USE:	Electrooculography
water index	Normalized difference	<i>EON</i>	
		USE:	Elastic optical networks
<i>Environmental metrics</i>		<i>EOS</i>	
USE:	Environmental	USE:	Earth Observing System
measurement			
Environmental monitoring		<i>Eosinophiles</i>	
BT:	Environmental factors	USE:	Eosinophils
	Monitoring		
RT:	Decontamination	Eosinophils	
	Environmental science	UF:	Eosinophiles
	Green communications	BT:	White blood cells
	Pollution control		
	Pollution measurement	Epidemics	
	Sentinel-3	BT:	Epidemiology
	Soil remediation	RT:	Coronaviruses
			Diseases
<i>Environmental noise</i>			Influenza
USE:	Working environment noise		Pathogens
			Virology
<i>Environmental problems</i>		NT:	Pandemics
USE:	Environmental factors		
		Epidemiology	
Environmental science		BT:	Science - general
BT:	Science - general	RT:	Biomedical monitoring
RT:	Biology		Contact tracing
	Environmental engineering		Environmental factors
	Environmental factors		Public healthcare
	Environmental	NT:	Epidemics
management			
	Environmental monitoring	<i>Epidermal</i>	
	Geography	USE:	Epidermis
	Physics		
<i>Environmental, social, and governance</i>		Epidermis	
USE:	ESG	UF:	Epidermal
		BT:	Skin
Environmentally friendly manufacturing techniques		Epigenetics	
BT:	Components, packaging, and manufacturing technology	BT:	Genetics
		RT:	DNA



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Epilepsy

BT: Diseases

Epitaxial growth

UF: Epitaxy
BT: Thin films
RT: Crystal growth
Crystals
Gallium
Germanium
Molecular beams
Nanotechnology
Photonics
Semiconductor devices
Semiconductor thin films
Silicon
Substrates
NT: Molecular beam epitaxial

growth

Epitaxial layers

BT: Coatings
Films
RT: Chemical vapor deposition
Semiconductor growth
Thin films
NT: Superconducting epitaxial

layers

Epitaxy

USE: Epitaxial growth

EPON

UF: Ethernet passive optical
networks
BT: Ethernet
RT: Passive optical networks

Epoxy resins

BT: Dielectric materials
Plastics
Resins

EPROM

UF: EAROM
EEPROM
Electrically alterable read
only memory
Electrically erasable
programmable read only memory
Erasable programmable
read only memory
BT: PROM

Epublishing

USE: Electronic publishing

Equal opportunities

BT: Human resource
management
RT: Industrial relations
Labor resources
Personnel
Recruitment
NT: Gender equity
Gender issues

Equalisers

USE: Equalizers

Equalizers

UF: Equalisers
BT: Filters
RT: Channel estimation
Impedance matching
Intersymbol interference
NT: Adaptive equalizers
Blind equalizers
Decision feedback

equalizers

Equations

BT: Mathematics
NT: Boltzmann equation
Difference equations
Integrodifferential equations
Maxwell equations
Nonlinear equations
Polynomials
Riccati equations

Equator

BT: Geoscience

Equipment failure

BT: Failure analysis

Equipment under test

USE: Device under test

Equivalent circuits

BT: Circuits

Er

USE: Erbium

Er-doped fiber amplifier

USE: Erbium-doped fiber
amplifiers



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Er-doped fiber lasers

USE: Erbium-doped fiber lasers

Erasable programmable read only memory

USE: EPROM

Erbium

UF: Er
BT: Metals
RT: Erbium-doped fiber

amplifiers

Lasers and electrooptics
Optical amplifiers
Optics

Erbium-doped fiber amplifiers

UF: EDFA
Er-doped fiber amplifier
BT: Optical amplifiers
RT: Erbium

Erbium-doped fiber laser

USE: Erbium-doped fiber lasers

Erbium-doped fiber lasers

UF: Er-doped fiber lasers
Erbium-doped fiber laser
Erbiumdoped fiber laser
BT: Fiber lasers

Erbiumdoped fiber laser

USE: Erbium-doped fiber lasers

Erection

USE: Construction

Ergonomics

UF: Human engineering
Human factors engineering
BT: Systems, man, and

cybernetics

RT: Anthropometry
Behavioral sciences
Cybernetics
Design methodology
Human factors
Human-machine systems
Keyboards
Occupational health
Working environment noise
NT: Job design
Smart spaces
User experience

Erosion

BT: Geology
RT: Sedimentation
Weathering

Error analysis

UF: Error estimation
Error rate
Error rates
Error statistics
BT: Testing
RT: Cyclic redundancy check
Cyclic redundancy check

codes

Error correction
Estimation
Mean square error methods
Measurement errors
Numerical analysis
Roundoff errors
NT: Bit error rate
Finite wordlength effects

Error compensation

BT: Information theory
RT: Error correction

Error correcting codes

USE: Error correction codes

Error correction

BT: Signal processing
RT: Codes
Convolutional codes
Cyclic redundancy check
Cyclic redundancy check

codes

Error analysis
Error compensation
Error correction codes
Linear codes
Power system faults
Product codes
Turbo codes
NT: Forward error correction

Error correction codes

UF: ECC
Error correcting codes
Error-correcting codes
Error-correction codes
Errorcorrection codes
BT: Codes
RT: Convolutional codes
Error correction



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	NT:	Polar codes Reed-Muller codes Reed-Solomon codes		<i>ESD protection</i>	USE:	Electrostatic discharge protection
<i>Error estimation</i>	USE:	Error analysis		ESG	UF:	Environmental, social, and governance
<i>Error free operation</i>	USE:	Error-free operations			BT:	Corporate social responsibility
Error probability	BT:	Probability		Esophagus	BT:	Digestive system
<i>Error rate</i>	USE:	Error analysis		Estimation	UF:	Signal estimation
<i>Error rates</i>	USE:	Error analysis			BT:	Mathematics
<i>Error recovery (computers)</i>	USE:	System recovery			RT:	Control systems
<i>Error statistics</i>	USE:	Error analysis				Error analysis
<i>Error-correcting codes</i>	USE:	Error correction codes				Filtering theory
<i>Error-correction codes</i>	USE:	Error correction codes				Kalman filters
						Measurement uncertainty
						Prediction methods
						Prediction theory
						Reduced order systems
						Signal processing
						Spectral analysis
					NT:	Estimation error
						Estimation theory
						Functional point analysis
						Life estimation
						Maximum likelihood
Error-free operations	UF:	Error free operation		estimation		Pose estimation
	BT:	Testing				State estimation
<i>Errorcorrection codes</i>	USE:	Error correction codes				Yield estimation
<i>ESA</i>	USE:	European Space Agency		Estimation error	BT:	Estimation
Escalators	BT:	Transportation		<i>Estimation of the direction of arrival</i>	USE:	Direction-of-arrival
	RT:	Buildings		estimation		
		Mechanical products		Estimation theory	BT:	Estimation
		Stairs			RT:	Mean square error methods
Escherichia coli	UF:	E-coli				Signal processing
		E. coli			NT:	Statistics
	BT:	Bacteria				Cramer-Rao bounds
	RT:	Bacterial infections		estimation		Maximum a posteriori
<i>ESD</i>	USE:	Electrostatic discharges		Estuaries	BT:	Ecosystems
						Geoscience



January 2026 IEEE Thesaurus

	RT:	Wetlands		Regulation	General Data Protection
<i>Etalons</i>					
	USE:	Interferometers		NT:	Neuromarketing Cyberethics Machine ethics
Etching				<i>Ethnic group</i>	
	UF:	Deep etching		USE:	Ethnicity
	BT:	Materials processing Surface treatment			
	RT:	Fabrication		Ethnicity	
		Micromachining		UF:	Ethnic group
	NT:	Dry etching Wet etching		BT:	Anthropology
				RT:	Cultural differences
Ethanol				<i>Ethyl alcohol</i>	
	UF:	Ethyl alcohol Grain alcohol		USE:	Ethanol
	BT:	Chemical compounds			
	NT:	Alcoholic beverages		ETSI	
<i>EtherEEE</i>				UF:	European
	USE:	Energy efficient ethernet		Telecommunications Standards Institute	
				BT:	Standards organizations
<i>Ethereum</i>				ETSI Standards	
	USE:	Blockchains		BT:	Standards publications
				NT:	HbbTV Standards SONET Synchronous digital
Ethernet				hierarchy	
	BT:	Computer networks		Euclidean distance	
	RT:	IEEE 802.3 Standard Local area networks Virtual links		UF:	Euclidean measurement Euclidean metric
	NT:	EPON Energy efficient ethernet		BT:	Distance measurement Mathematics
<i>Ethernet passive optical networks</i>				NT:	Hilbert space
	USE:	EPON		<i>Euclidean measurement</i>	
Ethical aspects				USE:	Euclidean distance
	BT:	Social implications of		<i>Euclidean metric</i>	
technology				USE:	Euclidean distance
	RT:	Digital divide Ethics Genetic engineering Legal factors Management Philosophical		Europe	
				BT:	Continents
considerations				<i>European Organization for Nuclear Research</i>	
				USE:	CERN
Ethics				European Space Agency	
	UF:	Morals		UF:	ESA
	BT:	Social implications of		BT:	Space agencies
technology				<i>European Telecommunications Standards Institute</i>	
	RT:	Digital intelligence Engineering profession Ethical aspects		USE:	ETSI



January 2026 IEEE Thesaurus

Europium

BT: Chemical elements

Eutrophication

BT: Aquaculture
Water resources
RT: Algae
Plankton

EUV Lithography

USE: Extreme ultraviolet
lithography

EV charging

USE: Electric vehicle charging

Evaluation models

BT: Modeling
Performance evaluation

Evaporation

BT: Liquids
Water cycle
RT: Transpiration
Water
NT: Evaporative cooling

Evaporative coolers

USE: Evaporative cooling

Evaporative cooling

UF: Evaporative coolers
BT: Cooling
Evaporation

Event detection

UF: Event-based
Event-triggered
BT: Wireless sensor networks
RT: Security information and
event management

Event-based

USE: Event detection

Event-triggered

USE: Event detection

Everyware

USE: Pervasive computing

EVI

USE: Enhanced vegetation index

Evidence theory

UF: Belief functions
Dempster?Shafer theory
BT: Uncertainty
RT: Belief propagation

Evolution (biology)

BT: Biology
NT: Memetics
Phylogeny

Evolutionary algorithm

USE: Evolutionary computation

Evolutionary computation

UF: Evolutionary algorithm
BT: Computational intelligence
RT: Evolutionary dynamics
Metaheuristics
NT: Evolutionary robotics
Particle swarm optimization

Evolutionary dynamics

BT: Computational modeling
RT: Evolutionary computation

Evolutionary robotics

BT: Evolutionary computation
Robots

Exascale computing

BT: High performance
computing
Supercomputers

Excavation

UF: Dredging
BT: Geotechnical engineering
RT: Construction industry
Marine technology
Mining industry
Rivers
Roads
Soil

Exchange rates

BT: Economics
RT: Costs
Economic indicators
International trade

Excitation of lasers

USE: Laser excitation



January 2026 IEEE Thesaurus

Excitons

BT: Charge carrier processes
RT: Quantum materials
Semiconductor materials

Executive programs

USE: Operating systems

Exhaust gases

BT: Gases
RT: Air pollution
Ash
Combustion
Exhaust systems
Flue gases
Internal combustion

engines

NT: Jet engines
Contrails

Exhaust manifolds

USE: Manifolds

Exhaust systems

UF: Catalytic converters
Catalytic convertors
Mufflers
BT: Machine components
Production systems
RT: Engines
Exhaust gases
Manifolds

Exo planets

USE: Extrasolar planets

Exo-planets

USE: Extrasolar planets

Exocrine glands

USE: Glands

Exoplanet

UF: Extrasolar planetary mass
BT: Extrasolar planets

Exoplanets

USE: Extrasolar planets

Exoskeletons

BT: User interfaces
RT: Assistive robots

Expanded dynamic range

USE: High dynamic range

Expectation-maximisation algorithms

USE: Expectation-maximization algorithms

Expectation-maximization algorithms

UF: Expectation-maximisation algorithms
BT: Iterative methods

Expert systems

BT: Knowledge based systems
RT: Cause effect analysis
Decision making
Intelligent systems
Knowledge acquisition
Knowledge representation
NT: Diagnostic expert systems
Medical expert systems

Explainable AI

UF: XAI
Xplainable AI
BT: Artificial intelligence
Machine learning

Explosion protection

BT: Protection
Safety
RT: Accident prevention
Flammability
Hazards
Military equipment

Explosions

BT: Hazards
RT: Accidents
Chemical hazards
Flammability
Hazardous areas
Safety
Seismic waves
NT: Explosives

Explosives

BT: Explosions

Exponential distribution

BT: Probability distribution

Extended definition TV

USE: HDTV

Extended dynamic range

USE: High dynamic range



January 2026 IEEE Thesaurus

Extended reality

UF: HumanXR
X reality
XR
BT: Human computer
interaction
Human-machine systems
Virtual reality
RT: Augmented reality

Extensibility

BT: Design methodology
RT: Adaptive systems
Scalability

Extensible Markup Language

USE: XML

External stimuli

UF: PhysiStimuli
BT: Interactive systems
Physiology

Extinction coefficients

BT: Optics
NT: Extinction ratio

Extinction ratio

BT: Extinction coefficients

Extra solar planets

USE: Extrasolar planets

Extra-solar planets

USE: Extrasolar planets

Extracellular

BT: Cells (biology)
NT: Local field potentials

Extranets

BT: Virtual private networks
RT: Data communication
Information systems
Internet
Web sites

Extraordinary magnetoresistance

BT: Magnetoresistance

Extrapolation

BT: Approximation methods
RT: Statistics

Extrasolar planetary mass

USE: Exoplanet

Extrasolar planets

UF: Exo planets
Exo-planets
Exoplanets
Extra solar planets
Extra-solar planets
Super earths

BT: Astronomy
RT: Extraterrestrial

measurements

Extraterrestrial phenomena
NT: Exoplanet

Extraterrestrial measurements

UF: Planetary composition
Space measurements
BT: Measurement
RT: Astronomy
Extrasolar planets
Extraterrestrial phenomena
Interstellar chemistry

Extraterrestrial phenomena

UF: Space phenomena
BT: Geophysics
RT: Extrasolar planets
Extraterrestrial

measurements

Planets
Space technology
NT: Cosmic rays
Solar radiation

Extreme learning machines

BT: Feedforward neural
networks
RT: Clustering methods

Extreme ultraviolet lithography

UF: EUV Lithography
BT: Lithography

Extremities

BT: Body regions
NT: Buttocks
Fingers
Foot
Hip
Knee
Limbs
Shoulder
Thigh



January 2026 IEEE Thesaurus

Eye diseases

BT: Diseases
RT: Vision defects
Visual impairment
NT: Glaucoma
Macular degeneration
Retinopathy

Eye protection

UF: Goggles
BT: Safety devices
RT: Occupational health
Occupational safety
Protective clothing
Safety

Eyebrows

BT: Hair

Eyelashes

BT: Eyes
Hair

Eyelids

BT: Eyes

Eyes

BT: Sense organs
RT: Electrooculography
Gaze tracking
Ophthalmology
Optical coherence

tomography

NT: Cataracts
Cornea
Eyelashes
Eyelids
Iris
Pupils
Retina

FAA

UF: Federal Aviation
Administration
BT: US Government agencies

Fabrication

UF: Fabrication process
BT: Manufacturing
RT: Etching
Materials processing
NT: Bonding processes
Microfabrication
Optical device fabrication
Soldering

Welding

Fabrication process

USE: Fabrication

Fabrics

UF: Knitted fabric composites
Woven fabric composites
BT: Textiles
RT: Clothing
Weaving
Wool

Fabry-Perot

BT: Interferometry
NT: Fabry-Perot interferometers

Fabry-Perot interferometers

BT: Fabry-Perot

Face detection

BT: Computer vision
RT: Facial animation
NT: Facial features

Face masks

UF: Cloth face masks
Dust masks
FFP2
N95
Surgical masks
BT: Protective clothing
RT: Personal protective

equipment

Face recognition

UF: Facial recognition
BT: Biometrics
Identification of persons
Pattern recognition
Gaze tracking
RT: Image recognition

Facebook

USE: Social networking (online)

Faces

BT: Head
RT: Stomatognathic system
NT: Facial muscles

Facial animation

BT: Animation
RT: Face detection



January 2026 IEEE Thesaurus

Facial attributes

USE: Facial features

Facial expressions

BT: Facial muscles

RT: Action units
Emotion recognition

Facial features

UF: Facial attributes

BT: Face detection

Facial muscles

BT: Faces

NT: Action units
Facial expressions

Facial recognition

USE: Face recognition

Facilities management

BT: Building services
Management
Organizational aspects
RT: Building information

management

Facsimile

BT: Communication systems
Image communication

Factories

USE: Production facilities

Factory automation

USE: Manufacturing automation

FACTS

USE: Flexible AC transmission

systems

Fading channels

BT: Signal processing
RT: Diversity methods
Diversity schemes
Intercell interference
Meteorological factors
Multipath channels
Radio propagation
NT: Frequency-selective fading

channels

Rayleigh channels
Weibull fading channels

Failure analysis

UF: Failure analytics
Failure management
Failure mechanisms
BT: Testing
RT: Cause effect analysis
Diagnostic expert systems
Electrostatic discharge

protection

Failure state
Fatigue
Fault diagnosis
Fault trees
Green's function methods
Life estimation
Mortality
Quality control
Reliability
Remaining life assessment
Root cause analysis
Shear testing
Weibull distribution
NT: Equipment failure
Semiconductor device

breakdown

Failure analytics

USE: Failure analysis

Failure management

USE: Failure analysis

Failure mechanisms

USE: Failure analysis

Failure state

BT: Systems engineering and
theory
RT: Failure analysis

Fair trade

BT: Environmental economics

Fake content

USE: Fake news

Fake news

UF: Disinformation
Fake content
Fake reviews
False information
Misinformation
BT: Media
RT: Information diffusion
Information integrity



January 2026 IEEE Thesaurus

NT:	Deepfakes	Fascia	BT:	Musculoskeletal system
<i>Fake reviews</i>				
USE:	Fake news	<i>Fashion industry</i>	USE:	Clothing industry
Fall detection		Fast charging		
BT:	Biomechanics	UF:	DC fast charging	
	Domestic safety	BT:	Charging stations	
RT:	Accelerometers	RT:	Electric vehicles	
	Alarm systems			
	Assisted living	Fast Fourier transforms		
	Biomedical communication	BT:	Fourier transforms	
	Biomedical signal	RT:	Digital signal processing	
processing			Harmonic analysis	
	Body sensor networks			
	Detectors	Fast light		
	Geriatrics	BT:	Light sources	
	Home automation			
	Human activity recognition	Fastbus		
	Image motion analysis	BT:	Data acquisition	
	Image recognition			
	Injuries	Fasteners		
	Patient monitoring	UF:	Bolts	
	Video signal processing		Hinges	
	Wearable devices		Nuts (fasteners)	
			Screws	
<i>False information</i>			Zip fasteners	
USE:	Fake news	BT:	Control equipment	
Fans			Mechanical products	
BT:	Machinery	RT:	Belts	
	Ventilation		Couplings	
RT:	Air conditioning		Joining processes	
	Blades		Welding	
	Jet engines			
Faraday effect		<i>Fastening</i>		
UF:	Faraday rotation	USE:	Joining processes	
BT:	Magnetooptic effects			
RT:	Gyromagnetism	Fatigue		
	Gyrotropism	BT:	Mechanical factors	
		RT:	Failure analysis	
			Life estimation	
<i>Faraday rotation</i>				
USE:	Faraday effect	Fats		
Farming		BT:	Chemical products	
BT:	Industries		Food products	
RT:	Agricultural engineering	RT:	Biological materials	
	Agricultural products		Lipidomics	
	Agriculture		Oils	
	Crop yield			
	Food products	Fault current limiters		
	Precision agriculture	BT:	Current limiters	



January 2026 IEEE Thesaurus

Fault currents

BT: Current
RT: Electric breakdown
Grounding
Leakage currents
NT: Fault protection

Fault detection

BT: System analysis and design

Fault diagnosis

BT: Reliability
RT: Cause effect analysis
Diagnostic expert systems
Failure analysis
Maintenance engineering
Testing
NT: Dissolved gas analysis
Fault location

Fault location

BT: Fault diagnosis
RT: Cables
Communication cables
Fault trees
Insulation testing

Fault protection

BT: Electrical safety
Fault currents

Fault tolerance

UF: System resilience
BT: Reliability
NT: Fault tolerant computer

networks

Fault tolerant computing
Fault tolerant control
Redundancy

Fault tolerant computer networks

BT: Computer network reliability
Fault tolerance
RT: Fault tolerant computing
Fault tolerant control
Fault tolerant systems

Fault tolerant computing

BT: Fault tolerance
RT: Fault tolerant computer

networks

Fault tolerant control
Fault tolerant systems

Fault tolerant control

BT: Control systems
Fault tolerance
RT: Fault tolerant computer

networks

Fault tolerant computing

Fault tolerant systems

BT: System analysis and design
RT: Fault tolerant computer

networks

Fault tolerant computing

Fault trees

BT: Risk analysis
RT: Boolean functions
Failure analysis
Fault location

Fauna

BT: Plants (biology)

FBAR

USE: Film bulk acoustic resonators

FBARs

USE: Film bulk acoustic resonators

FBT

USE: Flyback transformers

FCC

UF: Federal Communications Commission
BT: US Government agencies

FDA

UF: Food and Drug Administration
BT: US Government agencies

FDDI

UF: Fiber distributed data interface
BT: Communication systems
Optical fiber communication
RT: Communication standards
Local area networks

FDM

USE: Frequency division multiplexing



January 2026 IEEE Thesaurus

FDMA

USE: Frequency division
multiaccess

FDTD

USE: Finite difference methods
AND
Time-domain analysis

FDX

USE: Full-duplex system

Fe

USE: Iron

Feathers

BT: Animal structures

Feature detection

BT: Computer vision
Image processing
RT: Feature extraction
Normalized difference

vegetation index

Normalized difference

water index

Saliency detection

Feature extraction

UF: Feature selection
BT: Image processing
RT: Blob detection
Conditional random fields
Deep learning
Feature detection
Graph neural networks
Image annotation
Image edge detection
Image recognition
Independent component

analysis

License plate recognition
Mixture models
Motion capture
Pattern classification
Pattern recognition
Principal component

analysis

Saliency detection
Signal processing
Speech recognition

Feature learning

USE: Representation learning

Feature selection

USE: Feature extraction

Federal Aviation Administration

USE: FAA

Federal Communications Commission

USE: FCC

Federated identity

UF: FIM
BT: Identity management
systems
RT: Authentication
Blockchains
Token networks
Tokenization

Federated learning

UF: Collaborative learning
Collaborative networking
Collaborative problem

solving

Collaborative work

Cooperative work

Federated-learning

BT: Machine learning

RT: Collaborative software

Communication

effectiveness

Multimedia computing

Professional

communication

NT: Collaborative intelligence

Cooperative communication

Crowdsourcing

Social computing

Federated search

USE: Metasearch

Federated searching

USE: Metasearch

Federated-learning

USE: Federated learning

Feed in tariff

UF: Advanced renewable tariff
FIT
Feed-in-tariff
Renewable energy

payments

BT: Government policies
Tariffs



January 2026 IEEE Thesaurus

RT: Electricity supply industry
Energy conservation
Power generation
Pricing
Renewable energy sources

Control systems

Feedback loop

BT: Feedback communications
NT: Negative feedback loops

Feed-forward neural networks

USE: Feedforward neural networks

Feedforward neural nets

USE: Feedforward neural networks

Feed-in-tariff

USE: Feed in tariff

Feedforward neural networks

UF: Feed-forward neural networks

Feedback

UF: Saturation detection
BT: Circuits

BT: Feedforward neural nets
RT: Neural networks
RT: Artificial intelligence
Pattern recognition
Self-organizing feature

RT: Control systems
Control design
Positive train control
SIMO
Scrum (Software

maps

development)

NT: Support vector machines
Extreme learning machines
Multilayer perceptrons

System dynamics
Time invariant systems

NT: Feedback circuits
Negative feedback
Neurofeedback

Feedforward systems

BT: Intelligent control
RT: Forward error correction
Open loop systems

Feedback amplifiers

UF: NFB
Negative feedback amplifier
BT: Operational amplifiers

Feeds

BT: Antennas
NT: Antenna feeds

Feedback circuits

UF: Circuit feedback
BT: Feedback
RT: Control theory
Distributed feedback

FeFETs

UF: Ferroelectric FETs
BT: Field effect transistors

devices

Felines

USE: Cats

NT: Feedback communications
Output feedback

FEM

USE: Finite element analysis

Feedback communications

BT: Telecommunications
RT: Feedback circuits
NT: Automatic repeat request
Feedback control
Feedback loop

Femtocell networks

UF: Access point base station
BT: Cellular networks
RT: Base stations

Feedback control

BT: Feedback communications
NT: Windup

Femtocells

BT: Base stations

Feedback linearization

BT: Control nonlinearities

Fermentation

BT: Chemical processes
RT: Alcoholic beverages
Dairy products
Pickling



January 2026 IEEE Thesaurus

	Sewage treatment	<i>Ferroelectric FETs</i>	USE: FeFETs
<i>Fermi energy</i>	USE: Fermi level AND Quasi-Fermi level	Ferroelectric films	BT: Ferroelectric materials RT: Magnetic field induced strain
Fermi level	UF: Fermi energy Fermi-level BT: Energy states Thermodynamics RT: Quasi-Fermi level	Ferroelectric materials	BT: Ultrasonics, ferroelectrics, and frequency control RT: Dielectric materials Ferroelectric devices Magnetic field induced strain
<i>Fermi-level</i>	USE: Fermi level		NT: Pyroelectricity Ferroelectric films Lead zirconate titanate Relaxor ferroelectrics
Ferrimagnetic films	BT: Ferrimagnetic materials Films Magnetic films Magnetic materials	Ferrofluid	UF: LiquiFerrofluid BT: Fluids Magnetic materials
Ferrimagnetic materials	BT: Magnetic materials NT: Ferrimagnetic films Ferrite films Ferrites Garnet films Garnets	Ferromagnetic materials	BT: Magnetic materials
Ferrite devices	BT: Magnetic devices RT: Ferrites Flyback transformers Gyrators NT: Circulators	Ferromagnetic resonance	BT: Magnetic resonance
Ferrite films	BT: Ferrimagnetic materials Ferrites Films Magnetic films Magnetic materials	Ferroresonance	BT: Power engineering Resonance RT: Magnetic resonance Nonlinear magnetics
Ferrites	BT: Ferrimagnetic materials Magnetic materials RT: Ferrite devices Gyromagnetism NT: Ferrite films	<i>Fertilisers</i>	USE: Fertilizers
Ferroelectric devices	BT: Dielectric devices RT: Ferroelectric materials	Fertilizers	UF: Fertilisers BT: Agrochemicals RT: Boron Composting Crops
		FET circuits	BT: Solid state circuits RT: Nuclear electronics Operational amplifiers FET integrated circuits JFET circuits MESFET circuits MODFET circuits



January 2026 IEEE Thesaurus

	MOSFET circuits	NT:	Erbium-doped fiber lasers High power fiber lasers	
FET integrated circuits				
	BT:	FET circuits Integrated circuits	Fiber nonlinear optics	
	RT:	Field effect transistors	BT:	Fiber optics Nonlinear optics
	NT:	Field effect MMIC MESFET integrated circuits	Fiber optic sensors	
			USE:	Optical fiber sensors
Fetal heart				
	BT:	Heart	Fiber optics	
Fetal heart rate			UF:	Fibre optics
	BT:	Heart rate	BT:	Optics
			NT:	Fiber nonlinear optics Optical fibers
FETs				
	USE:	Field effect transistors	Fiber reinforced plastics	
Fetus			UF:	Fibre reinforced plastics
	BT:	Embryonic structures	BT:	Plastics
			RT:	Plastic insulators
Few shot learning			Fiber-Bragg gratings	
	UF:	Few-shot learning	USE:	Bragg gratings
	BT:	Deep learning Machine learning	Fiber-in-the-loop	
	RT:	Artificial intelligence One shot learning Training data	USE:	Optical fiber subscriber loops
Few-shot learning			Fibers	
	USE:	Few shot learning	USE:	Textile fibers
FFF			Fibre gratings	
	USE:	Field-flow fractionation	USE:	Fiber gratings
FFP2			Fibre lasers	
	USE:	Face masks	USE:	Fiber lasers
FGIR			Fibre optic sensors	
	USE:	Fine-grained image recognition	USE:	Optical fiber sensors
Fiber Bragg gratings			Fibre optics	
	USE:	Bragg gratings	USE:	Fiber optics
Fiber distributed data interface			Fibre reinforced plastics	
	USE:	FDDI	USE:	Fiber reinforced plastics
Fiber gratings			Fibrillation	
	UF:	Fibre gratings	BT:	Medical treatment
	BT:	Bragg gratings	RT:	Defibrillation
			NT:	Atrial fibrillation
Fiber lasers			Fibroblasts	
	UF:	Fibre lasers	BT:	Biological cells
	BT:	Ring lasers		



January 2026 IEEE Thesaurus

Fiducial markers

UF: Imagimarkers
BT: Image processing
RT: Microfabrication
Semiconductor device

manufacture

Field assisted sintering

USE: Spark Plasma sintering

Field buses

UF: Instrumentation buses
BT: Computer interfaces
RT: Industrial control
Local area networks

Field effect MMIC

BT: FET integrated circuits

Field effect transistors

UF: FETs
BT: Transistors
RT: FET integrated circuits
Graphene devices
Semiconductor devices
NT: CNTFETs
Double-gate FETs
FeFETs
HEMTs
JFETs
MESFETs
MISFETs
MODFETs
MOSFET
MOSHFETs
OFETs
Schottky gate field effect

transistors

TFETs
Thin film transistors

Field electron emission

USE: Electron emission

Field emitter arrays

BT: Electron tubes
RT: Vacuum technology

Field flow fractionation

USE: Field-flow fractionation

Field ion emission

USE: Ion emission

Field multiplication

USE: Galois fields

Field programmable analog arrays

UF: FPAA
Field programmable

analogue arrays

BT: Analog integrated circuits
Programmable circuits
RT: Application specific

integrated circuits

Field programmable gate

arrays

Space technology

Field programmable analogue arrays

USE: Field programmable analog
arrays

Field programmable gate arrays

UF: FPGA
Field-programmable gate

arrays

BT: Integrated circuits
RT: AI accelerators
Field programmable analog

arrays

Reconfigurable devices
VHDL

Field robots

BT: Mobile robots
RT: Assistive robots
Autonomous robots
Collaborative robots
Humanoid robots
Service robots

Field-flow fractionation

UF: EFFF
FFF
Field flow fractionation
BT: Fractionation

Field-programmable gate arrays

USE: Field programmable gate
arrays

Fifth Industrial Revolution

UF: 5IR
IR 5.0
Industrial Revolution 5.0
Industry 5.0
BT: Automation
Digital transformation



January 2026 IEEE Thesaurus

RT: Brain-computer interfaces
Fourth Industrial Revolution
Human computer
interaction
Internet of Things

Filament lamps

BT: Lamps
RT: Lighting

File servers

BT: Computers and information
processing
RT: Computer networks
Data communication
Local area networks

File sharing

USE: Peer-to-peer computing

File system permissions

USE: Permission

File systems

BT: System software
RT: Audio databases
Data structures
Database systems
Information systems

Fill factor (solar cell)

BT: Photovoltaic systems

Filler metals

BT: Joining materials
RT: Metals

Filling

BT: Freight handling
RT: Containers
Loading
Packaging

Film bulk acoustic resonators

UF: FBAR
FBARs
BT: Acoustic devices
Thin film devices
RT: Bulk acoustic wave devices
Cellular radio
Mobile communication
Radio communication
Resonance
Telecommunications

Films

BT: Materials
RT: Chemical vapor deposition
Coatings
Entertainment industry
Human image synthesis
Sputtering
NT: Conductive films
Dielectric films
Epitaxial layers
Ferrimagnetic films
Ferrite films
Garnet films
Magnetic films
Nanocomposite films
Optical films
Piezoelectric films
Plastic films
Polymer films
Semiconductor films
Thick films
Thin films

Films (Motion pictures)

USE: Motion pictures

Filter banks

UF: Filterbank
BT: Band-pass filters

Filter-theory

USE: Filtering theory

Filterbank

USE: Filter banks

Filtering

BT: Circuits and systems
RT: Noise cancellation
NT: Filters
Information filtering

Filtering algorithms

UF: Loop-filtering algorithm
Post-filtering algorithm
BT: Algorithms

Filtering theory

UF: Filter-theory
BT: Filters
RT: Estimation
Line enhancers
Matched filters
Maximum likelihood

detection



January 2026 IEEE Thesaurus

Filters	NT:	Transversal filters Collaborative filtering Image filtering	<i>Finance sector</i>	USE:	Financial industry
	Financial industry				
	BT:	Filtering	UF:	Finance sector	
		Signal processing	BT:	Industries	
	RT:	Passive filters	NT:	Banking	
		Signal to noise ratio		Financial services	
	NT:	Active filters	Financial management		
		Anisotropic	UF:	Financial planning	
		Bragg gratings		Money management	
		Channel bank filters	BT:	Management	
		Comb filters	RT:	Electronic commerce	
		Digital filters		Profitability	
		Equalizers		Public finance	
		Filtering theory	NT:	Costing	
		Gabor filters		Credit cards	
		Harmonic filters		Finance	
		IIR filters		Insurance	
		Kalman filters		Investment	
		Low-pass filters		Loans and mortgages	
		Matched filters		Management accounting	
		Microstrip filters		Mutual funds	
		Nonlinear filters		Pricing	
		Notch filters		Venture capital	
		Particle filters	<i>Financial planning</i>		
		Power filters	USE:	Financial management	
		Resonator filters	Financial services		
		Spatial filters	BT:	Financial industry	
		Superconducting filters	NT:	Automated teller machine	
		Transversal filters		Fintech	
				Micropayments	
Filtration		UF:	Nanofiltration	<i>Financial technology</i>	
technology	BT:	Materials science and	USE:	Fintech	
	RT:	Sieving	Fine-grained image recognition		
	NT:	Microfiltration	UF:	FGIR	
<i>FIM</i>		USE:	Federated identity	BT:	Image recognition
Finance				RT:	Computer vision
	UF:	Taxes		Deep learning	
	BT:	Financial management		Object detection	
	RT:	Banking	FinFETs		
		Bitcoin	BT:	MOSFET	
		Business	<i>Fingerprint identification</i>		
		Cryptocurrency	USE:	Fingerprint recognition	
		Economics	<i>Fingerprint images</i>		
		Open banking	USE:	Image matching	
	NT:	Bankruptcy			
		Crowdfunding			
		Currencies			



January 2026 IEEE Thesaurus

Fingerprint indexing

USE: Fingerprint recognition

Fingerprint matching

USE: Fingerprint recognition

Fingerprint modality

USE: Fingerprint recognition

Fingerprint recognition

UF: Fingerprint identification
Fingerprint indexing
Fingerprint matching
Fingerprint modality
Fingerprint sensing
Fingerprint sensors
Fingerprint verification
Fingerprinting

BT: Biometrics
Identification of persons
Pattern recognition

RT: Image matching

Fingerprint sensing

USE: Fingerprint recognition

Fingerprint sensors

USE: Fingerprint recognition

Fingerprint verification

USE: Fingerprint recognition

Fingerprinting

USE: Fingerprint recognition

Fingers

BT: Extremities
NT: Thumb

Finishing

BT: Surface treatment
RT: Machining
Materials processing
Planing
NT: Surface finishing

Finite difference methods

UF: FDTD
Finite difference time

domain analysis

Finite difference time

domain methods

Finite-difference methods
Finite-difference time-

domain methods

BT: Mathematics

Numerical analysis

RT: Computational

electromagnetics

Perfectly matched layers

Finite difference time domain analysis

USE: Finite difference methods

Finite difference time domain methods

USE: Finite difference methods

Finite element analysis

UF: Discrete element method
FEM

Finite element methods

Finite element modeling

Finite element modelling

Finite-element analysis

Finite-element methods

Finite-element modeling

Finite-element modelling

BT: Mathematics

Numerical analysis

RT: Eddy current testing

Perfectly matched layers

Finite element methods

USE: Finite element analysis

Finite element modeling

USE: Finite element analysis

Finite element modelling

USE: Finite element analysis

Finite fields

USE: Galois fields

Finite impulse response filters

UF: FIR
FIR filters

BT: Digital filters

RT: Discrete wavelet transforms
Frequency response

Finite state machines

USE: Automata

Finite volume methods

UF: Finite-volume method

BT: Numerical analysis

RT: Navier-Stokes equations



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Finite wordlength effects

UF: Overflow oscillations
Truncation errors
BT: Error analysis
RT: Quantization (signal)
NT: Roundoff errors

Finite-difference methods

USE: Finite difference methods

Finite-difference time-domain methods

USE: Finite difference methods

Finite-element analysis

USE: Finite element analysis

Finite-element methods

USE: Finite element analysis

Finite-element modeling

USE: Finite element analysis

Finite-element modelling

USE: Finite element analysis

Finite-volume method

USE: Finite volume methods

Finline

BT: Planar transmission lines

Fintech

UF: Financial technology
BT: Financial services
Technology
RT: Cryptocurrency
Investment
Mobile applications
Online banking
Smart phones

FIR

USE: Finite impulse response
filters

FIR filters

USE: Finite impulse response
filters

Fire extinguishers

BT: Safety devices
RT: Fire safety

Fire retardants

USE: Flame retardants

Fire safety

BT: Safety
RT: Fire extinguishers

Fireproofing

USE: Flame retardants

Fires

UF: Flames
BT: Hazards
RT: Accidents
Emergency services
Flammability
Hazardous areas
Safety
Smoke detectors
NT: Wildfires

Firewalls (computing)

BT: Computer security
RT: Computer networks
Hardware
Software

FireWire

USE: Firewire

Firewire

UF: FireWire
BT: Computer interfaces
RT: Computer peripherals
Consumer electronics
Data communication
Home computing
IEEE 1394 Standard
Video signal processing

Fireworks algorithm

BT: Optimization methods
RT: Particle swarm optimization

Firing

BT: Materials preparation
RT: Ceramics
Heat treatment
Kilns

Firmware

USE: Microprogramming

Fish

BT: Organisms
NT: Fisheries
Sharks



January 2026 IEEE Thesaurus

Fish schooling

USE: Fish schools

Fish schools

UF: Fish schooling

BT: Collective intelligence

Ensemble learning

RT: Algorithm design and

analysis

Artificial neural networks

Behavioral sciences

Clustering methods

Computational modeling

Computer simulation

Decision making

Neural networks

Neuroscience

Particle swarm optimization

Predictive models

Random forests

Training data

Fishbone diagrams

USE: Cause effect analysis

Fisheries

UF: Fishery

BT: Aquaculture

Fish

Fishery

USE: Fisheries

Fission reactors

UF: Nuclear fission

Nuclear reactors (fission)

BT: Nuclear power generation

RT: Pressure vessels

Radiation protection

FIT

USE: Feed in tariff

Fitbit

USE: Wearable Health Monitoring

Systems

Fitting

BT: Assembly

RT: Assembly systems

Fixed point arithmetic

USE: Fixed-point arithmetic

Fixed-point arithmetic

UF: Fixed point arithmetic

BT: Arithmetic

Fixtures

UF: Fixturing

Jigs

BT: Production equipment

RT: Machine tools

Fixturing

USE: Fixtures

Flame retardants

UF: Fire retardants

Fireproofing

BT: Retardants

RT: Bromine compounds

Flammability

Materials preparation

Flames

USE: Fires

Flammability

UF: Inflammability

BT: Hazards

RT: Explosion protection

Explosions

Fires

Flame retardants

Hazardous materials

Flanges

BT: Mechanical products

RT: Rails

Structural plates

Wheels

Flash memories

UF: NAND flash

BT: Memory

RT: Automation

Computer peripherals

Solid state drives

NT: Flash memory cells

Flash memory cells

BT: Flash memories

NT: Split gate flash memory

cells

Flashover

BT: Dielectric breakdown



January 2026 IEEE Thesaurus

Flat panel displays

UF: Plasma display panel
BT: Displays
RT: Consumer electronics
TV

Flex

USE: Flexible printed circuits

Flexible AC transmission systems

UF: FACTS
BT: AC power transmission
Power transmission

Flexible electronics

BT: Assembly systems
RT: Graphene devices
Soft electronics
NT: Electronic skin

Flexible fuel vehicles

USE: Land vehicles

Flexible manufacturing systems

BT: Manufacturing automation
RT: Agile manufacturing
Cellular manufacturing
Computer applications

Flexible printed circuits

UF: Flex
BT: Printed circuits

Flexible structures

UF: Deployable structures
BT: Buildings
Structural shapes
RT: Control systems
Decentralized control
Mechanical variables

control

Soft robotics
Structural engineering

Flicker

USE: Fluctuations

Flight control

USE: Aerospace control

Flight dynamics

BT: Aerospace engineering
RT: Aerodynamics

Flight recorder

USE: Flight recording

Flight recording

UF: Black box
Flight recorder
BT: Recording

Flight simulation

USE: Aerospace simulation

Flip chip

USE: Flip-chip devices

Flip chip solder joints

UF: Castellations
BT: Soldering
RT: Flip-chip devices

Flip-chip

USE: Flip-chip devices

Flip-chip devices

UF: Flip chip
Flip-chip
BT: Semiconductor device
manufacture
Semiconductor devices
RT: Flip chip solder joints
Microassembly
Microprocessor chips
Microprocessors

Flip-flops

BT: Pulse circuits
RT: Logic circuits

FLL

USE: Frequency locked loops

Floating point arithmetic

USE: Floating-point arithmetic

Floating-point arithmetic

UF: Floating point arithmetic
BT: Arithmetic

Floods

BT: Hazards
Hydrology
RT: Land use planning
Rain
Rivers
Stormwater
Structural engineering



January 2026 IEEE Thesaurus

Floors

BT: Building materials
RT: Construction industry
Tiles

Floppy disks

BT: Magnetic memory

Flora

BT: Plants (biology)

Flotation devices

USE: Underwater equipment

Flow

USE: Fluid flow

Flow batteries

USE: Batteries

Flow control

USE: Flow production systems

Flow cytometry

BT: Cytometry

Flow graphs

UF: Data flow graphs
Signal flow graphs
BT: Programmable control
RT: Circuits

Flow meters

USE: Flowmeters

Flow modeling

USE: Flow production systems

Flow production systems

UF: Flow control
Flow modeling
Sequential production
BT: Manufacturing systems
RT: System dynamics
NT: Continuous production

Flowcharts

BT: Engineering drawings
RT: Programming
System analysis and design

Flower pollination algorithm

BT: Algorithm design and
theory

Flowering plants

UF: Angiosperms
BT: Plants (biology)
NT: Bamboo
Pollination (plants)

Flowmeters

UF: Flow meters
BT: Meters
RT: Automatic meter reading
Fluid flow
Velocity measurement

Fluctuations

UF: Flicker
BT: Reliability

Flue ash

USE: Fly ash

Flue gases

BT: Gases
RT: Air pollution
Effluents
Exhaust gases

Fluid dynamics

BT: Fluid flow
Fluids
RT: Hydrodynamics
Lattice Boltzmann methods
NT: Buoyancy
Computational fluid

dynamics

Drag
Navier-Stokes equations
Rheology

Fluid flow

UF: Flow
Gas flow
Liquid flow
Smoothed particle

hydrodynamics

BT: Physics
RT: Electrohydraulics
Flowmeters
Fluid flow measurement
Fluidics
Hydraulic systems
Hydrodynamics
Magnetohydrodynamics
NT: Fluid dynamics
Hydraulic diameter
Hydrology



January 2026 IEEE Thesaurus

Pipelines
Supersonic flow
Valves

RT: Fluorescent lamps
Judd-Ofelt theory
NT: Immunofluorescence

Fluid flow control

BT: Control systems
RT: Valves

Fluid flow measurement

UF: Anemometers
BT: Measurement
RT: Fluid flow
Hydrologic measurements
Pressure gauges

Fluidic microsystems

BT: Micromechanical devices
RT: Microfluidics

Fluidics

BT: Control systems
RT: Fluid flow
Nanotechnology
Pneumatic systems
NT: Microfluidics
Nanofluidics

Fluidisation

USE: Fluidization

Fluidization

UF: Fluidisation
BT: Chemical technology
RT: Fluids

Fluids

BT: Materials
RT: Buoyancy
Fluidization
Oils
NT: Ferrofluid
Fluid dynamics
Gases
Hydraulic fluids
Liquids
Viscosity

Fluids and secretions

BT: Anatomy
NT: Amniotic fluid
Cerebrospinal fluid

Fluorescence

BT: Luminescence
Optics

Fluorescent lamps

BT: Lamps
RT: Fluorescence
Lighting

Fluorine

BT: Chemical elements
NT: Fluorine compounds

Fluorine compounds

BT: Fluorine
NT: Chlorofluorocarbons
Hydrogen fluoride

Flux pinning

BT: Magnetic flux
Superconductivity
RT: Type II superconductors

Fluxtronics

USE: Spintronics

Fly ash

UF: Coal ash
Flue ash
Fly-ash
BT: Ash
RT: Slag

Fly-ash

USE: Fly ash

Flyback transformers

UF: FBT
LOPT
Line output transformer
BT: Transformers
RT: Cathode ray tubes
Electron beam applications
Electron beams
Ferrite devices

Flying robots

USE: Drones

Flywheels

BT: Energy storage

FM

USE: Frequency modulation



January 2026 IEEE Thesaurus

fMRI

USE: Functional magnetic resonance imaging

fnirs

USE: Functional near-infrared spectroscopy

Focusing

BT: Imaging
RT: Lenses

Fog computing

USE: Edge computing

Food and Drug Administration

USE: FDA

Food hygiene

USE: Food safety

Food industry

BT: Manufacturing industries
RT: Beverage industry
Consumer products
Food preservation
Food products
Food safety
Food technology
Smart agriculture
Sugar industry
Sugar refining
NT: Food manufacturing
Food processing

Food loss

USE: Food waste

Food manufacturing

UF: Food production
BT: Food industry
Manufacturing systems
RT: Consumer products
Food preservation
Food products
Food technology
Food waste
NT: Food packaging

Food packaging

BT: Food manufacturing
Packaging
RT: Food products
Food technology
Food waste

Food preservation

UF: Food preservatives
BT: Food technology
RT: Food industry
Food manufacturing
Food products

Food preservatives

USE: Food preservation

Food processing

BT: Food industry

Food production

USE: Food manufacturing

Food products

BT: Manufactured products
Production
RT: Agricultural products
Consumer products
Farming
Food industry
Food manufacturing
Food packaging
Food preservation
Food safety
Seeds (agriculture)
Smart agriculture
Sugar industry
Sugar refining
Vegetable oils
NT: Dairy products
Fats
Food security
Food waste
Nutrients
Sugar

Food safety

UF: Food hygiene
BT: Safety
RT: Food industry
Food products

Food security

BT: Food products
Security
RT: Agricultural products
Economics
Emergency services
Environmental factors
Measurement
Social factors
Supply chain management



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	Sustainable development		Force control	
			BT:	Mechanical variables
Food technology			control	
BT:	Industry applications		RT:	Control systems
RT:	Food industry			Force
	Food manufacturing			Robot control
	Food packaging			
	Smart agriculture		Force feedback	
	Sugar refining		BT:	Haptic interfaces
NT:	Food preservation			
Food waste			Force measurement	
UF:	Food loss		BT:	Mechanical variables
BT:	Food products		measurement	
	Waste materials		RT:	Gravity
RT:	Agriculture			Pressure gauges
	Biofuels		NT:	Dynamometers
	Consumer behavior			Force myography
	Food manufacturing			Gravity measurement
	Food packaging			
	Government policies		Force myography	
	Recycling		BT:	Force measurement
Foot				
BT:	Extremities		Force sensors	
RT:	Ankle		BT:	Sensors
<i>Football</i>			Forebrain	
USE:	Sports		UF:	Prosencephalon
			BT:	Brain
<i>Footprinting</i>			RT:	Hindbrain
USE:	Network reconnaissance			Midbrain
			NT:	Olfactory bulb
Footwear			Forecast uncertainty	
UF:	Shoes		BT:	Forecasting
BT:	Clothing			Uncertainty
RT:	Clothing industry			
	Footwear industry		Forecasting	
Footwear industry			BT:	Probability
UF:	Shoe manufacture		RT:	Hindcasting
BT:	Manufacturing industries			Prediction methods
RT:	Clothing industry		NT:	Demand forecasting
	Consumer products			Economic forecasting
	Footwear			Forecast uncertainty
				Technology forecasting
Force			Forehead	
BT:	Mechanical factors		BT:	Head
RT:	Dynamics			
	Force control		<i>Forensic photography</i>	
	Magnetic forces		USE:	Image forensics
NT:	Axial force			
	Gravity		Forensics	
			BT:	Law
			NT:	Digital forensics
				Image forensics



January 2026 IEEE Thesaurus

Forest clearance

USE: Deforestation

Forest clearing

USE: Deforestation

Forestry

BT: Geoscience
RT: Agribusiness
Pulp and paper industry
Rainforests
Resource management
Vegetation
Vegetation mapping
Wood industry
NT: Afforestation
Deforestation
Forests
REDD+
Reforestation
Remineralize

Forests

BT: Forestry
Vegetation

Forgery

UF: Imposter signature
generation
BT: Handwriting recognition

Forging

UF: Cogging
BT: Manufacturing systems

Formal concept analysis

BT: Mathematical analysis
RT: Classification tree analysis
Data analysis
Knowledge representation
Unsupervised learning

Formal languages

BT: Computer science
NT: Computer languages
Runtime library

Formal logic

USE: Logic

Formal specifications

BT: Standardization
RT: Service-oriented systems
engineering

Formal verification

BT: Software engineering
RT: Circuits and systems
Model checking

Formaldehyde

BT: Organic compounds

Formation control

BT: Motion control
RT: Multi-agent systems
Robot control

Forward contracts

BT: Contracts

Forward error correction

BT: Error correction
RT: Feedforward systems

Fossil fuels

BT: Fuels
RT: Air pollution
NT: Natural gas

Foundation models

UF: Large AI models
BT: Artificial intelligence
Modeling
RT: Chatbots
Machine learning

Foundries

BT: Production facilities
RT: Casting
Furnaces
Heat treatment
Materials processing

Four wave mixing

USE: Four-wave mixing

Four-wave mixing

UF: Four wave mixing
BT: Distortion
Optics
RT: Multiwave mixing

Fourier series

BT: Mathematics
RT: Data compression
Signal processing
Spectroscopy



January 2026 IEEE Thesaurus

Fourier transform infrared spectra

USE: Fourier transform infrared spectroscopy

Fourier transform infrared spectroscopy

UF: Fourier transform infrared spectra

BT: Fourier transforms
Spectroscopy

Fourier transforms

BT: Transforms
RT: Acoustics
Cepstrum
Diffraction
Harmonic analysis
Optics
Partial differential equations
Probability
Statistics
Time-frequency analysis
NT: Discrete Fourier transforms
Fast Fourier transforms
Fourier transform infrared spectroscopy

Fourth Industrial Revolution

UF: Industrial Revolution 4.0
Industry 4.0
BT: Automation
Digital transformation
Manufacturing
RT: Fifth Industrial Revolution
Internet of Things
Machine-to-machine

communications

Smart devices
Smart manufacturing

FPAAs

USE: Field programmable analog

arrays

FPGA

USE: Field programmable gate

arrays

Fracking

USE: Fracking

Fracking

UF: Fracking
Hydraulic fracking
BT: Hydrological techniques
RT: Mining industry

Natural gas

Fractal antennas

BT: Antennas

Fractal art

BT: Art

Fractals

BT: Computational geometry
RT: Antennas
Chaos
Computer graphics
Econophysics

Fractional brownian motion

USE: Brownian motion

Fractional calculus

BT: Mathematical analysis

Fractionation

BT: Separation processes
RT: Chemical analysis
Oils
Petroleum industry
Field-flow fractionation
NT:

Frame relay

BT: Communication switching
Packet switching
RT: B-ISDN
Computer networks
ISDN
Protocols
Wide area networks

Franchises

USE: Franchising

Franchising

UF: Franchises
BT: Business

Franz-Keldysh effect

USE: Electro-absorption modulators

Fraud

BT: Law
NT: Identity theft

Free economic territories

USE: Free economic zones



January 2026 IEEE Thesaurus

Free economic zones

UF: Free economic territories
Free zones
BT: Commerce and trade
Economics
RT: International trade
Tariffs
Trade agreements
NT: Freeports

Free electron lasers

UF: Cerenkov lasers
BT: Lasers
RT: Electron beams
Relativistic effects
Undulators

Free ports

USE: Freeports

Free trade

USE: Trade agreements

Free zones

USE: Free economic zones

Free-space optical communication

BT: Optical fiber communication
RT: NOMA

Freeports

UF: Free ports
BT: Free economic zones
RT: Airports
Economics
International trade
Seaports
Trade agreements

Freeware

UF: WhatsApp
BT: Software

Freight containers

BT: Containers
RT: Freight handling
Transportation

Freight handling

UF: Cargo handling
BT: Materials handling
RT: Freight containers
Lifting equipment
Pulleys

discontinuities

NT: Filling
Loading

Frequency

BT: Electromagnetic radiation
RT: Band-pass filters
Electric variables
Transmission line theory
NT: Bandwidth
Frequency dependence
Frequency diversity
Frequency synchronization
Resonant frequency

Frequency allocation

USE: Radio spectrum
management

Frequency control

UF: Frequency regulation
BT: Ultrasonics, ferroelectrics,
and frequency control
RT: Electric variables control
Frequency locked loops
Mechanical variables

control

Optical variables control
Ring oscillators
Tuners

NT: Automatic frequency control
Tunable circuits and

devices

Tuning

Frequency conversion

UF: Frequency division
Frequency multiplication
Harmonic generation
BT: Converters
RT: Image converters
Image intensifiers
NT: Mixers
Optical frequency

conversion

Frequency dependence

UF: Frequency dependent
BT: Frequency

Frequency dependent

USE: Frequency dependence



January 2026 IEEE Thesaurus

Frequency diversity

BT: Frequency

measurements

Electromagnetic

Frequency division

USE: Frequency conversion

measurement

Frequency response
Mechanical variables

Frequency division multiaccess

UF: FDMA

BT: Multiaccess communication

RT: Broadband communication

measurement

Optical variables

Frequency division multiplexing

UF: FDM

Frequency multiplexing

BT: Multiplexing

RT: Layered division

multiplexing

NT:

Phase frequency detectors
Time-frequency analysis
Wavelength measurement
Frequency estimation
Frequency-domain analysis

Frequency domain

USE: Frequency-domain analysis

Frequency domain analysis

USE: Frequency-domain analysis

Frequency modulation

UF: FM

BT: Modulation

Radio broadcasting

RT: Demodulation

NT: Frequency shift keying

Frequency multiplexing

USE: Frequency division

multiplexing

Frequency estimation

BT: Frequency measurement

Parameter estimation

RT: Spectral analysis

Speech analysis

Frequency multiplication

USE: Frequency conversion

Frequency hop communication

USE: Spread spectrum

communication

Frequency regulation

USE: Frequency control

Frequency hop radar

USE: Spread spectrum radar

Frequency response

BT: Testing

RT: Amplifiers

Channel impulse response

Digital filters

Finite impulse response

filters

Frequency measurement

Impulse testing

Frequency locked loops

UF: FLL

BT: Linear feedback control

systems

RT: Signal processing

Frequency control

Frequency synthesizers

Phase locked loops

Synchronization

Frequency selective surfaces

BT: Antenna theory

Frequency shift keying

UF: FSK

Frequency-shift modulation

Frequency-shift signaling

BT: Frequency modulation

Frequency measurement

BT: Measurement

RT: Acoustic measurements

Atomic clocks

Doppler measurement

Electric variables

measurement

Frequency synchronization

BT: Frequency

NT: Frequency synthesizers



January 2026 IEEE Thesaurus

Frequency synthesizers

BT: Frequency synchronization
RT: Frequency locked loops
Tuners

Frequency-domain analysis

UF: Frequency domain
Frequency domain analysis
BT: Frequency measurement
RT: Circuit analysis
Functional analysis
Signal analysis
NT: Time-frequency analysis

Frequency-hop communication

USE: Spread spectrum
communication

Frequency-selective fading channels

BT: Fading channels

Frequency-shift modulation

USE: Frequency shift keying

Frequency-shift signaling

USE: Frequency shift keying

Freshwater

BT: Water

Fresnel integral

USE: Fresnel reflection

Fresnel lenses

USE: Fresnel reflection

Fresnel reflection

UF: Fresnel integral
Fresnel lenses
Fresnel zones
BT: Reflection

Fresnel zones

USE: Fresnel reflection

Friction

BT: Mechanical factors
RT: Drag
Dynamics
Lubrication
NT: Mechanical bearings

Friction stir processing

USE: Strain control

Frittage

USE: Sintering

Frontal lobe

UF: BrainLobe
BT: Brain

Froth flotation

USE: Manufacturing processes

FSK

USE: Frequency shift keying

FTTH

USE: Optical fiber subscriber
loops

Fuel additives

USE: Additives

Fuel cell vehicles

BT: Electric vehicles
RT: Energy storage
Fuel cells
Traction motors
Vehicle-to-grid

Fuel cells

UF: Enzymatic fuel cells
Microbial eletrolysis cells
Microbial fuel cells
Solid oxide electrolyzer
cells
BT: Electrochemical devices
Energy conversion
Energy storage
RT: Fuel cell vehicles
Fuel storage
Power generation

Fuel economy

BT: Economics
Fuels

Fuel processing industries

UF: Coal tar
BT: Manufacturing industries
RT: Fuel storage
Fuels
Mining industry
Oil drilling
Oils



January 2026 IEEE Thesaurus

	Petroleum		Fullerenes	
	Petroleum industry		UF:	Buckeyballs
Fuel pumps				Buckminsterfullerene
BT:	Pumps			Buckyballs
RT:	Engines			Buckytubes
	Fuels			CarboFullerene
			BT:	Fullerites
				Carbon
Fuel storage			<i>Fullerites</i>	
UF:	Fuel tanks		USE:	Fullerenes
	Oil tanks			
BT:	Material storage		Function approximation	
RT:	Containers		BT:	Approximation methods
	Energy storage		RT:	Computer science
	Fuel cells			
	Fuel processing industries		<i>Function generators</i>	
	Fuels		USE:	Signal generators
<i>Fuel tanks</i>			Functional analysis	
USE:	Fuel storage		BT:	Mathematics
Fuels			RT:	Eigenvalues and
BT:	Energy resources		eigenfunctions	Frequency-domain analysis
	Manufactured products			Functional data analysis
RT:	Coal gas			Inverse problems
	Fuel processing industries			Lyapunov methods
	Fuel pumps			Wave functions
	Fuel storage			
	Methanol		Functional connectivity	
	Petrochemicals		BT:	Neuroscience
	Waste materials		RT:	Functional neuroimaging
NT:	Biofuels			
	Coal		Functional data analysis	
	Fossil fuels		BT:	Data analysis
	Fuel economy			Statistical analysis
	Petroleum		RT:	Functional analysis
	Synthetic fuels			
<i>Full duplex system</i>			<i>Functional electrical stimulation</i>	
USE:	Full-duplex system		USE:	Neuromuscular stimulation
Full stack			Functional magnetic resonance imaging	
UF:	Software stack		UF:	fMRI
	Solution stack		BT:	Magnetic resonance
BT:	Software engineering		imaging	
			RT:	Biomedical image
<i>Full virtualization VM</i>			processing	
USE:	System virtual machines		Functional near-infrared spectroscopy	
Full-duplex system			UF:	fnirs
UF:	FDX		BT:	Functional neuroimaging
	Full duplex system			Spectroscopy
BT:	Duplex communication		RT:	Infrared imaging
systems				Infrared spectra
				Neuroimaging



January 2026 IEEE Thesaurus

Functional neuroimaging

BT: Neuroimaging
RT: Functional connectivity
NT: Functional near-infrared spectroscopy

Tokamaks

Fusion splicing

USE: Splicing

Futurism

USE: Technology forecasting

Functional point analysis

BT: Estimation
Size measurement
RT: Cost benefit analysis
Software engineering

Fuzz testing

USE: Fuzzing

Functional programming

BT: Programming
RT: Python

Fuzzing

UF: Fuzz testing
BT: Software testing

Fungi

BT: Microorganisms

Fuzzy cognitive maps

BT: Directed graphs
Knowledge representation
RT: Cognitive systems
Decision making
Fuzzy logic
Fuzzy reasoning
Fuzzy set theory
Fuzzy systems
Inference mechanisms
Learning (artificial

Furnaces

BT: Machinery
RT: Building services
Foundries
Gas appliances
Heating systems
NT: Blast furnaces
Kilns

intelligence)

Neural networks

Further education

USE: Continuing education

Fuzzy control

BT: Fuzzy systems
RT: Fuzzy logic
Fuzzy sets
Takagi-Sugeno model

Fuses

BT: Electronic components
RT: Interrupters
Power system protection
Protection
Switchgear

Fuzzy inference

USE: Fuzzy logic

Fusion power generation

BT: Nuclear and plasma sciences
RT: Nuclear power generation
Fusion reactors
Magnetic confinement

Fuzzy logic

UF: Fuzzy inference
Logic
BT: Fuzzy cognitive maps
Fuzzy control
Fuzzy reasoning
Fuzzy sets
Fuzzy systems
Possibility theory
NT: Takagi-Sugeno model

Fusion reactor design

BT: Fusion reactors

Fusion reactors

UF: Nuclear reactors (fusion)
Thermonuclear fusion
BT: Nuclear and plasma sciences
RT: Fusion power generation
NT: Fusion reactor design

Fuzzy neural nets

USE: Fuzzy neural networks

Fuzzy neural networks

UF: Fuzzy neural nets
Neuro fuzzy networks



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	BT:	Neuro-fuzzy networks Fuzzy systems
Fuzzy reasoning		
	BT:	Inference mechanisms
	RT:	Fuzzy cognitive maps Fuzzy logic
Fuzzy set theory		
	BT:	Set theory
	RT:	Fuzzy cognitive maps Fuzzy sets Fuzzy systems Power system faults TOPSIS
Fuzzy sets		
	BT:	Set theory
	RT:	Fuzzy control Fuzzy logic Fuzzy set theory Fuzzy systems Nonlinear dynamical
systems		Uncertainty
Fuzzy systems		
	BT:	Computational intelligence
	RT:	Fuzzy cognitive maps Fuzzy logic Fuzzy set theory Fuzzy sets Large-scale systems Soft sensors Takagi-Sugeno model
	NT:	Fuzzy control Fuzzy neural networks Hybrid intelligent systems
Ga		
	USE:	Gallium
GaAs		
	USE:	Gallium arsenide
Gabor filters		
	BT:	Filters
	RT:	Image processing
Gadolinium		
	BT:	Chemical elements
	NT:	Gadolinium oxide
Gadolinium oxide		
	BT:	Gadolinium

Gain		
	BT:	Electric variables
<i>Gain analysis</i>		
	USE:	Gain measurement
Gain control		
	UF:	Automatic gain control
	BT:	Electric variables control
Gain measurement		
	UF:	Gain analysis
	BT:	Measurement
	RT:	Electric variables
measurement		Refractive index
<i>Gait analysis</i>		
	USE:	Gait recognition
<i>Gait assessment</i>		
	USE:	Legged locomotion
<i>Gait control</i>		
	USE:	Legged locomotion
<i>Gait disorders</i>		
	USE:	Legged locomotion
Gait recognition		
	UF:	Gait analysis
	BT:	Biometrics
	RT:	Computer vision Motion capture
<i>Galerkin method</i>		
	USE:	Method of moments
Gallbladder		
	BT:	Digestive system
Gallium		
	UF:	Ga
	BT:	Metals
	RT:	Semiconductor materials Epitaxial growth Gallium compounds Molecular beam epitaxial
growth		Semiconductor thin films
	NT:	Gallium alloys
Gallium alloys		
	BT:	Gallium
	RT:	Alloying



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semiconductors	Wide band gap	Game theory	BT: Decision making RT: Control systems Games Minimax techniques Multi-armed bandit problem Oligopoly Optimal control Predator prey systems
Gallium arsenide	UF: GaAs Gallium-arsenide Gallium-arsenide (GaAs) BT: Gallium compounds Semiconductor materials	NT: Branching factor Differential games Mechanism design Nash equilibrium	
Gallium compounds	UF: Gallium devices Gallium materials BT: Compounds RT: Alloying Gallium NT: Aluminum gallium nitride Gallium arsenide Gallium nitride Gallium oxide Indium gallium arsenide Indium gallium nitride	Games	BT: Consumer products RT: Entertainment industry Game theory Metaverse Sports NT: Cloud gaming Procedural generation Serious games
<i>Gallium devices</i>	USE: Gallium compounds	<i>Gaming on demand</i>	USE: Cloud gaming
<i>Gallium materials</i>	USE: Gallium compounds	Gamma distribution	BT: Statistics RT: Probability
Gallium nitride	BT: Gallium compounds	<i>Gamma phase iron</i>	USE: Austenite
Gallium oxide	BT: Gallium compounds	<i>Gamma radiation detectors</i>	USE: Gamma-ray detectors
<i>Gallium-arsenide</i>	USE: Gallium arsenide	<i>Gamma ray bursters</i>	USE: Gamma-ray bursts
<i>Gallium-arsenide (GaAs)</i>	USE: Gallium arsenide	<i>Gamma ray bursts</i>	USE: Gamma-ray bursts
Galois fields	UF: Field multiplication Finite fields BT: Abstract algebra	<i>Gamma ray detection</i>	USE: Gamma-ray detection
<i>Galvanising</i>	USE: Galvanizing	<i>Gamma ray detectors</i>	USE: Gamma-ray detectors
Galvanizing	UF: Galvanising BT: Surface treatment RT: Corrosion Corrosion inhibitors Protection	<i>Gamma ray effects</i>	USE: Gamma-ray effects
		<i>Gamma ray telescopes</i>	USE: Gamma-ray telescopes



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Gamma rays

USE: Gamma-rays

Gamma-ray bursts

UF: Cosmic gamma ray bursts
Gamma ray bursters
Gamma ray bursts
BT: Gamma-rays

Gamma-ray detection

UF: Gamma ray detection
BT: Gamma-rays

Gamma-ray detectors

UF: Gamma radiation detectors
Gamma ray detectors
BT: Radiation detectors
RT: Astronomy
Biomedical applications of

radiation

X-ray detectors
X-ray imaging

Gamma-ray effects

UF: Gamma ray effects
BT: Gamma-rays
Radiation effects

Gamma-ray imaging

USE: Nuclear imaging

Gamma-ray telescopes

UF: Gamma ray telescopes
BT: Telescopes
RT: Aerospace electronics

Gamma-rays

UF: Gamma rays
BT: Electromagnetic radiation
Nuclear and plasma

sciences

RT: Collimators
Nuclear medicine
NT: Gamma-ray bursts
Gamma-ray detection
Gamma-ray effects

GAN

USE: Generative adversarial

networks

Ganglia

UF: Ganglion
BT: Cells (biology)
Nervous system

Ganglion

USE: Ganglia

Gap waveguide

BT: Electromagnetic
waveguides
RT: Waveguide components

Garbage collection (computers)

USE: Memory management

Garment industry

USE: Clothing industry

Garments

USE: Clothing

Garnet films

BT: Ferrimagnetic materials
Films
Garnets
Magnetic films
Magnetic materials

Garnets

BT: Ferrimagnetic materials
Magnetic materials
NT: Garnet films

Gas appliances

BT: Home appliances
RT: Furnaces
Space heating

Gas chromatography

BT: Measurement

Gas detectors

UF: Gas sensors
BT: Chemical and biological
sensors
RT: Chemical transducers
NT: Amperometric sensors

Gas discharge devices

BT: Nuclear and plasma
sciences
RT: Discharges (electric)
Electrophotography
Gases
Lighting
Plasma devices
Thyratrons
NT: Glow discharge devices



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<i>Gas discharges</i>	USE:	Discharges (electric)	technology	Gas lasers	Materials science and
<i>Gas flow</i>	USE:	Fluid flow	NT:	Natural gas	Argon
Gas industry	BT:	Industries		Carbon emissions	Coal gas
	RT:	Petroleum industry		Exhaust gases	Flue gases
<i>Gas insulated switchgear</i>	USE:	Gas insulation		Greenhouse gases	Helium
Gas insulated transmission lines	UF:	GITL		Hydrogen	Nitrogen
		Gas-insulated lines		Oxygen	Syngas
		Gas-insulated transmission		Xenon	
	BT:	Power transmission lines	Gaskets	BT:	Seals
	RT:	Gas insulation		RT:	Engine cylinders
Gas insulation	UF:	Gas insulated switchgear			Engines
	BT:	Insulation			Pistons
	RT:	Gas insulated transmission	<i>Gasoline</i>	USE:	Petroleum
lines		Gases		Gastritis	BT:
	NT:	Sulfur hexafluoride			RT:
Gas lasers	UF:	Atomic lasers			Medical conditions
		Metal vapor lasers			Stomach
	BT:	Lasers	<i>Gastroenterologists</i>	USE:	Gastroenterology
	RT:	Atom lasers		Gastroenterology	UF:
		Chemical lasers			Gastroenterologists
		Gases			BT:
<i>Gas platforms</i>	USE:	Offshore installations			Medical specialties
<i>Gas sensors</i>	USE:	Gas detectors	<i>Gastrointestinal</i>	USE:	Gastrointestinal tract
<i>Gas-insulated lines</i>	USE:	Gas insulated transmission		Gastrointestinal tract	UF:
lines					Gastrointestinal
<i>Gas-insulated transmission</i>	USE:	Gas insulated transmission			BT:
lines			Gate drivers	BT:	Power electronics
Gases	BT:	Fluids		RT:	High power amplifiers
	RT:	Discharges (electric)			MOSFET
		Gas discharge devices	Gate leakage	BT:	Leakage currents
		Gas insulation			Solid state circuits
					Tunneling



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Gate leakage current

USE: Leakage currents

Gated recurrent units

BT: Recurrent neural networks

GATT

USE: Trade agreements

Gaussian approximation

BT: Gaussian distribution

Gaussian channels

BT: Communication channels

RT: Intersymbol interference

NT: AWGN channels

Gaussian distribution

UF: Normal distribution

BT: Statistical distributions

NT: Gaussian approximation

Gaussian mixture model

BT: Gaussian processes
Statistics

Gaussian noise

BT: Noise

RT: Additive white noise

Image denoising

Signal processing

TV interference

NT: AWGN

Gaussian processes

BT: Stochastic processes

RT: Inference mechanisms

Learning (artificial

intelligence)

Prediction methods

NT: Gaussian mixture model

Gaze tracking

BT: Control systems
Human computer

interaction

RT: Assistive technologies

Computer vision

Eyes

Face recognition

Motion measurement

Position measurement

User interfaces

Video signal processing

NT: Electrooculography

GDP

USE: Economic indicators

GDPR

USE: General Data Protection

Regulation

Ge Si

USE: Germanium silicon alloys

Gears

UF: Bevel gears

Differential gears

Helical gears

Spur gears

Worm gears

BT: Machinery

Mechanical products

RT: Automotive components

Machine components

Machine tools

Mechanical power

transmission

Mechanical splines

Mechanical systems

Production equipment

Shafts

Torque converters

NT: Magnetic gears

Gender equality

USE: Gender issues

Gender equity

BT: Equal opportunities

RT: Transgender issues

Gender issues

UF: Gender equality

Women's issues

BT: Equal opportunities

RT: Digital divide

NT: Transgender issues

Gene expression

BT: Gene therapy

Gene therapy

BT: Genetics

RT: CRISPR

Transfection

NT: Gene expression

General agreement on tariffs and trade

USE: Trade agreements



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General Data Protection Regulation

UF: GDPR
BT: Data protection
Government policies
Legislation
RT: Data handling
Ethics
Privacy

Generation of electric power

USE: Power generation

Generation Y

USE: Millennials

Generative adversarial networks

UF: GAN
BT: Algorithm design and
analysis
RT: Artificial intelligence
Convolutional neural
networks
Machine learning
Neural networks
Unsupervised learning

Generative AI

UF: AI generated content
AI-generated content
Generative artificial
intelligence
BT: Artificial intelligence
RT: Neural networks
Pattern analysis
NT: Prompt engineering

Generative artificial intelligence

USE: Generative AI

Generative Pre-Trained Transformer

UF: GPT
BT: Large language models
Machine learning
RT: Chatbots
Deep learning
Natural language
processing
Training data

Generators

UF: Dynamo
BT: Electric machines
Rotating machines
RT: Coils
Islanding

Power generation
NT: AC generators
DC generators
Electric generators
Radioisotope thermoelectric
generators
Standby generators

Genetic algorithms

BT: Algorithms
Computational intelligence
RT: Job shop scheduling
Metaheuristics
Pareto optimization
Search methods
NT: Genetic operators

Genetic communication

BT: Genetics
Information theory
RT: Biological information
theory
Biomedical engineering
DNA

Genetic engineering

UF: Genetically modified crops
BT: Engineering in medicine
and biology
RT: Agriculture
Biomedical engineering
Biotechnology
Ethical aspects
Genetics
Molecular biophysics
Seeds (agriculture)
Tissue engineering
NT: CRISPR
Transfection

Genetic expression

BT: Genetics

Genetic mutations

BT: DNA
RT: CRISPR

Genetic operators

BT: Genetic algorithms

Genetic programming

BT: Genetics

Genetically modified crops

USE: Genetic engineering



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Genetics

BT: Biology
 RT: Amniocentesis
 Genetic engineering
 Memetics
 Molecular biophysics
 NT: DNA
 Epigenetics
 Gene therapy
 Genetic communication
 Genetic expression
 Genetic programming
 Genomics
 Genotypes
 Optogenetics
 Phenotypes

Genital system

USE: Reproductive system

Genitourinary surgery

USE: Urology

Genomes

USE: Genomics

Genomics

UF: Genomes
 BT: Genetics
 Molecular biomarkers
 RT: High dimensional data
 NT: Codons
 Metagenomics
 Transcriptomics

Genotypes

BT: Genetics
 RT: Phenotypes

Geo tagging

USE: Location awareness

Geoacoustic inversion

BT: Sea measurements

Geochemistry

UF: Hydrochemistry
 BT: Chemistry
 Geoscience
 RT: Geophysics
 Salinity (geophysical)

Geodesy

BT: Geophysical measurements
 RT: Position measurement

Theodolites

NT: Level measurement

Geodynamics

BT: Geophysics

Geoengineering

UF: Engineering geology
 Geological engineering
 BT: Geoscience
 RT: Drilling
 Geology
 Geophysics
 Hydrological techniques
 Mining industry
 NT: Solar geoengineering

Geographic information systems

UF: GIS
 Geographical information

systems

BT: Geoscience and remote

sensing

Global communication
 Intelligent transportation

systems

RT: Cartography
 Image databases
 NT: Geospatial analysis
 Gunshot detection systems

Geographical information systems

USE: Geographic information

systems

Geography

BT: Geoscience
 RT: Environmental science
 Geospatial analysis
 NT: Cartography
 Deserts
 Rural areas
 Urban areas

Geologic measurements

BT: Measurement
 RT: Geology
 Geophysical measurements
 Hyperspectral sensors
 Remote sensing
 Terrain mapping
 Theodolites



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Geologic processes

USE: Geological processes

Geological engineering

USE: Geoengineering

Geological processes

UF: Geologic processes

BT: Geology

NT: Sedimentation

Subduction

Weathering

Geology

BT: Geoscience

RT: Geoengineering

Geologic measurements

Geophysics

Paleontology

NT: Biogeochemistry

Continental crust

Erosion

Geological processes

Landslides

Minerals

Oceanic crust

Rocks

Stratigraphy

Tectonics

Geomagnetic disturbances

USE: Geomagnetic storms

Geomagnetic navigation

USE: Geomagnetism AND

Navigation

Geomagnetic storms

UF: Geomagnetic disturbances

BT: Geomagnetism

Weather

RT: Coronal mass ejections

Electromagnetic pulses

Solar wind

Sunspots

Geomagnetism

UF: Geomagnetic navigation

Geomatics

BT: Magnetic fields

RT: Electromagnetic induction

Geophysical measurements

Geophysics

Magnetosphere

NT: Geomagnetic storms

Geomatics

USE: Geomagnetism

Geometric modeling

BT: Computational geometry

Mathematical models

Geometrical optics

BT: Optics

RT: Reflectivity

NT: Ray tracing

Geometry

BT: Mathematics

RT: Layout

Lie groups

Shape

NT: Computational geometry

Elliptic curves

Elliptic design

Information geometry

Projective geometry

Solid geometry

Surface topography

Geophysical image processing

BT: Geophysical measurement techniques

Image processing

Geophysical measurement techniques

BT: Geoscience and remote sensing

RT: Laser radar

Magneto electrical

resistivity imaging technique

Remote sensing

Theodolites

Tomography

Vegetation mapping

NT: Geophysical image processing

Geophysical measurements

UF: Geophysical techniques

BT: Geoscience and remote

sensing

Measurement

RT: Atmospheric

measurements

Buried object detection

Depth measurement

Geologic measurements

Geomagnetism



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processing	Geophysical signal	Arctic
	Geophysics	Atmosphere
	Gravity measurement	Biosphere
	Pressure gauges	Continents
	Remote sensing	Cyclones
	Soil measurements	Earth
	Terrain mapping	Earthquakes
NT:	Geodesy	Equator
	Sea measurements	Estuaries
	Seismic measurements	Forestry
Geophysical signal processing		Geochemistry
BT:	Geoscience and remote	Geoengineering
sensing	Signal processing	Geography
RT:	Geophysical measurements	Geology
		Geophysics
<i>Geophysical techniques</i>		Glaciology
USE:	Geophysical measurements	Hydrography
		Ice
Geophysics		Lakes
BT:	Geoscience	Land surface
	Physics	Levee
RT:	Earth	Meteorological factors
	Geochemistry	Natural resources
	Geoengineering	Oceanography
	Geology	Oceans
	Geomagnetism	Rivers
	Geophysical measurements	Sediments
	Hydrologic measurements	Soil
	Hydrological techniques	Tornadoes
	Oceans	Tsunami
	Terrestrial atmosphere	Volcanoes
NT:	EMTDC	Wetlands
	Extraterrestrial phenomena	
	Geodynamics	Geoscience and remote sensing
	Geophysics computing	NT:
	Meteorology	Environmental factors
	Moisture	Geographic information
	Seismology	
	Surface waves	systems
	Well logging	techniques
Geophysics computing		processing
BT:	Geophysics	Geoscience
RT:	Computer aided analysis	Land surface temperature
Geoscience		Photometry
UF:	Earth science	Radiometry
BT:	Geoscience and remote	Remote sensing
sensing	Science - general	Terrain mapping
RT:	Hydrological techniques	Terrestrial atmosphere
NT:	Antarctica	Vegetation mapping
		Geospatial analysis
		BT:
		Geographic information
		systems
		RT:
		Geography



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Software
GEOSS
USE: Global Earth Observation
System of Systems

Geostationary communication satellites
USE: Geostationary satellites

Geostationary satellites
UF: Geostationary
communication satellites
Geosynchronous
communication satellites
Geosynchronous satellites
BT: Satellites
RT: Orbits

Geosynchronous communication satellites
USE: Geostationary satellites

Geosynchronous satellites
USE: Geostationary satellites

Geotechnical engineering
BT: Civil engineering
NT: Excavation

Geotechnical structures
BT: Civil engineering
NT: Dams

Geothermal energy
BT: Energy resources
RT: Geothermal power
generation
Renewable Portfolio
Standard

Geothermal power generation
BT: Power generation
RT: Geothermal energy

Geriatrics
BT: Medical treatment
RT: Assisted living
Assistive robots
Fall detection
Gerontology
Older adults

Germ warfare
USE: Biohazards AND
Biological weapons

Germanium
BT: Metals
Semiconductor materials
RT: Epitaxial growth
Semiconductor thin films
Silicon germanium
NT: Germanium alloys

Germanium alloys
BT: Germanium
RT: Alloying
NT: Germanium silicon alloys

Germanium silicon alloys
UF: Ge Si
BT: Germanium alloys
Silicon alloys

Germs
USE: Pathogens

Gerontechnology
BT: Biomedical equipment
Gerontology
RT: Assistive technologies

Gerontology
BT: Medical specialties
RT: Aging
Alzheimer's disease
Geriatrics
Older adults
NT: Gerontechnology

Gestational diabetes
BT: Diabetes
RT: Pregnancy

Gesture recognition
BT: Pattern recognition
NT: Sign language

Gettering
UF: Getters
BT: Vacuum systems
RT: Electron tubes
Integrated circuit
manufacture
Semiconductor device
manufacture
Vacuum technology

Getters
USE: Gettering



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<i>GFL</i>	USE:	Grid following	Glass	BT:	Amorphous materials
					Glass products
<i>GFM</i>	USE:	Grid forming		RT:	Ceramics
					Dielectric materials
<i>GHGP</i>	USE:	Greenhouse gas protocol			Glass industry
					Insulation
					Optical materials
<i>GHZ transverse electromagnetic cells</i>	USE:	TEM cells	<i>Glass bottles</i>	USE:	Glass products
Giant magnetoresistance	BT:	Magnetoresistance	Glass box	UF:	Glass-box
	RT:	Hard disks			White box
		Magnetoresistive devices			White-box
		Thin film devices			Whitebox
<i>Girders</i>	USE:	Structural beams		BT:	Software
					System analysis and design
<i>GIS</i>	USE:	Geographic information		RT:	Software testing
systems					System testing
<i>Github</i>	USE:	Software development	<i>Glass ceramics</i>	USE:	Ceramics
management					
<i>GITL</i>	USE:	Gas insulated transmission	<i>Glass furnaces</i>	USE:	Glass manufacturing
lines					
Glaciers	BT:	Glaciology	Glass industry	BT:	Manufacturing industries
		Ice		RT:	Glass
	RT:	Snow			Glass manufacturing
	NT:	Ice sheets			Glass products
Glaciology	BT:	Geoscience	Glass manufacturing	UF:	Glass furnaces
	NT:	Glaciers		BT:	Manufacturing systems
Glands	UF:	Endocrine glands		RT:	Glass industry
		Exocrine glands	Glass products	UF:	Glass bottles
	BT:	Biological tissues		BT:	Manufactured products
	NT:	Adrenal glands		RT:	Bottling
		Mammary glands			Ceramic products
		Pituitary gland			Ceramics
		Salivary glands			Chemical products
		Sebaceous glands			Glass industry
		Sweat glands			Windows
		Thyroid		NT:	Glass
			<i>Glass-box</i>	USE:	Glass box
			Glaucoma	BT:	Eye diseases



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Glazes

BT: Coatings
RT: Ceramics
NT: Ceramic glazes

Marine transportation
Military satellites
Road transportation
Satellite broadcasting
Satellite communications
Telecommunications
Terrain mapping

Glial cells

UF: Neuroglia
BT: Cells (biology)
Nervous system

Global System for Mobile Communications
USE: GSM

Global communication

UF: Global groups
Global teams
BT: Professional

Global teams
USE: Global communication

communication

NT: Cross-cultural

communication

Geographic information

systems

Global warming

UF: Global dimming
BT: Climate change
Temperature measurement
Terrestrial atmosphere
RT: Air pollution
Atmospheric

Global dimming

USE: Global warming

measurements

Global Earth Observation System of Systems

UF: GEOSS
BT: Earth Observing System

Carbon emissions
Carbon sequestration
Energy informatics
Environmental

Global groups

USE: Global communication

management

Global markets

USE: Globalization

Greenhouse effect
Land surface temperature
Low carbon economy
Net zero
Ocean temperature
Sea level rise
Thermal pollution
Global warming potential
Paris Agreement
Tipping point

Global navigation satellite system

UF: GNSS
BT: Satellite navigation systems
RT: Global Positioning System

NT: Global warming potential
Paris Agreement
Tipping point

Global Navigational Positioning System

USE: Global Positioning System

Global warming potential

UF: GWP
BT: Global warming
Greenhouse gases
NT: Green manufacturing

Global Positioning System

UF: DGPS
Differential GPS
GPS
Global Navigational

Globalisation

USE: Globalization

Positioning System

BT: Satellite navigation systems
RT: Air to ground

Globalization

UF: Global markets
Globalisation
BT: Social implications of

communication

Air transportation
Global navigation satellite

technology

system

Indoor navigation
Land transportation

RT: Developing countries
International collaboration
International relations



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	International trade		<i>Google</i>	USE:	Internet AND Search engines
	Trade agreements				
<i>Glossaries</i>			<i>Google Chrome</i>	USE:	Browsers
	USE:	Terminology			
Glow discharge devices			GOSAT	UF:	Greenhouse Gases
	BT:	Gas discharge devices	Observing Satellite		
Glow discharges				BT:	Ibuki Earth Observing System Greenhouse gases Satellites
	BT:	Dielectric breakdown		RT:	Carbon dioxide Methane
Glowworm swarm optimization					
	BT:	Algorithms Optimization methods			
Glucose			Government		
	BT:	Sugar		BT:	Organizations
	NT:	Continuous glucose monitoring		RT:	Governmental factors Leadership Macroeconomics Public finance
Glucose sensors				NT:	Electronic government Government policies Legislation Local government US Government Voting
	BT:	Sensors			
Glycomics					
	BT:	Molecular biomarkers			
<i>GNN</i>					
	USE:	Graph neural networks			
<i>GNP</i>			<i>Government borrowing</i>	USE:	Public finance
	USE:	Economic indicators			
<i>GNSS</i>			<i>Government expenditure</i>	USE:	Public finance
	USE:	Global navigation satellite system			
<i>Goal recognition</i>			Government policies		
	USE:	Intent recognition		BT:	Data governance Government Censorship
<i>Goggles</i>				RT:	Developing countries Digital policy Food waste Open data Public infrastructure
	USE:	Eye protection		NT:	Feed in tariff General Data Protection
Gold					
	UF:	Au			
	BT:	Metals			
	NT:	Gold alloys			
Gold alloys			Regulation		Must-carry regulations Public policy Regulation
	BT:	Gold			
	RT:	Alloying			
Goniometers			Governmental factors		
	BT:	Meters		BT:	Management
				RT:	Government



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	Legal factors		Granular superconductors	
	Social factors		BT:	Superconducting materials
NT:	Public finance		RT:	High-temperature superconductors
<i>GPR</i>			<i>Graph neural nets</i>	
USE:	Ground penetrating radar		USE:	Graph neural networks
<i>GPS</i>			Graph convolutional networks	
USE:	Global Positioning System		BT:	Convolutional neural networks
<i>GPT</i>				Graph neural networks
USE:	Generative Pre-Trained Transformer		Graph drawing	
<i>GPU</i>			BT:	Data visualization
USE:	Graphics processing units		RT:	Computational complexity
				Graph theory
Gradient methods				Knowledge representation
BT:	Mathematics			Mathematics computing
	Numerical analysis			Social sciences
	Optimization methods		Graph neural networks	
RT:	Level set		UF:	GNN
	Search methods			Graph neural nets
<i>Grain alcohol</i>			BT:	Graph theory
USE:	Ethanol			Neural networks
			RT:	Convolutional neural networks
Grain boundaries				Data visualization
BT:	Crystals			Deep learning
RT:	Conductivity			Feature extraction
	Corrosion			Pattern classification
	Electron backscatter			Recommender systems
diffraction				Supervised learning
	Grain size		NT:	Graph convolutional networks
	Thermal conductivity			
Grain size			Graph theory	
BT:	Crystals		BT:	Combinatorial mathematics
RT:	Electron backscatter			Mathematics
diffraction			RT:	Ant colony optimization
	Grain boundaries			Belief propagation
				Circuit topology
Grammar				Graph drawing
BT:	Professional			Topology
communication			NT:	Bipartite graph
	Writing			Conditional random fields
RT:	Syntactics			Directed acyclic graph
				Directed graphs
Granular computing				Graph neural networks
BT:	Programming			Network motifs
RT:	Concurrent computing			Optimal matching
	Information processing			Reachability analysis
	Quantization (signal)			Shortest path problem
				Tree graphs



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Graphene

BT: Carbon
NT: Graphene devices

Graphene devices

BT: Graphene
RT: Field effect transistors
Flexible electronics
Molecular electronics
Nanoelectronics

Graphic user interfaces

USE: Graphical user interfaces

Graphical models

BT: Modeling

Graphical user interfaces

UF: GUI
Graphic user interfaces
BT: Product development
User interfaces
NT: Avatars

Graphics

BT: Design methodology
RT: Art
Displays
Technical drawing
NT: Animation
Character generation
Computer graphics
Engineering drawings
Layout
Shape
Symbols
Virtual reality
Visualization

Graphics processing units

UF: GPU
VPU
BT: Program processors
RT: AI accelerators
Computer graphics
Hardware acceleration

Graphite

UF: Plumbago
BT: Carbon
RT: Lead

Grasping

BT: Haptic interfaces

Grasslands

UF: Prairie
Savanna
BT: Ecosystems

Gratings

UF: Optical gratings
Transient gratings
BT: Periodic structures
RT: Optical devices

Gravimeter

USE: Gravity measurement

Gravitational force

USE: Gravity

Gravitational waves

BT: Astronomy
Gravity

Gravimeter

USE: Gravity measurement

Gravity

UF: Gravitational force
BT: Force
RT: Force measurement
NT: Gravitational waves

Gravity measurement

UF: Gravimeter
Gravimeter
BT: Force measurement
RT: Astrophysics
Geophysical measurements

Gray codes

USE: Reflective binary codes

Gray matter

USE: Grey matter

Gray-scale

UF: Grayscale
BT: Image processing

Grayscale

USE: Gray-scale

Greedy algorithms

BT: Computation theory

Green (sustainability)

BT: Sustainable development



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Green buildings

BT: Construction
Green products
RT: Environmental factors
Green design

Green cleaning

BT: Green products

Green communications

UF: Green networking
BT: Communications
technology
RT: Carbon emissions
Environmental monitoring
Net zero
Renewable energy sources

Green computing

BT: Computer applications
Energy conservation
Green design
RT: Environmental factors
Sustainable development

Green design

UF: Environmental design
BT: Design methodology
RT: Energy informatics
Green buildings
NT: Ecodesign
Green computing

Green electronics

BT: Green products

Green energy

UF: Clean energy
BT: Renewable energy sources
RT: Anaerobic digestion
Environmental factors
Recycling
Sustainable development

Green function

USE: Green's function methods

Green hydrogen

BT: Hydrogen
Renewable energy sources
RT: Carbon emissions
Green manufacturing
Low carbon economy

Green jobs

BT: Employment
Environmental economics

Green manufacturing

BT: Global warming potential
Manufacturing
RT: Green hydrogen

Green networking

USE: Green communications

Green products

BT: Environmental factors
Biohazards
Pollution
NT: Green buildings
Green cleaning
Green electronics

Green transportation

BT: Environmental factors
Transportation

Green's function

USE: Green's function methods

Green's function methods

UF: Green function
Green's function
Green's functions
BT: Modeling
RT: Failure analysis
Materials reliability

Green's functions

USE: Green's function methods

Greenhouse effect

BT: Infrared heating
RT: Carbon emissions
Carbon footprint
Carbon sequestration
Environmental factors
Global warming
Low carbon economy
Pollution control
NT: Greenhouse gases

Greenhouse gas protocol

UF: GHGP
BT: Greenhouse gases



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Greenhouse gases

- BT: Gases
Greenhouse effect
RT: Carbon dioxide
Carbon dioxide equivalent
Chlorofluorocarbons
Emissions factor
Methane
Net zero
Nitrous oxide
NT: Decarbonization
GOSAT
Global warming potential
Greenhouse gas protocol

Greenhouse Gases Observing Satellite
USE: GOSAT

Greenhouses

- BT: Agriculture
Production facilities
RT: Crops

Greenwashing

BT: Corporate social responsibility

Grey codes

USE: Reflective binary codes

Grey matter

- UF: Gray matter
BT: Central nervous system

Greywater

- BT: Wastewater treatment
RT: Blackwater

Grid computing

- UF: Grid maze
BT: Metacomputing
RT: Cloud computing

Grid decarbonization

BT: Energy conservation

Grid following

- UF: GFL
Grid-following
BT: Inverters
RT: Synchronous generators

Grid forming

- UF: GFM
Grid-forming
BT: Inverters
Power conversion
RT: Inverter-based resource
Synchronous generators

Grid maze

USE: Grid computing

Grid-following

USE: Grid following

Grid-forming

USE: Grid forming

Grinding machines

BT: Machine tools

Grippers

UF: Microgrippers
Worms
BT: Materials handling equipment
RT: End effectors
Loading
Soft robotics

Gross domestic product

USE: Economic indicators

Gross national product

USE: Economic indicators

Ground penetrating radar

- UF: GPR
Ground-penetrating radar
BT: Imaging
Radar
RT: Buried object detection
Radar detection
Radar imaging
Synthetic aperture radar
Ultra wideband radar

Ground source heat pumps

USE: Heat pumps

Ground state

USE: Stationary state



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Ground support

UF: Aerospace ground equipment
 BT: Aerospace ground services
 RT: Aerospace control
 RT: Aircraft
 RT: Military equipment
 RT: Missiles
 RT: Navigation
 RT: Rockets
 RT: Space vehicles

Ground temperature

USE: Land surface temperature

Ground transportation

USE: Land transportation

Ground vehicles

USE: Land vehicles

Ground-penetrating radar

USE: Ground penetrating radar

Grounding

UF: earthing
 BT: Electrical safety
 RT: Circuit stability
 RT: Electric shock
 RT: Fault currents
 RT: Power system protection
 RT: Protection

Groundwater

BT: Water

Group technology

BT: Production
 RT: Product design
 RT: Production control

Groupware

USE: Collaborative software

GSM

UF: Global System for Mobile Communications
 BT: Cellular technology
 RT: Wireless communication
 RT: Dual band
 RT: Roaming
 NT: GSM-R

GSM-R

UF: GSM-Railway
 BT: GSM
 RT: Railway communication

GSM-Railway

USE: GSM-R

GTEM cells

USE: TEM cells

GUI

USE: Graphical user interfaces

Guided electromagnetic wave propagation

USE: Waveguide theory

Guidelines

BT: Standardization
 RT: IEEE publishing
 RT: Publishing
 NT: Digital policy

Guideways (mechanical)

USE: Mechanical guides

Gunn devices

UF: Transferred electron devices
 BT: Semiconductor devices

Guns

BT: Weapons

Gunshot detection systems

BT: Geographic information systems
 RT: Sensor systems

GWP

USE: Global warming potential

Gymnosperms

USE: Non-flowering plants

Gynaecology

USE: Gynecology

Gynecology

UF: Gynaecology
 BT: Medical specialties

Gyrators

BT: Active circuits
 RT: Active inductors



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	Ferrite devices	Habitats	BT: Ecology
<i>Gyroklystrons</i>		NT: Habitat loss	
USE: Klystrons		Hackathon	
Gyromagnetism		BT: Collaboration	
BT: Magnetics		Computer hacking	
RT: Faraday effect		RT: Agile software development	
Ferrites			
Gyrotropism		<i>Hacker</i>	
		USE: Computer hacking	
<i>Gyros</i>		<i>Hacking</i>	
USE: Gyroscopes		USE: Computer crime	
Gyroscopes		<i>Hacks</i>	
UF: Gyros		USE: Computer hacking	
Non-gyroscopes			
Nongyroscopes		<i>Haematology</i>	
BT: Level control		USE: Hematology	
RT: Laser applications			
Ring lasers		<i>Haemorrhaging</i>	
Gyrotrons		USE: Hemorrhaging	
BT: Masers		Hafnium	
RT: Electron beams		UF: Hf	
Gyrotropism		BT: Chemical elements	
BT: Magneto optic effects		Metals	
RT: Faraday effect		RT: Nuclear physics	
Gyromagnetism		NT: Hafnium compounds	
H infinity control		Hafnium compounds	
UF: H-infinity control		BT: Hafnium	
BT: Optimization methods		RT: Alloying	
RT: Closed loop systems		NT: Hafnium oxide	
Control systems		Hafnium oxide	
Intelligent control		BT: Hafnium compounds	
Optimal control		Hair	
<i>H-infinity control</i>		BT: Integumentary system	
USE: H infinity control		NT: Eyebrows	
<i>H.264</i>		Eyelashes	
USE: MPEG 4 Standard		Hair follicle	
<i>H2O</i>		Hair follicle	
USE: Water		BT: Hair	
Habitat loss		<i>Half duplex system</i>	
BT: Environmental degradation		USE: Half-duplex system	
Environmental factors		Half-duplex system	
Habitats		UF: HDX	
		Half duplex system	
		Semi-duplex systems	



January 2026 IEEE Thesaurus

BT: Duplex communication
systems

Half-wave plates

USE: Optical retarders

Hall effect

UF: Hall mobility
BT: Magnetoelectric effects
RT: Hall effect devices

Hall effect devices

BT: Semiconductor devices
RT: Hall effect

Hall mobility

USE: Hall effect

Ham radios

BT: Radio communication
equipment

Hamming distances

BT: Information theory

Hamming weight

BT: Information theory

HAMR

USE: Heat-assisted magnetic
recording

Hand held computers

USE: Personal digital devices

Hand tools

BT: Tools
RT: Machine tools

Handheld computers

USE: Personal digital devices

Handicapped aids

USE: Assistive technologies

Handover

BT: Communication switching
Data transfer
RT: Cellular networks
Communication system

signaling

Satellite communications

Hands

BT: Limbs

Handsets

USE: Telephone sets

Handwriting recognition

UF: Signature detection
Signature verification
Written character

recognition

Written characters
Written-character

recognition

BT: Identification of persons
Pattern recognition
RT: Biometrics
NT: Forgery

Haptic interfaces

UF: Haptic systems
Haptics
BT: Computer interfaces
RT: Braille
Electronic skin
Modeling
Touch sensitive screens
NT: Data gloves
Force feedback
Grasping
Tactile Internet

Haptic systems

USE: Haptic interfaces

Haptics

USE: Haptic interfaces

HAR

USE: Human activity recognition

Hard amorphous carbon

USE: Diamond-like carbon

Hard disc drives

USE: Hard disks

Hard discs

USE: Hard disks

Hard disk drives

USE: Hard disks

Hard disks

UF: Hard disc drives
Hard discs
Hard disk drives
Hard-disc drives



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	Hard-disk drives	BT:	Hardware
	Magnetic memory	RT:	Security
	Giant magnetoresistance	RT:	Communication system
<i>Hard-disc drives</i>		security	
USE:	Hard disks		Computer crime
			Computer security
<i>Hard-disk drives</i>			Information security
USE:	Hard disks		Mobile security
			Network security
<i>Hardsuit</i>			Physical layer security
USE:	Wearable robots		Power system security
			Security management
Hardware		Hardware-in-the-loop simulation	
UF:	Computer hardware	UF:	HIL simulation
BT:	Computers and information	BT:	Simulation
processing		RT:	Aerospace control
RT:	Firewalls (computing)		Control engineering
	Ports (computers)	computing	
NT:	Hardware acceleration		Control system synthesis
	Hardware security		Embedded systems
	Input devices		Real-time systems
	Open source hardware		Testing
	Reconfigurable devices		Vehicle dynamics
	Wireless access points		
Hardware acceleration		Harmonic analysis	
UF:	3D accelerators	UF:	Harmonics
	Accelerated computing	BT:	Mathematics
	Cryptographic accelerators		Signal analysis
	Hardware accelerators	RT:	Fast Fourier transforms
BT:	Computer performance		Fourier transforms
	Hardware		Spectral analysis
RT:	Central Processing Unit		Wavelet transforms
	Graphics processing units		
<i>Hardware accelerators</i>		Harmonic distortion	
USE:	Hardware acceleration	BT:	Nonlinear distortion
		RT:	Power conversion
<i>Hardware description languages</i>		harmonics	
USE:	Hardware design	NT:	Harmonics suppression
languages			Total harmonic distortion
Hardware design languages		Harmonic filters	
UF:	HDL	BT:	Filters
	HDVL		
	Hardware description	<i>Harmonic generation</i>	
languages		USE:	Frequency conversion
	IEEE 1364	<i>Harmonics</i>	
BT:	Computer languages	USE:	Harmonic analysis
RT:	Design automation	Harmonics suppression	
NT:	VHDL	BT:	Harmonic distortion
Hardware security		<i>Harmonised index of consumer prices</i>	
UF:	Security of hardware	USE:	Economic indicators



January 2026 IEEE Thesaurus

Harmonized index of consumer prices

USE: Economic indicators

Hash functions

BT: Algorithms
RT: Cryptography
Encoding
NT: Cryptographic hash

function

Hashtag

USE: Tagging

Hate speech

BT: Speech

Hazardous areas

BT: Hazards
RT: Accidents
Explosions
Fires
Hazardous materials
Industrial accidents
Protection
Radioactive pollution
Radioactive waste
Safety
Surveillance

Hazardous materials

UF: Hazmat
BT: Hazards
Materials
RT: Chemical hazards
Flammability
Hazardous areas
Radioactive waste
Toxicology

Hazards

BT: Safety
RT: Contamination
Disasters
Explosion protection
Landslides
Occupational stress
Pest control
Rescue robots
Soil remediation
Working environment noise
NT: Biohazards
Chemical hazards
Explosions
Fires
Flammability

Floods
Hazardous areas
Hazardous materials
Toxicology

Hazmat

USE: Hazardous materials

HbbTV Standards

BT: ETSI Standards
RT: Digital TV
TV

HBT

USE: Heterojunction bipolar transistors

HCCI engines

USE: Internal combustion engines

HCI

USE: Human computer interaction

HD

USE: High definition video

HD video

USE: High definition video

HDL

USE: Hardware design languages

HDMI

UF: High Definition Multimedia Interface
High-Definition Multimedia Interface

BT: Audio-visual systems
Computer interfaces
RT: Audio compression
Digital audio broadcasting
Multimedia communication

HDTV

UF: ATV
Advanced TV
EDTV
Extended definition TV
High definition television
High-definition TV
IDTV
Improved definition TV



January 2026 IEEE Thesaurus

	BT:	Digital TV	<i>Headsets</i>	
	NT:	UHDTV	USE:	Headphones
<i>HDVL</i>			<i>Health (occupational)</i>	
	USE:	Hardware design	USE:	Occupational health
languages				
<i>HDX</i>			Health and safety	
	USE:	Half-duplex system	BT:	Safety
			RT:	Environmental factors
Head			NT:	Occupational health
	BT:	Body regions		Occupational safety
	RT:	Auditory system		Personal protective
		Brain	equipment	
		Otorhinolaryngology	<i>Health care</i>	
		Visual systems	USE:	Medical services
	NT:	Cranium	<i>Health informatics</i>	
		Ear	USE:	Bioinformatics
		Faces	Health information management	
		Forehead	BT:	Medical services
		Lips	<i>Health physics</i>	
		Mouth	USE:	Radiation monitoring
		Nose	<i>Healthcare</i>	
		Scalp	USE:	Medical services
		Skull	<i>Hearing</i>	
<i>Head sets</i>			USE:	Auditory system
USE:		Headphones	Hearing aids	
			BT:	Sensory aids
Head-mounted displays			RT:	Auditory system
UF:		Head-worn displays		Speech enhancement
		Helmet mounted displays	Heart	
BT:		Displays	BT:	Cardiovascular system
		Human computer	RT:	Cardiac catheterization
interaction				Cardiology
			NT:	Cardiac function
Head-up displays				Fetal heart
UF:		Heads up displays		Heart rate
BT:		Displays		Heart transplantation
		Human computer		Heart valves
interaction				Heart ventricles
<i>Head-worn displays</i>			<i>Heart arrest</i>	
USE:		Head-mounted displays	USE:	Cardiac arrest
Headphones			<i>Heart attack</i>	
UF:		Earphones	USE:	Myocardial infarction
		Head sets		
		Headsets		
BT:		Audio systems		
<i>Heads up displays</i>				
USE:		Head-up displays		



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Heart beat

UF: Heartbeat
BT: Heart rate
RT: Arrhythmia

BT: Heart

Heart ventricles

BT: Heart

Heart cath

USE: Cardiac catheterization

Heartbeat

USE: Heart beat

Heart failure

USE: Cardiovascular diseases

Heat engines

BT: Engines
NT: Steam engines
Stirling engines

Heart rate

UF: HR
BT: Heart
NT: Bradycardia
Fetal heart rate
Heart beat
Heart rate detection
Heart rate interval
Heart rate measurement
Heart rate variability
Tachycardia

Heat islands

USE: Thermal pollution

Heat maps

BT: Data visualization

Heat networks

USE: District heating

Heart rate detection

BT: Heart rate

Heat pipes

BT: Heating systems
RT: Cooling
Heat transfer

Heart rate interval

BT: Heart rate

Heat pumps

UF: Ground source heat pumps
BT: Pumps
RT: Refrigerants

Heart rate measurement

BT: Heart rate
RT: Phonocardiography

Heat recovery

UF: Industrial heat recovery
BT: Heating systems
RT: Boilers
Thermal engineering

Heart rate variability

UF: HRV
BT: Heart rate

Heart transplantation

UF: Cardiac transplants
Heart transplants
BT: Heart
Organ transplantation

Heat sinks

UF: Heatsinks
BT: Cooling

Heart transplants

USE: Heart transplantation

Heat transfer

BT: Thermal conductivity
RT: Heat pipes
NT: Convection

Heart valves

UF: Aortic semilunar valves
Atrioventricular valves
Bicuspid valves
Mitral valves
Pulmonary semilunar

Heat treatment

BT: Materials processing
RT: Curing
Firing
Foundries
Kilns
Smelting
Thermal factors

valves

Semilunar valves
Tricuspid valves



January 2026 IEEE Thesaurus

processes
 NT: Thermomechanical
 Annealing
 Calcination

Heat-assisted magnetic recording

UF: HAMR
 BT: Magnetic recording

Heating systems

BT: Temperature control
 RT: Entropy
 Furnaces
 HVAC
 High-temperature

techniques

NT: Laser applications
 Rapid thermal processing
 Thermal engineering
 Boilers
 Cogeneration
 District heating
 Electromagnetic heating
 Heat pipes
 Heat recovery
 Induction heating
 Infrared heating
 Resistance heating
 Solar heating
 Space heating
 Thermal energy
 Trigeneration
 Water heating

Heating, ventilation, and air conditioning
 USE: HVAC

Heatsinks

USE: Heat sinks

Heavily tailed distribution

USE: Heavily-tailed distribution

Heavily-tailed distribution

UF: Heavily tailed distribution
 BT: Probability distribution
 RT: Kurtosis

Hebb's methods

USE: Hebbian theory

Hebb's rule

USE: Hebbian theory

Hebbian learning

USE: Hebbian theory

Hebbian principle

USE: Hebbian theory

Hebbian theory

UF: Hebb's methods
 Hebb's rule
 Hebbian learning
 Hebbian principle
 BT: Artificial neural networks

Height measurement

BT: Measurement

Helical antennas

BT: Antennas
 RT: Electromagnetic

waveguides

Telecommunications
 Transmission lines
 VHF circuits
 Waveguide components

Helical gears

USE: Gears

Helicopters

BT: Vertical takeoff and landing

aircraft

RT: Military systems
 Urban air mobility
 NT: Quadrotors

Helioseismology

BT: Asteroseismology
 Nuclear and plasma

sciences

Helium

BT: Chemical elements
 Gases

Helmet mounted displays

USE: Head-mounted displays

Helmets

BT: Personal protective

equipment

RT: Safety devices

Hematology

UF: Haematology
 BT: Medical specialties



January 2026 IEEE Thesaurus

RT: Blood flow
Blood platelets
Blood vessels
Coagulation

NT: Integrated optoelectronics
Semiconductor devices
Double heterojunction

Hemodynamics

UF: Hemorheology
BT: Blood flow

Hemorheology

USE: Hemodynamics

Hemorrhaging

UF: Bleeding
Haemorrhaging
BT: Medical conditions
RT: Blood flow

HEMTs

UF: Heterostructure FETs
High electron mobility
transistors
High electron-mobility
transistors
High-electron mobility
transistors
High-electron-mobility
transistors
BT: Field effect transistors
RT: MODFETs
NT: D-HEMTs
DH-HEMTs
PHEMTs
mHEMTs

Hepatectomy

BT: Medical treatment
Surgery

Hermetic seals

BT: Seals

Hetero-nanocrystal memory

BT: Single electron memory

Heterogeneous integration

USE: Multichip modules

Heterogeneous networks

BT: Computer networks

Heterojunction bipolar transistors

UF: HBT
BT: Transistors
RT: Heterojunctions

bipolar transistors

Heterojunctions

BT: Junctions
RT: Heterojunction bipolar
transistors

Heterostructure FETs

USE: HEMTs AND
MODFETs

Heuristic algorithms

UF: Dynamic algorithms
BT: Algorithms
RT: Metaheuristics

HEV

USE: Hybrid electric vehicles

HEVC

USE: High efficiency video coding

Hf

USE: Hafnium

HF radar

USE: High frequency radar

HFC

USE: Hybrid fiber coaxial cables

Hg

USE: Mercury (metals)

Hidden Markov models

BT: Modeling
RT: Markov processes
Pattern recognition

Hierarchical learning

USE: Deep learning

Hierarchical systems

BT: Systems engineering and
theory
NT: Multilevel systems

High Definition Multimedia Interface

USE: HDMI

High definition television

USE: HDTV



January 2026 IEEE Thesaurus

High definition video

UF: HD
HD video
BT: Video recording
NT: Ultra-high definition video

Real-time systems
Synchrotrons
NT: Linear particle accelerator

High dimensional data

UF: High-dimensional data
BT: Data models
Machine learning
RT: Genomics
Natural language

processing

High dynamic range

UF: Expanded dynamic range
Extended dynamic range
Wide dynamic range
BT: Dynamic range
RT: Photography

High efficiency video coding

UF: HEVC
High-efficiency video

coding

BT: Video coding
RT: MPEG 4 Standard

High electron mobility transistors

USE: HEMTs

High electron-mobility transistors

USE: HEMTs

High energy physics

UF: Particle physics
BT: Physics
RT: CERN

High energy physics instrumentation computing

BT: Computer applications
Instrumentation and

measurement

sciences

Nuclear and plasma

RT: Data acquisition
Elementary particles
Nuclear electronics
Particle measurements
Particle tracking
Position sensitive particle

detectors

Proton effects
Radiation effects

High frequency

BT: Radio frequency

High frequency radar

UF: HF radar
BT: Radar

High frequency transformers

USE: High-frequency transformers

High intensity discharge lamps

BT: Discharge lamps
RT: Arc discharges
Electrical ballasts
Emergency lighting
Light sources
Lighting
Lighting control

High K

USE: High-k dielectric materials

High level languages

BT: Computer languages
RT: Page description languages
NT: Java
Linux
Parallel languages

High level synthesis

BT: Circuit synthesis
RT: Programmable logic

devices

High performance computing

UF: HPC
High-performance

computing

BT: Computers and information

processing

NT: Exascale computing

High power amplifiers

BT: Power amplifiers
RT: Gate drivers

High power fiber lasers

UF: HPFL
High-power fiber lasers
BT: Fiber lasers



January 2026 IEEE Thesaurus

High power microwave generation

UF: HPM generation
High-power microwave
generation
BT: Microwave generation

High resolution imaging

USE: High-resolution imaging

High school engineering

USE: Pre-college engineering

High speed electronics

USE: High-speed electronics

High speed integrated circuits

USE: High-speed integrated
circuits

High speed networking

USE: High-speed networks

High speed networks

USE: High-speed networks

High speed optical methods

USE: High-speed optical
techniques

High speed optical techniques

USE: High-speed optical
techniques

High speed rail transportation

USE: High-speed rail
transportation

High speed techniques

USE: High-speed electronics

High T_c superconductors

USE: High-temperature
superconductors

High T_c superconductors

USE: High-temperature
superconductors

High temperature superconductors

USE: High-temperature
superconductors

High voltage

USE: High-voltage techniques

High-Definition Multimedia Interface

USE: HDMI

High-definition TV

USE: HDTV

High-dimensional data

USE: High dimensional data

High-efficiency video coding

USE: High efficiency video coding

High-electron mobility transistors

USE: HEMTs

High-electron-mobility transistors

USE: HEMTs

High-frequency transformers

UF: High frequency
transformers
BT: Transformers

High-K

USE: High-k dielectric materials

High-k dielectric materials

UF: High K
High-K
BT: Dielectric materials
RT: Semiconductor materials

High-k gate dielectrics

BT: Dielectric constant
Semiconductor device
manufacture

High-performance computing

USE: High performance
computing

High-power fiber lasers

USE: High power fiber lasers

High-power microwave generation

USE: High power microwave
generation

High-resolution imaging

UF: High resolution imaging
BT: Image resolution



January 2026 IEEE Thesaurus

High-resolution transmission electron microscopy

UF: HRTEM
BT: Transmission electron microscopy

High-speed electronics

UF: High speed electronics
High speed techniques
High-speed techniques
BT: Communications
technology
NT: High-speed integrated circuits
High-speed networks
Ultrafast electronics

High-speed integrated circuits

UF: High speed integrated circuits
BT: High-speed electronics
RT: Integrated circuit
technology
Microcontrollers

High-speed networking

USE: High-speed networks

High-speed networks

UF: High speed networking
High speed networks
High-speed networking
BT: High-speed electronics
RT: Long Term Evolution

High-speed optical methods

USE: High-speed optical techniques

High-speed optical techniques

UF: High speed optical methods
High speed optical techniques
High-speed optical methods
BT: Optical design techniques
RT: Laser pulses
Light fidelity

High-speed rail transportation

UF: High speed rail
transportation
BT: Rail transportation
RT: Magnetic levitation vehicles

High-speed techniques

USE: High-speed electronics

High- T_c superconductors

USE: High-temperature superconductors

High- T_c superconductors

USE: High-temperature superconductors

High-temperature effects

USE: Thermal factors

High-temperature superconductors

UF: HTS
High T_c superconductors
High T_c superconductors
High temperature superconductors
High- T_c superconductors
High- T_c superconductors
Superconductors (high temperature)
BT: Superconducting materials
RT: Ceramics
Granular superconductors
Persistent currents
Superconducting devices
Superconducting films
Superconducting transition temperature
Surface impedance
Surface resistance
NT: Yttrium barium copper oxide

High-temperature techniques

BT: Industry applications
RT: Heating systems
NT: Rapid thermal processing

High-voltage techniques

UF: High voltage
BT: Power engineering
RT: Power electronics
Pulsed power systems

Higher order statistics

BT: Statistics
RT: Differential equations

Highways

USE: Road transportation



January 2026 IEEE Thesaurus

HIL simulation

USE: Hardware-in-the-loop
simulation

Hilbert space

BT: Euclidean distance

Hilbert?Huang transforms

USE: Empirical mode
decomposition

Hindbrain

UF: Rhombencephalon
BT: Brain
RT: Forebrain
Midbrain

Hindcasting

BT: Probability
RT: Forecasting
Mathematical models
Prediction methods

Hinges

USE: Fasteners

Hip

BT: Extremities

Hip joint replacements

USE: Prosthetics

Hippocampus

BT: Temporal lobe
RT: Alzheimer's disease

Histograms

BT: Statistics

Histology

USE: Histopathology

Histopathology

UF: Histology
BT: Pathology

History

BT: Humanities

HITL

USE: Human in the loop

HIV

USE: Human immunodeficiency
virus

Hobbing machines

BT: Machining

Hockey

USE: Sports

Hoists

USE: Lifting equipment

Hole carriers

USE: Charge carrier processes

Holey fibers

UF: Holey fibres
BT: Photonic crystal fibers

Holey fibres

USE: Holey fibers

Hollow waveguides

BT: Electromagnetic
waveguides
NT: Liquid waveguides

Holmium

BT: Chemical elements

Holographic optical components

BT: Optical devices
RT: Holography

Holography

BT: Imaging
RT: Holographic optical
components
Image reconstruction
Laser applications
Photorefractive materials

Home appliances

UF: Appliances
Domestic appliances
Domestic induction
appliances
Household appliances
BT: Consumer products
NT: Gas appliances
Microwave ovens
Ovens
Refrigerators
Washing machines



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Home automation

UF: Home networks
BT: Consumer electronics
RT: Automation
Fall detection
Service robots
NT: Portable media players
Refrigerators
Smart homes
Washing machines

Home computing

BT: Consumer electronics
RT: Computers and information processing
Firewire
Microcomputers

Home networks

USE: Home automation

Home phone

USE: Landline

Home shopping

USE: Electronic commerce

Homeostasis

BT: Biology
Control systems

Homomorphic encryption

BT: Encryption
RT: Privacy
Public key cryptography

Homopolar machines

BT: DC machines

Honey pot (computing)

BT: Computer security

Honeycomb structures

BT: Structural shapes
RT: Lightweight structures
Sandwich structures
Structural panels
Thin wall structures

Hopfield networks

USE: Hopfield neural networks

Hopfield neural networks

UF: Hopfield networks
BT: Recurrent neural networks

Hormones

USE: Biochemistry

Horn antennas

BT: Antennas

Horses

BT: Animals

Horticulture

BT: Industries
NT: Hydroponics

Hoses

BT: Mechanical products
RT: Automotive components
Rubber products

Hospitals

BT: Medical services
Medical treatment
RT: Biomedical engineering

Hot carrier effects

BT: Hot carriers

Hot carrier injection

UF: Hot-carrier injection
BT: Hot carriers
NT: Channel hot electron

injection

Secondary generated hot

electron injection

Substrate hot electron

injection

Hot carriers

BT: Charge carriers
RT: Semiconductor devices
NT: Hot carrier effects
Hot carrier injection

Hot-carrier injection

USE: Hot carrier injection

Household appliances

USE: Home appliances

HPC

USE: High performance computing

HPFL

USE: High power fiber lasers



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HPM generation

USE: High power microwave generation

HR

USE: Heart rate

HRTEM

USE: High-resolution transmission electron microscopy

HRV

USE: Heart rate variability

HTML

BT: Markup languages

HTS

USE: High-temperature superconductors

HTTP

UF: Hypertext Transfer Protocol
BT: Hypertext systems
Protocols
RT: Data communication

Huffman coding

BT: Data compression
Entropy coding
RT: Algorithms
Communication systems
Multimedia communication
Multimedia databases
Multimedia systems
Symbols

Human action recognition

USE: Human activity recognition

Human activity recognition

UF: HAR
Human action recognition
Human motion recognition
BT: Activity recognition
RT: Biomechanics
Body area networks
Body sensor networks
Fall detection
Image motion analysis
Image recognition
Learning (artificial

intelligence)

Pattern recognition

Human anatomy

BT: Anatomy

Human augmentation

UF: Human enhancement
BT: Augmented reality
Human factors

Human cloning

USE: Cloning

Human computer interaction

UF: HCI
Human machine interaction
Human-centered computing
Human-computer

interaction

Human-computer interfaces
Human-machine interaction
User friendliness

BT: User interfaces
RT: Adaptive learning
Cyber-physical systems
Digital humans
Driver behavior
Fifth Industrial Revolution
Human factors

Human vehicle systems
Human-machine systems
Immersive audio
User experience

NT: Affective computing
Chatbots
Extended reality
Gaze tracking
Head-mounted displays
Head-up displays
Human in the loop
Immersive experience
Spatial computing
Telepresence
Telexistence

Human disease markers

USE: Biomarkers

Human engineering

USE: Ergonomics

Human enhancement

USE: Human augmentation



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Human factors

UF: Human factors engineering
Stress (psychological)
BT: Systems, man, and
cybernetics
RT: Aerospace biophysics
Affective computing
Androids
Anthropometry
Behavioral sciences
Chatbots
Cognitive science
Digital humans
Digital intelligence
Ergonomics
Human computer

interaction

Human-machine systems
Inclusive language
Persuasive systems
Problem-solving
Productivity
Social engineering

(security)

Socioeconomics
Telerobotics
User experience
NT: Anthropomorphism
Human augmentation
Human image synthesis
Human intelligence
Livelihood
Mental health
Pedestrians
Technology acceptance

model

Human factors engineering

USE: Ergonomics AND
Human factors

Human image synthesis

BT: Human factors
Image synthesis
RT: Films
Photorealism

Human immunodeficiency virus

UF: HIV
BT: Diseases
RT: Acquired immune
deficiency syndrome

Human in the loop

UF: HITL
Human-in-the-loop
BT: Human computer
interaction
Simulation
RT: Computer simulation
Modeling

Human intelligence

BT: Cognitive science
Human factors
NT: Digital intelligence
Hyper-intelligence

Human machine interaction

USE: Human computer
interaction

Human machine interface

USE: Human-machine systems

Human motion recognition

USE: Human activity recognition

Human resource management

BT: Management
RT: Industrial psychology
NT: Appraisal
Continuing professional

development

Employee welfare
Employment
Equal opportunities
Incentive schemes
Job specification
Labor resources
Leadership
Multiskilling
Personnel
Recruitment
Remuneration
Retirement
Termination of employment
Unemployment

Human robot interaction

USE: Human-robot interaction

Human vehicle systems

UF: Human-vehicle interaction
Human-vehicle systems
BT: User interfaces
RT: Driver behavior



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interaction	Human computer	NT:	Wearable robots Social robots
Human voice			<i>Human-vehicle interaction</i>
BT:	Speech processing	USE:	Human vehicle systems
<i>Human-centered computing</i>			<i>Human-vehicle systems</i>
USE:	Human computer	USE:	Human vehicle systems
interaction			
<i>Human-computer interaction</i>			Humanitarian activities
USE:	Human computer	UF:	Humanitarian aid
interaction		BT:	IEEE Corporate activities
<i>Human-computer interfaces</i>			<i>Humanitarian aid</i>
USE:	Human computer	USE:	Humanitarian activities
interaction			Humanities
<i>Human-in-the-loop</i>		UF:	Dance Drama Humanities data processing
USE:	Human in the loop	BT:	Education
<i>Human-machine interaction</i>		RT:	Anthropology Computer applications Journalism Museums Philosophical
USE:	Human computer		considerations
interaction		NT:	Archeology Art History Music Natural languages Social sciences
Human-machine systems			
UF:	Cyborgs Human machine interface Man machine systems Man-machine systems		
BT:	Systems, man, and		
cybernetics			
RT:	Androids Cybernetics Ergonomics Human computer		
interaction			
generation	Human factors Natural language		
	Persuasive systems Tactile Internet		
NT:	Digital intelligence Extended reality Interactive systems		
<i>Human-robot collaboration</i>			
USE:	Human-robot interaction		
Human-robot interaction			
UF:	Human robot interaction Human-robot collaboration		
BT:	User interfaces		
RT:	Admittance control Tactile Internet		



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Humidity

BT: Meteorology
RT: Humidity control
Humidity measurement
Trees - insulation
NT: Humidity effects

Humidity control

BT: Moisture control
RT: Humidity
Humidity effects

Humidity effects

BT: Humidity
RT: Humidity control
Moisture control

Humidity measurement

BT: Moisture measurement
RT: Humidity

Hurricanes

BT: Cyclones

HVAC

UF: Heating, ventilation, and air conditioning
BT: Thermal variables control
RT: Air conditioning
Cooling
Heating systems
Ventilation

HVDC transmission

BT: Power transmission
RT: DC distribution systems
Voltage-source converters

Hybrid automobiles

USE: Hybrid electric vehicles

Hybrid cars

USE: Hybrid electric vehicles

Hybrid electric vehicles

UF: HEV
Hybrid automobiles
Hybrid cars
BT: Electric vehicles
RT: Battery powered vehicles
Charging stations
Energy storage
Internal combustion

engines

Traction motors

Vehicle-to-grid
NT: Plug-in hybrid electric vehicles

Hybrid fiber coaxial cables

UF: HFC
Hybrid fibre coaxial cables
BT: Coaxial cables

Hybrid fibre coaxial cables

USE: Hybrid fiber coaxial cables

Hybrid integrated circuits

BT: Circuits
Integrated circuits
RT: Thick film circuits
Thin film circuits

Hybrid intelligent systems

BT: Fuzzy systems
RT: Intelligent systems

Hybrid junctions

BT: Junctions
RT: Directional couplers

Hybrid learning

UF: Blended learning
Distributed learning
BT: Computer aided instruction
Learning systems
RT: Courseware
Distance learning
Education
Educational courses
Educational institutions
Internet
Mobile learning

Hybrid office

USE: Remote working

Hybrid power systems

BT: Power systems
RT: Distributed power generation
Photovoltaic systems

Hybrid working

USE: Remote working

Hydraulic actuators

BT: Actuators
RT: Hydraulic drives



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Hydraulic diameter

BT: Fluid flow
NT: Microchannels

Hydraulic drives

BT: Drives
RT: Hydraulic actuators

Hydraulic equipment

BT: Hydraulic systems
RT: Water pumps
NT: Valves

Hydraulic fluids

UF: Hydraulic liquids
Hydraulic oils
BT: Fluids
Hydraulic systems
RT: Production materials

Hydraulic fracking

USE: Fracking

Hydraulic liquids

USE: Hydraulic fluids

Hydraulic oils

USE: Hydraulic fluids

Hydraulic systems

UF: Hydraulics
BT: Machinery
RT: Fluid flow
Irrigation
NT: Electrohydraulics
Hydraulic equipment
Hydraulic fluids

Hydraulic turbines

BT: Turbines
RT: Hydroelectric power

generation

Hydraulics

USE: Hydraulic systems

Hydrocarbon reservoirs

BT: Hydrocarbons

Hydrocarbons

UF: Oil sands
Oil shale
BT: Organic chemicals
Petroleum
RT: Natural gas

NT: Hydrocarbon reservoirs

Hydrocephalus

BT: Brain
Medical conditions

Hydrochemistry

USE: Geochemistry

Hydrodynamics

UF: Smoothed particle
hydrodynamics
BT: Dynamics
Mechanical factors
RT: Fluid dynamics
Fluid flow
Microfluidics
Water
NT: Electrohydrodynamics
Magnetohydrodynamics

Hydroelectric power

USE: Hydroelectric power
generation

Hydroelectric power generation

UF: Hydroelectric power
Hydroelectricity
Hydropower
Hydroelectric
Water energy
Water power
BT: Power generation
Renewable electricity
RT: Dams
Hydraulic turbines
NT: Hydroelectric-thermal
power generation
Microhydro power
Picohydro power
Wave energy conversion

Hydroelectric-thermal power generation

BT: Hydroelectric power
generation

Hydroelectricity

USE: Hydroelectric power
generation

Hydrogels

BT: Polymers
RT: Drug delivery
Tissue engineering



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Hydrogen

BT: Chemical elements
Gases
RT: Water splitting
NT: Deuterium
Green hydrogen

Biotechnology
Catalysis
Enzymes
Hydrophobicity
Solvation

Hydrogen chloride

USE: Chlorine compounds

Hydrogen fluoride

BT: Fluorine compounds

Hydrogen powered vehicles

BT: Vehicles

Hydrogen storage

BT: Energy storage

Hydrography

BT: Geoscience
RT: Bathymetry
Oceanography

Hydrologic measurements

BT: Hydrology
RT: Fluid flow measurement
Geophysics
Hydrological techniques
Oceanographic techniques
Water

Hydrological techniques

BT: Hydrology
RT: Geoengineering
Geophysics
Geoscience
Hydrologic measurements
NT: Fracking

Hydrology

BT: Fluid flow
RT: Water
Wetlands
NT: Aquifers
Floods
Hydrologic measurements
Hydrological techniques
Ocean waves
Stormwater
Watersheds

Hydrolysis

BT: Chemical reactions
RT: Biochemistry

Hydromagnetics

USE: Magnetohydrodynamics

Hydrometers

BT: Density measurement

Hydrophobicity

BT: Chemical processes
Measurement
RT: Hydrolysis

Hydrophones

BT: Microphones
Underwater equipment
RT: Echo sounders
Sonar detection

Hydroponics

BT: Horticulture

Hydropower

USE: Hydroelectric power
generation

Hydrosphere

USE: Aquatic ecosystems

Hyper intelligent systems

USE: Hyper-intelligent systems

Hyper ledger

USE: Distributed ledger

Hyper-intelligence

UF: Super intelligence
Superintelligence
BT: Human intelligence

Hyper-intelligent systems

UF: Hyper intelligent systems
Hyperintelligent systems
BT: Intelligent systems

Hypercomplex

BT: Mathematics

Hypercubes

BT: Multiprocessor
interconnection



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RT: Computer networks	Hyperthermia	BT: Medical conditions
<i>Hyperdermic needles</i>		Medical treatment
USE: Hypodermic needles		RT: Electromagnetic heating
<i>Hyperintelligent systems</i>	<i>Hypervisors</i>	USE: Virtual machine monitors
USE: Hyper-intelligent systems		
<i>Hyperledger</i>	Hypodermic needles	UF: Hyperdermic needles
USE: Distributed ledger		BT: Biomedical equipment
<i>Hyperlinks</i>	Hypotension	UF: Low blood pressure
USE: Hypertext systems		BT: Blood pressure
Hypermedia		RT: Hypertension
BT: Multimedia communication		
Hyperparameter optimization	Hypothalamus	BT: Brain
BT: Machine learning		RT: Central nervous system
Hypersonic vehicles	Hypoxia	BT: Medical conditions
BT: Vehicles		RT: Oxygen
RT: Aerospace control	<i>Hydroelectric</i>	USE: Hydroelectric power
Missiles		generation
Space vehicles	Hysteresis	BT: Materials science and
Hyperspectral imaging		technology
BT: Hyperspectral sensors		RT: Damping
Hyperspectral sensors		Magnetic hysteresis
BT: Remote sensing		Magnetization processes
RT: Geologic measurements		Spin valves
Military aircraft	Hysteresis motors	BT: AC motors
Military communication		Motors
Military satellites		Rotating machines
Mining industry		Synchronous machines
Submillimeter wave		Synchronous motors
measurements	<i>I/O programs</i>	USE: Input-output programs
NT: Wavelength measurement		
NT: Hyperspectral imaging	<i>IBR</i>	USE: Inverter-based resource
Hypertension		
BT: Blood pressure	<i>Ibuki</i>	USE: GOSAT
Medical conditions		
RT: Hypotension	<i>IC</i>	USE: Integrated circuits
Hypertext systems		
UF: Hyperlinks		
BT: Computer interfaces		
Information retrieval		
RT: Database systems		
NT: HTTP		
<i>Hypertext Transfer Protocol</i>		
USE: HTTP		



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IC packaging

USE: Integrated circuit packaging

Ice

BT: Geoscience
RT: Ice accretion
Ice sheets
Meteorology
Permafrost
Snow
NT: Glaciers
Ice cap
Ice clouds
Ice shelf
Ice surface
Ice thickness
Icebergs
Sea ice

Ice accretion

BT: Meteorology
RT: Ice

Ice cap

BT: Ice

Ice clouds

BT: Colloidal crystals
Ice
NT: Contrails

Ice sheets

BT: Glaciers
RT: Ice

Ice shelf

BT: Ice
RT: Icebergs

Ice surface

BT: Ice

Ice thickness

BT: Ice

Icebergs

BT: Ice
RT: Ice shelf

ICIC

USE: Intercell interference

ICP

USE: Iterative closest point algorithm

ICs

USE: Integrated circuits

ICT

USE: Information and communication technology

Icterus

USE: Jaundice

ID-based encryption

USE: Identity-based encryption

Identification of persons

UF: Person identification
Person re-identification
BT: Systems, man, and

cybernetics

RT: Access control
Palmprint recognition
Security
NT: Biometric identification
Biometrics
Face recognition
Fingerprint recognition
Handwriting recognition
Identity theft
Speaker recognition
Speech recognition

Identity infringement

USE: Identity theft

Identity management systems

BT: Computer security
Information systems
RT: Decentralized identity
NT: Federated identity

Identity piracy

USE: Identity theft

Identity theft

UF: Identity infringement
Identity piracy
BT: Fraud
Identification of persons
RT: Computer crime

Identity-based cryptography

USE: Identity-based encryption



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Identity-based encryption

UF: ID-based encryption
Identity-based cryptography
BT: Public key cryptography

BT: WiGig
IEEE 802 LAN-MAN
RT: Bluetooth

IDTV

USE: HDTV

IEC

UF: International
electrotechnical commission
BT: Standards organizations
RT: Communication standards
IEC Standards
Standardization
Standards
NT: Moving Pictures Experts
Group

RT: Bluetooth
Butler matrices
Computer networks
MIMO
Modulation
Protocols
Radio communication
Wireless LAN
Wireless access points
Wireless communication
Wireless fidelity
NT: IEEE 802.11ax Standard
IEEE 802.11e Standard
IEEE 802.11g Standard
IEEE 802.11n Standard
IEEE 802.11p Standard

IEC publications

USE: IEC Standards

IEC Standards

UF: IEC publications
BT: Standards publications
RT: Common Information Model
(electricity)
IEC
NT: MPEG standards

IEEE 802.11ax Standard

UF: 802.11ax
Wi-Fi 6
BT: IEEE 802.11 Standard

IEEE 802.11e Standard

UF: 802.11e
BT: IEEE 802.11 Standard
RT: Communication channels
Protocols
Quality assurance
Quality control
Quality of service
Streaming media
Wireless LAN

IEEE 1364

USE: Hardware design
languages

IEEE 1394 Standard

UF: P1394
BT: IEEE Standards
RT: Data buses
Data communication
Firewire
Machine vision
Video signal processing

IEEE 802.11g Standard

UF: 802.11g
BT: IEEE 802.11 Standard
RT: Bluetooth
Computer networks
Modulation
Protocols
Radio communication
Wireless LAN

IEEE 802 LAN-MAN Standards

BT: IEEE Standards
NT: IEEE 802.11 Standard
IEEE 802.15 Standard
IEEE 802.16 Standard
IEEE 802.19 Standard
IEEE 802.22 Standard
IEEE 802.3 Standard

IEEE 802.11n Standard

UF: 802.11n
BT: IEEE 802.11 Standard
RT: Antennas
Bluetooth
Communication channels
Computer networks
MIMO
Modulation

IEEE 802.11 Standard

UF: 802.11
P802.11



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Protocols
Radio communication
Wireless LAN

Ethernet
Local area networks
Packet switching
Switches
Wide area networks

IEEE 802.11p Standard

BT: IEEE 802.11 Standard
RT: Intelligent vehicles
Wireless Access in

Vehicular Environments

Wireless communication
Wireless networks

IEEE 802.15 Standard

UF: 802.15
BT: IEEE 802 LAN-MAN

Standards

RT: Bluetooth
Light fidelity
Personal communication

networks

Radio communication
Wireless LAN
Zigbee

IEEE 802.16 Standard

UF: 802.16
BT: IEEE 802 LAN-MAN

Standards

RT: Broadband communication
Computer networks
Cross layer design
Internet
MIMO
Metropolitan area networks
Multimedia communication
WiMAX

IEEE 802.19 Standard

BT: IEEE 802 LAN-MAN

Standards

IEEE 802.22 Standard

BT: IEEE 802 LAN-MAN

Standards

RT: Regional area networks
WRAN
Wireless communication
Wireless networks

IEEE 802.3 Standard

UF: 802.3
BT: IEEE 802 LAN-MAN

Standards

RT: Communication switching
Computer networks

IEEE 982.1 Standard

BT: IEEE Standards

IEEE activities

BT: IEEE organization
RT: IEEE Boards
NT: IEEE Awards activities
IEEE Conference activities
IEEE Corporate activities
IEEE Educational activities
IEEE Member and

Geographic activities

IEEE Technical activities
IEEE United States

activities

IEEE professional activities
IEEE publishing

IEEE Arc Flash Standards

USE: Power and energy

standards

IEEE Associate Members

BT: IEEE members

IEEE Awards activities

BT: IEEE activities
RT: Awards
IEEE Educational activities
IEEE Foundation
IEEE Technical activities
IEEE United States

activities

NT: IEEE professional activities
IEEE Corporate awards
IEEE Society awards
IEEE Standards awards
National Society Agreement

awards

IEEE Boards

BT: IEEE entities
RT: IEEE activities

IEEE bylaws

BT: IEEE governance



January 2026 IEEE Thesaurus

IEEE Chapters

BT: IEEE entities

IEEE Committees

BT: IEEE entities

IEEE Communities

BT: IEEE entities

IEEE Conference activities

BT: IEEE activities

IEEE Constitution

BT: IEEE governance

IEEE Corporate activities

BT: IEEE activities
RT: IEEE Corporate awards
IEEE professional activities
IEEE staff
Legal factors
NT: Humanitarian activities

IEEE Corporate awards

BT: IEEE Awards activities
RT: IEEE Corporate activities
NT: IEEE Recognitions
IEEE Technical Field

awards

IEEE Corporate recognitions

BT: IEEE Recognitions

IEEE Councils

BT: IEEE entities

IEEE Educational activities

BT: IEEE activities
RT: IEEE Awards activities
IEEE Foundation
IEEE professional activities

IEEE Electric Machinery Standards

USE: Power and energy
standards

IEEE employees

USE: IEEE staff

IEEE entities

BT: IEEE organization
NT: IEEE Boards
IEEE Chapters
IEEE Committees
IEEE Communities

IEEE Councils

IEEE Foundation
IEEE History Center
IEEE Regions
IEEE Societies

IEEE Foundation

BT: IEEE entities
RT: IEEE Awards activities
IEEE Educational activities

IEEE governance

BT: IEEE organization
NT: IEEE Constitution
IEEE bylaws
IEEE policy and procedures
IEEE staff

IEEE History Center

BT: IEEE entities

IEEE indexing

BT: IEEE organization
NT: Awards
Book reviews
Interviews
Obituaries
Software reviews
Special issues and sections
Tutorials
Video reviews

IEEE Member and Geographic activities

BT: IEEE activities

IEEE members

BT: IEEE organization
NT: IEEE Associate Members
IEEE Senior Members

IEEE merchandise

BT: IEEE products

IEEE organization

NT: IEEE activities
IEEE entities
IEEE governance
IEEE indexing
IEEE members
IEEE products



January 2026 IEEE Thesaurus

IEEE policy and procedures

BT: IEEE governance

IEEE Power Substations Standards

USE: Power and energy standards

IEEE Prize Paper awards

BT: IEEE Recognitions

IEEE products

BT: IEEE organization

NT: IEEE Xplore

IEEE merchandise

IEEE professional activities

UF: Non-united-states activities

BT: IEEE activities

RT: IEEE Awards activities

IEEE Corporate activities

IEEE Educational activities

IEEE Technical activities

IEEE United States

activities

IEEE publishing

BT: IEEE activities

RT: Guidelines

IEEE Recognitions

BT: IEEE Corporate awards

NT: IEEE Corporate

recognitions

IEEE Prize Paper awards

IEEE Service awards

IEEE Staff recognitions

IEEE Video awards

IEEE Regions

BT: IEEE entities

IEEE Senior Members

BT: IEEE members

IEEE Service awards

BT: IEEE Recognitions

IEEE Societies

BT: IEEE entities

IEEE Society awards

BT: IEEE Awards activities

IEEE staff

UF: IEEE employees

BT: IEEE governance

RT: IEEE Corporate activities

IEEE Staff recognitions

BT: IEEE Recognitions

IEEE Standard 1349

BT: IEEE Standards

IEEE Standards

BT: Standards publications

RT: ANSI Standards

NT: AIEE Standards

IEEE 1394 Standard

IEEE 802 LAN-MAN

Standards

IEEE 982.1 Standard

IEEE Standard 1349

IRE Standards

IEEE Standards Association

BT: Standards organizations

IEEE Standards awards

BT: IEEE Awards activities

IEEE Surge Protective Devices Standards

USE: Power and energy standards

IEEE Technical activities

BT: IEEE activities

RT: IEEE Awards activities

IEEE professional activities

IEEE Technical Field awards

BT: IEEE Corporate awards

IEEE United States activities

UF: US activities

BT: IEEE activities

RT: IEEE Awards activities

IEEE professional activities

IEEE Video awards

BT: IEEE Recognitions

IEEE Xplore

BT: IEEE products

RT: Information services

IGBT



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USE: Insulated gate bipolar
transistors

Ignition

BT: Chemical reactions
Internal combustion

engines

Nuclear physics
Plasma materials

processing

II-VI semiconductor materials

BT: Semiconductor materials

III-V semiconductor materials

BT: Semiconductor materials
RT: Aluminum gallium nitride

IIOT

USE: Industrial Internet of Things

IIR filters

UF: Infinite impulse response

filters

BT: Filters

Illumination

USE: Lighting

Illumination control

USE: Lighting control

Illumination gas

USE: Coal gas

IM

USE: Instant messaging

Image analysis

UF: Scene analysis
BT: Image processing
RT: Image recognition
Machine vision
NT: Image categorization
Image classification
Image motion analysis
Image quality
Image sequence analysis
Image texture analysis
Neural radiance field
Object detection
Subtraction techniques

Image annotation

UF: Image tagging
Linguistic indexing
Video annotation
BT: Image processing
RT: Feature extraction
Image classification
Image retrieval
Learning (artificial)
Metadata
Video signal processing

Image augmentation

BT: Image transformation
RT: Convolutional neural
networks
Data augmentation
Image classification
Machine learning

Image capture

BT: Image processing
RT: Cameras
Computer vision
Image sensors
Photography

Image categorization

BT: Image analysis

Image classification

BT: Image analysis
RT: Deep learning
Image annotation
Image augmentation
NT: Scene classification

Image coding

UF: Image compression
BT: Image processing
RT: Image communication
Image databases
Image storage
MPEG standards
Rate distortion theory
Steganography
Transcoding
Vector quantization
Video codecs
Video coding



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Image color analysis

UF: Image colour analysis
BT: Image processing
RT: Color gamut
Image filtering

Image colour analysis

USE: Image color analysis

Image communication

UF: Image transmission
BT: Communications
technology
RT: B-ISDN
Cable TV
ISDN
Image coding
Motion compensation
TV
Teleconferencing
Videophone systems
Visual communication
NT: Facsimile
Picture archiving and
communication systems

Image compression

USE: Image coding

Image converters

BT: Imaging
RT: Frequency conversion
Image sensors
NT: Image intensifiers

Image databases

BT: Database systems
Databases
RT: Geographic information
systems
Image coding
Image storage
Video sequences
NT: Image retrieval

Image de-noising

USE: Image denoising

Image deblurring

USE: Image restoration

Image decomposition

BT: Image processing

Image dehazing

BT: Image processing
RT: Deblurring

Image denoising

UF: Image de-noising
BT: Image processing
RT: Diffusion processes
Gaussian noise
Image enhancement
Image filtering
Image reconstruction
Image resolution
Image restoration

Image edge analysis

USE: Image edge detection

Image edge detection

UF: Edge detection
Image edge analysis
BT: Image recognition
RT: Corner detection
Feature extraction
Image segmentation
License plate recognition
Thresholding (Imaging)

Image enhancement

BT: Image processing
RT: Image denoising
Image intensifiers
Image restoration

Image filtering

BT: Filtering theory
Image processing
RT: Image color analysis
Image denoising
Image segmentation

Image forensics

UF: Forensic photography
BT: Forensics
Photography
RT: Image processing
Law enforcement
Visualization

Image fusion

BT: Image processing

Image generation

USE: Image processing AND
Image synthesis



January 2026 IEEE Thesaurus

Image intensifiers

BT: Image converters
RT: Frequency conversion
Image enhancement
Image sensors

processing

Multidimensional signal

Optical projectors
Photogrammetry
Reconstruction algorithms
Time-frequency analysis
Video sequences

Image matching

UF: Appearance matching
Fingerprint images
BT: Pattern matching
RT: Fingerprint recognition
Image recognition
Object detection
Object recognition
Stereo vision
Visual information retrieval
NT: Text to image
Visual place recognition

NT:

processing

Vision sensors
Active shape model
Blob detection
Corner detection
Feature detection
Feature extraction
Fiducial markers
Geophysical image

Image morphing

BT: Image transformation
Morphological operations

Image motion analysis

BT: Image analysis
RT: Fall detection
Human activity recognition
Motion capture
Motion detection
Object tracking
Robotics and automation
NT: Video tracking

Image object detection

USE: Object detection

Image object recognition

USE: Object recognition

Image pattern recognition

USE: Pattern recognition

Image preprocessing

BT: Image processing

Image processing

UF: Image generation
Picture processing
BT: Computers and information
processing
RT: Authentication
Diffusion processes
Gabor filters
Image forensics

Gray-scale
Image analysis
Image annotation
Image capture
Image coding
Image color analysis
Image decomposition
Image dehazing
Image denoising
Image enhancement
Image filtering
Image fusion
Image preprocessing
Image recognition
Image reconstruction
Image registration
Image representation
Image resolution
Image restoration
Image sampling
Image segmentation
Image sequences
Image stitching
Image synthesis
Image texture
Image transformation
Machine vision
Morphological operations
Optical feedback
Pansharpening
Saliency detection
Smart pixels
Spatial coherence
Structure from motion
Table lookup
Text detection
Thresholding (Imaging)



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Image quality

BT: Image analysis
RT: Deblurring
Pansharpening
Spatial resolution

Image enhancement
Video restoration
Deblurring

Image retrieval

BT: Image databases
RT: Cross modal retrieval
Image annotation

Image recognition

BT: Image processing
RT: Emotion recognition
Face recognition
Fall detection
Feature extraction
Human activity recognition
Image analysis
Image matching
License plate recognition
Machine vision
Object recognition
Video signal processing
Visual place recognition
NT: Fine-grained image

recognition

Image edge detection

Image sampling

BT: Image processing

Image segmentation

BT: Image processing
RT: Conditional random fields
Deep learning
Image edge detection
Image filtering
Mixture models
Object tracking
Text detection
NT: Instance segmentation
Semantic segmentation

Image reconstruction

BT: Image processing
RT: Holography
Image denoising
Inverse problems
Magnetic resonance

imaging

Pattern clustering
Tomography

Image sensors

UF: Sensors (image)
BT: Imaging
RT: Cameras
Endoscopes
Image capture
Image converters
Image intensifiers
Motion capture
Motion detection
Night vision
Optical sensors
Photodetectors
Robot vision systems
Wearable sensors
NT: Active pixel sensors
CCD image sensors
CMOS image sensors
Charge-coupled image

sensors

Infrared image sensors

Image registration

BT: Image processing

Image representation

BT: Image processing
NT: Digital representation

Image resolution

BT: Image processing
RT: Image denoising
Visual communication
NT: Contrast resolution
High-resolution imaging
Spatial resolution
Superresolution

Image restoration

UF: Image deblurring
BT: Image processing
RT: Distortion
Image denoising

Image sequence analysis

BT: Image analysis

Image sequences

BT: Image processing

Image skeletonization

BT: Image transformation
RT: Image thinning



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Image stitching

BT: Image processing

Image storage

BT: Imaging
RT: Image coding
Image databases
Photography
Video recording

Image synthesis

UF: Image generation
BT: Image processing
RT: Rendering (computer graphics)
NT: Human image synthesis
Neural radiance field

Image tagging

USE: Image annotation

Image textural analysis

USE: Image texture analysis

Image texture

BT: Image processing

Image texture analysis

UF: Image textural analysis
BT: Image analysis

Image thinning

BT: Image transformation
Morphological operations
RT: Image skeletonization

Image transformation

UF: Image transforming
BT: Image processing
RT: Morphological operations
NT: Image augmentation
Image morphing
Image skeletonization
Image thinning
Neural style transfer

Image transforming

USE: Image transformation

Image transmission

USE: Image communication

Image watermarking

USE: Watermarking

Imagimarkers

USE: Fiducial markers

Imaging

RT: Color
Motion pictures
Radiometry
Remote sensing
Robot vision systems
NT: Biomedical imaging
Cameras
Focusing
Ground penetrating radar
Holography
Image converters
Image sensors
Image storage
Infrared imaging
Magnetic resonance

imaging

Magneto electrical
resistivity imaging technique
Microscopy
Microwave imaging
Multispectral imaging
Nuclear imaging
Optical imaging
Photography
Radiation imaging
Radiography
Stereo vision
Terahertz wave imaging
Tomography

Imaging phantoms

BT: Biomedical image
processing

IMC

USE: In-memory computing

Imitation learning

BT: Machine learning

Immersion cooling

BT: Cooling

Immersive audio

BT: Audio systems
Augmented reality
RT: Ambisonics
Human computer

interaction

Virtual reality



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Immersive experience

BT: Augmented reality
Human computer
interaction
Virtual reality

Immersive learning

BT: Learning systems
Virtual reality
RT: Education

Immune reaction

USE: Immune response

Immune response

UF: Immune reaction
BT: Immune system
RT: Antigens
Vaccines

Immune system

BT: Anatomy
RT: Biological control systems
Biology
Immunology
Immunotherapy
Lymphocytes
Microbiome
Microorganisms
Rheumatology
Sepsis
NT: Allergies
Antibodies
Artificial immune systems
Immune response

Immunity testing

BT: Electromagnetic
compatibility
Electronic equipment
testing
Electrostatic interference
RT: Anechoic chambers
Electromagnetic
interference
Open area test sites

Immunofluorescence

BT: Fluorescence
Immunology
Microscopy
RT: Antigens

Immunoglobulin

USE: Antibodies

Immunology

BT: Medical specialties
RT: Immune system
NT: Antigens
Immunofluorescence

Immunotherapy

UF: Biological therapy
BT: Medical treatment
RT: Immune system
Monoclonal antibodies

Impact ionisation

USE: Impact ionization

Impact ionization

UF: Impact ionisation
BT: Ionization
RT: Charge carriers
Conductivity
Electrons
Insulators

Impedance

UF: Electric impedance
BT: Electric variables
RT: Admittance
Damping
Impedance matching
Impedance measurement

Impedance matching

BT: Electric variables
RT: Circuits
Equalizers
Impedance
NT: Baluns

Impedance measurement

UF: Impedance methods
Impedance performance
BT: Electric variables
measurement
RT: Admittance measurement
Impedance
Transmission line
measurements

Impedance methods

USE: Impedance measurement

Impedance performance

USE: Impedance measurement



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Impellers

BT: Machine components
RT: Blades
Propellers
Pumps

RT: Contamination
NT: Semiconductor impurities

Impersonation attacks

BT: Communication system
security

Implantable biomedical devices

USE: Implants

Implantable devices

USE: Implants

Implantable electronics

USE: Implants

Implants

UF: Implantable biomedical
devices
Implantable devices
Implantable electronics
BT: Biomedical equipment
NT: Auditory implants
Brainstem implants
Cochlear implants
Microelectronic implants
Neural implants

Importance sampling

USE: Monte Carlo methods

Imposter signature generation

USE: Forgery

Improved definition TV

USE: HDTV

Impulse generation

USE: Pulse generation

Impulse measurements

USE: Pulse measurements

Impulse testing

BT: Testing
RT: Channel impulse response
Frequency response
Insulation testing

Impurities

BT: Materials science and
technology

Imputation

BT: Statistics

Imref

USE: Quasi-Fermi level

IMT-2000

USE: 3G mobile communication

In memory computing

USE: In-memory computing

In memory processing

USE: In-memory computing

In vitro

BT: Medical services
NT: In vitro fertilization

In vitro fertilization

BT: In vitro

In vivo

BT: Medical services

In-memory computing

UF: CIM
Compute-in-memory
IMC
In memory computing
In memory processing
In-memory processing
BT: Memory management
RT: Buffer storage
Cache memory

In-memory processing

USE: In-memory computing

Incentive schemes

UF: Bonuses
Merit pay
Performance related pay
Profit sharing schemes
Human resource
BT: management
Remuneration
RT: Appraisal
Employee welfare
Productivity



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Incineration

UF: Afterburners
Incinerators
Refuse incineration
Waste incineration
BT: Waste disposal
RT: Air pollution
Ash
Radioactive pollution
Radioactive waste
Radioactive waste disposal

Incinerators

USE: Incineration

Inclusive language

BT: Professional communication
RT: Terminology
Cultural aspects
Human factors
Social factors

Incremental learning

BT: Machine learning

Independent component analysis

BT: Numerical analysis
RT: Artificial intelligence
Blind source separation
Computer aided analysis
Feature extraction
Principal component

analysis

Signal processing

Index of production

USE: Economic indicators

Indexes

BT: Database systems
RT: Information retrieval
Information systems
NT: Enhanced vegetation index
Indexing
Machine assisted indexing
Spatial indexes

Indexing

UF: Online indexing
BT: Indexes
Information analysis
RT: Keyword search
Machine assisted indexing
Tagging

Indian Ocean

BT: Oceans

Indigenous knowledge

BT: Subject matter experts

Indirect liquid cooling

BT: Liquid cooling

Indium

BT: Metals
RT: Indium compounds

Indium compounds

BT: Compounds
RT: Alloying
Indium
NT: Indium gallium arsenide
Indium tin oxide

Indium gallium arsenide

UF: InGaAs
BT: Gallium compounds
Indium compounds
Semiconductor materials

Indium gallium nitride

BT: Gallium compounds

Indium gallium zinc oxide

BT: Zinc oxide

Indium phosphide

UF: InP
BT: Semiconductor materials
RT: Phonons

Indium tin oxide

BT: Indium compounds
RT: Optical materials

Indoor air quality

BT: Air quality

Indoor communication

BT: Communication systems
RT: Indoor positioning systems
Mobile communication
Optical fiber communication
Optical modulation
NT: Indoor environment

Indoor environment

BT: Indoor communication
NT: Indoor positioning systems



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Indoor navigation

BT: Navigation
RT: Computer vision
Global Positioning System
Indoor positioning systems
Land mobile radio
Path planning
Radio navigation

Indoor positioning systems

BT: Indoor environment
RT: Indoor communication
Indoor navigation

Indoor radio

USE: Indoor radio communication

Indoor radio communication

UF: Indoor radio
Indoor radio
communications
BT: Radio communication

Indoor radio communications

USE: Indoor radio communication

Inductance

BT: Electric variables
RT: Coils
Inductance measurement
Inductors
Transmission line theory

Inductance measurement

BT: Electric variables
measurement
RT: Inductance

Induction (electromagnetic)

USE: Electromagnetic induction

Induction (electrostatic)

USE: Electrostatic induction

Induction generators

BT: AC generators
Induction machines
NT: Doubly fed induction
generators

Induction heating

BT: Heating systems
RT: Electromagnetic heating

Induction machines

BT: AC machines
Rotating machines
NT: Induction generators
Induction motors

Induction motor drives

BT: Induction motors

Induction motors

UF: Asynchronous machines
BT: AC motors
Induction machines
Motors
Rotating machines
RT: Sensorless control
NT: Induction motor drives

Inductive charging

UF: Wireless charging
BT: Energy exchange
Power supplies
RT: Wireless communication

Inductive coupling

BT: Circuit theory
Mutual coupling

Inductive energy transfer

USE: Inductive power
transmission

Inductive power transmission

UF: Inductive energy transfer
BT: Electromagnetic induction
Power transmission
RT: Inductors
Sensorless control
Transformers

Inductive transducers

BT: Transducers

Inductors

UF: Chokes
Reactors
BT: Electronic components
RT: Coils
Electrical ballasts
Inductance
Inductive power
transmission
Magnetic cores
Tunable circuits and
devices



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NT: Active inductors
Thick film inductors
Thin film inductors

Industrial accidents

BT: Accidents
RT: Hazardous areas
Occupational safety

Industrial communication

UF: Organizational
communication
BT: Communication networks
Industrial engineering
RT: Business
Organizational aspects

Industrial control

BT: Industrial electronics
RT: Assembly systems
Computer numerical control
Control system security
Field buses
Industrial engineering
Industrial plants
Manipulators
Manufacturing automation
Mobile robots
Motor drives
Programmable control
Robots
NT: Process control
Production control

Industrial democracy

USE: Industrial relations

Industrial economics

UF: Manufacturing economics
Production economics
BT: Microeconomics
RT: Economies of scale
Privatization

Industrial electronics

NT: Assembly systems
Computer aided
manufacturing
Cryogenic electronics
Industrial control
Integrated manufacturing
systems
Machine control
Manufacturing automation
Testing

Industrial engineering

BT: Industry applications
RT: Design methodology
Industrial control
Industrial plants
Industrial training
Precision engineering
Production engineering
Production management
Research and development
NT: Industrial communication

Industrial facilities

BT: Production facilities
RT: Industrial plants
Manufacturing systems
NT: Seaports

Industrial heat recovery

USE: Heat recovery

Industrial Internet of Things

UF: IIOT
Industrial IoT
BT: Internet of Things
RT: Machine-to-machine
communications
Robotics and automation
Industrial IoT
USE: Industrial Internet of Things

Industrial plants

UF: Plants (industrial)
BT: Production facilities
RT: Industrial control
Industrial engineering
Industrial facilities
Industrial power systems
Industries
Manufacturing
Paper mills
Production systems

Industrial pollution

BT: Pollution
RT: Air pollution
Industrial waste
Land pollution
Radioactive pollution
Thermal pollution
Water pollution



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Industrial power systems

UF: Commercial power systems
BT: Power systems
RT: Buildings
Cogeneration
Industrial plants
Power distribution

Industrial psychology

BT: Psychology
RT: Employee welfare
Human resource

management

Productivity
Psychometric testing

Industrial relations

UF: Collective bargaining
Industrial democracy
Trade unions
BT: Business
RT: Equal opportunities

Industrial Revolution 4.0

USE: Fourth Industrial Revolution

Industrial Revolution 5.0

USE: Fifth Industrial Revolution

Industrial robots

BT: Robots
RT: Manufacturing automation

Industrial training

BT: Training
RT: Industrial engineering
Multiskilling
On the job training
Vocational training

Industrial waste

BT: Waste materials
RT: Effluents
Industrial pollution
Slurries
Waste heat
Wastewater
NT: Ash
Slag

Industries

BT: Industry applications
RT: Business
Industrial plants
NT: Agriculture

Airline industry
Architecture
Beverage industry
Chemical industry
Coal industry
Communication industry
Computer industry
Construction
Construction industry
Defense industry
Electrical engineering

industry

Entertainment industry
Farming
Financial industry
Gas industry
Horticulture
Information industry
Lumber industry
Manufacturing industries
Metals industry
Mining industry
Natural gas industry
Petroleum industry
Pharmaceutical industry
Power industry
Real estate industry
Steel industry
Sugar industry
Textile technology
Tourism industry
Toy industry
Transportation industry
Wood industry

Industry 4.0

USE: Fourth Industrial Revolution

Industry 5.0

USE: Fifth Industrial Revolution

Industry applications

NT: Accident prevention
Chemical technology
Cryogenics
Electrochemical devices
Electrochemical processes
Electromechanical systems
Electrostatic devices
Electrostatic precipitators
Electrostatic processes
Engines
Environmental
Food technology

management



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techniques	High-temperature	Infertility	BT: Medical conditions RT: Pregnancy
	Industrial engineering	Infinite horizon	BT: Optimal control RT: Markov processes Optimization methods
	Industries		
	Inspection	<i>Infinite impulse response filters</i>	USE: IIR filters
	Machinery	<i>Inflammability</i>	USE: Flammability
	Manufacturing		
	Packaging	Inflammatory bowel disease	BT: Medical conditions RT: Digestive system NT: Crohn's disease Ulcerative colitis
	Paper technology		
	Production	Influenza	BT: Diseases Viruses (medical) RT: Epidemics Pandemics
	Safety		
	Security	Informatics	BT: Information processing Information systems RT: Computational materials science NT: Bioinformatics Cognitive informatics Energy informatics Neuroinformatics
	Wine industry		
Inertial confinement		Information age	UF: AOI Age of information Digital age New media age BT: Information technology
BT: Plasma confinement			
Inertial navigation		Information analysis	BT: Professional communication RT: Big Data applications Sentiment analysis NT: Decision analysis Indexing
BT: Navigation			
Inertial sensors			
BT: Sensors			
<i>Infant</i>	USE: Pediatrics		
<i>Infants</i>	USE: Pediatrics		
Infectious diseases	UF: Communicable disease Transmissible disease BT: Diseases RT: Contact tracing NT: Measles		
Inference algorithms	BT: Algorithms RT: Model compression		
Inference mechanisms	UF: Model-based reasoning BT: Knowledge engineering RT: Cognitive science Fuzzy cognitive maps Gaussian processes Intent recognition Learning systems NT: Belief propagation Fuzzy reasoning		
Inferential statistics	BT: Statistical analysis RT: Descriptive statistics		



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Information and communication technology

UF: ICT
 BT: Communications
 technology
 RT: Information technology
 NT: Energy informatics
 NT: Ambient assisted living

Information architecture

BT: Information systems
 RT: Database systems
 NT: Enterprise architecture
 management

Information centric networking

USE: Information-centric
 networking

Information centric networks

USE: Information-centric
 networking

Information diffusion

BT: Information exchange
 Social networking (online)
 RT: Fake news
 Information sharing
 Knowledge transfer
 Privacy

Information entropy

BT: Information theory

Information exchange

BT: Data processing
 Information processing
 RT: Border Gateway Protocol
 Common Information Model

(computing)

(electricity) Common Information Model

Information leakage
 Information management
 Information sharing
 Ports (computers)
 Tactile Internet
 NT: Information diffusion

Information extraction

USE: Information retrieval

Information filtering

BT: Filtering
 Information retrieval
 NT: Information filters

Recommender systems

Information filters

UF: Web filters
 BT: Information filtering
 RT: Information retrieval
 NT: Accesslists
 Blocklists

Information geometry

BT: Geometry
 RT: Probability

Information industry

BT: Industries

Information inequality

USE: Cramer-Rao bounds

Information integrity

UF: Information quality
 BT: Professional
 communication
 RT: Cyber threat intelligence
 Deepfakes
 Fake news
 Information leakage
 Synthetic data

Information leakage

BT: Information security
 RT: Eavesdropping
 Encryption
 Information exchange
 Information integrity
 Privacy
 Security

Information management

BT: Information systems
 Management
 RT: Big Data
 Data aggregation
 Decentralized identity
 Information exchange
 Information services
 Knowledge management
 Common Information Model
 NT: Common Information Model
 (computing)
 (electricity)
 Competitive intelligence
 Digital information
 Digital preservation
 Document handling



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management	Enterprise architecture	Persistent identifiers
	Information security	Portals
	Information sharing	Ranking (statistics)
	Knowledge transfer	Symbols
		Triples (Data structure)
Information processing		NT: Blogs
BT: Information systems		Content-based retrieval
RT: Big Data		Cross modal retrieval
Business Process		Dimensionality reduction
Execution Language		Hypertext systems
	Business data processing	Information filtering
	Data collection	Information rates
	Granular computing	Music information retrieval
	Information sharing	Online services
	Software as a service	Search engines
	Spectral efficiency	Search methods
NT: Electronic healthcare		Semantic communication
Informatics		Social networking (online)
Information exchange		Tagging
Smart agriculture		Taxonomy
Sonification		Terminology
		Visual information retrieval
		Vocabulary
		Web sites
<i>Information quality</i>		
USE: Information integrity		
Information rates		Information science
UF: Throughput		BT: Professional
(communication systems)		communication
BT: Information retrieval		NT: Quantum information
		science
Information representation		Information security
BT: Information technology		BT: Information management
RT: Visual analytics		Security
NT: Digital representation		RT: Cyber threat intelligence
		Data protection
Information resources		Decentralized identity
BT: Professional		Differential privacy
communication		Digital policy
RT: Information retrieval		Hardware security
Information systems		Internet security
		Security information and
Information retrieval		event management
UF: Information extraction		NT: Cyber espionage
BT: Professional		Data breach
communication		Information leakage
RT: Big Data		Intrusion detection
Cross lingual		Phishing
Data retrieval		Privacy breach
Document handling		SQL injection
Indexes		Social engineering
Information filters	(security)	
Information resources		Trust management
Knowledge discovery		
Named entity recognition		



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Information services

BT: Professional
communication
RT: IEEE Xplore
Information management
Journalism
NT: Dictionaries
Document delivery
Encyclopedias
Libraries
Teletext
Videotex

Information sharing

BT: Information management
RT: Collaboration
Cyber threat intelligence
Information diffusion
Information exchange
Information processing
NT: Data dissemination

Information systems

BT: Professional
communication
RT: Big Data applications
CD-ROMs
Computers and information
processing
Database machines
Digital information
Extranets
File systems
Indexes
Information resources
Information technology
Management information
base
Multimedia computing
Office automation
Strategic planning
Technology acceptance
model
NT: Data systems
Database systems
Decision support systems
Distributed information
systems
Identity management
systems
Informatics
Information architecture
Information management
Information processing

systems

Management information

Medical information

systems

Information technology

BT: Professional
communication
RT: Automation
Biometrics
Computer applications
Digital economy
Information systems
Bring your own device
DevOps
Information age
Information and
communication technology
Information representation
Printing
Semantic technology
Service computing
Telematics
Universal Serial Bus

Information theory

UF: Coding theory
Information-theoretic
Informationtheoretic
RT: Bandwidth
Code refractoring
Communication systems
Cybernetics
Cyclic redundancy check
Econophysics
Modulation coding
Quantum communication
Statistics
Teleportation
Viterbi algorithm
NT: Audio coding
Biological information
theory
Channel coding
Codes
Communication channels
Decoding
Encoding
Error compensation
Genetic communication
Hamming distances
Hamming weight
Information entropy
Mutual information
Network coding



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model	Rate distortion theory Rate-distortion Source coding Speech coding Technology acceptance	spectroscopy	Functional near-infrared Infrared surveillance Optical imaging Remote sensing Night vision
		NT:	
Information-centric networking		<i>Infrared lasers</i>	
UF:	Content-centric networking Information centric	USE:	Lasers
networking	Information centric	<i>Infrared propagation</i>	
networks	Information-centric	USE:	Optical propagation
networks	Information-centric	Infrared sensors	
BT:	Network architecture	BT:	Sensors
RT:	Telecommunication	Infrared spectra	
computing		UF:	IR Spectra
		BT:	Spectral analysis
<i>Information-centric networks</i>		RT:	Functional near-infrared
USE:	Information-centric	spectroscopy	Spectroscopy
networking			
<i>Information-theoretic</i>		Infrared surveillance	
USE:	Information theory	BT:	Surveillance
<i>Informationtheoretic</i>		RT:	Infrared detectors Infrared imaging
USE:	Information theory		
<i>Infrared communication</i>		<i>Infrared thermography</i>	
USE:	Optical fiber communication	USE:	Infrared imaging
Infrared detectors		<i>InGaAs</i>	
BT:	Radiation detectors	USE:	Indium gallium arsenide
RT:	Bolometers Infrared surveillance Motion detection Photodetectors Superconducting	Inhibitors	
photodetectors		BT:	Chemical products Production materials
	Wearable sensors	RT:	Retardants
		NT:	Corrosion inhibitors
Infrared heating		<i>Inhomogeneous media</i>	
BT:	Heating systems	USE:	Nonhomogeneous media
NT:	Greenhouse effect	<i>Injected beams</i>	
		USE:	Particle beam injection
Infrared image sensors		<i>Injection lasers</i>	
BT:	Image sensors	USE:	Semiconductor lasers
Infrared imaging		<i>Injection locked oscillators</i>	
UF:	Infrared thermography Thermal video	USE:	Injection-locked oscillators
BT:	Imaging	<i>Injection locking</i>	
RT:	Biomedical optical imaging	USE:	Injection-locked oscillators



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Injection molding

UF: Injection moulding
Power injection molding
Power injection moulding
BT: Production
RT: Compression molding
Embossing

Injection moulding

USE: Injection molding

Injection-locked oscillators

UF: Injection locked oscillators
Injection locking
BT: Oscillators

Injuries

UF: Injury
BT: Medical conditions
RT: Fall detection
NT: Amputation
Brain injuries
Pain
Skin burns
Wounds

Injury

USE: Injuries

Ink

BT: Production materials
RT: Paints
Printing
NT: Ink jet printing

Ink jet printing

UF: Ink-jet printers
Ink-jet printing
Inkjet printing
BT: Ink
Printing
RT: Three-dimensional printing

Ink-jet printers

USE: Ink jet printing

Ink-jet printing

USE: Ink jet printing

Inkjet printing

USE: Ink jet printing

Innovation

USE: Technological innovation

Innovation management

UF: Management of innovation
BT: Engineering management
Research and development
management
RT: Entrepreneurship
Technology management
NT: Creativity

Inorganic chemicals

BT: Chemistry

Inorganic compounds

BT: Compounds
RT: Metals
Organic inorganic hybrid

materials

NT: Transition metal

compounds

Inorganic LEDs

USE: Inorganic light emitting
diodes

Inorganic light emitting diodes

UF: Inorganic LEDs
BT: Light emitting diodes

Inorganic materials

BT: Materials
RT: Soft electronics

Inorganic organic hybrid materials

USE: Organic inorganic hybrid
materials

Inorganic-organic hybrid materials

USE: Organic inorganic hybrid
materials

InP

USE: Indium phosphide

Input devices

BT: Computer interfaces
Hardware

Input surge current

USE: Inrush current

Input variables

UF: Variable selection
BT: Modeling



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Input-output programs

UF: I/O programs
BT: Operating systems
RT: Program processors
NT: Device drivers

RT: Turing completeness
NT: Complex instruction set
Out of order
Prefetching
Reduced instruction set

computer

computing

Inrush current

UF: Input surge current
Switch on surge
BT: Current
Surges
RT: Converters
Current measurement
Electric current control
Electric fields

Instructional aids

USE: Educational technology

Instrument landing systems

BT: Aircraft navigation

Instrument transformers

BT: Transformers
RT: Protective relaying
NT: Voltage transformers

Insect control

USE: Pest control

Insects

BT: Animals
RT: Apiculture
Entomology

Instrumentation and measurement

NT: Computerized

instrumentation

Electric variables
High energy physics

instrumentation computing

Instruments
Measurement
Monitoring
Pulse oximetry
Testing

Insertion loss

BT: Propagation
RT: Attenuation

Inspection

BT: Industry applications
RT: Coordinate measuring

machines

Maintenance engineering
Testing

NT: Automatic optical inspection

Instrumentation buses

USE: Field buses

Instruments

BT: Instrumentation and
measurement

RT: Design tools
Measurement
Barometers
Compass
Medical instruments
Meters
Microscopy
Network analyzers
Odometers
Optical instruments
Oscilloscopes
Pressure gauges
Probes
Telescopes
Theodolites
Tuners

Instance segmentation

BT: Image segmentation
RT: Object detection
Semantic segmentation

Instant messaging

UF: IM
BT: Electronic messaging
Internet

Instanton vacuum

USE: Elementary particle vacuum

Instruction repertory

USE: Instruction sets

Instruction sets

UF: Instruction repertory
BT: Program processors



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Insulated gate bipolar transistors

UF: IGBT
BT: Bipolar transistors

Insulation

BT: Dielectrics and electrical
insulation

RT: Dielectric breakdown
Dielectric losses
Dielectric materials
Glass
Insulation life
Insulation testing
Oils
Polymer foams
Power transformer

insulation

NT: Spark gaps
Cable insulation
Ceramics
Gas insulation
Insulators
Isolation technology
Oil insulation
Plastic insulation
Solid insulation

Insulation life

BT: Insulation testing
RT: Aging
Insulation
Life estimation
Partial discharge

measurement

Trees - insulation

Insulation testing

BT: Insulator testing
RT: Fault location
Impulse testing
Insulation
Partial discharge
Pulsed electroacoustic

measurement

methods

NT: Insulation life

Insulator testing

BT: Testing
RT: Insulators
Surface discharges
NT: Insulation testing

Insulators

UF: Bushings
BT: Insulation
RT: Breakdown voltage
Ceramic products
Impact ionization
Insulator testing
Polymer foams
Temperature distribution
NT: Metal-insulator structures
Plastic insulators
Rubber
Topological insulators
Trees - insulation

Insulin

BT: Drugs

Insulin pumps

BT: Pumps
RT: Biomedical equipment
Diabetes

Insurance

BT: Financial management

Intake systems

UF: Data intake
BT: Machine components
RT: Data ingestion

Integer linear programming

BT: Programming
NT: Constraint theory
Mixed integer linear

programming

Integer programming

BT: Linear programming

Integral equations

UF: Antiderivatives
BT: Calculus
RT: Boundary-element methods
Deconvolution
Integrodifferential equations
Inverse problems
Method of moments
Numerical analysis
NT: Probability density function

Integrated circuit design

USE: Integrated circuit synthesis



January 2026 IEEE Thesaurus

Integrated circuit interconnections

BT: Integrated circuits

Integrated circuits
Plastic packaging
Semiconductor device

Integrated circuit layout

BT: Integrated circuit synthesis
RT: Layout
Physical design
Printed circuits

packaging

NT: Multichip modules
Plastic integrated circuit

packaging

Integrated circuit manufacture

BT: Components, packaging,
and manufacturing technology

RT: Gettering
Integrated circuits
Microassembly
Micromachining
Silicon compiler

NT: Surface mount technology

Integrated circuit reliability

BT: Reliability
RT: Electrostatic discharge

protection

Integrated circuit testing
Thermal stability

Integrated circuit synthesis

UF: Integrated circuit design
BT: Circuit synthesis
Integrated circuits
NT: Integrated circuit layout

Integrated circuit measurements

BT: Circuit testing
RT: Electric variables

measurement

Integrated circuit metallisation

USE: Integrated circuit
metallization

Integrated circuit metallization

UF: Integrated circuit
metallisation

BT: Metallization

Integrated circuit technology

BT: Circuits and systems
RT: Electrostatic discharge

protection

High-speed integrated

circuits

NT: Beyond CMOS
CMOS technology
Moore's Law

Integrated circuit modeling

UF: Integrated circuit modelling
BT: Integrated circuits
Modeling
NT: Cutoff frequency

Integrated circuit testing

BT: Testing
RT: Integrated circuit reliability
NT: Integrated circuit yield
Logic testing

Integrated circuit modelling

USE: Integrated circuit modeling

Integrated circuit yield

BT: Integrated circuit testing

Integrated circuit noise

BT: Integrated circuits
RT: Semiconductor device

noise

NT: Threshold voltage
Optical noise

Integrated circuits

UF: IC
ICs
Microchips
BT: Circuits
RT: Active inductors
Integrated circuit

manufacture

Integrated circuit packaging
Integrated optoelectronics
Memory modules
Microelectronics
Neural network hardware
Planarization
SPICE

Integrated circuit packaging

UF: IC packaging
BT: Components, packaging,
and manufacturing technology
RT: Chip scale packaging
Encapsulation



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	Semiconductor devices	<i>Integrated control</i>	
	Semiconductor memory	USE:	Centralized control
	Silicon-on-insulator		
	VHDL	Integrated design	
NT:	Analog integrated circuits	BT:	Design methodology
	Analog-digital integrated		Systems engineering and
circuits		theory	
integrated circuits	Application specific	Integrated manufacturing systems	
	CMOS integrated circuits	BT:	Industrial electronics
	Chiplets		Manufacturing systems
	Coprocessors	RT:	CAD/CAM
	Current-mode circuits		Computer aided
	Digital integrated circuits	manufacturing	System integration
	FET integrated circuits		
	Field programmable gate		
arrays		Integrated memory circuits	
	Hybrid integrated circuits	BT:	Digital integrated circuits
	Integrated circuit		Semiconductor memory
interconnections		RT:	Solid state drives
	Integrated circuit modeling	Integrated optics	
	Integrated circuit noise	BT:	Optics
	Integrated circuit synthesis	RT:	Arrayed waveguide gratings
	Large scale integration		Distributed Bragg reflectors
	MESFET integrated circuits		Electro-optic modulators
	Microprocessors		Integrated optoelectronics
	Microwave integrated		Microoptics
circuits			Nanoantennas
	Millimeter wave integrated		Optical films
circuits			Optical waveguides
	Monolithic integrated		Synapses
circuits			Thermo-optical devices
	Photonic integrated circuits	Integrated optoelectronics	
	Power integrated circuits	BT:	Optoelectronic devices
	Radiofrequency integrated	RT:	Heterojunction bipolar
circuits		transistors	
	SIM card		Integrated circuits
	Submillimeter wave		Integrated optics
integrated circuits			Liquid crystal on silicon
	Superconducting integrated		Microoptics
circuits			Microwave photonics
	Thick film circuits		Nanoantennas
	Thin film circuits		Smart pixels
	Three-dimensional		
integrated circuits		Integrated sensing and communication	
	Through-silicon vias	UF:	ISAC
	UHF integrated circuits	BT:	Communication systems
	Ultra large scale integration		Sensors
	Very high speed integrated		
circuits		<i>Integrated services digital networks</i>	
	Very large scale integration	USE:	ISDN
	Wafer scale integration		
<i>Integrated circuits industry</i>			
USE:	Electronics industry		



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Integrated services networks

USE: Intserv networks

Production systems

RT: Smart manufacturing

Integrodifferential equations

BT: Equations
RT: Differential equations
Integral equations

Intelligent materials

BT: Materials
RT: Biomimetics
Composite materials
Intelligent structures
Polymer gels
Smart materials

Integumentary system

BT: Anatomy
NT: Hair
Nails
Skin

Intelligent networks

BT: Telecommunication
network topology
RT: Software defined
networking

Intellectual capital

USE: Knowledge management

Intelligent reconfigurable surface

USE: Reconfigurable intelligent surfaces

Intellectual property

UF: IP rights
IPR
BT: Copyright protection
RT: Cyberethics
Patents
Software protection
NT: Digital rights management

Intelligent robots

BT: Intelligent systems
Robots
RT: Autonomous robots
Intelligent automation
Robot vision systems

Intelligent actuators

UF: Smart actuators
BT: Actuators

Intelligent sensors

UF: Smart sensors
BT: Sensors
RT: Electronic noses
Mechatronics
Soft sensors

Intelligent agents

BT: Software agents

Intelligent agriculture

USE: Smart agriculture

Intelligent structures

UF: Smart structures
BT: Buildings
RT: Intelligent materials
Intelligent systems
Structural engineering
NT: Smart cities

Intelligent automation

UF: Robotic process automation
BT: Automation
Intelligent systems
RT: Intelligent robots

Intelligent control

BT: Cybernetics
RT: Context awareness
H infinity control
Mechatronics
NT: Feedforward systems
Neurocontrollers

Intelligent systems

BT: Artificial intelligence
RT: Ambient intelligence
Artificial general intelligence
Automata
Collaborative intelligence
Context awareness
Controller area networks
Expert systems
Hybrid intelligent systems
Intelligent structures
Knowledge based systems

Intelligent databases

USE: Deductive databases

Intelligent manufacturing systems

BT: Manufacturing systems



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communications		Massive machine type		<i>Inter Planetary File System</i>
			USE:	InterPlanetary File System
		Mobile agents		
		Software agents		<i>Inter-cell interference</i>
	NT:	Autonomous systems	USE:	Intercell interference
		Collective intelligence		
		Hyper-intelligent systems		<i>Inter-cell interference coordination</i>
Intelligent transportation systems		Intelligent automation	USE:	Intercell interference
		Intelligent robots		
				<i>Inter-symbol interference</i>
	RT:	Advanced driver assistance	USE:	Intersymbol interference
				Interactive systems
systems		Automotive control	BT:	Human-machine systems
		Smart transportation	RT:	Affordances
	NT:	Automated highways		Authentication
		Geographic information		Spatial computing
			NT:	External stimuli
systems		Intelligent vehicles		
		Navigation		<i>Intercalibration</i>
		Transportation	USE:	Calibration
Intelligent vehicles				Inter-cell interference
	BT:	Intelligent transportation	UF:	ICIC
				Inter-cell interference
				Inter-cell interference
	RT:	Vehicles		
		Advanced driver assistance	coordination	
			BT:	Network resource
systems			management	
		Connected vehicles		Radiofrequency
		Dedicated short range		
			interference	
			RT:	Cellular radio
				Fading channels
communication		IEEE 802.11p Standard		Location awareness
		Smart transportation		Radio communication
		Vehicle routing		
		Vehicular automation		
		Wireless Access in		
Vehicular Environments				Interchannel interference
	NT:	Autonomous vehicles	UF:	Adjacent channel
		Vehicle-to-everything		
				Co-channel interference
				Cochannel interference
				Intersystem interference
Intensity modulation			BT:	Interference
	BT:	Optical modulation	RT:	Crosstalk
	RT:	Amplitude modulation		
		Electro-optic modulators		
<i>Intent classification</i>				Interconnected systems
	USE:	Intent recognition	UF:	Composite systems
			BT:	System analysis and design
			RT:	Control systems
			NT:	Botnet
Intent recognition				<i>Interconnection networks</i>
	UF:	Goal recognition	USE:	Multiprocessor
		Intent classification		
	BT:	User interfaces		
	RT:	Behavioral sciences		
		Inference mechanisms		



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Intercultural competence

USE: Cultural competence

Interest point detection

BT: Computer vision

Interest rates

USE: Economic indicators

Interface management

BT: Management
Systems engineering and

theory

RT: Computer interfaces
Network interfaces

Interface phenomena

BT: Computer interfaces
RT: Adsorption
NT: Network interfaces

Interface states

BT: Computer interfaces
MOSFET
RT: Silicon-on-insulator

Interference

UF: Interference (signal)
BT: Electromagnetic
compatibility and interference
RT: Coherence

Distortion
Diversity schemes
Noise
NT: Clutter
Crosstalk
Diffraction
Echo interference
Electromagnetic

interference

interference

Electromagnetic radiative

Electrostatic interference
Interchannel interference
Interference cancellation
Interference channels
Interference constraints
Interference elimination
Interference suppression
Intersymbol interference
Rain fading
TV interference
Terrain factors

Interference (signal)

USE: Interference

Interference cancellation

BT: Interference

Interference channels

BT: Interference

Interference constraints

BT: Interference

Interference elimination

BT: Interference

Interference suppression

BT: Interference

Interferometers

UF: Etalons
BT: Interferometry
NT: Mach-Zehnder

interferometers

Interferometric lithography

BT: Lithography

Interferometry

BT: Measurement
RT: Micrometers
Talbot effect
NT: Fabry-Perot
Interferometers
Optical interferometry
Phase shifting
interferometry
Radar interferometry
Radio interferometry
Sagnac interferometers

Intergovernmental Panel on Climate Change

UF: IPCC
BT: Climate change

Interleaved codes

UF: Bit interleaved coded
Bit-interleaved coded
BT: Modulation coding

Intermetallic

BT: Alloying

Intermodulation distortion

BT: Nonlinear distortion



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Internal combustion engines

UF: HCCI engines
BT: Engines
RT: Automotive components
Exhaust gases
Hybrid electric vehicles
NT: Diesel engines
Ignition

Internal stresses

BT: Stress
RT: Surface stress

International Atomic Time

UF: TAI
BT: Standards categories
RT: Atomic clocks

International collaboration

UF: Joint ventures
BT: Management
RT: Environmental factors
Environmental

management

Globalization
International relations
International trade
Research and development
Social factors
Standards
Trade agreements

International electrotechnical commission

USE: IEC

International organization for standardization

USE: ISO

International relations

BT: Social implications of
technology
RT: Globalization
International collaboration
Social factors

International Space Station

BT: Space stations

International standards organization

USE: ISO

International System of Units

BT: Measurement units

International Telecommunications Union

USE: ITU

International trade

UF: Trade (international)
BT: Economics
RT: Business
Developing countries
Exchange rates
Free economic zones
Freeports
Globalization
International collaboration
Macroeconomics
Trade agreements

Internet

UF: Google
BT: Communication systems
Computer networks
Digital systems
Distributed computing
RT: ARPANET
Blogs
Crowdfunding
Cyberspace
Diffserv networks
Digital economy
Electronic commerce
Electronic learning
Energy Internet
Extranets
Hybrid learning
IEEE 802.16 Standard
IP networks
IPTV
Internetworking
Intserv networks
Multicast protocols
Multiprotocol label
switching
Next generation networking
Online services
Open data
Point-to-multipoint
communications
Routing protocols
Service level agreements
Smart TV
Social networking (online)
Streaming media
TCPIP
Virtual enterprises
Virtual private networks
Web sites



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NT:	World Wide Web	Internet of Things	
	Bot (Internet)	UF:	IOT
	Botnet		Internet of Everything
	Cloud computing	BT:	Internet
	Crowdsourcing	RT:	Aerial computing
	Dark Web		Ambient intelligence
	Instant messaging		Bar codes
	Internet of Things		Cloud computing
	Internet privacy		Cyber-physical systems
	Internet security		Digital economy
	Internet telephony		Digital transformation
	Internet topology		Digital twins
	Linked data		Edge computing
	Middleboxes		Energy Internet
	Semantic Web		Fifth Industrial Revolution
	Social computing		Fourth Industrial Revolution
	Web 2.0		Machine-to-machine
	Web conferencing	communications	
	Web services		Middleware
<i>Internet banking</i>			Object detection
USE:	Online banking		Precision agriculture
<i>Internet bullying</i>			Protocols
USE:	Cyberbullying		Radiofrequency
<i>Internet governance</i>		identification	
USE:	Digital policy		Simultaneous wireless
<i>Internet neutrality</i>		information and power transfer	
USE:	Network neutrality		Tagging
<i>Internet of Energy</i>			Virtual environments
USE:	Energy Internet		Watermarking
<i>Internet of Everything</i>			Wireless sensor networks
USE:	Internet of Things	Things	NT: Agriculture Internet of
<i>Internet of Health Things</i>			Industrial Internet of Things
USE:	Internet of Medical Things		Internet of Medical Things
Internet of Medical Things			Internet of Vehicles
UF:	Internet of Health Things		Social Internet of Things
	IoHT		Tactile Internet
BT:	Internet of Things	Internet of Vehicles	
	Medical services	UF:	IOV
RT:	Biomedical monitoring		Internet-of-Vehicles
	Emergency services	BT:	Internet of Things
	Medical treatment		Vehicles
	Remote monitoring	RT:	Vehicular ad hoc networks
	Smart healthcare	Internet privacy	
	Wearable devices	BT:	Internet
			Privacy
		RT:	Internet security
		<i>Internet Protocol networks</i>	
		USE:	IP networks
		<i>Internet protocol television</i>	
		USE:	IPTV



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Internet security

BT: Computer security
Internet
RT: Information security
Internet privacy

Internet services

USE: Web and internet services

Internet telephony

UF: IP telephony
VOIP
Voice over IP
Voice over Internet protocol
Voice-over-Internet protocol
WhatsApp
BT: Internet
RT: Call admission control
Point-to-multipoint

communications

Internet topology

BT: Internet

Internet-of-Vehicles

USE: Internet of Vehicles

Internetworking

BT: Telecommunication
computing
RT: Computer networks
Internet
Local area networks
Metropolitan area networks
Open systems
Wide area networks
NT: Interoperability
LAN interconnection

Interoperability

UF: Service composability
BT: Internetworking
RT: Collaboration
Common Information Model
(electricity)
Open systems

Interplanetary exploration

UF: Emirates Mars Mission
Mars Express
Mars Odyssey
Tianwen-1
BT: Space exploration
RT: Space missions

InterPlanetary File System

UF: IPFS
Inter Planetary File System
BT: Peer-to-peer computing
Protocols
RT: World Wide Web

Interpolating

USE: Interpolation

Interpolation

UF: Interpolating
BT: Approximation methods
RT: Curve fitting
Digital-analog conversion
Radial basis function

networks

Statistics
Surface fitting

Interpreters (program)

USE: Program processors

Interrupters

UF: Interruption
BT: Switchgear
RT: Circuit breakers
Fuses

Interruption

USE: Interrupters

Interstellar chemistry

BT: Chemistry
RT: Extraterrestrial
measurements

Intersymbol interference

UF: Inter-symbol interference
BT: Interference
RT: AWGN channels
Equalizers
Gaussian channels

Intersystem interference

USE: Interchannel interference

Interviews

BT: IEEE indexing

Intestines

BT: Digestive system



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Intracranial pressure sensors

BT: Biomedical equipment
Sensors
RT: Brain
Neural engineering

Intracranial system

BT: Cranial pressure

Intrusion detection

BT: Information security
RT: Network function

virtualization

NT: Network intrusion detection

Intserv networks

UF: Integrated services
networks
BT: Computer networks
RT: Internet
Multimedia communication
NT: Space-air-ground
integrated networks

Intubation

UF: Tracheal intubation
BT: Medical treatment
RT: Respiratory system
Ventilators

Invasive software

USE: Privacy-invasive software

Invasive species

BT: Organisms

Invention

USE: Technological innovation

Inventory control

BT: Operations research
RT: Control systems
Production control
Production management

Inventory management

BT: Production management
RT: Bar codes
Production engineering
NT: Bills of materials

Inverse design

BT: Design methodology
RT: Reverse engineering

Inverse distortion

USE: Predistortion

Inverse method

USE: Inverse problems

Inverse methods

USE: Inverse problems

Inverse modeling

USE: Inverse problems

Inverse problem

USE: Inverse problems

Inverse problems

UF: Inverse method
Inverse methods
Inverse modeling
Inverse problem
Inverse scattering
BT: Modeling
RT: Functional analysis
Image reconstruction
Integral equations
Numerical analysis
Signal reconstruction
NT: Deconvolution

Inverse scattering

USE: Inverse problems

Inverse synthetic aperture radar

BT: Synthetic aperture radar

Inverse transforms

USE: Laplace equations

Inverted classroom

USE: Education AND
Online services

Inverter-based resource

UF: IBR
BT: Inverters
Power conversion
RT: Grid forming

Inverters

BT: Power electronics
RT: Maximum power point
trackers
Zero current switching
Zero voltage switching
NT: Grid following



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Grid forming
Inverter-based resource
Microinverters
Multilevel inverters
Pulse inverters
Resonant inverters
Voltage source inverters

RT: Electrodynamics
Ion accelerators
Ion beam applications
Ion emission
Ion sources
Ions
NT: Ion beam effects

Investment

BT: Financial management
RT: Crowdfunding
Developing countries
Fintech
Open banking

Ion emission

UF: Field ion emission
Secondary ion emission
BT: Nuclear imaging
RT: Ion beam effects
Ion beams
Ion sources
Ions
Thermionic emission

Iodides

USE: Iodine compounds

Iodine

BT: Chemical elements
NT: Iodine compounds

Iodine compounds

UF: Iodides
Iodine monofluoride
BT: Iodine

Iodine monofluoride

USE: Iodine compounds

IoHT

USE: Internet of Medical Things

Ion accelerators

BT: Particle accelerators
RT: Ion beam effects
Ion beams
Ion sources
Ions
Proton accelerators

Ion implantation

BT: Ion beam applications
Materials preparation
RT: Plasma sources
Semiconductor device

manufacture

NT: Plasma immersion ion
implantation

Ion optics

USE: Particle beam optics

Ion radiation effects

BT: Radiation effects
RT: Ionizing radiation
Proton radiation effects

Ion sources

BT: Ions
Nuclear physics
RT: Ion accelerators
Ion beams
Ion emission
Plasma sources

Ion beam applications

BT: Nuclear and plasma

sciences

RT: Ion beams
NT: Ion implantation

Ion trapping

USE: Trapped ions

Ion beam effects

BT: Ion beams
RT: Aerospace safety
Ion accelerators
Ion emission

Ionisation

USE: Ionization

Ionisation chambers

USE: Ionization chambers

Ionising radiation

USE: Ionizing radiation

Ion beams

BT: Particle beams



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Ionization

UF: Ionisation
Photoionisation
Photoionization
BT: Ions
RT: Discharges (electric)
Plasmas
Space radiation
NT: Impact ionization
Ionization chambers
Ionizing radiation
Single event transients
Single event upsets

Protons
Storage rings
NT: Ion sources
Ionization
Trapped ions

IOT

USE: Internet of Things

IOV

USE: Internet of Vehicles

IP

USE: IP networks

Ionization chambers

UF: Ionisation chambers
BT: Ionization
RT: Ionizing radiation
Smoke detectors

IP networks

UF: IP
IP-networks
Internet Protocol networks
BT: Communication systems
Computer networks
Telecommunications
RT: IPTV
Internet
Machine-to-machine

Ionized jet deposition

USE: Physical vapor deposition

communications

Ionizing radiation

UF: Ionising radiation
BT: Ionization
RT: Ion radiation effects
Ionization chambers
Radiation hardening

(electronics)

Silicon radiation detectors

Next generation networking
Quality of service
Transport protocols
TCP/IP

NT:

Ionizing radiation sensors

BT: Sensors
NT: Position sensitive particle
detectors
Radiation detectors
X-ray detectors

IP rights

USE: Intellectual property

IP telephony

USE: Internet telephony

IP-networks

USE: IP networks

Ionomeric polymer-metal composite actuators

USE: Actuators

IPCC

USE: Intergovernmental Panel on
Climate Change

Ionosphere

BT: Terrestrial atmosphere
RT: Meteorology
Plasmas

IPFS

USE: InterPlanetary File System

Ions

BT: Elementary particles
RT: Alpha particles
Elementary particle
exchange interactions
Ion accelerators
Ion beams
Ion emission

iPOD

USE: Portable media players

IPR

USE: Intellectual property



January 2026 IEEE Thesaurus

IPTV

UF: Internet protocol television
BT: Digital TV
RT: Broadband communication
Computer networks
IP networks
Internet
Local area networks
Protocols
Streaming media

Crops
Hydraulic systems
Water pumps

Itran 5

USE: Magnesium oxide

ISAC

USE: Integrated sensing and communication

IR 5.0

USE: Fifth Industrial Revolution

Ischemic pain

BT: Pain

IR Spectra

USE: Infrared spectra

ISDN

UF: Integrated services digital networks

BT: Communication systems
Digital communication
Digital systems

RT: Asynchronous transfer

IRE Standards

BT: IEEE Standards

Iridium

BT: Chemical elements

mode

Iris

BT: Eyes
RT: Ophthalmology

Data communication
Frame relay
Image communication
Multimedia communication
B-ISDN

NT:

Iris recognition

BT: Biometrics

Ishikawa diagrams

USE: Cause effect analysis

Irises

USE: Waveguide discontinuities

Islanding

BT: Power supplies
RT: Electrical safety
Generators

Iron

UF: Fe
BT: Metals
NT: Cast iron
Iron alloys
Iron oxide

ISO

UF: International organization for standardization

International standards organization

BT: Standards organizations
RT: Communication standards
ISO Standards
Measurement standards
Software standards
Standardization
Standards

NT: Moving Pictures Experts

Iron alloys

BT: Iron
RT: Alloying
Metallurgy
NT: Austenite

Iron oxide

BT: Iron

Irradiation

USE: Radiation effects

Group

ISO 9000

USE: Quality management

Irrigation

BT: Agriculture
RT: Agricultural products



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ISO Standards

BT: Standards publications
RT: ANSI Standards
Communication standards
ISO
Quality management
Software standards
Standardization
X3D
NT: MPEG standards

Isobaric processes

BT: Thermodynamics

Isolation technology

BT: Insulation
RT: Vibration control

Isolators

BT: Circuits

Isosurfaces

BT: Data visualization
RT: Biomedical imaging
Computational fluid

dynamics

Isothermal processes

BT: Thermodynamics

Isotopes

BT: Chemical elements
Nuclear physics
RT: Radioactive materials

Itemsets

BT: Data analysis
Transaction databases

Iterative algorithms

BT: Iterative methods
NT: Iterative closest point
algorithm
Sum product algorithm

Iterative closest point algorithm

UF: ICP
BT: Iterative algorithms

Iterative decoding

BT: Parity check codes

Iterative learning control

BT: Control theory
Iterative methods

RT: Adaptive control
Learning systems
Tracking

Iterative methods

BT: Mathematics
Numerical analysis
RT: Belief propagation
NT: Expectation-maximization

algorithms

Iterative algorithms
Iterative learning control

ITU

UF: International
Telecommunications Union
BT: Standards organizations

ITU Standards

BT: Standards publications
RT: UHDTV

J2EE

USE: Java

Jacks

USE: Lifting equipment

Jacobian matrices

UF: Jacobian matrix
BT: Matrices

Jacobian matrix

USE: Jacobian matrices

Jakarta EE

USE: Java

Jamming

BT: Electronic warfare
RT: Electronic countermeasures
Radar clutter
Radar countermeasures
Radio communication

countermeasures

Jaundice

UF: Icterus
BT: Medical conditions
RT: Liver diseases

Java

UF: J2EE
Jakarta EE
BT: High level languages



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	RT:	Webassembly		BT:	Human resource management
Jellyfish				RT:	Multiskilling Recruitment
	UF:	Sea jellies			
	BT:	Marine animals			
Jet engines			Jobs listings		
	BT:	Engines		BT:	Career development
	RT:	Aircraft propulsion Exhaust gases Fans		RT:	Employment
JFET circuits			<i>Johnson Nyquist noise</i>		
	BT:	FET circuits		USE:	Thermal noise
	NT:	JFET integrated circuits	Joining materials		
<i>JFET integrated circuit technology</i>				BT:	Production materials
USE:		JFET integrated circuits		RT:	Joining processes Soldering equipment
				NT:	Filler metals Sealing materials
JFET integrated circuits			Joining processes		
	UF:	JFET integrated circuit technology		UF:	Connecting Coupling (process) Fastening Linking
	BT:	JFET circuits		BT:	Manufacturing systems Materials processing
	RT:	JFETs		RT:	Couplings Fasteners Joining materials Plasma welding Soldering equipment
JFETs				NT:	Bonding processes Crimping Soldering Splicing Welding
	UF:	Junction FETs			
	BT:	Field effect transistors	<i>Joint ventures</i>		
	RT:	JFET integrated circuits		USE:	International collaboration
<i>Jigs</i>			Joints		
	USE:	Fixtures		BT:	Skeleton
Jitter				NT:	Ankle
	BT:	Distortion	<i>Josephson devices</i>		
	RT:	Circuit stability Quality of transmission Ring oscillators		USE:	Superconducting devices
	NT:	Timing jitter	Josephson effect		
Job design				BT:	Tunneling
	BT:	Ergonomics		RT:	Josephson junctions
Job production systems			Josephson junctions		
	UF:	Bespoke production		UF:	Josephson logic Superconducting junction devices
	BT:	Manufacturing systems			
<i>Job rotation</i>					
	USE:	Multiskilling			
Job shop scheduling					
	BT:	Scheduling			
	RT:	Genetic algorithms			
Job specification					



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BT:	Superconducting devices	Jupiter	BT:	Planets
RT:	Josephson effect			
<i>Josephson logic</i>		Just transition	BT:	Environmental economics
USE:	Josephson junctions			
Journalism		<i>k neighbor methods</i>	USE:	Nearest neighbor methods
BT:	Publishing			
RT:	Broadcasting	<i>k neighbour methods</i>	USE:	Nearest neighbor methods
	Electronic publishing			
	Humanities	<i>K-armed bandit problem</i>	USE:	Multi-armed bandit problem
	Information services			
	Multimedia communication	K-band	BT:	Microwave bands
	Radio broadcasting			
	Social networking (online)	<i>K-nearest neighbor</i>	USE:	Nearest neighbor methods
<i>JPEG</i>		<i>K-NN methods</i>	USE:	Nearest neighbor methods
USE:	Transform coding	<i>Ka band</i>	USE:	Ka-band
<i>JPEG2000</i>		Ka-band	UF:	Ka band
USE:	Transform coding	BT:	Microwave bands	
Judd-Ofelt theory		<i>Kaizen</i>	USE:	Continuous improvement
BT:	Spectral analysis	<i>Kalman filtering</i>	USE:	Kalman filters
RT:	Fluorescence			
	Photoluminescence	Kalman filters	UF:	Kalman filtering
<i>Junction detectors</i>			BT:	Filters
USE:	Semiconductor counters		RT:	Estimation
<i>Junction FETs</i>				Nonlinear dynamical
USE:	JFETs			systems
<i>Junction lasers</i>				Prediction methods
USE:	Semiconductor lasers			Sensor fusion
				Soft sensors
Junctionless nanowire transistors		<i>Kaons</i>	USE:	Mesons
BT:	MOSFET			
	Nanoelectronics	Karhunen-Loeve transforms	BT:	Transforms
RT:	Nanowires			
	Silicon-on-insulator	<i>KBO</i>	USE:	Kuiper belt
Junctions				
BT:	Semiconductor devices			
NT:	Heterojunctions			
	Hybrid junctions			
	P-n junctions			
	Waveguide junctions			
<i>Junk e-mail</i>				
USE:	Unsolicited e-mail			
<i>Junk email</i>				
USE:	Unsolicited e-mail			



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Kelvin

BT: Temperature measurement

Kernel

BT: Mathematics
Operating systems
NT: Null space
System kernels

Kernel machines

BT: Machine learning
algorithms
RT: Pattern analysis
Support vector machines

Kerr effect

BT: Electro-optic effects
RT: Cross-phase modulation
Magnetooptic effects

Key performance indicator

UF: KPI
BT: Measurement
Performance evaluation

Keyboards

BT: Computer peripherals
RT: Ergonomics

Keystroke dynamics

BT: Biometrics

Keyways

BT: Plugs

Keyword search

UF: Keyword searches
Keyword searching
BT: Search methods
RT: Indexing

Keyword searches

USE: Keyword search

Keyword searching

USE: Keyword search

Kidney

BT: Urogenital system
RT: Nephrology
NT: Chronic kidney disease
Kidney stones

Kidney stones

UF: Nephrolithiasis
Renal calculi
Urinary calculosis
BT: Kidney
Medical conditions
RT: Lithotripsy
Nephrology

Kilns

BT: Furnaces
RT: Calcination
Curing
Firing
Heat treatment

Kilowatt hour

USE: kWh

Kindle

USE: Consumer electronics AND
Electronic publishing

Kinematic analysis

USE: Kinematics

Kinematic faults

USE: Kinematics

Kinematic model

USE: Kinematics

Kinematic noise

USE: Kinematics

Kinematics

UF: Kinematic analysis
Kinematic faults
Kinematic model
Kinematic noise
BT: Mechanical factors
RT: Medical robotics
NT: Motion capture
Motor coordination

Kinetic energy

BT: Kinetic theory
RT: Mechanical energy
Potential energy
Thermal energy

Kinetic molecular theory

USE: Kinetic theory



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Kinetic theory

UF: Collision theory
Kinetic molecular theory
Kinetic-molecular theory
Kinetics
BT: Motion control
Physics
NT: Kinetic energy

Kinetic-molecular theory

USE: Kinetic theory

Kinetics

USE: Kinetic theory

Kirchhoff approximation

USE: Kirchhoff's Law

Kirchhoff current law

USE: Kirchhoff's Law

Kirchhoff law

USE: Kirchhoff's Law

Kirchhoff scattering

USE: Kirchhoff's Law

Kirchhoff's Law

UF: Kirchhoff approximation
Kirchhoff current law
Kirchhoff law
Kirchhoff scattering
BT: Spectroscopy

Kirk effect

USE: Kirk field collapse effect

Kirk field collapse effect

UF: Kirk effect
BT: Bipolar transistors

Klystrons

UF: Gyroklystrons
BT: Electron tubes
RT: Amplifiers
Cavity resonators
Colliding beam accelerators
Oscillators
Relativistic effects
Ring oscillators

Knee

BT: Extremities

Knee arthroplasty

USE: Knee replacement

Knee joint replacements

USE: Prosthetics

Knee replacement

UF: Knee arthroplasty
BT: Prosthetics

Knitted fabric composites

USE: Fabrics

KNN methods

USE: Nearest neighbor methods

Knowledge acquisition

BT: Knowledge engineering
RT: Context awareness
Econophysics
Expert systems
Knowledge based systems
Self-organizing feature

maps

NT: Subject matter experts

Knowledge based systems

UF: Knowledge systems
Knowledge-based systems
Rule based systems
BT: Artificial intelligence
RT: Decision support systems
Deductive databases
Intelligent systems
Knowledge acquisition
Knowledge representation
Linked data
Software agents
NT: Expert systems
Mobile agents

Knowledge discovery

BT: Knowledge engineering
RT: Data mining
Data science
Information retrieval
Knowledge management
NSL-KDD
Subject matter experts



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Knowledge engineering

BT: Artificial intelligence
RT: Knowledge management
NT: Inference mechanisms
Knowledge acquisition
Knowledge discovery
Knowledge representation

Knowledge graphs

BT: Knowledge representation

Knowledge management

UF: Intellectual capital
BT: Computer applications
Management
RT: Competitive intelligence
Information management
Knowledge discovery
Knowledge engineering
Management information

systems

NT: Semantic Web
Knowledge transfer

Knowledge representation

BT: Knowledge engineering
RT: Answer set programming
Expert systems
Formal concept analysis
Graph drawing
Knowledge based systems
Lexicon
Linked data
OWL
NT: Description logic
Fuzzy cognitive maps
Knowledge graphs
Ontologies
Thesauri

Knowledge systems

USE: Knowledge based systems

Knowledge transfer

BT: Information management
Knowledge management
RT: Cross lingual
Information diffusion

Knowledge-based systems

USE: Knowledge based systems

Kohonen maps

USE: Self-organizing feature

maps

KPI

USE: Key performance indicator

Krill herd algorithm

UF: Krill Herd optimization
algorithm
BT: Algorithms
Optimization
RT: Particle swarm optimization

Krill Herd optimization algorithm

USE: Krill herd algorithm

Krypton

BT: Chemical elements

Ku band

USE: Ku-band

Ku-band

UF: Ku band
BT: Microwave bands

Kuiper belt

UF: KBO
Kuiper belt objects
Kuiper belts
BT: Solar system

Kuiper belt objects

USE: Kuiper belt

Kuiper belts

USE: Kuiper belt

Kurtosis

BT: Probability distribution
RT: Heavily-tailed distribution

kWh

UF: Kilowatt hour
BT: Electricity

L-band

BT: Microwave bands

Lab-on-a-chip

UF: Lab-on-chip
BT: System-on-chip

Lab-on-chip

USE: Lab-on-a-chip



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Label swapping

USE: Multiprotocol label switching

Labeling

UF: Labelling
BT: Packaging
RT: Applicators
Packaging machines

Labelling

USE: Labeling

Labor productivity

USE: Productivity

Labor resources

UF: Labor supply
Labour resources
Labour supply
Manpower planning
BT: Human resource
management
RT: Personnel
Equal opportunities
Recruitment

Labor supply

USE: Labor resources

Laboratories

BT: Test facilities
RT: CERN
Engineering education
Research and development
Student experiments
NT: Remote laboratories

Labour productivity

USE: Productivity

Labour resources

USE: Labor resources

Labour supply

USE: Labor resources

Lacquers

BT: Chemical products
Coatings
Materials
RT: Paints

Lagrange duality

USE: Lagrangian functions

Lagrange functions

USE: Lagrangian functions

Lagrange relaxation

USE: Lagrangian functions

Lagrangian functions

UF: Lagrange duality
Lagrange functions
Lagrange relaxation
BT: Optimization methods
RT: Quantum mechanics

Lakes

BT: Geoscience
RT: Aquatic ecosystems
Reservoirs
Rivers
Sediments
Water
Water pollution
Water resources
Water storage
Wetlands

Laminates

BT: Materials
RT: Lamination

Lamination

BT: Materials processing
RT: Laminates

Lamps

BT: Lighting
RT: Light sources
Lighting control
Ultraviolet sources
NT: Discharge lamps
Electrodeless lamps
Filament lamps
Fluorescent lamps
LED lamps

LAN

USE: Local area networks

LAN emulation

UF: Local area network
emulation
BT: Optical fiber networks

LAN interconnection

BT: Internetworking
RT: Computer networks



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	Local area networks		Transceivers
	Metropolitan area networks	NT:	Mobile antennas
	Wide area networks		
	Wireless LAN		
<i>Land cover</i>		Land mobile radio networks	
USE:	Land surface	BT:	Cellular technology
<i>Land development</i>		<i>Land mobile radio service</i>	
USE:	Land use planning	USE:	Land mobile radio
<i>Land impacts</i>		Land pollution	
USE:	Land use planning	BT:	Pollution
		RT:	Industrial pollution
			Land use planning
			Oil pollution
			Pesticides
<i>Land mine detection</i>			Radioactive pollution
USE:	Landmine detection	NT:	Soil pollution
<i>Land mines</i>		Land surface	
USE:	Landmine detection	UF:	Land cover
Land mobile radio		BT:	Geoscience
UF:	Land mobile radio service	RT:	Land surface temperature
	Land-mobile radio	NT:	Watersheds
	Mobile radio		
BT:	Mobile communication	Land surface temperature	
	Radio communication	UF:	Ground temperature
RT:	5G mobile communication		Land temperature
	Ad hoc networks	BT:	Geoscience and remote
	Bluetooth	sensing	
	Channel estimation	RT:	Global warming
	Indoor navigation		Land surface
	Land mobile radio		Ocean temperature
equipment			Remote sensing
	Location awareness	<i>Land temperature</i>	
	Mobile antennas	USE:	Land surface temperature
	Mobile handsets		
	Multiuser detection	Land transportation	
	Personal area networks	UF:	Ground transportation
	Radio access networks		Planetary landers
	Routing protocols	BT:	Transportation
	Software radio	RT:	Global Positioning System
NT:	Cellular radio		Land vehicles
		NT:	Rail transportation
<i>Land mobile radio cellular systems</i>			Road transportation
USE:	Cellular radio		
Land mobile radio equipment		<i>Land use management</i>	
UF:	Land-mobile radio	USE:	Land use planning
equipment		Land use planning	
BT:	Radio communication	UF:	Land development
equipment			Land impacts
	Vehicular and wireless		Land use management
technologies		BT:	Environmental
RT:	Land mobile radio	management	
	Telephone equipment		



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RT: Floods
Land pollution
Reservoirs
Water storage
NT: Deforestation
Protected areas

Land vehicles

UF: Flexible fuel vehicles
Ground vehicles
BT: Vehicles
RT: Electric vehicles
Land transportation
Rail transportation
NT: Bicycles
Road vehicles

Land-mobile radio

USE: Land mobile radio

Land-mobile radio cellular systems

USE: Cellular radio

Land-mobile radio equipment

USE: Land mobile radio equipment

Landfills

BT: Waste disposal

Landline

UF: Home phone
Landline telephone
Main line
POTS
BT: Telephone equipment

Landline telephone

USE: Landline

Landmine detection

UF: Land mine detection
Land mines
Landmines
BT: Buried object detection
RT: Military equipment
Radar imaging
Remote sensing

Landmines

USE: Landmine detection

Landsat

BT: Earth Observing System
RT: NASA

Landslides

BT: Geology
RT: Hazards

Lane departure warning systems

BT: Road safety
Vehicle safety
RT: Collision avoidance
Lane detection

Lane detection

BT: Vehicle safety
RT: Collision avoidance
Lane departure warning

systems

Road safety

Lanthanum

BT: Metals
NT: Lanthanum compounds

Lanthanum compounds

BT: Lanthanum

Laparoscopes

BT: Surgical instruments
RT: Minimally invasive surgery

Laparoscopic surgery

USE: Minimally invasive surgery

Laplace equations

UF: Inverse transforms
Laplace operator
Laplace transform
Laplacian
BT: Mathematics

Laplace operator

USE: Laplace equations

Laplace transform

USE: Laplace equations

Laplacian

USE: Laplace equations

Lapping

BT: Machining
RT: Surface finishing

Laptops

USE: Microcomputers AND
Portable computers



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Large AI models

USE: Foundation models

Large Hadron Collider

UF: LHC
BT: Test facilities
RT: Particle accelerators

Large language models

UF: LLM
BT: Artificial intelligence
Text recognition
RT: Artificial neural networks
Deep learning
Neural networks
Prompt engineering
Recurrent neural networks
Self-supervised learning
Semisupervised learning
NT: Generative Pre-Trained
Transformer

Large scale integration

UF: LSI
Large-scale integration
BT: Circuits
Integrated circuits
NT: Ultra large scale integration
Very large scale integration
Wafer scale integration

Large screen displays

UF: Large-screen displays
BT: TV equipment

Large-scale integration

USE: Large scale integration

Large-scale systems

BT: System analysis and design
RT: Complex systems
Fuzzy systems

Large-screen displays

USE: Large screen displays

Larynx

UF: Voice tract
BT: Respiratory system

Laser ablation

BT: Laser applications

Laser applications

BT: Lasers
RT: CD recording
Endoscopes
Gyroscopes
Heating systems
Holography
Measurement by laser

beam

Optical recording
Photoacoustic effects
Stereolithography
NT: Dark states
Distributed feedback

devices

Laser ablation
Laser beam cutting
Laser beam machining
Laser fusion
Laser theory
Magnetooptic recording

Laser beam cutting

BT: Laser applications
RT: Laser beam machining

Laser beam machining

BT: Laser applications
Machining
RT: Laser beam cutting

Laser beams

UF: Laser guide stars
BT: Beams
RT: Bragg gratings
Electro-absorption

modulators

Electro-optic modulators
Laser theory
Lasers
Optical beams
Optical vortices
Refractive index
Supercontinuum generation
Thermal lensing
NT: Laser cooling
Laser pulses

Laser cavity resonators

BT: Cavity resonators
RT: Optical resonators
Surface emitting lasers



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Laser cladding

BT: Claddings
RT: Pulsed laser deposition

Laser cooling

BT: Cooling
Laser beams
RT: Trapped ions
Ultracold atoms

Laser deposition

USE: Pulsed laser deposition

Laser diodes

USE: Diode lasers AND
Semiconductor lasers

Laser excitation

UF: Electron beam pumping
Excitation of lasers
Pumping of lasers
BT: Lasers
NT: Optical pumping

Laser feedback

BT: Laser noise

Laser fusion

BT: Laser applications

Laser guide stars

USE: Laser beams

Laser materials processing

BT: Materials processing

Laser mode locking

BT: Laser modes

Laser modes

BT: Lasers
NT: Laser mode locking

Laser noise

BT: Noise
Optical signal processing
RT: Lasers and electrooptics
NT: Laser feedback

Laser physics

USE: Laser theory

Laser printers

BT: Printers

Laser pulses

UF: Laser pulsing
Pulsed lasers
BT: Laser beams
RT: High-speed optical
techniques
Lasers

Laser pulsing

USE: Laser pulses

Laser radar

UF: Lidar
Optical radar
BT: Radar
RT: Geophysical measurement
techniques
Optical scattering

Laser science

USE: Laser theory

Laser sintering

UF: Selective laser sintering
BT: Materials preparation
RT: Design automation
Prototypes
Stereolithography

Laser stability

BT: Lasers
RT: Stability analysis

Laser surgery

BT: Surgery

Laser theory

UF: Laser physics
Laser science
BT: Laser applications
RT: Laser beams
Lasers
Optical beams
Optical design
Optics
Particle beams
Quantum mechanics

Laser transitions

BT: Lasers

Laser tuning

BT: Semiconductor lasers
Tuning
RT: Optical tuning



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Laser velocimetry

BT: Measurement by laser
beam

Lasers

UF: Infrared lasers
BT: Lasers and electrooptics
RT: Laser beams
Laser pulses
Laser theory
Light sources
Nanobiophotonics
Optical distortion
Oscillators
Stereolithography
Stimulated emission
Superluminescent diodes
Threshold current
Ultraviolet sources
Waveguide lasers
NT: Atom lasers
Chemical lasers
Diode lasers
Free electron lasers
Gas lasers
Laser applications
Laser excitation
Laser modes
Laser stability
Laser transitions
Power lasers
Pump lasers
Quantum well lasers
Ring lasers
Semiconductor lasers
Solid lasers
Surface emitting lasers
X-ray lasers

Lasers and electrooptics

RT: Erbium
Laser noise
NT: Electro-optic devices
Electro-optic effects
Lasers
Optics
Optoelectronic devices
Photonics

Latches

BT: Bistable circuits

Lattice Boltzmann

USE: Lattice Boltzmann methods

Lattice Boltzmann methods

UF: Lattice Boltzmann
BT: Boltzmann distribution
Lattices
RT: Fluid dynamics

Lattices

UF: Optical lattices
BT: Mathematics
NT: Lattice Boltzmann methods

Launching (electromagnetic)

USE: Electromagnetic launching

Launching (electrothermal)

USE: Electrothermal launching

Lava

BT: Rocks
Volcanoes
RT: Magma
Volcanic ash

Law

UF: Legal aspects
BT: Legal factors
NT: Censorship
Commercial law
Consumer protection
Contract law
Criminal law
Employment law
Forensics
Fraud
Law enforcement
Patent law
Trademarks

Law enforcement

UF: Police
BT: Law
RT: Censorship
Digital forensics
Image forensics
Legal factors
Threat assessment

Layered division multiplexing

UF: LDM
Layered-division-
multiplexing
BT: Multiplexing
RT: Frequency division
multiplexing



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Layered manufacturing

BT: Manufacturing systems
RT: Computational geometry
Stereolithography

Layered media

USE: Nonhomogeneous media

Layered-division-multiplexing

USE: Layered division
multiplexing

Layout

BT: Graphics
RT: Art
Geometry
Integrated circuit layout
Wiring

LCA

USE: Life cycle assessment

LCD

USE: Liquid crystal displays

LCDs

USE: Liquid crystal displays

LCOE

USE: Levelized cost of energy

Lcos

USE: Liquid crystal on silicon

LDM

USE: Layered division
multiplexing

LDPA

USE: Log-periodic dipole
antennas

LDPC

USE: Parity check codes

Ldpc codes

USE: Parity check codes

Leaching

BT: Chemical processes

Lead

UF: Pb
BT: Metals
RT: Graphite

Lead compounds

NT: Lead isotopes

Lead acid batteries

UF: Lead-acid batteries
BT: Batteries

Lead compounds

BT: Compounds
RT: Lead

Lead isotopes

BT: Lead

Lead time reduction

BT: Production management
RT: Production planning
Project management

Lead zirconate titanate

UF: Lead zirconium titanate
PZT
BT: Ferroelectric materials
Piezoelectric materials

Lead zirconium titanate

USE: Lead zirconate titanate

Lead-acid batteries

USE: Lead acid batteries

Leadership

BT: Human resource
management
RT: Business
Government
Organizations

Leak detection

BT: Sensor systems and
applications
RT: Packaging
Testing
Vacuum systems

Leakage currents

UF: Gate leakage current
BT: Current
RT: Electron traps
Fault currents
NT: Gate leakage

Leaky wave antennas

UF: Leaky-wave antennas
BT: Antennas



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Leaky-wave antennas

USE: Leaky wave antennas

Lean production

BT: Manufacturing systems
Production systems
RT: Production management

Learning (artificial intelligence)

BT: Artificial intelligence
RT: AI accelerators
Conditional random fields
Data augmentation
Electronic learning
Fuzzy cognitive maps
Gaussian processes
Human activity recognition
Image annotation
Logistic regression
Manifold learning
Soft sensors
NT: Distance learning
Naive Bayes methods
Nearest neighbor methods

Learning automata

UF: Learning automaton
BT: Learning systems

Learning automaton

USE: Learning automata

Learning management systems

BT: Computer aided instruction
Learning systems
RT: Computer applications
Electronic learning
Management
Training

Learning mechanisms

USE: Learning systems

Learning methods

USE: Learning systems

Learning systems

UF: Learning mechanisms
Learning methods
Learning-based method
BT: Artificial intelligence
RT: Adaptive systems
Context awareness
Cybernetics
Deep learning

Dyslexia

Inference mechanisms
Iterative learning control
Mobile agents
Pattern recognition
Software agents
White matter
NT: Active learning
Backpropagation
Cognitive systems
Collective learning
Electronic learning
Hybrid learning
Immersive learning
Learning automata
Learning management

systems

Self-supervised learning
Semisupervised learning
Stability plasticity
Supervised learning
Unsupervised learning
Weak supervision

Learning-based method

USE: Learning systems

Least mean squares methods

BT: Least squares
approximations
Mean square error methods

Least squares approximations

UF: Least-squares
approximations
BT: Numerical analysis
RT: Approximation methods
Curve fitting
Mean square error methods
Optimization
Recursive estimation
NT: Least mean squares
methods

Least-squares approximations

USE: Least squares
approximations

LED

USE: Light emitting diodes

LED lamps

UF: AC light emitting diode
lamps
AC-LED lamps



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	Light emitting diode lamps		RT:	Micrometers
BT:	Lamps			Size measurement
	Light emitting diodes			
RT:	Light sources	<i>Lens</i>	USE:	Lenses
<i>LEDs</i>				
USE:	Light emitting diodes	Lenses	UF:	Lens
<i>Left handed materials</i>			BT:	Optical devices
USE:	Metamaterials		RT:	Focusing
				Optical materials
<i>Left-handed materials</i>				
USE:	Metamaterials	<i>LEO</i>	USE:	Low earth orbit satellites
<i>Legal aspects</i>				
USE:	Law	Lesions	BT:	Tumors
			NT:	Tissue damage
Legal factors				
BT:	Engineering management	Leukemia	BT:	Cancer
RT:	Censorship		RT:	Blood
	Ethical aspects			White blood cells
	Governmental factors			
	IEEE Corporate activities	Levee	UF:	Dike
	Law enforcement		BT:	Levee system
NT:	Copyright protection			Geoscience
	Law	<i>Levee system</i>	USE:	Levee
	Patents			
	Product liability			
	Software protection			
	Trademarks			
Legged locomotion				
UF:	Biped locomotion	Level control	UF:	Liquid level control
	Gait assessment		BT:	Mechanical variables
	Gait control	control	NT:	Gyroscopes
	Gait disorders			
	Walking			
BT:	Mobile robots	Level measurement	UF:	Liquid level measurement
RT:	Biological control systems		BT:	Geodesy
	Control systems			
	Motion control	Level set	BT:	Calculus
	Pedestrians		RT:	Gradient methods
	Stairs	<i>Levelized cost of electricity</i>	USE:	Levelized cost of energy
Legislation				
BT:	Government			
NT:	General Data Protection			
Regulation				
Legs		Levelized cost of energy	UF:	LCOE
BT:	Limbs			Levelized Energy Cost
RT:	Ankle			Levelized cost of electricity
			BT:	Power system
Length measurement		measurements		
BT:	Measurement			



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RT:	Costs	VLPR
	Power generation	Vehicle license plate
<i>Levelized Energy Cost</i>		recognition
USE:	Levelized cost of energy	BT:
		Vehicle plate detection
<i>Levels, energy</i>		Optical character
USE:	Energy states	recognition
		RT:
Levitation		Vehicle detection
BT:	Physics	Character recognition
NT:	Electrostatic levitation	Feature extraction
	Magnetic levitation	Image edge detection
		Image recognition
Lexicon		Licenses
BT:	Natural languages	UF:
RT:	Knowledge representation	BT:
NT:	Vocabulary	RT:
		Must-carry regulations
<i>LF noise</i>		<i>Licensing</i>
USE:	Low frequency noise	USE:
		Licenses
<i>LHC</i>		<i>Licensing (nuclear facilities)</i>
USE:	Large Hadron Collider	USE:
		Nuclear facility regulation
<i>Li</i>		<i>Lidar</i>
USE:	Lithium	USE:
		Laser radar
<i>li-fi</i>		Lie groups
USE:	Light fidelity	BT:
		Mathematics
		RT:
		Algebra
		Geometry
<i>Li-ion batteries</i>		<i>Life cycle analysis</i>
USE:	Lithium-ion batteries	USE:
		Life cycle assessment
<i>Li-S batteries</i>		Life cycle assessment
USE:	Lithium-sulfur batteries	UF:
		LCA
Libraries		Life cycle analysis
BT:	Information services	BT:
NT:	Software libraries	Environmental factors
		Product lifecycle
<i>License plate detection</i>		management
USE:	License plate recognition	Life estimation
		UF:
License plate recognition		Accelerated testing
UF:	ANPR	BT:
	Automatic number plate	Estimation
recognition		RT:
	Automatic-number plate	Aging
recognition		Failure analysis
	Car plate recognition	Fatigue
	LPD	Insulation life
	LPR	
	License plate detection	<i>Life long learning</i>
	Number plate recognition	USE:
		Continuing professional
		development



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Life sciences

UF: Computational life sciences
BT: Science - general
RT: Animals
Biology
Plants (biology)

Life sciences computing

USE: Computational modeling

Life testing

BT: Testing
RT: Reliability

Lifetime estimation

UF: Lifetime management
Lifetime measurement
Lifetime tests
BT: Measurement

Lifetime management

USE: Lifetime estimation

Lifetime measurement

USE: Lifetime estimation

Lifetime tests

USE: Lifetime estimation

lifi

USE: Light fidelity

Lifting equipment

UF: Hoists
Jacks
BT: Materials handling
equipment
RT: Freight handling
Materials handling
Pulleys
Winches
NT: Cranes

Ligaments

BT: Musculoskeletal system

Light attenuation

USE: Attenuation

Light deflectors

BT: Optical devices

Light emitters

USE: Light emitting diodes

Light emitting diode lamps

USE: LED lamps

Light emitting diodes

UF: LED
LEDs
Light emitters
Light-emitting diodes
BT: Diodes
Optoelectronic devices
RT: Microcavities
Molecular beam
applications
P-n junctions
Visible light communication
Inorganic light emitting
NT: Inorganic light emitting
diodes
LED lamps
Organic light emitting
diodes
Superluminescent diodes

Light fidelity

UF: li-fi
lifi
BT: Wireless LAN
RT: High-speed optical
techniques
IEEE 802.15 Standard
Optical fiber networks
Radio frequency
Visible light communication
Wireless communication
Wireless fidelity

Light fields

BT: Optics

Light interferometry

USE: Optical interferometry

Light polarisation

USE: Optical polarization

Light polarization

USE: Optical polarization

Light pollution

USE: Artificial light

Light rail systems

UF: Light railways
Streetcars
BT: Rail transportation
RT: Public transportation



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Light railways

USE: Light rail systems

lamps

High intensity discharge

Light scattering

BT: Scattering

RT: Optical scattering
Resonance light scattering
Turbidity

NT:

Light sources
Lighting control
Photometry
Quantum radar
Visible light communication
Artificial light
Daylighting
Electrical ballasts
Emergency lighting
Lamps
Lumen
Smart lighting
Solid state lighting

Light sources

BT: Optics

RT: Arc discharges
Color gamut
High intensity discharge

lamps

LED lamps
Lamps
Lasers
Lighting
Lighting control
Photometry
Supercontinuum generation
Synchrotron radiation
NT: Electroluminescent devices
Fast light
Luminescent devices
Phosphors
Slow light
Stray light
Superluminescent diodes
Ultraviolet sources

Lighting control

UF: Illumination control

BT: Optical control

RT: High intensity discharge

lamps

Lamps
Light sources
Lighting

Lightly tailed distribution

USE: Lightly-tailed distribution

Lightly-tailed distribution

UF: Light tailed distribution

Light-tailed distribution

Lightly tailed distribution

BT: Probability distribution

Light tailed distribution

USE: Lightly-tailed distribution

Light trapping

UF: Plasmonic solar cells

BT: Photovoltaic cells

RT: Reflectivity

Lightning

BT: Meteorology

RT: Dielectric breakdown

Electrostatic processes

Storms

NT: Lightning protection

Light-emitting diodes

USE: Light emitting diodes

Lightning protection

BT: Lightning

Protection

Light-tailed distribution

USE: Lightly-tailed distribution

Lightweight structures

BT: Structural shapes

RT: Aerospace engineering

Aerospace industry

Aerospace materials

Honeycomb structures

Metal foam

Sandwich structures

Thin wall structures

Lighting

UF: Arc lamps

Illumination

BT: Optical devices

RT: Building services

Buildings

Filament lamps

Fluorescent lamps

Gas discharge devices



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Limbic system

BT: Brain

Matrices

Vectors

Limbs

BT: Extremities
RT: Artificial limbs
Supernumery robotic limbs
NT: Arm
Hands
Legs

Linear antenna arrays

BT: Antenna arrays

Linear approximation

BT: Approximation methods
RT: Nonlinear equations
Nonlinear systems

Limit cycle

USE: Limit-cycles

Limit-cycle

USE: Limit-cycles

Limit-cycles

UF: Limit cycle
Limit-cycle
BT: Mathematics

Limiting

BT: Signal processing
RT: Nonlinear distortion
Voltage control

Linac

USE: Linear particle accelerator

LINACS

USE: Linear accelerators

Line enhancers

BT: Adaptive systems
RT: Digital filters
Filtering theory

Line output transformer

USE: Flyback transformers

Line-of-sight propagation

BT: Electromagnetic radiation

Linear accelerators

UF: LINACS
BT: Particle accelerators
RT: Collimators

Linear algebra

UF: Linear systems (algebraic)
BT: Algebra
RT: Eigenvalues and
eigenfunctions
NT: Linear programming

Linear circuits

BT: Circuits
RT: Ohmic contacts

Linear codes

BT: Block codes
RT: Error correction
NT: Polar codes

Linear discriminant analysis

UF: Linear discriminant
classification
BT: Statistics
RT: Machine learning

Linear discriminant classification

USE: Linear discriminant analysis

Linear feedback control systems

BT: Control systems
Cybernetics
RT: Linear systems
NT: Frequency locked loops
Phase locked loops
State feedback
Tracking loops

Linear feedback shift registers

BT: Shift registers

Linear filtering

USE: Maximum likelihood
detection

Linear frequency modulation

USE: Chirp modulation

Linear integrated circuits

USE: Analog integrated circuits

Linear matrix inequalities

BT: Mathematics
RT: Linear systems
Uncertain systems



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Linear parameter varying systems

USE: Linear systems

Linear particle accelerator

UF: Linac
Particle accelerator
BT: High energy physics

instrumentation computing

Linear predictive coding

BT: Prediction methods

Linear programming

UF: Linear-programming
BT: Linear algebra
RT: Algorithms
Microeconomics
Operations research
Optimization methods
NT: Data envelopment analysis
Integer programming

Linear regression

BT: Regression analysis
NT: Maximum likelihood linear

regression

Linear systems

UF: Linear parameter varying

systems

BT: Mathematics
RT: Control systems
Linear feedback control

systems

Linear matrix inequalities
Principal component

analysis

Time invariant systems
Transfer functions

NT: Sparse approximation

Linear systems (algebraic)

USE: Linear algebra

Linear-programming

USE: Linear programming

Linearisation techniques

USE: Linearization techniques

Linearity

BT: Electromagnetic

measurements

Linearization techniques

UF: Linearisation techniques
BT: Mathematics
RT: Control system synthesis
Control systems
MOSFET circuits
Modulation
Operational amplifiers
Transmitters

Linguistic indexing

USE: Image annotation

Linguistics

BT: Natural languages
RT: Anthropology
Cross lingual
Cultural competence
Semiotics
Translation
NT: Multilingual
Phonetics
Pragmatics

Link aggregation

BT: Telecommunication
network topology

Linkages

USE: Couplings

Linked data

BT: Internet
RT: Big Data
Database systems
Knowledge based systems
Knowledge representation
Metadata
NoSQL databases
Ontologies
Open data
Query processing
Semantic Web

Linkedin

USE: Social networking (online)

Linking

USE: Joining processes

Linux

BT: High level languages

Lipid bilayers

USE: Lipidomics



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Lipidomics

UF: Lipid bilayers
Lipids
BT: Molecular biomarkers
RT: Fats

Lipids

USE: Lipidomics

Lips

BT: Head
RT: Stomatognathic system

Liquefied natural gas

UF: LNG
BT: Natural gas

Liquid cooling

BT: Cooling
NT: Indirect liquid cooling

Liquid crystal devices

UF: Liquid-crystal devices
BT: Displays
RT: Electro-optic devices
Liquid crystals
Microdisplays
Thin film transistors
NT: Liquid crystal displays
Liquid crystal on silicon

Liquid crystal displays

UF: LCD
LCDs
Liquid-crystal displays
BT: Liquid crystal devices
NT: Active matrix liquid crystal

displays

Liquid crystal on silicon

UF: Lcos
BT: Liquid crystal devices
RT: Integrated optoelectronics
Microdisplays

Liquid crystal polymers

UF: Liquid-crystal polymers
BT: Polymers

Liquid crystals

BT: Crystals
RT: Liquid crystal devices

Liquid flow

USE: Fluid flow

Liquid insulation

USE: Dielectric liquids

Liquid level control

USE: Level control

Liquid level measurement

USE: Level measurement

Liquid nitrogen

BT: Cryogenics
Refrigerants

Liquid waveguides

BT: Hollow waveguides

Liquid-crystal devices

USE: Liquid crystal devices

Liquid-crystal displays

USE: Liquid crystal displays

Liquid-crystal polymers

USE: Liquid crystal polymers

Liquids

BT: Fluids
RT: Aerosols
Electrohydraulics
Materials science and

technology

NT: Spraying
Evaporation
Water

LiquiFerrofluid

USE: Ferrofluid

Literature review

USE: Systematic literature review

Lithium

UF: Li
BT: Metals
RT: Alloying
Batteries
NT: Lithium compounds

Lithium batteries

BT: Batteries
Lithium compounds

Lithium compounds

BT: Lithium
RT: Alloying



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		Batteries			Liver diseases		
	NT:	Lithium batteries			BT:	Liver	
		Lithium niobate			RT:	Jaundice	
<i>Lithium ion batteries</i>							
	USE:	Lithium-ion batteries			Liver neoplasms		
					BT:	Neoplasms	
Lithium niobate							
	BT:	Lithium compounds			Livestock		
					BT:	Animals	
					RT:	Agriculture	
Lithium-ion batteries							
	UF:	Li-ion batteries			<i>LLM</i>		
		Lithium ion batteries			USE:	Large language models	
	BT:	Batteries					
Lithium-sulfur batteries					<i>LNG</i>		
	UF:	Li-S batteries			USE:	Liquefied natural gas	
	BT:	Batteries					
Lithography					<i>Load balancing</i>		
	UF:	Photolithography			USE:	Load management	
	BT:	Manufacturing					
	RT:	Nanotechnology			<i>Load compensation</i>		
		Printing			USE:	Load management	
		Proximity effects			<i>Load composition</i>		
	NT:	Colloidal lithography			USE:	Load management	
		Extreme ultraviolet					
lithography		Interferometric lithography			Load flow		
		Nanolithography			UF:	Power flow	
		Soft lithography			BT:	Power system management	
		Stereolithography			NT:	Load flow analysis	
		X-ray lithography					
					Load flow analysis		
				UF:	Power flow analysis		
				BT:	Load flow		
				RT:	Power system security		
					Power transmission		
Lithotripsy							
	BT:	Medical treatment			Load flow control		
	RT:	Kidney stones			UF:	Power flow control	
		Lithotriptors			BT:	Power system control	
					NT:	Power factor correction	
Lithotriptors							
	BT:	Biomedical equipment			Load forecasting		
	RT:	Lithotripsy			BT:	Power demand	
					RT:	Load management	
						Load monitoring	
Livelihood							
	BT:	Employment			Load management		
		Human factors			UF:	Load balancing	
		Social factors				Load compensation	
						Load composition	
						Load variations	
Liver					BT:	Energy management	
	BT:	Digestive system			RT:	Energy storage	
	NT:	Liver diseases				Load forecasting	
Liver cancer							
	BT:	Cancer					



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	Pallets	Local area networks	
	Power demand	UF:	LAN
	Vehicle-to-grid	BT:	Communication systems
NT:	Load monitoring		Digital systems
	Load shedding	RT:	Distributed computing
Load modeling			Ethernet
BT:	Modeling		FDDI
	Power system modeling		Field buses
RT:	Power demand		File servers
			IEEE 802.3 Standard
Load monitoring			IPTV
BT:	Load management		Internetworking
	Monitoring		LAN interconnection
RT:	Energy consumption		Media Access Control
	Load forecasting	interconnection	Multiprocessor
	Power system control		Office automation
measurements	Power system		Open systems
	SCADA systems		Protocols
			Regional area networks
Load shedding			Storage area networks
BT:	Load management		Token networks
RT:	Power distribution	NT:	Virtual private networks
			Virtual LAN
			Wireless LAN
<i>Load tap changers</i>			
USE:	On load tap changers	<i>Local authorities</i>	
		USE:	Local government
<i>Load variations</i>			
USE:	Load management	Local field potentials	
		BT:	Extracellular
Loaded antennas			Signal processing
BT:	Antennas	RT:	Synapses
Loaded waveguides		Local government	
BT:	Electromagnetic	UF:	Local authorities
waveguides		BT:	Government
RT:	Dielectric materials		
	Waveguide discontinuities	Local oscillators	
		BT:	Oscillators
Loading			
BT:	Freight handling	<i>Localization</i>	
RT:	Containers	USE:	Location awareness
	Filling		
	Grippers	Location awareness	
	Pulleys	UF:	Geo tagging
			Localization
Loans and mortgages			Location based services
UF:	Mortgages		Location metadata
BT:	Financial management		Mobile location
		management	Mobile radio mobility
<i>Local area network emulation</i>		management	
USE:	LAN emulation	BT:	Mobile communication
		RT:	Cellular radio



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networks		Inter-cell interference		<i>Logic circuit testing</i>	
		Land mobile radio		USE:	Logic testing
		Mobile computing			
		Navigation		Logic circuits	
		Odometry		BT:	Circuits
		Personal communication			Switching circuits
				RT:	Adders
					Computers and information
		Position measurement		processing	
		Visual place recognition			Counting circuits
		Wireless communication			Digital circuits
	NT:	Network location			Digital integrated circuits
awareness					Flip-flops
					Logic
					Logic design
					Logic devices
					Logic functions
					Multiplying circuits
					Pulse inverters
					Shift registers
				NT:	Combinational circuits
					Logic arrays
					Programmable logic arrays
					Superconducting logic
				circuits	
				Logic design	
				UF:	Circuit design (logic)
					Logic CAD
				BT:	Design automation
				RT:	Circuit synthesis
					Design for testability
					Design methodology
					Engineering education
					Logic circuits
					Timing
				NT:	Reconfigurable logic
				Logic devices	
				BT:	Circuits and systems
				RT:	Logic circuits
				NT:	Logic gates
					Programmable logic
				devices	
				Logic functions	
				BT:	Boolean functions
				RT:	Logic
					Logic circuits
					Multivalued logic
				Logic gates	
				BT:	Logic devices
				RT:	Boolean algebra
					Ring oscillators



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Logic inverters

USE: Pulse inverters

Logic programming

BT: Programming
NT: Constraint handling

Logic test

USE: Logic testing

Logic testing

UF: Logic circuit testing
Logic test
BT: Integrated circuit testing
RT: Design for testability

Logical decomposition

USE: System analysis and design

Logistic regression

UF: Logit models
Ordered logit models
BT: Regression analysis
RT: Learning (artificial intelligence)
Pattern classification
Support vector machines

Logistics

UF: Physical distribution
management
BT: Production management
RT: Procurement
NT: Reverse logistics
Supply chains

Logit models

USE: Logistic regression

Long range

USE: LoRa

Long range WAN

USE: LoRaWAN

Long short term memory

UF: LSTM
Long short-term memory
BT: Artificial neural networks
RT: Machine learning
NT: Bidirectional long short term

memory

Long short-term memory

USE: Long short term memory

Long Term Evolution

UF: LTE
LTE advanced
BT: 3GPP Standards
Communication standards
RT: 4G mobile communication
High-speed networks
Mobile communication
Mobile handsets
Wireless communication

Long-term climate strategies

BT: Corporate social responsibility
RT: Climate change

Look-up table

USE: Table lookup

Lookup table

USE: Table lookup

Loop antennas

BT: Antennas

Loop-filtering algorithm

USE: Filtering algorithms

LOPT

USE: Flyback transformers

LoRa

UF: Long range
BT: Data communication
Radio communication
RT: Wide area networks
NT: LoRaWAN

Loran

USE: Radio navigation

LoRaWAN

UF: Long range WAN
BT: LoRa
Protocols

Lorentz covariance

UF: Lorentz force
Lorentz invariance
BT: Physics

Lorentz force

USE: Lorentz covariance



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Lorentz invariance

USE: Lorentz covariance

Loss and damage

BT: Environmental economics

RT: Climate change

Loss measurement

BT: Measurement

RT: Attenuation measurement

Magnetic losses

Optical losses

NT: Packet loss

Lot sizing

BT: Production control

RT: Materials requirements

planning

Loudspeakers

BT: Audio systems

RT: Acoustic distortion

Low blood pressure

USE: Hypotension

Low carbon economy

UF: Low fossile fuel economy
Low-carbon economy

BT: Power system economics

RT: Carbon

Decarbonization

Ecosystems

Electrification

Environmental

management

Global warming

Green hydrogen

Greenhouse effect

Renewable energy sources

Low density parity check codes

USE: Parity check codes

Low earth orbit satellites

UF: LEO

Low-earth-orbit

BT: Artificial satellites

Low fossile fuel economy

USE: Low carbon economy

Low frequency noise

UF: LF noise

Low-frequency noise

BT: Noise

Low latency communication

UF: Low-latency communication

BT: Communication systems

NT: Ultra reliable low latency

communication

Low noise amplifiers

USE: Low-noise amplifiers

Low pass filters

USE: Low-pass filters

Low power electronics

UF: Low-power electronics

Ultra low power*

Ultra-low power*

BT: Consumer electronics

RT: Electronic equipment

Nanogenerators

Simultaneous wireless

information and power transfer

Low temperature plasmas

USE: Low-temperature plasmas

Low voltage

UF: Low-voltage

BT: Voltage measurement

Low-carbon economy

USE: Low carbon economy

Low-earth-orbit

USE: Low earth orbit satellites

Low-frequency noise

USE: Low frequency noise

Low-latency communication

USE: Low latency communication

Low-noise amplifiers

UF: Low noise amplifiers

BT: Amplifiers

Low-pass filters

UF: Low pass filters

Lowpass filters

BT: Filters

Low-power electronics

USE: Low power electronics



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Low-power wide area networks

UF: LPWAN
BT: Wide area networks

Low-temperature plasmas

UF: Low temperature plasmas
BT: Plasmas
RT: Plasma applications

Low-voltage

USE: Low voltage

Lower bound

BT: Boundary conditions
RT: Upper bound

Lowpass filters

USE: Low-pass filters

LPD

USE: License plate recognition

LPR

USE: License plate recognition

LPWAN

USE: Low-power wide area networks

LSI

USE: Large scale integration

LSTM

USE: Long short term memory

LTE

USE: Long Term Evolution

LTE advanced

USE: Long Term Evolution

Lubricants

UF: Cutting fluids
BT: Production materials
RT: Lubrication

Lubricating oils

UF: Oil filters
Oiling (lubrication)
BT: Oils

Lubrication

BT: Mechanical factors
RT: Friction
Lubricants

Mechanical bearings

Lumber industry

UF: Timber industry
BT: Industries

Lumen

BT: Lighting
Measurement

Luminescence

BT: Optics
RT: Luminescent devices
Muon colliders
Scintillators
NT: Chemiluminescence
Electroluminescence
Fluorescence
Phosphorescence
Photoluminescence
Thermoluminescence

Luminescent devices

BT: Light sources
Optical devices
RT: Luminescence
Muon colliders
NT: Electroluminescent devices

Lunar

USE: Moon

Lung cancer

BT: Cancer

Lung diseases

USE: Pulmonary diseases

Lung neoplasms

BT: Neoplasms

Lungs

BT: Respiratory system
RT: Acute respiratory distress
syndrome
Pulmonary diseases
Pulmonology
Spirometry

LUT

USE: Table lookup

Lutecium

USE: Lutetium



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Lutetium

UF: Lutecium
BT: Chemical elements

Lyapunov function

USE: Lyapunov methods

Lyapunov methods

UF: Lyapunov function
Lyapunov stability
BT: System analysis and design
RT: Control design
Functional analysis
Stability

Lyapunov stability

USE: Lyapunov methods

Lymph nodes

BT: Lymphatic system

Lymphatic system

BT: Anatomy
NT: Lymph nodes
Spleen

Lymphocytes

UF: B-cells
T-cells
BT: White blood cells
RT: Immune system

M2M

USE: Machine-to-machine communications

MaaS

USE: Mobility as a service

MAC

USE: Media Access Control

MAC protocol

USE: Media Access Control

Mach-Zehnder interferometers

UF: Mach-Zehnder modulation
BT: Interferometers

Mach-Zehnder modulation

USE: Mach-Zehnder interferometers

Machine added indexing

USE: Machine assisted indexing

Machine aided indexing

USE: Machine assisted indexing

Machine assisted indexing

UF: Automated indexing
Automatic indexing
MAI
Machine added indexing
Machine aided indexing
Machine indexing
Machine-added indexing
Machine-aided indexing
Machine-assisted indexing
BT: Indexes
RT: Indexing

Machine components

BT: Machinery
Mechanical products
RT: Couplings
Engines
Gears
Turbomachinery
Wheels
NT: Air cleaners
Belts
Cams
Engine cylinders
Exhaust systems
Impellers
Intake systems
Manifolds
Mechanical splines
Pistons
Rotors
Shafts
Valves

Machine control

BT: Industrial electronics
NT: Machine vector control

Machine ethics

UF: Computational ethics
Computational morality
Machine morality
BT: Ethics
Technology
RT: Artificial intelligence
Philosophical

considerations

Machine indexing

USE: Machine assisted indexing



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Machine intelligence

BT: Computational and artificial
intelligence
RT: Machine-to-machine
communications
NT: Pattern analysis

Machine learning

UF: Dictionary learning
Machine-learning
BT: Artificial intelligence
RT: Behavior recognition
Bio-inspired computing
Cognitive systems
Convolutional neural

networks

Cross lingual
Data augmentation
Deep architecture
Energy informatics
Foundation models
Generative adversarial

networks

Image augmentation
Linear discriminant analysis
Long short term memory
Naive Bayes methods
Photorealistic images
Predictive analytics
Radiomics
Text detection
Text summarization
Text to image
Text to video
Weak supervision
NT: Adversarial machine

learning

Association rule learning
Attention mechanisms
Automated machine

learning

Boosting
Concept drift
Deep learning
Deep reinforcement

learning

Diffusion models
Dimensionality reduction
Ensemble learning
Explainable AI
Federated learning
Few shot learning
Generative Pre-Trained

Transformer

High dimensional data

optimization

Hyperparameter

Imitation learning
Incremental learning
Model compression
Multi-armed bandit problem
One shot learning
Random forests
Reinforcement learning
Relevance vector machines
Representation learning
Robot learning
Statistical learning
Tiny machine learning
Transfer learning

Machine learning algorithms

BT: Algorithms
NT: Kernel machines

Machine listening

UF: Audio signal processing
Computational auditory

scene analysis

BT: Artificial intelligence

Machine morality

USE: Machine ethics

Machine shops

BT: Production facilities
RT: Machine tools
Machinery production

industries

Machining

Machine tool spindles

UF: Spindle bearings
BT: Machine tools
RT: Mechanical splines
Shafts

Machine tools

BT: Production equipment
RT: Clamps
Coordinate measuring

machines

Cutting tools
Fixtures
Gears
Hand tools
Machine shops
Machining
Manufacturing
Mechanical guides



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NT: Turning
Dies
Drilling machines
Grinding machines
Machine tool spindles
Metalworking machines
Milling machines
Presses
Sawing machines

Machine translation

BT: Computational linguistics
Natural language
processing
RT: Translation
NT: Neural machine translation

Machine vector control

BT: Machine control
RT: AC-DC power converters
DC-DC power converters

Machine vision

UF: Vision systems
(nonbiological)
BT: Image processing
RT: Automatic optical inspection
IEEE 1394 Standard
Image analysis
Image recognition
Manufacturing automation
Observers
Pattern recognition
Stereo vision
Visual systems
NT: Object recognition
Object segmentation

Machine windings

BT: Windings

Machine-added indexing

USE: Machine assisted indexing

Machine-aided indexing

USE: Machine assisted indexing

Machine-assisted indexing

USE: Machine assisted indexing

Machine-learning

USE: Machine learning

Machine-to-machine communications

UF: M2M
BT: Communication systems
RT: Fourth Industrial Revolution
IP networks
Industrial Internet of Things
Internet of Things
Machine intelligence
Remote monitoring
Tactile Internet
Wireless communication
Wireless sensor networks
NT: Massive machine type
communications

Machinery

BT: Industry applications
RT: Machinery production
industries

Machining
Materials handling

equipment

NT: Production equipment
Agricultural machinery
Ball bearings
Belts
Drives
Electric machines
Fans
Furnaces
Gears
Hydraulic systems
Machine components
Motors
Printing machinery
Pumps
Textile machinery

Machinery production industries

BT: Manufacturing industries
RT: Machine shops
Machinery

Machining

BT: Materials processing
RT: Burnishing
Clamps
Deburring
Finishing
Machine shops
Machine tools
Machinery
Manufacturing
NT: Boring
Drilling



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Electrochemical machining
Hobbing machines
Lapping
Laser beam machining
Milling
Planing
Sawing
Turning
Virtual machining

Macrocell networks

UF: Macrocells
BT: Cellular networks
RT: Rural areas

Macrocells

USE: Macrocell networks

Macroeconomics

BT: Economics
RT: Government
International trade
Public finance
NT: Privatization

Macular degeneration

BT: Eye diseases
Visual impairment

Maglev

USE: Magnetic levitation

Maglev trains

USE: Magnetic levitation vehicles

Maglev transportation

USE: Magnetic levitation vehicles

Maglev vehicles

USE: Magnetic levitation vehicles

Magma

BT: Rocks
RT: Lava

Magnesium

UF: Mg
BT: Metals
NT: Magnesium compounds

Magnesium compounds

BT: Magnesium
NT: Magnesium oxide

Magnesium oxide

UF: Irtran 5
MgO
BT: Magnesium compounds
RT: Ceramics
Optical materials

Magnetic analysis

BT: Magnetics
RT: Electromagnetic analysis
Magnetic fields
NT: Magnetization

Magnetic anisotropy

BT: Magnetics
NT: Magnetic domain walls
Magnetic domains
Magnetic moments
Perpendicular magnetic

anisotropy

Magnetic anomaly detection

BT: Magnetic variables
measurement
RT: Magnetic fields
Object detection

Magnetic anomaly detectors

BT: Magnetometers
RT: Magnetic fields
Military equipment
Object detection

Magnetic bearings

USE: Magnetic levitation

Magnetic circuits

BT: Circuits
RT: Coils
Magnetic devices
Windings

Magnetic communication

BT: Communication systems
RT: Electromagnetic induction
Near field communication

Magnetic confinement

BT: Plasma confinement
RT: Electromagnets
Fusion power generation
Tokamak devices
Tokamaks



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Magnetic core losses

USE: Magnetic losses

Magnetic cores

BT: Magnetic devices
RT: Inductors
NT: Transformer cores

Magnetic devices

BT: Magnetics
RT: Magnetic circuits
Magnetic materials
NT: Accelerator magnets
Ferrite devices
Magnetic cores
Magnetic gears
Magnetic heads
Magnetic memory
Magnetic modulators
Magneto optic devices
Magnetoresistive devices
Magnetostriuctive devices
Solenoids
Transformer cores
Undulators

Magnetic domain walls

BT: Magnetic anisotropy

Magnetic domains

BT: Magnetic anisotropy

Magnetic field induced strain

BT: Magnetomechanical effects
RT: Ferroelectric films
Ferroelectric materials
MISFETs
Semiconductor diodes
Semiconductor films
Semiconductor-metal

interfaces

Magnetic field measurement

BT: Magnetic variables

measurement

RT: Magnetic fields

Magnetic fields

BT: Magnetics
RT: Biomagnetics
Compass
Electromagnetic fields
Magnetic analysis
Magnetic anomaly

detection

detectors

measurement

Magnetic anomaly

Magnetic field

Magnetic levitation vehicles
Maxwell equations
Remanence
SQUID magnetometers
Saturation magnetization
Synchrotrons
NT: Geomagnetism
Magnetic reconnection
Magnetic separation
Magnetostatics
Toroidal magnetic fields

Magnetic films

UF: Magnetic thin films
BT: Films
Magnetic materials
RT: Thin films
NT: Ferrimagnetic films
Ferrite films
Garnet films

Magnetic filters

USE: Magnetic separation

Magnetic fluids

USE: Magnetic liquids

Magnetic flux

BT: Magnetics
RT: Remanence
NT: Flux pinning
Magnetic flux density
Magnetic flux leakage

Magnetic flux density

BT: Magnetic flux

Magnetic flux leakage

BT: Magnetic flux
Nondestructive testing
RT: Corrosion
Pipelines

Magnetic force microscopy

BT: Magnetics
RT: Atomic force microscopy
Magnetic forces

Magnetic forces

BT: Magnetics
RT: Electromagnetic forces



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	Force	RT:	Eddy currents
	Magnetic force microscopy		Loss measurement
	Magnetic levitation		
NT:	Coercive force		
Magnetic gears		Magnetic materials	
BT:	Gears	BT:	Magnetics
	Magnetic devices		Materials
RT:	Electromagnetic devices	RT:	Biomagnetics
	Magnetic levitation		Boron alloys
	Permanent magnets		Magnetic devices
	Power transmission		Magnetoelasticity
	Variable speed drives		Magnetostriction
		NT:	Permeability
			Amorphous magnetic
Magnetic heads		materials	
BT:	Magnetic devices		Antiferromagnetic materials
RT:	Magnetic recording		Diamagnetic materials
	Magnetoresistive devices		Ferrimagnetic films
			Ferrimagnetic materials
			Ferrite films
Magnetic hysteresis			Ferrites
BT:	Magnetics		Ferrofluid
RT:	Hysteresis		Ferromagnetic materials
	Remanence		Garnet films
			Garnets
Magnetic levitation			Magnetic films
UF:	Maglev		Magnetic liquids
	Magnetic bearings		Magnetic semiconductors
BT:	Levitation		Magnetic superlattices
	Magnetics		Paramagnetic materials
RT:	Electromagnets		Soft magnetic materials
	Magnetic forces		
	Magnetic gears		
	Rail transportation		
NT:	Magnetic levitation vehicles	Magnetic measurements	
		USE:	Magnetic variables
		measurement	
Magnetic levitation vehicles		Magnetic memory	
UF:	Maglev trains	UF:	Magnetic storage
	Maglev transportation	BT:	Magnetic devices
	Maglev vehicles		Memory
BT:	Magnetic levitation	RT:	Magnetic recording
	Rail transportation	NT:	Floppy disks
RT:	Electromagnets		Hard disks
	High-speed rail		
transportation		Magnetic modulators	
	Magnetic fields	BT:	Magnetic devices
	Permanent magnets		Modulation
	Superconducting magnets		
Magnetic liquids		Magnetic moments	
UF:	Magnetic fluids	BT:	Magnetic anisotropy
BT:	Magnetic materials		
Magnetic losses		Magnetic multilayers	
UF:	Magnetic core losses	BT:	Magnetics
BT:	Magnetics	RT:	Coatings



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Magnetic nanoparticles

BT: Nanoparticles

Magnetic noise

BT: Magnetic recording

Magnetic particle imaging

BT: Tomography

RT: Medical diagnostic imaging

Magnetic particles

BT: Magnetics

RT: Biomagnetics

devices

Microelectromechanical

Nanomagnetics

Magnetic permeability

USE: Permeability

Magnetic properties

BT: Magnetics

Magnetic reconnection

BT: Magnetic fields

Magnetic recording

UF: Perpendicular recording

BT: Recording

RT: Magnetic heads

Magnetic memory

NT: Digital magnetic recording

Heat-assisted magnetic

recording

Magnetic noise

Magnetooptic recording

Microwave-assisted

magnetic recording

Perpendicular magnetic

recording

Shingled magnetic

recording

Magnetic resonance

BT: Resonance

Resonant frequency

RT: Ferroresonance

Magnetic resonance

imaging

NT: Antiferromagnetic

resonance

Ferromagnetic resonance

Nuclear magnetic

resonance

Paramagnetic resonance

Magnetic resonance elastography

BT: Magnetic resonance
imaging

Magnetic resonance fingerprinting

BT: Magnetic resonance
imaging

Magnetic resonance imaging

UF: Biomedical MRI

MRI

NMR imaging

Nuclear magnetic

resonance imaging

BT: Imaging

RT: Diagnostic radiography

Image reconstruction

Magnetic resonance

NT: Diffusion tensor imaging

Functional magnetic

resonance imaging

Magnetic resonance

elastography

Magnetic resonance

fingerprinting

Quantitative susceptibility

mapping

Magnetic semiconductors

BT: Magnetic materials

Semiconductor materials

Magnetic sensors

BT: Magnetics

Sensors

RT: Wearable sensors

NT: Spin valves

Magnetic separation

UF: Magnetic filters

BT: Magnetic fields

RT: Particle separators

Magnetic shielding

BT: Electromagnetic shielding

Magnetic stimulation

BT: Medical treatment

Magnetic storage

USE: Magnetic memory

Magnetic superlattices

BT: Magnetic materials

Superlattices



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Magnetic susceptibility

BT: Magnetics

Magnetic switching

BT: Magnetics

Magnetic thin films

USE: Magnetic films

Magnetic tunnel junctions

USE: Magnetic tunneling

Magnetic tunneling

UF: Magnetic tunnel junctions
Magnetic tunnelling
Spin-dependent tunneling
Spin-dependent tunnelling
BT: Magnetoelectric effects
Tunneling
RT: Magnetoelectronics
Spin polarized transport

Magnetic tunnelling

USE: Magnetic tunneling

Magnetic variables control

BT: Control systems

Magnetic variables measurement

UF: Magnetic measurements
BT: Measurement
NT: Magnetic anomaly

detection

measurement

Magnetic field
Magnetometers
Permeability measurement

Magnetics

NT: Biomagnetics
Demagnetization
Gyromagnetism
Magnetic analysis
Magnetic anisotropy
Magnetic devices
Magnetic fields
Magnetic flux
Magnetic force microscopy
Magnetic forces
Magnetic hysteresis
Magnetic levitation
Magnetic losses
Magnetic materials
Magnetic multilayers
Magnetic particles

Magnetic properties
Magnetic sensors
Magnetic susceptibility
Magnetic switching
Magnetization processes
Magnetoacoustic effects
Magnetoelectric effects
Magnetomechanical effects
Magneto optic effects
Magnets
Magnonics
Microwave magnetics
Nanomagnetics
Nonlinear magnetics
Remanence
Spin systems

Magnetisation

USE: Magnetization

Magnetisation processes

USE: Magnetization processes

Magnetisation reversal

USE: Magnetization reversal

Magnetization

UF: Magnetisation
BT: Magnetic analysis

Magnetization processes

UF: Magnetisation processes
BT: Magnetics
RT: Hysteresis
NT: Magnetization reversal
Saturation magnetization

Magnetization reversal

UF: Magnetisation reversal
BT: Magnetization processes

Magneto electrical resistivity imaging technique

UF: MERIT
BT: Imaging
RT: Geophysical measurement

techniques

Magneto-electronics

USE: Magnetoelectronics

Magneto-optical effects

USE: Magneto optic effects



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Magnetoacoustic effects

UF: Acoustomagnetic effects
BT: Magnetics
RT: Acoustics
Magnetomechanical effects

Magnetoelasticity

BT: Magnetomechanical effects
RT: Magnetic materials
NT: Magnetostriction

Magnetoelectric effects

BT: Magnetics
NT: Hall effect
Magnetic tunneling
Magnetoelectronics
Magnetoresistance
Spintronics

Magnetoelectronic devices

USE: Magnetoelectronics

Magnetoelectronics

UF: Magneto-electronics
Magnetoelectronic devices
BT: Magnetoelectric effects
RT: Magnetic tunneling
Magnetoresistive devices
NT: Spin polarized transport

Magnetoencephalography

BT: Biomagnetics
RT: Biomedical image
processing
Brain

Magnetofluid dynamics

USE: Magnetohydrodynamics

Magnetofluid dynamics

USE: Magnetohydrodynamics

Magnetohydrodynamic power generation

BT: Power generation
RT: Magnetohydrodynamics

Magnetohydrodynamics

UF: Hydromagnetics
MHD
Magnetofluid dynamics
Magnetofluid dynamics
BT: Dynamics
Hydrodynamics
Mechanical factors
RT: Electrohydraulics

Fluid flow
Magnetohydrodynamic

power generation

Magnetomechanical effects

UF: Piezomagnetic effects
BT: Magnetics
RT: Magnetoacoustic effects
Mechanical factors
Stress
NT: Magnetic field induced

strain

Magnetoelasticity
Magnetostriction

Magnetometers

UF: Magnetometry
BT: Magnetic variables
measurement
NT: Magnetic anomaly
detectors
SQUID magnetometers

Magnetometry

USE: Magnetometers

Magneto optic devices

UF: Photomagnetic devices
BT: Magnetic devices
RT: Magneto optic effects

Magneto optic effects

UF: Magneto-optical effects
Photomagnetic effects
BT: Magnetics
RT: Kerr effect
Magneto optic devices
Magneto optic recording
Optics
Ultracold atoms
NT: Faraday effect
Gyrotropism

Magneto optic recording

BT: Laser applications
Magnetic recording
RT: Magneto optic effects

Magnetoresistance

UF: Magnetoresistivity
BT: Magnetoelectric effects
RT: Magnetoresistive devices
Nanocontacts
Spin polarized transport



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NT: Anisotropic
magnetoresistance Colossal
magnetoresistance Enhanced
magnetoresistance Extraordinary
magnetoresistance Giant magnetoresistance
magnetoresistance Ordinary
magnetoresistance Tunneling
magnetoresistance

Magnetoresistive devices

UF: Magnetoresistors
BT: Magnetic devices
RT: Giant magnetoresistance
Magnetic heads
Magnetoelectronics
Magnetoresistance
Tunneling
magnetoresistance
Magnetoresistivity
USE: Magnetoresistance
Magnetoresistors
USE: Magnetoresistive devices

Magnetosphere

UF: Magnetotail
BT: Terrestrial atmosphere
RT: Geomagnetism

Magnetostatic waves

BT: Waves
RT: Electromagnetic
propagation
Magnetostatics

Magnetostatics

BT: Magnetic fields
RT: Magnetostatic waves

Magnetostriction

BT: Magnetoelasticity
Magnetomechanical effects
RT: Magnetic materials
Magnetostrictive devices

Magnetostrictive devices

BT: Magnetic devices
RT: Magnetostriction
Sensors

Magnetotail
USE: Magnetosphere

Magnetrons

BT: Electron tubes
RT: Relativistic effects
Sputtering

Magnets

BT: Magnetism
RT: Cobalt
Saturation magnetization
NT: Electromagnets
Micromagnetics
Permanent magnets

Magnonics

BT: Magnetism

MAI

USE: Machine assisted indexing

Mail

USE: Postal services

Mail (electronic)

USE: Electronic mail

Main frames

USE: Mainframes

Main line

USE: Landline

main secondary

USE: Main-secondary

Main-secondary

UF: main secondary
BT: Protocols

Mainframes

UF: Main frames
BT: Digital computers
RT: Microcomputers
Time sharing computer

systems

Maintenance

BT: Reliability

Maintenance engineering

UF: Repair
BT: Engineering - general
RT: Automatic testing



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	Availability		<i>Mammary neoplasms</i>
	Configuration management	USE:	Breast neoplasms
	Fault diagnosis		
	Inspection		Mammography
	Monitoring	BT:	Biomedical imaging
	Remaining life assessment	RT:	Medical tests
	Testing		
NT:	Maintenance management		<i>Man machine systems</i>
	Predictive maintenance	USE:	Human-machine systems
	Preventive maintenance		
	Systems support		<i>Man-machine interfaces</i>
		USE:	User interfaces
Maintenance management			
BT:	Maintenance engineering		<i>Man-machine systems</i>
	Technical management	USE:	Human-machine systems
<i>Major depressive disorder</i>			Management
USE:	Depression	UF:	Reliability management
		BT:	Business
Malaria		RT:	Analytic hierarchy process
BT:	Diseases		Ethical aspects
			Learning management
Maldistribution			systems
BT:	Reliability		Management information
			systems
<i>Malicious software</i>			Management training
USE:	Malware		Operations research
			Personnel
<i>Malignancy</i>			Productivity
USE:	Cancer	NT:	Asset management
			Best practices
<i>Malignant</i>			Building management
USE:	Cancer		systems
			Business continuity
Malignant tumors			Business process
BT:	Tumors		management
			Business process re-
Malware			engineering
UF:	Malicious software		Communication system
BT:	Software		operations and management
RT:	Anti-virus software		Conference management
	Cyber espionage		Content management
	Phishing		Contingency management
	Privacy		Contract management
	Security		Contracts
NT:	Computer viruses		Customer relationship
	Computer worms		management
	Ransomware		Dependability management
	Rootkit		Digital management
	Trojan horses		Distributed management
			Enterprise resource
Mammary glands			planning
BT:	Glands		Facilities management
			Financial management
			Governmental factors



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management	Human resource	Management training	BT: Training
	Information management	RT: Continuing education	Management
	Interface management		
	International collaboration	<i>MANET</i>	USE: Mobile ad hoc networks
	Knowledge management		
	Marketing management	Manganese	BT: Metals
	Organizational aspects	NT: Manganese alloys	Manganese oxide
	Outsourcing		
	Precision agriculture	Manganese alloys	BT: Manganese
	Process planning		
	Production management	Manganese oxide	BT: Manganese
	Program management		
	Project management		
	Public relations		
	Quality management		
	Requirements management		
	Research and development		
management	Resource management	Manifold learning	BT: Dimensionality reduction
	Risk analysis	RT: Learning (artificial intelligence)	
	Safety management		
	Security management	Manifolds	UF: Exhaust manifolds
	Storage management	BT: Machine components	
	Supply chain management	RT: Engines	Exhaust systems
	Technical management		Valves
	Technology management		
Management accounting		Manipulator dynamics	BT: Manipulators
BT: Financial management			
NT: Cost accounting		<i>Manipulator sensing systems</i>	USE: Robot sensing systems
Management information base		<i>Manipulator vision systems</i>	USE: Robot vision systems
UF: MIB			
BT: Computer network		Manipulators	BT: Robots
management			RT: Assembly
RT: Information systems			Assembly systems
Telecommunication			Control equipment
network management			Control systems
Virtual environments			Industrial control
Management information systems			Manufacturing automation
BT: Information systems			Materials handling
RT: Customer relationship			Mechanical variables
management			
Knowledge management			
Management			
Supply chain management			
NT: Portals			
<i>Management of innovation</i>		control	Medical robotics
USE: Innovation management			Motion control
			Nonlinear systems
			Position control



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	Service robots		Materials processing
	Servomechanisms		Production control
	Servosystems		Production engineering
	Telerobotics		Production facilities
NT:	End effectors		Production systems
	Manipulator dynamics		Productivity
	Micromanipulators		Soldering
			Stereolithography
			Welding
			Wheels
			Wire drawing
<i>Manipulators (nonrobotic)</i>		NT:	Assembly
USE:	Remote handling		Assembly systems
			Embossing
			Fabrication
<i>Manpower planning</i>			Fourth Industrial Revolution
USE:	Labor resources		Green manufacturing
			Lithography
Manuals			Manufactured products
UF:	Technical manuals		Manufacturing systems
BT:	Professional		Mass customization
communication			Smart manufacturing
RT:	Documentation		Tolerance analysis
	Training		
	Writing		
Manufactured products			
UF:	Counterfeit goods		
BT:	Manufacturing	Manufacturing automation	
RT:	Product customization	UF:	Factory automation
	Product design	BT:	Automation
	Product development		Industrial electronics
NT:	Ceramic products	RT:	Assembly
	Chemical products		Assembly systems
	Consumer products		Automatic optical inspection
	Electrical products		Industrial control
	Food products		Industrial robots
	Fuels		Machine vision
	Glass products		Manipulators
	Mechanical products		Mobile robots
	Metal products		Process control
	Paper products		Programmable control
	Paper pulp		Robots
	Plastic products	NT:	Computer aided
	Rubber products	manufacturing	
	Sports equipment		Computer integrated
	Textile products	manufacturing	
	Tools		Computer numerical control
	Windows	systems	Flexible manufacturing
Manufacturing			
BT:	Industry applications	<i>Manufacturing economics</i>	
RT:	Bonding	USE:	Industrial economics
	Business		
	Discrete-event systems	<i>Manufacturing facilities</i>	
	Industrial plants	USE:	Production facilities
	Machine tools		
	Machining		
	Materials handling		



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Manufacturing industries

BT: Industries
 NT: Aerospace industry
 Cement industry
 Ceramics industry
 Clothing industry
 Electrical products industry
 Electronics industry
 Food industry
 Footwear industry
 Fuel processing industries
 Glass industry
 Machinery production

industries

Metal product industries
 Plastics industry
 Pulp and paper industry
 Rubber industry
 Shipbuilding industry
 Textile industry
 Toy manufacturing industry

Manufacturing management

USE: Production management

Manufacturing process

USE: Manufacturing processes

Manufacturing processes

UF: Compliant mechanisms
 Froth flotation
 Manufacturing process
 BT: Manufacturing systems
 RT: Rapid prototyping
 NT: Sintering
 Social manufacturing

Manufacturing systems

BT: Manufacturing
 Production systems
 RT: Bleaching
 Industrial facilities
 Production engineering
 NT: Agile manufacturing
 Automobile manufacture
 Batch production systems
 Blanking
 Cellular manufacturing
 Flow production systems
 Food manufacturing
 Forging
 Glass manufacturing
 Integrated manufacturing

systems

systems

Intelligent manufacturing

Job production systems
 Joining processes
 Layered manufacturing
 Lean production
 Manufacturing processes
 Mass production
 Melt processing
 Pulp manufacturing
 Sheet metal processing
 Thermoforming
 Three-dimensional printing

Many core processing

USE: Multicore processing

Many core systems

USE: Multicore processing

Many valued logic

USE: Multivalued logic

Manycore computing

USE: Manycore processors

Manycore processors

UF: Manycore computing
 BT: Multicore processing
 RT: Program processors

Marine accidents

BT: Accidents
 RT: Marine pollution
 Marine safety
 Marine vehicles

Marine animals

UF: Ocean animals
 Sea animals
 Sea lions
 BT: Animals
 Marine ecosystems
 RT: Aquaculture
 NT: Dolphins
 Jellyfish
 Whales

Marine cables

USE: Underwater cables

Marine communications

USE: Maritime communications



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Marine ecosystems

BT: Aquatic ecosystems
NT: Marine animals
Marine vegetation

Marine engineering

BT: Engineering - general

Marine equipment

UF: Maritime technology
BT: Marine technology
RT: Marine robots

Marine navigation

BT: Navigation
Oceanic engineering and

marine technology

RT: Marine robots
Sea state
Underwater navigation

Marine pollution

BT: Water pollution
RT: Marine accidents
Oil pollution
Oils
Thermal pollution

Marine robots

UF: Underwater robots
BT: Marine vehicles
Robots
Underwater technology
RT: Autonomous underwater

vehicles

Boats
Marine equipment
Marine navigation
Military vehicles
Mobile robots
Oceanography
Rescue robots
Underwater vehicles

Marine safety

BT: Safety
RT: Marine accidents
Marine technology
Marine vehicles

Marine science

USE: Oceanography

Marine technology

UF: Ocean technology
BT: Oceanic engineering and

marine technology

RT: Excavation
Marine safety
Military systems
Oceans
Underwater vehicles
NT: Marine equipment
Marine transportation
Underwater cables
Underwater communication
Underwater equipment
Underwater structures
Underwater technology

Marine transportation

BT: Marine technology
RT: Global Positioning System
Seaports
NT: Marine vehicles

Marine vegetation

UF: Ocean vegetation
Sea vegetation
BT: Marine ecosystems
Vegetation
NT: Seaweed

Marine vehicles

UF: Maritime vehicles
Ships
BT: Marine transportation
RT: Marine accidents
Marine safety
Propellers
Seaports
NT: Amphibious robots
Boats
Marine robots
Underactuated surface

vessels

Underwater vehicles

Maritime communications

UF: Marine communications
BT: Communication networks

Maritime technology

USE: Marine equipment

Maritime vehicles

USE: Marine vehicles



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Market opportunities

BT: Marketing management
RT: Consumer behavior
Customer profiles
Disruptive innovation
Disruptive technologies
Emissions trading
Neuromarketing

NT: HTML
OWL
Page description languages
SGML
XML

Market research

BT: Customer relationship
management
Engineering management
RT: Brand management
Competitive intelligence
Consumer products
Customer satisfaction
Neuromarketing

Mars

BT: Planets

Mars Express

USE: Interplanetary exploration

Mars Odyssey

USE: Interplanetary exploration

Martensite

BT: Crystalline materials
Steel
RT: Smart materials

Marketing management

BT: Management
RT: Electronic commerce
Public relations
NT: Advertising
Brand management
Distribution strategy
Market opportunities
Mass customization
Promotion - marketing

Masers

UF: Microwave lasers
BT: Microwave devices
Microwave technology
RT: Atomic clocks
Relativistic effects
Ring oscillators
Stimulated emission
NT: Gyrotrons

Markov decision processes

BT: Markov processes

Markov network

USE: Markov random fields

MASH

USE: Multi-stage noise shaping

Mashing

USE: Mashups

Markov processes

BT: Stochastic processes
RT: Belief propagation
Dynamic programming
Hidden Markov models
Infinite horizon
Multi-armed bandit problem
Q-learning
NT: Markov decision processes
Markov random fields

Mashups

UF: Mashing
BT: Web services
World Wide Web

Markov random fields

UF: Markov network
BT: Markov processes

Masking threshold

BT: Psychoacoustic models

Mass customization

BT: Manufacturing
Marketing management

Markup languages

BT: Computer languages
RT: Cascading style sheets
Semantic Web

Mass production

BT: Manufacturing systems
RT: Production management

Mass spectrometry

USE: Mass spectroscopy



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Mass spectroscopy

UF: Mass spectrometry
BT: Spectroscopy

Massive machine type communications

UF: MMTC
mmtc
BT: Machine-to-machine communications
RT: Intelligent systems

Massive MIMO

BT: MIMO

Mast cells

BT: White blood cells
RT: Parasitic diseases

Mastectomy

BT: Surgery
RT: Breast cancer

Mastercard

USE: Credit cards

Masticatory muscles

BT: Stomatognathic system

Matched filters

BT: Filters
RT: Filtering theory

Matching pursuit algorithms

BT: Algorithms

Material properties

BT: Materials
NT: Biocompatibility
Creep
Elasticity
Elongation
Resilience
Rigidity

Material storage

BT: Materials, elements, and compounds
RT: Canning
Energy storage
NT: Bulk storage
Containers
Fuel storage
Secure storage
Stacking
Storage automation

Materials

BT: Materials, elements, and compounds
RT: Materials handling
Oxidation
NT: Acoustic materials
Additives
Aggregates
Amorphous materials
Auxetic materials
Bio-inspired materials
Biological materials
Biomedical materials
Building materials
Catalysts
Ceramics
Composite materials
Conducting materials
Corrosion inhibitors
Crystalline materials
Crystals
Dielectric materials
Films
Fluids
Hazardous materials
Inorganic materials
Intelligent materials
Lacquers
Laminates
Magnetic materials
Material properties
Media
Mesoporous materials
Metal foam
Metallic materials
Metamaterials
Nanostructured materials
Oils
Optical materials
Organic inorganic hybrid materials
Organic materials
Paints
Paper pulp
Petrochemicals
Phase change materials
Photoconducting materials
Plastics
Polymer foams
Polymer gels
Polymers
Production materials



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Quantum materials			Remote handling
Radioactive materials		equipment	
Raw materials			Winches
Resins			
Resists		<i>Materials handling systems</i>	
Semiconductor materials		USE:	Materials handling
Sheet materials			
Smart materials		Materials preparation	
Solids		UF:	Atmospheric sintering
Superconducting materials		BT:	Materials science and
Surfactants		technology	
Terahertz materials		RT:	Corrosion inhibitors
Textiles			Flame retardants
Thermoelectric materials		NT:	Doping
Waste materials			Firing
Wire			Ion implantation
			Laser sintering
			Sputtering
Materials handling			
UF:	Materials handling systems		
	Scrubbers		
BT:	Materials science and	Materials processing	
technology		UF:	Mineral processing
RT:	Canning	BT:	Production
	Compaction	RT:	Bonding
	Die casting		Canning
	Dispatching		Corrosion inhibitors
	Lifting equipment		Electrochemical deposition
	Manipulators		Fabrication
	Manufacturing		Finishing
	Materials		Foundries
	Mobile robots		Manufacturing
	Pipelines		Materials science and
	Radioactive waste	technology	
	Radioactive waste disposal		Metalworking machines
	Robots		Plasma welding
	Stacking		Soldering
NT:	Cleaning	NT:	Welding
	Decontamination		Annealing
	Freight handling		Bleaching
	Materials handling		Casting
			Coatings
equipment			Curing
	Pallets		Etching
	Remote handling		Heat treatment
	Sieving		Joining processes
			Lamination
Materials handling equipment			Laser materials processing
BT:	Materials handling		Machining
RT:	Machinery		Melt processing
	Production equipment		Plasma materials
	Waste handling equipment	processing	
NT:	Containers		Plating
	Grippers		Pressing
	Lifting equipment		Punching
	Pulleys		Refining
			Shearing



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	Smelting		Surfaces
	Softening		
	Swaging		
	Vapor deposition		
Materials reliability		Materials testing	
BT: Materials science and technology		BT: Materials science and technology	
RT: Reliability		NT: Testing	
RT: Green's function methods		NT: Accelerated aging	
		NT: Acoustic testing	
		NT: Adhesive strength	
		NT: Bonding forces	
		NT: Delamination	
		NT: Elastic recovery	
		NT: Nondestructive testing	
Materials requirements planning		Materials, elements, and compounds	
UF: MRP		NT: Atoms	
BT: Production planning		NT: Chemical elements	
RT: Lot sizing		NT: Compounds	
RT: Scheduling		NT: Material storage	
RT: Supply chain management		NT: Materials	
RT: Supply chains		NT: Materials science and technology	
NT: Bills of materials		NT: Metals	
<i>Materials science</i>			
USE: Materials science and technology			
Materials science and technology		<i>Maternity benefits</i>	
UF: Materials science		USE: Employee welfare	
BT: Materials, elements, and compounds			
RT: Austenite		Mathematical analysis	
RT: Crystals		BT: Mathematical models	
RT: Gases		NT: Accuracy	
RT: Liquids		NT: Formal concept analysis	
RT: Materials processing		NT: Fractional calculus	
RT: Solid-state physics		NT: Modal analysis	
RT: Solids			
NT: Absorption		Mathematical models	
NT: Aging		BT: Mathematics	
NT: Chemical analysis		RT: Modeling	
NT: Computational materials		RT: Artificial neural networks	
science		NT: Hindcasting	
		NT: Geometric modeling	
		NT: Mathematical analysis	
		NT: Optimization models	
		NT: Overfitting	
	Contamination	Mathematical programming	
	Degradation	BT: Mathematics	
	Filtration	RT: Optimization methods	
	Hysteresis		
	Impurities	Mathematics	
	Materials handling	RT: Bio-inspired computing	
	Materials preparation	RT: Econometrics	
	Materials reliability	RT: Maximum likelihood	
	Materials testing	detection	
	Metallurgy		
	Microstructure		
	Periodic structures		
	Pigmentation		
	Separation processes		
	Surface engineering		
			Metaheuristics
			STEM



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NT:	Viterbi algorithm	Random processes
	Waveguide theory	Root mean square
theory	Algebra	Sequences
	Algorithms	Set theory
	Arithmetic	Simulated annealing
	Azimuth	Smoothing methods
	Boundary value problems	Spirals
	Calculus	Statistics
	Chemical reaction network	Stochastic processes
		Superposition calculus
		Taylor series
		Tensors
eigenfunctions	Closed-form solutions	Topology
	Combinatorial mathematics	Transforms
	Computational efficiency	Transmission line matrix
	Conformal mapping	
	Convergence	
	Convex functions	Uncertain systems
	Cyclic redundancy check	Utility theory
	Dynamical systems	
	Eigenvalues and	
	Equations	Mathematics computing
	Estimation	BT: Computer applications
	Euclidean distance	RT: Graph drawing
	Finite difference methods	NT: MATLAB
	Finite element analysis	
	Fourier series	MATLAB
	Functional analysis	UF: Matrix laboratory
	Geometry	BT: Mathematics computing
	Gradient methods	RT: Computer aided instruction
	Graph theory	Numerical analysis
	Harmonic analysis	Simulation
	Hypercomplex	Software libraries
	Iterative methods	
	Kernel	Matrices
	Laplace equations	UF: Matrix algebra
	Lattices	BT: Linear algebra
	Lie groups	RT: Method of moments
	Limit-cycles	Mode matching methods
	Linear matrix inequalities	NT: Jacobian matrices
	Linear systems	Matrix decomposition
	Linearization techniques	Singular value
	Mathematical models	
	Mathematical programming	decomposition
	Method of moments	<i>Matrix algebra</i>
	Minimization	USE: Matrices
	Mode matching methods	
	Network theory (graphs)	Matrix converters
	Nonlinear equations	UF: Matrix convertors
	Nonlinear systems	BT: Power conversion
	Numerical analysis	RT: Power electronics
	Optimization	
	Piecewise linear techniques	<i>Matrix convertors</i>
	Predator prey systems	USE: Matrix converters
	Probability	
	Quaternions	



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Matrix decomposition

BT: Matrices
RT: Signal processing
Statistics

Matrix laboratory

USE: MATLAB

Matter waves

UF: De Broglie methods
De broglie hypothesis
BT: Waves

Max-min composition

USE: Minimax techniques

Maximum a posteriori estimation

UF: Maximum a posteriori
estimator
Maximum a posteriori
framework
Maximum a posteriori
method
Maximum a posteriori
probability
Maximum a-posteriori
Maximum aposteriori
BT: Estimation theory

Maximum a posteriori estimator

USE: Maximum a posteriori
estimation

Maximum a posteriori framework

USE: Maximum a posteriori
estimation

Maximum a posteriori method

USE: Maximum a posteriori
estimation

Maximum a posteriori probability

USE: Maximum a posteriori
estimation

Maximum a-posteriori

USE: Maximum a posteriori
estimation

Maximum aposteriori

USE: Maximum a posteriori
estimation

Maximum likelihood decoding

BT: Decoding

RT: Algorithms

Maximum likelihood detection

UF: Additive metric
Complexity constrained
detection
Linear filtering
BT: Algorithms
RT: Filtering theory
Mathematics
Probability
Statistics

Maximum likelihood estimation

UF: MLE
BT: Estimation
Statistics
RT: Set theory
Tracking

Maximum likelihood linear regression

BT: Linear regression

Maximum power point trackers

UF: MPPT
Maximum power point
tracking
BT: Solar power generation
RT: Inverters
Power conversion
Solar energy

Maximum power point tracking

USE: Maximum power point
trackers

Maxwell equations

BT: Equations
RT: Electric fields
Magnetic fields
Perfectly matched layers

Maxwell-Boltzmann distribution

UF: Maxwell-Boltzmann
statistics
BT: Probability distribution

Maxwell-Boltzmann statistics

USE: Maxwell-Boltzmann
distribution

MC-CDMA

USE: Multicarrier code division
multiple access



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<i>MCCDMA</i>				Testing
USE:	Multicarrier code division			Transducers
multiple access			NT:	Accelerometers
				Acoustic measurements
<i>MCDA</i>				Antenna measurements
USE:	MCDM			Anthropometry
				Area measurement
MCDM				Atmospheric
UF:	MCDA	measurements		Atomic measurements
	Multi-criteria decision			Bathymetry
analysis				Biomedical measurement
	Multi-criteria decision			Calorimetry
making				Coordinate measuring
BT:	Decision making			
RT:	Decision analysis	machines		Cytometry
	Decision support systems			Density measurement
	Operations research			Depth measurement
<i>MDDI</i>				Distance measurement
USE:	Musical instrument digital			Distortion measurement
interfaces				Doppler measurement
				Dosimetry
Mean field theory				Dynamic range
UF:	MFT			Electric variables
	Self-consistent field theory	measurement		
BT:	Statistical analysis			Electromagnetic
RT:	Probability	measurements		
	Random processes			Environmental
	Stochastic processes	measurement		
	Stochastic systems	measurements		Extraterrestrial
Mean square error methods				Fluid flow measurement
BT:	Approximation methods			Frequency measurement
RT:	Error analysis			Gain measurement
	Estimation theory			Gas chromatography
	Least squares			Geologic measurements
approximations				Geophysical measurements
NT:	Least mean squares			Height measurement
methods				Hydrophobicity
				Interferometry
Measles				Key performance indicator
BT:	Infectious diseases			Length measurement
RT:	Viruses (medical)			Lifetime estimation
				Loss measurement
Measurement				Lumen
UF:	Metrics			Magnetic variables
BT:	Instrumentation and	measurement		
measurement				Measurement by laser
RT:	Containers	beam		
	Data acquisition			Measurement errors
	Food security			Measurement techniques
	Instruments			Measurement uncertainty
	Measurement standards			Measurement units
	Phase frequency detectors			Mechanical variables
	Telemetry	measurement		



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	Micrometers		Measurement techniques
	Moisture measurement	BT:	Measurement
	Noise measurement	RT:	Measurement uncertainty
	Nuclear measurements	NT:	Calibration
	Optical variables		Dynamic equilibrium
measurement			
	Particle beam		Measurement uncertainty
measurements		BT:	Measurement
	Particle measurements	RT:	Estimation
	Performance evaluation		Measurement techniques
	Phase measurement		
	Plasma measurements	Measurement units	
	Pollution measurement	UF:	Units (measurement)
	Pressure measurement	BT:	Measurement
	Pulse measurements	RT:	Measurement standards
	Reflectometry	NT:	Crop yield
	Replicability		International System of
	Reproducibility of results	Units	
	Scintillation counters		Nanometers
	Sea state		
	Semiconductor device	Mechanical bearings	
measurement		BT:	Friction
	Sensitivity	RT:	Ball bearings
	Shape measurement		Lubrication
	Size measurement		Mechanical factors
	Software measurement	NT:	Rolling bearings
	Soil measurements		
	Spectral efficiency	Mechanical cables	
	Spectroscopy	UF:	Cables (mechanical)
	Thermal variables	BT:	Cables
measurement			
	Time measurement	Mechanical energy	
	UHF measurements	BT:	Mechanical systems
	Ultrasonic variables	RT:	Kinetic energy
measurement			Potential energy
	Viscosity		
	Wavelength measurement	Mechanical engineering	
	Wide area measurements	BT:	Engineering - general
	pH measurement	RT:	Mechanical products
			Precision engineering
			Pressure vessels
Measurement by laser beam		NT:	Dynamic response
BT:	Measurement		Mechanical power
RT:	Laser applications		
NT:	Laser velocimetry	transmission	
			Mechanical systems
Measurement errors			
BT:	Measurement	Mechanical factors	
RT:	Error analysis	UF:	Mechanical properties
	Statistical analysis	BT:	Physics
		RT:	Acoustic noise
Measurement standards			Electrostriction
BT:	Standards categories		Magnetomechanical effects
RT:	ISO		Mechanical bearings
	Measurement		Mechanical variables
	Measurement units	control	



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NT: Oils
Structural engineering
Aerodynamics
Bending
Biomechanics
Damping
Deformation
Dynamics
Fatigue
Force
Friction
Hydrodynamics
Kinematics
Lubrication
Magnetohydrodynamics
Photoelasticity
Pressure effects
Shock (mechanics)
Strain
Stress
Surface cracks
Surface stress
Torque
Vibrations
Volume relaxation
Workability

Mechanical guides

UF: Guideways (mechanical)
Slideways (mechanical)
BT: Mechanical products
RT: Machine tools
Position control

Mechanical power transmission

UF: Continuously variable transmission
BT: Powertrain
Mechanical engineering
RT: Cams
Drives
Engines
Gears
Oils
Power systems
Shafts
NT: Torque converters

Mechanical products

UF: Ball screws
BT: Manufactured products
Production
RT: Escalators
Mechanical engineering
Structural rings

NT: Automotive components
Axles
Bellows
Blades
Brakes
Couplings
Fasteners
Flanges
Gears
Hoses
Machine components
Mechanical guides
Needles
Orifices
Pistons
Pressure vessels
Seals
Springs
Steering systems
Structural shapes
Tires
Vents
Wheels

Mechanical properties

USE: Mechanical factors

Mechanical sensors

BT: Sensors
NT: Capacitive sensors

Mechanical splines

BT: Machine components
RT: Gears
Machine tool spindles
Shafts

Mechanical stress

USE: Stress

Mechanical systems

BT: Mechanical engineering
RT: Gears
Mechatronics
Microelectromechanical

devices

Pneumatic systems
Turbomachinery
NT: Automated parking
Mechanical energy
Micromechanical devices
Suspensions (mechanical

systems)



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Mechanical variables control

BT: Control systems
 RT: Flexible structures
 Frequency control
 Manipulators
 Mechanical factors
 Mobile robots
 Motor drives
 Robots
 NT: Displacement control
 Force control
 Level control
 Motion control
 Pitch control (position)
 Position control
 Shape control
 Size control
 Strain control
 Stress control
 Thickness control
 Torque control
 Velocity control
 Vibration control
 Weight control

Mechanical variables measurement

BT: Measurement
 RT: Frequency measurement
 Transducers
 NT: Angular velocity
 Displacement

measurement

Force measurement
 Motion measurement
 Position measurement
 Rotation measurement
 Strain measurement
 Stress measurement
 Thickness measurement
 Torque measurement
 Velocity measurement
 Vibration measurement
 Volume measurement
 Weight measurement

Mechanical vibrations

USE: Vibrations

Mechanism design

UF: Reverse game theory
 BT: Economics
 Game theory

Mechanobiology

BT: Biology

RT: Biological system modeling
 Biomechanics
 Cell signaling
 Nanomedicine

Mechatronics

BT: Electron devices
 RT: Autonomous vehicles
 Control equipment
 Intelligent control
 Intelligent sensors
 Mechanical systems
 Microelectromechanical

devices

systems

Microelectromechanical

Robots
 Vehicular automation
 Biomechatronics

NT:

Media

BT: Materials
 RT: Closed captioning
 Design tools
 NT: Fake news
 Nonhomogeneous media
 Photorealism
 Random media

Media Access Control

UF: MAC
 MAC protocol
 Media Access Protocol
 Medium access control
 BT: Access protocols
 RT: Local area networks
 Metropolitan area networks
 Source address validation

Media Access Protocol

USE: Media Access Control

Media streaming

USE: Streaming media

Mediated communication

USE: Computer mediated communication

Mediated discourse

USE: Computer mediated communication



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Mediated interaction

USE: Computer mediated communication

Mediated technology

USE: Computer mediated communication

Mediation

BT: Middleware

Medical computing

USE: Biomedical computing

Medical conditions

UF: Medical disorders
BT: Medical services
RT: Allergies
Anxiety disorders
NT: Acidosis
Acute respiratory distress

syndrome

Addiction
Anemia
Aneurysm
Aphasia
Arrhythmia
Atrophy
Autism
Blindness
Carpal tunnel syndrome
Cataracts
Cerebral palsy
Chronic kidney disease
Congestive heart failure
Cybersickness
Deafness
Degenerative disc disease
Dementia
Depression
Diabetes
Diseases
Drug resistance
Dyslexia
Eczema
Gastritis
Hemorrhaging
Hydrocephalus
Hypertension
Hyperthermia
Hypoxia
Infertility
Inflammatory bowel disease
Injuries
Jaundice

Kidney stones
Motion sickness
Muscular dystrophy
Obesity
Paralysis
People with disabilities
Pregnancy
Psoriasis
Scoliosis
Sepsis
Sleep apnea
Spasticity
Stroke (medical condition)
Thrombosis
Tumors
Visual impairment

Medical control systems

BT: Control systems
RT: Assistive technologies
Biomedical equipment
Orthotics
Prosthetics

Medical devices

BT: Biomedical equipment
RT: Biocompatibility
Biomedical communication
Continuous glucose

monitoring

Medical diagnosis

UF: Diagnosis (medical)
Medical diagnostics
Patient diagnosis
BT: Medical services
RT: Biomedical imaging
Diagnostic radiography
Diseases
Electroencephalography
Electrophoresis
Medical expert systems
Occupational medicine
Radiography
Translational research
NT: Autopsy
Bronchoscopy
Cardiac catheterization
Colonography
Computer aided diagnosis
Differential diagnosis
Medical signal detection
Nanomedicine
Plethysmography
Sensitivity and specificity



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Medical diagnostic imaging

BT: Biomedical imaging
RT: Cancer
Magnetic particle imaging
Positron emission

tomography

Solid scintillation detectors
Tumors
NT: Anatomical structure
Radionuclide imaging

Endomicroscopy
Kinematics
Manipulators
Motion control
Position control
Soft robotics
Supernumery robotic limbs
Telerobotics
Wearable robots
NT: Assistive robots

Medical diagnostics

USE: Medical diagnosis

Medical robots

USE: Medical robotics

Medical disorders

USE: Medical conditions

Medical services

UF: Doctor
Emergency medical

Medical equipment

USE: Biomedical equipment AND
Medical instruments

services

Health care
Healthcare
Nursing
Physician
BT: Engineering in medicine

Medical expert systems

BT: Biomedical computing
Expert systems
RT: Medical diagnosis
Medical treatment

BT: and biology
RT:

Behavioral sciences
Chemotherapy
Emergency services
Translational research
NT: Assisted living
Catheterization
Clinical diagnosis
Cybercare
Electronic healthcare
Health information

Medical image processing

USE: Biomedical image

processing

NT:

Medical imaging

USE: Biomedical imaging

Medical information systems

BT: Biomedical computing
Computer applications
Information systems
NT: Electronic medical records

management

Hospitals
In vitro
In vivo
Internet of Medical Things
Medical conditions
Medical diagnosis
Medical tests
Medical treatment
Occupational medicine
Organ transplantation
Point of care
Prosthetics
Public healthcare
Sensory aids
Smart healthcare
Vaccines
X-rays

Medical instruments

UF: Medical equipment
BT: Biomedical equipment
Instruments

Medical robotics

UF: Medical robots
Robot assisted surgery
Robot-assisted surgery
Surgical robotics
Surgical robots
BT: Robots
RT: Biomedical equipment
Biomedical image

processing



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Medical signal detection

BT: Medical diagnosis

Medical simulation

BT: Simulation

Medical specialties

BT: Engineering in medicine
and biology

NT: Anesthesiology
Cardiology
Dermatology
Embryology
Emergency medicine
Endocrinology
Gastroenterology
Gerontology
Gynecology
Hematology
Immunology
Neonatology
Nephrology
Neurology
Obstetrics
Oncology
Ophthalmology
Otorhinolaryngology
Pathology
Pediatrics
Pharmacology
Psychiatry
Pulmonology
Radiology
Rheumatology
Urology

prescriptions

Medical expert systems
Medicinal plants
Occupational medicine
Psychiatry
Acupuncture
Amputation
Anesthesia
Angioplasty
Brachytherapy
Brain stimulation
Cancer treatment
Chemotherapy
Clinical trials
Cryotherapy
Defibrillation
Dentistry
Electrical stimulation
Electronic medical

Electroporation
Electrosurgery
Embolization
Fibrillation
Geriatrics
Hepatectomy
Hospitals
Hyperthermia
Immunotherapy
Intubation
Lithotripsy
Magnetic stimulation
Neuromuscular stimulation
Neutron capture therapy
Noninvasive treatment
Orthopedic procedures
Orthotics
Palliative care
Patient rehabilitation
Pharmaceuticals
Phototherapy
Precision medicine
Proton therapy
Surgery

Medical tests

BT: Medical services
RT: Cytometry
Mammography
NT: Amniocentesis
Biopsy
Cancer detection
Colonoscopy
Pregnancy test

Medicinal plants

BT: Plants (biology)
RT: Drugs
Medical treatment

Medical treatment

UF: Patient identification
Patient treatment
Therapy
BT: Medical services
RT: Biohazards
Biomedical applications of
radiation
Internet of Medical Things

Medium access control

USE: Media Access Control

Medium resolution imaging spectrometer

USE: MERIS



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Medium voltage

UF: Medium-voltage
BT: Voltage measurement

Medium-voltage

USE: Medium voltage

Meeting planning

BT: Planning

Meetings

UF: Technical meetings
BT: Professional communication
RT: Public speaking
Teleconferencing
Video conferencing
NT: Conferences

Meetings (technical)

USE: Conferences

Mel frequency cepstral coefficient

UF: MFCC
Mel-frequency cepstral coefficient
BT: Cepstral analysis

Mel-frequency cepstral coefficient

USE: Mel frequency cepstral coefficient

Melanoma

BT: Skin cancer

Melittology

USE: Apicology

Melt processing

BT: Manufacturing systems
Materials processing
RT: Die casting
Smelting
NT: Vacuum arc remelting

Membrane potentials

UF: Membrane voltage
Transmembrane potential
BT: Cells (biology)
RT: Action potentials
Neurons

Membrane voltage

USE: Membrane potentials

Membranes

USE: Biomembranes

Memetics

BT: Evolution (biology)
RT: Cultural differences
Genetics

Memoirs

USE: Autobiographies

Memory

UF: Data storage
BT: Computers and information processing
RT: CMOS memory circuits
Cognitive load
Memory architecture
Phase change materials
Recording
NT: Analog memory
Associative memory
Cache memory
Content addressable storage
Flash memories
Magnetic memory
Memory management
Nonvolatile memory
Phase change memory
Random access memory
Read only memory
Read-write memory
Registers
Scanning probe data
Semiconductor memory

Memory architecture

BT: Computer architecture
RT: Memory
Memory management

Memory management

UF: Garbage collection
(computers)
BT: Computer architecture
Memory
RT: Memory architecture
Memory modules
Storage management
NT: In-memory computing
Neural network

compression



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Memory modules

BT: Printed circuits
RT: Integrated circuits
Memory management

Memory resistors

USE: Memristors

Memoryless channel

USE: Memoryless systems

Memoryless systems

UF: Memoryless channel
BT: Probability

Memristor circuits

USE: Memristors

Memristors

UF: Memory resistors
Memristor circuits
BT: Resistors
RT: Neuromorphic engineering
Resistive RAM

MEMS

USE: Microelectromechanical
systems

MEMS switches

USE: Microswitches

Mental disorders

UF: Mental illness
BT: Psychiatry
RT: People with disabilities
Schizophrenia
NT: Anxiety disorders
Eating disorders

Mental health

BT: Human factors
Psychology
RT: Anxiety disorders
Behavioral sciences

Mental illness

USE: Mental disorders

Mental models

USE: Cognitive science

Mentoring

BT: Career development
RT: Training

Mercury (metals)

UF: Hg
BT: Chemical elements
Metals

Mercury (planets)

BT: Planets

Mergers

USE: Corporate acquisitions

Merging

BT: Data handling
RT: Sorting

MERIS

UF: Medium resolution imaging
spectrometer
BT: Spectroscopy

MERIT

USE: Magneto electrical
resistivity imaging technique

Merit pay

USE: Incentive schemes

Mesencephalon

USE: Midbrain

MESFET circuits

BT: FET circuits
RT: MESFETs
NT: MESFET integrated circuits

MESFET integrated circuits

BT: FET integrated circuits
Integrated circuits
MESFET circuits
RT: MESFETs
NT: Microwave FET integrated
circuits

MESFETs

UF: Schottky FETs
BT: Field effect transistors
RT: MESFET circuits
MESFET integrated circuits
Schottky barriers
NT: Microwave FETs

Mesh generation

BT: Computer displays
RT: Computer graphics



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Mesh networks

BT: Ad hoc networks

Mesons

UF: Kaons
Muons
Pions
BT: Elementary particles
RT: Cosmic rays

Mesoporous

USE: Mesoporous materials

Mesoporous materials

UF: Mesoporous
BT: Materials
RT: Electrocatalysts

Message authentication

UF: Data authentication
BT: Data security
RT: Cryptography
Digital signatures
Message systems
Steganography

Message passing

BT: Distributed processing
RT: Belief propagation

Message services

UF: Messaging service
WeChat
BT: Web services

Message systems

BT: Communications
technology
RT: Cluster computing
Digital signatures
Message authentication
NT: Electronic mail
Electronic messaging
Postal services
Publish subscribe systems
Voice mail

Message-oriented middleware

BT: Middleware

Messaging service

USE: Message services

Messenger RNA

USE: mRNA

Meta cognition

USE: Metacognition

Meta data

USE: Metadata

Meta heuristics

USE: Metaheuristics

Meta search

USE: Metasearch

Meta verse

USE: Metaverse

Meta-cognition

USE: Metacognition

Meta-heuristics

USE: Metaheuristics

Meta-modeling

USE: Metamodeling

Meta-search

USE: Metasearch

Metabolic networks

USE: Biochemistry

Metabolism

BT: Biochemistry
Physiology

Metabolomics

BT: Molecular biomarkers

Metacognition

UF: Meta cognition
Meta-cognition
BT: Cognition

Metacomputing

BT: Distributed computing
NT: Grid computing

Metadata

UF: Meta data
BT: Data models
RT: Image annotation
Linked data
NT: Annotations



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Metagenomics

BT: Genomics

RT: Electrodes
MIS devices
MOS integrated circuits
NT: MIM capacitors
MIM devices

Metaheuristic algorithms

USE: Metaheuristics

Metaheuristics

UF: Meta heuristics
Meta-heuristics
Metaheuristic algorithms
BT: Algorithmic efficiency
Optimization
RT: Ant colony optimization
Evolutionary computation
Genetic algorithms
Heuristic algorithms
Mathematics
Particle swarm optimization
Search problems
Simulated annealing
NT: Quantum annealing

Metal-insulator-metal capacitors

USE: MIM capacitors

Metal-insulator-metal devices

USE: MIM devices

Metal-insulator-semiconductor devices

USE: MIS devices

Metal-oxide semiconductor field effect transistor

USE: MOSFET circuits

Metal-oxide semiconductors

USE: MOS devices

Metal-oxide-nitride-oxide-semiconductors

USE: MONOS devices

Metal-oxide-nitride-oxide-silicon

USE: MONOS devices

Metal-oxide-semiconductor devices

USE: MOS devices

Metal-semiconductor interfaces

USE: Semiconductor-metal
interfaces

Metal cutting tools

BT: Cutting tools

Metal enclosures

BT: Metal products

Metal foam

BT: Materials
RT: Lightweight structures
Metals
Polymer foams

Metal oxide semiconductor heterojunction FETs

USE: MOSHFETs

Metal product industries

BT: Manufacturing industries
RT: Metal products

Metal products

BT: Manufactured products
RT: Ball bearings
Blanking
Metal product industries
Metals industry
Swaging
NT: Metal enclosures

Metal vapor lasers

USE: Gas lasers

Metal-insulator structures

BT: Insulators

Metallic materials

BT: Materials
RT: Cermet

Metallic superlattices

BT: Superlattices

Metallisation

USE: Metallization

Metallization

UF: Metallisation
BT: Metals
RT: Wiring
NT: Integrated circuit
metallization



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Metallurgy

BT: Materials science and technology
RT: Iron alloys
Metals

Tin
Titanium
Tungsten
Yttrium
Zinc

Metalorganic vapor deposition

USE: MOCVD

Metals

UF: Alloys
BT: Materials, elements, and compounds
RT:

Blanking
Die casting
Filler metals
Inorganic compounds
Metal foam
Metallurgy
Metalworking machines

NT: Alloying
Aluminum
Barium
Bismuth
Boron
Cadmium
Calcium
Chromium
Cobalt
Copper
Digital alloys
Erbium
Gallium
Germanium
Gold
Hafnium
Indium
Iron
Lanthanum
Lead
Lithium
Magnesium
Manganese
Mercury (metals)
Metallization
Neodymium
Nickel
Niobium
Palladium
Platinum
Rare earth metals
Samarium
Silver
Steel
Strontium

Metals industry

UF: Aluminium industry
Aluminum industry
BT: Industries
RT: Metal products
Smelting

Metalworking machines

BT: Machine tools
RT: Cutting tools
Materials processing
Metals

Metamaterial cloaking

USE: Optical cloaking

Metamaterials

UF: Acoustic metamaterials
Left handed materials
Left-handed materials
BT: Materials
RT: Metasurfaces
Nanocomposites
Optical materials
Refractive index
Smart materials
Split ring resonators
Electromagnetic

NT: metamaterials

Optical cloaking
Optical metamaterials

Metamodeling

UF: Meta-modeling
BT: Modeling

Metamorphic HEMTs

USE: mHEMTs

Metasearch

UF: Federated search
Federated searching
Meta search
Meta-search
BT: Search methods
RT: Triples (Data structure)
Web search



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Metastasis

UF: Metastatic disease
BT: Cancer
RT: Diseases

Metastatic disease

USE: Metastasis

Metasurfaces

BT: Surfaces
RT: Metamaterials
Thin films

Metaverse

UF: Meta verse
BT: Virtual environments
RT: Augmented reality
Avatars
Blockchains
Games
Three-dimensional displays
User experience
User interfaces
Virtual reality

Meteorites

BT: Meteoroids
RT: Space debris

Meteoroids

BT: Asteroids
Comets
RT: Space debris
NT: Meteorites
Meteors

Meteorological factors

BT: Geoscience
RT: Fading channels
Multipath channels

Meteorological radar

UF: Radar meteorology
BT: Radar
RT: Backscatter
Radar imaging

Meteorology

BT: Geophysics
RT: Air pollution
Atmosphere
Atmospheric

measurements

Barometers
Climatology

Data assimilation
Environmental factors
Ice
Ionosphere
Pressure effects
Remote sensing
Terrestrial atmosphere
Climate
Droughts
Humidity
Ice accretion
Lightning
Monsoons
Precipitation
Storms
Weather
Weather forecasting
Weather modification
Wind

NT:

Meteors

BT: Meteoroids
RT: Comets

Meter reading

BT: Power system
measurements
NT: Automatic meter reading
Smart meters

Meters

BT: Instruments
NT: Dynamometers
Flowmeters
Goniometers
Potentiometers
Radiometers
Tachometers
Vibrometers
Voltmeters
Wattmeters

Methane

BT: Natural gas
RT: Anaerobic digestion
Carbon emissions
GOSAT
Greenhouse gases

Methanol

UF: Carbinol
Methyl alcohol
Wood alcohol
Wood naphtha
Wood spirits



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BT:	Chemical compounds	<i>MFT</i>	USE:	Mean field theory
RT:	Anti-freeze			
	Fuels	<i>Mg</i>	USE:	Magnesium
	Solvents			
Method of moments				
UF:	Galerkin method	<i>MgO</i>	USE:	Magnesium oxide
	Method-of-moments			
	MoM	<i>MHD</i>	USE:	Magnetohydrodynamics
	Moment methods			
BT:	Mathematics			
	Numerical analysis	mHEMTs	UF:	Metamorphic HEMTs
RT:	Boundary-element methods		BT:	HEMTs
	Integral equations			
	Matrices			
<i>Method-of-moments</i>				
USE:	Method of moments	<i>MIB</i>	USE:	Management information
<i>Methyl alcohol</i>				
USE:	Methanol	base		
<i>Metrics</i>				
USE:	Measurement	<i>MIC</i>	USE:	Microwave integrated
<i>Metro area networks</i>				
USE:	Metropolitan area networks	circuits		
Metrology				
BT:	Science - general	Mice	UF:	Mouse
NT:	Optical metrology		BT:	Animals
Metropolitan area networks				
UF:	Metro area networks	<i>Mice flows</i>	USE:	Communication system
BT:	Communication systems	traffic AND		Computer networks
	Computer networks			
	Digital systems	<i>Micro air vehicles</i>	USE:	Autonomous aerial vehicles
RT:	Distributed computing			
	IEEE 802.16 Standard	<i>Micro computers</i>	USE:	Microcomputers
	Internetworking			
	LAN interconnection	<i>Micro-computers</i>	USE:	Microcomputers
	Media Access Control			
	Multiprocessor	<i>Micro-electro-mechanical devices</i>	USE:	Microelectromechanical
interconnection	Open systems	devices		
	Protocols	<i>Micro-electro-mechanical systems</i>	USE:	Microelectromechanical
	Regional area networks	systems		
	Token networks			
<i>Metropolitan areas</i>				
USE:	Urban areas	<i>Micro-electromechanical devices</i>	USE:	Microelectromechanical
<i>MFCC</i>				
USE:	Mel frequency cepstral	devices		
coefficient		<i>Micro-electromechanical systems</i>		



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USE: Microelectromechanical systems

Micro-hydro

USE: Microhydro power

Micro-mirrors

USE: Micromirrors

Micro-optical components

USE: Microoptics

Micro-opticalmechanical devices

USE: Microoptics

Microactuators

BT: Actuators
Microelectromechanical

devices

RT: Microrelays

Microarchitecture

BT: Software architecture

Microassembly

UF: Die attach
Die bonding

BT: Assembly
RT: Flip-chip devices
Integrated circuit

manufacture

Micromachining
Semiconductor device

manufacture

Microbial eletrolysis cells

USE: Fuel cells

Microbial fuel cells

USE: Fuel cells

Microbiology

BT: Biology
NT: Antimicrobials
Electroporation
Virology
Virulence

Microbiome

BT: Microorganisms
RT: Digestive system
Immune system
Pathogens

Microcavities

BT: Optical resonators
RT: Cavity resonators
Light emitting diodes
Microoptics
Photoluminescence
Spontaneous emission
Whispering gallery modes

Microcell networks

UF: Microcells
Small cell networks
BT: Cellular networks
RT: Ultra-dense networks

Microcells

USE: Microcell networks

Microchannels

BT: Hydraulic diameter

Microchip lasers

BT: Solid lasers

Microchips

USE: Integrated circuits

Microcomputers

UF: Laptops
Micro computers
Micro-computers
Minicomputers
Personal computers
BT: Computers
RT: Consumer electronics
Home computing
Mainframes
Microprocessors
Office automation
NT: Portable computers
Workstations

Microcontact printing

USE: Soft lithography

Microcontrollers

BT: Control equipment
Microprocessors
RT: CMOS technology
Control systems
High-speed integrated

circuits

Neurocontrollers
System-on-chip
Tiny machine learning



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Microdisplays

BT: Displays
RT: Liquid crystal devices
Liquid crystal on silicon
Microoptics

Microeconomics

BT: Economics
RT: Linear programming
Monopoly
Oligopoly
Supply and demand
NT: Economies of scale
Industrial economics

Microelectrodes

BT: Electrodes
RT: Neurophysiology
Neurostimulation

Microelectromechanical devices

UF: Micro-electro-mechanical
devices
Micro-electromechanical
devices
BT: Microelectromechanical
systems
RT: Magnetic particles
Mechanical systems
Mechatronics
Micromachining
Microsensors
NT: Microactuators
Micromotors
Micropumps
Microvalves

Microelectromechanical systems

UF: MEMS
Micro-electro-mechanical
systems
Micro-electromechanical
systems
BT: Electron devices
RT: Mechatronics
Nanoelectromechanical
systems
NT: Microelectromechanical
devices
Radiofrequency
microelectromechanical systems

Microelectronic implants

BT: Implants
Microelectronic stimulation
USE: Electrical stimulation

Microelectronics

BT: Electronic equipment
RT: Integrated circuits

Microfabrication

BT: Fabrication
Micromechanical devices
RT: Fiducial markers
Nanotechnology

Microfiltration

BT: Filtration
RT: Contamination

Microfluidics

BT: Electron devices
Fluidics
RT: Biochips
Biomedical engineering
Fluidic microsystems
Hydrodynamics

Micrographs

USE: Photomicrography

Microgrids

BT: Power grids
RT: Distributed power
generation
Power distribution networks
Power generation
Power system management
Power system reliability
Smart grids

Microgrippers

USE: Grippers

Microhydro power

UF: Micro-hydro
BT: Hydroelectric power
generation
RT: Appropriate technology

Microinjection

BT: Biology



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Microinverters

BT: Inverters
RT: DC-AC power converters
Solar panels
Solar power generation

Micromachining

BT: Electronic equipment
manufacture
RT: Electrochemical machining
Embossing
Etching
Integrated circuit
manufacture
Microassembly
Microelectromechanical
devices
Semiconductor device
manufacture

Micromagnetics

BT: Magnets

Micromanipulators

BT: Manipulators

Micromechanical devices

UF: Micromechanical systems
Microsystems
BT: Electron devices
Mechanical systems
RT: Microoptics
Nanogenerators
NT: Biomedical
microelectromechanical systems
Fluidic microsystems
Microfabrication

Micromechanical systems

USE: Micromechanical devices

Micrometers

UF: Micrometres
Micrometry
Microns
BT: Measurement
RT: Distance measurement
Interferometry
Length measurement
Strain measurement
Thickness measurement

Micrometres

USE: Micrometers

Micrometry

USE: Micrometers

Micromirrors

UF: Digital micromirror devices
Micro-mirrors
BT: Microoptics
Mirrors
RT: Optical arrays
Optical projectors

Micromobility

BT: Shared transport
RT: Bicycles
Motorcycles

Micromotors

BT: Microelectromechanical
devices
Motors
Rotating machines

Microns

USE: Micrometers

Microoptical components

USE: Microoptics

Microoptics

UF: Micro-optical components
Micro-opticalmechanical
devices
Microoptical components
BT: Optics
RT: Integrated optics
Integrated optoelectronics
Microcavities
Microdisplays
Micromechanical devices
Microswitches
NT: Micromirrors

Microorganisms

UF: Viruses (microorganisms)
BT: Organisms
RT: Antibodies
Biological cells
Immune system
Molecular biophysics
Paleontology
NT: Adenoviruses
Anaerobic digestion
Bacteria
Biochemical oxygen

demand



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	Biofilms		Software
	Fungi		
	Microbiome	Micropumps	
	Viruses (medical)	BT:	Microelectromechanical
Micropayments		devices	
BT:	Financial services		Pumps
	Online banking	Microrelays	
RT:	Blockchains	BT:	Relays
	Cryptocurrency	RT:	Microactuators
Microphone arrays		<i>Microsatellites</i>	
BT:	Microphones	USE:	Small satellites
Microphones		Microscopy	
BT:	Audio systems	BT:	Imaging
NT:	Hydrophones		Instruments
	Microphone arrays	RT:	Optical imaging
<i>Microphotographs</i>		NT:	Atomic force microscopy
USE:	Photomicrography		Electron microscopy
<i>Microphotography</i>			Endomicroscopy
USE:	Photomicrography		Immunofluorescence
		microscopy	Scanning microwave
Microplastics			Scanning probe microscopy
BT:	Plastics	Microsensors	
Microprocessor chips		BT:	Electromechanical sensors
BT:	Microprocessors	RT:	Control systems
RT:	Flip-chip devices		Microelectromechanical
	Substrates	devices	
	Yield estimation		Wireless sensor networks
NT:	AI accelerators	Microservice architectures	
Microprocessors		BT:	Service-oriented
BT:	Circuits	architecture	
	Integrated circuits	<i>Microsoft Excel</i>	
RT:	CMOS technology	USE:	Spreadsheet programs
	Embedded systems	<i>Microsoft Windows</i>	
	Flip-chip devices	USE:	Operating systems
	Microcomputers	Microstrip	
	Processor scheduling	UF:	Microstrip lines
	System-on-chip	BT:	Planar transmission lines
NT:	Automatic logic units	RT:	Broadband antennas
	Biomimetics	NT:	Microstrip components
	Coprocessors	Microstrip antenna arrays	
	Microcontrollers	UF:	Microstrip arrays
	Microprocessor chips	BT:	Antenna arrays
	Vector processors	RT:	Antennas
Microprogramming			Aperture coupled antennas
UF:	Firmware		Microstrip antennas
BT:	Programming		
RT:	Computer architecture		



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Microstrip antennas

BT: Antennas
RT: Aperture coupled antennas
Microstrip antenna arrays
Patch antennas

Microstrip arrays

USE: Microstrip antenna arrays

Microstrip components

BT: Microstrip
RT: Power combiners
Power dividers
Thick film inductors
NT: Microstrip resonators

Microstrip filters

BT: Filters
RT: Microwave communication

Microstrip lines

USE: Microstrip

Microstrip resonators

BT: Microstrip components
RT: Resonance

Microstructure

BT: Materials science and
technology
RT: Crystal microstructure

Microstructured fibers

USE: Photonic crystal fibers

Microstructured fibres

USE: Photonic crystal fibers

Microsurgery

BT: Surgery

Microswitches

UF: MEMS switches
BT: Switches
RT: Microoptics

Microsystems

USE: Micromechanical devices

Microtransit

USE: Shared transport

Microtubules

BT: Cells (biology)

Microvalves

BT: Microelectromechanical
devices
Valves

Microwave ablation

BT: Cancer treatment
Microwave technology

Microwave amplifiers

BT: Microwave devices

Microwave antenna arrays

BT: Antenna arrays

Microwave antennas

BT: Antennas

Microwave bands

BT: Microwave technology
NT: C-band
K-band
Ka-band
Ku-band
L-band

Microwave circuits

BT: Circuits
Microwave technology
RT: Analog circuits
Distributed parameter
circuits
Microwave devices
Microwave integrated
circuits
Microwave photonics

Microwave communication

BT: Communication systems
Microwave technology
RT: Microstrip filters
Microwave photonics
NT: Rectennas

Microwave devices

BT: Microwave technology
RT: Electromagnetic
waveguides
Microwave circuits
Microwave photonics
Photonic crystals
Superconducting
microwave devices
NT: Masers
Microwave amplifiers



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Microwave filters
Microwave transistors

Microwave FET integrated circuits

BT: MESFET integrated circuits
RT: Microwave FETs

Microwave FETs

BT: MESFETs
Microwave transistors
RT: Microwave FET integrated
circuits

Microwave filters

BT: Microwave devices

Microwave frequencies

BT: Microwave measurement

Microwave generation

BT: Microwave technology
NT: High power microwave
generation

Microwave heating

USE: Electromagnetic heating

Microwave imaging

BT: Imaging
RT: Remote sensing

Microwave integrated circuits

UF: MIC
BT: Integrated circuits
RT: Analog integrated circuits
Microwave circuits
NT: MMICs

Microwave lasers

USE: Masers

Microwave magnetics

BT: Magnetics

Microwave measurement

BT: Electromagnetic
measurements
RT: Microwave technology
NT: Microwave frequencies

Microwave metamaterials

BT: Electromagnetic
metamaterials
RT: Split ring resonators

Microwave oscillators

BT: Oscillators

Microwave ovens

BT: Consumer electronics
Consumer products
Home appliances
Ovens

Microwave photonics

BT: Microwave technology
Photonics
RT: Electro-optic modulators
Elementary particles
Integrated optoelectronics
Microwave circuits
Microwave communication
Microwave devices
Optical modulation

Microwave plasmas

BT: Electromagnetic scattering
Plasmas

Microwave propagation

BT: Electromagnetic
propagation
RT: Broadband antennas
Electromagnetic
waveguides

Microwave radar

USE: Radar

Microwave radiometry

BT: Radiometry

Microwave sensors

BT: Microwave technology

Microwave technology

BT: Microwave theory and
techniques
RT: Microwave measurement
Radar
NT: Baluns
Beam steering
Circulators
Masers
Microwave ablation
Microwave bands
Microwave circuits
Microwave communication
Microwave devices
Microwave generation



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	Microwave photonics		Military aircraft	
	Microwave sensors		BT:	Military equipment
			RT:	Aircraft
Microwave theory and techniques				Hyperspectral sensors
NT:	Microwave technology			Military robotics
	Millimeter wave technology		NT:	Payloads
	Submillimeter wave			
technology			<i>Military command and control</i>	
			USE:	Command and control
Microwave transistors			systems	
BT:	Microwave devices		Military communication	
NT:	Microwave FETs		BT:	Communication systems
Microwave-assisted magnetic recording			RT:	Air to ground
BT:	Magnetic recording		communication	
				Command and control
Midbrain			systems	
UF:	Mesencephalon			Cross layer design
BT:	Brain			Electronic countermeasures
	Central nervous system			Hyperspectral sensors
RT:	Forebrain			Military satellites
	Hindbrain			Ultra wideband
			communication	
Middleboxes			NT:	Reconnaissance
BT:	Computer network			
management			Military computing	
	Internet		BT:	Computer applications
			RT:	Military systems
				Mobile computing
Middleware			Military control	
BT:	Client-server systems		BT:	Military systems
	Software			
RT:	Computer applications		Military equipment	
	Computer networks		BT:	Aerospace and electronic
	Internet of Things		systems	
	Publish subscribe systems		RT:	Defense industry
NT:	Mediation			Explosion protection
	Message-oriented			Ground support
middleware				Landmine detection
	Web services			Magnetic anomaly
Mie scattering			detectors	
BT:	Electromagnetic scattering			Military systems
RT:	Electromagnetic analysis			Night vision
	Electromagnetic fields			Open area test sites
	Electromagnetic forces			Wearable robots
measurements	Electromagnetic		NT:	Military aircraft
				Military satellites
propagation	Electromagnetic			Military vehicles
				Weapons
Migraine			Military robotics	
BT:	Neurological diseases		BT:	Military systems
<i>Mil standards</i>				Robots
USE:	Military standards		RT:	Autonomous aerial vehicles



January 2026 IEEE Thesaurus

vehicles

- Autonomous underwater
- Military aircraft
- Military vehicles
- Mobile robots
- Underwater vehicles
- Wearable robots

Military satellites

BT: Artificial satellites
Military equipment

RT: Global Positioning System
Hyperspectral sensors
Military communication

Military standards

UF: Mil standards

BT: Standards categories

Military systems

BT: Systems engineering and

theory

RT: Aerospace control
Autonomous aerial vehicles
Command and control

systems

Helicopters

Marine technology

Military computing

Military equipment

Military vehicles

Weapons

Wearable robots

NT: Military control
Military robotics

Military vehicles

BT: Military equipment
Vehicles

RT: Marine robots
Military robotics
Military systems

Milk

USE: Dairy products

Millennial generation

USE: Millennials

Millennials

UF: Generation Y
Millennial generation

BT: Social groups

Millimeter wave circuits

UF: Millimeter-wave circuits

BT: Circuits
Millimeter wave technology

RT: Analog circuits
Distributed parameter

circuits

Millimeter wave devices

NT: Millimeter wave integrated

circuits

Millimeter wave communication

UF: Millimeter-wave

communication

BT: Radio communication

RT: Millimeter wave

propagation

Millimeter wave devices

UF: Millimeter-wave devices

BT: Millimeter wave technology

RT: Millimeter wave circuits
Millimeter wave integrated

circuits

NT: Millimeter wave transistors

Millimeter wave integrated circuits

UF: Millimeter-wave integrated

circuits

BT: Circuits
Integrated circuits
Millimeter wave circuits
Millimeter wave technology

RT: Analog integrated circuits
Millimeter wave devices

NT: MIMICs

Millimeter wave measurements

UF: Millimeter-wave

measurements

BT: Electromagnetic

measurements

RT: Millimeter wave technology

Millimeter wave monolithic integrated circuits

USE: MIMICs

Millimeter wave propagation

UF: Millimeter-wave

propagation

BT: Electromagnetic

propagation

RT: Millimeter wave

communication



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Millimeter wave radar

UF: Millimeter-wave radar
BT: Millimeter wave technology
Radar

Millimeter wave technology

UF: Millimeter-wave technology
BT: Microwave theory and techniques
RT: Millimeter wave measurements
NT: Millimeter wave circuits
Millimeter wave devices
Millimeter wave integrated circuits
Millimeter wave radar

Millimeter wave transistors

UF: Millimeter-wave transistors
BT: Millimeter wave devices
Transistors

Millimeter-wave circuits

USE: Millimeter wave circuits

Millimeter-wave communication

USE: Millimeter wave communication

Millimeter-wave devices

USE: Millimeter wave devices

Millimeter-wave integrated circuits

USE: Millimeter wave integrated circuits

Millimeter-wave measurements

USE: Millimeter wave measurements

Millimeter-wave monolithic integrated circuits

USE: MIMICs

Millimeter-wave propagation

USE: Millimeter wave propagation

Millimeter-wave radar

USE: Millimeter wave radar

Millimeter-wave technology

USE: Millimeter wave technology

Millimeter-wave transistors

USE: Millimeter wave transistors

Millimicron

USE: Nanometers

Milling

BT: Machining
RT: Boring
Milling machines

Milling machines

BT: Machine tools
RT: Ball milling
Cutting tools
Milling

MIM capacitors

UF: Metal-insulator-metal capacitors
BT: Metal-insulator structures

MIM devices

UF: Metal-insulator-metal devices
BT: Metal-insulator structures
RT: Semiconductor-insulator interfaces

MIMICs

UF: Millimeter wave monolithic integrated circuits
Millimeter-wave monolithic integrated circuits
BT: Millimeter wave integrated circuits
Monolithic integrated circuits
RT: MMICs
Radiofrequency integrated circuits

MIMO

UF: MIMO systems
Multiple antenna systems
Multiple input multiple output
Multiple input multiple output systems
Multiple-input multiple-output
Multiple-input multiple-output systems
Multiple-input-multiple-output
Multivariable systems
BT: Communications technology



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	RT:	Wireless communication			Minimax techniques	UF:	Max-min composition
		3G mobile communication					Minmax techniques
		Control systems					BT:
		IEEE 802.11 Standard					Statistics
		IEEE 802.11n Standard					RT:
		IEEE 802.16 Standard					Artificial intelligence
		MISO					Game theory
		Multipath channels					Minimization methods
		NOMA					
		OFDM					
	NT:	Optimization methods			<i>Minimisation</i>	USE:	Minimization
		Radio communication					
		SIMO					
		SISO					
		Massive MIMO					
		Rician channels					
	BT:	Multistatic radar			Minimization	UF:	Minimisation
							BT:
							Mathematics
							RT:
							Optimization
							NT:
							Minimization methods
	USE:	MIMO			Minimization methods	UF:	Minimisation methods
							BT:
							Minimization
							RT:
							Approximation methods
							Minimax techniques
	USE:	Brain-computer interfaces			<i>Minimum analog-digital integrated circuits</i>	USE:	Analog-digital integrated
							circuits
	BT:	Minerals			Mining equipment	BT:	Production equipment
							RT:
							Mining industry
	BT:	Geology			Mining industry	BT:	Industries
		Mineral resources					Excavation
		Mineralization					Fracking
		Ores					Fuel processing industries
		Salt					Geoengineering
							Hyperspectral sensors
							Mining equipment
							Raw materials
							NT:
							Coal mining
	USE:	Small satellites			<i>Minmax techniques</i>	USE:	Minimax techniques
	USE:	Microcomputers			Mirrors	BT:	Optical devices
							RT:
							Optical materials
							Optical reflection
							Reflection
							NT:
							Distributed Bragg reflectors
							Micromirrors
	UF:	Laparoscopic surgery					
		Surgery					
		Concentric tube robots					
		Laparoscopes					



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MIS devices

UF: Metal-insulator-semiconductor devices
 BT: Semiconductor devices
 RT: Metal-insulator structures
 Semiconductor-insulator
 interfaces
 NT: Charge coupled devices
 MOS devices

MISFETs

BT: Field effect transistors
 RT: CMOSFET logic devices
 Magnetic field induced
 strain

Misinformation

USE: Fake news

MISO

UF: multiple input single-output
 multiple-input single output
 multiple-input single-output
 BT: Communications
 technology
 RT: Wireless communication
 MIMO
 SIMO
 SISO

Missile control

BT: Control systems

Missile guidance

BT: Missiles
 RT: Target recognition

Missiles

UF: Torpedoes
 BT: Weapons
 RT: Aerospace control
 Ground support
 Hypersonic vehicles
 NT: Missile guidance

Mission critical systems

UF: Mission-critical systems
 BT: Contingency management

Mission-critical systems

USE: Mission critical systems

Mitochondria

UF: Mitochondrion
 BT: Cells (biology)

Mitochondrion

USE: Mitochondria

Mitral valves

USE: Heart valves

Mixed analog digital integrated circuits

USE: Mixed analog-digital integrated circuits

Mixed analog-digital integrated circuits

UF: Mixed analog digital integrated circuits
 BT: Analog-digital integrated circuits
 RT: Analog processing circuits
 System-on-chip

Mixed convection

USE: Convection

Mixed integer linear programming

UF: Mixed-integer linear programming
 BT: Integer linear programming

Mixed reality

BT: Simulation
 RT: Augmented reality
 Multimedia systems
 Spatial computing
 Virtual reality

Mixed source separation

USE: Blind source separation

Mixed-integer linear programming

USE: Mixed integer linear programming

Mixers

BT: Frequency conversion
 RT: Demodulation
 Modulation
 Nonlinear circuits

Mixture models

BT: Statistics
 RT: Feature extraction
 Image segmentation
 Probabilistic logic

MLE

USE: Maximum likelihood estimation



January 2026 IEEE Thesaurus

MLFMA		RT:	Fintech
algorithm	UF: Multilevel fast multipole		Mobile communication
	BT: Algorithms		Mobile handsets
			Ride hailing
			Wireless communication
MMICs		NT:	Mobility as a service
integrated circuits	UF: Monolithic microwave		
circuits	BT: Microwave integrated	<i>Mobile apps</i>	
		USE:	Mobile applications
circuits	Monolithic integrated		
	RT: Analog integrated circuits	Mobile communication	
	MIMICs	BT:	Communication systems
circuits	Radiofrequency integrated	RT:	Near field communication
		(telecommunication)	Acoustic communication
			Block codes
MMTC			Film bulk acoustic
communications	USE: Massive machine type	resonators	
			Indoor communication
<i>mmtc</i>			Long Term Evolution
communications	USE: Massive machine type		Mobile applications
			Mobile handsets
			Mobility as a service
			Multiuser detection
MNN		management	Network resource
	USE: Multi-layer neural network		
			Radio communication
Mobile ad hoc networks			Routing protocols
UF: MANET			Time-varying channels
Wireless ad hoc network			Transceivers
BT: Ad hoc networks		NT:	Vehicular ad hoc networks
			3G mobile communication
Mobile agents			4G mobile communication
BT: Knowledge based systems			5G mobile communication
RT: Computer applications			6G mobile communication
Distributed computing			Ambient networks
Intelligent systems			Carrier aggregation
Learning systems			Cellular technology
Mobile computing			Dual band
Software agents			Land mobile radio
			Location awareness
Mobile antennas			Mobile learning
BT: Antennas			Mobile nodes
Land mobile radio			Mobile security
equipment			Mobile video
	RT: Land mobile radio		Mobility models
			Open RAN
<i>Mobile application development</i>			SIM card
USE: Application programming			Software radio
interfaces			Ultra-dense networks
		Mobile computing	
Mobile applications		BT:	Computers and information
UF: Mobile apps		processing	
BT: Computer applications		RT:	Ad hoc networks



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	Bring your own device	Mobile nodes	
	Crowdsensing	BT:	Mobile communication
	Crowdsourcing		Telecommunication
	Data dissemination	network management	
	Edge computing	<i>Mobile office</i>	
	Location awareness	USE:	Remote working
	Military computing	<i>Mobile payment</i>	
	Mobile agents	USE:	Online banking
	Mobile learning	<i>Mobile phones</i>	
	Software defined	USE:	Mobile handsets
networking	Telecommunication	<i>Mobile radio</i>	
computing	User experience	USE:	Land mobile radio
NT:	Multi-access edge	<i>Mobile radio mobility management</i>	
computing	Wireless access points	USE:	Location awareness
<i>Mobile device security</i>		<i>Mobile robot sensing systems</i>	
USE:	Mobile security	USE:	Robot sensing systems
<i>Mobile devices</i>		<i>Mobile robot vision systems</i>	
USE:	Mobile handsets	USE:	Robot vision systems
<i>Mobile edge computing</i>		Mobile robots	
USE:	Multi-access edge	BT:	Robots
computing		RT:	Advanced driver assistance
Mobile handsets		systems	
UF:	Mobile devices		Agricultural robots
	Mobile phones		Amphibious robots
BT:	Telephone sets		Assembly systems
RT:	Dual band		Control systems
	Land mobile radio		Drones
	Long Term Evolution		Humanoid robots
	Mobile applications		Industrial control
	Mobile communication		Manufacturing automation
	Personal communication		Marine robots
networks			Materials handling
	Tablet computers		Mechanical variables
	Transceivers	control	
	UHF communication		Military robotics
NT:	SIM card		Motion control
	Smart phones		Motion detection
Mobile learning			Nonlinear systems
BT:	Electronic learning		Service robots
	Mobile communication		Social robots
RT:	Distance learning		Soft robotics
	Hybrid learning		Stairs
	Mobile computing		Telerobotics
<i>Mobile location management</i>			Vehicles
USE:	Location awareness		Vehicular automation
		NT:	Wearable robots
			Climbing robots



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	Field robots		Model checking
	Legged locomotion	BT:	System testing
		RT:	Algorithms
Mobile security			Concurrent computing
UF:	Mobile device security		Formal verification
BT:	Computer security		Static analysis
	Mobile communication		
RT:	Hardware security		
<i>Mobile television</i>		Model compression	
USE:	TV	BT:	Algorithm design and analysis
Mobile TV			Machine learning
BT:	TV	RT:	Artificial intelligence
			Deep learning
			Inference algorithms
Mobile video		Model driven engineering	
BT:	Mobile communication	UF:	Model-driven engineering
	Streaming media	BT:	Software design
RT:	Video recording		
Mobility as a service		<i>Model predictive control</i>	
UF:	MaaS	USE:	Predictive control
	Mobility on demand		
BT:	Mobile applications	<i>Model reduction</i>	
	Online services	USE:	Reduced order systems
RT:	Mobile communication		
	Ride hailing	<i>Model-based reasoning</i>	
	Shared transport	USE:	Inference mechanisms
Mobility models		Model-driven development	
BT:	Mobile communication	BT:	Software development management
	Modeling		
<i>Mobility on demand</i>		<i>Model-driven engineering</i>	
USE:	Mobility as a service	USE:	Model driven engineering
MOCVD		<i>Model-predictive control</i>	
UF:	Metalorganic vapor	USE:	Predictive control
deposition			
BT:	Chemical vapor deposition	Modeling	
RT:	Vapor deposition	UF:	Modelling
			System modeling
Modal analysis		BT:	Systems engineering and theory
BT:	Mathematical analysis		
RT:	Vibration measurement	RT:	Computer graphics
			Data visualization
Mode matching methods			Digital simulation
BT:	Mathematics		Haptic interfaces
	Numerical analysis		Human in the loop
	Statistical analysis		Monte Carlo methods
RT:	Antenna theory		Numerical simulation
	Matrices		Petri nets
	Waveguide theory		Plasma simulation
			Power system analysis
		computing	



January 2026 IEEE Thesaurus

Language	Systems Modeling	<i>Moderate resolution imaging spectroradiometer</i>
	USE: MODIS	
NT:	Time series analysis	
	Analytical models	<i>Moderate-resolution Imaging Spectroradiometer</i>
management	Atmospheric modeling	USE: MODIS
	Brain modeling	
	Building information	MODFET circuits
	Computational modeling	BT: FET circuits
	Context modeling	<i>modfet integrated circuits</i>
	Data models	USE: MODFETs
	Data-driven modeling	
	Deformable models	MODFETs
	Digital elevation models	UF: Heterostructure FETs
	Emulation	SDHTs
	Evaluation models	Selectively doped
	Foundation models	heterojunction transistors
	Graphical models	TEGFETs
	Green's function methods	Two-dimensional electron
	Hidden Markov models	gas FETs
	Input variables	modfet integrated circuits
	Integrated circuit modeling	BT: Field effect transistors
	Inverse problems	RT: HEMTs
	Load modeling	
	Mathematical models	MODIS
	Metamodeling	UF: Moderate resolution
	Mobility models	imaging spectroradiometer
	Numerical models	Moderate-resolution
	Object oriented modeling	Imaging Spectroradiometer
	Power system modeling	BT: Payloads
	Process modeling	Spectroradiometers
	Semiconductor device	
	Semiconductor process	Modular construction
modeling	Signal representation	BT: Construction
	Simulation	RT: Buildings
modeling	Solid modeling	Prefabricated construction
	Space mapping	
	System identification	<i>Modular multi-level converters</i>
	Systems modeling	USE: Multilevel converters AND
Modelling		Voltage-source converters
	USE: Modeling	<i>Modular multilevel converters</i>
Models of emotion		USE: Multilevel converters
	USE: Emotion recognition	
Modems	UF: Modulator-demodulators	Modulation
	BT: Communication equipment	UF: Modulation format
	Computer peripherals	Modulation index
	Data communication	Modulation-coding
	Demodulation	Modulators
	Modulation	BT: Communications
		technology
		Signal processing
		RT: Direct sequence spread
		spectrum communication
		Encoding



January 2026 IEEE Thesaurus

Proteomics

Molecular biophysics

UF: Biological macromolecules
Biomolecules
BT: Biophysics
RT: Biochemistry
Biomedical equipment
Biomedical imaging
Biomedical materials
Cellular biophysics
DNA
Genetic engineering
Genetics
Microorganisms
NT: Nucleic acids

Molecular clones

USE: Cloning

Molecular communication

BT: Biological systems
Communication systems
RT: Nanoantennas
Nanocommunication

Molecular computing

BT: Computers and information processing
RT: Nanotechnology
DNA computing
Quantum chemistry

Molecular electronics

UF: Biomolecular electronics
BT: Nanotechnology
RT: Graphene devices
Nanoelectronics
Organic light emitting

diodes

Molecular imaging

BT: Biomedical imaging

Molecular sieves

BT: Chemical processes
Sieving
RT: Adsorption

Molybdenum

BT: Chemical elements

MoM

USE: Method of moments

Moment methods

USE: Method of moments

Money management

USE: Financial management

Monitoring

BT: Instrumentation and measurement
RT: Alarm systems
Maintenance engineering
Power system management
NT: Computerized monitoring
Environmental monitoring
Load monitoring
Patient monitoring
Process monitoring
Radiation monitoring
Remote monitoring
Surveillance
Water monitoring

Monoclonal antibodies

BT: Antibodies
RT: Cancer treatment
Immunotherapy
Proteins

Monolithic integrated circuits

UF: Monolithic integration
BT: Circuits
Integrated circuits
NT: MIMICs
MMICs

Monolithic integration

USE: Monolithic integrated circuits

Monolithic microwave integrated circuits

USE: MMICs

Monopoly

BT: Economics
RT: Microeconomics

MONOS devices

UF: Metal-oxide-nitride-oxide-semiconductors
Metal-oxide-nitride-oxide-silicon
BT: Semiconductor devices



January 2026 IEEE Thesaurus

Monsoons

BT: Meteorology
RT: Rain
Storms

Monte Carlo methods

UF: Importance sampling
Monte Carlo simulations
Monte-Carlo methods
Monte-Carlo simulations

BT: Statistical analysis
RT: Computational

electromagnetics

Modeling
Probability
Simulated annealing
Simulation
Temporal difference

learning

Monte Carlo simulations

USE: Monte Carlo methods

Monte-Carlo methods

USE: Monte Carlo methods

Monte-Carlo simulations

USE: Monte Carlo methods

Mood

BT: Psychology

Moon

UF: Lunar
BT: Satellites

Moore's Law

BT: Integrated circuit
technology

Mopeds

USE: Motorcycles

Morals

USE: Ethics

Morphological operations

BT: Image processing
RT: Image transformation
Topology
NT: Image morphing
Image thinning

Morphology

BT: Natural language
processing
RT: Entomology

Mortality

UF: Death rate
BT: Statistics
RT: Failure analysis

Mortar

BT: Building materials
Chemical products
RT: Construction industry

Mortgages

USE: Loans and mortgages

MOS capacitors

BT: MOS devices
RT: Capacitors

MOS devices

UF: Metal-oxide
semiconductors
Metal-oxide-semiconductor

devices

BT: MIS devices
RT: Semiconductor-insulator

interfaces

NT: MOS capacitors
MOSFET
Negative bias temperature

instability

MOS integrated circuits

BT: MOSFET circuits
RT: Metal-insulator structures

MOS transistors

USE: MOSFET

MOSFET

UF: MOS transistors
nMOSFETs
pMOSFETs
BT: Field effect transistors
MOS devices
RT: CMOS technology
CMOSFET logic devices
Gate drivers
TFETs
NT: CMOSFETs
FinFETs
Interface states



January 2026 IEEE Thesaurus

transistors	Junctionless nanowire	Motion control	
		BT:	Mechanical variables
MOSFET circuits		control	
UF:	Metal-oxide semiconductor	RT:	Aerospace control
field effect transistor			Legged locomotion
BT:	Circuits		Manipulators
	FET circuits		Medical robotics
RT:	Active inductors		Mobile robots
	Linearization techniques		Motion detection
	Operational amplifiers		Motor coordination
	Power dissipation		Servosystems
	Rail to rail amplifiers		Structure from motion
	Rail to rail operation		Trajectory
	Threshold voltage		Trajectory tracking
NT:	CMOSFET circuits	NT:	Velocity control
	MOS integrated circuits		Collision avoidance
	Power MOSFET		Collision mitigation
			Formation control
MOSHFETs			Kinetic theory
UF:	Metal oxide semiconductor		Motion planning
heterojunction FETs			Path planning
BT:	Field effect transistors		Visual servoing
		Motion detection	
Motion analysis		UF:	Motion sensors
BT:	Robot kinematics	BT:	Signal detection
RT:	3-DOF	RT:	Alarm systems
	5-DOF		Corner detection
	6-DOF		Image motion analysis
NT:	Active contours		Image sensors
	Motion segmentation		Infrared detectors
	Robot localization		Mobile robots
Motion artifacts			Motion control
BT:	Biomedical image		Surveillance
processing			Video signal processing
	Video signal processing	NT:	Video surveillance
			Odometry
Motion capture		Motion estimation	
BT:	Kinematics	BT:	Parameter estimation
RT:	Animation	RT:	Motion capture
	Biomechanics		Object tracking
	Feature extraction		
	Gait recognition	Motion measurement	
	Image motion analysis	BT:	Mechanical variables
	Image sensors	measurement	
	Motion compensation	RT:	Doppler measurement
	Motion estimation		Gaze tracking
	Object detection		Velocity measurement
	Object tracking	NT:	Tracking
Motion compensation			Tribology
BT:	Control systems		
RT:	Image communication		
	Motion capture		



January 2026 IEEE Thesaurus

Motion pictures

UF: Cinema
Films (Motion pictures)
Movies
Moving pictures
BT: Broadcasting
RT: Cameras
Cinematography
Entertainment industry
Imaging
Optical projectors

Motion planning

BT: Motion control
RT: Navigation
Robot control

Motion segmentation

BT: Motion analysis

Motion sensors

USE: Motion detection

Motion sickness

BT: Medical conditions

Motor control

USE: Motor drives

Motor coordination

BT: Kinematics
RT: Motion control
Paralysis

Motor drives

UF: Motor control
BT: Drives
RT: Industrial control
Mechanical variables

control

Motors
Sensorless control
Servosystems
Torque control
Variable speed drives
Velocity control
Voltage control

Motor vehicles

USE: Automobiles

Motorbikes

USE: Motorcycles

Motorcycles

UF: Mopeds
Motorbikes
Scooters
BT: Road vehicles
RT: Micromobility

Motors

BT: Energy conversion
Machinery
RT: Coils
Motor drives
Sensorless control
NT: AC motors
Brushless motors
Commutation
DC motors
Electric motors
Hysteresis motors
Induction motors
Micromotors
Permanent magnet motors
Servomotors
Traction motors
Universal motors

Moulding equipment

USE: Molding equipment

Mouse

USE: Mice

Mouth

BT: Digestive system
Head
RT: Stomatognathic system
NT: Teeth

Movies

USE: Motion pictures

Moving object databases

USE: Visual databases

Moving picture experts group

USE: MPEG standards

Moving pictures

USE: Motion pictures

Moving Pictures Experts Group

BT: IEC
ISO



January 2026 IEEE Thesaurus

<i>MP3</i>	USE:	Digital audio players AND Portable media players	MPEG 21 Standard MPEG 4 Standard MPEG 7 Standard
<i>MPEG</i>	USE:	Transform coding	<i>MPEG-2</i> USE: MPEG 2 Standard
MPEG 1 Standard	BT:	MPEG standards	<i>MPEG4</i> USE: MPEG 4 Standard
MPEG 2 Standard	UF: BT:	MPEG-2 MPEG standards	<i>MPEG7</i> USE: MPEG 7 Standard
MPEG 21 Standard	BT:	MPEG standards	<i>MPLS</i> USE: Multiprotocol label switching
MPEG 4 Standard	UF: BT: RT:	H.264 MPEG4 MPEG standards Digital multimedia	<i>MPPT</i> USE: Maximum power point trackers
broadcasting		High efficiency video coding Streaming media Vector quantization Video codecs Video coding Video signal processing	MPTCP UF: Multipath TCP BT: TCP RT: Multipath channels
MPEG 7 Standard	UF: BT: RT:	MPEG7 MPEG standards Audio coding Content management Digital multimedia	<i>MRI</i> USE: Magnetic resonance imaging
broadcasting		Multimedia communication Multimedia systems	mRNA UF: Messenger RNA BT: RNA
MPEG standards	UF:	Moving picture experts	<i>MRP</i> USE: Materials requirements planning
group	BT:	IEC Standards ISO Standards	<i>Mud</i> USE: Sediments
broadcasting	RT:	Digital multimedia	<i>Mufflers</i> USE: Exhaust systems
		Image coding Streaming media Transform coding Video codecs Video coding Video signal processing	<i>Multi access edge computing</i> USE: Multi-access edge computing
NT:		MPEG 1 Standard MPEG 2 Standard	Multi label classification UF: Multi-label classification Multilabel classification BT: Classification algorithms
			<i>Multi lingual</i> USE: Multilingual



January 2026 IEEE Thesaurus

Multi modal integration

USE: Multisensory integration

Multi party computation

USE: Multi-party computation

Multi sensory integration

USE: Multisensory integration

Multi stage noise shaping

USE: Multi-stage noise shaping

Multi-access edge computing

UF: Mobile edge computing
Multi access edge

computing

BT: Edge computing
Mobile computing

Multi-agent models

USE: Multi-agent systems

Multi-agent systems

UF: Multi-agent models
Multiagent models
Multiagent systems
BT: Adaptive systems
RT: Agent-based modeling
Autonomous vehicles
Formation control
System analysis and design
Vehicular automation
NT: Collaborative intelligence

Multi-armed bandit problem

UF: K-armed bandit problem
N-armed bandit problem
BT: Machine learning
Probability
RT: Game theory
Markov processes
Optimization
Reinforcement learning
Resource management
Statistical distributions
Stochastic processes

Multi-attribute optimization

USE: Pareto optimization

Multi-carrier code division multiple access

USE: Multicarrier code division
multiple access

Multi-cast addressing

USE: Multicast communication

Multi-casting

USE: Multicast communication

Multi-core processing

USE: Multicore processing

Multi-core processors

USE: Multicore processing

Multi-criteria decision analysis

USE: MCDM

Multi-criteria decision making

USE: MCDM

Multi-factor authentication

BT: Access control
Authentication

Multi-hop

USE: Spread spectrum
communication

Multi-label classification

USE: Multi label classification

Multi-layer neural network

UF: MNN
BT: Neural networks

Multi-level converters

USE: Multilevel converters

Multi-level inverters

USE: Multilevel inverters

Multi-level invertors

USE: Multilevel inverters

Multi-lingual

USE: Multilingual

Multi-modal sensors

USE: Multimodal sensors

Multi-objective programming

USE: Pareto optimization

Multi-party computation

UF: Multi party computation
BT: Cryptography



January 2026 IEEE Thesaurus

Multi-physics

USE: Multiphysics

Multi-resolution

USE: Multiresolution analysis

Multi-robot systems

UF: Multirobot systems
BT: Robotics and automation
NT: Swarm robotics

Multi-spectral imaging

USE: Multispectral imaging

Multi-stage noise shaping

UF: MASH
Multi stage noise shaping
Multistage noise shaping
BT: Noise shaping

Multi-tasking

USE: Multitasking

Multi-threaded systems

USE: Program processors

Multi-threading systems

USE: Program processors

Multi-user channels

USE: Multiuser channels

Multi-user detection

USE: Multiuser detection

Multi-vibrators

USE: Multivibrators

Multi-wave mixing

USE: Multiwave mixing

Multiaccess communication

UF: CDMA
CSMA
Carrier sense multiaccess
Code division multiaccess
Code division multiple

access

Code-division multiple

access

Code-division multiple-

access

Multiaccess systems
Multiple access

communication

communication

BT: Cellular technology
Communication systems
RT: 3G mobile communication
Delay estimation
Multiplexing
OFDM
Telecommunications
Viterbi algorithm
NT: Access charges
Direct-sequence code-

division multiple access

Frequency division

multiaccess

Multicarrier code division

multiple access

Subscriber loops
Time division multiple

access

Time division synchronous

code division multiple access

Zero correlation zone

Multiaccess systems

USE: Multiaccess communication

Multiagent models

USE: Multi-agent systems

Multiagent systems

USE: Multi-agent systems

Multicarrier code division multiple access

UF: MC-CDMA
MCCDMA
Multi-carrier code division

multiple access

BT: Multiaccess communication
RT: Code division multiplexing
Communication channels
OFDM
Protocols
Spread spectrum

communication

Telecommunications
Time division synchronous
code division multiple access

Multicast addressing

USE: Multicast communication



January 2026 IEEE Thesaurus

Multicast algorithms

BT: Algorithms

Multicast communication

UF: Multi-cast addressing
Multi-casting
Multicast addressing
Multicasting
BT: Communication systems
RT: Ad hoc networks
Multicast protocols
Optical wavelength

conversion

Routing
Telecommunications
Wavelength division

multiplexing

NT: Multicast VPN

Multicast protocols

BT: Protocols
RT: Internet
Multicast communication
Routing protocols

Multicast VPN

BT: Multicast communication

Multicasting

USE: Multicast communication

Multichip modules

UF: Heterogeneous integration
BT: Integrated circuit packaging
Packaging

Multiconductor transmission lines

BT: Transmission lines
RT: Coupled mode analysis

Multicore

USE: Multicore processing

Multicore processing

UF: Many core processing
Many core systems
Multi-core processing
Multi-core processors
Multicore
BT: Parallel architectures
NT: Embedded multicore
processing
Manycore processors

Multidimensional signal processing

BT: Signal processing
RT: Image processing
NT: Video signal processing

Multidimensional systems

BT: Systems engineering and theory

Multifilamentary superconductors

BT: Superconducting materials
Superconducting wires

Multifrequency antennas

BT: Antennas

Multigrid methods

BT: Numerical analysis

Multihop

USE: Spread spectrum communication

Multilabel classification

USE: Multi label classification

Multilayer perceptrons

BT: Feedforward neural networks

Multilayers

USE: Nonhomogeneous media

Multileaf collimators

USE: Collimators

Multilevel converters

UF: Modular multi-level converters
Modular multilevel converters
Multi-level converters
BT: Converters

Multilevel fast multipole algorithm

USE: MLFMA

Multilevel inverters

UF: Multi-level inverters
Multi-level invertors
Multilevel invertors
BT: Inverters

Multilevel invertors

USE: Multilevel inverters



January 2026 IEEE Thesaurus

Multilevel systems

BT: Hierarchical systems

Multilinear systems

USE: Nonlinear systems

Multilingual

UF: Multi lingual
Multi-lingual
Multilingualism
Multilinguistics

BT: Linguistics

RT: Natural language

processing

Natural languages
Semantics

Multilingualism

USE: Multilingual

Multilinguistics

USE: Multilingual

Multimedia communication

BT: Communication systems
Multimedia systems

RT: B-ISDN
Broadband communication
Diffserv networks
Digital multimedia

broadcasting

HDMI
Huffman coding
IEEE 802.16 Standard
ISDN
Intserv networks
Journalism
MPEG 7 Standard
Multimedia computing
Streaming media
Transcoding
NT: Hypermedia

Multimedia computing

BT: Multimedia systems
RT: Audio user interfaces
Computer graphics
Computers and information

processing

Content management
Federated learning
Information systems
Multimedia communication
Multimedia databases

Video sequences

Multimedia databases

BT: Database systems
Databases
Multimedia systems
RT: Audio databases
Huffman coding
Multimedia computing

Multimedia products

USE: Videos

Multimedia systems

BT: Consumer electronics
RT: Authoring systems
Electronic publishing
Huffman coding
MPEG 7 Standard
Mixed reality
NT: Multimedia communication
Multimedia computing
Multimedia databases

Multimodal integration

USE: Multisensory integration

Multimodal sensing

USE: Multimodal sensors

Multimodal sensors

UF: Multi-modal sensors
Multimodal sensing
BT: Sensors
RT: Sensor fusion

Multiobjective programming

USE: Pareto optimization

Multipath channels

BT: Communication channels
RT: Channel estimation
Diversity methods
Fading channels
MIMO
MPTCP
Meteorological factors
Multiuser detection
Radio propagation
Terrain factors
Ultra wideband

communication



January 2026 IEEE Thesaurus

Multipath TCP

USE: MPTCP

Multiphysics

UF: Multi-physics
BT: Computational modeling
RT: Computer simulation
Physics computing

Multiple access communication

USE: Multiaccess communication

Multiple access interference

BT: OFDM

Multiple antenna systems

USE: MIMO

Multiple input multiple output

USE: MIMO

Multiple input multiple output systems

USE: MIMO

multiple input single-output

USE: MISO

Multiple sclerosis

BT: Diseases
RT: Spasticity

Multiple signal classification

UF: MUSIC
BT: Noise measurement

Multiple-input multiple-output

USE: MIMO

Multiple-input multiple-output systems

USE: MIMO

multiple-input single output

USE: MISO

multiple-input single-output

USE: MISO

Multiple-input-multiple-output

USE: MIMO

Multiplexed

USE: Multiplexing

Multiplexing

UF: Multiplexed
BT: Communications

technology

RT: Arrayed waveguide gratings
Multiaccess communication
NT: Code division multiplexing
Demultiplexing
Frequency division

multiplexing

Layered division

multiplexing

Multiplexing equipment
OFDM
Space division multiplexing
Time division multiplexing
Wavelength division

multiplexing

Multiplexing equipment

BT: Multiplexing
RT: Communication equipment
NT: Add-drop multiplexers

Multiplying circuits

BT: Circuits
RT: Digital integrated circuits
Logic circuits

Multiprocessing systems

BT: Parallel processing
RT: AI accelerators
Computer networks
Computers and information

processing

Concurrency control
Distributed computing
Parallel languages
Parallel programming
Pipeline processing
NT: Data flow computing
Processor scheduling
Systolic arrays

Multiprocessor interconnection

UF: Interconnection networks
Parallel processor

interconnection

BT: Computer architecture
RT: Computer networks
Data communication
Local area networks
Metropolitan area networks
Wide area networks
NT: Hypercubes



January 2026 IEEE Thesaurus

Multiprocessor interconnection networks

BT: Computer networks

Multiprocessor scheduling

USE: Processor scheduling

Multiprotocol label switching

UF: Label swapping
MPLS

BT: Communication switching
Packet switching
Protocols

RT: Asynchronous transfer

mode

Internet
Routing protocols

Multiresolution analysis

UF: Multi-resolution
BT: Wavelet analysis

Multirobot systems

USE: Multi-robot systems

Multisensor systems

BT: Sensor fusion
RT: Robot sensing systems

Multisensory integration

UF: Multi modal integration
Multi sensory integration
Multimodal integration
BT: Sense organs
RT: Somatosensory

Multiskilling

UF: Job rotation
BT: Human resource

management

RT: Industrial training
Job specification
Vocational training

Multispectral imaging

UF: Multi-spectral imaging
BT: Imaging

Multistage noise shaping

USE: Multi-stage noise shaping

Multistatic radar

BT: Radar
NT: MIMO radar

Multitasking

UF: Multi-tasking
BT: Computers and information
processing
NT: Parametric study

Multithreaded systems

USE: Program processors

Multithreading

BT: Parallel processing

Multithreading systems

USE: Program processors

Multiusers channels

UF: Multi-user channels
BT: Communication channels

Multiusers detection

UF: Multi-user detection
BT: Signal detection
RT: Cellular radio
Land mobile radio
Mobile communication
Multipath channels
Spread spectrum

communication

Multivalued logic

UF: Many valued logic
Ternary logic
BT: Logic
RT: Logic functions

Multivariable systems

USE: MIMO

Multivariate regression

BT: Regression analysis

Multivibrators

UF: Multi-vibrators
BT: Electronic circuits

Multiwave mixing

UF: Multi-wave mixing
BT: Optical mixing
RT: Four-wave mixing

Munitions

USE: Weapons



January 2026 IEEE Thesaurus

Muon colliders

UF: Muon sources
BT: Colliding beam devices
RT: Luminescence
Luminescent devices
Storage rings

Muon sources

USE: Muon colliders

Muons

USE: Mesons

Muscle-like actuators

USE: Artificial muscles

Muscles

BT: Musculoskeletal system
RT: Myoelectric control
Spasticity
NT: Myocardium
Neuromuscular

Muscular dystrophy

BT: Diseases
Medical conditions
RT: Musculoskeletal system

Musculoskeletal system

BT: Anatomy
RT: Muscular dystrophy
Rheumatology
NT: Cartilage
Fascia
Ligaments
Muscles
Skeleton
Tendons

Museums

BT: Educational institutions
RT: Art
Cultural aspects
Cultural differences
Humanities
Research and development
Technology
NT: Virtual museums

MUSIC

USE: Multiple signal classification

Music

UF: Computer music
Musical
BT: Humanities
RT: Audio systems
White noise
NT: Acoustics
Computer generated music
Electronic music
Musical instrument digital

interfaces

Musical instruments
Rhythm
Timbre

Music information retrieval

BT: Information retrieval
RT: Cepstral analysis
Cross modal retrieval

Music instruments

USE: Musical instruments

Music recommendation

USE: Recommender systems

Musical

USE: Music

Musical instrument digital interfaces

UF: MDDI
BT: Computer interfaces
Music
RT: Digital communication

Musical instruments

UF: Music instruments
BT: Music

Must carry regulations

USE: Must-carry regulations

Must-carry regulations

UF: Must carry regulations
BT: Cable TV
Government policies
RT: Licenses
TV

Mutual conductance

USE: Transconductance

Mutual coupling

BT: Electromagnetic coupling
NT: Inductive coupling



January 2026 IEEE Thesaurus

Mutual funds

BT: Financial management

Mutual information

UF: Transinformation
BT: Information theory

Myelin

BT: Nerve fibers
RT: Axons

Myocardial

USE: Myocardium

Myocardial infarction

UF: Heart attack
BT: Cardiovascular diseases
RT: Myocardium

Myocardium

UF: Myocardial
BT: Muscles
RT: Myocardial infarction

Myoelectric control

BT: Prosthetics
RT: Muscles
Robot kinematics

Myopia

USE: Vision defects

Myspace

USE: Social networking (online)

N-armed bandit problem

USE: Multi-armed bandit problem

N95

USE: Face masks

NACE International

UF: National Association of Corrosion Engineers
BT: Standards organizations

NACE Standards

BT: Standards publications

Nails

BT: Integumentary system

Naive Bayes algorithms

USE: Naive Bayes methods

Naive Bayes classification

USE: Naive Bayes methods

Naive Bayes classifiers

USE: Naive Bayes methods

Naive Bayes methods

UF: Naive Bayes algorithms
Naive Bayes classification
Naive Bayes classifiers
BT: Bayes methods
Learning (artificial intelligence)
RT: Pattern classification
Data mining
Machine learning
Probability
Supervised learning
Support vector machines
Text analysis

Nakagami distribution

BT: Probability distribution

Named entity recognition

BT: Natural language processing
RT: Information retrieval

NAND flash

USE: Flash memories

Nano biophotonics

USE: Nanobiophotonics

Nano communication

USE: Nanocommunication

Nano devices

USE: Nanoscale devices

Nano generators

USE: Nanogenerators

Nano packaging

USE: Nanopackaging

Nano ribbons

USE: Nanoribbons

Nano-magnetics

USE: Nanomagnetism

Nano-rods

USE: Nanorods



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<i>Nanoactuators</i>		Wireless sensor networks
USE:	Actuators AND Nanoelectronics	
Nanoantennas		Nanocomposite films
BT:	Nanoscale devices	BT: Films
	Optical antennas	RT: Nanoparticles
RT:	Integrated optics	Nanocomposites
	Integrated optoelectronics	BT: Nanostructured materials
	Molecular communication	RT: Metamaterials
	Nanoelectromechanical	
systems		Nanocontacts
	Nanoelectronics	BT: Nanoscale devices
	Nanophotonics	RT: Enhanced
	Nanotubes	magnetoresistance
	Optical communication	Magnetoresistance
equipment		Nanoelectronics
	Optical metamaterials	Nanowires
	Plasmons	
Nanobiophotonics		<i>Nanocrystal</i>
UF:	Nano biophotonics	USE: Nanocrystals
BT:	Nanobiotechnology	
	Photonics	Nanocrystals
RT:	Biomedical imaging	UF: Nanocrystal
	Biosensors	BT: Crystalline materials
	Lasers	Nanoparticles
	Nanoparticles	RT: Quantum dots
	Nanoscale devices	NT: Colloidal nanocrystals
Nanobioscience		<i>Nanodevices</i>
BT:	Biology	USE: Nanoscale devices
	Nanotechnology	
RT:	Colloidal lithography	Nanoelectromechanical systems
	Nanofluidics	UF: NEMS
NT:	DNA computing	BT: Nanotechnology
	Nanobiotechnology	RT: Microelectromechanical
Nanobiotechnology		systems
BT:	Nanobioscience	Nanoantennas
RT:	Nanomedicine	
	Nanopores	Nanoelectronics
NT:	Nanobiophotonics	UF: Nanoactuators
Nanocarriers		BT: Nanotechnology
BT:	Nanomaterials	RT: Graphene devices
RT:	Drug delivery	Molecular electronics
		Nanoantennas
Nanocommunication		Nanocontacts
UF:	Nano communication	NT: Junctionless nanowire
BT:	Communication systems	transistors
	Nanotechnology	
RT:	Biomedical communication	Nanofabrication
	Molecular communication	BT: Nanotechnology
	Wireless networks	
		<i>Nanofiltration</i>
		USE: Filtration



January 2026 IEEE Thesaurus

Nanofluidics

UF: Nanofluids
BT: Fluidics
RT: Nanotechnology
Nanobioscience

Nanofluids

USE: Nanofluidics

Nanogenerators

UF: Nano generators
BT: Energy harvesting
RT: Electric generators
Low power electronics
Micromechanical devices
Nanowires
Piezoelectric devices
Triboelectricity
Vibrations

Nanoimprint lithography

USE: Nanolithography

Nanolithography

UF: Nanoimprint lithography
BT: Lithography
Nanotechnology
RT: Nanopatterning
Soft lithography

Nanomagnetics

UF: Nano-magnetics
BT: Magnetism
Nanotechnology
RT: Magnetic particles
Nanostructured materials

Nanomaterials

BT: Nanotechnology
RT: Nanopackaging
NT: Nanocarriers

Nanomedicine

BT: Biomedical monitoring
Medical diagnosis
RT: Cellular biophysics
Mechanobiology
Nanobiotechnology
Nanoparticles

Nanometers

UF: Millimicron
Nanometres
BT: Measurement units

Nanometres

USE: Nanometers

Nanopackaging

UF: Nano packaging
BT: Nanotechnology
Packaging
RT: Nanomaterials

Nanoparticles

UF: Nanopowders
BT: Nanostructures
RT: Nanobiophotonics
Nanocomposite films
Nanomedicine
Nanosensors
NT: Magnetic nanoparticles
Nanocrystals

Nanopatterning

BT: Nanotechnology
RT: Nanolithography
Nanotopography
Soft lithography
NT: Colloidal lithography

Nanophotonics

BT: Nanotechnology
Photonics
RT: Nanoantennas
Optical antennas

Nanoplasmonics

BT: Nanotechnology
Plasmons
NT: Surface plasmon
resonance

Nanopores

BT: Nanoscale technology
RT: Nanobiotechnology
Nanoporous materials
Nanostructures

Nanoporous materials

BT: Nanostructured materials
RT: Nanopores

Nanopositioning

BT: Nanotechnology
Position control

Nanopowders

USE: Nanoparticles



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Nanoribbons

UF: Nano ribbons
BT: Nanostructures

Nanorods

UF: Nano-rods
BT: Nanostructures

Nanosatellites

USE: Small satellites

Nanoscale devices

UF: Nano devices
Nanodevices
BT: Nanoscale technology
RT: Nanobiophotonics
Single electron devices
NT: Nanoantennas
Nanocontacts
Nanotube devices

Nanoscale technology

BT: Nanotechnology
NT: Nanopores
Nanoscale devices

Nanosensors

BT: Nanotechnology
Sensors
RT: Biomedical equipment
Nanoparticles
Nanostructures
Wearable sensors

Nanostructured materials

UF: Core-shelf nanostructures
BT: Materials
Nanotechnology
RT: Nanomagnetism
NT: Nanocomposites
Nanoporous materials

Nanostructures

BT: Nanotechnology
RT: Nanopores
Nanosensors
NT: Nanoparticles
Nanoribbons
Nanorods
Nanotubes
Nanowires
Semiconductor

nanostructures

Nanotechnology

RT: Atomic force microscopy
Bio-inspired materials
Epitaxial growth
Fluidics
Lithography
Microfabrication
Nanotube devices
Power dissipation
Quantum mechanics
Semiconductor device

manufacture

NT: Single electron devices
Very large scale integration
Bionanotechnology
Casimir effect
Molecular computing
Molecular electronics
Nanobioscience
Nanocommunication
Nanoelectromechanical

systems

Nanoelectronics
Nanofabrication
Nanofluidics
Nanolithography
Nanomagnetics
Nanomaterials
Nanopackaging
Nanopatterning
Nanophotonics
Nanoplasmonics
Nanopositioning
Nanoscale technology
Nanosensors
Nanostructured materials
Nanostructures
Self-assembly
Self-replicating machines

Nanotopography

BT: Surface topography
RT: Colloidal lithography
Nanopatterning

Nanotube devices

BT: Nanoscale devices
RT: Nanotechnology

Nanotubes

BT: Nanostructures
RT: Nanoantennas
NT: Carbon nanotubes
Semiconductor nanotubes



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Nanowires

BT: Nanostructures
RT: Junctionless nanowire

transistors

Nanocontacts
Nanogenerators
Wires

NAP

USE: National Adaptation Plans

Narrowband

BT: Bandwidth
Communication systems
RT: Wideband

NASA

UF: National Aeronautics &
Space Administration
National Aeronautics and
Space Administration
BT: Space agencies
US Government agencies
RT: Landsat
Space exploration
Space missions
Space technology

Nash equilibrium

BT: Game theory

National Adaptation Plans

UF: NAP
BT: Environmental
management

National Aeronautics & Space Administration

USE: NASA

National Aeronautics and Space Administration

USE: NASA

National Association of Corrosion Engineers

USE: NACE International

National Bureau of Standards

USE: NIST

National Electric Code

UF: National electric safety
code
BT: ANSI Standards

National electric safety code

USE: National Electric Code

National Fire Protection Agency

USE: NFPA

National Fire Protection Association

USE: NFPA

National Institute of Standards & Technology

USE: NIST

National Institute of Standards and Technology

USE: NIST

National Institutes of Health

UF: NIH
BT: US Department of Health
and Human Services

National Oceanic and Atmospheric Administration

UF: NOAA
BT: US Department of
Commerce

National security

BT: Terrorism
RT: Biological weapons
Chemical weapons
Control system security
Cyber warfare
Nuclear weapons
Weapons of mass
destruction

*National Security Laboratory-Knowledge
Discovery and Data Mining*

USE: NSL-KDD

National Society Agreement awards

BT: IEEE Awards activities

*National Telecommunications and Information
Administration*

USE: NTIA

National vocational qualification

USE: Vocational training

Nationally Determined Contributions

UF: NDC
BT: Climate change

Natural capital

BT: Environmental economics



January 2026 IEEE Thesaurus

Natural fibers

UF: Natural fibres
BT: Textile fibers
RT: Cotton
Wool

Natural fibres

USE: Natural fibers

Natural gas

BT: Fossil fuels
RT: Air pollution
Energy resources
Fracking
Gases
Hydrocarbons
Natural gas industry
NT: Liquefied natural gas
Methane

Natural gas industry

BT: Industries
RT: Natural gas
Petroleum industry
Pipelines

Natural language generation

UF: NLG
BT: Natural language

processing

RT: Computational linguistics
Human-machine systems
Semantic technology
Speech synthesis

Natural language processing

UF: NLP
BT: Natural languages
RT: Bidirectional long short term

memory

Conditional random fields
Digital humans
Generative Pre-Trained

Transformer

High dimensional data
Multilingual
Personal voice assistants
Phonetics
Pragmatics
Semantic search
Semantic technology
Semantics
Semiotics
Social robots
Syntactics

Text summarization
Translation
Vision language model
Bag of words model
Chatbots
Cross lingual
Machine translation
Morphology
Named entity recognition
Natural language

generation

(information retrieval)

Question answering
Question generation
Sentiment analysis
Tokenization

Natural languages

UF: Natural speech
BT: Humanities
RT: Artificial intelligence
Computer languages
Multilingual
NT: Lexicon
Linguistics
Natural language

processing

Translation

Natural resources

BT: Geoscience

Natural response

USE: Transient response

Natural speech

USE: Natural languages

Nature-based solutions

BT: Environmental
management

Navier-Stokes equations

BT: Differential equations
Fluid dynamics
RT: Finite volume methods
Viscosity

Navigation

UF: Direction-finding
Geomagnetic navigation
BT: Intelligent transportation

systems

Vehicular and wireless

technologies



January 2026 IEEE Thesaurus

	RT:	Compass	BT:	Communication standards
		Ground support		Radio communication
		Location awareness	RT:	Magnetic communication
		Motion planning	NT:	Mobile communication
		Position measurement		
		Sensor systems		
	NT:	Aircraft navigation		<i>Near field radiation pattern</i>
		Course correction	USE:	Near-field radiation pattern
		Dead reckoning		
		Indoor navigation		<i>Near vertical incidence skywave</i>
		Inertial navigation	USE:	NVIS
		Marine navigation		
		Radio navigation		<i>Near-field communication</i>
		Satellite navigation systems	USE:	Near field communication
		Sonar navigation		
		Underwater navigation		
				Near-field radiation pattern
			UF:	Near field radiation pattern
			BT:	Antenna radiation patterns
<i>Nb</i>				Nearest neighbor methods
	USE:	Niobium	UF:	K-NN methods
<i>Nb3Sn</i>				K-nearest neighbor
	USE:	Niobium-tin		KNN methods
<i>NBS</i>				Nearest neighbor searches
	USE:	NIST		Nearest neighbour methods
<i>NBTI</i>				k neighbor methods
	USE:	Negative bias temperature	BT:	k neighbour methods
instability			intelligence)	Learning (artificial
<i>NC machines</i>				Nonparametric statistics
	USE:	Computer numerical control	RT:	Pattern recognition
<i>Nd</i>				Data mining
	USE:	Neodymium		Pattern classification
<i>Nd2O3</i>				Pattern clustering
	USE:	Neodymium compounds		Regression analysis
<i>NDC</i>				Search methods
	USE:	Nationally Determined		Statistical analysis
Contributions				
<i>NDVI</i>				<i>Nearest neighbor searches</i>
	USE:	Normalized difference	USE:	Nearest neighbor methods
vegetation index				
<i>NDWI</i>				<i>Nearest neighbour methods</i>
	USE:	Normalized difference	USE:	Nearest neighbor methods
water index				
				<i>Nearfield communication</i>
Near field communication			USE:	Near field communication
	UF:	NFC		
		Near-field communication		
		Nearfield communication		
				Neck
			BT:	Body regions
			RT:	Otorhinolaryngology
				Needles
			BT:	Mechanical products
			RT:	Biomedical equipment
				Textile machinery



January 2026 IEEE Thesaurus

Negative bias temperature instability

UF: NBTI
BT: MOS devices

Liver neoplasms
Lung neoplasms
Skin neoplasms

Negative feedback

BT: Feedback

Nephrolithiasis

USE: Kidney stones

Negative feedback amplifier

USE: Feedback amplifiers

Nephrology

BT: Medical specialties
RT: Kidney
Kidney stones

Negative feedback loops

BT: Feedback loop

Neptunium

BT: Chemical elements

NEMA

BT: Standards organizations

NeRF

USE: Neural radiance field

NEMS

systems

USE: Nanoelectromechanical

Nerve cells

USE: Neurons

Neodymium

UF: Nd
BT: Metals
NT: Neodymium alloys
Neodymium compounds

Nerve endings

BT: Nervous system

Neodymium alloys

BT: Neodymium
RT: Alloying

Nerve fibers

BT: Neurons
NT: Axons
Myelin

Neodymium compounds

UF: Nd₂O₃
Neodymium oxide
BT: Neodymium

Nerve tissues

BT: Nervous system

Neodymium oxide

USE: Neodymium compounds

Nervous system

BT: Anatomy
RT: Bioelectric phenomena
Computational

Neon

BT: Chemical elements

neuroscience

Neural networks
Neurological diseases
Neurology
Neuromuscular stimulation
Neuropathology
NT: Autonomic nervous system
Brain
Brain mapping
Central nervous system
Cranial
Ganglia
Glial cells
Nerve endings
Nerve tissues
Neural pathways
Neuroanatomy
Neurons
Neuroradiology

Neonatal

BT: Pediatrics

Neonatology

BT: Medical specialties
RT: Pediatrics

Neoplasia

USE: Neoplasms

Neoplasms

UF: Neoplasia
BT: Biological tissues
NT: Breast neoplasms



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	Neuroscience		Network function virtualization
	Peripheral nervous system	UF:	NFV
	Pituitary gland	BT:	Computer networks
	Spinal cord		Network architecture
	Spine	RT:	Application virtualization
	Synapses		Cloud computing
<i>Net neutrality</i>			Computer network
USE:	Network neutrality	management	Intrusion detection
			Routing
Net zero			Servers
UF:	Net-Zero		Software defined
BT:	Environmental	networking	
management			Virtual machining
RT:	Carbon dioxide		Virtualization
	Global warming	NT:	Cloud radio access
	Green communications	networks	
	Greenhouse gases		Virtual LAN
NT:	Net zero water		
	Zero waste		
Net zero water		Network interfaces	
BT:	Net zero	BT:	Interface phenomena
		RT:	Interface management
<i>Net-Zero</i>		Network intrusion	
USE:	Net zero	BT:	Data breach
		RT:	Data security
			Network security
Network address translation			Privacy
BT:	Computer network	NT:	Network intrusion detection
management			
<i>Network analysis</i>		Network intrusion detection	
USE:	Network analyzers	BT:	Intrusion detection
			Network intrusion
Network analyzers		NT:	NSL-KDD
UF:	Network analysis		
BT:	Instruments	Network location awareness	
		BT:	Location awareness
Network architecture			
BT:	Network topology	Network motifs	
	Telecommunication	BT:	Graph theory
network management		RT:	Pattern recognition
NT:	Active networking		
	Information-centric	Network neutrality	
networking		UF:	Internet neutrality
	Network function		Net neutrality
virtualization		BT:	Telecommunication
	Network slicing	network management	
Network coding			
BT:	Information theory	<i>Network of workstations</i>	
RT:	Network security	USE:	Cluster computing
<i>Network control systems</i>		Network operating systems	
USE:	Networked control systems	BT:	Operating systems
		RT:	Software defined
		networking	



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NT:	Autonomic systems	network topology	Telecommunication
Network reconnaissance		NT:	Complex networks
UF:	Footprinting		Computer network reliability
BT:	Network security		Network architecture
Network resource management		<i>Network traffic</i>	
UF:	Dynamic service delivery	USE:	Telecommunication traffic
BT:	Resource management		
	Telecommunication	<i>Network-on-a-chip</i>	
network management		USE:	Network-on-chip
RT:	Cellular radio		
	Mobile communication	Network-on-chip	
NT:	Intercell interference	UF:	Network-on-a-chip
		BT:	System-on-chip
Network security		Networked control systems	
BT:	Security	UF:	Network control systems
RT:	Communication networks	BT:	Control systems
	Computer networks	RT:	Real-time systems
	Hardware security		System of systems
	NSL-KDD		
	Network coding	Neural activity	
	Network intrusion	UF:	Neural oscillation
NT:	Network reconnaissance	BT:	Brain
		RT:	Stability plasticity
Network servers		Neural architecture search	
BT:	Computer networks	UF:	Neural architectures search
Network slicing		BT:	Artificial neural networks
BT:	Network architecture	RT:	Automated machine
RT:	Augmented reality	learning	
Network synthesis		<i>Neural architectures search</i>	
BT:	Computer network	USE:	Neural architecture search
management			
Network systems		<i>Neural chips</i>	
BT:	Systems engineering and	USE:	Neural network hardware
theory			
	NT:	Neural circuits	
	Autonomous networks	BT:	Circuits
	DC distribution systems		Neural engineering
		RT:	Neurons
Network theory (graphs)		Neural engineering	
BT:	Computer science	UF:	Neuro engineering
	Mathematics		Neuroengineering
	Physics	BT:	Biomedical engineering
RT:	Social sciences	RT:	Brain-computer interfaces
			Intracranial pressure
<i>Network throughput</i>		sensors	
USE:	Throughput	NT:	Neural circuits
			Neural microtechnology
Network topology			Neural nanotechnology
BT:	Communications		Neural prosthesis
technology			
	RT:		
	Overlay networks		



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Neural implants

UF: Brain implants
BT: Brain
Implants
RT: Deep brain stimulation

systems

Nonlinear dynamical

Pattern classification
Reinforcement learning
Semisupervised learning
Soft sensors

Neural machine translation

BT: Machine translation
RT: Artificial neural networks

NT:

Systems neuroscience
Artificial neural networks
Biological neural networks
Cellular neural networks
Feedforward neural

Neural microtechnology

BT: Neural engineering

networks

Neural nanotechnology

BT: Neural engineering

compression

Graph neural networks
Multi-layer neural network
Neural network

Neural nets

USE: Neural networks

networks

Neural network hardware
Perception evolution

Neural network compression

BT: Data compression
Memory management
Neural networks

networks

Radial basis function

Recurrent neural networks
Stability plasticity
Vision language model

Neural network hardware

UF: Neural chips
BT: Neural networks
RT: AI accelerators
Analog integrated circuits
Integrated circuits
Neurocontrollers

Neural oscillation

USE: Neural activity

Neural pathways

BT: Nervous system

Neural prostheses

USE: Prosthetics

Neural networks

UF: Neural nets
Wavelet neural networks
BT: Computational and artificial
intelligence
RT: AI accelerators
Adaptive systems
Artificial intelligence
Associative memory
Backpropagation
Bio-inspired computing
Cybernetics
Deep architecture
Deep learning
Dynamic programming
Fish schools
Fuzzy cognitive maps
Generative AI
Generative adversarial

Neural prosthesis

BT: Neural engineering

Neural prosthetics

USE: Neuroprostheses

Neural radiance field

UF: NeRF
BT: Deep learning
Image analysis
Image synthesis
RT: Object detection
Synthetic data
Three-dimensional displays
Two-dimensional displays

networks

Large language models
Nervous system
Neurophysiology

Neural style transfer

BT: Image transformation



January 2026 IEEE Thesaurus

Neurites

UF: Neuronal process
BT: Neurons

Neuro engineering

USE: Neural engineering

Neuro fuzzy networks

USE: Fuzzy neural networks

Neuro imaging

USE: Neuroimaging

Neuro marketing

USE: Neuromarketing

Neuro transmitters

USE: Neurotransmitters

Neuro-feedback

USE: Neurofeedback

Neuro-fuzzy networks

USE: Fuzzy neural networks

Neuro-imaging

USE: Neuroimaging

Neuro-modulation

USE: Neuromodulation

Neuro-rehabilitation

USE: Neurorehabilitation

Neuro-transmitters

USE: Neurotransmitters

Neuroanatomy

BT: Anatomy
Nervous system

Neurocontrollers

BT: Intelligent control
RT: Artificial intelligence
Microcontrollers
Neural network hardware

Neurodynamics

BT: Brain
RT: Neurophysiology

Neuroengineering

USE: Neural engineering

Neurofeedback

UF: Neuro-feedback
BT: Feedback

Neuroglia

USE: Glial cells

Neuroimaging

UF: Brain imaging
Neuro imaging
Neuro-imaging
BT: Biomedical image
processing
RT: Brain mapping
Functional near-infrared
spectroscopy
NT: Neuroradiology
Functional neuroimaging

Neuroinformatics

BT: Bioinformatics
Informatics
Neuroscience
RT: Analytical models
Big Data
Computational modeling
Data science
Synapses

Neurological diseases

UF: Neurological disorders
BT: Diseases
RT: Nervous system
Spinal cord injury
NT: Migraine

Neurological disorders

USE: Neurological diseases

Neurology

BT: Medical specialties
RT: Nervous system
NT: Dyslexia
Neurorehabilitation

Neuromarketing

UF: Neuro marketing
BT: Consumer behavior
Neuroscience
RT: Electronic commerce
Ethics
Market opportunities
Market research



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Neuromodulation

UF: Neuro-modulation
BT: Neurons
Physiology
RT: Control systems
NT: Neurostimulation
Optogenetics

Neurites
Neuromodulation
Photoreceptors
Soma

Neuromorphic computing

USE: Neuromorphic engineering

Neuromorphic engineering

UF: Neuromorphic computing
BT: Neuromorphics
RT: AI accelerators
Artificial neural networks
CMOS integrated circuits
Memristors
Neurophysiology
Synapses

Neuromorphics

BT: Very large scale integration
RT: Analog circuits
Biological system modeling
NT: Neuromorphic engineering

Neuromuscular

BT: Muscles

Neuromuscular stimulation

UF: Functional electrical stimulation
BT: Medical treatment
RT: Nervous system

Neuronal networks

USE: Biological neural networks

Neuronal process

USE: Neurites

Neurone

USE: Neurons

Neurons

UF: Nerve cells
Neurone
BT: Nervous system
RT: Action potentials
Membrane potentials
Neural circuits
Synapses
NT: Dendrites (neurons)
Nerve fibers

Neuropathic pain

BT: Pain

Neuropathology

BT: Pathology
RT: Nervous system

Neurophysiology

BT: Brain
RT: Biomedical signal processing
Microelectrodes
Neural networks
Neurodynamics
Neuromorphic engineering
Science - general
NT: Biological neural networks
Neuroplasticity

Neuroplasticity

UF: Brain plasticity
Cortical plasticity
BT: Neurophysiology
RT: Stability plasticity

Neuroprostheses

UF: Neural prosthetics
Neuroprosthetics
BT: Brain
Neuroscience
Prosthetics
RT: Brain-computer interfaces

Neuroprosthetics

USE: Neuroprostheses

Neuropsychology

BT: Brain
Psychology

Neuroradiology

BT: Nervous system
Radiology
RT: Electromagnetics
Neuroimaging

Neurorehabilitation

UF: Neuro-rehabilitation
BT: Neurology
Patient rehabilitation



January 2026 IEEE Thesaurus

Neuroscience

BT: Nervous system
Science - general
RT: Fish schools
NT: Clinical neuroscience
Cognitive neuroscience
Computational

neuroscience

Functional connectivity
Neuroinformatics
Neuromarketing
Neuroprostheses
Systems neuroscience
Transcranial direct current

stimulation

Transcranial magnetic

stimulation

Neurostimulation

UF: Electroceuticals
BT: Neuromodulation
RT: Deep brain stimulation
Microelectrodes
NT: Transcranial direct current

stimulation

Transcranial magnetic

stimulation

Neurosurgery

BT: Surgery
NT: Deep brain stimulation

Neurotechnology

BT: Brain
Technology

Neurotransmission

USE: Neurotransmitters

Neurotransmitters

UF: Neuro transmitters
Neuro-transmitters
Neurotransmission
Synaptic transmission
BT: Transmitters
RT: Synapses

Neutrino

USE: Neutrino sources

Neutrino sources

UF: Neutrino
Neutrinos
BT: Elementary particles
RT: Radioactive materials

Neutrinos

USE: Neutrino sources

Neutron beams

USE: Particle beams

Neutron capture therapy

UF: BNCT
Boron neutron capture

therapy

BT: Medical treatment
RT: Biological effects of

radiation

Dosimetry

Neutron radiation effects

BT: Radiation effects

Neutron scattering

USE: Neutron spin echo

Neutron spin echo

UF: Neutron scattering
BT: Spectroscopy

Neutron stars

BT: Stars

Neutrons

BT: Elementary particles
RT: Cosmic rays

New media age

USE: Information age

New Radio

BT: 5G mobile communication
Radio access technologies
RT: 3GPP

Newborns

USE: Pediatrics

Newton Fourier method

USE: Newton method

Newton method

UF: Newton Fourier method
Newton Raphson method
Newton's method
Newton-Fourier method
Newton-Raphson method
Newtons method
BT: Numerical analysis



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RT: Optimization methods
Poles and zeros

Newton Raphson method
USE: Newton method

Newton's method
USE: Newton method

Newton-Fourier method
USE: Newton method

Newton-Raphson method
USE: Newton method

Newtons method
USE: Newton method

Next generation network architecture
USE: Next generation networking

Next generation networking
UF: 21CN
21st century networks
NGN
NGNA
Next generation network architecture
BT: Next generation networks
Next-generation networks
Computer networks
RT: 3G mobile communication
4G mobile communication
5G mobile communication
IP networks
Internet
Packet switching
Pervasive computing
Quality of service
Telecommunications

Next generation networks
USE: Next generation networking

Next-generation networks
USE: Next generation networking

NFB
USE: Feedback amplifiers

NFC
USE: Near field communication

NFPA
UF: National Fire Protection Agency
National Fire Protection Association
BT: Standards organizations

NFT
USE: Nonfungible tokens

NFV
USE: Network function virtualization

NGN
USE: Next generation networking

NGNA
USE: Next generation networking

NGO
USE: Non-governmental organizations

Ni
USE: Nickel

Nickel
UF: Ni
BT: Metals
NT: Nickel alloys
Nickel compounds

Nickel alloys
BT: Nickel
RT: Alloying

Nickel cadmium batteries
UF: Nickel-cadmium batteries
BT: Batteries

Nickel compounds
BT: Nickel

Nickel-cadmium batteries
USE: Nickel cadmium batteries

Night vision
BT: Infrared imaging
RT: Image sensors
Military equipment

NIH
USE: National Institutes of Health



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Niobium

UF: Nb
BT: Metals
RT: Type II superconductors
NT: Niobium alloys
Niobium compounds

Niobium alloys

BT: Niobium
RT: Alloying
NT: Niobium-tin

Niobium compounds

BT: Niobium

Niobium-tin

UF: Nb₃Sn
BT: Niobium alloys
Superconducting materials
Tin alloys

NISO Standards

BT: Standards publications

NIST

UF: NBS
National Bureau of
Standards
National Institute of
Standards & Technology
National Institute of
Standards and Technology
BT: Standards organizations
US Department of
Commerce

NIST Standards

BT: Standards publications

Nitrification

BT: Oxidation

Nitrogen

BT: Chemical elements
Gases
NT: Denitrification
Nitrogen compounds
Silicon nitride

Nitrogen compounds

UF: Potassium nitrate
Sodium nitrate
BT: Nitrogen
NT: Ammonia
Nitrous oxide

Nitrous oxide

BT: Nitrogen compounds
RT: Greenhouse gases

NLG

USE: Natural language
generation

NLP

USE: Natural language
processing

nMOSFETs

USE: MOSFET

NMR

USE: Nuclear magnetic
resonance

NMR imaging

USE: Magnetic resonance
imaging

NOAA

USE: National Oceanic and
Atmospheric Administration

Nobel Prize

BT: Awards

Noise

BT: Signal processing
RT: Autoregressive processes
Cyclic redundancy check
Distortion
Electromagnetic

interference

Interference
Noise generators
Noise measurement
Roundoff errors
NT: 1/f noise
Additive noise
Colored noise
Gaussian noise
Laser noise
Low frequency noise
Noise cancellation
Phase noise
Signal to noise ratio
Superconducting device

noise

White noise



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Noise abatement

USE: Noise reduction

Noise cancellation

UF: Noise cancellers

BT: Acoustic noise

Noise

RT: Filtering

Noise cancellers

USE: Noise cancellation

Noise figure

BT: Noise measurement

RT: Signal to noise ratio

Noise generators

BT: Signal generators

RT: Noise

Noise level

BT: Acoustic noise

Noise measurement

UF: Noisy

BT: Measurement

RT: Distortion measurement

Electric variables

measurement

Noise

Packet loss

NT: Multiple signal classification

Noise figure

Noise shaping

Noise reduction

UF: Audio enhancement

De-noising

Denoising

Noise abatement

Noise suppression

BT: Acoustic noise

NT: Active noise reduction

Noise robustness

Wiener filters

Noise robust

USE: Noise robustness

Noise robustness

UF: Noise robust

BT: Noise reduction

Noise shaping

UF: Noise-shaping

BT: Noise measurement

NT: Multi-stage noise shaping

Noise suppression

USE: Noise reduction

Noise-shaping

USE: Noise shaping

Noisy

USE: Noise measurement

NOMA

UF: Nonorthogonal multiple

access

non-orthogonal multiple

access

BT: Access protocols

Communication systems

RT: Cellular radio

Free-space optical

communication

MIMO

OFDM

Radio communication

Radiofrequency

interference

Non fungible tokens

USE: Nonfungible tokens

Non relational databases

USE: NoSQL databases

Non repudiation

USE: Non-repudiation

Non-flowering plants

UF: Conifers

Gymnosperms

BT: Plants (biology)

Non-fungible tokens

USE: Nonfungible tokens

Non-government organizations

USE: Non-governmental

organizations

Non-governmental organizations

UF: NGO

Non-government

organizations



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organizations
 BT: Organizations
 RT: Professional societies

Non-gyroscopes
 USE: Gyroscopes

non-orthogonal multiple access
 USE: NOMA

Non-parametric statistics
 USE: Nonparametric statistics

Non-repudiation
 UF: Non repudiation
 BT: Access control
 Data integrity
 RT: Digital forensics

Non-united-states activities
 USE: IEEE professional activities

Non-volatile memory
 USE: Nonvolatile memory

Nonce
 BT: Cryptography
 Random variables

Nonconductive adhesives
 BT: Adhesives

Nondestructive testing
 BT: Materials testing
 RT: Acoustic emission
 Ultrasonic transducers
 NT: Magnetic flux leakage

Nondeterministic polynomial-time hard
 USE: NP-hard problem

Nonfungible tokens
 UF: NFT
 Non fungible tokens
 Non-fungible tokens
 BT: Authentication
 Blockchains
 RT: Bitcoin
 Cryptocurrency
 Distributed ledger

Nongovernmental organizations
 USE: Non-governmental
 organizations

Nongyroscopes
 USE: Gyroscopes

Nonhomogeneous media
 UF: Composite media
 Inhomogeneous media
 Layered media
 Multilayers
 Periodic media
 Stratified media
 BT: Media
 RT: Random media

Noninvasive diagnosis
 USE: Noninvasive treatment

Noninvasive measurement
 USE: Noninvasive treatment

Noninvasive surgery
 USE: Noninvasive treatment

Noninvasive technique
 USE: Noninvasive treatment

Noninvasive therapy
 USE: Noninvasive treatment

Noninvasive treatment
 UF: Noninvasive diagnosis
 Noninvasive measurement
 Noninvasive surgery
 Noninvasive technique
 Noninvasive therapy
 BT: Medical treatment
 RT: Pulse oximeter
 NT: Embolization
 Pulse oximetry

Nonlinear acoustics
 BT: Acoustics
 RT: Acoustic distortion
 Nonlinear wave

propagation

Nonlinear circuits
 BT: Circuits
 RT: Chaos
 Mixers
 Power conversion
 Rail to rail inputs
 Rail to rail outputs
 NT: Nonlinear network analysis



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Nonlinear control systems

BT: Control systems
RT: Control nonlinearities
Piecewise linear techniques

Nonlinear distortion

BT: Distortion
RT: Limiting
Predistortion
NT: Harmonic distortion
Intermodulation distortion

Nonlinear dynamical systems

UF: Nonlinear dynamics
BT: Dynamical systems
Nonlinear systems
RT: Chaos
Econophysics
Fuzzy sets
Kalman filters
Neural networks
Pattern formation
Possibility theory
Predator prey systems
Spatiotemporal phenomena
Uncertainty

Nonlinear dynamics

USE: Nonlinear dynamical systems

Nonlinear equations

BT: Equations
Mathematics
RT: Algebra
Chaotic mapping
Linear approximation
Nonlinear systems
Numerical analysis
NT: Bifurcation

Nonlinear filters

BT: Filters
RT: Detectors
Phase locked loops

Nonlinear magnetics

BT: Magnetics
RT: Ferroresonance

Nonlinear network analysis

BT: Circuit analysis
Nonlinear circuits

Nonlinear optical devices

BT: Nonlinear optics
RT: Optical detectors

Nonlinear optics

BT: Optics
RT: Cross-phase modulation
Electro-optic effects
Pattern formation
Photonic crystals
Thermal lensing
NT: Fiber nonlinear optics
Nonlinear optical devices
Optical mixing
Optical saturation
Photorefractive effect
Raman scattering
Supercontinuum generation

Nonlinear systems

UF: Bilinear systems
Multilinear systems
BT: Mathematics
RT: Control systems
Linear approximation
Manipulators
Mobile robots
Nonlinear equations
Robots
NT: Chaos
Nonlinear dynamical

systems

Nonlinear wave propagation

BT: Propagation
RT: Nonlinear acoustics

Nonorthogonal multiple access

USE: NOMA

Nonparametric statistics

UF: Non-parametric statistics
BT: Statistics
NT: Nearest neighbor methods

Nonrelational databases

USE: NoSQL databases

Nonuniform electric fields

BT: Electric fields

Nonuniform sampling

BT: Sampling methods



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Nonuniform transmission lines

USE: Distributed parameter circuits

Nonvolatile memories

USE: Nonvolatile memory

Nonvolatile memory

UF: Non-volatile memory
Nonvolatile memories
BT: Memory
NT: Racetrack memory

Normal distribution

USE: Gaussian distribution

Normalized difference vegetation index

UF: NDVI
BT: Environmental measurement
RT: Vegetation
Feature detection

Normalized difference water index

UF: NDWI
BT: Environmental measurement
RT: Water
Feature detection

North America

BT: Continents

North Pole

BT: Arctic

Nose

BT: Head
Sense organs
NT: Olfactory

NoSQL databases

UF: Non relational databases
Nonrelational databases
BT: Database systems
RT: Big Data
Data structures
Data warehouses
Distributed databases
Linked data
Query processing

Notch filters

UF: Band-stop filters
BT: Filters

Novelty detection

USE: Anomaly detection

NP hard problem

USE: NP-hard problem

NP-C

USE: NP-complete problem

NP-complete problem

UF: NP-C
BT: Complexity theory

NP-hard problem

UF: NP hard problem
Nondeterministic
polynomial-time hard
BT: Complexity theory
NT: Traveling salesman problems

NSL-KDD

UF: National Security Laboratory-Knowledge Discovery and Data Mining
BT: Data models
Network intrusion detection
RT: Benchmark testing
Data mining
Knowledge discovery
Network security

NTIA

UF: National Telecommunications and Information Administration
BT: US Department of Commerce

Nuclear and plasma sciences

NT: Biomedical applications of radiation
Colliding beam devices
Electron emission
Elementary particles
Fusion power generation
Fusion reactors
Gamma-rays
Gas discharge devices
Helioseismology
High energy physics
instrumentation computing
Ion beam applications
Nuclear electronics
Nuclear imaging



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(electronics)
Nuclear medicine
Nuclear physics
Particle accelerators
Particle beam handling
Particle beam injection
Plasmas
Radiation effects
Radiation hardening

Radiation monitoring
Radiation safety
Reactor instrumentation
Scintillation counters
Thermionic emission

Nuclear bombs

USE: Nuclear weapons

Nuclear electronics

BT: Nuclear and plasma sciences

RT: FET circuits
High energy physics instrumentation computing

Nuclear energy

UF: Nuclear fuels
BT: Energy resources
RT: Nuclear power generation
Radioactive waste

Nuclear facility licensing

USE: Nuclear facility regulation

Nuclear facility regulation

UF: Licensing (nuclear facilities)
Nuclear facility licensing
BT: Power industry
RT: Radioactive waste

Nuclear fission

USE: Fission reactors

Nuclear fuels

USE: Nuclear energy

Nuclear imaging

UF: Gamma-ray imaging
BT: Imaging
Nuclear and plasma sciences
RT: Nuclear medicine
Radiography
NT: Energy resolution
Ion emission

Nuclear magnetic resonance

UF: NMR
BT: Magnetic resonance

Nuclear magnetic resonance imaging

USE: Magnetic resonance imaging

Nuclear measurements

BT: Measurement
RT: Atomic measurements
CAMAC
Position sensitive particle detectors
Radiation detectors
Spectroscopy
NT: Particle tracking

Nuclear medicine

BT: Engineering in medicine and biology
Nuclear and plasma sciences
RT: Energy resolution
Gamma-rays
Nuclear imaging
Positron emission tomography

Nuclear phase transformations

UF: Nuclear phase transitions
Phase transformations, nuclear
BT: Phase transitions, nuclear physics
RT: Nuclear thermodynamics

Nuclear phase transitions

USE: Nuclear phase transformations

Nuclear physics

BT: Nuclear and plasma sciences
RT: CERN
Hafnium
NT: Alpha particles
Beta rays
Ignition
Ion sources
Isotopes
Nuclear phase transformations
Nuclear thermodynamics
Relativistic effects



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Nuclear Power Generating Stations

USE: Nuclear power generation

Nuclear power generation

UF: Atomic energy
Nuclear Power Generating

Stations

BT: Power generation
RT: Nuclear energy
NT: Atomic batteries
Fission reactors
Fusion power generation

Nuclear reactors (fission)

USE: Fission reactors

Nuclear reactors (fusion)

USE: Fusion reactors

Nuclear thermodynamics

BT: Nuclear physics
RT: Elementary particles
Entropy
Nuclear phase

transformations

Phase change materials

Nuclear warfare

USE: Nuclear weapons

Nuclear warheads

USE: Nuclear weapons

Nuclear wastes

USE: Radioactive pollution

Nuclear weapons

UF: A-bomb
Atom bomb
Atomic warfare
Atomic weapons
Nuclear bombs
Nuclear warfare
Nuclear warheads
Nuke
BT: Weapons
RT: Biological weapons
Chemical weapons
Electromagnetic pulses
National security
Terrorism
US Department of

Homeland Security

Nucleic acids

BT: Molecular biophysics
RT: DNA
DNA sequencing
RNA

Nuke

USE: Nuclear weapons

Null space

BT: Kernel

Null value

UF: Nullvalue
BT: Data structures
RT: Programming

Nullvalue

USE: Null value

Number plate recognition

USE: License plate recognition

Number portability

BT: Telecommunication
services

Numerical analysis

BT: Mathematics
RT: Convolution
Deconvolution
Difference equations
Differential equations
Error analysis
Integral equations
Inverse problems
MATLAB
Nonlinear equations
Numerical models
Transforms
NT: Adaptive mesh refinement
Approximation methods
Convergence of numerical

methods

Finite difference methods
Finite element analysis
Finite volume methods
Gradient methods
Independent component

analysis

Iterative methods
Least squares

approximations

Method of moments
Mode matching methods



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	Multigrid methods				
	Newton method			Obesity	
	Numerical simulation		BT:		Medical conditions
	Numerical stability			Obituaries	
	Relaxation methods		BT:		IEEE indexing
	Sparse matrices			Object detection	
	Splines (mathematics)		UF:		Image object detection
	Surface fitting				Target detection
	Symmetric matrices		BT:		Image analysis
	Transmission line matrix		RT:		Advanced driver assistance
methods				systems	
					Fine-grained image
Numerical models				recognition	
	BT:	Modeling			Image matching
	RT:	Numerical analysis			Instance segmentation
					Internet of Things
Numerical simulation					Magnetic anomaly
	BT:	Numerical analysis		detection	
	RT:	Modeling			Magnetic anomaly
		Plasma simulation		detectors	
		Simulation			Motion capture
					Neural radiance field
Numerical stability					Photogrammetry
	BT:	Numerical analysis			Robot vision systems
	RT:	Algorithms		NT:	Buried object detection
					Time difference of arrival
<i>Nursing</i>					YOLO
	USE:	Medical services			
				Object oriented databases	
Nutrients				BT:	Database systems
	BT:	Food products			Databases
				RT:	Object oriented methods
<i>Nuts (fasteners)</i>					
	USE:	Fasteners		Object oriented methods	
				BT:	Programming
NVIS				RT:	Object oriented databases
	UF:	Near vertical incidence			Object oriented
skywave				programming	
	BT:	Broadcasting			
		Radiowave propagation		Object oriented modeling	
				BT:	Modeling
<i>NVQ</i>					
	USE:	Vocational training		Object oriented programming	
				UF:	Object-oriented
<i>Nylon fiber</i>				programming	
	USE:	Synthetic fibers		BT:	Programming
				RT:	C languages
<i>O-RAN</i>					C# languages
	USE:	Open RAN			Object oriented methods
					Python
<i>O-rings</i>					Software libraries
	USE:	Structural rings			Software reusability
				NT:	Dispatching
<i>OATS</i>					
	USE:	Open area test sites			



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Object recognition

UF: Image object recognition
BT: Machine vision
RT: Image matching
Image recognition
Object tracking
Robot vision systems
NT: Affordances
Target recognition

RT: Accidents
Biological effects of
radiation
Domestic safety
Electric shock
Employee welfare
Environmental factors
Ergonomics
Eye protection
Occupational medicine
Occupational safety
Pollution
Protective clothing
Radioactive materials
Risk analysis
Safety
Toxicology
Working environment noise
NT: Occupational stress

Object segmentation

BT: Machine vision
NT: Subspace constraints

Object tracking

BT: Tracking
RT: Cinematography
Image motion analysis
Image segmentation
Motion capture
Motion estimation
Object recognition
Trajectory
Video signal processing

Occupational medicine

BT: Medical services
RT: Medical diagnosis
Medical treatment
Occupational health

Object-oriented programming

USE: Object oriented
programming

Occupational pensions

USE: Pensions

Observability

BT: Control theory

Observatories

BT: Astronomy
RT: Telescopes

Observers

BT: State estimation
RT: Machine vision
NT: Disturbance observers
Earth Observing System

Occupational safety

UF: OSHA
BT: Health and safety
RT: Accidents
Domestic safety
Electric shock
Employee welfare
Eye protection
Industrial accidents
Occupational health
Protection
Protective clothing
Radioactive materials
Risk analysis
Working environment noise

Obstacle avoidance

USE: Collision avoidance

Obstetrics

BT: Medical specialties
RT: Pregnancy

Occupational stress

BT: Occupational health
RT: Employee welfare
Hazards

Occipital Lobe

BT: Brain

OCDM

USE: Code division multiplexing

Occupational health

UF: Health (occupational)
BT: Health and safety



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Ocean acidification

BT: Acidification
Oceans

Ocean animals

USE: Marine animals

Ocean circulation

BT: Ocean dynamics
Oceanography
RT: Ocean waves
Sea level
Tides

Ocean composition

USE: Oceans

Ocean dynamics

BT: Oceanography
RT: Aquatic ecosystems
Ocean temperature
Ocean waves
Tides
NT: Ocean circulation
Storm surge

Ocean salinity

BT: Oceans
RT: Aquatic ecosystems
Salinity (geophysical)
NT: SMOS mission

Ocean surface temperature

USE: Sea surface temperature

Ocean technology

USE: Marine technology

Ocean temperature

BT: Oceanic engineering and
marine technology
Oceans
RT: Global warming
Land surface temperature
Ocean dynamics
NT: Ocean thermal energy
conversion
Sea surface temperature

Ocean thermal energy conversion

UF: OTEC
BT: Energy conversion
Ocean temperature
RT: Thermal energy

Ocean vegetation

USE: Marine vegetation

Ocean waves

BT: Hydrology
RT: Ocean circulation
Ocean dynamics
NT: Sea state
Wave power

Oceanic crust

BT: Geology
RT: Continental crust
Rocks
Tectonics

Oceanic engineering and marine technology

NT: Marine navigation
Marine technology
Ocean temperature
Oceanographic techniques
Water pollution

Oceanographic techniques

BT: Oceanic engineering and
marine technology
RT: Acoustic imaging
Hydrologic measurements
Radar applications
Remote sensing

Oceanography

UF: Marine science
Oceanology
BT: Geoscience
RT: Hydrography
Marine robots
Oceans
NT: Ocean circulation
Ocean dynamics

Oceanology

USE: Oceanography

Oceans

UF: Ocean composition
Planetary oceans
BT: Geoscience
RT: Aquatic ecosystems
Geophysics
Marine technology
Oceanography
Sea ice
Sea measurements
Water



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NT:	Antarctic Ocean	OFDM modulation	BT:	OFDM	
	Arctic Ocean				
	Atlantic Ocean				
	Indian Ocean				
	Ocean acidification				
	Ocean salinity				
	Ocean temperature				
	Pacific Ocean				
	Sea coast				
	Sea floor				
	Sea level				
	Sea surface				
Tides					
OCR	USE:	Optical character			
recognition					
Octrees	BT:	Data structures			
Odometers	BT:	Distance measurement			
					Instruments
Odometry	BT:	Motion detection			
					Position measurement
	RT:	Calibration			
					Location awareness
		Robot control			
					Sensors
OFDM	UF:	Orthogonal frequency			
					division multiple access
		Orthogonal frequency			
					division multiplexing
	BT:	Multiplexing			
					RT:
(telecommunication)		3G mobile communication			
					Acoustic communication
		Communication channels			
					Digital signal processing
		MIMO			
					Modulation
		Multiaccess communication			
					Multicarrier code division
multiple access		NOMA			
					NT:
		Multiple access interference			
					OFDM modulation
		Partial transmit sequences			
					Peak to average power
ratio					
OFETs	UF:	Organic FETS			
					transistors
					transistors
					transistors
					transistors
					transistors
					transistors
					transistors
					transistors
					transistors
					transistors
					transistors
Office automation	BT:	Automation			
					RT:
					Bring your own device
					Communication systems
					Data communication
					Desktop publishing
					Document handling
					Electronic mail
					Information systems
					Local area networks
					Microcomputers
					Teleconferencing
Text processing					
Unsolicited e-mail					
Voice mail					
Workflow management					
software	NT:				
<i>Offshore distribution systems</i>	USE:	Offshore installations			
Offshore installations	UF:	Gas platforms			
					Offshore distribution
systems		Offshore platforms			
					Offshore power plants
		Oil platforms			
					Structural engineering
	BT:	Oil drilling			
					Petroleum industry
	RT:	Power industry			
<i>Offshore platforms</i>	USE:	Offshore installations			
<i>Offshore power plants</i>	USE:	Offshore installations			
OGC	USE:	Open Geospatial			
					Consortium



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Ohmic contacts

BT: Contacts
RT: Linear circuits

Ohmmeters

USE: Electrical resistance measurement

Oil drilling

UF: Drilling oil
BT: Petroleum industry
RT: Drilling
Fuel processing industries
Offshore installations
Technology
Well logging

Oil filled cables

BT: Oil insulation
RT: Cable insulation

Oil filters

USE: Lubricating oils

Oil industry

USE: Petroleum industry

Oil insulation

UF: Transformer oil
BT: Insulation
RT: Oils
NT: Oil filled cables

Oil platforms

USE: Offshore installations

Oil pollution

BT: Pollution
RT: Accidents
Land pollution
Marine pollution
Oils
Petroleum
Petroleum industry

Oil refineries

BT: Petroleum industry

Oil sands

USE: Hydrocarbons

Oil shale

USE: Hydrocarbons

Oil tanks

USE: Fuel storage

Oiling (lubrication)

USE: Lubricating oils

Oils

BT: Materials
RT: Engines
Fats
Fluids
Fractionation
Fuel processing industries
Insulation
Marine pollution
Mechanical factors
Mechanical power

transmission

Oil insulation
Oil pollution
Petroleum
Petroleum industry
Pipelines
Water pollution
Lubricating oils
Vegetable oils

NT:

Older adults

UF: Elderly
Senior citizens
BT: Social groups
RT: Aging
Alzheimer's disease
Assisted living
Assistive robots
Geriatrics
Gerontology

OLED

USE: Organic light emitting

diodes

Olfactory

BT: Nose

Olfactory bulb

BT: Forebrain
Sense organs

Oligopoly

BT: Economics
RT: Game theory
Microeconomics



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Omnidirectional antennas

BT: Antennas

NT: Radiation therapy

On board unit

BT: Communication equipment

RT: Dedicated short range

communication

Vehicle-to-everything

One shot learning

UF: One-shot learning

BT: Machine learning

RT: Computer vision

Few shot learning

Training data

On demand software

USE: Software as a service

One-shot learning

USE: One shot learning

On load tap changers

UF: Load tap changers
On-load tap changers
Onload tap changers

BT: Power transformers
Tap changers

RT: Voltage control

Online banking

UF: Digital currency
E-banking
E-currency
E-wallets
Electronic banking
Electronic currency
Electronic wallets
Internet banking
Mobile payment
Virtual currency
BT: Banking
Online services
RT: Bitcoin
Cryptocurrency
Digital economy
Electronic commerce
Fintech
Open banking
Distributed ledger
Micropayments
NT:

On the job training

UF: On-the-job training

BT: Training

RT: Industrial training

On-chip

USE: System-on-chip

On-demand software

USE: Software AND
Software as a service

On-line services

USE: Online services

On-load tap changers

USE: On load tap changers

On-the-job training

USE: On the job training

Online bullying

USE: Cyberbullying

Online conferencing

USE: Web conferencing

Online indexing

USE: Indexing

Online learning

USE: Electronic learning

Oncological surgery

UF: Otologic surgery
Surgery oncology

BT: Surgery

RT: Cancer

Oncology

Online services

UF: Inverted classroom
On-line services
Reverse teaching
BT: Information retrieval
RT: Cloud gaming
Electronic learning
Internet
NT: Mobility as a service

Oncology

BT: Medical specialties

RT: Cancer

Cell therapy

Chemotherapy

Oncological surgery

Tumors



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Online banking
Web conferencing

Online shopping
USE: Electronic commerce

Online social networks
USE: Social networking (online)

Online voting
USE: Electronic voting

Onload tap changers
USE: On load tap changers

Ontologies

UF: Ontology
BT: Knowledge representation
RT: Linked data
Open data
Ranking (statistics)
Semantic Web
Semantic search
Thesauri
NT: Description logic

Ontology
USE: Ontologies

ONU
USE: Optical network units

OOO
USE: Out of order

Op amp
USE: Operational amplifiers

Open Access

BT: Open systems
Publishing
NT: Public domain software

Open AI
USE: Artificial intelligence

Open area test sites

UF: OATS
BT: Test facilities
RT: Electromagnetic
compatibility and interference
Electromagnetic
interference
Immunity testing
Military equipment

Open banking

BT: Banking
Open source software
RT: Finance
Investment
Online banking
Open systems

Open data

BT: Data handling
RT: Electronic publishing
Government policies
Internet
Linked data
Ontologies
Open systems
Public domain software
Semantic Web

Open Educational Resources

BT: Educational courses
Open systems

Open Geospatial Consortium

UF: OGC
BT: Standards organizations

Open loop control

USE: Open loop systems

Open loop systems

UF: Open loop control
BT: Control systems
RT: Feedforward systems
Operational amplifiers

Open radio access networks

USE: Open RAN

Open RAN

UF: O-RAN
Open radio access
networks
BT: Mobile communication
Radio access networks
RT: Open systems

Open source hardware

UF: Open-source hardware
BT: Hardware

Open source software

UF: Open-source software
BT: Software
RT: Public domain software



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NT:	Open banking	Kernel
Open systems		Network operating systems
		System kernels
	UF:	OSI
	BT:	Computers and information
		System analysis and design
	RT:	Common Information Model
	(electricity)	
		Computer networks
		Internetworking
	Interoperability	
	Local area networks	
	Metropolitan area networks	
	Open RAN	
	Open banking	
	Open data	
	Standards	
	Wide area networks	
NT:	Open Access	
	Open Educational	
Resources		
	Physical layer	
Open wireless architecture		
UF:	OWA	
BT:	Wireless communication	
<i>Open-source hardware</i>		
USE:	Open source hardware	
<i>Open-source software</i>		
USE:	Open source software	
<i>Operating cost reduction</i>		
USE:	Costing	
Operating systems		
UF:	Android (operating system)	
	Computer operating	
systems		
	Executive programs	
	Microsoft Windows	
	Robot operating systems	
	Supervisory programs	
BT:	System software	
RT:	Computer security	
	Cyber-physical systems	
	Program processors	
	Software defined	
networking		
	System recovery	
NT:	Bootimg	
	Embedded systems	
	Input-output programs	

Operational amplifiers	
UF:	Op amp
BT:	Active circuits
	Amplifiers
RT:	FET circuits
	Linearization techniques
	MOSFET circuits
	Open loop systems
NT:	Feedback amplifiers
Operations research	
BT:	Business
RT:	Linear programming
	MCDM
	Management
	Optimization methods
	Principal component
analysis	
	Resource management
	Statistics
	TOPSIS
NT:	Inventory control
	Virtual enterprises
Ophthalmology	
BT:	Medical specialties
RT:	Cornea
	Eyes
	Iris
	Pupils
	Retina
Opinion dynamics	
BT:	Social dynamics
RT:	Computational modeling
<i>Opinion mining</i>	
USE:	Sentiment analysis
Opioids	
BT:	Drugs
Opportunistic software systems development	
BT:	Programming
<i>Optic flow</i>	
USE:	Optical flow



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Optical add-drop multiplexers

UF: ROADMS
BT: Add-drop multiplexers

Optical amplification

USE: Stimulated emission

Optical amplifiers

BT: Optics
RT: Erbium
NT: Doped fiber amplifiers
Erbium-doped fiber

amplifiers

amplifiers

Semiconductor optical

Optical antennas

BT: Antennas
Optics
RT: Nanophotonics
Optical communication

equipment

Optical metamaterials
Optoelectronic devices
Plasmons
Receivers
Terahertz materials
Transceivers
Transmitters
NT: Nanoantennas

Optical arrays

BT: Optical devices
RT: Micromirrors
Phased arrays

Optical attenuators

UF: Variable optical attenuators
BT: Attenuators
Optical devices
RT: Optical communication

equipment

Optical losses

Optical beam splitting

BT: Optical beams

Optical beams

BT: Beams
RT: Bragg gratings
Laser beams
Laser theory
NT: Optical beam splitting

Optical bistability

UF: Bistability (optical)
BT: Electro-optic effects
RT: Electro-optic devices
Optical switches

Optical buffering

BT: Optical fiber communication

Optical burst switching

BT: Burst switching

Optical character recognition

UF: OCR
BT: Software
NT: License plate recognition

Optical cloaking

UF: Metamaterial cloaking
BT: Metamaterials
Optical materials

Optical code division multiplexing

USE: Code division multiplexing

Optical coherence tomography

BT: Tomography
RT: Eyes

Optical collimators

BT: Optical devices

Optical communication

USE: Optical fiber communication

Optical communication equipment

BT: Communication equipment
RT: Biomedical optical imaging
Nanoantennas
Optical antennas
Optical attenuators
Optical switches
NT: Optical transmitters

Optical components

USE: Optical devices

Optical computing

BT: Computers and information processing

Optical control

BT: Control systems
RT: Optical switches
NT: Lighting control



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	Optical variables control	Optical attenuators	
	Optogenetics	Optical collimators	
Optical coupling		Optical device fabrication	
BT:	Electromagnetic coupling	Optical filters	
RT:	Optical fiber couplers	Optical modulators	
		Optical resonators	
Optical crosstalk		Optical sensors	
BT:	Optics	Optical tweezers	
RT:	Optical fiber communication	Retroreflectors	
		Thermooptical devices	
Optical design		Optical diffraction	
BT:	Optics	BT:	Electromagnetic diffraction
RT:	Laser theory	RT:	Photonic band gap
NT:	Optical design techniques	NT:	Diffraction gratings
Optical design techniques		Optical distortion	
BT:	Optical design	BT:	Optics
RT:	Design methodology	RT:	Lasers
NT:	High-speed optical		Optical noise
techniques			Thermal lensing
Optical detectors		<i>Optical distortion measurement</i>	
BT:	Optical sensors	USE:	Distortion measurement
RT:	Nonlinear optical devices		
NT:	Bar codes	Optical emission spectroscopy	
		BT:	Spectroscopy
Optical device fabrication		RT:	Plasmas
UF:	Optical device manufacture		
BT:	Fabrication	Optical engineering	
	Optical devices	BT:	Engineering - general
RT:	Electronic equipment		Optics
manufacture		RT:	Optical materials
<i>Optical device manufacture</i>		Optical feedback	
USE:	Optical device fabrication	BT:	Image processing
		RT:	Distributed feedback
Optical devices		devices	
UF:	Optical components	Optical fiber amplifiers	
BT:	Optics	UF:	Optical fibre amplifiers
RT:	Biomedical optical imaging	BT:	Optical fibers
	Endomicroscopy	RT:	Amplifiers
	Gratings		
	Optical materials	Optical fiber applications	
NT:	Bragg gratings	UF:	Optical fibre applications
	Collimators	BT:	Optics
	Displays	RT:	Channel spacing
	Holographic optical		Code division multiplexing
components			Optical fiber cables
	Lenses		Optical fiber communication
	Light deflectors		Optical fibers
	Lighting	NT:	Optical fiber devices
	Luminescent devices		
	Mirrors		
	Optical arrays		



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Optical fiber cables

UF: Communication cables
(optical)
BT: Optical fibre cables
RT: Cables
Optical fiber applications
Splicing
NT: Claddings

Optical fiber communication

UF: Infrared communication
Optical communication
Optical fibre communication
Optical links
BT: Communication systems
RT: Avalanche photodiodes
Broadband communication
Indoor communication
Optical crosstalk
Optical fiber applications
Optical transmitters
Quantum communication
Silicon photonics
Synchronous digital
hierarchy
NT: FDDI
Free-space optical
communication
Optical buffering
Optical fiber networks
Optical fiber subscriber
loops
Optical interconnections
Optical packet switching
Optical wavelength
conversion
SONET
Scheduling algorithms
Visible light communication

Optical fiber couplers

UF: Optical fibre couplers
BT: Optical fibers
RT: Optical coupling

Optical fiber devices

UF: Optical fibre devices
BT: Optical fiber applications
RT: Optical fibers
NT: Optical fiber sensors

Optical fiber dispersion

UF: Optical fibre dispersion
BT: Dispersion

Optical fiber filters

BT: Optical filters

Optical fiber LAN

UF: Optical fiber local area
network
Optical fibre LAN
Optical fibre local area
network
BT: Optical fiber networks

Optical fiber local area network

USE: Optical fiber LAN

Optical fiber loss

USE: Optical fiber losses

Optical fiber losses

UF: Optical fiber loss
Optical fibre losses
BT: Optical fibers

Optical fiber networks

UF: Optical fibre networks
Optical networks
Optical-fiber networks
Optical-fibre networks
BT: Optical fiber communication
RT: Light fidelity
NT: All-optical networks
Elastic optical networks
LAN emulation
Optical fiber LAN
Optical network units
Passive optical networks
Protection switching
Wavelength assignment

Optical fiber polarization

UF: Optical fibre polarisation
Polarization-maintaining
optical fibers
BT: Optical fibers
RT: Optical fiber sensors
Optical interferometry
NT: Polarization mode
dispersion

Optical fiber sensors

UF: Fiber optic sensors
Fibre optic sensors
Optical fibre sensors
BT: Optical fiber devices
Optical sensors
RT: Optical fiber polarization



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measurement	Partial discharge	<i>Optical fibre couplers</i> USE: Optical fiber couplers
Optical fiber subscriber loops	UF: FTTH Fiber-in-the-loop Optical fibre subscriber	<i>Optical fibre devices</i> USE: Optical fiber devices
loops	BT: Optical fiber communication	<i>Optical fibre dispersion</i> USE: Optical fiber dispersion
Optical fiber testing	UF: Optical fibre testing BT: Testing RT: Optical fibers	<i>Optical fibre LAN</i> USE: Optical fiber LAN
Optical fiber theory	UF: Optical fibre theory BT: Optical fibers RT: Electromagnetic field theory	<i>Optical fibre local area network</i> USE: Optical fiber LAN
Optical fibers	UF: Optical fibres BT: Fiber optics Optical waveguides RT: Electromagnetic	<i>Optical fibre losses</i> USE: Optical fiber losses
waveguides	Optical fiber applications Optical fiber devices Optical fiber testing Optical materials Optical propagation Optical waveguide theory Optical wavelength	<i>Optical fibre networks</i> USE: Optical fiber networks
conversion	Supercontinuum generation Temperature sensors NT: Optical fiber amplifiers Optical fiber couplers Optical fiber losses Optical fiber polarization Optical fiber theory Plastic optical fiber Wavelength conversion	<i>Optical fibre polarisation</i> USE: Optical fiber polarization
		<i>Optical fibre sensors</i> USE: Optical fiber sensors
		<i>Optical fibre subscriber loops</i> USE: Optical fiber subscriber loops
		<i>Optical fibre testing</i> USE: Optical fiber testing
		<i>Optical fibre theory</i> USE: Optical fiber theory
		<i>Optical fibres</i> USE: Optical fibers
		Optical films BT: Films RT: Integrated optics Optical materials
<i>Optical fibre amplifiers</i> USE: Optical fiber amplifiers		Optical filters BT: Optical devices RT: Photography NT: Optical fiber filters
<i>Optical fibre applications</i> USE: Optical fiber applications		Optical flow UF: Optic flow BT: Optical imaging Relativistic effects
<i>Optical fibre cables</i> USE: Optical fiber cables		
<i>Optical fibre communication</i> USE: Optical fiber communication		



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Optical frequency combs

USE: Optical harmonic generation

Optical frequency conversion

BT: Frequency conversion

Optical gratings

USE: Gratings

Optical harmonic generation

UF: Optical frequency combs

BT: Optics

Optical heterodyning

USE: Optical mixing

Optical imaging

BT: Imaging

RT: Infrared imaging

Microscopy

Remote sensing

Sentinel-2

Sentinel-3

Sentinel-5P

NT: Optical flow

Optical projectors

Talbot effect

Thermoreflectance imaging

Optical instruments

BT: Instruments

Optical interconnections

UF: Optical interconnects

BT: Optical fiber communication

Optical interconnects

USE: Optical interconnections

Optical interferometry

UF: Light interferometry

BT: Interferometry

RT: Optical fiber polarization

Speckle

Talbot effect

Optical lattices

USE: Lattices

Optical links

USE: Optical fiber communication

Optical losses

BT: Optics

RT: Loss measurement

Optical attenuators

Optical scattering

Optical materials

BT: Materials

RT: Glass

Indium tin oxide

Lenses

Magnesium oxide

Metamaterials

Mirrors

Optical devices

Optical engineering

Optical fibers

Optical films

Optics

Organic inorganic hybrid

materials

Phase change materials

Photonic crystals

SIMO

NT: Colloidal nanocrystals

Optical cloaking

Optical polymers

Optical retarders

Optical superlattices

Photorefractive materials

Optical measurements

USE: Optical variables

measurement

Optical metamaterials

UF: Photonic metamaterials

BT: Metamaterials

RT: Electromagnetic

metamaterials

Nanoantennas

Optical antennas

Optical metrology

BT: Metrology

Optical microscopy

BT: Optics

Optical mixing

UF: Optical heterodyning

BT: Nonlinear optics

Optics

RT: Photorefractive materials

NT: Multiwave mixing



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Optical modulation

BT: Modulation
RT: Indoor communication
Microwave photonics
Optical transmitters
NT: Cross-phase modulation
Intensity modulation

Video equipment
RT: Image processing
Micromirrors
Motion pictures

Optical modulators

BT: Modulation
Optical devices
NT: Electro-absorption
modulators
Electro-optic modulators

Optical multilayers

USE: Optical superlattices

Optical network units

UF: ONU
BT: Optical fiber networks

Optical networks

USE: Optical fiber networks

Optical noise

BT: Integrated circuit noise
RT: Optical distortion
NT: Speckle

Optical packet switching

BT: Optical fiber communication

Optical planar waveguides

BT: Optical waveguides

Optical polarisation

USE: Optical polarization

Optical polarization

UF: Light polarisation
Light polarization
Optical polarisation
BT: Optics
RT: Photoelasticity
NT: Polarization shift keying
Stokes parameters

Optical polymers

BT: Optical materials
Polymers

Optical projectors

UF: projectors (optical)
BT: Optical imaging

Optical propagation

UF: Infrared propagation
BT: Electromagnetic
propagation
RT: Optical fibers
Thermooptic effects
NT: Optical surface waves
Optical waveguides

Optical pulse compression

BT: Pulse compression
methods

Optical pulse generation

BT: Pulse generation
RT: Optical pulse shaping

Optical pulse shaping

BT: Pulse shaping methods
RT: Optical pulse generation

Optical pulses

BT: Optics

Optical pumping

BT: Laser excitation

Optical radar

USE: Laser radar

Optical receivers

BT: Receivers

Optical recording

BT: Recording
RT: Laser applications
NT: CD recording

Optical reflection

BT: Electromagnetic reflection
RT: Antireflection coatings
Mirrors
Optical scattering
Reflectivity
Reflectometry
Thermooptic effects

Optical refraction

BT: Physical optics
RT: Photorefractive effect



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Photorefractive materials
Refractive index
Thermooptic effects

Optical regenerators

USE: Repeaters

Optical resonators

BT: Optical devices
RT: Digital filters
Laser cavity resonators
Resonance
Split ring resonators
NT: Microcavities
Optical ring resonators

Optical retarders

UF: Half-wave plates
Quarter-wave plates
BT: Optical materials
Optics
RT: Polarimetry

Optical ring resonators

UF: Ring resonators
BT: Optical resonators

Optical saturation

BT: Nonlinear optics
Optics

Optical scattering

BT: Electromagnetic scattering
RT: Laser radar
Light scattering
Optical losses
Optical reflection
Speckle

Optical sensors

BT: Optical devices
Sensors
RT: Image sensors
Wearable sensors
NT: Optical detectors
Optical fiber sensors

Optical signal detection

BT: Signal detection
RT: Photodetectors

Optical signal processing

BT: Signal processing
NT: Laser noise

Optical solitons

BT: Optics
Solitons
RT: Optical vortices

Optical squeezing

UF: Squeezed light
Squeezed states
BT: Quantum optics

Optical superlattices

UF: Optical multilayers
BT: Optical materials
Superlattices

Optical surface waves

BT: Optical propagation

Optical switch

USE: Optical switches

Optical switches

UF: Optical switch
BT: Switches
RT: Optical bistability
Optical communication

equipment

Optical control
Photothyristors
Smart pixels
Thermooptical devices

Optical transmitters

BT: Optical communication
equipment

Transmitters
RT: Bragg gratings
Diodes
Optical fiber communication
Optical modulation
Photodiodes
Semiconductor lasers
Semiconductor optical

amplifiers

Optical tuning

BT: Optics
Tuning
RT: Laser tuning

Optical tweezers

BT: Optical devices



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Optical variables control

BT: Optical control
RT: Frequency control
Phase control

Optical variables measurement

UF: Optical measurements
BT: Measurement
RT: Frequency measurement
Phase measurement
Reflectometry
Wavelength measurement
NT: Ellipsometry
Photometry
Reflection coefficient
Refractive index

Optical vortex

USE: Optical vortices

Optical vortices

UF: Optical vortex
Vortices, optical
BT: Physical optics
RT: Laser beams
Optical solitons

Optical waveguide components

BT: Optical waveguides

Optical waveguide theory

BT: Optical waveguides
RT: Optical fibers

Optical waveguides

BT: Optical propagation
Waveguide components
RT: Electro-optic modulators
Integrated optics
Photonic crystals
NT: Arrayed waveguide gratings
Electro-optical waveguides
Optical fibers
Optical planar waveguides
Optical waveguide

components

Optical waveguide theory

Optical wavelength conversion

BT: Optical fiber communication
Signal processing
RT: Multicast communication
Optical fibers
Telecommunications

Optical-fiber networks

USE: Optical fiber networks

Optical-fibre networks

USE: Optical fiber networks

Optics

BT: Lasers and electrooptics
RT: Erbium
Fourier transforms
Laser theory
Magnetooptic effects
Optical materials
NT: Adaptive optics
Aspherical optics
Birefringence
Brightness
Color
Electron optics
Extinction coefficients
Fiber optics
Fluorescence
Four-wave mixing
Geometrical optics
Integrated optics
Light fields
Light sources
Luminescence
Microoptics
Nonlinear optics
Optical amplifiers
Optical antennas
Optical crosstalk
Optical design
Optical devices
Optical distortion
Optical engineering
Optical fiber applications
Optical harmonic
generation
Optical losses
Optical microscopy
Optical mixing
Optical polarization
Optical pulses
Optical retarders
Optical saturation
Optical solitons
Optical tuning
Optogenetics
Particle beam optics
Photoluminescence
Physical optics
Ray tracing
Stray light



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	Ultrafast optics		Optimization methods
	Whispering gallery modes	UF:	Optimisation methods
Optimal control		BT:	Optimization
BT:	Control systems	RT:	Infinite horizon
RT:	Game theory		Linear programming
	H infinity control		MIMO
NT:	Bang-bang control		Newton method
	Infinite horizon		Operations research
			Processor scheduling
			Response surface
Optimal matching		methodology	
BT:	Graph theory		Search methods
Optimal scheduling			Single machine scheduling
BT:	Optimization		Traveling salesman
<i>Optimisation</i>		problems	
USE:	Optimization	NT:	Affordances
<i>Optimisation methods</i>			Circuit optimization
USE:	Optimization methods		Concave programming
<i>Optimisation models</i>			Design optimization
USE:	Optimization models		Fireworks algorithm
<i>Optimised production technology</i>			Glowworm swarm
USE:	Optimized production	optimization	
technology			Gradient methods
<i>Optimising compilers</i>			H infinity control
USE:	Optimizing compilers		Lagrangian functions
			Mathematical programming
		technology	Optimization models
			Optimized production
			Pareto optimization
			Quadratic programming
			Simulated annealing
Optimization		Optimization models	
UF:	Optimisation	UF:	Optimisation models
	Performance optimisation	BT:	Mathematical models
	Performance optimization		Optimization methods
BT:	Mathematics		
RT:	Artificial bee colony	Optimized production technology	
algorithm		UF:	Optimised production
	Doping profiles	technology	
	Least squares	BT:	Optimization methods
approximations			Production control
	Minimization	RT:	Production planning
	Multi-armed bandit problem		Production systems
	Parametric study		
	Performance analysis	Optimizing compilers	
	TOPSIS	UF:	Optimising compilers
NT:	Bees algorithm	BT:	Program processors
	Cost function		
	Krill herd algorithm	<i>Opto-electronic devices</i>	
	Metaheuristics	USE:	Optoelectronic devices
	Optimal scheduling		
	Optimization methods	<i>Optoacoustic effects</i>	
	Trajectory optimization	USE:	Photoacoustic effects



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Optoelectronic and photonic sensors

BT: Sensors

Optoelectronic devices

UF: Opto-electronic devices

BT: Lasers and electrooptics

RT: Electro-optic devices

Optical antennas

Phototransistors

NT: Charge-coupled image

sensors

Integrated optoelectronics

Light emitting diodes

Photoconducting devices

Photodetectors

Superluminescent diodes

Optogenetics

BT: Genetics

Neuromodulation

Optical control

Optics

Optothermal effects

USE: Photothermal effects

Optothyristors

USE: Photothyristors

Oral communication

UF: Speech communication

BT: Professional

communication

NT: Public speaking

Speech

Orange technology

USE: Social implications of technology

Orbital calculations

BT: Energy states

Orbital debris

USE: Space debris

Orbital robotics

BT: Robots

Orbits

BT: Astrophysics

RT: Geostationary satellites

NT: Orbits (stellar)

Planetary orbits

Orbits (stellar)

BT: Orbits

Ordered logit models

USE: Logistic regression

Ordinance

USE: Weapons

Ordinary differential equations

BT: Differential equations

Ordinary magnetoresistance

BT: Magnetoresistance

Ores

BT: Minerals

Organ transplantation

UF: Transplants

BT: Medical services

NT: Heart transplantation

Organelles

BT: Cells (biology)

NT: Ribosomes

Organic chemicals

BT: Chemistry

NT: Hydrocarbons

Organic compounds

BT: Compounds

RT: Carbon

NT: Carbon compounds

Formaldehyde

Organic semiconductors

Volatile organic compounds

Organic electronics

UF: Paper electronics

BT: Electronic equipment

RT: Synapses

Organic FETS

USE: OFETs

Organic field effect transistors

USE: OFETs

Organic field-effect transistors

USE: OFETs



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Organic inorganic hybrid materials

UF: Inorganic organic hybrid materials
Inorganic-organic hybrid materials
Organic-inorganic hybrid materials
Organically modified silicates
BT: Materials
RT: Bio-inspired materials
Inorganic compounds
Optical materials

Organic light emitting diodes

UF: OLED
Organic light-emitting diodes
BT: Polymer led
Diodes
Light emitting diodes
RT: Electroluminescence
Molecular electronics
NT: Active matrix organic light emitting diodes

Organic light-emitting diodes

USE: Organic light emitting diodes

Organic materials

BT: Materials

Organic semiconductors

BT: Organic compounds
Semiconductor materials
NT: Pentacene

Organic thin film transistors

UF: OTFT
Organic thin-film transistors
BT: Thin film transistors

Organic thin-film transistors

USE: Organic thin film transistors

Organic-inorganic hybrid materials

USE: Organic inorganic hybrid materials

Organically modified silicates

USE: Organic inorganic hybrid materials

Organisational aspects

USE: Organizational aspects

Organisational culture

USE: Organizational aspects

Organisational structure

USE: Organizational aspects

Organisms

BT: Biological systems
NT: Algae
Animals
Archaea
Fish
Invasive species
Microorganisms
Plankton
Plants (biology)

Organizational aspects

UF: Business organisation
Business organization
Organisational aspects
Organisational culture
Organisational structure
Organizational culture
Organizational structure
BT: Management
RT: Business process re-

engineering

Industrial communication
Teamwork
NT: Business communication
Corporate acquisitions
Data governance
Facilities management
Role transfer
Scheduling
Stakeholders

Organizational communication

USE: Industrial communication

Organizational culture

USE: Organizational aspects

Organizational structure

USE: Organizational aspects

Organizations

BT: Business
RT: Leadership
NT: BNSC
CERN



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organization	Companies	Biomedical equipment	
	Decentralized autonomous	Medical control systems	
organizations	Government	Prosthetics	
	Non-governmental	Sensory aids	
		Wearable robots	
	Sociotechnical systems	<i>Oscillations</i>	
	Space agencies	USE: Oscillators	
<i>Organobromine compounds</i>		Oscillators	
USE: Bromine compounds		UF: Oscillations	
Organoids		BT: Circuits and systems	
	BT: Cells (biology)	RT: Circuits	
	RT: Biological systems		Damping
			Klystrons
<i>Organs (biological)</i>		Lasers	
USE: Biological systems		Resonant frequency	
		Vibrations	
<i>Orientation control</i>		NT: Digital-controlled oscillators	
USE: Position control		Injection-locked oscillators	
		Local oscillators	
<i>Orientation determination</i>		Microwave oscillators	
USE: Position measurement		Phase noise	
		Ring oscillators	
		Voltage-controlled	
<i>Orientation measurement</i>		oscillators	
USE: Position measurement		Oscilloscopes	
Orifices		UF: Cathode-ray oscilloscopes	
	BT: Mechanical products	BT: Instruments	
		RT: Electric variables	
<i>Ormosils</i>		measurement	
USE: Organic inorganic hybrid		Test equipment	
materials			
Ornithology		<i>OSHA</i>	
	BT: Zoology	USE: Occupational safety	
RT: Birds			
		<i>OSI</i>	
<i>Orthogonal frequency division multiple access</i>		USE: Open systems	
USE: OFDM			
<i>Orthogonal frequency division multiplexing</i>		Osmium	
USE: OFDM		BT: Chemical elements	
Orthopedic procedures		Osmosis	
	BT: Medical treatment	BT: Chemical processes	
		NT: Electro-osmosis	
Orthopedic surgery		Osteoarthritis	
BT: Surgery		BT: Bone diseases	
Orthotics		Osteoporosis	
	BT: Medical treatment	BT: Bone diseases	
	RT: Assistive technologies	RT: Cancellous bone	
	Biomedical engineering		



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OTEC

USE: Ocean thermal energy conversion

OTFT

USE: Organic thin film transistors

Otolaryngology

USE: Otorhinolaryngology

Otologic surgery

USE: Oncological surgery

Otorhinolaryngology

UF: Otolaryngology
BT: Medical specialties
RT: Head
Neck

OTT

USE: Over-the-top media services

Out of order

UF: OOO
BT: Instruction sets

Outlier detection

USE: Anomaly detection

Output feedback

BT: Feedback circuits

Output power

USE: Power generation

Outsourcing

BT: Management
RT: Crowdsourcing

Ovarian cancer

BT: Cancer

Ovens

BT: Home appliances
NT: Microwave ovens

Over-the-air

USE: Wireless networks

Over-the-top media services

UF: OTT
BT: Streaming media

Overfitting

BT: Mathematical models

Overflow oscillations

USE: Finite wordlength effects

Overhead distribution lines

USE: Power distribution lines

Overhead transmission lines

USE: Power transmission lines

Overlay networks

BT: Computer networks
RT: Network topology
Transport protocols
NT: Dark Web

OWA

USE: Open wireless architecture

OWL

UF: Web ontology language
BT: Markup languages
Semantic Web
RT: Knowledge representation

Oxidation

BT: Chemical processes
RT: Antioxidants
Materials
NT: Combustion
Nitrification
Redox

Oxygen

BT: Chemical elements
Gases
RT: Hypoxia
Pulse oximetry

Ozonation

UF: Ozone treatment
BT: Wastewater treatment
RT: Environmental factors
Pollution control

Ozone generators

USE: Discharges (electric)

Ozone treatment

USE: Ozonation

Ozonizers

USE: Discharges (electric)



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<i>P-I-N</i>	USE:	PIN photodiodes	Labeling Multichip modules Nanopackaging Plastic packaging Wrapping
P-i-n diodes	BT:	Diodes Semiconductor devices Semiconductor diodes	
	RT:	CMOSFET logic devices Electro-optic modulators Vertical cavity surface emitting lasers	
P-n junctions	BT:	Junctions	
	RT:	Light emitting diodes Photodiodes Semiconductor diodes	
<i>P1394</i>	USE:	IEEE 1394 Standard	
<i>P2MP</i>	USE:	Point-to-multipoint communications	
<i>P2P</i>	USE:	Peer-to-peer computing	
<i>P802.11</i>	USE:	IEEE 802.11 Standard	
<i>PAAS</i>	USE:	Platform as a service	
Pacemakers	BT:	Biomedical equipment	
	RT:	Cardiology	
Pacific Ocean	BT:	Oceans	
<i>Package delivery</i>	USE:	Product delivery	
Packaging	BT:	Industry applications	
	RT:	Filling Leak detection Packaging machines Seals	
	NT:	Bagging Bottling Canning Encapsulation Food packaging	
		Packaging machines	
	BT:	Production equipment	
	RT:	Bagging Bottling Labeling Packaging Wrapping	
		Packet loss	
	BT:	Loss measurement Packet switching	
	RT:	Data communication Noise measurement	
		<i>Packet radio</i>	
	USE:	Packet radio networks	
		Packet radio networks	
	UF:	Packet radio	
	BT:	Radio communication	
		Packet switching	
	BT:	Communication switching	
	RT:	ARPANET Data transfer IEEE 802.3 Standard Next generation networking	
	NT:	Burst switching Frame relay Multiprotocol label	
		switching	Packet loss
		<i>Pacs</i>	
	USE:	Picture archiving and communication systems	
		<i>Paediatrics</i>	
	USE:	Pediatrics	
		Page description languages	
	UF:	Postscript	
	BT:	Markup languages	
	RT:	Desktop publishing High level languages	
		<i>Paging strategies</i>	
	USE:	Paging systems	



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Paging systems

UF: Paging strategies
BT: Cellular radio
RT: Wireless communication

Pain

BT: Injuries
NT: Ischemic pain
Neuropathic pain

Painting

BT: Surface finishing
Surface treatment
RT: Coatings
Paints
Photorealism

Paints

BT: Chemical products
Coatings
Materials
RT: Ink
Lacquers
Painting

Pair-wise error probability

USE: Pairwise error probability

Pairwise correlations

USE: Pairwise error probability

Pairwise error probability

UF: Pair-wise error probability
Pairwise correlations
BT: Probability

Palaeontology

USE: Paleontology

Paleontology

UF: Palaeontology
BT: Science - general
RT: Biology
Geology
Microorganisms

Palladium

BT: Metals

Palletising

USE: Pallets

Palletizing

USE: Pallets

Pallets

UF: Palletising
Palletizing
BT: Materials handling
RT: Containers
Load management

Palliative care

BT: Medical treatment

Palm print recognition

USE: Palmprint recognition

Palmprint identification

USE: Palmprint recognition

Palmprint recognition

UF: Palm print recognition
Palmprint identification
BT: Biometrics
RT: Identification of persons

Palmtop computers

USE: Personal digital devices

Pancreas

BT: Digestive system
RT: Pancreatic cancer

Pancreatic cancer

BT: Cancer
RT: Pancreas

Pandemics

BT: Epidemics
COVID-19
Coronaviruses
Diseases
Influenza

Pansharpened

USE: Pansharpening

Pansharpening

UF: Pansharpened
BT: Image processing
RT: Image quality

Paper electronics

USE: Organic electronics

Paper industry

USE: Pulp and paper industry



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Paper making

BT: Pulp and paper industry
RT: Bleaching
Paper making machines
Paper products
Paper pulp
Paper technology
Pulp manufacturing
Spinning machines

Paper making machines

BT: Production equipment
Pulp and paper industry
RT: Paper making
Paper products
Paper pulp
Paper technology
Pulp manufacturing
Spinning machines

Paper mills

BT: Production facilities
Pulp and paper industry
RT: Industrial plants
Paper products
Paper pulp
Pulp manufacturing
Spinning machines

Paper products

BT: Manufactured products
RT: Paper making
Paper making machines
Paper mills
Paper pulp
Paper technology
Pulp and paper industry

Paper pulp

BT: Manufactured products
Materials
RT: Cellulose
Paper making
Paper making machines
Paper mills
Paper products
Pulp and paper industry
Pulp manufacturing

Paper technology

BT: Industry applications
RT: Paper making
Paper making machines
Paper products
Pulp and paper industry

PAPR

USE: Peak to average power ratio

Paraelectric materials

USE: Dielectric materials

Parallel algorithms

BT: Algorithms
Parallel processing

Parallel architectures

BT: Computer architecture
RT: Parallel machines
Parallel processing
NT: Multicore processing

Parallel computing

USE: Parallel processing

Parallel languages

BT: High level languages
RT: Multiprocessing systems
Parallel processing
Parallel programming

Parallel machines

BT: Computers
RT: Parallel architectures
Parallel processing

Parallel processing

UF: Array processing
Parallel computing
Parallelism
BT: Computers and information processing
RT: Cluster computing
Concurrency control
Digital computers
Parallel architectures
Parallel languages
Parallel machines
Parallel programming
NT: Multiprocessing systems
Multithreading
Parallel algorithms
Pipeline processing
Single instruction multiple

data

Parallel processor interconnection

USE: Multiprocessor interconnection



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Parallel programming

BT: Programming
RT: Multiprocessing systems
Parallel languages
Parallel processing
VHDL

Parallel robots

BT: Robots

Parallelism

USE: Parallel processing

Paralysis

BT: Medical conditions
RT: Motor coordination

Paramagnetic materials

BT: Magnetic materials
RT: Paramagnetic resonance
NT: Superparamagnetic iron

oxide nanoparticles

Paramagnetic resonance

BT: Magnetic resonance
RT: Paramagnetic materials

Parameter estimation

UF: Parameter identification
BT: Signal analysis
Statistical analysis
RT: Control systems
Power system analysis

computing

NT: Spectral analysis
Amplitude estimation
Direction-of-arrival

estimation

Frequency estimation
Motion estimation
Phase estimation
Time of arrival estimation

Parameter extraction

BT: Electromagnetic
measurements
RT: Bipolar transistor circuits
Very large scale integration

Parameter identification

USE: Parameter estimation

Parameter uncertainty

USE: Uncertain systems

Parametric model

USE: Parametric statistics

Parametric statistics

UF: Parametric model
BT: Statistics

Parametric study

BT: Multitasking
RT: Optimization

Parasitic capacitance

BT: Capacitance

Parasitic diseases

BT: Diseases
RT: Mast cells

Parasympathetic nervous system

BT: Autonomic nervous system

Pareto analysis

BT: Statistical analysis
RT: Cause effect analysis
Quality management
NT: Pareto optimization

Pareto optimisation

USE: Pareto optimization

Pareto optimization

UF: Multi-attribute optimization
Multi-objective

programming

Multiobjective programming
Pareto optimisation
Vector optimization
BT: Optimization methods
Pareto analysis
RT: Genetic algorithms

Parietal lobe

BT: Brain

Paris Agreement

BT: Global warming
RT: Climate change

Parity check

USE: Codes

Parity check codes

UF: LDPC
Ldpc codes



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codes	Low density parity check	NT:	Accelerator magnets Colliding beam accelerators Cyclotrons Electron accelerators Ion accelerators Linear accelerators Photon collider Plasma accelerators Proton accelerators Storage rings Synchrocyclotrons Synchrotrons
	Parity-check codes		
	BT: Codes		
	RT: Decoding		
	NT: Iterative decoding		
<i>Parity-check codes</i>			
	USE: Parity check codes		
Parkinson's disease			
	BT: Diseases		
<i>Parotid</i>			
	USE: Salivary glands		
Partial differential equations		Particle beam bunching	
	BT: Differential equations		UF: Beam bunches
	RT: Boundary value problems		BT: Particle beam handling
	Fourier transforms		
	NT: Boundary-element methods		
	Poisson equations		
Partial discharge measurement		Particle beam handling	
	BT: Electric variables		BT: Nuclear and plasma
measurement		sciences	
	RT: Electrical safety		RT: Particle beams
	Insulation life		NT: Particle beam bunching
	Insulation testing		
	Optical fiber sensors		
Partial discharges		Particle beam injection	
	BT: Dielectric breakdown		UF: Injected beams
	RT: Corona		BT: Nuclear and plasma
Partial response channels		sciences	
	BT: Communication channels		RT: Particle beams
Partial response signaling		Particle beam measurements	
	BT: Digital modulation		BT: Measurement
			RT: Particle beams
Partial transmit sequences		Particle beam optics	
	BT: OFDM		UF: Ion optics
<i>Particle accelerator</i>			BT: Optics
	USE: Linear particle accelerator		RT: Electrodynamics
Particle accelerators			Particle beams
	BT: Nuclear and plasma		NT: Atom optics
sciences			Electron optics
	RT: CERN		Stimulated emission
	Colliding beam devices		
	Large Hadron Collider		
	Particle beams		
	Voltage multipliers	Particle beams	
			UF: Accelerator beams
			Neutron beams
			Proton beams
			BT: Beams
			Elementary particles
			RT: Colliding beam accelerators
			Laser theory
			Particle accelerators
			Particle beam handling
			Particle beam injection
			Particle beam
		measurements	



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	Particle beam optics		Swarm intelligence
	Storage rings	RT:	Artificial bee colony
	Synchrotrons	algorithm	
NT:	Atomic beams		Fireworks algorithm
	Electron beams		Fish schools
	Ion beams		Krill herd algorithm
			Metaheuristics
			Stochastic processes
Particle charging		Particle tracking	
BT:	Electrostatic processes	BT:	Nuclear measurements
RT:	Semiconductor detectors	RT:	High energy physics
Particle collisions		instrumentation computing	
BT:	Elementary particles		Tracking
<i>Particle detectors</i>		<i>Particle-swarm optimization</i>	
USE:	Radiation detectors	USE:	Particle swarm optimization
Particle filters		<i>Particles (elementary)</i>	
BT:	Filters	USE:	Elementary particles
Particle measurements		<i>Particulate measurements</i>	
UF:	Particulate measurements	USE:	Particle measurements
BT:	Measurement		
RT:	Current density	Partitioning algorithms	
	High energy physics	BT:	Algorithms
instrumentation computing		Passband	
	Position sensitive particle	BT:	Digital communication
detectors			Radio communication
<i>Particle physics</i>		RT:	Baseband
USE:	High energy physics	Passivation	
Particle production		BT:	Surface treatment
BT:	Electrostatic processes	RT:	Corrosion
RT:	Aerosols	Passive circuits	
	Spraying	BT:	Circuits
Particle scattering		Passive filters	
BT:	Scattering	BT:	Passive networks
RT:	Scanning electron	RT:	Filters
microscopy		Passive microwave remote sensing	
Particle separators		BT:	Remote sensing
UF:	Separators	Passive networks	
BT:	Separation processes	BT:	Telecommunication
RT:	Magnetic separation	network topology	
<i>Particle swarm</i>		NT:	Passive filters
USE:	Particle swarm optimization	Passive optical networks	
Particle swarm optimization		UF:	PON
UF:	Particle swarm		Passive-optical-network
	Particle-swarm optimization	BT:	Optical fiber networks
BT:	Algorithms		
	Evolutionary computation		



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RT:	EPON	Death	Pathogenesis
Passive radar		Pathology	
BT:	Radar	UF:	Pathological
RT:	Radar detection	BT:	Medical specialties
	Radar imaging	RT:	Autopsy
Passive RFID tags			Diseases
BT:	RFID tags	NT:	Histopathology
<i>Passive-optical-network</i>			Neuropathology
USE:	Passive optical networks		Pathological processes
Passwords		<i>Patient diagnosis</i>	
BT:	Access control	USE:	Medical diagnosis
	Computer security	<i>Patient identification</i>	
RT:	Authentication	USE:	Medical treatment
Patch antennas		Patient monitoring	
BT:	Antennas	BT:	Monitoring
RT:	Microstrip antennas	RT:	Assistive robots
Patent law			Chemotherapy
BT:	Law		Electronic medical records
Patents			Fall detection
BT:	Legal factors		Point of care
RT:	Intellectual property	Patient rehabilitation	
	US Government agencies	BT:	Medical treatment
Path planning		RT:	Aphasia
UF:	piano mover's problem		Assistive robots
BT:	Motion control		Supernumery robotic limbs
RT:	Course correction	NT:	Neurorehabilitation
	Indoor navigation	<i>Patient treatment</i>	
	Vehicle routing	USE:	Medical treatment
NT:	Trajectory	Pattern analysis	
	Trajectory planning	BT:	Machine intelligence
	Trajectory tracking	RT:	Generative AI
Pathogenesis			Kernel machines
BT:	Pathological processes		Surface reconstruction
Pathogens			Trend analysis
UF:	Germes	Pattern classification	
BT:	Diseases	UF:	Signal classification
RT:	Epidemics	BT:	Decision making
	Microbiome	RT:	Feature extraction
<i>Pathological</i>			Graph neural networks
USE:	Pathology		Logistic regression
Pathological processes			Nearest neighbor methods
BT:	Pathology		Neural networks
NT:	Cadaver		Pattern recognition
			Reinforcement learning
			Semisupervised learning
			Support vector machines



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	NT:	Naive Bayes methods			Face recognition																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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PDF

USE: Portable document format

Peace technology

BT: Social implications of technology

Peak signal to noise ratio

USE: PSNR

Peak signal-to-noise ratio

USE: PSNR

Peak to average power ratio

UF: PAPR
Peak-to-average power ratio

BT: Peak-to-average ratio
OFDM

Peak-to-average power ratio

USE: Peak to average power ratio

Peak-to-average ratio

USE: Peak to average power ratio

Peat

BT: Soil

Pedestrians

BT: Human factors
Road traffic
RT: Legged locomotion
Road safety
Road traffic control
Vehicle safety

Pediatrics

UF: Babies
Baby
Child
Children
Infant
Infants
Newborns
Paediatrics
Toddler
BT: Medical specialties
RT: Neonatology
NT: Neonatal

Peer to peer communications

USE: Peer-to-peer computing

Peer to peer computing

USE: Peer-to-peer computing

Peer to peer exchange

USE: Peer-to-peer computing

Peer to peer network

USE: Peer-to-peer computing

Peer-to-peer communications

USE: Peer-to-peer computing

Peer-to-peer computing

UF: File sharing
P2P
Peer to peer communications
Peer to peer computing
Peer to peer exchange
Peer to peer network
Peer-to-peer communications
Peer-to-peer exchange
Peer-to-peer network
BT: Computer networks
Distributed computing
RT: Border Gateway Protocol
Cluster computing
Decentralized applications
Distributed ledger
Sidelink
Workstations
NT: InterPlanetary File System

Peer-to-peer exchange

USE: Peer-to-peer computing

Peer-to-peer network

USE: Peer-to-peer computing

Peltier effect

BT: Thermoelectricity

Pelvic bones

BT: Bones

Pelvis

BT: Body regions

Pen test

USE: Penetration testing

Penetration testing

UF: Pen test
BT: Computer security



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Pensions

UF: Occupational pensions
Personal pensions
Stakeholder pensions
State pensions
BT: Remuneration
RT: Employee welfare
Termination of employment

Pentacene

BT: Organic semiconductors

People with disabilities

UF: Disabled people
Disabled persons
BT: Medical conditions
RT: Assistive technologies
Mental disorders
Physiology
Psychology

Peptides

BT: Biochemistry

Perception evolution networks

BT: Neural networks
RT: Unsupervised learning

Perfectly matched layers

BT: Propagation
RT: Finite difference methods
Finite element analysis
Maxwell equations

Performance analysis

UF: Dynamic program analysis
Performance index
BT: Programming
RT: Optimization
NT: Performance gain

Performance evaluation

UF: Performance measurement
BT: Measurement
RT: Benchmark testing
NT: Evaluation models
Key performance indicator
Performance metrics

Performance gain

BT: Performance analysis

Performance index

USE: Performance analysis

Performance loss

BT: Computer performance

Performance measurement

USE: Performance evaluation

Performance metrics

BT: Performance evaluation

Performance optimisation

USE: Optimization

Performance optimization

USE: Optimization

Performance related pay

USE: Incentive schemes

Perineum

BT: Body regions

Periodic media

USE: Nonhomogeneous media

Periodic structures

BT: Materials science and
technology
NT: Gratings
Photonic crystals

Peripheral equipment

USE: Computer peripherals

Peripheral nervous system

BT: Nervous system

Permafrost

BT: Soil
RT: Ice
Tundra

Permanent magnet generators

BT: Permanent magnet
machines

Permanent magnet machines

UF: Permanent magnet
synchronous machines
BT: Electric machines
Rotating machines
RT: Permanent magnet motors
Permanent magnets
NT: Permanent magnet
generators



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Permanent magnet motors

UF: Permanent magnet
synchronous motors
Permanent-magnet
generators
Permanent-magnet motors
BT: Motors
RT: Permanent magnet
machines

Permanent magnet synchronous machines

USE: Permanent magnet
machines

Permanent magnet synchronous motors

USE: Permanent magnet motors

Permanent magnets

BT: Magnets
RT: Magnetic gears
Magnetic levitation vehicles
Permanent magnet
machines
Remanence

Permanent-magnet generators

USE: Permanent magnet motors

Permanent-magnet motors

USE: Permanent magnet motors

Permeability

UF: Magnetic permeability
BT: Electromagnetic analysis
RT: Magnetic materials
Permeability measurement

Permeability measurement

BT: Magnetic variables
measurement
RT: Permeability

Permission

UF: Access rights
File system permissions
BT: Computer security
RT: Zero Trust

Permissioned blockchains

USE: Blockchains

Permittivity

BT: Electric variables
RT: Dielectric constant
Dielectric materials

Permittivity measurement

Permittivity measurement

BT: Dielectric measurement
RT: Permittivity

Perovskites

BT: Crystalline materials

Perpendicular magnetic anisotropy

BT: Magnetic anisotropy

Perpendicular magnetic recording

UF: Vertical recording
BT: Magnetic recording
RT: Disk drives

Perpendicular recording

USE: Magnetic recording

Persistent currents

BT: Current
RT: High-temperature
superconductors
Superconducting magnets

Persistent identifiers

BT: Data structures
RT: Digital systems
Information retrieval

Person identification

USE: Identification of persons

Person re-identification

USE: Identification of persons

Personal area networks

UF: Piconets
Scatternets
BT: Radio communication
RT: Computer networks
Data communication
Land mobile radio
Personal communication
networks
Wireless LAN
Zigbee
NT: Bluetooth
Body area networks
Body sensor networks
Wireless personal area
networks



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Personal communication networks

UF: Personal communication services
BT: Communication systems
RT: Cellular radio
Digital systems
IEEE 802.15 Standard
Location awareness
Mobile handsets
Personal area networks
Zigbee

Personal communication services

USE: Personal communication networks

Personal computers

USE: Microcomputers

Personal digital assistants

USE: Personal digital devices

Personal digital devices

UF: Blackberry
Digital devices
Hand held computers
Handheld computers
PD
PDA
Palmtop computers
Personal digital assistants
Wireless digital devices
BT: Portable computers

Personal pensions

USE: Pensions

Personal protective equipment

BT: Health and safety
RT: Face masks
Protective clothing
NT: Helmets

Personal voice assistants

UF: PVA
BT: Speech recognition
Virtual assistants
RT: Natural language
processing
Smart devices
Speech synthesis
Voice activity detection

Personalized medicine

USE: Precision medicine

Personnel

BT: Human resource
management
RT: Appraisal
Bring your own device
Education
Employment
Equal opportunities
Management
Productivity
Training
NT: Labor resources

Persuasive systems

BT: Decision making
Social computing
RT: Behavioral sciences
Human factors
Human-machine systems
Psychology

Perturbation methods

UF: Perturbation techniques
BT: Approximation methods
NT: Cavity perturbation
methods

Perturbation techniques

USE: Perturbation methods

Pervasive computing

UF: Everywhere
Ubicomp
BT: Computers and information
processing
Systems, man, and
cybernetics
RT: Artificial intelligence
Context awareness
Next generation networking
NT: Ubiquitous computing
Wearable devices

Pest control

UF: Insect control
Vermin control
BT: Environmental
management
RT: Agriculture
Hazards
Pesticides

Pesticides

BT: Agrochemicals
RT: Agricultural products



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		Chemical industry		Fractionation
		Chemical products		Fuel processing industries
		Land pollution		Gas industry
		Pest control		Natural gas industry
		Plant diseases		Offshore installations
		Soil pollution		Oil pollution
		Water pollution		Oils
<i>PET</i>				Petrochemicals
	USE:	Positron emission		Petroleum
tomography			NT:	Pipelines
				Oil drilling
				Oil refineries
				Well logging
Petascale computing				
	BT:	Computers and information		
processing			<i>PETs</i>	
	RT:	Supercomputers		USE: Privacy enhancing
			technologies	
Petri dishes				
	BT:	Biomedical equipment	<i>PFD</i>	
				USE: Phase frequency detectors
Petri nets				
	BT:	System analysis and design	<i>PGA</i>	
	RT:	Discrete-event systems		USE: Electronics packaging
		Modeling		
			pH measurement	
Petrochemicals				BT: Chemical analysis
	BT:	Chemical products		Measurement
		Materials		
	RT:	Chemical industry	<i>Phages</i>	
		Chemistry		USE: Bacteriophages
		Fuels		
		Petroleum	Phantoms	
		Petroleum industry		BT: Biomedical imaging
		Plastic products		RT: Dosimetry
		Plastics		Positron emission
			tomography	
<i>Petrol</i>				Single photon emission
	USE:	Petroleum	computed tomography	
				X-ray applications
Petroleum				X-ray detection
	UF:	Gasoline		X-ray imaging
		Petrol		
	BT:	Chemical products	Pharmaceutical industry	
		Fuels		BT: Industries
	RT:	Fuel processing industries		RT: Drug discovery
		Oil pollution		Drugs
		Oils		
		Petrochemicals	Pharmaceutical technology	
		Petroleum industry		BT: Chemical technology
	NT:	Hydrocarbons		RT: Biochemistry
				Chemistry
Petroleum industry				Pharmaceuticals
	UF:	Oil industry		
	BT:	Industries	Pharmaceuticals	
	RT:	Chemical industry		BT: Chemical products



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RT: Medical treatment
Biochemistry
Chemistry
Pharmaceutical technology
NT: Drugs

Phase change RAM
Phase-change RAM
Phase-change random
access memory
BT: Phase change memory
Random access memory

Pharmacodynamics

BT: Pharmacology
RT: Biochemistry
Drug resistance
Drugs
Pharmacokinetics
NT: Drug interactions

Phase control

UF: Phase-control
BT: Power electronics
RT: Electric variables control
Optical variables control
Phase transformers

Pharmacokinetics

BT: Pharmacology
RT: Biochemistry
Drugs
Pharmacodynamics

Phase detection

BT: Signal detection
NT: Phase frequency detectors

Pharmacology

BT: Medical specialties
RT: Biochemistry
Biology
Drugs
NT: Drug discovery
Pharmacodynamics
Pharmacokinetics

Phase distortion

BT: Distortion
RT: Delay effects

Phase estimation

BT: Parameter estimation

Phase frequency detectors

UF: PFD
BT: Phase detection
RT: Frequency measurement
Measurement
Voltage
Voltage control

Pharynx

BT: Digestive system
RT: Stomatognathic system

Phase change materials

UF: PCM
BT: Materials
RT: Memory
Nuclear thermodynamics
Optical materials
Phase change memory
Solar heating

Phase locked loops

UF: PLL
Phase locked-loops
Phase-locked loops
Phase-locked-loops
BT: Linear feedback control

systems

Signal processing
RT: Frequency locked loops
Modulation
Nonlinear filters
Ring oscillators

Phase change memory

BT: Memory
RT: Phase change materials
Resistive RAM
NT: Phase change random

access memory

Phase locked-loops

USE: Phase locked loops

Phase change RAM

USE: Phase change random
access memory

Phase measurement

BT: Measurement
RT: Acoustic measurements
Electric variables

Phase change random access memory

UF: PCRAM

measurement



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measurement	Optical variables	<i>Phase-change random access memory</i>	USE: Phase change random access memory
Phase modulation		<i>Phase-control</i>	USE: Phase control
BT:	Modulation		
RT:	Demodulation		
	Electro-optic modulators		
NT:	Continuous phase	<i>Phase-locked loops</i>	USE: Phase locked loops
modulation	Cross-phase modulation		
	Differential phase shift	<i>Phase-locked-loops</i>	USE: Phase locked loops
keying	Phase shift keying	<i>Phase-shift keying</i>	USE: Phase shift keying
Phase noise		<i>Phase-shift-keying</i>	USE: Phase shift keying
BT:	Noise		
	Oscillators		
RT:	Time-domain analysis		
Phase shift keying		Phased arrays	
UF:	PSK	UF:	Antenna phased arrays
	Phase-shift keying	BT:	Antenna arrays
	Phase-shift-keying	RT:	Optical arrays
	QPSK	NT:	Steerable antennas
BT:	Phase modulation	Phasor measurement units	
NT:	Binary phase shift keying	UF:	PMU
	Quadrature phase shift		PMUs
keying			Synchrophasors
Phase shifters		BT:	Electric variables
BT:	Circuits	measurement	
RT:	Butler matrices	PHEMTs	
NT:	Phase transformers	UF:	Pseudomorphic HEMTs
Phase shifting interferometry		BT:	HEMTs
BT:	Interferometry	Phenology	
<i>Phase transformations, nuclear</i>		BT:	Biology
USE:	Nuclear phase	RT:	Biological processes
transformations			Chronobiology
			Climate change
Phase transformers		Phenotypes	
BT:	Phase shifters	BT:	Genetics
	Transformers	RT:	Genotypes
RT:	Circuits	Philosophical considerations	
	Phase control	BT:	Social implications of
<i>Phase transitions, nuclear</i>		technology	
USE:	Nuclear phase	RT:	Econophysics
transformations			Ethical aspects
			Humanities
<i>Phase-change RAM</i>			Machine ethics
USE:	Phase change random		Quantum mechanics
access memory			Social factors



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	Technology	<i>Photo-cathodes</i>	
	Technology social factors	USE:	Photocathodes
Phishing		<i>Photo-realistic images</i>	
BT:	Computer security	USE:	Photorealistic images
	Information security		
RT:	Malware	Photoacoustic effects	
<i>PHM</i>		UF:	Optoacoustic effects
USE:	Prognostics and health	BT:	Spectroscopy
management		RT:	Acoustic testing
			Laser applications
Phonetics			Photothermal effects
BT:	Linguistics	NT:	Photoacoustic imaging
RT:	Natural language		
processing		Photoacoustic imaging	
	Semiotics	BT:	Biomedical imaging
	Speech processing		Photoacoustic effects
NT:	Acoustic phonetics	Photobleaching	
<i>Phonocardiogram</i>		BT:	Photochemistry
USE:	Phonocardiography	Photocatalysis	
Phonocardiography		UF:	Photo-catalysis
UF:	PCG	BT:	Catalysis
	Phonocardiogram		Photochemistry
BT:	Cardiography	RT:	Photocatalysts
RT:	Biomedical monitoring	Photocatalysts	
	Cardiology	BT:	Catalysts
	Heart rate measurement	RT:	Photocatalysis
<i>Phonographs</i>		Photocathodes	
USE:	Audio systems	UF:	Photo-cathodes
Phonons		BT:	Cathodes
BT:	Elementary particles	RT:	Photomultipliers
RT:	Acoustics	Photochemistry	
	Crystals	BT:	Chemistry
	Electrons	RT:	Water splitting
	Indium phosphide	NT:	Photobleaching
Phosphorescence			Photocatalysis
BT:	Luminescence		Photosynthesis
RT:	Phosphors	Photochromism	
Phosphors		UF:	Photodarkening
BT:	Light sources	BT:	Photonics
RT:	Phosphorescence	RT:	Color
Phosphorus		<i>Photocomposition</i>	
BT:	Chemical elements	USE:	Text processing
<i>Photo-catalysis</i>		Photoconducting devices	
USE:	Photocatalysis	BT:	Optoelectronic devices
		RT:	Photoconducting materials
			Photoconductivity



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NT: Photodetectors
Semiconductor devices
Electrophotography

RT: Optical polarization
Piezooptic effects
Stress

Photoconducting materials

UF: Photoconductors
BT: Materials
RT: Photoconducting devices
Photoconductivity
Photodetectors
Semiconductor materials

Photoconductivity

UF: Photocurrent
BT: Conductivity
RT: Photoconducting devices
Photoconducting materials

Photoconductors

USE: Photoconducting materials

Photocurrent

USE: Photoconductivity

Photodarkening

USE: Photochromism

Photodetector

USE: Photodetectors

Photodetectors

UF: Photodetector
BT: Optoelectronic devices
Radiation detectors
RT: Image sensors
Infrared detectors
Optical signal detection
Photoconducting devices
Photoconducting materials
Photoelectricity
Photon detectors
NT: Photodiodes
Phototransistors
Superconducting

photodetectors

Photodiodes

BT: Photodetectors
RT: Optical transmitters
P-n junctions
NT: Avalanche photodiodes
PIN photodiodes

Photoelasticity

BT: Mechanical factors

Photoelectricity

UF: Photoemission
Phototubes
BT: Electricity
Electron devices
RT: Electron emission
Photodetectors
Photomultipliers
Photovoltaic cells
NT: Photovoltaic effects

Photoelectron microscopy

UF: Photoemission electron
microscopy
BT: Electron microscopy

Photoemission

USE: Photoelectricity

Photoemission electron microscopy

USE: Photoelectron microscopy

Photogalvanic effects

USE: Photovoltaic effects

Photogrammetry

BT: Photometry
RT: Electromagnetic
measurements
Image processing
Object detection
Photography

Photography

BT: Imaging
RT: Cameras
Electrophotography
High dynamic range
Image capture
Image storage
Optical filters
Photogrammetry
Photorealistic images
NT: Cinematography
Digital photography
Image forensics
Photomicrography
Photorealism

Photoionisation

USE: Ionization



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Photoionization

USE: Ionization

Photolithography

USE: Lithography

Photoluminescence

UF: Electrophotoluminescence

BT: Luminescence

Optics

RT: Chemiluminescence

Judd-Ofelt theory

Microcavities

Photomagnetic devices

USE: Magneto optic devices

Photomagnetic effects

USE: Magneto optic effects

Photometry

BT: Geoscience and remote sensing

Optical variables

measurement

RT: Light sources

Lighting

Radiometry

NT: Photogrammetry

Photomicrographs

USE: Photomicrography

Photomicrography

UF: Micrographs

Microphotographs

Microphotography

Photomicrographs

BT: Photography

Photomultipliers

BT: Vacuum technology

RT: Avalanche photodiodes

Electron multipliers

Photocathodes

Photoelectricity

Photon collider

BT: Particle accelerators

Photon crystal fibers

USE: Photonic crystal fibers

Photon crystal fibres

USE: Photonic crystal fibers

Photon detectors

BT: Photonics

RT: Photodetectors

Photonic band gap

UF: Band gap

Band-gap

Bandgap

Photonic bandgap

BT: Photonic crystals

RT: Electromagnetic wave

polarization

Optical diffraction

Photonic bandgap

USE: Photonic band gap

Photonic bandgap fibers

UF: Photonic bandgap fibres

BT: Photonic crystal fibers

Photonic bandgap fibres

USE: Photonic bandgap fibers

Photonic crystal fibers

UF: Microstructured fibers

Microstructured fibres

Photon crystal fibers

Photon crystal fibres

Photonic crystal fibres

Photonic-crystal fibers

Photonic-crystal fibres

BT: Photonic crystals

NT: Holey fibers

Photonic bandgap fibers

Photonic crystal fibres

USE: Photonic crystal fibers

Photonic crystals

UF: 2-D photonic crystals

2D photonic crystals

Photonic crystal fibers

Two dimensional photonic

crystals

Two-dimensional photonic

crystals

BT: Periodic structures

RT: Microwave devices

Nonlinear optics

Optical materials

Optical waveguides

Spontaneous emission

NT: Photonic band gap



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	Photovoltaic systems		<i>Physical distribution management</i>
	Solar panels	USE:	Logistics
NT:	Light trapping		
Photovoltaic effects		Physical layer	
UF:	Photogalvanic effects	BT:	Open systems
BT:	Photoelectricity	NT:	Physical layer security
RT:	Photovoltaic cells		
	Photovoltaic systems	Physical layer security	
NT:	Shunts (electrical)	BT:	Physical layer
		RT:	Hardware security
<i>Photovoltaic modules</i>		Physical optics	
USE:	Solar panels	BT:	Optics
		NT:	Optical refraction
<i>Photovoltaic power systems</i>			Optical vortices
USE:	Photovoltaic systems		
Photovoltaic systems		Physical theory of diffraction	
UF:	Photovoltaic power systems	BT:	Electromagnetic diffraction
BT:	Solar power generation		
RT:	Hybrid power systems	Physical unclonable function	
	Photovoltaic cells	UF:	PUF
	Photovoltaic effects		Physically unclonable
	Water pumps	function	
NT:	Building integrated	BT:	Control system security
photovoltaics			Semiconductor device
	Fill factor (solar cell)	manufacture	
	Solar panels	RT:	Cryptography
			Security
			Semiconductor devices
			Smart devices
Phthalates		Physical vapor deposition	
BT:	Chemical products	UF:	Ionized jet deposition
RT:	Plastics		Physical vapor transport
			Physical vapour deposition
<i>Phylogenetic tree</i>			Physical vapour transport
USE:	Phylogeny	BT:	Plasma materials
		processing	
<i>Phylogenetics</i>		RT:	Sputtering
USE:	Phylogeny		
Phylogeny		<i>Physical vapor transport</i>	
UF:	Cladistics	USE:	Physical vapor deposition
	Phylogenetic tree		
	Phylogenetics	<i>Physical vapour deposition</i>	
BT:	Evolution (biology)	USE:	Physical vapor deposition
Physical chemistry		<i>Physical vapour transport</i>	
BT:	Chemistry	USE:	Physical vapor deposition
Physical design		<i>Physically unclonable function</i>	
BT:	System analysis and design	USE:	Physical unclonable
	Systems engineering and	function	
theory			
RT:	Integrated circuit layout	<i>Physician</i>	
		USE:	Medical services



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Physics

BT: Science - general
 RT: Buoyancy
 Environmental science
 NT: Acoustics
 Astrophysics
 Beams
 Biophysics
 Dark energy
 Entropy
 Fluid flow
 Geophysics
 High energy physics
 Kinetic theory
 Levitation
 Lorentz covariance
 Mechanical factors
 Network theory (graphs)
 Physics education
 Quantum mechanics
 Rydberg atoms
 Solid-state physics
 String theory
 Thermal factors
 Waves

Physics computing

BT: Computer applications
 RT: Multiphysics

Physics education

BT: Engineering education
 Physics

Physiology

BT: Biology
 RT: Entomology
 People with disabilities
 NT: Action potentials
 External stimuli
 Metabolism
 Neuromodulation
 Somatosensory

PhysiStimuli

USE: External stimuli

Phytoplankton

BT: Plankton

PI control

UF: PI controller
 Proportional + integral

control

Proportional-integral control

Proportional-integral
 controller
 Proportional-integral-
 derivative control
 Proportional-integral-
 derivative controller
 BT: Control systems

PI controller

USE: PI control

piano mover's problem

USE: Path planning

Pickling

BT: Surface treatment
 RT: Chemistry
 Fermentation

Pico-hydro

USE: Picohydro power

Picohydro power

UF: Pico-hydro
 BT: Hydroelectric power

generation

RT: Appropriate technology

Piconets

USE: Personal area networks

Picture archiving and communication systems

UF: Pacs
 BT: Image communication
 RT: Biomedical communication
 Biomedical computing
 Biomedical imaging

Picture phones

USE: Videophone systems

Picture processing

USE: Image processing

Picturephones

USE: Videophone systems

PID control

USE: PD control

Piecewise linear approximation

BT: Piecewise linear techniques



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Piecewise linear techniques

BT: Mathematics
RT: Control system analysis
Control system synthesis
Difference equations
Nonlinear control systems
NT: Piecewise linear

approximation

Piezoceramics

USE: Piezoelectric materials

Piezoelectric actuators

BT: Actuators

Piezoelectric devices

BT: Dielectric devices
RT: Acoustic devices
Acoustoelectric devices
Nanogenerators
Piezoelectric films
Piezoelectric materials
Piezoelectricity
Piezoresistive devices
Surface acoustic wave

devices

Piezoelectric effect

BT: Piezoelectricity

Piezoelectric effects

USE: Piezoelectricity

Piezoelectric films

BT: Dielectric films
Films
Piezoelectric materials
RT: Piezoelectric devices
Piezoelectricity

Piezoelectric materials

UF: Piezoceramics
BT: Dielectric materials
RT: Acoustic materials
Crystals
Piezoelectric devices
Piezoelectricity
NT: Lead zirconate titanate
Piezoelectric films

Piezoelectric polarization

BT: Piezoelectricity

Piezoelectric transducers

BT: Transducers

Piezoelectricity

UF: Piezoelectric effects
BT: Electricity
Ultrasonics, ferroelectrics,
and frequency control
RT: Electrostriction
Piezoelectric devices
Piezoelectric films
Piezoelectric materials
Piezoresistance
Pyroelectricity
Stress
Ultrasonic transducers
NT: Piezoelectric effect
Piezoelectric polarization

Piezomagnetic effects

USE: Magnetomechanical effects

Piezo optic effects

BT: Acoustooptic effects
RT: Photoelasticity
Pressure effects
Stress

Piezoresistance

UF: Piezoresistive
BT: Electric variables
Resistance
RT: Piezoelectricity
Piezoresistive devices
Pressure effects
Stress

Piezoresistive

USE: Piezoresistance

Piezoresistive devices

UF: Piezoresistors
BT: Semiconductor devices
RT: Piezoelectric devices
Piezoresistance
Pressure measurement

Piezoresistors

USE: Piezoresistive devices

Pigmentation

BT: Color
Materials science and
technology
NT: Pigments

Pigments

BT: Pigmentation



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PIN diodes

USE: PIN photodiodes

Pin grid arrays

USE: Electronics packaging

PIN photodiodes

UF: P-I-N
PIN diodes
BT: Photodiodes

Pink noise

USE: 1/f noise

Pins

BT: Plugs

Pions

USE: Mesons

Pipeline processing

UF: Computer pipeline processing
BT: Pipelining
Parallel processing
RT: Multiprocessing systems
Systolic arrays

Pipelines

BT: Fluid flow
RT: Chemical industry
Magnetic flux leakage
Materials handling
Natural gas industry
Oils
Petroleum industry

Pipelining

USE: Pipeline processing

Piracy (software)

USE: Computer crime

Pistons

BT: Machine components
Mechanical products
RT: Bellows
Engine cylinders
Engines
Gaskets
Shafts
Structural rings

Pitch control (audio)

BT: Audio systems
Variable speed drives

Pitch control (position)

BT: Mechanical variables control

Pituitary gland

BT: Glands
Nervous system

Pixel

BT: Digital images

PLA

USE: Programmable logic arrays

Placenta

BT: Reproductive system
RT: Pregnancy

Plagiarism

BT: Professional communication
RT: Copyright protection
Publishing

Planar antennas

USE: Planar arrays

Planar array

USE: Planar arrays

Planar arrays

UF: Planar antennas
Planar array
BT: Antenna arrays

Planar motors

BT: Electric motors

Planar transmission lines

BT: Transmission lines
RT: Spurline
NT: Coplanar transmission lines
Finline
Microstrip
Slot lines
Stripline

Planar waveguides

BT: Electromagnetic waveguides
RT: Rectangular waveguides



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Planarisation

USE: Planarization

Planarization

UF: Chemical mechanical
planarisation

Chemical mechanical
planarization

Planarisation
BT: Surface treatment
RT: Dielectric films
Integrated circuits

Planetary chemistry

USE: Astrochemistry

Planetary composition

USE: Extraterrestrial
measurements AND

Planets

Planetary landers

USE: Land transportation AND
Space vehicles

Planetary oceans

USE: Oceans AND
Planets

Planetary orbits

BT: Orbits

Planets

UF: Planetary composition
Planetary oceans
BT: Solar system
RT: Extraterrestrial phenomena
NT: Asteroids
Comets
Earth
Jupiter
Mars
Mercury (planets)
Pluto
Saturn
Venus

Planing

BT: Machining
RT: Finishing
Surface roughness
Surface treatment

Plankton

BT: Organisms

RT: Eutrophication
NT: Phytoplankton
Zooplankton

Planning

UF: System planning
BT: Engineering management
RT: Decision making
Economics
NT: Meeting planning
Schedules
Strategic planning
Technical planning
Technology planning

Plant diseases

BT: Diseases
RT: Pesticides

Plants (biology)

BT: Organisms
RT: Life sciences
NT: Chlorophyll
Fauna
Flora
Flowering plants
Medicinal plants
Non-flowering plants
Trees (botanical)

Plants (industrial)

USE: Industrial plants

Plasma accelerators

BT: Particle accelerators
Plasma devices

Plasma applications

BT: Plasmas
RT: Low-temperature plasmas
Plasma materials

processing

NT: Plasma devices
Plasma immersion ion

implantation

Plasma welding
Tokamaks

Plasma chemistry

BT: Plasma properties

Plasma confinement

BT: Plasmas
NT: Inertial confinement
Magnetic confinement



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Plasma density

BT: Plasma properties

Plasma temperature

Plasmons

Plasma devices

BT: Plasma applications

RT: Gas discharge devices

Plasmas

NT: Plasma accelerators

Plasma jets

Tokamaks

Plasma sheaths

BT: Plasma properties

Plasma simulation

BT: Plasmas

RT: Modeling

Numerical simulation

Tokamaks

Plasma diagnostics

BT: Plasmas

RT: Plasma measurements

Plasma sources

BT: Plasmas

RT: Ion implantation

Ion sources

Plasma display panel

USE: Flat panel displays

Plasma stability

BT: Plasma properties

Plasma immersion ion implantation

BT: Ion implantation

Plasma applications

RT: Semiconductor impurities

Plasma temperature

BT: Plasma properties

Plasma jets

BT: Plasma devices

RT: Propulsion

Plasma transport processes

BT: Plasmas

Plasma materials processing

BT: Materials processing

RT: Plasma applications

NT: Chemical vapor deposition

Ignition

Physical vapor deposition

Spark Plasma sintering

Plasma waves

BT: Waves

RT: Plasmas

Plasma welding

BT: Plasma applications

RT: Joining processes

Materials processing

Plasmas

Plasma measurements

BT: Measurement

RT: Plasma diagnostics

Plasmas

Plasma x-ray sources

BT: X-ray imaging

RT: X-ray lasers

Plasma membrane

USE: Cell membrane

Plasma-assisted combustion

BT: Combustion

Plasmas

Plasma pressure compaction

USE: Spark Plasma sintering

Plasmas

BT: Nuclear and plasma

sciences

RT: Arc discharges

Discharges (electric)

Ionization

Ionosphere

Optical emission

spectroscopy

Plasma devices

Plasma measurements

Plasma properties

BT: Plasmas

RT: Electron mobility

Stability analysis

NT: Dusty plasmas

Plasma chemistry

Plasma density

Plasma sheaths

Plasma stability



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plasmas	NT:	Plasma waves	Plastic insulators
		Plasma welding	Plastic insulators BT: Insulators RT: Fiber reinforced plastics Plastic films Plastic insulation Plastic packaging
		Relativistic effects	
		Atmospheric-pressure	
		Low-temperature plasmas	
		Microwave plasmas	
		Plasma applications	
		Plasma confinement	
		Plasma diagnostics	
		Plasma properties	
		Plasma simulation	
combustion		Plasma simulation	Plastic integrated circuit packaging UF: Plastic IC packaging BT: Integrated circuit packaging
		Plasma sources	
		Plasma transport processes	
		Plasma-assisted	
<i>Plasmon</i>	USE:	Plasmons	Plastic optical fiber BT: Optical fibers Plastics
<i>Plasmonic solar cells</i>	USE:	Light trapping	Plastic packaging BT: Packaging RT: Bagging Electronics packaging Encapsulation Integrated circuit packaging Plastic insulators Plastics
<i>Plasmonics</i>	USE:	Plasmons	<i>Plastic pollution</i> USE: Plastic waste
Plasmons	UF:	Plasmon	<i>Plastic pollution waste</i> USE: Plastic waste
		Plasmonics	
		Plasma properties	
		Nanoantennas	
		Optical antennas	
		Nanoplasmonics	
		Surface plasmons	
<i>Plastic bottles</i>	USE:	Plastic products	Plastic products UF: Plastic bottles Plastic containers BT: Manufactured products RT: Bottling Chemical industry Chemical products Chemistry Consumer products Petrochemicals Plastics Plastics industry
<i>Plastic containers</i>	USE:	Plastic products	Plastic waste UF: Plastic pollution Plastic pollution waste BT: Plastics Waste materials
Plastic films	BT:	Films	Plastics BT: Chemical products Materials RT: Chemicals
		Plastics	
		Plastic insulation	
		Plastic insulators	
<i>Plastic IC packaging</i>	USE:	Plastic integrated circuit packaging	
packaging			
Plastic insulation	BT:	Insulation	
		Dielectric materials	
		Plastic films	



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	Petrochemicals		<i>Plugboard</i>		
	Phthalates		USE:	Breadboard	
	Plastic packaging		Plugs		
	Plastic products		BT:	Connectors	
	Plastics industry		NT:	Keyways	
	Polymers			Pins	
	Resins		<i>Plumbago</i>		
NT:	Epoxy resins		USE:	Graphite	
	Fiber reinforced plastics		Pluto		
	Microplastics		BT:	Planets	
	Plastic films		Plutonium		
	Plastic optical fiber		BT:	Chemical elements	
	Plastic waste		<i>pMOSFETs</i>		
Plastics industry			USE:	MOSFET	
BT:	Manufacturing industries		<i>PMU</i>		
RT:	Chemical industry		USE:	Phasor measurement units	
	Plastic products		<i>PMUs</i>		
	Plastics		USE:	Phasor measurement units	
Platform as a service			<i>PNA</i> s		
UF:	PAAS		USE:	Presence network agents	
BT:	Cloud computing		Pneumatic actuators		
Platform virtualization			BT:	Actuators	
BT:	Computers and information		Pneumatic systems		
processing			BT:	Control systems	
RT:	Virtual machine monitors		RT:	Bellows	
Plating				Fluidics	
BT:	Materials processing			Mechanical systems	
NT:	Chrome plating		<i>Pneumology</i>		
Platinum			USE:	Pulmonology	
BT:	Metals		Pneumonia		
NT:	Platinum alloys		BT:	Pulmonary diseases	
Platinum alloys			RT:	Asthma	
BT:	Platinum			Chronic obstructive	
RT:	Alloying		pulmonary disease		
Plethysmography				Emphysema	
BT:	Biomedical measurement		<i>Pnictide superconductors</i>		
	Medical diagnosis		USE:	Superconducting materials	
NT:	Photoplethysmography		<i>Pockels readout optical modulator</i>		
<i>PLL</i>			USE:	Electro-optic modulators	
USE:	Phase locked loops				
Plug-in electric vehicles					
BT:	Electric vehicles				
Plug-in hybrid electric vehicles					
BT:	Hybrid electric vehicles				
RT:	Charging stations				



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Podcast

USE: Digital audio broadcasting

Poincare group

USE: Poincare invariance

Poincare invariance

UF: Poincare group

BT: Transforms

Point cloud compression

UF: PCC

BT: Data compression

Visual databases

RT: Three-dimensional displays

Point of care

BT: Documentation

Medical services

RT: Biomedical communication

Clinical diagnosis

Patient monitoring

Smart healthcare

Point-to-multipoint communications

UF: P2MP

PTMP

BT: Wireless communication

RT: Internet

Internet telephony

Poisons

USE: Toxicology

Poisson equation

USE: Poisson equations

Poisson equations

UF: Poisson equation

BT: Partial differential equations

RT: Electrostatics

Polar codes

BT: Block codes

Linear codes

RT: Channel coding

Error correction codes

Reed-Muller codes

Polar cyclones

USE: Cyclones

Polarimetric synthetic aperture radar

BT: Synthetic aperture radar

Polarimetry

UF: Solar polarimetry

BT: Electromagnetic

measurements

RT: Ellipsometry

Optical retarders

Polarisation

USE: Polarization

Polaritons

BT: Energy states

RT: Quantum materials

NT: Surface plasmon polaritons

Polarization

UF: Circular polarisation

Circular polarization

Polarisation

BT: Electromagnetic scattering

Polarization mode dispersion

UF: Polarization-mode

dispersion

BT: Optical fiber polarization

Polarization shift keying

BT: Optical polarization

Polarization-maintaining optical fibers

USE: Optical fiber polarization

Polarization-mode dispersion

USE: Polarization mode

dispersion

Poles & zeros

USE: Poles and zeros

Poles and towers

UF: Pylons

Towers

Wood poles

BT: Transmission lines

RT: Power distribution lines

Power transmission lines

NT: Telephone poles

Poles and zeros

UF: Poles & zeros

Roots

Zeros

BT: Transfer functions

RT: Circuits

Control systems



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	Newton method		Waste management
	Polynomials	NT:	Soil remediation
<i>Police</i>			
USE:	Law enforcement	Pollution measurement	
		BT:	Measurement
		RT:	Environmental monitoring
Polishing machines			Pollution
BT:	Production equipment		Pollution control
RT:	Deburring		
	Rough surfaces		
	Surface finishing		
	Surface roughness		
		Polonium	
		BT:	Chemical elements
Pollination (plants)			
BT:	Flowering plants	Polycaprolactone	
	Seeds (agriculture)	BT:	Polymers
		RT:	Smart materials
Pollution			
BT:	Environmental factors	Polyethylene	
RT:	Contamination	BT:	Polymers
	Design for disassembly	NT:	Thermoplastic polyethylene
	Environmental degradation		
	Environmental economics	Polyimides	
	Green products	BT:	Polymers
	Occupational health		
	Pollution control	<i>Polymer coatings</i>	
	Pollution measurement	USE:	Polymer films
	Sewage treatment		
	Toxicology	Polymer fibers	
	Waste disposal	BT:	Polymers
NT:	Acidification	RT:	Cellulose
	Agricultural pollution		
	Air pollution	Polymer films	
	Emissions trading	UF:	Polymer coatings
	Industrial pollution	BT:	Films
	Land pollution	RT:	Dielectric thin films
	Oil pollution		
	Radioactive pollution	Polymer foams	
	Thermal pollution	BT:	Materials
	Urban pollution	RT:	Insulation
	Water pollution		Insulators
			Metal foam
			Resins
Pollution control		Polymer gels	
BT:	Environmental	BT:	Materials
management		RT:	Intelligent materials
RT:	Carbon emissions		
	Decontamination	<i>Polymer led</i>	
	Electrostatic precipitators	USE:	Organic light emitting
	Environmental monitoring	diodes	
	Greenhouse effect		
	Ozonation	Polymers	
	Pollution	UF:	Electroactive polymers
	Pollution measurement	BT:	Materials
	Sewage treatment	RT:	Colloidal lithography
	Sludge treatment		Plastics



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NT:	Azobenzene Biopolymers Cellulose Elastomers Hydrogels Liquid crystal polymers Optical polymers Polycaprolactone Polyethylene Polyimides Polymer fibers	<i>Portable Multimedia players</i> USE: Portable media players
Polynomials	BT: Equations RT: Poles and zeros	<i>Portable PCs</i> USE: Portable computers
<i>PON</i>	USE: Passive optical networks	<i>Portable video players</i> USE: Portable media players
Population density	BT: Demography Density measurement	Portals BT: Management information systems RT: Information retrieval Web sites
Porcelain	BT: Ceramics RT: Ceramic products Ceramics industry	Portfolios UF: Electronic portfolios BT: Professional communication
Porous silicon	BT: Silicon	Ports (computers) UF: Computer ports BT: Computer interfaces RT: Computer networks Hardware Information exchange
Portable computers	UF: Laptops Portable PCs BT: Microcomputers NT: Personal digital devices	Pose estimation BT: Estimation RT: Computer vision
Portable document format	UF: PDF BT: Document handling RT: Document image processing	Position control UF: Orientation control BT: Mechanical variables control RT: Admittance control Attitude control Capacitive transducers Manipulators Mechanical guides Medical robotics Servosystems NT: Nanopositioning
Portable media players	UF: MP3 Portable Multimedia players Portable video players iPOD BT: Audio systems Digital communication Home automation RT: Digital audio broadcasting Tablet computers	Position measurement UF: Attitude determination Orientation determination Orientation measurement Source location BT: Mechanical variables measurement RT: Direction-of-arrival estimation Distance measurement Gaze tracking



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	Geodesy	<i>Post-wall waveguides</i>	
	Location awareness	USE:	Substrate integrated
	Navigation	waveguides	
	Tracking		
NT:	Odometry	Postal services	
		UF:	Mail
Position sensitive particle detectors		BT:	Message systems
BT:	Ionizing radiation sensors	RT:	Electronic mail
RT:	High energy physics		
instrumentation computing		Posterior probability	
	Nuclear measurements	BT:	Probability
	Particle measurements	RT:	Bayes methods
	Semiconductor counters		
		Posthuman	
Positive train control		UF:	Post human
BT:	Control systems		Post-human
	Rail transportation	BT:	Systems, man, and
RT:	Feedback	cybernetics	
	Railway accidents	RT:	Artificial intelligence
	Railway safety		Transhuman
		<i>Postscript</i>	
Positron emission tomography		USE:	Page description languages
UF:	PET		
BT:	Tomography	Potassium	
RT:	Biomedical applications of	BT:	Chemical elements
radiation			
	Medical diagnostic imaging	<i>Potassium nitrate</i>	
	Nuclear medicine	USE:	Nitrogen compounds
	Phantoms		
	Tumors	Potential energy	
NT:	Whole-body PET	BT:	Energy conservation
		RT:	Kinetic energy
Positrons			Mechanical energy
BT:	Elementary particles	<i>Potential transformers</i>	
		USE:	Voltage transformers
Possibility theory			
BT:	Probability	Potential well	
RT:	Fuzzy logic	UF:	Potential wells
	Nonlinear dynamical		Quantum confinement
systems		BT:	Energy conversion
<i>Post human</i>		<i>Potential wells</i>	
USE:	Posthuman	USE:	Potential well
Post-consumer waste		Potentiometers	
BT:	Recycling	BT:	Meters
	Waste materials	RT:	Resistors
			Voltage measurement
<i>Post-filtering algorithm</i>		<i>POTS</i>	
USE:	Filtering algorithms	USE:	Landline
<i>Post-human</i>			
USE:	Posthuman		



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Powders

BT: Coatings
RT: Ceramics
Spark Plasma sintering

Power amplifiers

UF: Radio frequency power
amplifiers
Radiofrequency power
amplifiers
BT: Amplifiers
NT: High power amplifiers
Predistortion

Power and energy standards

UF: IEEE Arc Flash Standards
IEEE Electric Machinery
Standards
IEEE Power Substations
Standards
IEEE Surge Protective
Devices Standards
BT: Standards categories

Power cable insulation

BT: Cable insulation
RT: Power cables

Power cables

BT: Cables
Power transmission lines
RT: Conductors
Power cable insulation
Power distribution lines
NT: Underground power cables

Power capacitors

BT: Capacitors
NT: Supercapacitors

Power combiners

BT: Waveguide components
RT: Microstrip components
Power dividers
Stripline components

Power conditioning

BT: Power electronics
RT: Power conversion
Pulse width modulation

converters

NT: Power smoothing

Power consumption

USE: Power demand

Power control

BT: Electric variables control
RT: Electric current control
Power factor correction
Pulse width modulation

converters

Power conversion

BT: Converters
RT: Choppers (circuits)
Maximum power point

trackers

Nonlinear circuits
Power conditioning
Power electronics
Power semiconductor

devices

Power supplies
Pulse width modulation

converters

Regulators
Switched systems
AC-AC converters
AC-DC power converters
DC-AC power converters
DC-DC power converters
Grid forming
Inverter-based resource
Matrix converters
Power conversion

harmonics

Voltage-source converters

Power conversion harmonics

BT: Power conversion
RT: Harmonic distortion

Power demand

UF: Power consumption
BT: Power supplies
Power system planning
RT: Electricity supply industry
Energy conservation
Energy resources
Load management
Load modeling
Power distribution
NT: Demand response
Load forecasting

Power dissipation

BT: Circuits
RT: CMOS logic circuits
MOSFET circuits
Nanotechnology



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	Power transmission		Power dividers	
	System-on-chip		BT:	Waveguide components
			RT:	Microstrip components
Power distribution				Power combiners
UF:	Distribution of electric			Stripline components
power			Power electronics	
BT:	Power systems		UF:	Electric power
RT:	Electricity supply industry		RT:	High-voltage techniques
	Energy Internet			Matrix converters
	Industrial power systems			Power conversion
	Load shedding			Power filters
	Power demand			Pulse width modulation
	Transactive energy			
NT:	DC distribution systems	converters		Rectifiers
	Electrical power distribution			Resonant inverters
	Power distribution control			Switching converters
	Power distribution faults			Voltage-source converters
	Power distribution lines		NT:	Converters
	Power distribution networks			Current limiters
	Power distribution planning			Gate drivers
	Power distribution reliability			Inverters
	Simultaneous wireless			Phase control
information and power transfer				Power conditioning
				Power semiconductor
Power distribution control				
BT:	Power distribution	devices		Power semiconductor
RT:	Voltage control	switches		
Power distribution faults				Snubbers
BT:	Power distribution			Three-phase electric power
Power distribution lines			Power engineering	
UF:	Overhead distribution lines		BT:	Power engineering and
BT:	Power distribution	energy		
RT:	Conductors	education	RT:	Power engineering
	Poles and towers		NT:	Electrification
	Power cables			Ferroresonance
				High-voltage techniques
Power distribution networks				Power engineering
BT:	Power distribution	computing		Power system simulation
RT:	Microgrids		Power engineering and energy	
	Power grids		RT:	Electrochemical devices
	Smart grids		NT:	Electric variables control
NT:	Active distribution networks			Energy
Power distribution planning				Power engineering
BT:	Power distribution			Power generation
	Power system planning			Power systems
Power distribution reliability		Power engineering computing		
BT:	Power distribution	BT:		Computer applications
	Power system reliability			Power engineering
<i>Power distribution transformers</i>		RT:		Energy Internet
USE:	Power transformers			



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computing	Power system analysis		Pulsed power systems
	Virtual power plants		Space power stations
		NT:	Automatic generation
Power engineering education		control	Cogeneration
BT:	Engineering education		Distributed power
RT:	Power engineering	generation	Geothermal power
<i>Power exchange</i>		generation	Hydroelectric power
USE:	Power markets	generation	Magnetohydrodynamic
<i>Power factor</i>		power generation	Nuclear power generation
USE:	Reactive power		Power generation control
Power factor correction			Power generation dispatch
BT:	Electric current control		Power generation planning
	Load flow control		Power generation reliability
RT:	Power control		Solar power generation
	Power transmission		Trigeneration
	Voltage control		Turbomachinery
Power filters			Wind energy generation
UF:	Power line filters		Wind power generation
BT:	Filters	Power generation control	
RT:	Power electronics	BT:	Automatic control
NT:	Spurline		Power generation
<i>Power flow</i>		Power generation dispatch	
USE:	Load flow	BT:	Power generation
<i>Power flow analysis</i>		Power generation economics	
USE:	Load flow analysis	BT:	Economics
<i>Power flow control</i>		RT:	Power generation
USE:	Load flow control	NT:	Electricity supply industry
Power generation		deregulation	
UF:	Generation of electric	Power generation planning	
power	Output power	BT:	Power generation
	Power plants	Power generation reliability	
	Power stations	BT:	Power generation
BT:	Power engineering and	RT:	Capacity factor
energy	Batteries	Power grids	
	Electrification	UF:	Electricity grids
	Energy Internet	BT:	Power systems
	Feed in tariff	RT:	Energy Internet
	Fuel cells		Power distribution networks
	Generators		Wind energy integration
	Levelized cost of energy	NT:	Microgrids
	Microgrids		Smart grids
	Power generation		
economics	Power supplies		



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Power harmonic filters

BT: Power system harmonics

Power transmission

Transactive energy

Power harvesting

USE: Energy harvesting

Power measurement

BT: Electric variables
measurement

RT: Wattmeters

NT: Dynamometers

Power industry

UF: Electric utilities

BT: Industries

RT: Offshore installations

Power system faults

Telecontrol equipment

NT: Electrical equipment

industry

Electricity supply industry

Nuclear facility regulation

Power system

interconnection

Power MOSFET

BT: MOSFET circuits

RT: Power semiconductor

devices

Power outages

USE: Power system reliability

Power overhead lines

BT: Power transmission lines

RT: Railway electrification

Power injection molding

USE: Injection molding

Power plants

USE: Power generation

Power injection moulding

USE: Injection molding

Power pools

USE: Power markets

Power integrated circuits

BT: Circuits

Integrated circuits

RT: Power semiconductor

devices

Power quality

UF: Power supply quality
Voltage sags

BT: Power supplies

RT: Electricity supply industry

Power system harmonics

Power system transients

Power lasers

BT: Lasers

RT: Power semiconductor

devices

Power semiconductor devices

BT: Power electronics

Semiconductor devices

RT: Power MOSFET

Power conversion

Power integrated circuits

Power lasers

NT: Power transistors

Power line communications

BT: Transmission lines

Power line filters

USE: Power filters

Power management

USE: Power system management

Power semiconductor switches

BT: Power electronics

Semiconductor devices

NT: Bipolar transistors

Thyristors

Power markets

UF: Electricity markets

Electricity trading

Power exchange

Power pools

Power trading

Power wheeling

BT: Electricity supply industry

deregulation

RT: Emissions trading

Power smoothing

BT: Power conditioning

Power spectra

USE: Spectral analysis



January 2026 IEEE Thesaurus

Power stations

USE: Power generation

Power stations (space)

USE: Space power stations

Power stations (substations)

USE: Substations

Power steering

BT: Automotive engineering

Power supplies

BT: Power systems
RT: Power conversion
Power generation
Pulsed power systems
Uninterruptible power

systems

NT: Battery chargers
Charging stations
Current supplies
Emergency power supplies
Inductive charging
Islanding
Power demand
Power quality
Power system restoration
Switched mode power

supplies

Traction power supplies
Umbilical cable

Power supplies to apparatus

USE: Umbilical cable

Power supply industry

USE: Electricity supply industry

Power supply quality

USE: Power quality

Power system analysis computing

BT: Computer applications
Power systems
RT: Digital simulation
Modeling
Parameter estimation
Power engineering

computing

Software packages

Power system control

BT: Electric variables control
RT: Control system security

Energy Internet

Load monitoring

Power systems

Virtual power plants

NT: Bidirectional power flow

Load flow control

SCADA systems

Power system dynamics

BT: Power systems

Power system economics

BT: Power systems
RT: Electricity supply industry

deregulation

NT: Transactive energy
Low carbon economy

Power system faults

BT: Power systems
RT: Electricity supply industry
Error correction
Fuzzy set theory
Power industry
Power system protection
Signal analysis

Power system harmonics

BT: Power systems
RT: Power quality
NT: Power harmonic filters

Power system interconnection

BT: Power industry
RT: Energy Internet
Power systems
Power transmission

Power system management

UF: Power management
Telecommunication power

management

BT: Power systems
RT: Energy Internet
Microgrids
Monitoring
Preventive maintenance
Virtual power plants
NT: Load flow

Power system measurements

BT: Power systems
RT: Load monitoring
NT: Capacity factor
Levelized cost of energy



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	Meter reading	RT:	Power systems
Power system modeling		Power system stability	
BT:	Modeling	BT:	Power systems
RT:	Power systems	RT:	Power system reliability
NT:	Load modeling		
Power system planning		Power system transients	
BT:	Power systems	BT:	Electromagnetic transients
RT:	Demand side management	RT:	Arresters
	Electricity supply industry		Power quality
NT:	Power demand	NT:	Power system protection
	Power distribution planning		Transient analysis
Power system protection		Power systems	
BT:	Power systems	UF:	Electric power
	Product safety engineering	BT:	Power engineering and
RT:	Arresters	energy	RT:
	Circuit breakers		Civil engineering
	Fuses		Mechanical power
	Grounding		
	Power system faults		Power system control
	Power system transients		Power system
	Protective relaying		
NT:	Electrical safety		Power system modeling
	Substation protection		Power system relaying
	Surge protection		Power system restoration
			Power system security
			Power system simulation
Power system relaying			Skin effect
BT:	Relays		Telecontrol equipment
RT:	Power systems		Time-frequency analysis
	Protective relaying		Voltage fluctuations
		NT:	Data center power
Power system reliability			Energy Internet
UF:	Power outages		Hybrid power systems
BT:	Power systems		Industrial power systems
RT:	Microgrids		PSCAD
	Power system stability		Power distribution
	Reliability		Power grids
NT:	Power distribution reliability		Power supplies
			Power system analysis
Power system restoration		computing	
BT:	Power supplies		Power system dynamics
RT:	Electricity supply industry		Power system economics
	Power systems		Power system faults
			Power system harmonics
Power system security			Power system management
BT:	Security		Power system
RT:	Hardware security	measurements	
	Load flow analysis		Power system planning
	Power systems		Power system protection
	Reactive power control		Power system reliability
			Power system stability
Power system simulation			Power transmission
BT:	Power engineering		Pulsed power systems



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	Reactive power	Static VAr compensators
	Substations	Transmission lines
	Transformers	Wireless power
	Uninterruptible power	transmission
systems	Wind energy integration	
		Power transmission lines
		UF: Overhead transmission
<i>Power Systems Computer Aided Design</i>		lines
USE: PSCAD		BT: Transmission lines
		RT: Conductors
<i>Power trading</i>		DC distribution systems
USE: Power markets		Poles and towers
		Power transmission
		Superconducting
Power transformer insulation		
BT: Power transformers		transmission lines
RT: Insulation		NT: Gas insulated transmission
		lines
		Power cables
		Power overhead lines
Power transformers		
UF: Power distribution		
transformers		
BT: Transformers		<i>Power wheeling</i>
RT: Transformer cores		USE: Power markets
Windings		
NT: On load tap changers		<i>Powered armor</i>
Power transformer		USE: Wearable robots
insulation		
		<i>Powered exoskeleton</i>
Power transistors		USE: Wearable robots
BT: Power semiconductor		
devices		<i>Powertrain</i>
RT: Driver circuits		USE: Mechanical power
		transmission
Power transmission		
UF: Transmission of electric		<i>Praeseodymium</i>
power		USE: Praseodymium
BT: Power systems		
RT: Electric current control		Pragmatics
Energy Internet		BT: Linguistics
Load flow analysis		Semiotics
Magnetic gears		RT: Communication symbols
Power dissipation		Context
Power factor correction		Natural language
Power markets		
Power system		processing
interconnection		communication
	Power transmission lines	
NT: AC power transmission		<i>Prairie</i>
Common Information Model		USE: Grasslands
(electricity)		
	DC power transmission	
	Flexible AC transmission	Praseodymium
systems		UF: Praeseodymium
	HVDC transmission	BT: Chemical elements
	Inductive power	
transmission		



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Pre-college engineering

UF: Elementary school
engineering
High school engineering
Precollege engineering
BT: Educational programs

Preamplifiers

BT: Amplifiers

Precipitation

BT: Meteorology
Water cycle
NT: Rain
Snow

Precision agriculture

BT: Agriculture
Management
RT: Farming
Internet of Things
Smart agriculture
Sustainable development

Precision engineering

BT: Engineering - general
RT: Industrial engineering
Mechanical engineering

Precision medicine

UF: Personalized medicine
BT: Medical treatment

Precoding

BT: Encoding

Precollege engineering

USE: Pre-college engineering

Predator prey systems

UF: Predator-prey models
Predator-prey systems
BT: Biology
Mathematics
RT: Chaos
Differential equations
Game theory
Nonlinear dynamical
systems
Stability

Predator-prey models

USE: Predator prey systems

Predator-prey systems

USE: Predator prey systems

Prediction algorithms

BT: Algorithms

Prediction methods

BT: Artificial intelligence
RT: Estimation
Forecasting
Gaussian processes
Hindcasting
Kalman filters
Prediction theory
Signal processing
Spectral analysis
Speech processing
NT: Linear predictive coding
Predictive coding
Predictive encoding
Predictive models

Prediction theory

BT: Statistics
RT: Artificial intelligence
Estimation
Prediction methods

Predictive analysis

USE: Predictive analytics

Predictive analytics

UF: Predictive analysis
BT: Statistical analysis
RT: Data mining
Machine learning
Predictive models
NT: Concept drift

Predictive coding

BT: Prediction methods

Predictive control

UF: Model predictive control
Model-predictive control
BT: Process control
RT: Control engineering

Predictive encoding

BT: Prediction methods

Predictive maintenance

BT: Maintenance engineering



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Predictive models

BT: Prediction methods
RT: Ensemble learning
Fish schools
Predictive analytics
NT: Temporal difference

learning

Predistortion

UF: Inverse distortion
BT: Power amplifiers
RT: Nonlinear distortion

Prefabricated buildings

USE: Prefabricated construction

Prefabricated construction

UF: Prefabricated buildings
BT: Construction
Construction industry
RT: Building materials
Buildings
Modular construction

Prefetching

BT: Instruction sets

Preforms

BT: Assembly

Pregnancy

UF: Pregnant
BT: Medical conditions
RT: Gestational diabetes
Infertility
Obstetrics
Placenta

Pregnancy test

BT: Medical tests

Pregnant

USE: Pregnancy

Presence network agents

UF: PNAs
BT: Communications

technology

Presses

BT: Machine tools
RT: Dies
Pressing

Pressing

BT: Materials processing
RT: Presses

Pressure control

BT: Control systems

Pressure effects

BT: Mechanical factors
RT: Meteorology
Piezooptic effects
Piezoresistance

Pressure gauges

BT: Instruments
RT: Atmospheric

measurements

Density measurement
Fluid flow measurement
Force measurement
Geophysical measurements
Pressure measurement
Pressure sensors
Torque measurement

Pressure measurement

BT: Measurement
RT: Piezoresistive devices
Pressure gauges
Pressure sensors
Tactile sensors
NT: Altimetry
Tire pressure

Pressure sensors

BT: Sensors
RT: Pressure gauges
Pressure measurement
Wearable sensors

Pressure vessels

BT: Mechanical products
RT: Concrete
Fission reactors
Mechanical engineering
Steel

Prevention and mitigation

BT: Disaster management
Risk management
RT: Risk minimization
Risk mitigation



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Preventive maintenance

BT: Maintenance engineering
RT: Accident prevention
Power system management
Reliability
Safety
NT: Condition monitoring

Pricing

BT: Financial management
RT: Emissions trading
Feed in tariff

Primary motor cortex

BT: Brain

Principal component analysis

UF: PCA
BT: Statistical analysis
RT: Feature extraction
Independent component

analysis

Linear systems
Operations research
Pattern recognition
Transform coding

Print readers

USE: Character recognition

Printed circuit boards

USE: Printed circuits

Printed circuits

UF: Circuit boards
Printed circuit boards
BT: Circuits
RT: Electronics packaging
Integrated circuit layout
Substrates
Wiring
NT: Flexible printed circuits
Memory modules
Surface mount technology

Printers

BT: Computer peripherals
RT: Printing
NT: Laser printers

Printing

BT: Information technology
RT: Character generation
Color gamut
Ink

Lithography
Printers
Printing machinery
Publishing
Typesetting
NT: Digital printing
Ink jet printing
Teleprinting
Three-dimensional printing

Printing machinery

UF: Printing presses
BT: Machinery
RT: Printing

Printing presses

USE: Printing machinery

Privacy

BT: Technology social factors
RT: Authorization
Communication system

security

Computer security
Cryptography
Cyberbullying
Cyberethics
Data privacy
Data protection
Data security
Decentralized identity
Differential privacy
Digital policy
General Data Protection

Regulation

Homomorphic encryption
Information diffusion
Information leakage
Malware
Network intrusion
Session hijacking
Trust management
NT: Eavesdropping
Internet privacy
Privacy breach
Privacy enhancing

technologies

Privacy breach

BT: Information security
Privacy
RT: Data breach
Data security



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Privacy enhancing technologies

UF: PETs
BT: Privacy
RT: Computer security
Data collection
Data privacy
Encryption

Privacy preserving data mining

USE: Data privacy

Privacy-invasive software

UF: Invasive software
BT: Software
RT: Computer crime
Computer security
Unsolicited e-mail
NT: Spyware

Private blockchains

USE: Blockchains

Privatisation

USE: Privatization

Privatization

UF: Privatisation
BT: Macroeconomics
RT: Industrial economics

Probabilistic computing

BT: Computers and information processing

Probabilistic logic

BT: Logic
RT: Mixture models

Probability

BT: Mathematics
RT: Belief propagation
Decision theory
Distribution functions
Fourier transforms
Gamma distribution
Information geometry
Maximum likelihood

detection

Mean field theory
Monte Carlo methods
Naive Bayes methods
Random processes
Statistical analysis
Statistics
Stochastic processes

Stochastic systems
Viterbi algorithm
Weibull distribution
NT: Ant colony optimization
Bayes methods
Error probability
Forecasting
Hindcasting
Memoryless systems
Multi-armed bandit problem
Pairwise error probability
Possibility theory
Posterior probability
Probability distribution
Random variables
Statistical distributions
Total variance
Uncertainty

Probability computing

BT: Computers and information processing

Probability density function

BT: Integral equations
RT: Distribution functions

Probability distribution

BT: Probability
NT: Exponential distribution
Heavily-tailed distribution
Kurtosis
Lightly-tailed distribution
Log-normal distribution
Maxwell-Boltzmann
distribution
Nakagami distribution

Probes

BT: Instruments

Problem-solving

BT: Cognitive science
RT: Human factors

Procedural content generation

USE: Procedural generation

Procedural generation

UF: Procedural content
generation
BT: Algorithms
Games



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Process control

BT: Industrial control
 RT: Bleaching
 Chemical reactions
 Continuous production
 Manufacturing automation
 Process design
 Process modeling
 Process planning
 Production control
 Soft sensors
 NT: Predictive control
 Three-term control
 Two-term control

Process design

BT: Design methodology
 RT: Chemical engineering
 Design for disassembly
 Design for quality
 Process control
 Process mining
 Process planning
 Product design
 Service computing
 NT: Pattern formation
 Process modeling

Process mining

BT: Data mining
 RT: Process design
 Process planning

Process modeling

BT: Modeling
 Process design
 RT: Process control

Process monitoring

BT: Monitoring
 RT: Soft sensors

Process planning

BT: Management
 Production
 Production management
 Production planning
 RT: Process control
 Process design
 Process mining
 NT: Business process
 integration Business process
 management Cause effect analysis

Root cause analysis

Process virtual machines

BT: Virtual machines

Processor scheduling

UF: Multiprocessor scheduling
 BT: Concurrency control
 Multiprocessing systems
 RT: Microprocessors
 Optimization methods
 NT: Scheduling algorithms

Processors (program)

USE: Program processors

Procurement

BT: Supply chain management
 RT: Contracts
 Logistics
 Proposals
 Supply chains

Product codes

BT: Codes
 RT: Decoding
 Error correction
 QR codes
 Radiofrequency

identification

NT: Bar codes

Product customisation

USE: Product customization

Product customization

UF: Product customisation
 BT: Product development
 RT: Customer satisfaction
 Manufactured products
 Product design

Product delivery

UF: Delivery of goods
 Package delivery
 BT: Commerce and trade
 RT: Electronic commerce

Product design

BT: Design methodology
 RT: Concurrent engineering
 Design for disassembly
 Design for quality
 Design tools
 Digital twins



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Group technology		Product safety engineering	
Manufactured products		RT:	Software safety
Process design		NT:	Consumer protection
Product customization			Power system protection
Product development			Safety
Prototypes			Vehicle crash testing
Requirements engineering		<i>Product warranties</i>	
Product development		USE:	Warranties
BT:	Crowdsourcing	<i>Product warranty</i>	
	Engineering management	USE:	Warranties
RT:	Brand management	Production	
	Manufactured products	BT:	Industry applications
	Product design	RT:	Containers
	Quality function deployment		Wheels
	Rapid prototyping		Wire drawing
	Reverse engineering	NT:	Ball milling
	Social manufacturing		Compression molding
NT:	Virtual prototyping		Embossing
	Graphical user interfaces		Food products
	Product customization		Group technology
	Product lifecycle		Injection molding
management	Software product lines		Materials processing
	Time to market		Mechanical products
Product liability			Process planning
BT:	Legal factors		Production control
RT:	Consumer products		Production engineering
	Product safety		Production equipment
	Quality assurance		Production facilities
	Quality management		Production management
NT:	Warranties		Production materials
<i>Product life cycle management</i>			Production systems
USE:	Product lifecycle		Productivity
management			Shafts
			Springs
			Transfer molding
Product lifecycle management		Production control	
UF:	Product life cycle	BT:	Industrial control
management			Production
	System lifecycle	RT:	Adaptive scheduling
management			Cellular manufacturing
BT:	Product development		Group technology
RT:	Release engineering		Inventory control
NT:	Life cycle assessment		Manufacturing
	Prognostics and health		Process control
management			Production systems
Product safety			Supply chain management
BT:	Safety	NT:	Continuous production
RT:	Accidents		Lot sizing
	Consumer products		Optimized production
	Product liability	technology	Scheduling



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Production economics

USE: Industrial economics

RT: Continuous improvement
Continuous production
Industrial engineering
Inventory control
Lean production
Mass production
Production engineering
Productivity
Research and development

Production engineering

BT: Engineering - general
Production
RT: Industrial engineering
Inventory management
Manufacturing
Manufacturing systems
Production equipment
Production management
Production materials
NT: Production planning

management

Social manufacturing
Technology management
Waste management
NT: Control charts
Inventory management
Lead time reduction
Logistics
Process planning
Production footprint
Production planning

Production equipment

BT: Production
RT: Gears
Machinery
Materials handling

equipment

NT: Production engineering
Applicators
Clamps
Cutting tools
Fixtures
Machine tools
Mining equipment
Molding equipment
Packaging machines
Paper making machines
Polishing machines
Soldering equipment

Production materials

BT: Materials
Production
RT: Additives
Cast iron
Chemical products
Hydraulic fluids
Production engineering
NT: Abrasives
Aerospace materials
Automotive materials
Inhibitors
Ink
Joining materials
Lubricants
Retardants

Production facilities

UF: Factories
Manufacturing facilities
BT: Production
RT: Manufacturing
Warehousing
NT: Foundries
Greenhouses
Industrial facilities
Industrial plants
Machine shops
Paper mills

Production planning

BT: Production engineering
Production management
RT: Demand forecasting
Lead time reduction
Optimized production

technology

NT: Capacity planning
Materials requirements

planning

Process planning

Production footprint

BT: Production management

Production management

UF: Manufacturing
management
BT: Management
Production

Production systems

BT: Production
RT: Discrete-event systems
Industrial plants
Manufacturing



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technology	Optimized production	Manuals
	Production control	Meetings
NT:	Assembly systems	Oral communication
	Exhaust systems	Plagiarism
systems	Intelligent manufacturing	Portfolios
	Lean production	Professional societies
	Manufacturing systems	Public speaking
	Steering systems	Rhetoric
		Writing
Productivity		Professional societies
UF:	Labor productivity	BT: Professional
	Labour productivity	communication
BT:	Production	RT: Non-governmental
RT:	Business	organizations
	Human factors	<i>Profit sharing schemes</i>
	Incentive schemes	USE: Incentive schemes
	Industrial psychology	
	Management	Profitability
	Manufacturing	BT: Economics
	Personnel	RT: Cost accounting
	Production management	Econometrics
	Profitability	Financial management
		Productivity
Professional aspects		Progenitor cells
BT:	Engineering profession	BT: Cells (biology)
RT:	Subject matter experts	RT: Stem cells
Professional communication		Prognostics and health management
UF:	Technical communication	UF: PHM
RT:	Cooperative communication	BT: Product lifecycle
	Federated learning	management
	Pragmatics	<i>Program generators</i>
	Semantics	USE: Automatic programming
	Semiotics	
	Syntactics	Program management
NT:	Collaboration	UF: Programme management
	Communication aids	BT: Management
effectiveness	Communication	RT: Project management
	Communication symbols	Technical management
	Context	
	Databases	Program processors
	Global communication	UF: Assemblers (program)
	Grammar	Compilers (program)
	Inclusive language	Interpreters (program)
	Information analysis	Multi-threaded systems
	Information integrity	Multi-threading systems
	Information resources	Multithreaded systems
	Information retrieval	Multithreading systems
	Information science	Processors (program)
	Information services	BT: System software
	Information systems	RT: Input-output programs
	Information technology	



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processors
NT: Manycore processors
Operating systems
Application specific
Graphics processing units
Instruction sets
Optimizing compilers

Program profiling

USE: Programming

Programmable circuits

arrays
BT: Circuits
NT: Field programmable analog
Programmable logic arrays
Programmable logic
devices

Programmable control

BT: Digital control
RT: Industrial control
Manufacturing automation
NT: Flow graphs

Programmable logic arrays

UF: PLA
BT: Circuits
Logic arrays
Logic circuits
Programmable circuits
RT: Programmable logic
devices

Programmable logic controllers

USE: Programmable logic
devices

Programmable logic devices

UF: Programmable logic
controllers
BT: Circuits
Logic devices
Programmable circuits
RT: High level synthesis
Programmable logic arrays

Programmable read only memory

USE: PROM

Programme management

USE: Program management

Programmed instruction

USE: Educational technology

Programming

UF: Computer programming
Program profiling
BT: Computer science
RT: Aerospace and electronic
systems
Digital computers
Flowcharts
Null value
Programming environments
Runtime
Self-assembly
Software
Software debugging
Software tools
Structured Query Language
Syntactics
NT: Answer set programming
Augmented reality
Automatic programming
Concatenated codes
Functional programming
Granular computing
Integer linear programming
Logic programming
Microprogramming
Object oriented methods
Object oriented
programming
Opportunistic software
systems development
Parallel programming
Performance analysis
Programming profession
Robot programming

Programming environments

BT: Software engineering
RT: Computer aided software
engineering
Programming
Software debugging
Software tools

Programming languages

USE: Computer languages

Programming profession

UF: Computer programming
profession
BT: Programming
RT: Employment
Engineering profession



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Project engineering

BT: Engineering management
NT: Scheduling
Turnkey project

Project management

BT: Management
RT: Building information
management
Concurrent engineering
Crowdfunding
Lead time reduction
Program management
Requirements engineering
Requirements management
Research and development
management
Scrum (Software
development)
System integration
Technology management
NT: Agile project management
Proposals
Turnkey project

Projectiles

BT: Weapons

Projection algorithms

BT: Algorithms

Projective geometry

BT: Geometry

Projective shadowing

USE: Shadow mapping

projectors (optical)

USE: Optical projectors

PROM

UF: Programmable read only
memory
BT: Read only memory
NT: EPROM

Promethium

BT: Chemical elements

Promotion - marketing

UF: Sales promotion
BT: Marketing management
RT: Public relations

Prompt engineering

BT: Generative AI
RT: Chatbots
Large language models

Proof of stake

UF: Proof-of-stake
BT: Blockchains
Protocols
RT: Cryptocurrency

Proof of Work

UF: proof-of-work
BT: Computer security
Protocols
RT: Blockchains
Denial-of-service attack

Proof-of-stake

USE: Proof of stake

proof-of-work

USE: Proof of Work

Propagation

UF: Wave equations
Wave propagation
BT: Waves
RT: Damping
Electromagnetic

waveguides

NT: Attenuation
Electromagnetic

propagation

Insertion loss
Nonlinear wave

propagation

Perfectly matched layers
Reflection
Scattering
Transient response

Propagation constant

BT: Electromagnetic
propagation

Propagation delay

BT: Delay effects

Propagation loss

USE: Propagation losses



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Propagation losses

UF: Propagation loss
BT: Electromagnetic
propagation

Propellants

BT: Chemical products

Propellers

BT: Aircraft propulsion
RT: Aircraft
Blades
Engines
Impellers
Marine vehicles
Shafts

Propioception

UF: proprioceptive
BT: Robot sensing systems
RT: Robot kinematics

Proportional + derivative control

USE: PD control

Proportional + integral control

USE: PI control

Proportional control

BT: Control systems

Proportional derivative control

USE: PD control

Proportional plus derivative control

USE: PD control

Proportional-derivative control

USE: PD control

Proportional-integral control

USE: PI control

Proportional-integral controller

USE: PI control

Proportional-integral-derivative

USE: PD control

Proportional-integral-derivative control

USE: PI control

Proportional-integral-derivative controller

USE: PI control

Proposals

UF: Technical proposals
BT: Project management
RT: Contracts
Procurement
Technical requirements
Writing

proprioceptive

USE: Proprioception

Propulsion

BT: Vehicular and wireless
technologies
RT: Engines
Plasma jets
Traction motors
Vehicle-to-grid
NT: Aerospace propulsion
Aircraft propulsion
Electromagnetic launching
Electrothermal launching
Rockets

Prosencephalon

USE: Forebrain

Prostate cancer

BT: Cancer

Prosthesis

USE: Prosthetics

Prosthetic hand

BT: Prosthetics

Prosthetic limbs

BT: Prosthetics

Prosthetics

UF: Hip joint replacements
Knee joint replacements
Neural prostheses
Prosthesis
BT: Medical services
RT: Assistive technologies
Bioceramics
Biological control systems
Biomedical equipment
Medical control systems
Orthotics
Sensory aids
Supernumery robotic limbs
Wearable robots
NT: Artificial biological organs



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	Artificial limbs		Power system relaying
	Knee replacement		
	Myoelectric control	<i>Protective relays</i>	
	Neuroprostheses	USE:	Protective relaying
	Prosthetic hand		
	Prosthetic limbs	Protein engineering	
	Visual prosthesis	BT:	Biomedical engineering
Protactinium		Protein sequence	
BT:	Chemical elements	BT:	Proteins
Protected areas		Proteins	
UF:	Conservation areas	BT:	Biochemistry
BT:	Land use planning	RT:	Monoclonal antibodies
			Ribosomes
Protection		NT:	Enzymes
BT:	Safety		Protein sequence
RT:	Circuit breakers		
	Fuses	Proteomics	
	Galvanizing	BT:	Molecular biomarkers
	Grounding		
	Hazardous areas	Protocols	
	Occupational safety	UF:	Communication protocols
	Security	BT:	Communication systems
	Uninterruptible power	RT:	Ad hoc networks
systems			Bluetooth
NT:	Electrostatic discharge		Concurrency control
protection			Frame relay
	Explosion protection		IEEE 802.11 Standard
	Lightning protection		IEEE 802.11e Standard
	Radiation protection		IEEE 802.11g Standard
			IEEE 802.11n Standard
Protection switching			IPTV
UF:	Automatic protection		Internet of Things
switching			Local area networks
BT:	Optical fiber networks		Metropolitan area networks
			Multicarrier code division
Protective clothing		multiple access	
BT:	Clothing		Software defined
	Safety devices	networking	
RT:	Clothing industry		TCPIP
	Eye protection		Wide area networks
	Occupational health	NT:	Access protocols
	Occupational safety		Asynchronous transfer
	Personal protective	mode	
equipment			Border Gateway Protocol
NT:	Safety		Consensus protocol
	Face masks		Cryptographic protocols
Protective relaying			HTTP
UF:	Distance relays		InterPlanetary File System
	Protective relays		LoRaWAN
BT:	Relays		Main-secondary
RT:	Instrument transformers		Multicast protocols
	Power system protection	switching	Multiprotocol label



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protocol		Proof of Work		Ions	
		Proof of stake		Proton accelerators	
		Routing protocols		Proton effects	
		Smart contracts		Proton radiation effects	
		Spanning tree protocol			
		Transport protocols	Prototypes		
		Wireless application		BT:	Design methodology
				RT:	Laser sintering
					Product design
					Stereolithography
Proton accelerators				Virtual prototyping	
	BT:	Particle accelerators	NT:	Breadboard	
	RT:	Ion accelerators		Rapid prototyping	
		Protons			
	Proton beam effects				
USE:	Proton effects				
Proton beams					
USE:	Particle beams				
Proton effects			Proximity effects		
	UF:	Biological effects of protons		UF:	Current crowding
		Proton beam effects		BT:	Electromagnetics
	BT:	Electrothermal effects		RT:	Conductors
		Quantum mechanics			Lithography
exchange interactions	RT:	Elementary particle			
		Elementary particles			
		High energy physics			
	instrumentation computing				
		Protons			
		Radiation effects			
		Space vehicles			
		Thermal factors			
	NT:	Proton radiation effects			
		Single event latchup			
Proton radiation effects					
	BT:	Proton effects			
	RT:	Bipolar transistors			
		Ion radiation effects			
		Protons			
		Radiation effects			
		Semiconductor devices			
		Silicon-on-insulator			
	Proton therapy				
		BT:	Medical treatment		
RT:		Biological effects of			
radiation					
Protons					
	BT:	Elementary particles			
	RT:	Cosmic rays			



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Psoriasis

BT: Medical conditions
RT: Dermatology
Skin

pSPICE

USE: SPICE

PSTN

USE: Communication networks

Psychiatry

BT: Behavioral sciences
Medical specialties
RT: Depression
Medical treatment
NT: Mental disorders

Psychoacoustic models

BT: Auditory system
NT: Masking threshold

Psychoacoustics

BT: Acoustics
RT: Auditory system

Psychology

BT: Behavioral sciences
Social sciences
RT: Affective computing
Behavioral economics
Cognition
Cognitive science
Digital intelligence
Emotion recognition
Employee welfare
People with disabilities
Persuasive systems
Social engineering

(security)

NT: Active perception
Crowd dynamics
Emotion regulation
Emotional responses
Industrial psychology
Mental health
Mood
Neuropsychology
Psychometric testing
Psychophysics

Psychometric testing

BT: Psychology
RT: Industrial psychology

Psychophysics

BT: Psychology

PTMP

USE: Point-to-multipoint
communications

Public blockchains

USE: Blockchains

Public domain software

BT: Open Access
Software
RT: Copyright protection
Open data
Open source software
NT: Python
R language

Public finance

UF: Government borrowing
Government expenditure
BT: Governmental factors
RT: Financial management
Government
Macroeconomics

Public health

USE: Public healthcare

Public healthcare

UF: Public health
BT: Medical services
RT: Epidemiology
NT: Contact tracing

Public infrastructure

BT: Asset management
RT: Electricity supply industry
Environmental

management

Government policies
Public policy
Rural areas
Urban areas
Urban planning
NT: Critical infrastructure

Public key

UF: Digital certificates
BT: Cryptography
NT: Public key cryptography



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Public key cryptography

UF: Public key cryptosystems
BT: Public key
RT: Homomorphic encryption
NT: Elliptic curve cryptography
Identity-based encryption

Public key cryptosystems

USE: Public key cryptography

Public policy

BT: Government policies
RT: Public infrastructure

Public relations

BT: Management
RT: Customer relationship
management
Marketing management
Promotion - marketing

Public safety

USE: Public security

Public security

UF: Public safety
BT: Safety
RT: Security
Terrorism
Threat assessment

Public speaking

UF: Speechmaking
BT: Oral communication
Professional
communication
RT: Meetings

Public switched telephone network

USE: Communication networks

Public transportation

UF: Subways
Taxi
Trolley cars
BT: Transportation
RT: Light rail systems
Rail transportation
Urban areas

Publish subscribe systems

UF: Publish-subscribe systems
Publish/subscribe systems
BT: Message systems
RT: Content management

Middleware
Pattern recognition
Queueing analysis

Publish-subscribe

BT: Distributed information
systems

Publish-subscribe systems

USE: Publish subscribe systems

Publish/subscribe systems

USE: Publish subscribe systems

Publishing

BT: Computer applications
RT: Bibliographies
Copyright protection
Digital printing
Document handling
Guidelines
Plagiarism
Printing
Systematic literature review
Text processing
NT: Bibliometrics
Desktop publishing
Electronic publishing
Journalism
Open Access
Scientific publishing

PUF

USE: Physical unclonable
function

Pulleys

BT: Materials handling
equipment
RT: Freight handling
Lifting equipment
Loading

Pulmonary diseases

UF: Lung diseases
Respiratory diseases
BT: Diseases
RT: Lungs
Respiratory system
Spirometry
NT: Asthma
Chronic obstructive
pulmonary disease
Emphysema
Pneumonia



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Pulmonary semilunar valves

USE: Heart valves

Pulmonology

UF: Chest medicine
Pneumology
Respiratory medicine
Respirology
BT: Medical specialties
RT: Lungs
Respiratory system
NT: Spirometry

Pulp and paper industry

UF: Paper industry
BT: Manufacturing industries
RT: Forestry
Paper products
Paper pulp
Paper technology
Pulp manufacturing
Spinning machines
Wood industry
NT: Paper making
Paper making machines
Paper mills

Pulp manufacturing

BT: Manufacturing systems
RT: Paper making
Paper making machines
Paper mills
Paper pulp
Pulp and paper industry
Wood industry

Pulse amplifiers

BT: Amplifiers

Pulse circuits

UF: Bistable multivibrator
BT: Circuits
RT: Digital circuits
NT: Flip-flops

Pulse compression methods

BT: Signal processing
NT: Optical pulse compression

Pulse generation

UF: Impulse generation
Pulse generators
BT: Signal generators
RT: Electromagnetic pulses
NT: Optical pulse generation

Pulse generators

USE: Pulse generation

Pulse inverters

UF: Logic inverters
BT: Inverters
RT: Logic circuits

Pulse measurements

UF: Impulse measurements
BT: Measurement
RT: Electric variables
measurement

Pulse modulation

BT: Modulation
RT: Demodulation

Pulse oximeter

BT: Biomedical equipment
Biomedical measurement
RT: Noninvasive treatment
Pulse oximetry

Pulse oximetry

BT: Instrumentation and
measurement
Noninvasive treatment
RT: Biomedical measurement
Oxygen
Pulse oximeter
Remote sensing

Pulse shaping

USE: Pulse shaping methods

Pulse shaping methods

UF: Pulse shaping
BT: Signal processing
NT: Optical pulse shaping

Pulse transformers

BT: Transformers

Pulse width modulated power converters

USE: Pulse width modulation
converters

Pulse width modulation

UF: PWM
Pulsewidth modulation
Pulsewidth-modulation
BT: Modulation
RT: AC generators
AC machines



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AC motors
 Converters
 DC generators
 DC machines
 DC motors
 Pulse width modulation
 converters
 NT: Pulse width modulation
 inverters
 Space vector pulse width
 modulation
Pulse width modulation converters
 UF: PWM converters
 PWM convertors
 Pulse width modulated
 power converters
 Pulse width modulation
 convertors
 Pulsewidth modulation
 converters
 Pulsewidth modulation
 convertors
 BT: Converters
 RT: Power conditioning
 Power control
 Power conversion
 Power electronics
 Pulse width modulation
 Voltage-source converters

Pulse width modulation convertors
 USE: Pulse width modulation
 converters

Pulse width modulation inverters
 UF: PWM inverters
 PWM invertors
 Pulse width modulation
 invertors
 BT: Pulse width modulation
 RT: AC motors
 AC-DC power converters
 Converters
 DC motors
 DC-DC power converters

Pulse width modulation invertors
 USE: Pulse width modulation
 inverters

Pulsed electric current sintering
 USE: Spark Plasma sintering

Pulsed electroacoustic methods
 BT: Acoustoelectric effects
 RT: Acoustoelectric devices
 Charge measurement
 Insulation testing
 Space charge
Pulsed laser deposition
 UF: Laser deposition
 BT: Chemical vapor deposition
 RT: Laser cladding
 Vapor deposition

Pulsed lasers
 USE: Laser pulses

Pulsed power supplies
 BT: Pulsed power systems

Pulsed power systems
 BT: Power systems
 RT: Energy storage
 High-voltage techniques
 Power generation
 Power supplies
 NT: Pulsed power supplies

Pulsewidth modulation
 USE: Pulse width modulation

Pulsewidth modulation convertors
 USE: Pulse width modulation
 converters

Pulsewidth modulation convertors
 USE: Pulse width modulation
 converters

Pulsewidth-modulation
 USE: Pulse width modulation

Pump lasers
 BT: Lasers

Pumping of lasers
 USE: Laser excitation

Pumps
 BT: Machinery
 RT: Bellows
 Compressors
 Impellers
 Turbomachinery
 NT: Fuel pumps
 Heat pumps



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	Insulin pumps Micropumps Water pumps		RT: Ferroelectric materials Piezoelectricity Pyroelectric devices Thermal factors
Punching	BT: Materials processing RT: Sheet metal processing	Python	BT: Computer languages Public domain software RT: Functional programming Object oriented
Pupils	BT: Eyes RT: Ophthalmology	programming	Software libraries
Purification	BT: Cleaning RT: Air cleaners Decontamination Refining Sugar refining	<i>PZT</i>	USE: Lead zirconate titanate
Pursuit algorithms	BT: Algorithms	<i>Q factor</i>	USE: Q-factor
<i>PVA</i>	USE: Personal voice assistants	<i>Q learning</i>	USE: Q-learning
<i>PWM</i>	USE: Pulse width modulation	Q measurement	BT: Electric variables RT: Q-factor
<i>PWM converters</i>	USE: Pulse width modulation converters	Q-factor	UF: Q factor Quality factor BT: Electric variables RT: Capacitors Q measurement
<i>PWM converters</i>	USE: Pulse width modulation converters	Q-learning	UF: Q learning BT: Reinforcement learning RT: Markov processes Stochastic processes
<i>PWM inverters</i>	USE: Pulse width modulation inverters	<i>QAM</i>	USE: Quadrature amplitude modulation
<i>PWM invertors</i>	USE: Pulse width modulation inverters	<i>qbit</i>	USE: Qubit
<i>Pylons</i>	USE: Poles and towers	<i>QCD vacuum</i>	USE: Elementary particle vacuum
Pyroelectric devices	BT: Dielectric devices RT: Pyroelectricity	<i>QFD</i>	USE: Quality function deployment
Pyroelectricity	BT: Electricity Ultrasonics, ferroelectrics, and frequency control	<i>QFP</i>	USE: Electronics packaging



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<i>QoE</i>	USE: Quality of experience	Qualifications BT: Training RT: Continuing professional development
<i>QoS</i>	USE: Quality of service	Standards
<i>QoT</i>	USE: Quality of transmission	Quality assessment BT: Quality management
<i>Qox</i>	USE: Quality of experience	Quality assurance BT: Quality management RT: Consumer protection
<i>QPSK</i>	USE: Phase shift keying	Data integrity Design for quality IEEE 802.11e Standard Product liability Quality awards Quality control Quality function deployment Six sigma Total quality management Best practices
QR codes	UF: Quick response codes BT: Bar codes RT: Product codes	
<i>QSM</i>	USE: Quantitative susceptibility mapping	NT: Best practices
<i>Quad flat packs</i>	USE: Electronics packaging	Quality awards BT: Quality management RT: Continuous improvement Quality assurance Quality function deployment Total quality management
<i>Quadcopters</i>	USE: Quadrotors	
Quadratic programming	BT: Optimization methods	Quality control BT: Quality management RT: Contamination Coordinate measuring machines
Quadrature amplitude modulation	UF: QAM BT: Amplitude modulation	
Quadrature phase shift keying	BT: Phase shift keying NT: Differential quadrature phase shift keying	Data integrity Design for quality Failure analysis IEEE 802.11e Standard Quality assurance Quality function deployment Reliability Six sigma Total quality management
Quadrotors	UF: Quadcopters BT: Helicopters RT: Autonomous aerial vehicles Drones	<i>Quality factor</i> USE: Q-factor
Quadrupedal robots	UF: Quadrupeds BT: Robots	Quality function deployment UF: QFD BT: Quality management RT: Concurrent engineering Product development Quality assurance
<i>Quadrupeds</i>	USE: Quadrupedal robots	



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	Quality awards		Quality of experience
	Quality control		Throughput
Quality management		<i>Quality-of-service</i>	
UF:	ISO 9000	USE:	Quality of service
BT:	Management		
RT:	Control charts	<i>Quantisation</i>	
	Customer relationship	USE:	Quantization (signal)
management			
	Customer satisfaction	Quantitative susceptibility mapping	
	Data governance	UF:	QSM
	Design for quality	BT:	Magnetic resonance
	ISO Standards	imaging	
	Pareto analysis		
	Product liability	Quantization (signal)	
	Reliability	UF:	Quantisation
	System improvement		Quantization effects
NT:	Quality assessment		Quantization errors
	Quality assurance		Signal quantisation
	Quality awards		Signal quantization
	Quality control	BT:	Signal processing
	Quality function deployment	RT:	Analog-digital conversion
	Total quality management		Data compression
			Digital representation
Quality of experience			Encoding
UF:	QoE		Finite wordlength effects
	Qox		Granular computing
BT:	Communication systems		Signal sampling
	Customer satisfaction	NT:	Vector quantization
RT:	Quality of service		
	Quality of transmission	<i>Quantization effects</i>	
	User experience	USE:	Quantization (signal)
Quality of service		<i>Quantization errors</i>	
UF:	QoS	USE:	Quantization (signal)
	Quality-of-service		
BT:	Communication systems	Quantum advantage	
	Customer satisfaction	UF:	Quantum supremacy
RT:	Conformance testing	BT:	Quantum computing
	IEEE 802.11e Standard		
	IP networks	Quantum algorithm	
	Next generation networking	BT:	Quantum computing
	Quality of experience		
	Service level agreements	Quantum annealing	
	Spatial diversity	BT:	Metaheuristics
	Telecommunication		Quantum computing
computing		RT:	Annealing
NT:	Admission control		
	Quality of transmission	Quantum capacitance	
Quality of transmission		BT:	Capacitance
UF:	QoT		Quantum mechanics
BT:	Quality of service	RT:	CNTFETs
RT:	Delays		
	Jitter		



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Quantum cascade lasers

UF: Cascade lasers
BT: Quantum well lasers
RT: Quantum mechanics

Quantum cellular automata

BT: Quantum computing

Quantum channels

BT: Communication channels
Quantum information

science

RT: Qubit

Quantum chemistry

BT: Chemistry
Quantum computing
RT: Molecular computing

Quantum circuit

BT: Circuits
Quantum communication
Quantum computing
Quantum information

science

Quantum communication

BT: Communication systems
RT: Channel capacity
Information theory
Optical fiber communication
Teleportation
NT: Quantum circuit
Quantum networks

Quantum computing

BT: Computers and information
processing

RT: Electron devices
Coherence time
Trapped ions
NT: Quantum advantage
Quantum algorithm
Quantum annealing
Quantum cellular automata
Quantum chemistry
Quantum circuit
Quantum networks
Quantum simulation
Qubit

Quantum confinement

USE: Potential well

Quantum cryptography

BT: Cryptography
Quantum mechanics

Quantum dash

USE: Quantum dots

Quantum decoherence

BT: Quantum mechanics
RT: Coherence

Quantum dot lasers

UF: Quantum-dot lasers
BT: Semiconductor lasers
RT: Quantum dots
Quantum mechanics
Quantum well lasers

Quantum dots

UF: Quantum dash
Quantum-dot
Quantum-dots
BT: Semiconductor devices
RT: Nanocrystals
Quantum dot lasers
Quantum mechanics

Quantum entanglement

UF: Entangled states
BT: Quantum mechanics
RT: Quantum radar
Quantum repeaters
Quantum state
Teleportation

Quantum information science

BT: Information science
Quantum mechanics
NT: Quantum channels
Quantum circuit

Quantum key distribution

BT: Communication system
security
Quantum mechanics
RT: Cryptography

Quantum materials

BT: Materials
Quantum mechanics
RT: Excitons
Polaritons
Topological insulators



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Quantum mechanics

UF: Quantum theory
BT: Physics
RT: Lagrangian functions
Laser theory
Nanotechnology
Philosophical

considerations

Quantum cascade lasers
Quantum dot lasers
Quantum dots
Quantum well devices
Quantum well lasers
Resonant frequency
Solid-state physics
String theory

NT: Coherence time
Density functional theory
Proton effects
Quantum capacitance
Quantum cryptography
Quantum decoherence
Quantum entanglement
Quantum information

science

Quantum key distribution
Quantum materials
Quantum optics
Quantum sensing
Quantum sensors
Quantum simulation
Quantum state
Quantum system
Relativistic quantum

mechanics

Schrodinger equation
Stationary state
Teleportation
Tunneling

Quantum networks

BT: Quantum communication
Quantum computing
NT: Quantum repeaters

Quantum optics

BT: Quantum mechanics
NT: Optical squeezing

Quantum radar

BT: Radar
Remote sensing
RT: Lighting
Quantum entanglement

Quantum repeaters

BT: Quantum networks
RT: Quantum entanglement

Quantum sensing

BT: Quantum mechanics
Sensors

Quantum sensors

BT: Quantum mechanics
Sensors

Quantum simulation

BT: Quantum computing
Quantum mechanics
Simulation

Quantum state

BT: Quantum mechanics
RT: Quantum entanglement

Quantum supremacy

USE: Quantum advantage

Quantum system

BT: Quantum mechanics

Quantum teleportation

USE: Teleportation

Quantum theory

USE: Quantum mechanics

Quantum vacuum

USE: Elementary particle vacuum

Quantum well devices

UF: Quantum-well devices
Quantumwell devices
BT: Electron devices
RT: Electro-optic modulators
Quantum mechanics
Resonant tunneling devices
Tunneling
NT: Quantum well lasers
Quantum wells
Two dimensional hole gas

Quantum well lasers

UF: Quantum-well lasers
BT: Lasers
Quantum well devices
Semiconductor devices
Semiconductor lasers
Solid lasers



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amplifiers	RT:	Quantum dot lasers	Quasi-resonant invertors	
		Quantum mechanics	USE:	Resonant inverters
		Quantum wells		
		Semiconductor optical	Quaternions	
		Surface emitting lasers	BT:	Mathematics
		Two dimensional hole gas	Qubit	
	NT:	Quantum cascade lasers	UF:	qbit
Quantum wells			BT:	Quantum computing
	UF:	Semiconductor quantum	RT:	Coherence time
wells				Quantum channels
	BT:	Electrons	NT:	Spin qubits
		Quantum well devices	Query evaluation	
	RT:	Quantum well lasers	USE:	Query processing
		Surface emitting lasers	Query languages	
		Two dimensional hole gas	USE:	Database languages
Quantum-dot			Query optimisation	
	USE:	Quantum dots	USE:	Query processing
Quantum-dot lasers			Query optimization	
	USE:	Quantum dot lasers	USE:	Query processing
Quantum-dots			Query pipeline	
	USE:	Quantum dots	USE:	Query processing
Quantum-well devices			Query process	
	USE:	Quantum well devices	USE:	Query processing
Quantum-well lasers			Query processing	
	USE:	Quantum well lasers		
Quantumwell devices			UF:	
	USE:	Quantum well devices	Query evaluation	
			Query optimisation	
			Query optimization	
			Query pipeline	
Quarter-wave plates			Query process	
	USE:	Optical retarders	Query routing	
Quartz crystals			BT:	Database systems
	BT:	Crystals	RT:	Cross modal retrieval
			Data retrieval	
			Linked data	
Quasi Fermi level			NoSQL databases	
	USE:	Quasi-Fermi level	Semantic search	
Quasi-Fermi level			Query routing	
	UF:	Fermi energy	USE:	Query processing
		Imref		
		Quasi Fermi level	Question answering (information retrieval)	
	BT:	Energy states	BT:	Natural language
		Thermodynamics	processing	
	RT:	Fermi level	RT:	Chatbots
			Question generation	
Quasi-resonant invertors				
	USE:	Resonant inverters		



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Question generation

UF: Automatic question generation
BT: Natural language processing
RT: Chatbots
Question answering (information retrieval)

Queueing analysis

UF: Queueing theory
BT: Traffic control
RT: Publish subscribe systems
Scheduling

Queueing theory

USE: Queueing analysis

Quick response codes

USE: QR codes

R & D management

USE: Research and development management

R language

BT: Computer languages
Public domain software
RT: Data analysis
Data mining
Data visualization
Statistical analysis

Rabbits

BT: Animals

Racetrack memory

UF: Domain wall memory
Domain-wall memory
BT: Nonvolatile memory

Rad hardened

USE: Radiation hardening (electronics)

Rad-hard

USE: Radiation hardening (electronics)

Radar

UF: Microwave radar
BT: Electromagnetic compatibility and interference
RT: Microwave technology
Radar detection

NT: Radar scattering
Airborne radar
Bistatic radar
Cognitive radar
Doppler radar
Ground penetrating radar
High frequency radar
Laser radar
Meteorological radar
Millimeter wave radar
Multistatic radar
Passive radar
Quantum radar
Radar applications
Radar clutter
Radar cross-sections
Radar equipment
Radar theory
Spaceborne radar
Spread spectrum radar
Synthetic aperture radar
Ultra wideband radar

Radar antennas

BT: Antennas

Radar applications

BT: Radar
RT: Oceanographic techniques
NT: Radar countermeasures
Radar detection
Radar imaging
Radar measurements
Radar polarimetry
Radar remote sensing
Radar tracking

Radar clutter

BT: Radar
RT: Jamming

Radar countermeasures

BT: Electronic warfare
Radar applications
RT: Adaptive arrays
Electronic countermeasures
Jamming
Spread spectrum radar

Radar cross section

USE: Radar cross-sections

Radar cross sections

USE: Radar cross-sections



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Radar cross-section

USE: Radar cross-sections

Radar cross-sections

UF: Radar cross section
Radar cross sections
Radar cross-section
BT: Radar
Radar measurements
Reflection

Radar detection

BT: Radar applications
Signal detection
RT: Ground penetrating radar
Passive radar
Radar
Ultra wideband radar

Radar equipment

BT: Radar

Radar imaging

BT: Radar applications
RT: Ground penetrating radar
Landmine detection
Meteorological radar
Passive radar
Remote sensing
Synthetic aperture radar
Ultra wideband radar

Radar interferometry

BT: Interferometry
NT: Synthetic aperture radar
interferometry

Radar measurements

BT: Radar applications
RT: Remote sensing
NT: Radar cross-sections

Radar meteorology

USE: Meteorological radar

Radar polarimetry

UF: SAR imaging
Synthetic aperture radar
imaging
BT: Radar applications

Radar remote sensing

BT: Radar applications
RT: Spaceborne radar

Radar scattering

BT: Electromagnetic scattering
RT: Radar

Radar signal processing

BT: Signal processing

Radar theory

BT: Radar

Radar tracking

BT: Radar applications
RT: Target tracking

Radial basis function networks

UF: RBF networks
Radial basis function neural
networks
BT: Neural networks
RT: Artificial intelligence
Computer networks
Cybernetics
Interpolation

Radial basis function neural networks

USE: Radial basis function
networks

Radiation counters

USE: Radiation detectors

Radiation detection

USE: Radiation detectors

Radiation detector circuits

BT: Circuits
RT: Counting circuits
Radiation detectors

Radiation detectors

UF: Counters
Particle detectors
Radiation counters
Radiation detection
Ratemeters
BT: Ionizing radiation sensors
RT: Atomic measurements
Dosimetry
Nuclear measurements
Phototransistors
Radiation detector circuits
Radiation monitoring
Radiofrequency exposure
Spectroscopy
X-ray detectors



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NT: Bolometers
Gamma-ray detectors
Infrared detectors
Photodetectors
Semiconductor radiation
detectors
Silicon radiation detectors

Radiation dosage

BT: Radiation monitoring

Radiation dosimetry

USE: Dosimetry

Radiation effects

UF: Irradiation
BT: Nuclear and plasma
sciences
RT: Biomedical applications of
radiation
Brachytherapy
High energy physics
instrumentation computing
Proton effects
Proton radiation effects
Radiation monitoring
Radiation protection
Safety
NT: Biological effects of
radiation
EMP radiation effects
Gamma-ray effects
Ion radiation effects
Neutron radiation effects
Scintillators
Single event latchup
Space radiation
Terahertz radiation
Total ionizing dose

Radiation hardening (electronics)

UF: Rad hardened
Rad-hard
BT: Electronic equipment
manufacture
Nuclear and plasma
sciences
RT: Ionizing radiation
Satellite communications
Total ionizing dose

Radiation imaging

BT: Imaging
RT: Biomedical imaging
NT: Radionuclide imaging

Radiation monitoring

UF: Health physics
BT: Monitoring
Nuclear and plasma
sciences
RT: Dosimetry
Radiation detectors
Radiation effects
Radiation protection
Radioactive pollution
Reactor instrumentation
NT: Radiation dosage

Radiation pattern

USE: Antenna radiation patterns

Radiation protection

UF: Radiation shielding
Radiological protection
BT: Protection
Radiation safety
RT: Biological effects of
radiation
Contamination
Dosimetry
Fission reactors
Radiation effects
Radiation monitoring
Radioactive pollution

Radiation safety

BT: Nuclear and plasma
sciences
Safety
NT: Radiation protection
Radiofrequency safety

Radiation shielding

USE: Radiation protection

Radiation therapy

UF: Radiotherapy
BT: Biomedical applications of
radiation
Oncology
RT: Cancer treatment

Radiative recombination

BT: Spontaneous emission
RT: Semiconductor materials

Radiators (automotive)

USE: Automotive components



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Radio access networks

UF: RAN
BT: Telecommunications
RT: 3G mobile communication
4G mobile communication
Aerial computing
Land mobile radio
Radio communication
Telecommunication

services

NT: Cloud radio access
networks Open RAN

Radio access technologies

UF: RATS
BT: Communication networks
NT: New Radio

Radio astronomy

UF: Radio telescopes
BT: Antennas and propagation
Astronomy
RT: Reflector antennas
Telescopes

Radio broadcasting

BT: Broadcasting
Radio communication
RT: Digital multimedia
broadcasting Journalism
NT: Frequency modulation
Radio networks

Radio communication

BT: Communication systems
RT: Bandwidth
Convolutional codes
Cross layer design
Diversity methods
Film bulk acoustic
resonators IEEE 802.11 Standard
IEEE 802.11g Standard
IEEE 802.11n Standard
IEEE 802.15 Standard
Inter-cell interference
MIMO
Mobile communication
NOMA
Radio access networks
Radio communication

equipment

Radio propagation

SIMO
SISO
Wireless LAN
Baseband
Bluetooth
Cellular technology
Indoor radio communication
Land mobile radio
LoRa
Millimeter wave

communication

Near field communication
Packet radio networks
Passband
Personal area networks
Radio broadcasting
Radio communication

countermeasures

Radio frequency
Radio links
Radio spectrum

management

Satellite communications
Satellite ground stations
Software radio
Zigbee

Radio communication countermeasures

BT: Communication system
security
RT: Radio communication
Adaptive arrays
Electronic countermeasures
Electronic warfare
Jamming
Spread spectrum

communication

Radio communication equipment

BT: Communication equipment
RT: Antennas
Radio communication
Telephone equipment
NT: Base stations
Ham radios
Land mobile radio
equipment
Radio transceivers
Transponders

Radio control

BT: Control systems



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Radio frequency

UF: RF
Radio-frequency
Radiofrequency
BT: Radio communication
RT: Electrosurgery
Light fidelity
Wireless fidelity
NT: High frequency
Radiofrequency exposure
Radiofrequency safety

Radio frequency exposure

USE: Radiofrequency exposure

Radio frequency identification

USE: Radiofrequency
identification

Radio frequency integrated circuits

USE: Radiofrequency integrated
circuits

Radio frequency interference

USE: Radiofrequency
interference

Radio frequency power amplifiers

USE: Power amplifiers

Radio frequency safety

USE: Radiofrequency safety

Radio interference

USE: Electromagnetic
interference

Radio interferometry

UF: Radiowave interferometry
BT: Interferometry

Radio LAN

USE: Wireless LAN

Radio links

BT: Radio communication
RT: Transport protocols

Radio navigation

UF: Loran
BT: Navigation
RT: Air traffic control
Indoor navigation
Satellite navigation systems
Transponders

Radio networks

BT: Radio broadcasting

Radio propagation

BT: Electromagnetic
propagation
RT: Fading channels
Multipath channels
Radio communication
Radiowave propagation
Rayleigh channels

Radio receivers

USE: Receivers

Radio resource management

USE: Resource management

Radio spectrum management

UF: Frequency allocation
Spectrum management
BT: Radio communication
RT: Communication standards
NT: Direct sequence spread
spectrum communication
White spaces

Radio telescopes

USE: Radio astronomy

Radio transceivers

BT: Radio communication
equipment
Transceivers
NT: Dynamic spectrum access

Radio transmitters

BT: Transmitters

Radio-frequency

USE: Radio frequency

Radio-frequency identification

USE: Radiofrequency
identification

Radio-frequency interference

USE: Radiofrequency
interference

Radio-frequency safety

USE: Radiofrequency safety



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Radioactive decay

BT: Radioactive materials

Radioactive label

USE: Radiotracer

Radioactive materials

UF: Alphavoltaic power sources
Betavoltaic power sources
Radioisotopes
BT: Materials
RT: Isotopes
Neutrino sources
Occupational health
Occupational safety
Radioactive pollution
Safety
NT: Radioactive decay
Radioactive waste

Radioactive pollution

UF: Nuclear wastes
BT: Pollution
RT: Hazardous areas
Incineration
Industrial pollution
Land pollution
Radiation monitoring
Radiation protection
Radioactive materials
Radioactive waste
Radioactive waste disposal
Safety
Waste materials

Radioactive tracer

USE: Radiotracer

Radioactive waste

BT: Radioactive materials
Waste materials
RT: Hazardous areas
Hazardous materials
Incineration
Materials handling
Nuclear energy
Nuclear facility regulation
Radioactive pollution
Waste disposal
Waste management
NT: Radioactive waste disposal

Radioactive waste disposal

BT: Radioactive waste
Waste disposal

RT: Incineration
Materials handling
Radioactive pollution
Vitrification
Waste handling

Radiofrequency

USE: Radio frequency

Radiofrequency amplifiers

UF: Radiofrequency power
amplifiers
BT: Amplifiers

Radiofrequency exposure

UF: RF exposure
Radio frequency exposure
BT: Radio frequency
RT: Electromagnetic pulses
Electromagnetic radiation
Radiation detectors

Radiofrequency identification

UF: RF identification
RFID
Radio frequency
identification
Radio-frequency
identification
BT: Sensor systems and
applications
RT: Internet of Things
Product codes
Radiofrequency integrated
circuits
NT: RFID tags

Radiofrequency integrated circuits

UF: RFIC
Radio frequency integrated
circuits
BT: Integrated circuits
RT: MIMICs
MMICs
Radiofrequency
identification

Radiofrequency interference

UF: Radio frequency
interference
Radio-frequency
interference
BT: Electromagnetic
interference
RT: NOMA



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NT: Superconducting filters
Inter-cell interference

Radiofrequency micro-electro-mechanical systems
USE: Radiofrequency
microelectromechanical systems

Radiofrequency micro-electromechanical systems
USE: Radiofrequency
microelectromechanical systems

Radiofrequency microelectromechanical systems
UF: RF micro-electro-mechanical systems
RF microelectromechanical systems
Radiofrequency micro-electro-mechanical systems
Radiofrequency micro-electromechanical systems
BT: Microelectromechanical systems

Radiofrequency power amplifiers
USE: Power amplifiers AND
Radiofrequency amplifiers

Radiofrequency safety
UF: RF safety
Radio frequency safety
Radio-frequency safety
BT: Radiation safety
Radio frequency
RT: Electromagnetic radiation

Radiographic image enhancement
BT: Biomedical image
processing

Radiography
BT: Imaging
RT: Biomedical applications of
radiation
Medical diagnosis
Nuclear imaging
X-ray detection
X-ray imaging
NT: Diagnostic radiography

Radioisotope thermoelectric generators
BT: Generators

Radioisotopes
USE: Radioactive materials

Radiological protection
USE: Radiation protection

Radiology
BT: Biomedical image
processing
Medical specialties
NT: Neuroradiology

Radiometers
BT: Meters
Radiometry
NT: Spectroradiometers

Radiometry
BT: Electromagnetic
measurements
Geoscience and remote
sensing
RT: Imaging
Photometry
Remote sensing
Temperature measurement
NT: Microwave radiometry
Radiometers

Radiomics
BT: Biomedical image
processing
RT: Machine learning

Radionuclide imaging
UF: Radionuclide scanning
Radionuclide scans
BT: Medical diagnostic imaging
Radiation imaging

Radionuclide scanning
USE: Radionuclide imaging

Radionuclide scans
USE: Radionuclide imaging

Radiotherapy
USE: Radiation therapy

Radiotracer
UF: Radioactive label
Radioactive tracer
BT: Chemical compounds



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Radiowave interferometry

USE: Radio interferometry

Radiowave propagation

BT: Electromagnetic
propagation

RT: Radio propagation

NT: NVIS

Radium

BT: Chemical elements

Radomes

BT: Antenna accessories

Radon

BT: Chemical elements

RAG

USE: Retrieval augmented
generation

Rail guns

USE: Railguns

Rail line

USE: Rail transportation

Rail lines

USE: Rail transportation

Rail to rail amplifiers

BT: Rail to rail operation

RT: Amplifiers
MOSFET circuits
Rail to rail inputs
Rail to rail outputs

Rail to rail inputs

UF: RRI

BT: Rail to rail operation

RT: Nonlinear circuits
Rail to rail amplifiers

Rail to rail operation

BT: Circuits

RT: Amplifiers
CMOSFET circuits
MOSFET circuits

NT: Rail to rail amplifiers
Rail to rail inputs
Rail to rail outputs

Rail to rail outputs

UF: RRO

BT: Rail to rail operation

RT: Nonlinear circuits
Rail to rail amplifiers

Rail traffic

USE: Rail transportation

Rail transportation

UF: Rail line

Rail lines

Rail traffic

Rail ways

Railways

BT: Land transportation

RT: Block signalling

Land vehicles

Magnetic levitation

Public transportation

NT: High-speed rail
transportation

Light rail systems

Magnetic levitation vehicles

Positive train control

Railway communication

Railway electrification

Rail ways

USE: Rail transportation

Rail-gun

USE: Railguns

Railguns

UF: Rail guns

Rail-gun

BT: Electromagnetic launching

RT: Rails

Rails

BT: Structural shapes

RT: Flanges

Railguns

Railway accidents

UF: Derailments

BT: Accidents

RT: Positive train control

Railway engineering

Railway safety

Railway bridges

USE: Structural panels



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Railway communication

BT: Rail transportation
Telecommunications
NT: Block signalling
GSM-R

Railway electrification

BT: Rail transportation
RT: Power overhead lines

Railway engineering

BT: Civil engineering
RT: Railway accidents
NT: Railway safety

Railway safety

BT: Railway engineering
RT: Positive train control
Railway accidents
Safety devices

Railways

USE: Rail transportation

Rain

UF: Rainfall
BT: Precipitation
RT: Floods
Monsoons
Stormwater

Rain fades

USE: Rain fading

Rain fading

UF: Rain fades
BT: Interference

Rain forests

USE: Rainforests

Rainfall

USE: Rain

Rainforests

UF: Rain forests
BT: Ecosystems
RT: Deforestation
Forestry

RAKE receivers

BT: Receivers
RT: Signal to noise ratio

Raleigh fading

USE: Rayleigh channels

Raleigh fading channels

USE: Rayleigh channels

RAM

USE: Random access memory

Raman effect

USE: Raman scattering

Raman scattering

UF: Raman effect
Raman spectroscopy
BT: Electromagnetic scattering
Nonlinear optics

Raman spectroscopy

USE: Raman scattering

RAN

USE: Radio access networks

AND

Regional area networks

Random access communication

USE: Multiaccess communication

Random access memory

UF: RAM
Random access storage
BT: Memory
RT: Buffer storage
NT: DRAM chips
Phase change random

access memory

Resistive RAM
SDRAM
SRAM cells
SRAM chips

Random access storage

USE: Random access memory

Random forests

BT: Machine learning
Random processes
RT: Decision trees
Fish schools
Pattern recognition
Regression analysis



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Random media

UF: Turbulent media
BT: Media
RT: Chaos
Nonhomogeneous media

Random number generation

BT: Cryptography
RT: Random sequences
Stochastic processes
White noise

Random processes

BT: Mathematics
RT: Algorithms
Mean field theory
Probability
Signal processing
Statistical analysis
Time series analysis
NT: Brownian motion
Conditional random fields
Random forests

Random sequences

UF: Pseudorandom sequences
BT: Sequences
RT: Chaotic mapping
Cryptography
Random number

generation

Random variables

BT: Probability
RT: Stochastic processes
Stochastic systems
NT: Nonce

Ranging

USE: Distance measurement

Ranking (statistics)

BT: Statistics
RT: Information retrieval
Ontologies
Search methods
Semantic Web
Vocabulary

Ransomware

BT: Malware

Rapid eye movement sleep

UF: REM sleep
BT: Sleep

Rapid prototyping

BT: Prototypes
RT: CAD/CAM
Design methodology
Manufacturing processes
Product development
Software engineering
Three-dimensional printing
Virtual prototyping

Rapid thermal annealing

BT: Annealing
RT: Semiconductor devices

Rapid thermal processing

BT: High-temperature
techniques
RT: Heating systems

Rare earth metals

UF: Rare-earth metals
BT: Metals

Rare-earth metals

USE: Rare earth metals

Rate distortion

USE: Rate-distortion

Rate distortion theory

BT: Information theory
RT: Audio coding
Channel coding
Channel spacing
Distortion
Image coding
Signal analysis
Signal processing
Source coding
Speech coding
Video coding
NT: Channel rate control

Rate-distortion

UF: Rate distortion
BT: Information theory
RT: Data compression

Ratemeters

USE: Radiation detectors

RATS

USE: Radio access technologies



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Rats

BT: Animals

Raw materials

BT: Materials
RT: Mining industry

Ray tracing

UF: Ray-tracing
BT: Geometrical optics
Optics
RT: Computer graphics
Stray light

Ray-tracing

USE: Ray tracing

Rayleigh channels

UF: Raleigh fading
Raleigh fading channels
Rayleigh-fading
BT: Fading channels
RT: Radio propagation

Rayleigh scattering

BT: Electromagnetic scattering

Rayleigh-Benard convection

USE: Convection

Rayleigh-fading

USE: Rayleigh channels

RBF networks

USE: Radial basis function

networks

RDF

USE: Resource description

framework

Re-configurable devices

USE: Reconfigurable devices

Reachability analysis

BT: Graph theory

Reactive power

UF: Power factor
VAR
BT: Power systems
RT: Reactive power control
Static VAR compensators

Reactive power control

BT: Electric variables control
RT: Power system security
Reactive power
Voltage control

Reactor instrumentation

BT: Nuclear and plasma
sciences
RT: Radiation monitoring

Reactors

USE: Inductors

Read only memory

UF: ROM
BT: Memory
RT: Read-write memory
NT: PROM

Read-write memory

BT: Memory
RT: Read only memory

Readability formulas

USE: Readability metrics

Readability metrics

UF: Readability formulas
Readability tests
BT: Writing

Readability tests

USE: Readability metrics

Readout electronics

BT: Displays
RT: Detectors
SQUIDS

Real estate industry

BT: Industries

Real time

USE: Real-time systems

Real time control

USE: Real-time systems

Real time monitoring

USE: Real-time systems

Real time processing

USE: Real-time systems



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Real time systems

USE: Real-time systems

Real-time control

USE: Real-time systems

Real-time monitoring

USE: Real-time systems

Real-time processing

USE: Real-time systems

Real-time systems

UF: Real time
Real time control
Real time monitoring
Real time processing
Real time systems
Real-time control
Real-time monitoring
Real-time processing
BT: Computers and information

processing

RT: Control systems
Endomicroscopy
Hardware-in-the-loop

simulation

High energy physics

instrumentation computing

Networked control systems
YOLO
NT: Telexistence
WebRTC

Reasoning

USE: Cognition AND
Cognitive systems

reasoning about programs

USE: Commonsense reasoning

Rebreathers

USE: Rebreathing equipment

Rebreathing equipment

UF: Rebreathers
BT: Underwater equipment

Receive antennas

USE: Receiving antennas

Received signal strength indicator

UF: RSSI
BT: Communication system

signaling

Signal processing

Receivers

UF: Radio receivers
BT: Communication equipment
RT: Demodulation
Optical antennas
Signal detection
NT: Optical receivers
RAKE receivers
Receiving antennas

Receiving antennas

UF: Receive antennas
BT: Antennas
Receivers
RT: Spatial diversity
Transmitting antennas

Receptor (biochemistry)

BT: Biochemistry

Reclaimed

BT: Recycling

Recommendation systems

USE: Recommender systems

Recommender systems

UF: Music recommendation
Recommendation systems
BT: Information filtering
RT: Collaborative filtering
Graph neural networks

Reconfigurable architectures

BT: Computer architecture
NT: Reconfigurable intelligent

surfaces

Reconfigurable devices

UF: Re-configurable devices
BT: Hardware
RT: Field programmable gate

arrays

Reconfigurable intelligent surfaces

UF: Intelligent reconfigurable

surface

RIS
BT: Reconfigurable

architectures



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Reconfigurable logic

BT: Logic design

RT: Bridge circuits
Power electronics
Voltage multipliers

Reconfigurable radio

USE: Software radio

Reconnaissance

BT: Military communication
Security
RT: Remote sensing
Surveillance

Rectifying circuits

BT: AC-DC power converters

Recurrent neural nets

USE: Recurrent neural networks

Recurrent neural networks

UF: RNN
Recurrent neural nets
BT: Neural networks
RT: Backpropagation through

Reconstruction algorithm

USE: Reconstruction algorithms

Reconstruction algorithms

UF: Reconstruction algorithm
Reconstruction methods
BT: Tomography
RT: Image processing
Three-dimensional displays

time

Deep learning
Large language models
Reservoir computing
NT: Bidirectional long short term

Reconstruction methods

USE: Reconstruction algorithms

memory

Echo state networks
Gated recurrent units
Hopfield neural networks

Recording

BT: Signal processing
RT: Memory
NT: Audio recording
Digital recording
Disk recording
Flight recording
Magnetic recording
Optical recording
Video recording

Recursive estimation

BT: Bayes methods
RT: Least squares

approximations

Recycle

USE: Recycling

Recycling

UF: Recycle
BT: Environmental
management
RT: Circular economy
Food waste
Green energy
NT: Byproduct
Composting
Post-consumer waste
Reclaimed
Upcycle
Wishcycling

Recruitment

BT: Human resource
management
RT: Equal opportunities
Job specification
Labor resources

Red blood cells

BT: Blood
RT: Anemia

Rectangular waveguides

BT: Electromagnetic
waveguides
RT: Planar waveguides

Rectennas

BT: Antennas
Microwave communication
RT: Converters

REDD+

UF: Reducing Emissions from
Deforestation and Forest Degradation
BT: Forestry

Rectifiers

BT: Circuits



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Reddit

USE: Social networking (online)
AND Web sites

Redox

UF: Reduction-oxidation
BT: Chemical reduction
Oxidation

Reduced instruction set computing

UF: RISC
BT: Instruction sets
RT: Complex instruction set
computer

Reduced order model

USE: Reduced order systems

Reduced order systems

UF: Model reduction
Reduced order model
Reduced-order model
Reduced-order systems
BT: Systems engineering and
theory
RT: Estimation
Simulation

Reduced-order model

USE: Reduced order systems

Reduced-order systems

USE: Reduced order systems

Reducing Emissions from Deforestation and Forest Degradation

USE: REDD+

Reduction-oxidation

USE: Redox

Redundancy

BT: Fault tolerance
RT: Codes
Reliability

Redundancy (employment)

USE: Termination of employment

Redundant array of independent disks

UF: DRAID
BT: Computer architecture

Reed Solomon codes

USE: Reed-Solomon codes

Reed-Muller codes

BT: Error correction codes
RT: Polar codes

Reed-Solomon codes

UF: Reed Solomon codes
BT: Error correction codes

Refining

BT: Materials processing
RT: Chemical technology
Cleaning
Purification
Smelting
Sugar refining

Reflectance

USE: Reflectivity

Reflection

BT: Propagation
RT: Mirrors
Scattering
NT: Acoustic reflection
Backscatter
Electromagnetic reflection
Fresnel reflection
Radar cross-sections

Reflection coefficient

BT: Optical variables
measurement
Waveguide discontinuities
RT: Amplitude estimation

Reflective binary codes

UF: Gray codes
Grey codes
BT: Binary codes

Reflectivity

UF: Reflectance
BT: Waves
RT: Geometrical optics
Light trapping
Optical reflection
Sonar detection
Telecommunications
NT: Diffuse reflectance

spectroscopy



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Reflectometry

BT: Measurement
RT: Electromagnetic
measurements
Electromagnetic reflection
Optical reflection
Optical variables
measurement

Reflector antennas

BT: Antennas
RT: Aperture antennas
Radio astronomy

Reflow soldering

BT: Soldering

Reforestation

BT: Forestry
RT: Deforestation
NT: Rewilding

Refractive index

UF: Refractivity
BT: Optical variables
measurement
RT: Birefringence
Cross-phase modulation
Dispersion
Gain measurement
Laser beams
Metamaterials
Optical refraction
Photorefractive effect
Semiconductor device
measurement
Semiconductor lasers

Refractivity

USE: Refractive index

Refracting

USE: Code refracting

Refrigerants

BT: Coolants
RT: Heat pumps
Space cooling
NT: Liquid nitrogen

Refrigeration

BT: Cooling

Refrigerators

BT: Home appliances
Home automation

Refuse

USE: Waste materials

Refuse incineration

USE: Incineration

Regeneration engineering

BT: Tissue engineering
RT: Circular economy

Regenerative agriculture

BT: Agriculture

Regional area networks

UF: RAN
BT: Communication systems
RT: IEEE 802.22 Standard
Local area networks
Metropolitan area networks
Wireless communication
NT: WRAN

Register transfer level

UF: RTL
Register-transfer level
BT: Circuit synthesis
RT: Digital circuits

Register-transfer level

USE: Register transfer level

Registers

BT: Memory
NT: Shift registers

Regression analysis

BT: Statistical analysis
RT: Correlation coefficient
Econometrics
Nearest neighbor methods
Random forests
NT: Linear regression
Logistic regression
Multivariate regression

Regression tree analysis

BT: Decision trees

Regulation

BT: Government policies
NT: Tariffs



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Regulators

BT: Control equipment
RT: Current control
Electric variables control
Power conversion
Voltage control

Rehabilitation robotics

USE: Assistive robots

Rehabilitation robots

USE: Assistive robots

Reinforcement learning

BT: Machine learning
RT: Artificial intelligence
Bayes methods
Deep learning
Multi-armed bandit problem
Neural networks
Pattern classification
Semisupervised learning
Support vector machines
Unsupervised learning
NT: Deep reinforcement

learning

Q-learning
Temporal difference

learning

Relational databases

BT: Databases
RT: Structured Query Language
Triples (Data structure)

Relativistic effects

BT: Nuclear physics
RT: Electron beams
Free electron lasers
Klystrons
Magnetrons
Masers
Plasmas
NT: Optical flow

Relativistic quantum mechanics

BT: Quantum mechanics

Relaxation methods

BT: Numerical analysis
RT: Simulated annealing

Relaxor ferroelectrics

BT: Ferroelectric materials

Relay networks

BT: Communication networks
Relays

Relays

BT: Switchgear
RT: Switching circuits
NT: Digital relays
Microrelays
Power system relaying
Protective relaying
Relay networks
Road side unit

Release engineering

UF: Releng
BT: Software engineering
RT: Product lifecycle
management
Software development
management
Source coding
Releng
USE: Release engineering

Relevance vector machines

BT: Classification algorithms
Machine learning
RT: Bayes methods
Support vector machines

Reliability

UF: Reliability management
System reliability
RT: Aging
Dependability management
Electron traps
Failure analysis
Life testing
Power system reliability
Preventive maintenance
Quality control
Quality management
Redundancy
Risk analysis
System improvement
System recovery
Thermal management
NT: Availability
Fault diagnosis
Fault tolerance
Fluctuations
Integrated circuit reliability
Maintenance



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reliability

- Maldistribution
- Materials reliability
- Reliability engineering
- Reliability theory
- Robustness
- Semiconductor device

- Software reliability
- Stability
- Telecommunication

network reliability

Reliability engineering

BT: Reliability
RT: Weibull distribution

Reliability management

USE: Management AND
Reliability

Reliability theory

BT: Reliability

Reluctance generators

BT: Synchronous generators

Reluctance machines

BT: Rotating machines
Synchronous machines
NT: Reluctance motors

Reluctance motors

BT: Reluctance machines
Synchronous motors
NT: Switched reluctance motors

REM sleep

USE: Rapid eye movement sleep

Remaining life assessment

BT: Testing
RT: Failure analysis
Maintenance engineering

Remanence

BT: Magnetism
RT: Magnetic fields
Magnetic flux
Magnetic hysteresis
Permanent magnets

Remineralize

BT: Forestry

Remote control

BT: Control equipment

Remote handling

UF: Manipulators (nonrobotic)
BT: Materials handling
RT: Remote handling
equipment
Telecontrol equipment

Remote handling equipment

BT: Materials handling
equipment
RT: Remote handling
Telerobotics
Waste handling equipment

Remote laboratories

BT: Laboratories

Remote learning

USE: Distance learning

Remote monitoring

BT: Monitoring
Remote sensing
RT: Internet of Medical Things
Machine-to-machine
communications

Remote piloted aircraft

USE: Remotely piloted aircraft

Remote sensing

BT: Geoscience and remote
sensing
RT: Atmospheric
measurements
Earth
Geologic measurements
Geophysical measurement
techniques
Geophysical measurements
Imaging
Infrared imaging
Land surface temperature
Landmine detection
Meteorology
Microwave imaging
Oceanographic techniques
Optical imaging
Pulse oximetry
Radar imaging
Radar measurements
Radiometry



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Reconnaissance
Sea measurements
Soil measurements
Sonar measurements
Surveillance
Terrain mapping
Vegetation mapping
Water resources
NT: Hyperspectral sensors
Passive microwave remote

sensing

Remote working

UF: Distance work
Hybrid office
Hybrid working
Mobile office
Telecommuting
Teleworking
Virtual office
BT: Systems, man, and

cybernetics

Remotely controlled aerial vehicles

USE: Remotely piloted aircraft

Remotely controlled vehicles

USE: Remotely guided vehicles

Remotely guided underwater vehicles

BT: Remotely guided vehicles
Underwater vehicles

Remotely guided vehicles

UF: Automated guided vehicles
Remotely controlled

vehicles

automobiles

Remotely operated

Remotely operated cars
Remotely operated vehicles

BT: Vehicles

NT: Drones

Remotely guided

underwater vehicles

Remotely piloted aircraft

Remotely operated automobiles

USE: Remotely guided vehicles

Remotely operated cars

USE: Remotely guided vehicles

Remotely operated vehicles

USE: Remotely guided vehicles

Remotely piloted aircraft

UF: RPA
Remote piloted aircraft
Remotely controlled aerial

vehicles

BT: Unmanned air vehicles
Remotely guided vehicles

Remuneration

BT: Human resource
management
RT: Employee welfare
NT: Incentive schemes
Pensions

Renal calculi

USE: Kidney stones

Rendering (computer graphics)

BT: Computer graphics
RT: Image synthesis

Renewable electricity

BT: Renewable energy sources
NT: Hydroelectric power

generation

Renewable Electricity Standard

USE: Renewable Portfolio
Standard

Renewable energy

BT: Environmental
management
NT: Renewable energy sources

Renewable energy payments

USE: Feed in tariff

Renewable energy resources

USE: Renewable energy sources

Renewable energy sources

UF: Alternative energy
resources

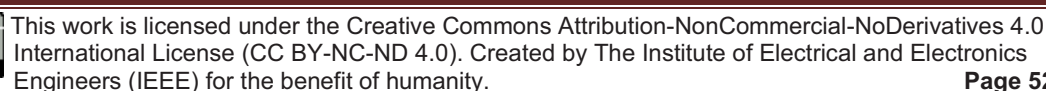
Alternative energy sources
Clean energy
Renewable energy

resources

BT: Renewables
Energy conservation
Renewable energy



Standard	RT:	Anaerobic digestion	NT:	Disentangled
		Circular economy	representation learning	
		Electrification		
		Feed in tariff	Representational state transfer	
		Green communications	BT:	Software architecture
		Low carbon economy		
		Solar panels	Reproducibility of results	
		Wave power	UF:	Reproducible research
	NT:	Biomass	BT:	Measurement
		Green energy		
Renewable Portfolio Standard		Green hydrogen	<i>Reproducible research</i>	
		Renewable Portfolio	USE:	Reproducibility of results
		Renewable electricity	<i>Reproductive cloning</i>	
		Tidal energy	USE:	Cloning
			Reproductive system	
Standard	UF:	RES	UF:	Genital system
		RPS	BT:	Urogenital system
		Renewable Electricity	NT:	Placenta
	BT:	Renewable energy sources		Uterus
Renewables		Standards	Requirements engineering	
	RT:	Biomass	BT:	Systems engineering and
		Electricity supply industry	theory	
		Geothermal energy	RT:	Product design
		Solar power generation		Project management
		Wind power generation		Requirements management
Repair	USE:	Renewable energy sources		Software engineering
			NT:	Stakeholders
Repeaters				Technical requirements
	USE:	Maintenance engineering	Requirements management	
Replica molding			BT:	Management
	USE:	Soft lithography		Systems engineering and
Replica moulding			theory	
	USE:	Soft lithography	RT:	Project management
Replicability				Requirements engineering
	BT:	Measurement	<i>ReRAM</i>	
Report writing			USE:	Resistive RAM
	USE:	Writing	<i>RES</i>	
Representation learning			USE:	Renewable Portfolio
	UF:	Feature learning	Standard	
	BT:	Machine learning	Rescue robots	
			UF:	Search and rescue robots
			BT:	Robots
			RT:	Emergency services
				Hazards
				Marine robots



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Research & development

USE: Research and development

Research and development

UF: Research & development

BT: Engineering - general

RT: CERN

Electrical engineering

Engineering profession

Industrial engineering

International collaboration

Laboratories

Museums

Research and development

management

Reverse engineering

Science - general

Technology

Virtual enterprises

Virtual manufacturing

Virtual prototyping

NT: Translational research

Research and development management

UF: R & D management

BT: Engineering management

Management

RT: Concurrent engineering

Engineering profession

Production management

Project management

Research and development

Technology management

Venture capital

NT: Innovation management

Research initiatives

BT: Engineering management

RT: Systematic literature review

Reservoir computing

BT: Computation theory

RT: Recurrent neural networks

Reservoirs

BT: Water resources

Water storage

RT: Dams

Lakes

Land use planning

Water

Residual networks

USE: Residual neural networks

Residual neural networks

UF: Residual networks

BT: Artificial neural networks

Residual stress

USE: Residual stresses

Residual stresses

UF: Residual stress

BT: Stress

Resilience

UF: Resiliency

Resilient systems

BT: Material properties

Resiliency

USE: Resilience

Resilient systems

USE: Resilience

Resin transfer molding

USE: Transfer molding

Resin transfer moulding

USE: Transfer molding

Resins

BT: Materials

RT: Plastics

Polymer foams

NT: Epoxy resins

Resistance

BT: Electric variables

RT: Electrical resistance

measurement

Resistance heating

Skin effect

NT: Electric resistance

Piezoresistance

Surface resistance

Thermal resistance

Viscosity

Resistance heating

UF: Electric heating

BT: Heating systems

RT: Electrothermal actuators

Resistance



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Resistive RAM

UF: RRAM
ReRAM
BT: Random access memory
RT: Memristors
Phase change memory

Resistive transducers

BT: Transducers

Resistivity

USE: Conductivity

Resistivity measurement

USE: Conductivity measurement

Resistors

BT: Electronic components
RT: Electrical ballasts
Potentiometers
NT: Memristors
Switched capacitor

networks

Varistors

Resists

UF: Photoresists
BT: Materials

Resonance

RT: Cavity resonators
Dielectric resonator

antennas

Film bulk acoustic

resonators

Microstrip resonators
Optical resonators
Resonant inverters
Resonant tunneling devices
Resonator filters
Resonators
Vibrations
NT: Ferroresonance
Magnetic resonance
Resonance light scattering
Stochastic resonance

Resonance frequency

USE: Resonant frequency

Resonance light scattering

BT: Resonance
Spectroscopy
RT: Light scattering

Resonant circuits

USE: RLC circuits

Resonant converters

BT: Converters

Resonant frequency

UF: Resonance frequency
BT: Frequency
RT: Oscillators
Quantum mechanics
NT: Magnetic resonance

Resonant inverters

UF: Quasi-resonant inverters
Quasi-resonant invertors
Resonant invertors
BT: Inverters
RT: Power electronics
Resonance

Resonant invertors

USE: Resonant inverters

Resonant tunneling devices

UF: Resonant tunnelling
devices
Resonant-tunneling devices
Resonant-tunnelling
devices
BT: Tunneling
RT: Quantum well devices
Resonance
Single electron devices

Resonant tunnelling devices

USE: Resonant tunneling devices

Resonant-tunneling devices

USE: Resonant tunneling devices

Resonant-tunnelling devices

USE: Resonant tunneling devices

Resonator filters

BT: Filters
RT: Resonance

Resonators

BT: Amplifiers
RT: Acoustics
Resonance
Ring oscillators
Tuners
NT: Cavity resonators



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Split ring resonators

Resource allocation

USE: Resource management

Resource description framework

UF: RDF

BT: Semantic Web

Resource distribution

USE: Resource management

Resource management

UF: Allocation

Radio resource

management

Resource allocation

Resource distribution

Resource sharing

Resource utilisation

Resource utilization

Resources management

BT: Management

RT: Business process

integration

Business process

management

Cluster computing

Edge computing

Environmental engineering

Forestry

Multi-armed bandit problem

Operations research

System integration

NT: Circular economy

Elastic computing

Network resource

management

Resource virtualization

Serverless computing

Resource sharing

USE: Resource management

Resource utilisation

USE: Resource management

Resource utilization

USE: Resource management

Resource virtualization

BT: Resource management

Resources management

USE: Resource management

Respiratory diseases

USE: Pulmonary diseases

Respiratory medicine

USE: Pulmonology

Respiratory system

UF: Bronchi

BT: Anatomy

RT: Intubation

Pulmonary diseases

Pulmonology

Ventilators

NT: Larynx

Lungs

Trachea

Respirology

USE: Pulmonology

Response surface methodology

BT: Surface fitting

RT: Optimization methods

Restful API

BT: Application programming

interfaces

Software architecture

Resumes

BT: Writing

Retail price index

USE: Economic indicators

Retardants

BT: Production materials

RT: Inhibitors

NT: Flame retardants

Retina

UF: Retinal

BT: Eyes

RT: Ophthalmology

NT: Retinal vessels

Retinal

USE: Retina

Retinal vessels

BT: Retina

RT: Diabetic retinopathy



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Retinopathy

BT: Eye diseases
NT: Diabetic retinopathy

Retirement

BT: Human resource management

Retrieval augmented generation

UF: RAG
Retrieval-augmented generation
BT: Artificial intelligence

Retrieval-augmented generation

USE: Retrieval augmented generation

Retrofit

BT: Assembly

Retroreflectors

UF: Corner cube prisms
Corner-cube prisms
BT: Optical devices
RT: Road safety

Reverberation

BT: Acoustics

Reverberation chambers

BT: Electromagnetic compatibility

Reverse engineering

BT: Engineering - general
RT: Inverse design
Product development
Research and development

Reverse game theory

USE: Mechanism design

Reverse logistics

BT: Logistics

Reverse osmosis

BT: Chemical processes
RT: Desalination

Reverse teaching

USE: Education AND
Online services

Reversible computing

BT: Computational modeling
Time complexity

Reviews

BT: Writing
NT: Systematic literature review

Rewilding

BT: Reforestation

RF

USE: Radio frequency

RF exposure

USE: Radiofrequency exposure

RF identification

USE: Radiofrequency identification

RF interference

USE: Electromagnetic interference

RF micro-electro-mechanical systems

USE: Radiofrequency microelectromechanical systems

RF microelectromechanical systems

USE: Radiofrequency microelectromechanical systems

RF safety

USE: Radiofrequency safety

RF signals

BT: Signal processing

RFIC

USE: Radiofrequency integrated circuits

RFID

USE: Radiofrequency identification

RFID tags

BT: Radiofrequency identification
NT: Active RFID tags
Chipless RFID
Passive RFID tags



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Rhenium	BT:	Chemical elements			Mobility as a service Shared transport
Rheology	BT:	Fluid dynamics	<i>Ride-hailing</i>	USE:	Ride hailing
	RT:	Viscosity	<i>Ridesharing</i>	USE:	Shared transport
Rhetoric	BT:	Professional	Rigidity	BT:	Material properties
communication			Ring generators	BT:	Automatic testing Testing
Rheumatology	BT:	Medical specialties	Ring lasers	BT:	Lasers
	RT:	Immune system		RT:	Gyroscopes
		Musculoskeletal system		NT:	Fiber lasers
Rhodium	BT:	Chemical elements	Ring oscillators	BT:	Oscillators
<i>Rhombencephalon</i>	USE:	Hindbrain		RT:	Frequency control Jitter Klystrons Logic gates Masers Phase locked loops Resonators Tuning Voltage-controlled
Rhythm	BT:	Music	oscillators		
<i>Ribonucleic acid</i>	USE:	RNA	<i>Ring resonators</i>	USE:	Optical ring resonators
Ribosomes	BT:	Organelles	<i>RIS</i>	USE:	Reconfigurable intelligent
	RT:	Proteins	surfaces		
Ribs	BT:	Thorax	<i>RISC</i>	USE:	Reduced instruction set computing
Riccati equations	BT:	Equations	Risk analysis	BT:	Management
Rician channels	UF:	Rician fading		RT:	Accident prevention Accidents Decision making Occupational health Occupational safety Reliability Safety Technology social factors
		Rician fading channels			
	BT:	MIMO			
<i>Rician fading</i>	USE:	Rician channels			
<i>Rician fading channels</i>	USE:	Rician channels			
Ride hailing	UF:	Didi Ride-hailing Uber			
	BT:	Transportation			
	RT:	Mobile applications			



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NT: Venture capital
Fault trees
Risk management
Threat assessment

Risk assessment
USE: Risk management

Risk handling
USE: Risk management

Risk management
UF: Risk assessment
Risk handling
BT: Risk analysis
RT: Contract management
NT: Prevention and mitigation
Risk minimization
Risk mitigation

Risk minimisation
USE: Risk minimization

Risk minimization
UF: Risk minimisation
BT: Risk management
RT: Prevention and mitigation

Risk mitigation
BT: Risk management
RT: Prevention and mitigation

Rivers
BT: Geoscience
RT: Aquatic ecosystems
Excavation
Floods
Lakes
Sediments
Stormwater
Water
Water pollution
Water resources
Watersheds
Wetlands

RLC circuits
UF: Resonant circuits
BT: Circuits
Tunable circuits and
devices

RMS
USE: Root mean square

RNA
UF: Ribonucleic acid
BT: Biological cells
RT: Codons
Nucleic acids
Transcriptomics
NT: mRNA

RNN
USE: Recurrent neural networks

Road accidents
BT: Accidents
RT: Road safety
Road vehicles

Road bridges
USE: Structural panels

Road safety
BT: Roads
RT: Automated highways
Automotive engineering
Lane detection
Pedestrians
Retroreflectors
Road accidents
Vehicle-to-everything
NT: Lane departure warning
systems

Road side unit
UF: Roadside unit
BT: Relays
Vehicular ad hoc networks

Road traffic
BT: Road traffic control
NT: Pedestrians

Road traffic control
BT: Road transportation
Traffic control
RT: Automotive control
Pedestrians
NT: Road traffic

Road transportation
UF: Highways
BT: Land transportation
RT: Civil engineering
Global Positioning System
NT: Road traffic control
Roads
Traffic congestion



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Road vehicles

BT: Land vehicles
RT: Automotive control
Road accidents
Roads
NT: Automobiles
Motorcycles

Roadmaps (technology planning)

BT: Strategic planning
Technology forecasting

ROADMS

USE: Optical add-drop
multiplexers

Roads

BT: Road transportation
RT: Civil engineering
Excavation
Road vehicles
NT: Road safety

Roadside unit

USE: Road side unit

Roaming

BT: Wireless communication
RT: Dual band
GSM

Robot assisted surgery

USE: Medical robotics

Robot automobiles

USE: Autonomous automobiles

Robot cars

USE: Autonomous automobiles

Robot control

UF: Robotic control
BT: Control systems
Robots
RT: Force control
Formation control
Motion planning
Odometry
Trajectory tracking
NT: Robot motion

Robot kinematics

BT: Robots
RT: Myoelectric control
Proprioception

NT: Motion analysis

Robot learning

BT: Machine learning
Robots
RT: Artificial intelligence

Robot localization

BT: Motion analysis
RT: Robot sensing systems

Robot motion

UF: Robotic motion
BT: Robot control
NT: 2-DOF
3-DOF
5-DOF
6-DOF

Robot operating systems

USE: Operating systems

Robot programming

UF: Robotic programming
BT: Programming
Robots

Robot sensing systems

UF: Manipulator sensing
systems
Mobile robot sensing
systems
Robot sensor networks
BT: Robots
Sensor systems and
applications
RT: Multisensor systems
Robot localization
NT: Proprioception
Robot vision systems
Simultaneous localization
and mapping
Stereognosis
Tactile sensors

Robot sensor networks

USE: Robot sensing systems

Robot vision systems

UF: Manipulator vision systems
Mobile robot vision systems
BT: Robot sensing systems
RT: Image sensors
Imaging
Intelligent robots



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	Object detection		Nonlinear systems
	Object recognition		Servosystems
	Pattern recognition		Underactuated surface
	Stereo vision	vessels	
NT:	Visual servoing	NT:	Agricultural robots
<i>Robot-assisted surgery</i>			Amphibious robots
USE:	Medical robotics AND Surgery		Androids
<i>Robot-based assembly</i>			Aquatic robots
USE:	Robotic assembly		Automata
Robotic assembly			Autonomous robots
UF:	Assembly robots		Bio-inspired robotics
	Robot-based assembly		Cognitive robotics
BT:	Assembly systems		Collaborative robots
RT:	Robotics and automation		Computer vision
<i>Robotic control</i>			Continuum robots
USE:	Robot control		Educational robots
<i>Robotic motion</i>			Evolutionary robotics
USE:	Robot motion		Humanoid robots
<i>Robotic process automation</i>			Industrial robots
USE:	Intelligent automation		Intelligent robots
<i>Robotic programming</i>			Manipulators
USE:	Robot programming		Marine robots
Robotics and automation			Medical robotics
RT:	Image motion analysis		Military robotics
	Industrial Internet of Things		Mobile robots
	Robotic assembly		Orbital robotics
NT:	Animatronics		Parallel robots
	Automation		Quadrupedal robots
	Autonomous systems		Rescue robots
	Cloud robotics		Robot control
	Multi-robot systems		Robot kinematics
	Robots		Robot learning
Robots			Robot programming
BT:	Robotics and automation		Robot sensing systems
RT:	Assembly systems		Service robots
	Botnet		Snake robots
	Control equipment		Social robots
	Control systems		Soft robotics
	Cybernetics		Telerobotics
	Industrial control		Visual odometry
	Manufacturing automation		Wearable robots
	Materials handling		
	Mechanical variables		
control			
	Mechatronics		
		Robust control	
		UF:	Active disturbance rejection
		control	
		BT:	System analysis and design
		RT:	Disturbance observers
		Robust stability	
		BT:	Stability
		Robustness	
		BT:	Reliability
		RT:	Control systems
			Sensitivity



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	Stability		<i>Roof mounted photovoltaics</i>
	Uncertain systems	USE:	Building integrated photovoltaics
Rockets			
	BT: Propulsion		<i>Roof mounted solar cell arrays</i>
	RT: Engines	USE:	Building integrated photovoltaics
	Ground support		
Rocks		Root cause analysis	
	BT: Geology	BT:	Process planning
	RT: Aquifers	RT:	Failure analysis
	Continental crust		
	Oceanic crust	<i>Root kit</i>	
	Stratigraphy	USE:	Rootkit
	Tectonics		
	NT: Lava	Root mean square	
	Magma	UF:	RMS
Rodents			Root mean square error
	BT: Animals	BT:	Root mean square value
			Mathematics
			Statistics
Roentgenium			
	BT: Chemical elements	<i>Root mean square error</i>	
		USE:	Root mean square
Role playing games		<i>Root mean square value</i>	
	UF: Role-playing games	USE:	Root mean square
	BT: Video games		
Role transfer		Rootkit	
	BT: Organizational aspects	UF:	Root kit
		BT:	Malware
<i>Role-playing games</i>			
USE:	Role playing games	<i>Roots</i>	
		USE:	Poles and zeros
<i>Roller bearings</i>		Rotating machines	
USE:	Rolling bearings	BT:	Electric machines
		RT:	Brushes
Rolling bearings			Coils
	UF: Roller bearings		DC generators
	Rolling contact bearings		Synchronous motors
	Rolling element bearings		Windings
	BT: Mechanical bearings	NT:	Generators
	RT: Ball bearings		Hysteresis motors
			Induction machines
<i>Rolling contact bearings</i>			Induction motors
USE:	Rolling bearings		Micromotors
			Permanent magnet
<i>Rolling element bearings</i>		machines	
USE:	Rolling bearings		Reluctance machines
			Servomotors
Rollover			Standby generators
	BT: Vehicle dynamics		
<i>ROM</i>			
USE:	Read only memory		



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Rotation measurement

UF: Rotation representation
BT: Mechanical variables
measurement

Rotation representation

USE: Rotation measurement

Rotational measurement

USE: Velocity control

Rotational speed

USE: Velocity control

Rotors

BT: Electric machines
Machine components

Rough sets

BT: Set theory

Rough surfaces

BT: Surfaces
RT: Polishing machines
Surface roughness
Terrain factors
NT: Corrugated surfaces

Round robin

BT: Scheduling algorithms

Roundoff errors

BT: Finite wordlength effects
RT: Error analysis
Noise

Routing

BT: Communication systems
RT: Multicast communication
Network function

virtualization

Routing protocols
Soft switching
Virtual links
NT: Wavelength routing

Routing protocols

BT: Protocols
RT: Border Gateway Protocol
Internet
Land mobile radio
Mobile communication
Multicast protocols
Multiprotocol label

switching

Routing

Wireless access points
Virtual links

RPA

USE: Remotely piloted aircraft

RPI

USE: Economic indicators

RPS

USE: Renewable Portfolio
Standard

RRAM

USE: Resistive RAM

RRI

USE: Rail to rail inputs

RRO

USE: Rail to rail outputs

RSSI

USE: Received signal strength
indicator

RTL

USE: Register transfer level

Rubber

BT: Insulators
RT: Rubber industry
Rubber products

Rubber industry

BT: Manufacturing industries
RT: Chemical industry
Rubber
Rubber products

Rubber products

BT: Manufactured products
RT: Hoses
Rubber
Rubber industry
Wastewater treatment
NT: Tires

Rubidium

BT: Chemical elements

Rule based systems

USE: Knowledge based systems



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Runtime

BT: Software engineering
RT: Programming
NT: Dynamic compiler
Runtime environment

Runtime environment

BT: Runtime

Runtime library

BT: Formal languages

Rural areas

BT: Geography
RT: Macrocell networks
Public infrastructure

Ruthenium

BT: Chemical elements

Rydberg atoms

UF: Rydberg sensors
BT: Electric variables
measurement
Physics

Rydberg sensors

USE: Rydberg atoms

S Box

UF: S-Box
Substitution box
BT: Cryptography
RT: Ciphers

S parameters

USE: Scattering parameters

S-Box

USE: S Box

S-parameters

USE: Scattering parameters

S/N

USE: Signal to noise ratio

SaaS

USE: Software as a service

Safety

BT: Industry applications
Product safety engineering
RT: Alarm systems
Control system security

Electric shock
Environmental factors
Explosions
Eye protection
Fires
Hazardous areas
Occupational health
Preventive maintenance
Protective clothing
Radiation effects
Radioactive materials
Radioactive pollution
Risk analysis
Aerospace safety
Disasters
Domestic safety
Emergency services
Explosion protection
Fire safety
Food safety
Hazards
Health and safety
Marine safety
Product safety
Protection
Public security
Radiation safety
Safety devices
Safety management
Vehicle safety

Safety devices

BT: Safety
RT: Accident prevention
Alarm systems
Helmets
Railway safety
Smoke detectors
Eye protection
Fire extinguishers
Protective clothing

Safety in the home

USE: Domestic safety

Safety management

BT: Management
Safety
RT: Dependability management

SAGIN

USE: Space-air-ground
integrated networks



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Sagnac interferometers

BT: Interferometry

Sales promotion

USE: Promotion - marketing

Saliency detection

BT: Image processing
RT: Feature detection
Feature extraction
Visual systems

Salinity (geophysical)

BT: Soil measurements
RT: Geochemistry
Ocean salinity
Sea measurements
Sea surface salinity

Salivary glands

UF: Parotid
BT: Glands
Stomatognathic system

Salmonella

BT: Bacteria

Salt

BT: Chlorine compounds
Minerals

Samarium

BT: Metals
NT: Samarium alloys
Samarium compounds

Samarium alloys

UF: Samarium cobalt
BT: Samarium

Samarium cobalt

USE: Samarium alloys

Samarium compounds

BT: Samarium

Sampled data circuits

UF: Sampled-data circuits
BT: Circuits

Sampled data systems

BT: Discrete-time systems

Sampled-data circuits

USE: Sampled data circuits

Sampling methods

BT: Statistics
RT: Signal sampling
Surveys
NT: Compressed sensing
Nonuniform sampling
Synthetic data

SAN

USE: Storage area networks

Sandblasting

BT: Surface treatment
RT: Surface roughness

Sandwich structures

BT: Structural shapes
RT: Honeycomb structures
Lightweight structures
Sheet materials
Structural panels
Thin wall structures

Sanitary engineering

BT: Engineering - general
RT: Environmental
management

Sewage treatment
Waste disposal
Waste management
Waste materials
Wastewater
Wastewater treatment
Water pollution

SAR

USE: Specific absorption rate

AND

Synthetic aperture radar

SAR imaging

USE: Radar polarimetry

SAS

USE: Synthetic aperture sonar

Satellite antennas

BT: Antennas

Satellite born radar

USE: Spaceborne radar

Satellite borne radar

USE: Spaceborne radar



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Satellite broadcasting

UF: DBS
Direct broadcast satellites
Satellite broadcasts
BT: Broadcasting
Satellite communications
RT: Global Positioning System

Satellite broadcasts

USE: Satellite broadcasting

Satellite communications

UF: Communication satellites
BT: Communication systems
Radio communication
RT: Artificial satellites
Convolutional codes
Global Positioning System
Handover
Radiation hardening

(electronics)

Space-air-ground

integrated networks

NT: Transponders
Downlink
Satellite broadcasting
Satellite ground stations
Uplink

Satellite constellations

UF: Starlink
BT: Satellite navigation systems

Satellite ground stations

BT: Communication systems
Radio communication
Satellite communications
RT: Communication equipment

Satellite images

UF: Aerial photography
Earth observation images
Satellite photography
Spaceborne photography
BT: Earth Observing System
RT: Sentinel-2
Sentinel-3
Sentinel-5P

Satellite navigation systems

BT: Navigation
RT: Radio navigation
Time dissemination
NT: Global Positioning System

Global navigation satellite
system
Satellite constellations

Satellite photography

USE: Satellite images

Satellite-terrestrial integrated networks

USE: Space-air-ground
integrated networks

Satellite-terrestrial networks

USE: Space-air-ground
integrated networks

Satellites

BT: Solar system
RT: Artificial satellites
NT: GOSAT
Geostationary satellites
Moon
Small satellites

Saturation detection

USE: Feedback

Saturation magnetisation

USE: Saturation magnetization

Saturation magnetization

UF: Saturation magnetisation
BT: Magnetization processes
RT: Magnetic fields
Magnets

Saturn

BT: Planets

Savanna

USE: Grasslands

SAW

USE: Weighted sum model

SAW filters

BT: Electromechanical devices

Sawing

BT: Machining
RT: Sawing machines

Sawing machines

BT: Machine tools
RT: Sawing



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<i>SBTi</i>		Electromagnetic scattering Light scattering Particle scattering
USE:	Science based targets	
SCADA systems		
UF:	Supervisory control and data acquisition systems	
	Supervisory control and data-acquisition systems	
BT:	Control systems Power system control Supervisory control	
RT:	Load monitoring Substation automation	
Scalability		
BT:	System analysis and design	
RT:	Extensibility	
Scalp		
BT:	Head	
Scandium		
BT:	Chemical elements	
Scanning electron microscopy		
UF:	SEM	
BT:	Electron microscopy	
RT:	Electron backscatter diffraction	
	Electron beam applications Particle scattering	
Scanning microwave microscopy		
BT:	Microscopy	
RT:	Atomic force microscopy	
Scanning probe data storage		
BT:	Memory	
Scanning probe microscopy		
BT:	Microscopy	
NT:	Scanning thermal microscopy	
Scanning thermal microscopy		
BT:	Scanning probe microscopy	
Scattering		
UF:	Backscattering Wave scattering	
BT:	Propagation	
RT:	Reflection Scattering parameters	
NT:	Acoustic scattering Brillouin scattering	
Scattering parameters		
UF:	S parameters S-parameters	
BT:	System analysis and design	
RT:	Circuits Scattering	
Scatternets		
USE:	Personal area networks	
Scenario generation		
UF:	Scenario generators	
BT:	Stochastic processes	
RT:	Autonomous driving	
Scenario generators		
USE:	Scenario generation	
Scene analysis		
USE:	Image analysis	
Scene classification		
BT:	Image classification	
Schedules		
BT:	Planning	
RT:	Scheduling	
Scheduling		
BT:	Organizational aspects Production control Project engineering Materials requirements	
RT:		
planning		
	Queueing analysis Schedules Statistics Synchronization	
NT:	Adaptive scheduling Dynamic scheduling Job shop scheduling Single machine scheduling	
Scheduling algorithms		
BT:	Optical fiber communication Processor scheduling	
NT:	Round robin	
Schizophrenia		
BT:	Diseases	
RT:	Mental disorders	



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Scholarships

BT: Educational programs

Schools

USE: Educational institutions

Schottky barriers

UF: Schottky contacts

BT: Semiconductor-metal

interfaces

RT: MESFETs
Schottky diodes

Schottky contacts

USE: Schottky barriers

Schottky diode

USE: Schottky diodes

Schottky diodes

UF: Schottky diode

BT: Diodes
Semiconductor devices
Semiconductor diodes
RT: Schottky barriers
Semiconductor-metal

interfaces

Schottky FETs

USE: MESFETs

Schottky gate FET

USE: Schottky gate field effect
transistors

Schottky gate field effect transistors

UF: Schottky gate FET
BT: Field effect transistors

Schrodinger equation

BT: Quantum mechanics
RT: Electrons

Science - general

RT: Econophysics
Neurophysiology
Research and development
STEM
NT: Astronomy
Atmospheric science
Biology
Chemistry
Electricity
Environmental science
Epidemiology

Geoscience
Life sciences
Metrology
Neuroscience
Paleontology
Physics
Social sciences
Thermodynamics

Science based targets

UF: SBTi
BT: Environmental economics
RT: Climate change

science technology engineering and math
USE: STEM

Science technology engineering mathematics
USE: STEM

science, technology, engineering, and math
USE: STEM

Scientific computing

UF: Computational science
BT: Computer applications

Scientific publishing

BT: Publishing
RT: Systematic literature review

Scintillation counters

BT: Measurement
Nuclear and plasma

sciences

NT: Solid scintillation detectors

Scintillators

BT: Radiation effects
RT: Luminescence

SCM supply chains

USE: Supply chain management

Scoliosis

BT: Medical conditions
Spine

Scooters

USE: Motorcycles

SCR

USE: Thyristors



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Screws

USE: Fasteners

Scrubbers

USE: Materials handling

Scrum (Software development)

BT: Agile software development
RT: Agile project management
Feedback
Project management
Software development

management

SDH

USE: Synchronous digital

hierarchy

SDHTs

USE: MODFETs

SDN

USE: Software defined

networking

SDRAM

UF: Synchronous DRAM
Synchronous dynamic

random access memory

BT: Random access memory

Sea animals

USE: Marine animals

Sea coast

BT: Oceans

Sea floor

UF: Seafloor
BT: Oceans
RT: Aquatic ecosystems
Sediments
NT: Bathymetry
Sea floor roughness

Sea floor roughness

BT: Sea floor

Sea ice

BT: Ice
RT: Oceans

Sea jellies

USE: Jellyfish

Sea level

BT: Oceans
RT: Ocean circulation
NT: Sea level rise

Sea level rise

BT: Sea level
RT: Global warming

Sea lions

USE: Marine animals

Sea measurements

UF: Current measurement
(water)
BT: Geophysical measurements
RT: Oceans
Remote sensing
Salinity (geophysical)
Sonar measurements
Storm surge
NT: Geoacoustic inversion

Sea ports

USE: Seaports

Sea state

BT: Measurement
Ocean waves
RT: Marine navigation
Sea surface roughness

Sea surface

BT: Oceans
RT: Surface waves
Wind
NT: Sea surface roughness
Sea surface salinity

Sea surface roughness

BT: Sea surface
RT: Sea state

Sea surface salinity

BT: Sea surface
RT: Salinity (geophysical)

Sea surface temperature

UF: Ocean surface temperature
BT: Ocean temperature
RT: El Nino
El Nino-Southern

Oscillation



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<i>Sea vegetation</i>		Metaheuristics	
USE:	Marine vegetation		
<i>Seafloor</i>		<i>Seat belts</i>	
USE:	Sea floor	USE:	Belts
<i>Sealants</i>		Seaweed	
USE:	Sealing materials	UF:	Seaweeds
		BT:	Algae
			Marine vegetation
Sealing materials		<i>Seaweeds</i>	
UF:	Sealants	USE:	Seaweed
BT:	Joining materials		
RT:	Seals	<i>Sebaceous glands</i>	
Seals		USE:	Sebaceous glands
BT:	Mechanical products	Sebaceous glands	
RT:	Packaging	UF:	Sebaceous glands
	Sealing materials	BT:	Glands
	Structural rings		Skin
NT:	Gaskets	Second Life	
	Hermetic seals	BT:	Social networking (online)
Seaports		<i>Secondary cells</i>	
UF:	Sea ports	USE:	Batteries
BT:	Industrial facilities	<i>Secondary electron emission</i>	
	Transportation	USE:	Electron emission
RT:	Freeports	Secondary generated hot electron injection	
	Marine transportation	UF:	Secondary generated hot-electron injection
	Marine vehicles	BT:	Hot carrier injection
<i>Search and rescue robots</i>		<i>Secondary generated hot-electron injection</i>	
USE:	Rescue robots	USE:	Secondary generated hot electron injection
Search engines		<i>Secondary ion emission</i>	
UF:	Google	USE:	Ion emission
BT:	Information retrieval	Secure storage	
Search methods		BT:	Material storage
BT:	Information retrieval	RT:	Security
RT:	Cross modal retrieval	Security	
	Genetic algorithms	BT:	Industry applications
	Gradient methods	RT:	Anti-virus software
	Nearest neighbor methods		Biometric authentication
	Optimization methods		Biometrics
	Ranking (statistics)		Bring your own device
NT:	Keyword search		Business continuity
	Metasearch		Identification of persons
	Search problems		Information leakage
	Semantic search		
	Web search		
Search problems			
BT:	Search methods		
RT:	Artificial bee colony		
algorithm			



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function	Malware	BT:	Geoscience
	Physical unclonable	RT:	Lakes
	Protection		Rivers
	Public security		Sea floor
	Secure storage		Sedimentation
	Surveillance		Soil
NT:	Access control	Seebeck effect	
	Alarm systems	USE:	Thermoelectricity
	Capability-based security		
	Computer security	Seeds (agriculture)	
	Control system security	BT:	Crops
	Cryptography	RT:	Agriculture
	Data security		Botany
	Digital signatures		Ecology
	Food security		Food products
	Hardware security		Genetic engineering
	Information security		Soil
	Network security	NT:	Pollination (plants)
	Power system security		
	Reconnaissance	Seismic measurements	
	Security management	UF:	Seismic visualization
	Terrorism	BT:	Geophysical measurements
	Watermarking	RT:	Acoustic measurements
	Zero Trust		Seismology
Security information and event management			
UF:	SIEM	Seismic retrofitting	
	Security management	USE:	Earthquake engineering
BT:	Cyberattack		
	Data security	Seismic visualization	
RT:	Event detection	USE:	Seismic measurements
	Information security		
	Threat assessment	Seismic waves	
		BT:	Waves
		RT:	Acoustic waves
			Earthquakes
			Elastodynamics
			Explosions
			Seismology
			Shock waves
Security management			
BT:	Management		
	Security		
RT:	Hardware security		
	Security information and		
NT:			
event management			
Security of data			
USE:	Data security	Seismocardiography	
		BT:	Cardiography
		RT:	Vibrations
Security of hardware			
USE:	Hardware security	Seismology	
		BT:	Geophysics
		RT:	Earthquake engineering
			Earthquakes
			Seismic measurements
			Seismic waves
Sedimentation			
BT:	Geological processes		
	Erosion		
RT:	Sediments		
			Well logging
Sediments			
UF:	Mud	Selective laser sintering	
		USE:	Laser sintering



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Selectively doped heterojunction transistors
USE: MODFETs

Selenium

BT: Chemical elements

Self managing systems

USE: Autonomous systems

Self organising feature maps

USE: Self-organizing feature maps

Self organizing feature maps

USE: Self-organizing feature maps

Self organizing maps

USE: Self-organizing feature maps

Self organizing networks

USE: Self-organizing networks

Self replicating machines

USE: Self-replicating machines

Self service

USE: Self-service

Self supervised learning

USE: Self-supervised learning

Self testing

USE: Automatic testing

Self-assembly

BT: Bio-inspired materials
Nanotechnology

RT: Biological cells
Programming
Semiconductor device

manufacture

NT: Thin films
Electrostatic self-assembly

Self-aware

BT: Cognition

Self-consistent field theory

USE: Mean field theory

Self-driving automobiles

USE: Autonomous automobiles

Self-driving car

USE: Autonomous automobiles

Self-dynamic voltage scaling

USE: Dynamic voltage scaling

Self-managing systems

USE: Autonomous systems

Self-organizing feature maps

UF: Kohonen maps
SOM
Self organising feature maps

maps

Self organizing feature maps

maps

Self organizing maps

Self-organizing maps

BT: Artificial neural networks

RT: Feedforward neural networks

networks

Knowledge acquisition

Self-organizing maps

USE: Self-organizing feature maps

Self-organizing networks

UF: Self organizing networks

BT: Wireless networks

Self-replicating machines

UF: Self replicating machines

BT: Nanotechnology

Self-service

UF: Self service

BT: Consumer behavior

Customer services

Self-study courses

BT: Educational programs

Self-supervised learning

UF: Self supervised learning

BT: Learning systems

RT: Large language models

Supervised learning

Unsupervised learning

NT: Contrastive learning

Self-testing

USE: Built-in self-test



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Self-tuning regulators

USE: Adaptive control

SEM

USE: Scanning electron microscopy

Semantic communication

BT: Communication systems
Information retrieval
Semantics

Semantic search

BT: Search methods
Semantics
RT: Context awareness
Natural language

processing

Ontologies
Query processing
Semantic Web

Semantic segmentation

BT: Image segmentation
RT: Deep learning
Instance segmentation

Semantic technology

BT: Information technology
Semantics
RT: Data models
Encoding
Natural language

generation

Natural language

processing

Semantic Web

Semantic triple

USE: Triples (Data structure)

Semantic Web

UF: Web 3.0
BT: Internet
RT: Artificial intelligence
Content management
Data models
Decentralized identity
Distributed computing
Document handling
Knowledge management
Linked data
Markup languages
Ontologies
Open data

Ranking (statistics)
Semantic search
Semantic technology
OWL
Resource description

NT:

framework

Semantics

BT: Semiotics
RT: Multilingual
Natural language

processing

Professional

communication

Sign language
Translation

NT: Semantic communication
Semantic search
Semantic technology

Semi-duplex systems

USE: Half-duplex system

Semi-insulating materials

USE: Semiconductor materials

Semi-supervised learning

USE: Semisupervised learning

Semiconductivity

BT: Conductivity
Electron devices
RT: Charge carriers

Semiconductor alloys

USE: Semiconductor materials

Semiconductor charge carriers

USE: Charge carrier processes

Semiconductor controlled rectifiers

USE: Thyristors

Semiconductor counters

UF: Junction detectors
BT: Semiconductor devices
RT: Position sensitive particle

detectors

Semiconductor detectors

BT: Detectors
Semiconductor devices
RT: Absorption
Particle charging



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Semiconductor device breakdown

BT: Failure analysis
 RT: Semiconductor device
 reliability
 testing
 Tolerance analysis

Semiconductor device doping

UF: Semiconductor doping
 BT: Semiconductor device
 manufacture
 RT: Doping
 Semiconductor materials

Semiconductor device manufacture

UF: Semiconductor
 manufacturing
 BT: Electronic equipment
 manufacture
 RT: Fiducial markers
 Gettering
 Ion implantation
 Microassembly
 Micromachining
 Nanotechnology
 Self-assembly
 Semiconductor devices
 Surface cleaning
 Surface contamination
 NT: Diffusion processes
 Flip-chip devices
 High-k gate dielectrics
 Physical unclonable
 function
 doping
 layers
 Semiconductor device
 Semiconductor epitaxial
 Semiconductor growth
 Silicidation
 Wafer bonding

Semiconductor device measurement

BT: Measurement
 RT: Depth measurement
 Refractive index
 Semiconductor device
 noise
 reliability
 testing
 Semiconductor device

Semiconductor device modeling

UF: Semiconductor device
 models
 BT: Modeling
 Semiconductor devices
 RT: Semiconductor device
 noise

Semiconductor device models

USE: Semiconductor device
 modeling

Semiconductor device noise

BT: Semiconductor devices
 RT: Integrated circuit noise
 Semiconductor device
 measurement
 modeling
 Semiconductor device

Semiconductor device packaging

BT: Components, packaging,
 and manufacturing technology
 RT: Integrated circuit packaging
 Semiconductor devices

Semiconductor device reliability

BT: Reliability
 RT: Electrostatic discharge
 protection
 breakdown
 measurement
 Semiconductor device
 Semiconductor device

Semiconductor device testing

BT: Testing
 RT: Semiconductor device
 breakdown
 measurement
 Semiconductor device

Semiconductor devices

UF: SIS devices
 (semiconductor)
 Semiconductor-insulator-
 semiconductor devices
 BT: Electron devices
 RT: Contacts
 Epitaxial growth
 Field effect transistors
 Heterojunction bipolar
 transistors
 Hot carriers
 Integrated circuits



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applications	Molecular beam	interfaces	Semiconductor-metal
	Photoconducting devices		Superluminescent diodes
	Physical unclonable		Varactors
function	Proton radiation effects	<i>Semiconductor doping</i>	
	Rapid thermal annealing	USE: Semiconductor device	
	Semiconductor device	doping	
manufacture	Semiconductor device	<i>Semiconductor electronics industry</i>	
packaging		USE: Electronics industry	
	Silicon-on-insulator		
	Varistors	Semiconductor epitaxial layers	
NT:	Flip-chip devices	UF: Silicon epitaxial layers	
	Gunn devices	BT: Semiconductor device	
	Hall effect devices	manufacture	
	Junctions	RT: Bipolar transistors	
	MIS devices		
	MONOS devices	Semiconductor films	
	P-i-n diodes	BT: Films	
	Piezoresistive devices	RT: Buffer layers	
	Power semiconductor	Dielectric thin films	
devices		Magnetic field induced	
	Power semiconductor	strain	
switches		Semiconductor growth	
	Quantum dots	Semiconductor materials	
	Quantum well lasers	Thick films	
	SONOS devices	Thin films	
	Schottky diodes		
	Semiconductor counters	Semiconductor growth	
	Semiconductor detectors	BT: Semiconductor device	
	Semiconductor device	manufacture	
modeling		RT: Buffer layers	
	Semiconductor device	Crystal growth	
noise		Epitaxial layers	
	Semiconductor diodes	Semiconductor films	
	Semiconductor lasers	Semiconductor materials	
	Semiconductor waveguides		
	Semiconductor-insulator	Semiconductor impurities	
interfaces		BT: Impurities	
	Silicon devices	RT: Charge carrier processes	
	Superluminescent diodes	Plasma immersion ion	
	Surface emitting lasers	implantation	
	Thermistors	Semiconductor materials	
	Transistors		
Semiconductor diodes		<i>Semiconductor industry</i>	
BT: Semiconductor devices		USE: Electronics industry	
RT: Diodes			
Magnetic field induced		Semiconductor laser arrays	
strain		BT: Semiconductor lasers	
	P-n junctions		
NT: P-i-n diodes		Semiconductor lasers	
Schottky diodes		UF: Injection lasers	
		Junction lasers	
		Laser diodes	



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applications	BT:	Diodes	Magnetic semiconductors
		Lasers	Organic semiconductors
		Semiconductor devices	Semiconductor
		Solid lasers	
	RT:	Molecular beam	superlattices
			Silicon
		Optical transmitters	Silicon germanium
		Refractive index	Substrates
	NT:	Laser tuning	Transition metal
		Quantum dot lasers	
		Quantum well lasers	Wide band gap
		Semiconductor laser arrays	
amplifiers		Semiconductor optical	
		Surface emitting lasers	
<i>Semiconductor manufacturing</i>			
manufacture	USE:	Semiconductor device	
Semiconductor materials			
semiconductors	UF:	Pseudobinary	
		Semi-insulating materials	
		Semiconductor alloys	
	BT:	Materials	
	RT:	Acoustoelectric effects	
		Charge carriers	
		Conducting materials	
		Crystals	
		Excitons	
		High-k dielectric materials	
		Photoconducting materials	
		Radiative recombination	
		Semiconductor device	
doping		Semiconductor films	
		Semiconductor growth	
		Semiconductor impurities	
		Semiconductor thin films	
		Silicon compounds	
		Tunneling	
	NT:	Amorphous semiconductors	
		Deep level transient	
spectroscopy			
materials		Elemental semiconductors	
		Gallium	
		Gallium arsenide	
		Germanium	
materials		II-VI semiconductor	
		III-V semiconductor	
materials		Indium gallium arsenide	
		Indium phosphide	
Semiconductor memory			
	UF:	Semiconductor storage	
	BT:	Memory	
	RT:	Integrated circuits	
	NT:	Integrated memory circuits	
Semiconductor nanostructures			
	BT:	Nanostructures	
Semiconductor nanotubes			
	BT:	Nanotubes	
Semiconductor optical amplifiers			
	UF:	SOA	
	BT:	Optical amplifiers	
		Semiconductor lasers	
	RT:	Optical transmitters	
		Quantum well lasers	
Semiconductor process modeling			
	BT:	Modeling	
	RT:	Circuit simulation	
<i>Semiconductor quantum wells</i>			
	USE:	Quantum wells	
Semiconductor radiation detectors			
	BT:	Radiation detectors	
<i>Semiconductor storage</i>			
	USE:	Semiconductor memory	
Semiconductor superlattices			
	BT:	Semiconductor materials	
		Superlattices	
Semiconductor thin films			
	BT:	Thin films	
	RT:	Epitaxial growth	
		Gallium	
		Germanium	
		Semiconductor materials	
		Silicon	



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dichalcogenides	Transition metal	RT:	Linguistics Natural language
Semiconductor waveguides		processing	Phonetics Professional
BT:	Semiconductor devices	communication	
Semiconductor-insulator interfaces		NT:	Pragmatics Semantics Syntactics
BT:	Semiconductor devices		
RT:	CMOSFETs MIM devices MIS devices MOS devices Silicon-on-insulator		
<i>Semiconductor-insulator-semiconductor devices</i>		Semisupervised learning	
USE:	Semiconductor devices	UF:	Semi-supervised learning
		BT:	Learning systems
Semiconductor-metal interfaces		RT:	Artificial intelligence Bayes methods Large language models Neural networks Pattern classification Reinforcement learning Supervised learning Unsupervised learning Weak supervision
UF:	Metal-semiconductor		
interfaces		<i>Senior citizens</i>	
BT:	Semiconductor diodes	USE:	Older adults
RT:	Magnetic field induced		
strain		<i>Sense and avoid</i>	
	Schottky diodes	USE:	Collision avoidance
	NT:		
	Schottky barriers		
<i>Semicustom integrated circuits</i>		Sense organs	
USE:	Application specific	BT:	Anatomy
integrated circuits		RT:	Somatosensory
		NT:	Ear Eyes Multisensory integration Nose Olfactory bulb Taste buds Visual systems
Semidefinite programming			
BT:	Convex functions		
RT:	Array signal processing Computational complexity		
<i>Semilunar valves</i>		Sensitivity	
USE:	Heart valves	BT:	Measurement
		RT:	Circuit analysis Control systems Robustness Tolerance analysis Sensitivity analysis
Seminars			
BT:	Educational programs		
<i>Semiology</i>		Sensitivity analysis	
USE:	Semiotics	BT:	Sensitivity
<i>Semiosis</i>			
USE:	Semiotics		
<i>Semiotic studies</i>		Sensitivity and specificity	
USE:	Semiotics	BT:	Biomedical measurement Medical diagnosis
Semiotics			
UF:	Semiology Semiosis Semiotic studies		
BT:	Communication symbols		



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Sensor arrays	BT:	Arrays	RT:	Capacitive transducers
applications	RT:	Sensor systems and	NT:	Magnetostrictive devices
	NT:	Wearable sensors		Odometry
		Sensor fusion		Wireless sensor networks
				Acoustic sensors
				Chemical and biological
Sensor fusion			sensors	
BT:	Sensor arrays			Electromechanical sensors
RT:	Active perception			Force sensors
	Kalman filters			Glucose sensors
	Multimodal sensors			Inertial sensors
	Wearable sensors			Infrared sensors
NT:	Multisensor systems	communication		Integrated sensing and
Sensor phenomena and characterization				Intelligent sensors
BT:	Sensors	sensors		Intracranial pressure
Sensor placement				Ionizing radiation sensors
BT:	Sensors			Magnetic sensors
				Mechanical sensors
Sensor systems				Multimodal sensors
BT:	Aerospace and electronic			Nanosensors
systems				Optical sensors
	Sensor systems and	photonic sensors		Optoelectronic and
applications				Pressure sensors
RT:	Navigation			Quantum sensing
	Tiny machine learning			Quantum sensors
NT:	Activity recognition			Sensor phenomena and
	Gunshot detection systems	characterization		
Sensor systems and applications				Sensor placement
BT:	Sensors	applications		Sensor systems and
NT:	Detectors			Soft sensors
	Electric sensing devices			Thermal sensors
	Leak detection			Thick film sensors
identification	Radiofrequency			Thin film sensors
	Robot sensing systems			Vision sensors
	Sensor arrays			Wearable sensors
	Sensor systems	<i>Sensors (image)</i>		
Sensorless control		USE:		Image sensors
BT:	Control systems	Sensory aids		
RT:	AC machines	BT:		Medical services
	DC machines	RT:		Assistive technologies
	Drives			Biomedical equipment
	Induction motors			Orthotics
transmission	Inductive power			Prosthetics
		NT:		Hearing aids
	Motor drives			
	Motors	Sentiment analysis		
Sensors		UF:		Opinion mining
UF:	Sun sensors	BT:		Computational linguistics



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processing
RT: Anxiety disorders
Emotion recognition
Information analysis

Sentinel-1
BT: Earth Observing System

Sentinel-2
BT: Earth Observing System
RT: Optical imaging
Satellite images
Spatial resolution

Sentinel-3
BT: Earth Observing System
RT: Atmospheric modeling
Environmental monitoring
Optical imaging
Satellite images

Sentinel-5 Precursor
USE: Sentinel-5P

Sentinel-5P
UF: Sentinel-5 Precursor
BT: Earth Observing System
RT: Atmospheric modeling
Optical imaging
Satellite images

Separation processes
BT: Materials science and
technology
NT: Electrophoresis
Fractionation
Particle separators

Separators
USE: Particle separators

Sepsis
BT: Medical conditions
RT: Immune system

September 11
USE: Terrorism

Sequence analysis
USE: Sequences

Sequences
UF: Digital sequences
Sequence analysis

BT: Mathematics
RT: Codes
NT: Binary sequences
Random sequences

Sequencing
USE: Sequential analysis

Sequential analysis
UF: Sequencing
BT: System analysis and design
RT: DNA sequencing
NT: Zero correlation zone

Sequential circuits
UF: Sequential logic circuits
BT: Circuits

Sequential diagnosis
BT: System analysis and design

Sequential logic circuits
USE: Sequential circuits

Sequential production
USE: Flow production systems

Serious games
BT: Games
Simulation

Serverless computing
BT: Cloud computing
Resource management
Servers

Servers
BT: Client-server systems
RT: Network function
virtualization
NT: Serverless computing
Web servers

Service composability
USE: Interoperability

Service computing
BT: Information technology
RT: Business
Cloud computing
Process design
Service-oriented
architecture
Service-oriented systems
engineering



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	Web services	Software engineering
NT:	Service level agreements	
Service function chaining		<i>Servo control</i>
UF:	Service function chains	USE: Servosystems
BT:	Software defined	<i>Servo-control</i>
networking		USE: Servosystems
<i>Service function chains</i>		Servomechanisms
USE:	Service function chaining	BT: Servomotors
		RT: Actuators
		Manipulators
Service level agreements		Servomotors
UF:	SLA	UF: Servos
BT:	Contracts	BT: Motors
	Service computing	Rotating machines
RT:	Internet	Servosystems
	Quality of service	NT: Servomechanisms
<i>Service oriented architecture</i>		<i>Servos</i>
USE:	Service-oriented	USE: Servomotors
architecture		
Service robots		Servosystems
BT:	Robots	UF: Servo control
RT:	Field robots	Servo-control
	Home automation	BT: Control equipment
	Manipulators	RT: Actuators
	Mobile robots	Manipulators
	Wearable robots	Motion control
NT:	Assistive robots	Motor drives
		Position control
Service-oriented architecture		Robots
UF:	SOA	Velocity control
	Service oriented	NT: Servomotors
architecture		
	Service-oriented	Session hijacking
architectures		UF: Cookie hijacking
BT:	Web services	Cookie side-jacking
RT:	Service computing	Cookie sidejacking
	Service-oriented systems	TCP session hijacking
engineering		BT: Cyberattack
NT:	Microservice architectures	RT: Computer security
		Privacy
<i>Service-oriented architectures</i>		TCP
USE:	Service-oriented	Web sites
architecture		
Service-oriented systems engineering		Set theory
UF:	SOSE	BT: Algebra
BT:	Systems engineering and	Mathematics
theory		RT: Boolean algebra
	Formal specifications	Maximum likelihood
	Service computing	estimation
	Service-oriented	NT: Fuzzy set theory
architecture		Fuzzy sets



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		Rough sets		Shape control	
<i>SEU</i>				BT:	Mechanical variables
	USE:	Single event upsets		control	
				RT:	Shape
Sewage treatment				Shape measurement	
	BT:	Waste handling		BT:	Measurement
	RT:	Fermentation		RT:	Shape
		Pollution			
		Pollution control		Shape memory alloys	
		Sanitary engineering		BT:	Alloying
		Sludge treatment		RT:	Actuators
		Water pollution			
<i>SF6</i>				<i>Shape memory material</i>	
	USE:	Sulfur hexafluoride		USE:	Smart materials
<i>SGIN</i>				<i>Shape memory technology</i>	
	USE:	Space-air-ground		USE:	Smart materials
integrated networks				Sharding	
SGML				BT:	Database systems
	UF:	Standard Generalized			Distributed processing
Markup Language				Share prices	
	BT:	Markup languages		BT:	Economic indicators
Shadow mapping				<i>Shared ledger</i>	
	UF:	Projective shadowing		USE:	Distributed ledger
		Shadowing			
	BT:	Computer graphics		<i>Shared mobility</i>	
	RT:	Three-dimensional displays		USE:	Shared transport
<i>Shadowing</i>				Shared transport	
	USE:	Shadow mapping		UF:	Bicycle sharing
Shafts					Bike sharing
	BT:	Machine components			Car pools
		Production			Car sharing
	RT:	Couplings			Car-sharing
		Gears			Carpools
		Machine tool spindles			Cycle sharing
		Mechanical power			Microtransit
transmission					Ridesharing
		Mechanical splines			Shared mobility
		Pistons			Vanpools
		Propellers		BT:	Sharing economy
		Torque converters			Transportation
	NT:	Camshafts		RT:	Mobility as a service
Shape					Ride hailing
	BT:	Graphics			Micromobility
	RT:	Geometry		Sharing economy	
		Pattern recognition		BT:	Economics
		Shape control		NT:	Shared transport
		Shape measurement			



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Sharks

BT: Fish

Shear testing

BT: Stress
Testing
RT: Failure analysis
Shearing

Shearing

BT: Materials processing
RT: Shear testing
Sheet metal processing

Sheet materials

BT: Materials
Structural shapes
RT: Sandwich structures
Sheet metal processing
Structural panels
Thin wall structures

Sheet metal processing

BT: Manufacturing systems
RT: Blanking
Embossing
Punching
Shearing
Sheet materials

Shewhart charts

USE: Control charts

Shift registers

BT: Registers
RT: Logic circuits
NT: Linear feedback shift

registers

Shingled magnetic recording

BT: Magnetic recording

Shipbuilding industry

UF: Boat building industry
BT: Manufacturing industries
RT: Construction industry

Ships

USE: Marine vehicles

Shock

USE: Electric shock

Shock (mechanics)

BT: Mechanical factors

NT: Thermal shock

Shock absorbers

UF: Dampers
BT: Suspensions (mechanical systems)
RT: Automotive components
Damping
Springs
Vibration control

Shock waves

BT: Waves
RT: Aerodynamics
Seismic waves

Shoe manufacture

USE: Footwear industry

Shoes

USE: Footwear

Short circuit currents

USE: Short-circuit currents

Short-circuit currents

UF: Short circuit currents
BT: Current
RT: Molded case circuit breakers

Shortest path problem

UF: Shortest-path-problem
BT: Graph theory
RT: Traveling salesman problems

Shortest-path-problem

USE: Shortest path problem

Shoulder

BT: Extremities
NT: Axilla

Shunts (electrical)

BT: Electric current control
Photovoltaic effects

Si

USE: Silicon

SiC

USE: Silicon carbide



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Sick pay

USE: Employee welfare

Side channel attacks

USE: Side-channel attacks

Side imaging sonar

USE: Side-scan sonar

Side-channel attacks

UF: Side channel attacks

BT: Cryptography

Side-scan sonar

UF: Side imaging sonar

Sidescan sonar

BT: Sonar

RT: Bathymetry

Sidelink

BT: 5G mobile communication

Telecommunication traffic

RT: Cellular networks

Device-to-device

communication

Peer-to-peer computing

Sidescan sonar

USE: Side-scan sonar

SIEM

USE: Security information and
event management

Sieving

BT: Materials handling

RT: Filtration

NT: Molecular sieves

SiGe

USE: Silicon germanium

Sigma delta

USE: Sigma-delta modulation

Sigma-delta modulation

UF: Delta sigma

Sigma delta

BT: Delta modulation

Sign language

BT: Gesture recognition

RT: Assistive technologies

Deafness

Semantics

Translation

Signal analysis

UF: Waveform analysis

BT: Signal processing

RT: Autocorrelation

Blind source separation

Frequency-domain analysis

Pattern clustering

Power system faults

Rate distortion theory

Signal resolution

Speech analysis

Total harmonic distortion

Transient analysis

Wavelet transforms

NT: Discrete-event systems

Harmonic analysis

Parameter estimation

Signal mapping

Sound recognition

Spectral analysis

Signal classification

USE: Pattern classification

Signal constellation

USE: Constellation diagram

Signal de-noising

USE: Signal denoising

Signal decomposition

USE: Signal resolution

Signal denoising

UF: Signal de-noising

BT: Signal reconstruction

RT: Signal resolution

Signal restoration

Signal to noise ratio

Signal design

BT: Signal processing

Signal detection

UF: Detection (signal)

BT: Signal processing

RT: Blind source separation

Channel estimation

Correlators

Decision making

Demodulation

Pattern clustering

Receivers



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	Signal resolution		Pattern clustering
	Source separation		Prediction methods
	Time of arrival estimation		Random processes
NT:	Acoustic signal detection		Rate distortion theory
	Motion detection		Stability analysis
	Multiuser detection		Structure from motion
	Optical signal detection		Synapses
	Phase detection		System-on-chip
	Radar detection		Transforms
			Transversal filters
<i>Signal estimation</i>			Vectors
USE:	Estimation		Wavelet transforms
		NT:	Acoustic signal processing
<i>Signal flow graphs</i>			Adaptive signal processing
USE:	Flow graphs		Amplifiers
			Array signal processing
Signal generators			Attenuators
UF:	Function generators		Chirp
	Waveform generators		Convolution
BT:	Signal processing		Cyclostationary process
NT:	Noise generators		Decorrelation
	Pulse generation		Digital signal processing
			Dispersion
Signal integrity			Distortion
BT:	Signal processing		Error correction
			Fading channels
Signal mapping			Filters
BT:	Signal analysis		Frequency locked loops
			Geophysical signal
Signal processing		processing	
UF:	Vibrational signal		Limiting
processing			Local field potentials
RT:	Analog processing circuits		Modulation
	Antennas and propagation		Multidimensional signal
	Band-pass filters	processing	
	Bandwidth		Noise
	Biomedical computing		Optical signal processing
	Bit rate		Optical wavelength
	Correlators	conversion	
	Data processing		Phase locked loops
	Decoding		Pulse compression
	Deconvolution	methods	
	Digital signal processors		Pulse shaping methods
	Discrete Fourier transforms		Quantization (signal)
	Empirical mode		RF signals
decomposition			Radar signal processing
	Encoding		Received signal strength
	Estimation	indicator	
	Estimation theory		Recording
	Feature extraction		Signal analysis
	Fourier series		Signal design
	Gaussian noise		Signal detection
	Independent component		Signal generators
analysis			Signal integrity
	Matrix decomposition		Signal reconstruction



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Signal resolution
Signal restoration
Signal sampling
Signal synthesis
Source separation
Spectrogram
Tracking loops

Signal processing algorithms

BT: Algorithms

Signal quantisation

USE: Quantization (signal)

Signal quantization

USE: Quantization (signal)

Signal reconstruction

BT: Signal processing
RT: Inverse problems
Signal sampling
Signal to noise ratio
NT: Signal denoising

Signal representation

BT: Modeling
RT: Approximation methods
Wavelet transforms

Signal resolution

UF: Signal decomposition
BT: Signal processing
RT: Array signal processing
Signal analysis
Signal denoising
Signal detection
Spectral analysis
NT: Diversity reception

Signal restoration

BT: Signal processing
RT: Deconvolution
Distortion
Signal denoising

Signal sampling

BT: Signal processing
RT: Quantization (signal)
Sampling methods
Signal reconstruction

Signal separation

USE: Source separation

Signal synthesis

BT: Signal processing
RT: Speech synthesis

Signal to noise ratio

UF: S/N
SNR
Signal-to-noise ratio
Signal-to-noise-ratio
BT: Noise
RT: Filters
Noise figure
RAKE receivers
Signal denoising
Signal reconstruction
NT: PSNR

Signal-to-noise ratio

USE: Signal to noise ratio

Signal-to-noise-ratio

USE: Signal to noise ratio

Signaling block systems

USE: Block signalling

Signaling systems

USE: Communication system

signaling

Signalling block systems

USE: Block signalling

Signature detection

USE: Handwriting recognition

Signature verification

USE: Handwriting recognition

Silica

USE: Silicon dioxide

Silicidation

BT: Semiconductor device
manufacture

Silicides

BT: Silicon compounds

Silicon

UF: Si
Silicon materials
Siliconization
BT: Semiconductor materials
RT: Amorphous semiconductors



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	Elemental semiconductors		Silicon
	Epitaxial growth		Silicon-on-insulator
	Semiconductor thin films		Substrates
	Silicon devices		Transistors
	Silicon germanium		
	Silicon-on-insulator		
NT:	Amorphous silicon	<i>Silicon materials</i>	
	Porous silicon	USE:	Silicon
	Silicon alloys		
	Silicon photonics	Silicon nitride	
		BT:	Nitrogen
			Silicon compounds
Silicon alloys			
BT:	Silicon	<i>Silicon on insulator</i>	
RT:	Alloying	USE:	Silicon-on-insulator
NT:	Germanium silicon alloys		
		<i>Silicon on insulator technology</i>	
Silicon carbide		USE:	Silicon-on-insulator
UF:	SiC		
BT:	Silicon compounds	Silicon on sapphire	
		UF:	Silicon-on-sapphire
Silicon compiler		BT:	CMOS technology
BT:	Computer aided		Silicon-on-insulator
manufacturing		RT:	Substrates
RT:	Integrated circuit		
manufacture		Silicon photonics	
		BT:	Photonics
Silicon compounds			Silicon
BT:	Compounds	RT:	Optical fiber communication
RT:	Semiconductor materials		
NT:	Silicides	Silicon radiation detectors	
	Silicon carbide	BT:	Radiation detectors
	Silicon dioxide	RT:	Ionizing radiation
	Silicon nitride		
<i>Silicon controlled rectifiers</i>		Silicon-on-insulator	
USE:	Thyristors	UF:	SOI
			SOS (silicon on sapphire)
Silicon devices			Silicon on insulator
BT:	Semiconductor devices		Silicon on insulator
RT:	Doping	technology	
	Photonics		Silicon-on-insulator
	Silicon	technology	
		BT:	Circuits
Silicon dioxide		RT:	Double-gate FETs
UF:	Silica		Integrated circuits
BT:	Silicon compounds		Interface states
			Junctionless nanowire
<i>Silicon epitaxial layers</i>		transistors	
USE:	Semiconductor epitaxial		Proton radiation effects
layers			Semiconductor devices
			Semiconductor-insulator
Silicon germanium		interfaces	
UF:	SiGe		Silicon
BT:	Semiconductor materials		Silicon germanium
RT:	Germanium		Thin film circuits
		NT:	Silicon on sapphire



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Silicon-on-insulator technology

USE: Silicon-on-insulator

Silicon-on-sapphire

USE: Silicon on sapphire

Silicon-oxide-nitride-oxide-silicon

USE: SONOS devices

Siliconization

USE: Silicon

Silver

UF: Ag

BT: Metals

SIM card

UF: Subscriber identification

module

BT: Subscriber identity module
Integrated circuits
Mobile communication
Mobile handsets

SIMD

USE: Single instruction multiple

data

Similarity learning

BT: Supervised learning

SIMO

UF: Single input multiple output

systems

BT: Communication systems
RT: Antenna arrays
Diversity reception
Feedback
MIMO
MISO
Optical materials
Radio communication
SISO

Simple additive weighting

USE: Weighted sum model

Simple object access protocol

UF: SOAP

BT: Web services

Simulated annealing

BT: Mathematics

Optimization methods

RT: Annealing

Metaheuristics

Monte Carlo methods

Relaxation methods

Simulation

UF: Simulation results

BT: Modeling

RT: Application virtualization

Computer aided analysis

Computer graphics

Emulation

MATLAB

Monte Carlo methods

Numerical simulation

Reduced order systems

NT: Computer simulation

Digital simulation

Hardware-in-the-loop

simulation

Human in the loop

Medical simulation

Mixed reality

Quantum simulation

Serious games

Systems simulation

Simulation Program with Integrated Circuit Emphasis

USE: SPICE

Simulation results

USE: Simulation

Simultaneous localization and mapping

UF: SLAM

BT: Robot sensing systems

Simultaneous wireless information and power transfer

UF: SWIPT

BT: Data transfer

Power distribution

Wireless power transfer

RT: Internet of Things

Low power electronics

Single atom lasers

USE: Atom lasers

Single electron devices

BT: Circuits and systems

Electron devices

RT: Nanoscale devices

Nanotechnology

Resonant tunneling devices



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NT:	Single electron memory Single electron transistors	<i>Single-electron transistors</i> USE: Single electron transistors
Single electron memory	BT: Single electron devices NT: Hetero-nanocrystal memory	<i>Single-event upsets</i> USE: Single event upsets
Single electron transistors	UF: Single-electron transistors BT: Single electron devices	Single-photon avalanche diodes UF: SPAD single photon avalanche diodes BT: Avalanche photodiodes
Single event latchup	BT: Proton effects Radiation effects	<i>Single-wall carbon nanotubes</i> USE: Carbon nanotubes
Single event transients	BT: Ionization	Singular value decomposition BT: Matrices
Single event upsets	UF: SEU Single-event upsets BT: Ionization	Sintering UF: Frittagage BT: Manufacturing processes NT: Spark Plasma sintering
<i>Single input multiple output systems</i> USE: SIMO		<i>siot</i> USE: Social Internet of Things
<i>Single input single output systems</i> USE: SISO		<i>Siri</i> USE: Virtual assistants
Single instruction multiple data UF: SIMD Single instruction, multiple data BT: Parallel processing		<i>SIS devices (semiconductor)</i> USE: Semiconductor devices <i>SIS devices (superconductor)</i> USE: Superconducting devices
<i>Single instruction, multiple data</i> USE: Single instruction multiple data		SISO UF: Single input single output systems BT: Communications technology RT: Wireless communication Antenna arrays Diversity reception MIMO MISO Radio communication SIMO Transmitters
Single machine scheduling BT: Scheduling RT: Optimization methods		Six sigma BT: Total quality management RT: Quality assurance Quality control
<i>single photon avalanche diodes</i> USE: Single-photon avalanche diodes		
Single photon emission computed tomography UF: SPECT BT: Computed tomography RT: Cancer Collimators Phantoms Tumors		



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Size control

BT: Mechanical variables
control
RT: Thickness control

Size measurement

BT: Measurement
RT: Area measurement
Length measurement
Thickness measurement
Volume measurement
NT: Functional point analysis

Skeleton

BT: Musculoskeletal system
NT: Bones
Joints
Spine
Thorax

Skin

BT: Integumentary system
RT: Phototherapy
Psoriasis
NT: Dermis
Electronic skin
Epidermis
Sebaceous glands
Skin burns
Sweat glands

Skin burns

UF: burn injuries
burn wounds
BT: Injuries
Skin

Skin cancer

BT: Cancer
NT: Basal cell carcinoma
Melanoma
Squamous cell carcinoma

Skin effect

BT: Current density
RT: Conductors
Power systems
Resistance
System analysis and design

Skin neoplasms

BT: Neoplasms

Skull

BT: Head
RT: Bones

Skymions

BT: Solitons

SLA

USE: Service level agreements

Slabs

BT: Structural shapes

Slag

BT: Industrial waste
RT: Fly ash
Waste disposal
Waste management

SLAM

USE: Simultaneous localization
and mapping

SLD

USE: Superluminescent diodes

Sleep

BT: Brain
NT: Rapid eye movement sleep
Sleep apnea

Sleep apnea

UF: Sleep apnoea
Snore activity
Snore signals
Snoring
BT: Medical conditions
Sleep

Sleep apnoea

USE: Sleep apnea

Slideways (mechanical)

USE: Mechanical guides

Sliding mode control

UF: Sliding-mode control
BT: Control systems

Sliding-mode control

USE: Sliding mode control



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Slot antennas

BT: Antennas

Slot line components

UF: Slotline components

BT: Slot lines

Slot lines

UF: Slotline

BT: Planar transmission lines

NT: Slot line components

Slotline

USE: Slot lines

Slotline components

USE: Slot line components

Slow light

BT: Light sources

RT: Velocity measurement

SLR

USE: Systematic literature review

Sludge treatment

UF: Activated sludge process

BT: Waste handling

RT: Pollution control

Sewage treatment

Wastewater

Wastewater treatment

Slurries

BT: Waste materials

RT: Industrial waste

Small business technology transfer

BT: Technology transfer

Small cell networks

USE: Microcell networks

Small satellites

UF: Microsatellites

Miniaturized satellites

Nanosatellites

Smallsats

BT: Satellites

NT: CubeSat

Smallsats

USE: Small satellites

SMAP mission

UF: Soil Moisture Active

Passive mission

BT: Soil moisture

Smart actuators

USE: Intelligent actuators

Smart agriculture

UF: Digital agriculture

Intelligent agriculture

Smart farming

e-agriculture

BT: Agriculture

Digital systems

Information processing

RT: Agribusiness

Anaerobic digestion

Food industry

Food products

Food technology

Precision agriculture

Smart assistants

USE: Virtual assistants

Smart automobile parking

USE: Automated parking

Smart buildings

BT: Buildings

Smart cameras

BT: Cameras

Computer vision

Smart car parking

USE: Automated parking

Smart cards

BT: User interfaces

RT: Access control

Data processing

Smart charging

BT: Electric vehicle charging

Smart devices

Smart cities

BT: Intelligent structures

Urban areas

RT: Buildings

Construction industry

Cyber-physical systems

Energy informatics



January 2026 IEEE Thesaurus

<i>Smart clothing</i>			Energy informatics
USE:	Smart textiles		Microgrids
			Power distribution networks
			Smart meters
Smart contracts			Transactive energy
BT:	Contracts	NT:	Vehicle-to-grid
	Protocols		
RT:	Decentralized applications	<i>Smart health</i>	
		USE:	Smart healthcare
Smart devices			
BT:	Electronic equipment	Smart healthcare	
	Wireless communication	UF:	Smart health
RT:	Fourth Industrial Revolution		Smart medical services
	Personal voice assistants	BT:	Medical services
	Physical unclonable	RT:	Electronic healthcare
function			Internet of Medical Things
	Smart healthcare		Point of care
	Smart manufacturing		Smart devices
	Tactile Internet		
	Virtual assistants	NT:	Wearable Health Monitoring
	Wearable Health Monitoring	Systems	
Systems		Smart homes	
NT:	Smart charging	BT:	Buildings
	Smart glasses		Home automation
	Smart lighting		
<i>Smart education</i>		<i>Smart learning</i>	
USE:	Educational technology	USE:	Educational technology
<i>Smart elastomers</i>		Smart lighting	
USE:	Dielectric elastomers	BT:	Lighting
			Smart devices
<i>Smart fabrics</i>		Smart manufacturing	
USE:	Smart textiles	UF:	Smart factories
		BT:	Manufacturing
<i>Smart factories</i>		RT:	Fourth Industrial Revolution
USE:	Smart manufacturing		Intelligent manufacturing
<i>Smart farming</i>		systems	
USE:	Smart agriculture		Smart devices
			Smart materials
<i>Smart garments</i>		Smart materials	
USE:	Smart textiles	UF:	Shape memory material
			Shape memory technology
Smart glasses		BT:	Materials
UF:	Smartglasses	RT:	Austenite
BT:	Smart devices		Azobenzene
	Wearable devices		Dielectric elastomers
			Intelligent materials
Smart grids			Martensite
UF:	Smart microgrids		Metamaterials
	Smart power grids		Polycaprolactone
BT:	Power grids		Smart manufacturing
RT:	Cyber-physical systems		Smart transportation
	Energy Internet		



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processes
NT: Thermomechanical
Biomimetic materials
Smart textiles

Smart medical services
USE: Smart healthcare

Smart meters
BT: Meter reading
RT: Automatic meter reading
Smart grids

Smart microgrids
USE: Smart grids

Smart parking
USE: Automated parking

Smart phones
UF: Smartphones
BT: Mobile handsets
RT: Bring your own device
Fintech

Smart pixels
BT: Image processing
RT: Integrated optoelectronics
Optical switches

Smart power grids
USE: Smart grids

Smart sensors
USE: Intelligent sensors

Smart spaces
BT: Ergonomics
Sociotechnical systems

Smart structures
USE: Intelligent structures

Smart systems
BT: Systems support

Smart textiles
UF: Electronic textiles
Smart clothing
Smart fabrics
Smart garments
BT: Smart materials
RT: Wearable devices
NT: Textile antennas

Smart transportation
BT: Transportation
RT: Automated highways
Automated parking
Intelligent transportation

systems
Intelligent vehicles
Smart materials

Smart TV
BT: TV
RT: Internet

Smartglasses
USE: Smart glasses

Smartphones
USE: Smart phones

Smartwatch
USE: Wearable Health Monitoring
Systems

Smelting
BT: Materials processing
RT: Blast furnaces
Heat treatment
Melt processing
Metals industry
Refining

SMES
USE: Superconducting magnetic
energy storage

Smoke alarms
USE: Smoke detectors

Smoke detectors
UF: Smoke alarms
BT: Alarm systems
RT: Domestic safety
Fires
Ionization chambers
Safety devices
Zigbee

Smooth methods
USE: Smoothing methods

Smoothed particle hydrodynamics
USE: Fluid flow AND
Hydrodynamics



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Smoothing methods

UF: Smooth methods
BT: Mathematics

SMOS mission

UF: Soil moisture and ocean salinity
Soil moisture and ocean salinity mission
BT: Ocean salinity
Soil moisture

SMPS

USE: Switched mode power supplies

SMPTE

UF: Society of Motion Picture and Television Engineers
BT: Standards organizations

SMPTE Standards

BT: Standards publications

Sn

USE: Tin

Snake bots

USE: Snake robots

Snake robots

UF: Snake bots
Snakebots
BT: Robots

Snakebots

USE: Snake robots

Snore activity

USE: Sleep apnea

Snore signals

USE: Sleep apnea

Snoring

USE: Sleep apnea

Snow

BT: Precipitation
RT: Glaciers
Ice

SNR

USE: Signal to noise ratio

SNS devices

USE: Superconducting devices

Snubbers

BT: Power electronics

SOA

USE: Semiconductor optical amplifiers AND
Service-oriented architecture

SOAP

USE: Simple object access protocol

SOC

USE: System-on-chip

Soccer

USE: Sports

Social computing

BT: Federated learning
Internet
RT: Behavioral sciences
Community networks
Crowdsourcing
Social computing
Social networking (online)
NT: Cyberbullying
Persuasive systems
Social Internet of Things
Social intelligence

Social dynamics

BT: Sociology
NT: Opinion dynamics

Social economics

USE: Socioeconomics

Social engineering (security)

BT: Information security
RT: Human factors
Psychology
Social factors

Social factors

BT: Social implications of technology
RT: Bio-inspired computing
Corporate social responsibility
Digital divide



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	Digital humans	RT:	Cultural differences
	Food security		Digital intelligence
	Governmental factors		Social factors
	Inclusive language		
	International collaboration	Social Internet of Things	
	International relations	UF:	SocialNet of Things
	Philosophical		sIoT
considerations		BT:	Internet of Things
	Social Internet of Things		Social computing
(security)	Social engineering	RT:	Behavioral sciences
			Collaboration
	Social intelligence		Social factors
	Technology planning		Social networking (online)
NT:	Demography		
	Developing countries	Social manufacturing	
	Livelihood	BT:	Manufacturing processes
	Social manufacturing		Social factors
	Socioeconomics	RT:	Product development
	Technology social factors		Production management
Social groups		<i>Social media</i>	
BT:	Sociology	USE:	Social networking (online)
NT:	Millennials		
	Older adults	Social network theory	
		BT:	Sociotechnical systems
Social implications of technology		RT:	Complex networks
UF:	Orange technology		Social networking (online)
RT:	Cyberbullying		
	Cyberethics	<i>Social networking</i>	
	Digital divide	USE:	Social networking (online)
	Social robots		
	Socioeconomics	Social networking (online)	
	Technology acceptance	UF:	Facebook
model			Linkedin
	NT:		Myspace
	Cultural aspects		Online social networks
	Cultural differences		Reddit
	Environmental factors		Social media
	Ethical aspects		Social networking
	Ethics		Social networking services
	Globalization		Social networks
	International relations		Social-networks
	Peace technology		TikTok
	Philosophical		Twitter
considerations			WeChat
	Social factors	BT:	Information retrieval
	Sustainable development	RT:	Blogs
	Technology		Crowdsourcing
Social inequality			Electronic mail
BT:	Sociology		Internet
			Journalism
Social intelligence			Social Internet of Things
BT:	Behavioral sciences		Social computing
	Social computing		Social network theory
	Sociology		Web sites



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communication	NT:	Community networks Computer mediated	<i>Socio-technical systems</i>	USE:	Sociotechnical systems
		Cyberbullying Information diffusion Second Life	Socioeconomics	UF:	Social economics Socio-economics
				BT:	Economics Social factors
<i>Social networking services</i>	USE:	Social networking (online)		RT:	Corporate social
<i>Social networks</i>	USE:	Social networking (online)	responsibility		Human factors Social implications of
Social robots			technology		Sociotechnical systems
BT:		Human-robot interaction Robots	Sociology	BT:	Social sciences
RT:		Assistive robots Assistive technologies Educational robots Emotion recognition Humanoid robots Mobile robots Natural language		RT:	Collective intelligence
processing				NT:	Crowd dynamics Digital divide Social dynamics Social groups Social inequality Social intelligence
technology		Social implications of	Sociotechnical systems	UF:	Socio-technical systems
Social sciences				BT:	Organizations
BT:		Humanities Science - general		RT:	Data mesh Socioeconomics
RT:		Complex networks Graph drawing Network theory (graphs)		NT:	Smart spaces Social network theory
NT:		Anthropology Behavioral sciences Psychology Sociology	Sockets	BT:	Connectors
<i>Social-networks</i>	USE:	Social networking (online)	Sodium	BT:	Chemical elements
Socially responsible investing			<i>Sodium chloride</i>	USE:	Chlorine compounds
UF:		SRI	<i>Sodium nitrate</i>	USE:	Nitrogen compounds
BT:		Corporate social			
responsibility			Soft electronics	BT:	Electronic equipment
<i>SocialNet of Things</i>	USE:	Social Internet of Things		RT:	Flexible electronics Inorganic materials Wearable devices
<i>Society of Motion Picture and Television Engineers</i>	USE:	SMPTE	Soft lithography	UF:	Microcontact printing Replica molding Replica moulding
<i>Socio-economics</i>	USE:	Socioeconomics			



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BT:	Lithography	<i>Softening (metallurgical)</i>	
RT:	Nanolithography	USE:	Softening
	Nanopatterning		
Soft magnetic materials		<i>Softswitch</i>	
BT:	Magnetic materials	USE:	Soft switching
Soft robotics		Software	
UF:	Soft robots	UF:	Computer software
BT:	Robots		On-demand software
RT:	Assistive robots	BT:	Computers and information
	Biomimetics	processing	
	Flexible structures	RT:	Algorithms
	Grippers		Computer languages
	Medical robotics		Computer science
	Mobile robots		Courseware
	Wearable robots		Documentation
			Enterprise resource
<i>Soft robots</i>		planning	
USE:	Soft robotics		Firewalls (computing)
<i>Soft sensing</i>			Geospatial analysis
USE:	Soft sensors		Microprogramming
			Programming
Soft sensors			Software engineering
UF:	Data sources	NT:	Software protection
	Soft sensing		Software standards
	Software sensors		Anti-virus software
	Virtual sensing		Application software
	Virtual sensors		Closed box
BT:	Sensors		Embedded software
	Software		Freeware
RT:	Fuzzy systems		Glass box
	Intelligent sensors		Malware
	Kalman filters		Middleware
	Learning (artificial		Open source software
intelligence)		recognition	Optical character
	Neural networks		Privacy-invasive software
	Process control		Public domain software
	Process monitoring		Soft sensors
<i>Soft switch</i>			Software agents
USE:	Soft switching		Software as a service
			Software debugging
Soft switching			Software maintenance
UF:	Soft switch		Software packages
	Softswitch		Software performance
BT:	Telecommunication		Software quality
computing			Software reusability
RT:	Routing		Software safety
			Software systems
			Software tools
			System software
Softening		Software agents	
UF:	Softening (metallurgical)	BT:	Software
BT:	Materials processing	RT:	Artificial intelligence
RT:	Annealing		



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	Computer applications		Intelligent networks
	Distributed computing		Mobile computing
	Intelligent systems		Network function
	Knowledge based systems	virtualization	
	Learning systems		Network operating systems
	Mobile agents		Operating systems
NT:	Agent-based modeling		Protocols
	Autonomous agents		Virtual machining
	Botnet		Virtualization
	Intelligent agents	NT:	Service function chaining
			Virtual LAN
Software algorithms			
BT:	Algorithms	<i>Software defined networks</i>	
		USE:	Software defined
Software architecture		networking	
BT:	Software engineering	<i>Software defined radio</i>	
RT:	Distributed computing	USE:	Software radio
NT:	Client-server systems		
	Deep architecture		
	Dew computing		
	Microarchitecture	Software design	
	Representational state	BT:	Software debugging
transfer		RT:	Web design
	Restful API	NT:	Model driven engineering
			Usability
Software as a service		Software development management	
UF:	On demand software	UF:	Github
	On-demand software	BT:	Engineering management
	SaaS	RT:	Release engineering
	Software-as-a-service		Scrum (Software
BT:	Software	development)	
RT:	Cloud computing		Software product lines
	Information processing	NT:	Agile project management
			Agile software development
<i>Software assurance</i>			Continuous integration
USE:	Software reliability		DevOps
			Model-driven development
Software debugging		<i>Software documentation</i>	
BT:	Software	USE:	Documentation
RT:	Programming		
	Programming environments	Software engineering	
NT:	Software design	BT:	Computers and information
Software defined networking		processing	
UF:	SDN	RT:	Code refractoring
	Software defined networks		Functional point analysis
	Software-defined		Rapid prototyping
networking			Requirements engineering
BT:	Computer networks		Service-oriented systems
RT:	Application programming	engineering	
interfaces			Software
	Cloud computing		Static analysis
	Computer network		Systems Modeling
management		Language	
	Distributed processing		Visual BASIC



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engineering	NT:	Capability maturity model	Software protection	BT:	Copyright protection	
		Computer aided software			Legal factors	
		Formal verification		RT:	Digital rights management	
		Full stack			Intellectual property	
		Programming environments			Software	
		Release engineering				
		Runtime				
		Software architecture		Software prototyping	BT:	System analysis and design
Software libraries		Software libraries	Software quality			
		Software product lines		BT:	Software	
	BT:	Libraries		RT:	Algorithmic efficiency	
		Software engineering				
	RT:	Algorithms		Software radio	UF:	Reconfigurable radio
		MATLAB				Software defined radio
		Object oriented				Software-defined radio
	programming				Python	BT:
		Software reusability			Radio communication	
			RT:		Cellular radio	
					Code division multiplexing	
			Land mobile radio			
			Telecommunication			
Software maintenance			computing			
BT:	Software					
RT:	Software product lines					
Software measurement						
BT:	Measurement					
Software packages						
BT:	Software					
RT:	Computer applications					
computing		Power system analysis				
	NT:	EMTDC				
		PSCAD				
		SPICE				
	Software performance					
	BT:	Software				
	RT:	Algorithmic efficiency				
		Capability maturity model				
<i>Software piracy</i>			Software reliability			
USE:	Computer crime					
Software product lines						
BT:	Product development					
	Software engineering					
RT:	Software development					
management		Software maintenance				



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Software standards

BT: Standards categories
RT: ISO
ISO Standards
Software

RT: Geophysical measurements
Moisture measurement
Remote sensing
Soil
NT: Salinity (geophysical)

Software systems

BT: Software

Soil moisture

BT: Soil
NT: SMAP mission
SMOS mission

Software testing

BT: Testing
RT: Closed box
Glass box
NT: Combinatorial testing
Fuzzing

Soil Moisture Active Passive mission

USE: SMAP mission

Soil moisture and ocean salinity

USE: SMOS mission

Software tools

BT: Software
RT: Computer aided software

Soil moisture and ocean salinity mission

USE: SMOS mission

engineering

Programming
Programming environments
Visual BASIC
NT: Authoring systems

Soil pollution

BT: Land pollution
RT: Agriculture
Pesticides
Soil

Software-as-a-service

USE: Software as a service

Soil properties

BT: Soil

Software-defined networking

USE: Software defined

Soil remediation

BT: Pollution control
Soil
RT: Anaerobic digestion
Contamination
Environmental monitoring
Hazards

networking

Software-defined radio

USE: Software radio

SOI

USE: Silicon-on-insulator

Soil

BT: Geoscience
RT: Earth
Excavation
Sediments
Seeds (agriculture)
Soil measurements
Soil pollution
NT: Peat
Permafrost
Soil moisture
Soil properties
Soil remediation
Soil texture

Soil texture

BT: Soil

Solar cells

USE: Photovoltaic cells

Solar cooling

BT: Cooling

Solar eclipses

BT: Sun

Solar energy

BT: Energy resources
RT: Maximum power point

trackers

Solar heating
Solar power generation

Soil measurements

BT: Measurement



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NT:	Solar radiation Solar geoengineering Solar irradiance	RT:	Building integrated photovoltaics Microinverters Renewable Portfolio
Solar farms		Standard	
BT:	Solar power generation		Solar energy Solar powered vehicles Maximum power point
RT:	Solar panels	NT:	
Solar flares		trackers	Photovoltaic systems Solar farms Solar panels
BT:	Sun		
RT:	Sunspots		
<i>Solar generation</i>		Solar powered vehicles	
USE:	Solar power generation	BT:	Electric vehicles
Solar geoengineering		RT:	Battery powered vehicles Energy storage Solar power generation Traction motors Vehicle-to-grid
UF:	Solar radiation modification		
BT:	Geoengineering Solar energy		
Solar heating		Solar radiation	
BT:	Energy conversion Heating systems	BT:	Extraterrestrial phenomena
RT:	Phase change materials Solar energy Space heating	RT:	Solar energy Solar irradiance Space radiation
Solar irradiance		<i>Solar radiation modification</i>	
BT:	Solar energy	USE:	Solar geoengineering
RT:	Solar radiation		
<i>Solar modules</i>		Solar system	
USE:	Solar panels	BT:	Astronomy Kuiper belt Planets Satellites Sun
Solar panels		NT:	
UF:	Photovoltaic modules Solar modules	<i>Solar trackers</i>	
BT:	Photovoltaic systems Solar power generation	USE:	Solar tracking
RT:	Building materials Building services Microinverters Photovoltaic cells Renewable energy sources Solar farms Solar tracking	Solar tracking	
NT:	Backsheets	UF:	Solar trackers Sun trackers Sun tracking
<i>Solar polarimetry</i>		BT:	Tracking
USE:	Polarimetry	RT:	Solar panels
Solar power generation		Solar wind	
UF:	Solar generation	BT:	Sun
BT:	Power generation	RT:	Geomagnetic storms
		<i>Solder joints</i>	
		USE:	Soldering



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Soldering

UF: Solder joints
 BT: Assembly
 Fabrication
 Joining processes
 RT: Bonding processes
 Manufacturing
 Materials processing
 Soldering equipment
 NT: Brazing
 Flip chip solder joints
 Reflow soldering

Soldering equipment

UF: Soldering irons
 BT: Production equipment
 RT: Joining materials
 Joining processes
 Soldering

Soldering irons

USE: Soldering equipment

Solderless breadboard

USE: Breadboard

Solenoids

BT: Magnetic devices
 RT: Switches
 Transducers

Solid geometry

UF: Stereometry
 BT: Geometry

Solid insulation

UF: Solid insulators
 BT: Insulation

Solid insulators

USE: Solid insulation

Solid lasers

UF: Color center lasers
 Solid state lasers
 Solid-state lasers
 BT: Lasers
 RT: Thermal lensing
 Thermo-optical devices
 NT: Microchip lasers
 Quantum well lasers
 Semiconductor lasers
 Surface emitting lasers

Solid modeling

BT: Modeling
 RT: Digital twins
 Solid-state physics
 Virtual reality

Solid oxide electrolyzer cells

USE: Fuel cells

Solid permittivity

USE: Dielectric constant

Solid scintillation detectors

BT: Scintillation counters
 RT: Energy resolution
 Medical diagnostic imaging
 Spectroscopy

Solid state batteries

UF: Solid-state batteries
 BT: Batteries

Solid state circuit design

UF: Solid-state circuit design
 BT: Solid state circuits
 RT: Circuit synthesis

Solid state circuits

UF: Solid-state circuits
 RT: Circuits and systems
 Solid state drives
 NT: Circuit subsystems
 Circuit theory
 FET circuits
 Gate leakage
 Solid state circuit design
 Transistors

Solid state drives

UF: Solid-state drives
 BT: Digital storage
 RT: DRAM chips
 Flash memories
 Integrated memory circuits
 Solid state circuits

Solid state lasers

USE: Solid lasers

Solid state lighting

UF: Solid-state lighting
 BT: Lighting

Solid waste

USE: Waste materials



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Solid-state batteries

USE: Solid state batteries

Solid-state circuit design

USE: Solid state circuit design

Solid-state circuits

USE: Solid state circuits

Solid-state drives

USE: Solid state drives

Solid-state lasers

USE: Solid lasers

Solid-state lighting

USE: Solid state lighting

Solid-state physics

BT: Physics

RT: Materials science and
technology
Quantum mechanics
Solid modeling

Solids

BT: Materials

RT: Crystals
Materials science and

technology

NT: Young's modulus

Solitons

BT: Waves

NT: Optical solitons
Skyrmions

Solution design

BT: Systems engineering and
theory

Solution stack

USE: Full stack

Solvation

BT: Chemical reactions

RT: Hydrolysis

Solvents

BT: Chemical processes

RT: Methanol

SOM

USE: Self-organizing feature
maps

Soma

UF: Somata

BT: Neurons

RT: Brain

Somata

USE: Soma

Somatosensory

BT: Physiology

RT: Multisensory integration
Sense organs

Sonar

BT: Aerospace and electronic
systems

RT: Acoustic arrays
Chirp modulation
Ultrasonic transducers

NT: Side-scan sonar
Sonar applications
Sonar equipment
Synthetic aperture sonar

Sonar applications

BT: Sonar

RT: Sonar navigation

NT: Sonar detection
Sonar measurements

Sonar detection

BT: Acoustic signal detection
Sonar applications

RT: Echo sounders
Hydrophones
Reflectivity

Sonar equipment

BT: Sonar

NT: Echo sounders

Sonar measurements

BT: Sonar applications

RT: Remote sensing
Sea measurements

Sonar navigation

BT: Navigation

RT: Sonar applications

SONET

UF: Synchronous optical

network
BT: Communication standards
Digital communication



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ETSI Standards
Optical fiber communication
RT: Asynchronous transfer
mode
Synchronous digital
hierarchy
Transport protocols

Sonification

BT: Audio systems
Information processing

Sonogram

BT: Ultrasonography
RT: Spectrogram

SONOS devices

UF: Silicon-oxide-nitride-oxide-
silicon
BT: Semiconductor devices

Sorting

BT: Data handling
RT: Merging

SOS (*silicon on sapphire*)

USE: Silicon-on-insulator

SOSE

USE: Service-oriented systems
engineering

Sound recognition

BT: Pattern recognition
Signal analysis

Sound systems

USE: Audio systems

Source address validation

BT: Computer network
management
RT: Media Access Control

Source coding

BT: Data compression
Encoding
Information theory
RT: Rate distortion theory
Release engineering

Source location

USE: Position measurement

Source separation

UF: Signal separation
BT: Signal processing
RT: Adaptive signal detection
Array signal processing
Signal detection
NT: Blind source separation

Source signal equalizers

USE: Blind equalizers

South America

BT: Continents

South Pole

BT: Antarctica

Southern Ocean

USE: Antarctic Ocean

Space agencies

UF: Space programs
BT: Organizations
RT: Space exploration
Space missions
NT: European Space Agency
NASA

Space based radar

USE: Spaceborne radar

Space born radar

USE: Spaceborne radar

Space charge

BT: Charge carrier processes
Electrostatic processes
RT: Pulsed electroacoustic
methods
Vacuum technology

Space communications

BT: Telecommunications
NT: Deep-space
communications

Space cooling

BT: Cooling
RT: Buildings
Coolants
Refrigerants

Space debris

UF: Orbital debris
Space junk



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BT: Space waste
RT: Space technology
Meteorites
Meteoroids

Space diversity

USE: Spatial diversity

Space division multiplexing

UF: Space-division multiplexing
Spatial division multiplexing
Spatial multiplexing
BT: Multiplexing

Space exploration

UF: Space travel
BT: Space technology
RT: NASA
Space agencies
NT: Interplanetary exploration
Space missions

Space habitats

USE: Buildings AND
Space technology

Space heating

BT: Heating systems
RT: Building services
District heating
Gas appliances
Solar heating
Temperature control
Vents

Space junk

USE: Space debris

Space mapping

BT: Design optimization
Modeling

Space measurements

USE: Extraterrestrial
measurements

Space missions

BT: Space exploration
RT: Interplanetary exploration
NASA
Space agencies

Space phenomena

USE: Extraterrestrial phenomena

Space power stations

UF: Power stations (space)
BT: Space stations
RT: Power generation

Space programs

USE: Space agencies

Space radiation

BT: Radiation effects
RT: Ionization
Solar radiation

Space shuttles

BT: Space vehicles
RT: Aerospace safety

Space stations

BT: Artificial satellites
NT: International Space Station
Space power stations

Space technology

UF: Space habitats
BT: Aerospace engineering
RT: Artificial satellites
Extraterrestrial phenomena
Field programmable analog

arrays

NASA
Space vehicles
NT: Payloads
Space debris
Space exploration

Space travel

USE: Space exploration

Space vector modulation

BT: Modulation

Space vector pulse width modulation

UF: SVPWM
BT: Pulse width modulation
RT: AC motors
Converters
DC motors

Space vehicle electronics

USE: Aerospace electronics

Space vehicle instrumentation

USE: Aerospace electronics



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Space vehicle navigation

USE: Space vehicles

Space vehicles

UF: Planetary landers
Space vehicle navigation
BT: Vehicles
RT: Aerospace accidents
Aerospace control
Aerospace electronics
Aerospace materials
Aerospace safety
Artificial satellites
Ground support
Hypersonic vehicles
Proton effects
Space technology
NT: Space shuttles

Space waste

USE: Space debris

Space-air-ground integrated networks

UF: SAGIN
SGIN
Satellite-terrestrial

integrated networks

Satellite-terrestrial networks
Space-ground

communications

BT: Intserv networks
RT: 6G mobile communication
Satellite communications
Terrestrial communications

Space-division multiplexing

USE: Space division multiplexing

Space-ground communications

USE: Space-air-ground

integrated networks

Space-time codes

BT: Codes
RT: Channel coding
Decoding

Spaceborn radar

USE: Spaceborne radar

Spaceborne photography

USE: Satellite images

Spaceborne radar

UF: Satellite born radar

Satellite borne radar

Space based radar
Space born radar
Spaceborn radar
BT: Radar
RT: Radar remote sensing
Synthetic aperture radar

Spacecraft materials

USE: Aerospace materials

Spacial indices

USE: Spatial indexes

SPAD

diodes
USE: Single-photon avalanche

Spam

USE: Unsolicited e-mail

Spamming

USE: Unsolicited e-mail

Spanning tree protocol

BT: Protocols
RT: Spanning trees

Spanning trees

BT: Tree graphs
RT: Spanning tree protocol

Spark gaps

BT: Electromagnetic analysis
RT: Air gaps
Electrodes
Insulation
Sparks
Switches

Spark Plasma sintering

UF: Field assisted sintering
Plasma pressure

compaction

Pulsed electric current

sintering

BT: Plasma materials

processing

Sintering
RT: Powders

Sparks

BT: Electric breakdown
RT: Spark gaps



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Sparse approximation

UF: Sparse representation
BT: Linear systems

Sparse matrices

UF: Sparse matrix
BT: Numerical analysis

Sparse matrix

USE: Sparse matrices

Sparse representation

USE: Sparse approximation

Spasticity

BT: Medical conditions
RT: Brain injuries
Cerebral palsy
Multiple sclerosis
Muscles
Spinal cord injury
Stroke (medical condition)

Spatial audio

UF: Surround sound
BT: Audio systems
RT: Ambisonics

Spatial augmented reality

BT: Augmented reality

Spatial coherence

BT: Image processing

Spatial computing

BT: Human computer
interaction
RT: Augmented reality
Interactive systems
Mixed reality
Virtual reality

Spatial databases

BT: Databases

Spatial diversity

UF: Antenna diversity
Space diversity
BT: Communication systems
Wireless communication
RT: Antennas
Quality of service
Receiving antennas

Spatial division multiplexing

USE: Space division multiplexing

Spatial filters

BT: Filters

Spatial indexes

UF: Spatial indices
BT: Indexes

Spatial multiplexing

USE: Space division multiplexing

Spatial resolution

BT: Image resolution
RT: Contrast resolution
Image quality
Sentinel-2

Spatial temporal resolution

UF: Spatial-temporal resolution
Spatio temporal resolution
BT: Spatiotemporal phenomena

Spatial-temporal resolution

USE: Spatial temporal resolution

Spatio temporal resolution

USE: Spatial temporal resolution

Spatio-temporal phenomena

USE: Spatiotemporal phenomena

Spatiotemporal phenomena

UF: Spatio-temporal
phenomena
BT: Chaos
RT: Nonlinear dynamical
systems
Pattern formation
Pattern matching
Pattern recognition
NT: Spatial temporal resolution

Speaker recognition

BT: Identification of persons
RT: Biometrics
Speech
Speech recognition
Viterbi algorithm

Special issues

USE: Special issues and sections



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Special issues and sections

UF: Special issues
Special sections
BT: IEEE indexing

BT: Channel allocation
Measurement
RT: Bandwidth
Information processing

Special sections

USE: Special issues and sections

Spectral shape

BT: Acoustics

Specific absorption rate

UF: SAR
BT: Electromagnetic

Spectral waterfall

USE: Spectrogram

interference

Spectral-domain

USE: Spectral analysis

Specification languages

BT: Computer languages
NT: Domain specific languages
Unified modeling language

Spectrogram

UF: Spectral waterfall
Voice print
Voicegram
Voiceprint
BT: Signal processing
RT: Sonogram

Speckle

BT: Optical noise
RT: Optical interferometry
Optical scattering

Spectrometry

USE: Spectroscopy

SPECT

USE: Single photon emission
computed tomography

Spectroradiometers

BT: Radiometers
Spectral analysis
NT: MODIS

Spectral analysis

UF: Power spectra
Spectral domain
Spectral-domain
Spectrum analysis
Spectrum estimation
BT: Signal analysis
RT: Direction-of-arrival

Spectroscopy

UF: Spectrometry
BT: Measurement
RT: Atomic measurements
Bandwidth
Fourier series
Infrared spectra
Nuclear measurements
Radiation detectors
Solid scintillation detectors
Spectral analysis
Thermoreflectance imaging
NT: Deep level transient

estimation

Estimation
Frequency estimation
Harmonic analysis
Parameter estimation
Prediction methods
Signal resolution
Spectroscopy
Speech analysis
Time series analysis
Transforms
NT: Infrared spectra
Judd-Ofelt theory
Spectroradiometers

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

spectroscopy

Spectral domain

USE: Spectral analysis

Spectral efficiency

UF: Bandwidth efficiency

Kirchhoff's Law



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spectroscopy	MERIS	<i>Speech communication</i>
	Mass spectroscopy	USE: Oral communication
	Neutron spin echo	<i>Speech detection</i>
	Optical emission	USE: Voice activity detection
	Photoacoustic effects	Speech enhancement
	Resonance light scattering	BT: Speech processing
<i>Spectrum analysis</i>		RT: Hearing aids
USE: Spectral analysis		Speech recognition
<i>Spectrum estimation</i>		Speech processing
USE: Spectral analysis		BT: Acoustic signal processing
<i>Spectrum management</i>		RT: Delay estimation
USE: Radio spectrum		Phonetics
management		Prediction methods
		NT: Human voice
		Speech enhancement
		Speech synthesis
		Voice activity detection
Speech		Speech recognition
BT: Oral communication		UF: Voice recognition
RT: Aphasia		BT: Identification of persons
Speaker recognition		Pattern recognition
NT: Hate speech		RT: Cepstral analysis
<i>Speech activity detection</i>		Emotion recognition
USE: Voice activity detection		Feature extraction
		Speaker recognition
Speech analysis		Speech enhancement
BT: Speech recognition		Text to speech
RT: Cepstral analysis		Voice activity detection
Frequency estimation		NT: Automatic speech
Signal analysis		recognition
Spectral analysis		Personal voice assistants
Speech coding		Speech analysis
Speech synthesis		Speech to text
Speech codecs		Speech synthesis
BT: Codecs		UF: Synthetic speech
Communication equipment		Voice response systems
RT: Decoding		BT: Speech processing
Speech coding		RT: Biomedical equipment
Vocoders		Natural language
Speech coding		generation
BT: Encoding		Personal voice assistants
Information theory		Signal synthesis
RT: Audio coding		Speech analysis
Rate distortion theory		Voice activity detection
Speech analysis		NT: Chatbots
Speech codecs		Text to speech
Vector quantization		
Vocoders		
Voice activity detection		



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Speech to text

UF: Computer speech
recognition
BT: Speech-to-text
RT: Speech recognition
RT: Automatic speech
recognition
Text to speech

Speech-to-text

USE: Speech to text

Speechmaking

USE: Public speaking

Speed control

USE: Velocity control

Speed measurement

USE: Velocity measurement

SPICE

UF: Simulation Program with
Integrated Circuit Emphasis
pSPICE
BT: Software packages
RT: Circuit analysis
Design automation
Integrated circuits

Spiking neural networks

BT: Artificial neural networks

Spin electronics

USE: Spintronics

Spin injection

USE: Spin polarized transport

Spin polarised transport

USE: Spin polarized transport

Spin polarized transport

UF: Spin injection
Spin polarised transport
BT: Magnetoelectronics
RT: Magnetic tunneling
Magnetoresistance

Spin qubits

BT: Qubit

Spin systems

BT: Magnetics

Spin valves

BT: Magnetic sensors
RT: Hysteresis

Spin-dependent tunneling

USE: Magnetic tunneling

Spin-dependent tunnelling

USE: Magnetic tunneling

Spinal cord

BT: Nervous system
NT: Cerebrospinal fluid
Spinal cord injury

Spinal cord injuries

USE: Spinal cord injury

Spinal cord injury

UF: Spinal cord injuries
BT: Spinal cord
RT: Neurological diseases
Spasticity

Spinal cord stimulation

USE: Electrical stimulation

Spindle bearings

USE: Machine tool spindles

Spine

BT: Nervous system
Skeleton
RT: Degenerative disc disease
NT: Scoliosis

Spinelectronics

USE: Spintronics

Spinning

BT: Textile technology
RT: Spinning machines
Textile fibers

Spinning machines

BT: Textile machinery
RT: Paper making
Paper making machines
Paper mills
Pulp and paper industry
Spinning
Textile industry
Textiles



January 2026 IEEE Thesaurus

Spintronics

UF: Fluxtronics
Spin electronics
Spinelectronics
BT: Magnetoelectric effects

SPIONs

USE: Superparamagnetic iron
oxide nanoparticles

Spirals

BT: Mathematics

Spirometry

BT: Biomedical measurement
Pulmonology
RT: Lungs
Pulmonary diseases

Spleen

BT: Lymphatic system

Splicing

UF: Cable splicing
Fusion splicing
BT: Joining processes
RT: Optical fiber cables
Transmission lines

Spline functions

USE: Splines (mathematics)

Splines (mathematics)

UF: B-Spline
Spline functions
BT: Numerical analysis
RT: Curve fitting

Split gate flash memory cells

UF: Split-gate flash memory
cells

BT: Flash memory cells

Split ring resonators

BT: Resonators
RT: Electromagnetic
metamaterials
Metamaterials
Microwave metamaterials
Optical resonators
Terahertz metamaterials

Split-gate flash memory cells

USE: Split gate flash memory
cells

SPO

USE: Triples (Data structure)

Spontaneous emission

UF: Superradiance
BT: Photonics
RT: Microcavities
Photonic crystals
NT: Radiative recombination

Sports

UF: Baseball
Football
Hockey
Soccer
Swimming
Tennis
BT: Entertainment industry
RT: Games
Sports equipment

Sports equipment

BT: Manufactured products
RT: Bicycles
Sports

Spot welding

BT: Welding

SPR

USE: Surface plasmon
resonance

Spraying

BT: Surface finishing
Aerosols
Coatings
Liquids
Particle production
Surface charging
NT: Thermal spraying

Spread spectrum communication

UF: Frequency hop
communication
Frequency-hop
communication
Multi-hop
Multihop
Pseudonoise coded
communication
BT: Digital communication
RT: 3G mobile communication
4G mobile communication

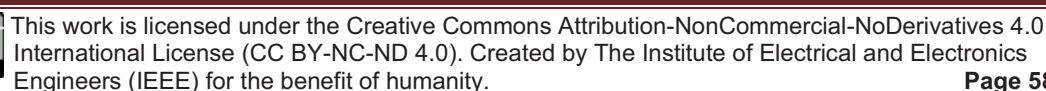


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multiple access	Bluetooth	Sputter etching	BT: Sputtering
	Channel estimation	RT: Cardiography	
	Chirp modulation		
	Code division multiplexing	Sputtering	UF: Sputter deposition
	Electronic countermeasures		Thin film deposition
	Electronic warfare	BT: Materials preparation	
	Multicarrier code division	RT: Coatings	Films
			Magnetrons
			Physical vapor deposition
		NT: Sputter etching	
countermeasures	Multiuser detection		
	Radio communication		
	Time division synchronous		
code division multiple access	Ultra wideband		
communication		Spyware	BT: Privacy-invasive software
Spread spectrum radar	UF: Frequency hop radar	SQL	USE: Structured Query Language
	Pseudonoise coded radar		
	BT: Radar		
	RT: Chirp modulation	SQL injection	BT: Computer crime
	Electronic countermeasures		Information security
	Electronic warfare		
	Radar countermeasures		
		Squamous cell carcinoma	BT: Skin cancer
		<i>Squeezed light</i>	USE: Optical squeezing
Spreadsheet programs	UF: Microsoft Excel		
	BT: Data processing		
Springs	BT: Mechanical products	SQUID magnetometers	BT: Magnetometers
	Production	RT: Magnetic fields	
	RT: Shock absorbers		
	Suspensions (mechanical		
	systems)		
Sprites (computer)	Wires	SQUIDs	UF: Superconducting quantum
			interference devices
			BT: Superconducting devices
			RT: Readout electronics
<i>Spur gears</i>	USE: Gears	<i>Squeezed states</i>	USE: Optical squeezing
Spurline		<i>Sr</i>	USE: Strontium
	BT: Power filters		
	RT: Planar transmission lines		
Spurline components	NT: Spurline components	SRAM	USE: SRAM chips
<i>Sputter deposition</i>		SRAM cells	BT: Random access memory
	USE: Sputtering		
		SRAM chips	UF: SRAM
			BT: Random access memory



	RT:	CMOS memory circuits			Escalators
					Legged locomotion
<i>SRI</i>					Mobile robots
	USE:	Socially responsible			
investing				<i>Stakeholder pensions</i>	
				USE:	Pensions
Stability				Stakeholders	
	BT:	Reliability		BT:	Customer relationship
	RT:	Asymptotic stability		management	
		Control systems			Organizational aspects
		Damping			Decision making
		Lyapunov methods		RT:	Requirements engineering
		Predator prey systems			Strategic planning
		Robustness			
		Stability plasticity			
		Time invariant systems		<i>Standard Generalized Markup Language</i>	
	NT:	Circuit stability		USE:	SGML
		Robust stability			
		Stability analysis		Standardization	
		Thermal stability		BT:	Engineering - general
				RT:	IEC
Stability analysis					ISO
	BT:	Stability			ISO Standards
	RT:	Algorithms		NT:	Formal specifications
		Differential equations			Guidelines
		Laser stability			Standards
		Plasma properties			
		Signal processing		Standards	
		System analysis and design		BT:	Standardization
	NT:	Stability criteria		RT:	Conformance testing
Stability criteria					IEC
	BT:	Stability analysis			ISO
Stability plasticity					International collaboration
					Open systems
	UF:	Stability-plasticity			Qualifications
	BT:	Learning systems		NT:	Renewable Portfolio
		Neural networks		Standard	
	RT:	Constraint theory			Standards categories
		Neural activity			Standards organizations
		Neuroplasticity			Standards publications
		Stability		Standards categories	
<i>Stability-plasticity</i>				BT:	Standards
	USE:	Stability plasticity		NT:	Communication standards
Stacking					International Atomic Time
					Measurement standards
	BT:	Material storage			Military standards
	RT:	Containers		standards	Power and energy
		Materials handling			Software standards
		Warehousing			
Stairs				Standards organizations	
				BT:	Standards
	BT:	Construction		NT:	3GPP
	RT:	Elevators			ANSI



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Association	American Standards	<i>Starter motors (automotive)</i>
	USE: Automotive components	
Association	CSA Group	STATCOM
	DMTF	UF: Static compensator
Association	ETSI	BT: Static VAR compensators
	IEC	
Association	IEEE Standards	State estimation
	ISO	UF: State observers
Association	ITU	BT: Estimation
	NACE International	RT: Control systems
Association	NEMA	NT: Observers
	NFPA	
Association	NIST	State feedback
	Open Geospatial	BT: Linear feedback control
Consortium	SMPTE	systems
	W3C	<i>State observers</i>
Consortium		USE: State estimation
Standards publications		State of charge
BT:	Standards	BT: Battery chargers
	3GPP Standards	
NT:	ANSI Standards	<i>State pensions</i>
	ASA Standards	USE: Pensions
NT:	CSA Group Standards	<i>State space theory</i>
	ETSI Standards	USE: State-space methods
NT:	IEC Standards	
	IEEE Standards	State-space methods
NT:	ISO Standards	UF: State space theory
	ITU Standards	State-space model
NT:	NACE Standards	BT: Control system analysis
	NISO Standards	RT: Time-domain analysis
NT:	NIST Standards	
	SMPTE Standards	<i>State-space model</i>
	W3C Standards	USE: State-space methods
Standby generators		Static analysis
UF:	Emergency power	UF: Static projection
	generators	Statis scoring
BT:	Generators	BT: Statistical analysis
	Rotating machines	System analysis and design
RT:	Emergency power supplies	RT: Model checking
		Software engineering
<i>Standby power supplies</i>		
USE:	Emergency power supplies	<i>Static compensator</i>
		USE: STATCOM
Stark effect		<i>Static converters</i>
BT:	Electro-optic effects	USE: Static power converters
<i>Starlink</i>		Static induction transistors
USE:	Satellite constellations	BT: Transistors
Stars		
BT:	Astronomy	
	Neutron stars	
NT:		



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Static power converters

UF: Static converters
BT: Converters

NT: Distribution functions
Gaussian distribution
Weibull distribution

Static projection

USE: Static analysis

Statistical learning

BT: Machine learning
RT: Decision theory
Pattern recognition
NT: Confusion matrices

Static VAR compensators

UF: SVC
BT: Power transmission
RT: Reactive power
NT: STATCOM

Statistical testing

USE: Statistical analysis

Stationary state

UF: Ground state
BT: Quantum mechanics

Statistics

BT: Mathematics
RT: Econometrics
Estimation theory
Extrapolation
Fourier transforms
Information theory
Interpolation
Matrix decomposition
Maximum likelihood

Statis scoring

USE: Static analysis

Statistical analysis

UF: Statistical testing
BT: Statistics
RT: Decision theory
Measurement errors
Nearest neighbor methods
Probability
R language
Random processes
Technology acceptance

detection

Operations research
Probability
Scheduling
Weibull distribution
NT: Adaptive estimation
Autoregressive processes
Biostatistics
Boltzmann distribution
Computational statistics
Correlation
Correlation coefficient
Covariance matrices
Cyclostationary process
Decision theory
Differential privacy
Dimensionality reduction
Ensemble learning
Gamma distribution
Gaussian mixture model
Higher order statistics
Histograms
Imputation
Linear discriminant analysis
Maximum likelihood

model

NT: Time series analysis
Analysis of variance
Conditional random fields
Descriptive statistics
Functional data analysis
Inferential statistics
Mean field theory
Mode matching methods
Monte Carlo methods
Parameter estimation
Pareto analysis
Predictive analytics
Principal component

analysis

Regression analysis
Static analysis
Trend analysis

estimation

Statistical computing

USE: Computational statistics

Statistical distributions

BT: Probability
RT: Multi-armed bandit problem

Minimax techniques
Mixture models
Mortality
Nonparametric statistics
Parametric statistics
Prediction theory



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	Ranking (statistics)		Wheels
	Root mean square		
	Sampling methods	Steganography	
	Statistical analysis	BT:	Cryptography
	Surveys	RT:	Image coding
	Time series analysis		Message authentication
	Total variance		
Stator bars		<i>Steiner points</i>	
BT:	Stators	USE:	Steiner trees
Stator cores		Steiner trees	
BT:	Stators	UF:	Steiner points
			Steiner vertices
		BT:	Combinatorial mathematics
Stator windings		<i>Steiner vertices</i>	
BT:	Stators	USE:	Steiner trees
Stators		Stellar dynamics	
BT:	Electric machines	BT:	Astrophysics
NT:	Stator bars	NT:	Stellar motion
	Stator cores		
	Stator windings	Stellar motion	
<i>Steady state</i>		BT:	Stellar dynamics
USE:	Steady-state		
Steady-state		STEM	
UF:	Steady state	UF:	Science technology
BT:	Dynamic equilibrium		engineering mathematics
RT:	Transient analysis		science technology
			engineering and math
			science, technology,
Steam engines			engineering, and math
BT:	Heat engines	BT:	Educational programs
RT:	Boilers	RT:	Curriculum development
	Water		Educational courses
			Engineering - general
Steel			Mathematics
BT:	Metals		Science - general
RT:	Pressure vessels		Technology
NT:	Martensite		
Steel industry		<i>Stem cell research</i>	
BT:	Industries	USE:	Stem cells
Steerable antennas		Stem cells	
BT:	Antennas	UF:	Stem cell research
	Beam steering	BT:	Biological cells
	Phased arrays	RT:	Cloning
			Progenitor cells
Steering systems		Stereo image processing	
BT:	Mechanical products	UF:	Stereoscopic
	Production systems		Stereoscopy
RT:	Advanced driver assistance	BT:	Stereo vision
systems			
	Automotive components		



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Stereo vision

UF: Stereovision
Three-dimensional vision
BT: Imaging
RT: Image matching
Machine vision
Robot vision systems
NT: Stereo image processing

Stereognosis

BT: Robot sensing systems

Stereolithography

BT: Lithography
RT: Laser applications
Laser sintering
Lasers
Layered manufacturing
Manufacturing
Prototypes

Stereometry

USE: Solid geometry

Stereophonic systems

USE: Audio systems

Stereoscopic

USE: Stereo image processing

Stereoscopy

USE: Stereo image processing

Stereovision

USE: Stereo vision

Sterilisation

USE: Sterilization

Sterilization

UF: Sterilisation
BT: Decontamination

Sternum

BT: Thorax

Stethoscope

BT: Biomedical equipment

Stimulated emission

UF: Optical amplification
BT: Particle beam optics
RT: Lasers
Masers

Stirling engines

BT: Heat engines

Stochastic distribution

USE: Stochastic processes

Stochastic prediction

USE: Stochastic processes

Stochastic processes

UF: Stochastic distribution
Stochastic prediction
Stochastic theory

BT: Mathematics
RT: Computational

electromagnetics

Diffusion processes
Mean field theory
Multi-armed bandit problem
Particle swarm optimization
Probability
Q-learning
Random number

generation

Random variables
Viterbi algorithm
NT: Gaussian processes
Markov processes
Scenario generation

Stochastic resonance

BT: Resonance

Stochastic systems

BT: Systems engineering and theory

RT: Control systems
Mean field theory
Probability
Random variables

Stochastic theory

USE: Stochastic processes

Stock exchanges

USE: Stock markets

Stock markets

UF: Stock exchanges
BT: Economics

Stokes parameters

BT: Optical polarization

S



January 2026 IEEE Thesaurus

Stomach

BT: Digestive system
RT: Gastritis

Tides
Tsunami

Stomatognathic system

UF: stomognathic system
BT: Anatomy
RT: Faces
Lips
Mouth
Pharynx
Tongue
NT: Masticatory muscles
Salivary glands

Storm systems

USE: Tropical cyclones

Storm water

USE: Stormwater

Storms

BT: Meteorology
RT: Lightning
Monsoons
Stormwater

stomognathic system

USE: Stomatognathic system

Stormwater

UF: Storm water
BT: Hydrology
Water resources
RT: Disaster management
Emergency services
Floods
Rain
Rivers
Storms

Storage area networks

UF: SAN
BT: Computer networks
RT: Buffer storage
Local area networks

Storage automation

UF: Automated storage and retrieval systems
BT: Automation
Material storage
RT: Warehousing

Strain

BT: Mechanical factors
RT: Elasticity
Elongation
Strain control
Strain measurement
NT: Tensile strain
Uniaxial strain

Storage batteries

USE: Batteries

Storage battery

USE: Batteries

Strain based sensors

USE: Capacitive sensors

Storage management

BT: Capacity planning
Management
RT: Memory management
NT: Digital storage

Strain control

UF: Friction stir processing
BT: Mechanical variables
control
RT: Strain

Storage rings

BT: Particle accelerators
RT: Ions
Muon colliders
Particle beams

Strain gauges

USE: Strain measurement

Stored energy

USE: Energy storage

Strain measurement

UF: Strain gauges
BT: Mechanical variables
measurement
RT: Micrometers
Strain

Storm surge

BT: Ocean dynamics
RT: Sea measurements



January 2026 IEEE Thesaurus

Strain sensors

USE: Capacitive sensors

Strategic planning

BT: Planning
RT: Analytic hierarchy process
Business intelligence
Decision making
Information systems
Stakeholders
NT: Roadmaps (technology planning)

Stratified media

USE: Nonhomogeneous media

Stratigraphy

BT: Geology
RT: Rocks

Stratosphere

USE: Terrestrial atmosphere

Stray light

BT: Light sources
Optics
RT: Ray tracing

Stream processing

USE: Data stream processing

Streaming media

UF: Media streaming
BT: Communication system
software
RT: Data compression
IEEE 802.11e Standard
IPTV
Internet
MPEG 4 Standard
MPEG standards
Multimedia communication
Unicast
Video coding
Video signal processing
NT: Audio streaming
Mobile video
Over-the-top media
services
Video on demand

Streams

BT: Water

Streetcars

USE: Light rail systems

Stress

UF: Mechanical stress
BT: Mechanical factors
RT: Magnetomechanical effects
Photoelasticity
Piezoelectricity
Piezooptic effects
Piezoresistance
Stress control
Stress measurement
NT: Compressive stress
Internal stresses
Residual stresses
Shear testing
Tensile stress
Wind stress

Stress (psychological)

USE: Human factors

Stress control

BT: Mechanical variables
control
RT: Stress
Surface stress

Stress measurement

BT: Mechanical variables
measurement
RT: Stress

String theory

BT: Physics
RT: Quantum mechanics

String vacuum

USE: Elementary particle vacuum

Stripboard circuit

UF: Veroboard
BT: Electronic circuits

Stripline

BT: Planar transmission lines
Transmission lines
NT: Stripline components

Stripline components

BT: Stripline
RT: Power combiners
Power dividers



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Strips

BT: Structural shapes

Structural parameter

USE: Structural engineering

Stroke (medical condition)

BT: Medical conditions
RT: Aphasia
Spasticity

Structural plates

BT: Electronic components
Structural shapes
RT: Flanges
Structural panels
Wheels

Strontium

UF: Sr
BT: Metals
NT: Strontium compounds

Structural rings

UF: O-rings
BT: Structural shapes
RT: Engine cylinders
Mechanical products
Pistons
Seals

Strontium compounds

BT: Strontium
RT: Alloying

Structural beams

UF: Cantilever beams
Girders
BT: Structural shapes
RT: Building materials

Structural rods

BT: Structural shapes

Structural biology

BT: Biology

Structural shapes

BT: Mechanical products
NT: Bars
Bridges
Ducts
Flexible structures
Honeycomb structures
Lightweight structures
Rails
Sandwich structures
Sheet materials
Slabs
Strips
Structural beams
Structural discs
Structural panels
Structural plates
Structural rings
Structural rods
Structural shells
Thin wall structures
Wires

Structural discs

UF: Disks (structures)
BT: Structural shapes

Structural engineering

UF: Structural parameter
Structural stability
BT: Civil engineering
RT: Architecture
Bridges
Building information

management

Construction
Flexible structures
Floods
Intelligent structures
Mechanical factors
NT: Offshore installations

Structural panels

UF: Railway bridges
Road bridges
Suspension bridges
BT: Structural shapes
RT: Honeycomb structures
Sandwich structures
Sheet materials
Structural plates
Thin wall structures

Structural shells

BT: Structural shapes
RT: Thin wall structures

Structural stability

USE: Structural engineering

Structure from motion

BT: Image processing
RT: Motion control
Signal processing



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Three-dimensional displays
Two-dimensional displays

Structured Query Language

UF: SQL
BT: Database languages
RT: Programming
Relational databases

Student engineers

USE: Engineering students

Student experiments

BT: Engineering education
RT: Laboratories

Style sheet languages

BT: Computer languages
NT: Cascading style sheets

Sub-mm wave filters

USE: Submillimeter wave filters

Sub-sea cables

USE: Underwater cables

Subcontracting

BT: Contracts

Subduction

UF: Subduction zones
BT: Geological processes
Tectonics

Subduction zones

USE: Subduction

Subject matter experts

BT: Knowledge acquisition
RT: Knowledge discovery
Professional aspects
NT: Indigenous knowledge

Subject predicate object

USE: Triples (Data structure)

Submarine cables

USE: Underwater cables

Submarine technology

USE: Underwater technology

Submarines

USE: Underwater vehicles

Submersible cables

USE: Underwater cables

Submersibles

USE: Underwater vehicles

Submillimeter wave circuits

BT: Circuits
Submillimeter wave

technology

RT: Analog circuits
Submillimeter wave devices
Submillimeter wave filters
Submillimeter wave

integrated circuits

Submillimeter wave communication

BT: Communication systems
Submillimeter wave

technology

Submillimeter wave devices

UF: Submillimeter wave

systems

BT: Submillimeter wave

technology

RT: Submillimeter wave circuits
Submillimeter wave

integrated circuits

NT: Submillimeter wave filters

Submillimeter wave filters

UF: Sub-mm wave filters
Submillimetre wave filters
BT: Submillimeter wave devices
RT: Submillimeter wave circuits
Terahertz wave absorption

Submillimeter wave integrated circuits

BT: Integrated circuits
Submillimeter wave circuits
Submillimeter wave

technology

RT: Analog integrated circuits
Submillimeter wave devices

Submillimeter wave measurements

BT: Electromagnetic

measurements

RT: Hyperspectral sensors
Submillimeter wave

technology



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Submillimeter wave propagation

BT: Electromagnetic propagation

Submillimeter wave systems

USE: Submillimeter wave devices

Submillimeter wave technology

BT: Microwave theory and techniques

RT: Submillimeter wave measurements

NT: Submillimeter wave circuits
Submillimeter wave

communication
Submillimeter wave devices
Submillimeter wave

integrated circuits

Submillimetre wave filters

USE: Submillimeter wave filters

Subroutines

USE: Algorithms

Subscriber identification module

USE: SIM card

Subscriber identity module

USE: SIM card

Subscriber loops

BT: Communication systems
Multiaccess communication

Subscriber sets

USE: Telephone sets

Subsea cables

USE: Underwater cables

Subspace constraints

BT: Object segmentation

Substance abuse

USE: Addiction

Substation automation

BT: Substations
RT: Automation
SCADA systems
Substation protection

Substation protection

BT: Power system protection

Substations

RT: Substation automation

Substations

UF: Power stations
(substations)

BT: Power systems
NT: Substation automation
Substation protection

Substitution box

USE: S Box

Substrate hot electron injection

UF: Substrate hot-electron injection

BT: Hot carrier injection

Substrate hot-electron injection

USE: Substrate hot electron injection

Substrate integrated waveguides

UF: Post-wall waveguides
BT: Waveguide lasers

Substrates

BT: Semiconductor materials
RT: Epitaxial growth
Microprocessor chips
Printed circuits
Silicon germanium
Silicon on sapphire
Vapor deposition

Subthreshold conduction

USE: Subthreshold current

Subthreshold current

UF: Subthreshold conduction
Subthreshold drain current
Subthreshold leakage
BT: Threshold voltage

Subthreshold drain current

USE: Subthreshold current

Subthreshold leakage

USE: Subthreshold current

Subtraction techniques

BT: Image analysis
RT: Biomedical image processing



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Subways

USE: Public transportation

Sucrose

USE: Sugar

Sufficient conditions

BT: Logic

Sugar

UF: Sucrose
BT: Agricultural products
Food products
RT: Sugar industry
Sugar refining
NT: Glucose

Sugar industry

BT: Industries
RT: Food industry
Food products
Sugar
NT: Sugar refining

Sugar refining

BT: Sugar industry
RT: Food industry
Food products
Food technology
Purification
Refining
Sugar

Sulfur

UF: Sulphur
BT: Chemical elements
NT: Sulfur compounds

Sulfur compounds

UF: Sulphur compounds
BT: Sulfur

Sulfur hexafluoride

UF: SF₆
BT: Gas insulation

Sulphur

USE: Sulfur

Sulphur compounds

USE: Sulfur compounds

Sum product algorithm

UF: Sum-product algorithm
BT: Iterative algorithms

Sum product message passing

USE: Belief propagation

Sum-product algorithm

USE: Sum product algorithm

Summing circuits

BT: Circuits
RT: Analog computers

Sun

BT: Solar system
NT: Coronal mass ejections
Solar eclipses
Solar flares
Solar wind
Sunspots

Sun sensors

USE: Sensors

Sun trackers

USE: Solar tracking

Sun tracking

USE: Solar tracking

Sunspots

BT: Sun
RT: Coronal mass ejections
Geomagnetic storms
Solar flares

Super capacitors

USE: Supercapacitors

Super earths

USE: Extrasolar planets

Super hi-vision

USE: UHDTV

Super intelligence

USE: Hyper-intelligence

Super resolution

USE: Superresolution

Super-resolution

USE: Superresolution

Supercapacitors

UF: Electrical double layer
capacitors
Super capacitors



January 2026 IEEE Thesaurus

BT: Ultracapacitors
Electrochemical devices
Energy storage
Power capacitors
RT: Capacitance
Capacitance measurement
Electrolytes

Supercomputers

BT: Computers
RT: Petascale computing
NT: Exascale computing

Superconducting cables

BT: Superconducting
transmission lines
RT: Superconducting coils
Superconducting magnets

Superconducting coils

BT: Coils
Superconducting devices
RT: Superconducting cables
Superconducting magnets
Superconducting wires

Superconducting device noise

BT: Noise
RT: Superconducting devices

Superconducting devices

UF: Josephson devices
SIS devices
(superconductor)
SNS devices
Superconductor-insulator-
superconductor devices
Superconductor-normal-
superconductor devices
BT: Superconductivity
RT: Cryogenic electronics
High-temperature
superconductors
Superconducting device
noise
Superconducting films
Thermal factors
NT: Josephson junctions
SQUIDs
Superconducting coils
Superconducting magnets
Superconducting
microwave devices
Superconducting
photodetectors

Superconducting epitaxial layers

BT: Epitaxial layers
RT: Superconducting materials

Superconducting filaments

BT: Superconducting materials

Superconducting films

UF: Superconducting tapes
BT: Superconductivity
RT: High-temperature
superconductors
Superconducting devices
Surface impedance
Surface resistance
Thick films
Thin films
NT: Superconducting thin films

Superconducting filters

BT: Filters
RT: Radiofrequency
interference

Superconducting infrared detectors

USE: Superconducting
photodetectors

Superconducting integrated circuits

BT: Integrated circuits
Superconductivity

Superconducting junction devices

USE: Josephson junctions

Superconducting logic circuits

BT: Logic circuits

Superconducting magnet energy storage

USE: Superconducting magnetic
energy storage

Superconducting magnetic energy storage

UF: SMES
Superconducting magnet
energy storage
BT: Energy storage
Superconductivity

Superconducting magnets

BT: Electromagnets
Superconducting devices
RT: Magnetic levitation vehicles
Persistent currents
Superconducting cables



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Superconducting coils

Superconducting materials

UF: Pnictide superconductors
BT: Materials
RT: Superconductivity
Critical current density
Cryogenic electronics
Superconducting epitaxial

layers

NT: Thermal factors
Granular superconductors
High-temperature

superconductors

superconductors

Multifilamentary
Niobium-tin
Superconducting filaments
Superconducting wires
Type II superconductors

Superconducting microwave devices

BT: Superconducting devices
RT: Microwave devices

Superconducting photodetectors

UF: Superconducting infrared
detectors
Superconducting ultraviolet
detectors
BT: Photodetectors
Superconducting devices
RT: Infrared detectors

Superconducting quantum interference devices

USE: SQUIDs

Superconducting tapes

USE: Superconducting films

Superconducting thin films

BT: Superconducting films
RT: Thin films

Superconducting transition temperature

BT: Superconductivity
RT: High-temperature
superconductors

Superconducting transmission lines

BT: Transmission lines
RT: Power transmission lines
NT: Superconducting cables

Superconducting ultraviolet detectors

USE: Superconducting
photodetectors

Superconducting wires

BT: Superconducting materials
RT: Superconducting coils
NT: Multifilamentary
superconductors

Superconductive tunneling

UF: Superconductive tunnelling
BT: Superconductivity
Tunneling

Superconductive tunnelling

USE: Superconductive tunneling

Superconductivity

NT: Bean model
Critical current density
Flux pinning
Superconducting devices
Superconducting films
Superconducting integrated

circuits

energy storage

temperature

Superconducting magnetic

Superconducting materials
Superconducting transition

Superconductive tunneling

Superconductor-insulator-superconductor devices

USE: Superconducting devices

Superconductor-normal-superconductor devices

USE: Superconducting devices

Superconductors (high temperature)

USE: High-temperature
superconductors

Supercontinuum generation

BT: Nonlinear optics
RT: Laser beams
Light sources
Optical fibers

Superintelligence

USE: Hyper-intelligence



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Superlattices

BT: Crystalline materials
NT: Magnetic superlattices
Metallic superlattices
Optical superlattices
Semiconductor

superlattices

Superluminescent diodes

UF: SLD
BT: Diodes
Light emitting diodes
Light sources
Optoelectronic devices
Semiconductor devices
Semiconductor diodes
RT: Lasers

Supernumery robotic limbs

BT: Wearable robots
RT: Limbs
Medical robotics
Patient rehabilitation
Prosthetics

Superparamagnetic iron oxide nanoparticles

UF: SPIONS
BT: Paramagnetic materials
RT: Targeted drug delivery

Superposition calculus

BT: Mathematics

Superradiance

USE: Spontaneous emission

Superresolution

UF: Super resolution
Super-resolution
BT: Image resolution
RT: Deblurring

Supersonic flow

BT: Fluid flow

Superstring vacuum

USE: Elementary particle vacuum

Supervised learning

BT: Learning systems
RT: Deep learning
Graph neural networks
Naive Bayes methods
Self-supervised learning
Semisupervised learning

Unsupervised learning
Weak supervision
Boosting
Similarity learning

Supervisory control

BT: Control systems
NT: SCADA systems

Supervisory control and data acquisition systems

USE: SCADA systems

Supervisory control and data-acquisition systems

USE: SCADA systems

Supervisory programs

USE: Operating systems

Supply and demand

BT: Economics
RT: Microeconomics
Utility theory

Supply chain management

UF: SCM supply chains
BT: Management
RT: Business process

integration

Business process

management

Capacity planning
Customer relationship

management

Electronic commerce
Food security
Management information

systems

Materials requirements

planning

Production control
Supply chains
Procurement

Supply chains

BT: Logistics
RT: Materials requirements

planning

Procurement
Supply chain management
Cold chain
Distribution networks



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Support vector machine classification

BT: Support vector machines

Support vector machines

UF: SVM
Support vector regression
BT: Computation theory
RT: Artificial intelligence
Feedforward neural

networks

Kernel machines
Logistic regression
Naive Bayes methods
Pattern classification
Reinforcement learning
Relevance vector machines
Support vector machine

NT: classification

Support vector regression

USE: Support vector machines

Surface acoustic wave devices

BT: Acoustic devices
RT: Acoustoelectric devices
Piezoelectric devices

Surface acoustic waves

UF: Acoustic surface waves
BT: Acoustic waves
Surface waves
Waves

Surface charging

BT: Electrostatic processes
RT: Spraying
NT: Triboelectricity

Surface cleaning

BT: Cleaning
Surface treatment
RT: Semiconductor device
manufacture
Surface contamination

Surface contamination

BT: Contamination
RT: Semiconductor device
manufacture
Surface cleaning
Surface treatment

Surface cracks

BT: Mechanical factors

Surface discharges

BT: Dielectric breakdown
RT: Insulator testing

Surface emitting lasers

BT: Lasers
Semiconductor devices
Semiconductor lasers
Solid lasers
RT: Laser cavity resonators
Quantum well lasers
Quantum wells
NT: Vertical cavity surface

emitting lasers

Surface engineering

BT: Materials science and
technology
RT: Surface treatment

Surface finishing

BT: Finishing
RT: Lapping
Polishing machines
NT: Burnishing
Deburring
Painting
Spraying

Surface fitting

BT: Numerical analysis
RT: Computational geometry
Computer graphics
Curve fitting
Interpolation
NT: Response surface

methodology

Surface impedance

BT: Surfaces
RT: High-temperature
superconductors
Superconducting films

Surface morphology

BT: Surfaces
RT: Surface roughness
NT: Adsorption

Surface mount technology

UF: Surface-mount technology
BT: Integrated circuit
manufacture
Printed circuits
NT: Ball grid arrays



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Surface plasmon polaritons

BT: Polaritons
Surface plasmons

Surface plasmon resonance

UF: SPR
BT: Nanoplasmonics
Surface plasmons

Surface plasmons

BT: Plasmons
NT: Surface plasmon polaritons
Surface plasmon

resonance

Surface reconstruction

BT: Visualization
RT: Pattern analysis

Surface resistance

BT: Resistance
Surfaces
RT: High-temperature
superconductors
Superconducting films

Surface roughness

BT: Surfaces
RT: Planing
Polishing machines
Rough surfaces
Sandblasting
Surface morphology

Surface soil

BT: Surfaces

Surface states

BT: Energy states
Surface structures
NT: Surfactants

Surface stress

BT: Mechanical factors
Surfaces
RT: Internal stresses
Stress control

Surface structures

BT: Surfaces
NT: Surface states

Surface tension

BT: Surfaces
RT: Surfactants

Surface texture

BT: Surfaces

Surface topography

BT: Geometry
Surfaces
NT: Nanotopography

Surface treatment

BT: Surfaces
RT: Colloidal lithography
Planing
Surface contamination
Surface engineering
Electrochemical deposition
Etching
Finishing
Galvanizing
Painting
Passivation
Pickling
Planarization
Sandblasting
Surface cleaning
Surfactants
Vapor deposition

Surface waves

BT: Geophysics
RT: Sea surface
NT: Surface acoustic waves

Surface-mount technology

USE: Surface mount technology

Surfaces

BT: Materials science and
technology
NT: Corrosion
Corrugated surfaces
Metasurfaces
Rough surfaces
Surface impedance
Surface morphology
Surface resistance
Surface roughness
Surface soil
Surface stress
Surface structures
Surface tension
Surface texture
Surface topography
Surface treatment



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Surfactants

BT: Materials
Surface states
Surface treatment
RT: Adsorption
Surface tension

Surge protection

BT: Power system protection
RT: Surges
NT: Arresters

Surgery

UF: Robot-assisted surgery
BT: Medical treatment
RT: Biomedical equipment
Catheters
Endoscopes
NT: Ambulatory surgery
Electrosurgery
Hepatectomy
Laser surgery
Mastectomy
Microsurgery
Minimally invasive surgery
Neurosurgery
Oncological surgery
Orthopedic surgery

Surgery oncology

USE: Oncological surgery

Surges

BT: Electromagnetic transients
RT: Surge protection
NT: Inrush current

Surgical instruments

BT: Biomedical equipment
NT: Laparoscopes

Surgical masks

USE: Face masks

Surgical robotics

USE: Medical robotics

Surgical robots

USE: Medical robotics

Surround sound

USE: Spatial audio

Surveillance

BT: Monitoring
RT: Air to ground
communication
Conformance testing
Hazardous areas
Motion detection
Reconnaissance
Remote sensing
Security
Terrorism
NT: Infrared surveillance
Video surveillance

Surveys

BT: Statistics
RT: Sampling methods

Suspension bridges

USE: Structural panels

Suspensions (mechanical systems)

BT: Mechanical systems
RT: Automotive components
Springs
NT: Shock absorbers

Sustainability

USE: Sustainable development

Sustainable development

UF: Sustainability
BT: Environmental
management
Social implications of
technology
RT: Anaerobic digestion
Circular economy
Electrification
Food security
Green computing
Green energy
Precision agriculture
NT: Green (sustainability)

SVC

USE: Static VAr compensators

SVM

USE: Support vector machines

SVPWM

USE: Space vector pulse width
modulation



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Swaging

BT: Materials processing
RT: Metal products

Swarm intelligence

UF: Swarm optimization
BT: Artificial intelligence
NT: Ant colony optimization
Particle swarm optimization

Swarm optimization

USE: Swarm intelligence

Swarm robotics

UF: Swarm robots
BT: Multi-robot systems
RT: Consensus control

Swarm robots

USE: Swarm robotics

Sweat glands

BT: Glands
Skin

Swimming

USE: Sports

Swimming robots

USE: Aquatic robots

SWIPT

USE: Simultaneous wireless
information and power transfer

Switch on surge

USE: Inrush current

Switched capacitor circuits

UF: Switched-capacitor circuit
BT: Switched circuits

Switched capacitor networks

UF: Switched-capacitor
networks
BT: Resistors
RT: Analog circuits
Capacitors

Switched circuits

BT: Circuits
RT: Telecommunications
NT: Switched capacitor circuits

Switched mode power

USE: Switched mode power
supplies

Switched mode power supplies

UF: SMPS
Switched mode power
BT: Power supplies

Switched reluctance motors

UF: Bridge converters
BT: Reluctance motors
RT: Brushless motors

Switched systems

BT: Time-varying systems
RT: Control systems
Power conversion

Switched-capacitor circuit

USE: Switched capacitor circuits

Switched-capacitor networks

USE: Switched capacitor
networks

Switches

BT: Control equipment
Electronic components
RT: Current control
IEEE 802.3 Standard
Solenoids
Spark gaps
Switchgear
Switching circuits
NT: Contactors
Microswitches
Optical switches

Switchgear

BT: Control equipment
RT: Current control
Fuses
Switches
NT: Circuit breakers
Interrupters
Relays

Switching circuits

BT: Circuits
RT: Circuit breakers
Digital circuits
Relays
Switches
NT: Choppers (circuits)



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Logic circuits
Switching converters
Zero current switching
Zero voltage switching

Switching converters

BT: Switching circuits
RT: Power electronics
Zero current switching
Zero voltage switching

Switching convertors

USE: Converters

Switching frequency

BT: Switching systems

Switching loss

UF: Switching losses
BT: Switching systems

Switching losses

USE: Switching loss

Switching systems

BT: Communication systems
RT: Communication switching
NT: Electronic switching

systems

Switching frequency
Switching loss
Telecommunication

switching

Symbiosis

UF: Symbiotic relationships
BT: Biological processes

Symbiotic relationships

USE: Symbiosis

Symbols

BT: Graphics
RT: Digital information
Huffman coding
Information retrieval
Pattern recognition
NT: CAPTCHAs
Communication symbols
Emojis

Symmetric matrices

UF: Symmetric matrix
BT: Numerical analysis

Symmetric matrix

USE: Symmetric matrices

Sympathetic nervous system

UF: Sympathetic outflow
BT: Autonomic nervous system

Sympathetic outflow

USE: Sympathetic nervous system

Symposia

USE: Conferences

Synapses

BT: Nervous system
RT: Artificial intelligence
Artificial neural networks
Brain
Communication channels
Computational intelligence
Electrochemical devices
Integrated optics
Local field potentials
Neuroinformatics
Neuromorphic engineering
Neurons
Neurotransmitters
Organic electronics
Photonics
Signal processing
Synaptic communication

Synaptic communication

BT: Biological processes
RT: Synapses

Synaptic transmission

USE: Neurotransmitters

Sync

USE: Synchronization

Synchrocyclotrons

BT: Particle accelerators

Synchronisation

USE: Synchronization

Synchronization

UF: Clock synchronization
Sync
Synchronisation
BT: Timing
RT: Chaotic communication



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Concurrence control
Frequency locked loops
Scheduling
Synchronous digital
hierarchy
Time dissemination
Tracking loops

Synchronous digital hierarchy

UF: SDH
BT: Communication standards
Communication systems
ETSI Standards
RT: Digital communication
Optical fiber communication
SONET
Synchronization
Transport protocols

Synchronous DRAM

USE: SDRAM

Synchronous dynamic random access memory

USE: SDRAM

Synchronous generators

BT: AC generators
Synchronous machines
RT: Alternators
Grid following
Grid forming
NT: Reluctance generators

Synchronous machines

BT: AC machines
NT: Hysteresis motors
Reluctance machines
Synchronous generators
Synchronous motors

Synchronous motors

BT: Synchronous machines
RT: Rotating machines
NT: Hysteresis motors
Reluctance motors

Synchronous optical network

USE: SONET

Synchrophasors

USE: Phasor measurement units

Synchrotron radiation

BT: Synchrotrons
RT: Biomedical applications of
radiation
Light sources
X-rays

Synchrotrons

BT: Particle accelerators
RT: Colliding beam accelerators
Electric fields
High energy physics
instrumentation computing
Magnetic fields
Particle beams
NT: Synchrotron radiation
Undulators

Synfuel

USE: Synthetic fuels

Syngas

UF: Synthesis gas
Synthetic gas
BT: Gases

Syntactics

UF: Syntax
BT: Semiotics
RT: Communication symbols
Grammar
Natural language

processing

Professional

communication

Programming

Syntax

USE: Syntactics

Synthesis gas

USE: Syngas

Synthesizers

USE: Synthesizers

Synthesizers

UF: Synthesizers
BT: Electronic music



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Synthetic aperture radar

UF: SAR
BT: Radar
RT: Airborne radar
Ground penetrating radar
Radar imaging
Spaceborne radar
Synthetic aperture sonar
Ultra wideband radar
NT: Inverse synthetic aperture

radar

Polarimetric synthetic

aperture radar

Synthetic aperture radar imaging

USE: Radar polarimetry

Synthetic aperture radar interferometry

BT: Radar interferometry

Synthetic aperture sonar

UF: SAS
BT: Sonar
RT: Synthetic aperture radar

Synthetic biology

UF: Synthetic life research
BT: Biology
Engineering in medicine

and biology

RT: Biological system modeling
Computational biology

Synthetic data

BT: Computer simulation
Sampling methods
RT: Data augmentation
Data privacy
Information integrity
Neural radiance field

Synthetic fibers

UF: Artificial fibers
Artificial fibres
Nylon fiber
Synthetic fibres
BT: Textile fibers

Synthetic fibres

USE: Synthetic fibers

Synthetic fuels

UF: Synfuel
BT: Fuels

Synthetic gas

USE: Syngas

Synthetic life research

USE: Synthetic biology

Synthetic speech

USE: Speech synthesis

SYSML

USE: Systems Modeling

Language

System analysis

USE: System analysis and design

System analysis and design

UF: Logical decomposition
System analysis
System design
System metrics

BT: Systems engineering and

theory

RT: Configuration management
Design methodology
Flowcharts
Multi-agent systems
Skin effect
Stability analysis
System improvement
System validation
System verification
Systems simulation
Threat modeling

NT: Asymptotic stability
Closed box
Control system analysis
Diakoptics
Distributed processing
Distributed vision networks
Fault detection
Fault tolerant systems
Glass box
Interconnected systems
Large-scale systems
Lyapunov methods
Open systems
Petri nets
Physical design
Robust control
Scalability
Scattering parameters
Sequential analysis
Sequential diagnosis
Software prototyping



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Language	Static analysis	<i>System level design</i>
	System dynamics	USE: System-level design
	System performance	
	System-level design	<i>System life cycle management</i>
	Systems Modeling	USE: Technical management
	Systems modeling	<i>System lifecycle management</i>
	Task analysis	USE: Product lifecycle management
	Time factors	
System buses		<i>System metrics</i>
BT:	Computer interfaces	USE: System analysis and design
<i>System design</i>		<i>System modeling</i>
USE:	System analysis and design	USE: Modeling
System dynamics		System of systems
BT:	System analysis and design	BT: Systems engineering and theory
RT:	Behavioral sciences	RT: Complex networks
	Complex networks	Emergent phenomena
	Feedback	Networked control systems
	Flow production systems	NT: Cyber-physical systems
	Timing	
System identification		<i>System on chip</i>
BT:	Modeling	USE: System-on-chip
System implementation		System performance
BT:	Systems engineering and theory	UF: Cooperative cache
		BT: System analysis and design
		NT: Cooperative caching
System improvement		<i>System planning</i>
BT:	Systems engineering and theory	USE: Planning
RT:	Quality management	<i>System privacy management</i>
	Reliability	USE: Data security
	System analysis and design	
	System testing	
<i>System in package</i>		System realization
USE:	System-in-package	BT: Systems engineering and theory
System integration		System recovery
BT:	Systems engineering and theory	UF: Deadlocks (computers)
RT:	Enterprise resource	Error recovery (computers)
planning	Integrated manufacturing	BT: Computers and information processing
systems	Project management	RT: Business continuity
	Resource management	Operating systems
		Reliability
System kernels		NT: Checkpointing
BT:	Kernel	Core dumps
	Operating systems	Debugging



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System reliability

USE: Reliability

System resilience

USE: Fault tolerance

System software

BT: Software
RT: Visual BASIC
NT: File systems
Operating systems
Program processors
Utility programs

System testing

BT: System validation
System verification
Testing
RT: Closed box
Glass box
System improvement
NT: Model checking

System validation

BT: Systems engineering and
theory
RT: System analysis and design
NT: System testing

System verification

BT: Systems engineering and
theory
RT: System analysis and design
NT: System testing

System virtual machines

UF: Full virtualization VM
BT: Virtual machines

System-in-package

UF: System in package
BT: Chip scale packaging
System-on-chip
NT: Antenna-in-package

System-level design

UF: System level design
BT: System analysis and design

System-on-a-chip

USE: System-on-chip

System-on-chip

UF: On-chip
SOC

System on chip
System-on-a-chip
Application specific

BT:
integrated circuits

RT: AI accelerators
Microcontrollers
Microprocessors
Mixed analog-digital

integrated circuits

NT: Power dissipation
Signal processing
Lab-on-a-chip
Network-on-chip
System-in-package

Systematic literature review

UF: Literature review
SLR
Systematic review
BT: Reviews
RT: Publishing
Research initiatives
Scientific publishing

Systematic review

USE: Systematic literature review

Systematics

UF: Biological systematics
BT: Biology

Systems architecture

BT: Systems engineering and
theory
NT: Deep architecture

Systems biology

BT: Biology

Systems engineering

USE: Systems engineering and
theory

Systems engineering and theory

UF: Systems engineering
RT: Aerospace and electronic
systems

Business process

integration

Business process

management

NT: Digital twins
Adaptive systems
Capability engineering
Complex systems



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engineering	Configuration management		Systems operation	
	Failure state		BT:	Systems engineering and
	Hierarchical systems		theory	
	Integrated design		Systems simulation	
	Interface management		BT:	Simulation
	Military systems			Systems engineering and
	Modeling		theory	
	Multidimensional systems		RT:	System analysis and design
	Network systems			Technical management
	Physical design			
education	Reduced order systems		Systems support	
	Requirements engineering		BT:	Maintenance engineering
	Requirements management		theory	Systems engineering and
	Service-oriented systems		NT:	Smart systems
	Solution design		Systems thinking	
	Stochastic systems		BT:	Systems engineering and
	System analysis and design		theory	
	System implementation		RT:	Systems, man, and
	System improvement		cybernetics	
	System integration		Systems, man, and cybernetics	
theory	System of systems		RT:	Systems thinking
	System realization		NT:	Behavioral sciences
	System validation			Biological control systems
	System verification			Computational linguistics
	Systems architecture			Cybernetics
	Systems engineering			Ergonomics
	Systems operation			Human factors
	Systems simulation			Human-machine systems
	Systems support			Identification of persons
	Systems thinking			Pervasive computing
theory	Task analysis			Posthuman
	Technical management			Remote working
				Transhuman
				User interfaces
theory			Systolic arrays	
			BT:	Multiprocessing systems
			RT:	Pipeline processing
			<i>T-cells</i>	
			USE:	Lymphocytes
			<i>T2V</i>	
			USE:	Text to video
theory			Table lookup	
			UF:	LUT
				Look-up table
				Lookup table
			BT:	Data structures
				Image processing



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Tablet computers

UF: Tablet PC
BT: Computers
RT: Mobile handsets
Portable media players

Tablet PC

USE: Tablet computers

Tachometers

BT: Meters

Tachycardia

BT: Heart rate
RT: Bradycardia

Tactile feedback

USE: Tactile sensors

Tactile Internet

BT: Haptic interfaces
Internet of Things
RT: 5G mobile communication
Human-machine systems
Human-robot interaction
Information exchange
Machine-to-machine

communications

Smart devices
Tactile sensors

Tactile sensors

UF: Tactile feedback
Touch sensors
BT: Robot sensing systems
RT: Braille
Electronic skin
Pressure measurement
Tactile Internet
Touch sensitive screens
Wearable sensors

Tag clouds

UF: Word cloud
BT: Tagging

Tagging

UF: Hashtag
BT: Information retrieval
RT: Indexing
Internet of Things
NT: Tag clouds

TAI

USE: International Atomic Time

Tail

BT: Animal structures

Takagi-Sugeno model

BT: Fuzzy logic
RT: Fuzzy control
Fuzzy systems
NT: Takagi-Sugeno-Kang

model

Takagi-Sugeno-Kang model

BT: Takagi-Sugeno model

Talbot effect

BT: Optical imaging
RT: Interferometry
Optical interferometry

TAM

USE: Technology acceptance
model

Tantalum

BT: Chemical elements

Tap changers

BT: Transformers
NT: On load tap changers

Tape casting

BT: Casting
RT: Ceramics

Target detection

USE: Object detection

Target recognition

BT: Object recognition
RT: Missile guidance

Target tracking

BT: Tracking
RT: Control systems
Radar tracking

Targeted drug delivery

BT: Drug delivery
RT: Superparamagnetic iron
oxide nanoparticles

Tariffs

BT: Regulation
RT: Free economic zones
Trade agreements



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NT:	Feed in tariff	NT:	Transport protocols
Task analysis			TCP
BT:	Business process	<i>TD learning</i>	
management		USE:	Temporal difference
	System analysis and design	learning	
theory	Systems engineering and	<i>TDM</i>	
		USE:	Time division multiplexing
Taste buds		<i>TDSCDMA</i>	
BT:	Sense organs	USE:	Time division synchronous
<i>Taxes</i>		code division multiple access	
USE:	Finance	<i>Teaching</i>	
<i>Taxi</i>		USE:	Education
USE:	Public transportation	<i>Teaching machines</i>	
Taxonomy		USE:	Computer aided instruction
BT:	Information retrieval	<i>Team work</i>	
<i>Taylor expansion</i>		USE:	Teamwork
USE:	Taylor series	<i>Team working</i>	
Taylor series		USE:	Teamwork
UF:	Taylor expansion	Teamwork	
BT:	Mathematics	UF:	Team work
TCP			Team working
BT:	TCPIP	BT:	Collaboration
RT:	Session hijacking	RT:	Organizational aspects
NT:	MPTCP	Technetium	
<i>TCP session hijacking</i>		BT:	Chemical elements
USE:	Session hijacking	<i>Technical assessment</i>	
<i>TCP/IP</i>		USE:	Technical management
USE:	TCPIP	<i>Technical communication</i>	
<i>TCP/IP protocol suite</i>		USE:	Professional
USE:	TCPIP	communication	
TCPIP		<i>Technical data management</i>	
UF:	TCP/IP	USE:	Database systems AND
	TCP/IP protocol suite		Technical management
	Transmission control	Technical drawing	
protocol-internet protocol		BT:	Design methodology
	Transmission control	RT:	Engineering drawings
protocol/internet protocol			Graphics
BT:	IP networks	Technical management	
RT:	Computer networks	UF:	System life cycle
	Data communication	management	
	Digital communication		Technical assessment
	Internet		
	Protocols		



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management	Technical data	Disruptive technologies	Technology social factors
	Technical risk management		
BT:	Management	Technology	
	Systems engineering and	BT:	Social implications of
theory		technology	
RT:	Program management	RT:	Engineering - general
	Systems simulation		Museums
NT:	Maintenance management		Oil drilling
	Technical planning		Philosophical
<i>Technical manuals</i>		considerations	
USE:	Manuals		Research and development
			STEM
<i>Technical meetings</i>			Technology forecasting
USE:	Meetings	NT:	Technology planning
			Appropriate technology
Technical planning			Disruptive technologies
BT:	Planning		Fintech
	Technical management		Machine ethics
			Neurotechnology
<i>Technical proposals</i>			Technological innovation
USE:	Proposals		Technology social factors
			Technology transfer
<i>Technical reports</i>			Telepresence
USE:	Writing		Telexistence
		Technology acceptance model	
Technical requirements		UF:	TAM
BT:	Requirements engineering	BT:	Human factors
RT:	Proposals		Information theory
		RT:	Computer aided instruction
<i>Technical risk management</i>			Consumer behavior
USE:	Technical management		Information systems
			Social implications of
<i>Technical textiles</i>		technology	
USE:	Textile products		Statistical analysis
			Technology transfer
<i>Technical writing</i>			User centered design
USE:	Writing		User experience
		Technology forecasting	
<i>Technician training</i>		UF:	Futurism
USE:	Training		Technological forecasting
		BT:	Forecasting
<i>Technique for order of preference by similarity to ideal solution</i>		RT:	Technology
USE:	TOPSIS		Technology social factors
		NT:	Roadmaps (technology
<i>Technological forecasting</i>		planning)	
USE:	Technology forecasting	Technology management	
		BT:	Management
Technological innovation		RT:	Data processing
UF:	Innovation		Innovation management
	Invention		Production management
BT:	Technology		Project management
RT:	Disruptive innovation		



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management Research and development
Technology transfer

Technology mediated communication
USE: Computer mediated communication

Technology planning

BT: Planning
RT: Social factors
Technology

Technology social factors

BT: Social factors
Technology
RT: Philosophical
considerations
Risk analysis
Technological innovation
Technology forecasting
NT: Privacy

Technology transfer

BT: Technology
RT: Technology acceptance
model
NT: Technology management
Small business technology
transfer

Technology-mediated communication
USE: Computer mediated communication

Tectonics

BT: Geology
RT: Continental crust
Oceanic crust
Rocks
NT: Subduction

Teeth

UF: Tooth
BT: Mouth

TEGFETs

USE: MODFETs

Telecom

USE: Telecommunications

Telecom channels

USE: Communication channels

Telecom computing

USE: Telecommunication computing

Telecom congestion control

USE: Telecommunication congestion control

Telecom control

USE: Telecommunication control

Telecom network management

USE: Telecommunication network management

Telecom network reliability

USE: Telecommunication network reliability

Telecom network topology

USE: Telecommunication network topology

Telecom services

USE: Telecommunication services

Telecom signaling

USE: Communication system signaling

Telecom switching

USE: Telecommunication switching

Telecom system signaling

USE: Communication system signaling

Telecom traffic

USE: Telecommunication traffic

Telecommunication buffers

BT: Data communication

Telecommunication channels

USE: Communication channels

Telecommunication computing

UF: Communications computing
Telecom computing
BT: Computer applications
Telecommunications
RT: 3G mobile communication
4G mobile communication



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networking	Information-centric	services	Telecommunication
	Mobile computing		Telecommunication traffic
	Quality of service		
	Software radio	Telecommunication network reliability	
	TV	UF:	Communication network
	Telecommunication control	reliability	
	Telecommunication		Telecom network reliability
network management	Telegraphy	BT:	Reliability
	Telephony		Telecommunication
NT:	Internetworking	network topology	
	Soft switching	RT:	Telecommunication
		network performance	
		NT:	Diversity schemes
Telecommunication congestion control		Telecommunication network topology	
UF:	Telecom congestion control	UF:	Telecom network topology
BT:	Telecommunication	BT:	Telecommunications
network topology		RT:	Dynamic spectrum access
NT:	Call admission control		Network topology
			Telecommunication
Telecommunication control		network performance	
UF:	Telecom control	NT:	Backhaul networks
BT:	Communication system		Intelligent networks
control			Link aggregation
RT:	Telecommunication		Passive networks
computing			Telecommunication
NT:	Virtual links	congestion control	
			Telecommunication
Telecommunication network management		network management	
UF:	Telecom network		Telecommunication
management		network reliability	
BT:	Telecommunication		Telecommunication traffic
network topology			
RT:	Border Gateway Protocol	<i>Telecommunication power management</i>	
	Management information	USE:	Power system management
base			
	Telecommunication	<i>Telecommunication security</i>	
computing		USE:	Communication system
	Telecommunication	security	
network performance			
NT:	Mobile nodes	Telecommunication services	
	Network architecture	UF:	Telecom services
	Network neutrality	BT:	Telecommunications
	Network resource	RT:	Radio access networks
management			Telecommunication
		network performance	
Telecommunication network performance		NT:	Acoustic communication
BT:	Communication networks	(telecommunication)	
RT:	Telecommunication		Number portability
network management			
	Telecommunication	<i>Telecommunication signalling</i>	
network reliability		USE:	Communication system
	Telecommunication	signaling	
network topology			



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Telecommunication standards

USE: Communication standards

Telecommunication switching

UF: Telecom switching

BT: Switching systems

Telecommunication traffic

UF: Network traffic

Telecom traffic

Traffic load

BT: Telecommunication

network topology

RT: Border Gateway Protocol

Communication system

traffic

Telecommunication

network performance

NT: Sidelink

Telecommunications

UF: Telecom

BT: Communication systems

RT: Convolutional codes

Diversity reception

Film bulk acoustic

resonators

Global Positioning System

Helical antennas

Multiaccess communication

Multicarrier code division

multiple access

Multicast communication

Next generation networking

Optical wavelength

conversion

Reflectivity

Switched circuits

Telecontrol equipment

NT: Ambient intelligence

Feedback communications

IP networks

Radio access networks

Railway communication

Space communications

Telecommunication

computing

Telecommunication

network topology

Telecommunication

services

Telematics

Telecommuting

USE: Remote working

Teleconferencing

BT: Communication systems

RT: Image communication

Meetings

Office automation

Web conferencing

Telecontrol equipment

BT: Control equipment

RT: Communication systems

Data communication

Power industry

Power systems

Remote handling

Telecommunications

Telegraphy

BT: Communication systems

RT: Telecommunication

computing

Telehealth

USE: Telemedicine

Telehealthcare

USE: Telemedicine

Teleheating

USE: District heating

Telematics

BT: Information technology

Telecommunications

RT: Cyberspace

Telemedicine

UF: Telehealth

Telehealthcare

BT: Biomedical communication

RT: Telepresence

Telemetry

BT: Aerospace and electronic

systems

Data communication

RT: Deep-space

communications

Measurement

NT: Biomedical telemetry

Teleoperators

BT: Telerobotics

Telephone equipment

BT: Communication equipment



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equipment	RT:	Land mobile radio		Mobile robots	
				Remote handling	
equipment		Radio communication	equipment		
			NT:	Teleoperators	
NT:		Telephony	Telescopes		
		Cellular phones			
		Landline		BT:	Instruments
		Telephone sets		RT:	Astronomy
		Vocoders			Observatories
Telephone poles				Radio astronomy	
			NT:	Gamma-ray telescopes	
	BT:	Poles and towers		X-ray telescopes	
Telephone sets			<i>Telesurgical robotics</i>		
	UF:	Handsets	USE:	Telerobotics	
		Subscriber sets			
	BT:	Telephone equipment	Teletext		
	RT:	Telephony		BT:	Communication systems
	NT:	Mobile handsets			Information services
Telephony			RT:	Data communication	
				Videotex	
	BT:	Communication systems			
	RT:	Telecommunication	<i>Teletype</i>		
computing			USE:	Teleprinting	
		Telephone equipment			
		Telephone sets	<i>Television</i>		
		Videophone systems	USE:	TV	
Teleportation			<i>Teleworking</i>		
	UF:	Quantum teleportation	USE:	Remote working	
	BT:	Quantum mechanics			
	RT:	Information theory	Teleexistence		
		Quantum communication		BT:	Human computer
		Quantum entanglement		interaction	
Telepresence				Real-time systems	
				Technology	
	BT:	Human computer			
interaction			Tellurium		
		Technology	BT:	Chemical elements	
	RT:	Telemedicine			
Teleprinting			<i>TEM</i>		
			USE:	Transmission electron	
	UF:	Teletype	microscopy		
	BT:	Communication systems			
		Data communication	TEM cells		
		Printing		UF:	GHZ transverse
	RT:	Digital communication		electromagnetic cells	
Telerobotics				GTEM cells	
				Transverse electromagnetic	
		Telesurgical robotics	cells		
		Robots	BT:	Test facilities	
		Delay systems	RT:	Anechoic chambers	
		Human factors		Electromagnetic	
		Manipulators	compatibility and interference		
		Medical robotics			



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interference	Electromagnetic	learning	Temporal-difference
testing	Electronic equipment	BT:	Predictive models
		RT:	Reinforcement learning
Temperature			Dynamic programming
BT:	Thermal factors		Monte Carlo methods
RT:	Temperature control		Unsupervised learning
	Temperature measurement	Temporal lobe	
NT:	Color temperature	BT:	Brain
	Temperature distribution	NT:	Hippocampus
Temperature control		<i>Temporal-difference learning</i>	
BT:	Thermal variables control	USE:	Temporal difference
RT:	Space heating	learning	
	Temperature	Tendons	
	Thermal factors	BT:	Musculoskeletal system
	Ventilation	NT:	Achilles tendon
NT:	Cooling	<i>Tennis</i>	
	Heating systems	USE:	Sports
Temperature dependence		<i>Tensile modulus</i>	
BT:	Thermal factors	USE:	Young's modulus
Temperature distribution		Tensile strain	
BT:	Temperature	BT:	Strain
RT:	Insulators	RT:	Tensile stress
Temperature measurement		Tensile stress	
BT:	Thermal variables	BT:	Stress
measurement		RT:	Tensile strain
RT:	Bolometers	Tensors	
	Radiometry	BT:	Mathematics
	Temperature	Terahertz communications	
	Temperature sensors	BT:	Communication systems
	Thermistors	Terahertz materials	
	Thermoresistivity	BT:	Materials
NT:	Cryobiology	RT:	Optical antennas
	Cryogenics		Terahertz wave absorption
	Cryotherapy	NT:	Terahertz metamaterials
	Global warming	Terahertz metamaterials	
	Kelvin	BT:	Electromagnetic
	Thomson effect	metamaterials	
Temperature sensors		RT:	Terahertz materials
BT:	Thermal sensors		Split ring resonators
RT:	Bragg gratings		
	Optical fibers		
	Temperature measurement		
	Transducers		
NT:	Thermocouples		
	Thermometers		
Temporal difference learning			
UF:	TD learning		



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Terahertz radiation

BT: Electromagnetic radiation
Radiation effects
RT: Terahertz wave absorption

Terahertz wave absorption

BT: Electromagnetic wave
absorption
RT: Submillimeter wave filters
Terahertz materials
Terahertz radiation

Terahertz wave imaging

BT: Imaging

Terbium

BT: Chemical elements

Terminal velocity

BT: Velocity measurement

Termination of employment

UF: Dismissal
Redundancy (employment)
BT: Employment
Human resource
management
RT: Pensions

Terminology

UF: Definitions
Glossaries
BT: Information retrieval
RT: Translation
NT: Dictionaries
Inclusive language

Ternary logic

USE: Multivalued logic

Terrain factors

BT: Interference
RT: Earth
Multipath channels
Rough surfaces

Terrain mapping

UF: Topography (earth)
BT: Geoscience and remote
sensing
RT: Earth
Geologic measurements
Geophysical measurements
Global Positioning System
Remote sensing

Vegetation mapping
Digital elevation models

Terrestrial atmosphere

UF: Earth atmosphere
Stratosphere
Troposphere
BT: Geoscience and remote
sensing
RT: Atmospheric
measurements
Geophysics
Meteorology
Clouds
Global warming
Ionosphere
Magnetosphere

Terrestrial communications

BT: Communication systems
RT: Space-air-ground
integrated networks

Terrorism

UF: 9/11
9/11 attack
911 attack
September 11
Terrorist
BT: Security
RT: Biohazards
Chemical weapons
Nuclear weapons
Public security
Surveillance
Threat assessment
US Department of
Homeland Security
Weapons
Weapons of mass
destruction
NT: Bioterrorism
Cyber terrorism
National security

Terrorist

USE: Terrorism

Test data compression

BT: Data compression

Test equipment

BT: Testing
RT: Oscilloscopes
NT: Automatic test equipment



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Test facilities

BT: Testing
NT: Anechoic chambers
Laboratories
Large Hadron Collider
Open area test sites
TEM cells
Wind tunnels

Test equipment
Test facilities

Test generation

USE: Test pattern generators

Test pattern generators

UF: Test generation
BT: Automatic test pattern

generation

Testing

BT: Industrial electronics
Instrumentation and

measurement

RT: Cause effect analysis
Fault diagnosis
Hardware-in-the-loop

simulation

Inspection
Leak detection
Maintenance engineering
Measurement
NT: Aerospace testing
Automatic testing
Benchmark testing
Built-in self-test
Circuit testing
Conformance testing
Device under test
Electronic equipment

testing

Error analysis
Error-free operations
Failure analysis
Frequency response
Impulse testing
Insulator testing
Integrated circuit testing
Life testing
Materials testing
Optical fiber testing
Remaining life assessment
Ring generators
Semiconductor device

testing

Shear testing
Software testing
System testing

Text analysis

BT: Data mining
RT: Annotations
Conditional random fields
Naive Bayes methods
Translation
NT: Text categorization
Text summarization

Text categorization

UF: Text classification
BT: Text analysis
RT: Data analysis

Text classification

USE: Text categorization

Text detection

BT: Image processing
Text processing
Video signal processing
RT: Discrete Fourier transforms
Discrete wavelet transforms
Image segmentation
Machine learning
Text to speech

Text messaging

USE: Electronic messaging

Text mining

BT: Data mining
RT: Triples (Data structure)

Text processing

UF: Photocomposition
Word processing
BT: Data processing
RT: Desktop publishing
Document handling
Office automation
Publishing
Text recognition
NT: Text detection
Text to image
Text to speech
Typesetting

Text recognition

BT: Pattern recognition
RT: Character recognition
Text processing



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	Text to image		
	Vision language model		<i>Textile based electrodes</i>
NT:	Large language models	USE:	Textile electrodes
	Text to video		
Text summarization		Textile electrodes	
UF:	Automatic summarization	UF:	Textile based electrodes
BT:	Abstracts		Textile-based electrodes
	Text analysis	BT:	Electrodes
RT:	Artificial intelligence	RT:	Textile technology
	Machine learning		Wearable devices
	Natural language		
processing		Textile fibers	
	Unsupervised learning	UF:	Fibers
			Textile fibres
		BT:	Textiles
		RT:	Cotton
Text to image			Spinning
UF:	Text-to-image		Textile products
BT:	Artificial intelligence		Textile technology
	Image matching		Weaving
	Text processing		Wool
RT:	Machine learning	NT:	Natural fibers
	Text recognition		Synthetic fibers
			Yarn
Text to speech			
UF:	Text-to-speech	<i>Textile fibres</i>	
BT:	Speech synthesis	USE:	Textile fibers
	Text processing		
RT:	Audio-visual systems	Textile industry	
	Digital TV	BT:	Manufacturing industries
	Speech recognition	RT:	Clothing industry
	Speech to text		Cotton
	Text detection		Spinning machines
			Textile machinery
Text to video			Textile products
UF:	T2V		Textile technology
	Text-to-video		Weaving
BT:	Artificial intelligence		
	Text recognition	Textile machinery	
	Videos	BT:	Machinery
RT:	Animation	RT:	Needles
	Machine learning		Textile industry
			Textile products
<i>Text-to-image</i>			Textile technology
USE:	Text to image		Textiles
		NT:	Spinning machines
<i>Text-to-speech</i>			
USE:	Text to speech	Textile products	
		UF:	Technical textiles
<i>Text-to-video</i>		BT:	Manufactured products
USE:	Text to video	RT:	Textile fibers
			Textile industry
Textile antennas			Textile machinery
BT:	Antennas		Textile technology
	Smart textiles		Textiles
RT:	Wearable antennas		



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Textile technology

BT: Industries
RT: Bleaching
Textile electrodes
Textile fibers
Textile industry
Textile machinery
Textile products
Textiles
NT: Spinning
Weaving

Textile-based electrodes

USE: Textile electrodes

Textiles

BT: Materials
RT: Spinning machines
Textile machinery
Textile products
Textile technology
Weaving
NT: Cotton
Fabrics
Textile fibers
Wool

TFETs

UF: Tunnel field effect
transistors
BT: Field effect transistors
RT: MOSFET

TFT

USE: Thin film transistors

Thalamus

BT: Brain

Thallium

BT: Chemical elements

Theodolites

BT: Instruments
RT: Geodesy
Geologic measurements
Geophysical measurement

techniques

Theoretical neuroscience

USE: Computational
neuroscience

Theory of constraints

USE: Constraint theory

Therapy

USE: Medical treatment

Thermal analysis

BT: Thermal variables control
NT: Thermomechanical
processes

Thermal conductivity

BT: Thermal factors
RT: Grain boundaries
Thermal resistance
NT: Heat transfer

Thermal decomposition

BT: Thermolysis

Thermal degradation

BT: Thermolysis

Thermal energy

BT: Heating systems
RT: Energy
Kinetic energy
Ocean thermal energy
conversion
Thermal engineering

Thermal engineering

BT: Engineering - general
RT: Cooling
Heat recovery
Heating systems
Thermal energy
Thermal factors
Thermal variables control
Thermal variables
measurement

Thermal expansion

BT: Thermal factors
RT: Electrothermal actuators
NT: Thermal force

Thermal factors

UF: High-temperature effects
BT: Physics
RT: Annealing
Critical current density
Heat treatment
Proton effects
Pyroelectricity
Superconducting devices
Superconducting materials
Temperature control



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	Thermal engineering	RT:	Air pollution
	Thermal stability		Global warming
	Thermal variables control		Industrial pollution
	Thermal variables		Marine pollution
measurement			Waste heat
NT:	Temperature		Water pollution
	Temperature dependence		
	Thermal conductivity	Thermal quenching	
	Thermal expansion	BT:	Cooling
	Thermal management	Thermal resistance	
	Thermal stresses	BT:	Resistance
	Thermoelasticity	RT:	Thermal conductivity
	Thermoelectricity	Thermal sensors	
	Thermolysis	BT:	Sensors
	Thermooptic effects	NT:	Electrothermal actuators
	Thermoresistivity		Temperature sensors
Thermal force			
BT:	Thermal expansion		
Thermal lensing		Thermal shock	
BT:	Thermooptic effects	BT:	Shock (mechanics)
RT:	Laser beams		Thermal stresses
	Nonlinear optics	Thermal spraying	
	Optical distortion	BT:	Spraying
	Solid lasers	Thermal stability	
Thermal loading		BT:	Stability
BT:	Thermal stresses	RT:	Integrated circuit reliability
			Thermal factors
Thermal management		Thermal stresses	
BT:	Thermal factors	BT:	Thermal factors
RT:	Enthalpy	NT:	Thermal loading
	Reliability		Thermal shock
Thermal management of electronics		Thermal variables control	
BT:	Components, packaging,	BT:	Control systems
and manufacturing technology		RT:	Thermal engineering
NT:	Electronic packaging		Thermal factors
thermal management		NT:	HVAC
	Electronics cooling		Temperature control
			Thermal analysis
Thermal noise		Thermal variables measurement	
UF:	Johnson Nyquist noise	BT:	Measurement
BT:	Circuit noise	RT:	Calorimetry
RT:	Conductors		Thermal engineering
<i>Thermal plumes</i>			Thermal factors
USE:	Thermal pollution		Transducers
Thermal pollution		NT:	Temperature measurement
UF:	Heat islands	<i>Thermal video</i>	
	Thermal plumes	USE:	Infrared imaging
	Urban heat islands		
BT:	Pollution		



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Thermal wave imaging

USE: Photothermal effects

Thermionic emission

BT: Nuclear and plasma sciences
RT: Electron emission
Ion emission
Transmission electron microscopy
Vacuum arcs

Thermionic valves

USE: Electron tubes

Thermistors

BT: Semiconductor devices
RT: Temperature measurement
Thermoresistivity

Thermo-mechanics

USE: Thermomechanical processes

Thermo-optic effects

USE: Thermooptic effects

Thermo-optical devices

USE: Thermooptical devices

Thermochromism

BT: Thermooptic effects

Thermocouples

BT: Temperature sensors

Thermodynamics

BT: Science - general
NT: Adiabatic processes
Enthalpy
Fermi level
Isobaric processes
Isothermal processes
Quasi-Fermi level

Thermoelasticity

BT: Thermal factors

Thermoelectric devices

BT: Thermoelectricity

Thermoelectric effect

USE: Thermoelectricity

Thermoelectric materials

BT: Materials
Thermoelectricity

Thermoelectricity

UF: Seebeck effect
Thermoelectric effect
BT: Electricity
Energy conversion
Thermal factors
NT: Electrothermal effects
Peltier effect
Thermoelectric devices
Thermoelectric materials

Thermoforming

BT: Manufacturing systems

Thermoluminescence

BT: Luminescence
RT: Chemiluminescence

Thermolysis

BT: Chemical processes
Thermal factors
NT: Thermal decomposition
Thermal degradation

Thermomechanical effects

USE: Thermomechanical processes

Thermomechanical processes

UF: Thermo-mechanics
Thermomechanical effects
Thermomechanics
BT: Thermal analysis
RT: Heat treatment
Smart materials

Thermomechanics

USE: Thermomechanical processes

Thermometers

BT: Temperature sensors

Thermonuclear fusion

USE: Fusion reactors

Thermooptic effects

UF: Thermo-optic effects
BT: Thermal factors
RT: Birefringence
Optical propagation



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Optical reflection
Optical refraction
Thermooptical devices
NT: Thermal lensing
Thermochromism
Thermoreflectance

Thermooptical devices

UF: Thermo-optical devices
BT: Optical devices
RT: Integrated optics
Optical switches
Solid lasers
Thermooptic effects

Thermoplastic polyethylene

BT: Polyethylene
NT: UHMWPE

Thermoreflectance

BT: Thermooptic effects

Thermoreflectance imaging

BT: Optical imaging
RT: Spectroscopy

Thermoresistivity

BT: Thermal factors
RT: Temperature measurement
Thermistors

Thermostats

BT: Control equipment

Thesauri

UF: Thesaurus
BT: Knowledge representation
Writing
RT: Ontologies

Thesaurus

USE: Thesauri

Thick film circuits

BT: Circuits
Integrated circuits
RT: Hybrid integrated circuits
Thick film devices
Thick film inductors

Thick film devices

BT: Electron devices
RT: Thick film circuits
Thick films
NT: Thick film inductors

Thick film inductors

BT: Inductors
Thick film devices
RT: Microstrip components
Thick film circuits
Thick films

Thick film sensors

UF: Thick-film sensors
BT: Sensors

Thick films

BT: Films
RT: Dielectric films
Semiconductor films
Superconducting films
Thick film devices
Thick film inductors

Thick-film sensors

USE: Thick film sensors

Thickness control

BT: Mechanical variables
control
RT: Size control

Thickness measurement

BT: Mechanical variables
measurement
RT: Micrometers
Size measurement

Thigh

BT: Extremities

Thin film circuits

BT: Circuits
Integrated circuits
RT: Hybrid integrated circuits
Silicon-on-insulator
Thin film devices
Thin film inductors

Thin film deposition

USE: Sputtering

Thin film devices

BT: Electron devices
RT: Amorphous semiconductors
Doping profiles
Giant magnetoresistance
Thin film circuits
Thin films



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NT: Film bulk acoustic resonators
Thin film inductors
Thin film transistors

Thin film inductors

BT: Inductors
Thin film devices
RT: Thin film circuits
Thin films

Thin film sensors

BT: Sensors

Thin film transistors

UF: TFT
Thin-film transistors
BT: Active matrix technology
Field effect transistors
Thin film devices
RT: Displays
Liquid crystal devices
NT: Organic thin film transistors

Thin films

BT: Films
RT: Diamond-like carbon
Dielectric films
Epitaxial layers
Magnetic films
Metasurfaces
Molecular beam epitaxial

growth

Molecular beams
Self-assembly
Semiconductor films
Superconducting films
Superconducting thin films
Thin film devices
Thin film inductors
Vapor deposition
NT: Buffer layers
Epitaxial growth
Semiconductor thin films

Thin wall structures

BT: Structural shapes
RT: Honeycomb structures
Lightweight structures
Sandwich structures
Sheet materials
Structural panels
Structural shells

Thin-film transistors

USE: Thin film transistors

Third generation mobile communication

USE: 3G mobile communication

Thomson effect

BT: Temperature measurement

Thorax

BT: Body regions
Skeleton
NT: Ribs
Sternum

Thorium

BT: Chemical elements

Threat assessment

UF: Threat detection
BT: Risk analysis
RT: Law enforcement
Public security
Security information and

event management

Terrorism
NT: Threat modeling

Threat detection

USE: Threat assessment

Threat modeling

UF: Threat modelling
BT: Systems modeling
Threat assessment
RT: Computer crime
Computer security
Cyber threat intelligence
Cyberattack
Data privacy
Data security
System analysis and design

Threat modelling

USE: Threat modeling

Three dimensional displays

USE: Three-dimensional displays

Three dimensional integrated circuits

USE: Three-dimensional integrated circuits



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Three-dimensional displays

UF: 3-D displays
3-D modeling
3-D modelling
3-D reconstruction
3D displays
3D modeling
3D modelling
3D reconstruction
Three dimensional displays
BT: Displays
RT: Metaverse
Neural radiance field
Point cloud compression
Reconstruction algorithms
Shadow mapping
Sprites (computer)
Structure from motion
NT: Bundle adjustment
X3D

Three-dimensional integrated circuits

UF: 3D integrated circuits
3D integration
3D interconnect
3DI
Three dimensional
integrated circuits
BT: Integrated circuits

Three-dimensional printing

UF: 3D printing
Additive manufacturing
BT: Manufacturing systems
Printing
RT: Ink jet printing
Rapid prototyping
NT: Bioprinting

Three-dimensional television

BT: TV

Three-dimensional vision

USE: Stereo vision

Three-phase electric power

BT: Power electronics
RT: Conductors
Voltage control

Three-term control

BT: Process control

Threshold current

BT: Current

RT: Electron devices
Lasers

Threshold voltage

BT: Voltage
RT: Integrated circuit noise
MOSFET circuits
Transistors
NT: Subthreshold current

Thresholding (Imaging)

BT: Image processing
RT: Image edge detection

Thrombosis

BT: Medical conditions

Through-silicon vias

UF: TSV
BT: Integrated circuits

Throughput

UF: Network throughput
BT: Communication channels
RT: Quality of transmission

Throughput (communication systems)

USE: Information rates

Thulium

BT: Chemical elements

Thumb

BT: Fingers

Thyratrons

BT: Electron tubes
RT: Gas discharge devices

Thyristor circuits

BT: Circuits
RT: Thyristors

Thyristors

UF: Diacs
SCR
Semiconductor controlled

rectifiers

Silicon controlled rectifiers
Triacs

BT: Power semiconductor

switches

RT: Thyristor circuits
NT: Photothyristors



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Thyroid

BT: Glands
RT: Endocrinology
Thyroid cancer

Thyroid cancer

BT: Cancer
RT: Thyroid

Ti

USE: Titanium

Tianwen-1

USE: Interplanetary exploration

Tidal energy

UF: Tidal power
BT: Renewable energy sources
Tides

Tidal power

USE: Tidal energy

Tides

BT: Oceans
RT: Ocean circulation
Ocean dynamics
Storm surge
NT: Tidal energy

TikTok

USE: Social networking (online)

Tiles

BT: Building materials
RT: Ceramic products
Ceramics
Floors

Timber industry

USE: Lumber industry

Timbre

BT: Music

Time complexity

BT: Computational complexity
RT: Computational modeling
NT: Reversible computing

Time delay

USE: Delay effects

Time difference of arrival

UF: Time-difference-of-arrival

BT: Object detection

Time dissemination

BT: Time measurement
RT: Satellite navigation systems
Synchronization

Time division multiaccess

USE: Time division multiple access

Time division multiple access

UF: Time division multiaccess
BT: Multiaccess communication

Time division multiplexed

USE: Time division multiplexing

Time division multiplexing

UF: TDM
BT: Time division multiplexed
Multiplexing

Time division synchronous code division multiple access

UF: TDSCDMA
BT: Multiaccess communication
RT: 3G mobile communication
4G mobile communication
Cellular radio
Multicarrier code division

multiple access

Spread spectrum

communication

Time domain analysis

USE: Time-domain analysis

Time factors

BT: System analysis and design
RT: Bang-bang control
NT: Continuous time systems
Discrete-time systems
Time invariant systems
Time-varying systems

Time frequency analysis

USE: Time-frequency analysis

Time invariant systems

BT: Time factors
RT: Differential equations
Discrete-time systems
Feedback
Linear systems



January 2026 IEEE Thesaurus

	Stability		Time-domain analysis
	Time-varying systems	UF:	FDTD
			Time domain analysis
Time measurement		BT:	Electromagnetic analysis
BT:	Measurement	RT:	Phase noise
RT:	Time-frequency analysis		State-space methods
	Watches		Waves
NT:	Clocks		
	Time dissemination	Time-frequency analysis	
	Timing	UF:	Time frequency analysis
		BT:	Frequency-domain analysis
Time of arrival estimation		RT:	Biomedical signal
UF:	TOA estimation		
	Time-of-arrival estimation	processing	
BT:	Parameter estimation		Fourier transforms
RT:	Array signal processing		Frequency measurement
	Direction-of-arrival		Image processing
estimation			Power systems
	Signal detection		Time measurement
			Video signal processing
Time series analysis			
UF:	time-series analysis	<i>Time-of-arrival estimation</i>	
BT:	Statistics	USE:	Time of arrival estimation
RT:	Autocorrelation		
	Autoregressive processes	<i>time-series analysis</i>	
	Chaotic communication	USE:	Time series analysis
	Modeling		
	Random processes	<i>Time-sharing computer systems</i>	
	Spectral analysis	USE:	Time sharing computer
	Statistical analysis	systems	
Time sharing computer systems		<i>Time-sharing systems</i>	
UF:	Time-sharing computer	USE:	Time sharing computer
systems		systems	
	Time-sharing systems		
BT:	Computers and information	Time-varying channels	
processing		BT:	Communication channels
RT:	Mainframes	RT:	Mobile communication
			Wireless LAN
Time to market		Time-varying systems	
BT:	Design methodology	UF:	Time varying systems
	Product development	BT:	Time factors
RT:	Concurrent engineering	RT:	Control systems
			Time invariant systems
Time varying circuits		NT:	Switched systems
BT:	Circuits		
<i>Time varying systems</i>		Timing	
USE:	Time-varying systems	BT:	Time measurement
		RT:	Clocks
Time warp simulation			Logic design
BT:	Discrete event simulation		System dynamics
			Timing jitter
<i>Time-difference-of-arrival</i>		NT:	Bit rate
USE:	Time difference of arrival		Delays
			Synchronization



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Timing jitter					Diamond-like carbon
BT:	Jitter				Genetic engineering
RT:	Timing				Hydrogels
				NT:	Regeneration engineering
Tin					
UF:	Sn		<i>Tissue scaffolds</i>		
BT:	Metals		USE:	Tissue engineering	
NT:	Tin alloys				
	Tin compounds		<i>Tissues</i>	USE:	Biological tissues
Tin alloys					
BT:	Tin		<i>Titanates</i>		
RT:	Alloying		USE:	Titanium compounds	
NT:	Niobium-tin				
Tin compounds			<i>Titania</i>	USE:	Titanium dioxide
BT:	Tin				
Tiny machine learning			Titanium		
UF:	TinyML		UF:	Ti	
BT:	Machine learning		BT:	Chemical elements	
RT:	Data analysis		NT:	Metals	
	Microcontrollers			Titanium alloys	
	Sensor systems			Titanium compounds	
				Titanium dioxide	
				Titanium nitride	
<i>TinyML</i>					
USE:	Tiny machine learning		Titanium alloys		
Tipping point			BT:	Titanium	
BT:	Global warming		RT:	Alloying	
RT:	Climate change				
Tire pressure			Titanium compounds		
BT:	Pressure measurement		UF:	Titanates	
	Tires		BT:	Titanium	
Tires			Titanium dioxide		
UF:	Tyres		UF:	Titania	
BT:	Mechanical products		BT:	Titanium	
	Rubber products				
RT:	Automobile manufacture		Titanium nitride		
	Automotive components		BT:	Titanium	
	Vehicles				
	Wheels		<i>TMR</i>	USE:	Tunneling
NT:	Tire pressure		magnetoresistance		
Tissue damage			<i>TOA estimation</i>		
BT:	Lesions		USE:	Time of arrival estimation	
Tissue engineering			<i>Toddler</i>	USE:	Pediatrics
UF:	Tissue scaffolds				
BT:	Biomedical engineering		Tokamak devices		
RT:	Biological materials		BT:	Tokamaks	
	Bioprinting		RT:	Magnetic confinement	
	Colloidal lithography			Toroidal magnetic fields	



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Tokamaks

BT: Fusion reactors
Plasma applications
Plasma devices
RT: Magnetic confinement
Plasma simulation
NT: Tokamak devices

Token networks

BT: Communication systems
Computer networks
Digital systems
RT: Federated identity
Local area networks
Metropolitan area networks
Wide area networks

Tokenization

BT: Data security
Natural language
processing
RT: Federated identity

Tolerance analysis

UF: Circuit tolerance analysis
Tolerating problems
BT: Manufacturing
RT: Circuit analysis
Circuit optimization
Semiconductor device
breakdown
Sensitivity

Tolerating problems

USE: Tolerance analysis

Tomographic

USE: Tomography

Tomography

UF: Tomographic
BT: Imaging
RT: Biomedical imaging
Geophysical measurement
techniques
Image reconstruction
NT: Computed tomography
Electrical capacitance
tomography
Electrical impedance
tomography
Magnetic particle imaging
Optical coherence
tomography

Positron emission
tomography
Reconstruction algorithms

Tomosynthesis

USE: Biomedical imaging

Tongue

BT: Digestive system
RT: Stomatognathic system

Tons of CO2

BT: Carbon dioxide

Tools

BT: Manufactured products
NT: Hand tools

Tooth

USE: Teeth

Topography (earth)

USE: Terrain mapping

Topological insulators

BT: Insulators
RT: Quantum materials

Topology

BT: Mathematics
RT: Graph theory
Morphological operations

TOPSIS

UF: Technique for order of
preference by similarity to ideal solution
BT: Decision theory
RT: Decision making
Fuzzy set theory
Operations research
Optimization

Tornado

USE: Tornadoes

Tornadoes

UF: Tornado
Tornados
BT: Geoscience

Tornados

USE: Tornadoes



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Toroidal magnetic fields

BT: Magnetic fields
RT: Tokamak devices

RT: Aerospace electronics
Radiation hardening

(electronics)

Torpedoes

USE: Missiles

Torque

BT: Mechanical factors
RT: Torque control
Torque converters
Torque measurement

Torque control

BT: Mechanical variables
control
RT: Admittance control
Motor drives
Torque

Torque converters

UF: Torque convertors
BT: Mechanical power
transmission
RT: Automotive components
Drives
Engines
Gears
Shafts
Torque

Torque convertors

USE: Torque converters

Torque measurement

UF: Torque ripple
BT: Mechanical variables
measurement
RT: Pressure gauges
Torque
NT: Dynamometers

Torque ripple

USE: Torque measurement

Torso

BT: Body regions

Total harmonic distortion

BT: Distortion measurement
Harmonic distortion
RT: Signal analysis

Total ionizing dose

BT: Radiation effects

Total quality management

UF: TQM
BT: Quality management
RT: Business process re-engineering
Design for quality
Quality assurance
Quality awards
Quality control
NT: Continuous improvement
Six sigma

Total variance

BT: Probability
Statistics

Touch screens

USE: Touch sensitive screens

Touch sensitive screens

UF: Touch screens
touchscreens
BT: Computer displays
RT: Haptic interfaces
Tactile sensors

Touch sensors

USE: Tactile sensors

touchscreens

USE: Touch sensitive screens

Tourism industry

BT: Industries

Towers

USE: Poles and towers

Town gas

USE: Coal gas

Toxic chemicals

BT: Chemical hazards
Toxicology

Toxicology

UF: Poisons
BT: Hazards
RT: Chemical hazards
Hazardous materials
Occupational health



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	NT:	Pollution Toxic chemicals			Solar powered vehicles
Toy industry			Traction power supplies		
	BT:	Industries		BT:	Power supplies
Toy manufacturing industry			<i>Tractors</i>		
	UF:	Toys		USE:	Agricultural machinery
	BT:	Manufacturing industries	<i>Trade (international)</i>		
	RT:	Electronics industry		USE:	International trade
<i>Toys</i>			Trade agreements		
	USE:	Toy manufacturing industry		UF:	Free trade GATT General agreement on
<i>TQM</i>					
	USE:	Total quality management	tariffs and trade		
Trachea				BT:	Economics
	UF:	Windpipe		RT:	Free economic zones Freeports Globalization International collaboration International trade Tariffs
	BT:	Respiratory system			
<i>Tracheal intubation</i>			<i>Trade unions</i>		
	USE:	Intubation		USE:	Industrial relations
Tracking			Trademarks		
	BT:	Motion measurement		BT:	Law Legal factors
	RT:	Iterative learning control Maximum likelihood		RT:	Copyright protection
estimation			Traffic congestion		
		Particle tracking Position measurement Tracking loops Velocity measurement		BT:	Road transportation
	NT:	Object tracking Solar tracking Target tracking Trajectory tracking Underwater tracking Video tracking		RT:	Traffic control
Tracking loops			Traffic control		
	UF:	Delay lock loops		UF:	Traffic pattern Traffic simulation
	BT:	Linear feedback control		BT:	Control systems
systems				RT:	Communication systems Computer network
		Signal processing Modulation Synchronization Tracking	management		
	RT:				Traffic congestion Advanced driver assistance
Traction motors			systems		
	BT:	Motors			Queueing analysis Road traffic control Vehicle routing
	RT:	Battery powered vehicles Fuel cell vehicles Hybrid electric vehicles Propulsion	<i>Traffic load</i>		
				USE:	Telecommunication traffic
			<i>Traffic pattern</i>		
				USE:	Traffic control



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Traffic simulation

USE: Traffic control

Training

UF: Technician training
BT: Education
RT: Accreditation
Continuing education
Continuing professional

development

Electronic learning
Learning management

systems

Manuals
Mentoring
Personnel
NT: Certification
Industrial training
Management training
On the job training
Qualifications
Vocational training

Training data

BT: Data analysis
RT: Few shot learning
Fish schools
Generative Pre-Trained

Transformer

One shot learning

Trajectory

BT: Path planning
RT: Motion control
Object tracking
NT: Trajectory optimization

Trajectory optimization

BT: Optimization
Trajectory

Trajectory planning

BT: Path planning

Trajectory tracking

BT: Path planning
Tracking
RT: Motion control
Robot control
Underactuated surface

vessels

Trans human

USE: Transhuman

Trans-human

USE: Transhuman

Transaction databases

BT: Databases
RT: Automated teller machine
NT: Itemsets

Transactive control

USE: Transactive energy

Transactive energy

UF: Transactive control
BT: Energy management
RT: Power distribution
Power markets
Power system economics
Smart grids

Transceivers

BT: Communication equipment
RT: Land mobile radio
equipment

Mobile communication
Mobile handsets
Optical antennas
Software radio
NT: Radio transceivers

Transcoding

BT: Encoding
RT: Data compression
Image coding
Multimedia communication
Video coding

Transconductance

UF: Mutual conductance
BT: Conductivity
RT: Transconductors

Transconductors

BT: CMOS integrated circuits
RT: Transconductance

Transcranial direct current stimulation

BT: Neuroscience
Neurostimulation

Transcranial magnetic stimulation

BT: Neuroscience
Neurostimulation

Transcriptomic technologies

USE: Transcriptomics



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Transcriptomics

UF: Transcriptomic technologies

technologies

BT: Genomics

RT: DNA
RNA

Transcriptomics technologies

USE: Transcriptomics

Transducers

BT: Electronic components

RT: Electric variables

measurement

Measurement

Mechanical variables

measurement

Solenoids

Temperature sensors

Thermal variables

measurement

NT: Acoustic transducers
Biomedical transducers
Capacitive transducers
Chemical transducers
Inductive transducers
Piezoelectric transducers
Resistive transducers
Ultrasonic transducer

arrays

Transfection

BT: Genetic engineering

RT: Gene therapy

Transfer function

USE: Transfer functions

Transfer functions

UF: Transfer function

BT: Differential equations

RT: Control systems

Damping

Linear systems

NT: Poles and zeros

Transfer learning

BT: Machine learning

Transfer molding

UF: Resin transfer molding
Resin transfer moulding
Transfer moulding

BT: Production

Transfer moulding

USE: Transfer molding

Transferred electron devices

USE: Gunn devices

Transform coding

UF: JPEG
JPEG2000
MPEG

BT: Data compression

RT: Digital photography

MPEG standards

Principal component

analysis

Vector quantization

Transformer cores

BT: Magnetic cores

Magnetic devices

RT: Power transformers

Transformers

Transformer oil

USE: Oil insulation

Transformer windings

USE: Windings

Transformers

BT: Power systems

RT: Coils

Core loss

Inductive power

transmission

Transformer cores

Voltage multipliers

Windings

NT:

Baluns

Current transformers

Flyback transformers

High-frequency

transformers

Instrument transformers

Phase transformers

Power transformers

Pulse transformers

Tap changers

Vision transformers

Transforms

BT: Mathematics

RT: Numerical analysis



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decomposition	NT:	Signal processing Spectral analysis Discrete transforms Empirical mode	transistors	Heterojunction bipolar Millimeter wave transistors Phototransistors Static induction transistors
		Fourier transforms Karhunen-Loeve transforms Poincare invariance Wavelet transforms		
Transgender issues			Transition metal compounds	
	BT:	Gender issues	BT:	Inorganic compounds
	RT:	Gender equity	NT:	Transition metal dichalcogenides
Transhuman			Transition metal dichalcogenides	
	UF:	Trans human Trans-human Transhumanism	BT:	Semiconductor materials Transition metal compounds
	BT:	Systems, man, and	RT:	Semiconductor thin films
cybernetics	RT:	Posthuman	Translation	
			BT:	Natural languages
			RT:	Linguistics Machine translation Natural language processing
<i>Transhumanism</i>				Semantics Sign language Terminology Text analysis Writing
USE:	Transhuman		<i>Translational medicine</i>	
			USE:	Translational research
Transient analysis			Translational research	
	UF:	Transients	UF:	Bench to bedside Translational medicine Translational science
	BT:	Power system transients	BT:	Biomedical engineering Research and development
	RT:	Electromagnetic transients Signal analysis Steady-state	RT:	Medical diagnosis Medical services
<i>Transient gratings</i>			<i>Translational science</i>	
USE:	Gratings		USE:	Translational research
Transient response			<i>Transmembrane potential</i>	
	UF:	Natural response	USE:	Membrane potentials
	BT:	Propagation	<i>Transmissible disease</i>	
	RT:	Damping	USE:	Infectious diseases
<i>Transients</i>			<i>Transmission control protocol-internet protocol</i>	
USE:	Transient analysis		USE:	TCP/IP
<i>Transinformation</i>			<i>Transmission control protocol/internet protocol</i>	
USE:	Mutual information		USE:	TCP/IP
Transistors				
	BT:	Semiconductor devices Solid state circuits		
	RT:	Aluminum gallium nitride Bipolar transistors CMOS technology Silicon germanium Threshold voltage		
	NT:	Field effect transistors		



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Transmission electron microscopy

UF: TEM
BT: Electron microscopy
RT: Electron beams
Thermionic emission
NT: High-resolution

transmission electron microscopy

Transmission line antennas

BT: Antennas
RT: Transmission lines

Transmission line circuits

USE: Distributed parameter

circuits

Transmission line discontinuities

BT: Transmission lines
RT: Freight handling
NT: Waveguide discontinuities

Transmission line matrix methods

BT: Mathematics
Numerical analysis

Transmission line measurements

BT: Electric variables
measurement
RT: Impedance measurement
Transmission lines

Transmission line theory

BT: Transmission lines
RT: Capacitance
Conductivity
Crosstalk
Frequency
Inductance

Transmission lines

UF: Transmission-line
BT: Power transmission
RT: Baluns
Circuit noise
Civil engineering
Coaxial cables
Distributed parameter

circuits

Helical antennas
Splicing
Transmission line antennas
Transmission line

measurements

NT: Cables

waveguides

lines

Electromagnetic

Multiconductor transmission

Planar transmission lines

Poles and towers

Power line communications

Power transmission lines

Stripline

Superconducting

transmission lines

Transmission line

discontinuities

Transmission line theory

Transmission of electric power

USE: Power transmission

Transmission-line

USE: Transmission lines

Transmit antennas

USE: Transmitting antennas

Transmitters

BT: Communication equipment
RT: Linearization techniques
Modulation
Optical antennas
SISO
NT: Auxiliary transmitters
Diversity methods
Neurotransmitters
Optical transmitters
Radio transmitters
Transmitting antennas

Transmitting antennas

UF: Transmit antennas
BT: Antennas
Transmitters
RT: Receiving antennas

Transparency (climate reporting)

BT: Environmental economics
RT: Climate change

Transpiration

BT: Water cycle
RT: Evaporation

Transplants

USE: Organ transplantation



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Transponders

BT: Communication equipment
Radio communication
equipment
RT: Radio navigation
Satellite communications

Transport protocols

BT: Protocols
RT: IP networks
Overlay networks
Radio links
SONET
Synchronous digital
hierarchy
TCPIP

Transportation

BT: Intelligent transportation
systems
RT: Bridges
Freight containers
NT: Air transportation
Escalators
Green transportation
Land transportation
Public transportation
Ride hailing
Seaports
Shared transport
Smart transportation
Vehicles

Transportation industry

BT: Industries

Transversal filters

BT: Filters
RT: Digital filters
Filtering theory
Signal processing

Transverse electromagnetic cells

USE: TEM cells

Trapped ions

UF: Ion trapping
BT: Ions
RT: Laser cooling
Quantum computing

Traveling salesman

USE: Traveling salesman
problems

Traveling salesman problems

UF: Traveling salesman
Travelling salesman
problem
BT: NP-hard problem
RT: Optimization methods
Shortest path problem

Traveling wave tubes

UF: Travelling wave tubes
BT: Electron tubes

Travelling salesman problem

USE: Traveling salesman
problems

Travelling wave tubes

USE: Traveling wave tubes

Tree data structures

BT: Data structures
NT: Binary trees
Branching factor
Tree traversal

Tree graphs

BT: Graph theory
RT: Circuit topology
NT: Spanning trees

Tree searching

USE: Decision trees

Tree traversal

BT: Tree data structures

Trees (botanical)

BT: Plants (biology)

Trees - insulation

UF: Water trees
BT: Insulators
RT: Humidity
Insulation life
Moisture

Trellis codes

USE: Convolutional codes

Trend analysis

BT: Statistical analysis
RT: Data analysis
Pattern analysis



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Triacs

USE: Thyristors

Triboelectricity

BT: Electricity
Electrostatic processes
Surface charging

RT: Electrostatic discharge

protection

Nanogenerators

Tribology

BT: Motion measurement

Tricuspid valves

USE: Heart valves

Trigen

USE: Trigenation

Trigenation

UF: CHCP
Combined heat, cooling

and power

Combined heat, cooling,

and power

BT: Trigen
Cooling
Heating systems
Power generation

RT: Cogeneration

Trigger circuits

BT: Circuits

Trions

BT: Electrons

Triple bottom line

BT: Environmental economics

Triples (Data structure)

UF: SPO
Semantic triple
Subject predicate object
Triplestore

BT: Buffer storage

RT: Database systems
Information retrieval
Metasearch
Relational databases
Text mining

Triplestore

USE: Triples (Data structure)

Trojan horses

UF: Trojans
BT: Malware
RT: Cyber espionage

Trojans

USE: Trojan horses

Trolley cars

USE: Public transportation

Tropical cyclones

UF: Cyclonic storms
Storm systems
Tropical depressions
Tropical storms
BT: Cyclones

Tropical depressions

USE: Tropical cyclones

Tropical storms

USE: Tropical cyclones

Troposphere

USE: Terrestrial atmosphere

Truncation errors

USE: Finite wordlength effects

Trust management

BT: Decision making
Information security

RT: Access control
Computer security
Cryptography
Privacy

NT: Trusted computing
Trustless services

Trusted computing

BT: Computer security
Trust management

Trustless applications

USE: Trustless services

Trustless services

UF: Trustless applications

BT: Trust management

RT: Blockchains

Tsunami

BT: Geoscience

RT: Storm surge



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<i>TSV</i>	USE:	Through-silicon vias	Tunable circuits and devices	BT:	Circuits and systems Frequency control
Tuberculosis	BT:	Diseases	RT:	Inductors Tuners Tuning	
<i>Tubes</i>	USE:	Electron tubes	NT:	RLC circuits Tuned circuits	
<i>Tumor</i>	USE:	Tumors	Tundra	BT:	Ecosystems
<i>Tumor cells</i>	USE:	Tumors	RT:	Permafrost	
<i>Tumor detection</i>	USE:	Tumors	Tuned circuits	BT:	Tunable circuits and devices
<i>Tumor micro-environment</i>	USE:	Tumor microenvironment	Tuners	BT:	Instruments Tuning
Tumor microenvironment	UF:	Tumor micro-environment Tumour microenvironment	RT:	Frequency control Frequency synthesizers Resonators Tunable circuits and devices	
	BT:	Cellular biophysics	Tungsten	UF:	Wolfram
	RT:	Cancer Tumors	BT:	Metals	
Tumors	UF:	Tumor Tumor cells Tumor detection Tumours	Tuning	BT:	Frequency control
	BT:	Medical conditions	RT:	Ring oscillators Tunable circuits and devices	
	RT:	Cancer Medical diagnostic imaging Oncology Positron emission	NT:	Laser tuning Optical tuning Tuners	
tomography		Single photon emission	<i>Tuning forks</i>	USE:	Vibrations
computed tomography	NT:	Tumor microenvironment Benign tumors Brain tumors Breast tumors Colonic polyps Lesions Malignant tumors	<i>Tunnel effect</i>	USE:	Tunneling
<i>Tumour microenvironment</i>	USE:	Tumor microenvironment	<i>Tunnel field effect transistors</i>	USE:	TFETs
<i>Tumours</i>	USE:	Tumors	Tunneling	UF:	Tunnel effect
			BT:	Electron devices Quantum mechanics	
			RT:	Quantum well devices Semiconductor materials	
			NT:	Gate leakage	



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	Josephson effect		Engines
	Magnetic tunneling		Machine components
	Resonant tunneling devices		Mechanical systems
	Superconductive tunneling		Pumps
	Tunneling	NT:	Turbines
magnetoresistance			Turbogenerators
Tunneling magnetoresistance		<i>Turbulent media</i>	
UF:	TMR	USE:	Random media
	Tunnelling		
magnetoresistance		Turing completeness	
BT:	Magnetoresistance	BT:	Computation theory
	Tunneling		Data handling
RT:	Magnetoresistive devices	RT:	Instruction sets
<i>Tunnelling magnetoresistance</i>		Turing machines	
USE:	Tunneling	BT:	Automata
magnetoresistance		RT:	Digital computers
Turbidity		Turning	
BT:	Water quality	BT:	Machining
RT:	Light scattering	RT:	Boring
			Machine tools
Turbines		Turnkey project	
BT:	Turbomachinery	BT:	Project engineering
RT:	Aircraft propulsion		Project management
	Boilers		
	Compressors		
	Turbogenerators		
	Wind energy	Tutorials	
NT:	Hydraulic turbines	BT:	Educational programs
	Wind turbines		IEEE indexing
Turbo codes		TV	
BT:	Channel coding	UF:	Mobile television
RT:	Error correction		TV broadcasting
	Viterbi algorithm		Television
<i>Turbo generators</i>		BT:	Communications
USE:	Turbogenerators	technology	
Turbogenerators		RT:	Closed captioning
UF:	Turbo generators		Electronic learning
BT:	Turbomachinery		Entertainment industry
RT:	Turbines		Flat panel displays
	Wind power generation		HbbTV Standards
<i>Turbomachine blades</i>			Image communication
USE:	Turbomachinery		Must-carry regulations
Turbomachinery			TV equipment
UF:	Turbomachine blades		Telecommunication
BT:	Power generation	computing	
RT:	Blades		UHDTV
	Compressors	NT:	Visual communication
			Analog TV
			Cable TV
			Color TV
			Digital TV
			Mobile TV



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television	Smart TV Three-dimensional	Two-term control BT: Process control
	Web TV	Type II superconductors BT: Superconducting materials RT: Flux pinning Niobium
<i>TV broadcasting</i> USE: TV		<i>Type testing</i> USE: Conformance testing
TV equipment BT: Communication equipment RT: TV Video equipment NT: Large screen displays TV receivers		Typesetting BT: Text processing RT: Printing
TV interference BT: Interference RT: Echo interference Gaussian noise		<i>Tyres</i> USE: Tires
TV receivers BT: TV equipment		<i>UAM</i> USE: Urban air mobility
<i>Twitter</i> USE: Blogs AND Social networking (online)		<i>UAV</i> USE: Autonomous aerial vehicles
<i>two degrees of freedom</i> USE: 2-DOF		<i>Uber</i> USE: Ride hailing
Two dimensional hole gas UF: 2-d hole gas 2d hole gas BT: Quantum well devices RT: Quantum well lasers Quantum wells		<i>Ubicom</i> USE: Pervasive computing
<i>Two dimensional photonic crystals</i> USE: Photonic crystals		Ubiquitous computing UF: Ubiquitous wireless* BT: Pervasive computing RT: Ambient intelligence NT: Context-aware services
Two-dimensional displays UF: 2-D displays 2D displays BT: Displays RT: Neural radiance field Sprites (computer) Structure from motion		<i>Ubiquitous wireless*</i> USE: Ubiquitous computing
<i>Two-dimensional electron gas FETs</i> USE: MODFETs		<i>UDN</i> USE: Ultra-dense networks
<i>Two-dimensional photonic crystals</i> USE: Photonic crystals		<i>UHD</i> USE: UHDTV
		UHDTV UF: 4K UHD 8K UHD Super hi-vision UHD Ultra HD Ultra HD TV Ultra-high definition TV Ultra-high definition
		television



January 2026 IEEE Thesaurus

BT:	HDTV	<i>UHF radio propagation</i>	
RT:	ITU Standards	USE:	UHF propagation
	TV		
UHF antennas		UHF technology	
BT:	Antennas	UF:	Ultra-high-frequency
	UHF technology	technology	
RT:	UHF devices	BT:	Communications
		technology	
UHF circuits		RT:	UHF measurements
UF:	Ultra-high-frequency	NT:	UHF antennas
circuits			UHF circuits
BT:	Circuits		UHF communication
	UHF technology		UHF devices
RT:	Analog circuits		UHF integrated circuits
NT:	UHF integrated circuits		
UHF communication		UHMWPE	
UF:	Ultra-high-frequency	UF:	Ultra high molecular weight
communication		polyethylene	
BT:	Communication systems	BT:	Thermoplastic polyethylene
	UHF technology		
RT:	Mobile handsets	Ulcerative colitis	
		BT:	Inflammatory bowel disease
UHF devices		RT:	Crohn's disease
UF:	Ultra-high-frequency	<i>ULSI</i>	
devices		USE:	Ultra large scale integration
BT:	UHF technology	<i>Ultra HD</i>	
RT:	UHF antennas	USE:	UHDTV
	UHF integrated circuits	<i>Ultra HD TV</i>	
UHF integrated circuits		USE:	UHDTV
UF:	Ultra-high-frequency	<i>Ultra high molecular weight polyethylene</i>	
integrated circuits		USE:	UHMWPE
BT:	Circuits	Ultra large scale integration	
	Integrated circuits	UF:	ULSI
	UHF circuits	BT:	Circuits
	UHF technology		Integrated circuits
RT:	Analog integrated circuits		Large scale integration
	UHF devices		
UHF measurements		<i>Ultra low power*</i>	
UF:	Ultra-high-frequency	USE:	Low power electronics
measurements		Ultra reliable low latency communication	
BT:	Measurement	UF:	URLLC
RT:	UHF technology		Ultra-reliable low latency
UHF propagation		communication	
UF:	UHF radio propagation		Ultra-reliable low-latency
	Ultra-high-frequency	communication	
propagation		BT:	Low latency communication
BT:	Electromagnetic	<i>Ultra violet</i>	
propagation		USE:	Ultraviolet sources
RT:	Broadband antennas		



January 2026 IEEE Thesaurus

Ultra wide-band

USE: Ultra wideband technology

Ultra wideband

USE: Ultra wideband technology

Ultra wideband antennas

UF: UWB antennas
Ultrawideband antennas
BT: Broadband antennas
Ultra wideband technology
RT: Ultra wideband radar

Ultra wideband communication

UF: UWB communication
Ultrawideband

communication

BT: Ultra wideband technology
RT: Broadband communication
Military communication
Multipath channels
Spread spectrum

communication

Ultra wideband radar

UF: UWB radar
Ultrawideband radar
BT: Radar
Ultra wideband technology
RT: Ground penetrating radar
Radar detection
Radar imaging
Synthetic aperture radar
Ultra wideband antennas

Ultra wideband technology

UF: UWB technology
Ultra wide-band
Ultra wideband
Ultra-wide-band
Ultrawideband
Ultrawideband technology
BT: Communications
NT: Ultra wideband antennas
Ultra wideband

technology

communication

Ultra wideband radar

Ultra-dense networks

UF: UDN
BT: Cellular networks
Mobile communication
RT: 5G mobile communication

Microcell networks

Ultra-fast optics

USE: Ultrafast optics

Ultra-high definition television

USE: UHDTV

Ultra-high definition TV

USE: UHDTV

Ultra-high definition video

BT: High definition video

Ultra-high-frequency circuits

USE: UHF circuits

Ultra-high-frequency communication

USE: UHF communication

Ultra-high-frequency devices

USE: UHF devices

Ultra-high-frequency integrated circuits

USE: UHF integrated circuits

Ultra-high-frequency measurements

USE: UHF measurements

Ultra-high-frequency propagation

USE: UHF propagation

Ultra-high-frequency technology

USE: UHF technology

*Ultra-low power**

USE: Low power electronics

Ultra-reliable low latency communication

USE: Ultra reliable low latency communication

Ultra-reliable low-latency communication

USE: Ultra reliable low latency communication

Ultra-violet

USE: Ultraviolet sources

Ultra-wide-band

USE: Ultra wideband technology

Ultra-wideband

USE: Ultra wideband technology



January 2026 IEEE Thesaurus

Ultracapacitors

USE: Supercapacitors

Ultracold atoms

BT: Atoms
Cooling
RT: Laser cooling
Magnetooptic effects

Ultrafast electronics

BT: High-speed electronics

Ultrafast optics

UF: Ultra-fast optics
BT: Optics

Ultrasonic applications

USE: Acoustic applications

Ultrasonic devices

USE: Acoustic devices

Ultrasonic imaging

UF: Ultrasonic techniques
Ultrasound
BT: Ultrasonics, ferroelectrics,
and frequency control
RT: Amniocentesis
Biomedical imaging
NT: Ultrasonography

Ultrasonic techniques

USE: Ultrasonic imaging

Ultrasonic transducer arrays

BT: Transducers

Ultrasonic transducers

BT: Ultrasonics, ferroelectrics,
and frequency control
RT: Nondestructive testing
Piezoelectricity
Sonar

Ultrasonic variables measurement

BT: Measurement

Ultrasonics

USE: Acoustics

Ultrasonics, ferroelectrics, and frequency control

NT: Ferroelectric materials
Frequency control
Piezoelectricity

Pyroelectricity

Ultrasonic imaging
Ultrasonic transducers

Ultrasonography

BT: Biomedical image
processing
NT: Ultrasonic imaging
Sonogram

Ultrasound

USE: Ultrasonic imaging

Ultraviolet sources

UF: UV sources
Ultra violet
Ultra-violet
BT: Light sources
RT: Lamps
Lasers

Ultrawideband

USE: Ultra wideband technology

Ultrawideband antennas

USE: Ultra wideband antennas

Ultrawideband communication

USE: Ultra wideband
communication

Ultrawideband radar

USE: Ultra wideband radar

Ultrawideband technology

USE: Ultra wideband technology

Umbilical cable

UF: Power supplies to
apparatus
BT: Power supplies

UML

USE: Unified modeling language

UMTS

USE: 3G mobile communication

Uncertain systems

UF: Parameter uncertainty
BT: Mathematics
RT: Control systems
Linear matrix inequalities
Robustness
Uncertainty



January 2026 IEEE Thesaurus

Uncertainty

BT: Probability
RT: Cognitive science
Fuzzy sets
Nonlinear dynamical

systems

NT: Uncertain systems
Evidence theory
Forecast uncertainty

Uncrewed

USE: Autonomous vehicles

Uncrewed aerial systems

USE: Autonomous aerial vehicles

Uncrewed aircraft

USE: Autonomous aerial vehicles

Underactuated marine surface vehicles

USE: Underactuated surface

vessels

Underactuated marine vehicles

USE: Underactuated surface

vessels

Underactuated surface vessels

UF: USV
Underactuated marine

surface vehicles

Underactuated marine

vehicles

Underactuated unmanned

surface vehicles

Underactuated unmanned

surface vessels

BT: Marine vehicles
RT: Control systems
Robots
Trajectory tracking

Underactuated unmanned surface vehicles

USE: Underactuated surface

vessels

Underactuated unmanned surface vessels

USE: Underactuated surface

vessels

Underarm

USE: Axilla

Underground communication cables

USE: Communication cables

Underground object detection

USE: Buried object detection

Underground objects

USE: Buried object detection

Underground power cables

BT: Power cables

Undersea communication

USE: Underwater communication

Underwater acoustics

BT: Acoustics

Underwater autonomous vehicles

USE: Autonomous underwater

vehicles

Underwater cables

UF: Marine cables
Sub-sea cables
Submarine cables
Submersible cables
Subsea cables
BT: Cables
Marine technology

Underwater communication

UF: Undersea communication
BT: Communication systems
Marine technology
Underwater technology

Underwater drones

USE: Autonomous underwater

vehicles

Underwater equipment

UF: Diving equipment
Flotation devices
BT: Marine technology
Underwater technology
RT: Underwater vehicles
NT: Hydrophones
Rebreathing equipment

Underwater exploration robots

USE: Autonomous underwater

vehicles

Underwater navigation

BT: Navigation
Underwater technology
RT: Marine navigation



January 2026 IEEE Thesaurus

Underwater vehicles
Underwater robots
USE: Autonomous underwater vehicles AND

Marine robots

Underwater sensor networks
USE: Wireless sensor networks

Underwater structures
BT: Marine technology
Underwater technology

Underwater technology
UF: Submarine technology
BT: Marine technology
RT: Underwater vehicles
NT: Marine robots
Underwater communication
Underwater equipment
Underwater navigation
Underwater structures

Underwater tracking
BT: Tracking

Underwater vehicles
UF: Aquatic vehicles
Submarines
Submersibles
BT: Marine vehicles
RT: Marine robots
Marine technology
Military robotics
Underwater equipment
Underwater navigation
Underwater technology
NT: Autonomous underwater

vehicles
Remotely guided

underwater vehicles

Undulators
UF: Wiggler magnets
BT: Magnetic devices
Synchrotrons
RT: Free electron lasers
X-rays

Unemployment
BT: Human resource
management

UNFCCC
USE: United Nations Framework Convention on Climate Change

Uniaxial strain
BT: Strain

Unicast
BT: Computer networks
RT: Streaming media

Unified messaging
BT: Electronic mail
Electronic messaging

Unified modeling language
UF: UML
BT: Specification languages
RT: Client-server systems
Common Information Model
(electricity)

Uniform Resource Identifier
USE: Uniform resource locators

Uniform resource locators
UF: URL
Uniform Resource Identifier
Uniform resource name
BT: Web sites

Uniform resource name
USE: Uniform resource locators

Uninterruptible power systems
BT: Power systems
RT: Batteries
Emergency power supplies
Power supplies
Protection

Unit under test
USE: Device under test

United Nations Framework Convention on Climate Change
UF: UNFCCC
BT: Climate change

Units (measurement)
USE: Measurement units

Universal mobile telecommunication service
USE: 3G mobile communication



January 2026 IEEE Thesaurus

Universal motors

BT: Motors

Universal Serial Bus

UF: USB

BT: Communication standards
Information technology

Universities

USE: Educational institutions

Unmanned aerial vehicles

USE: Autonomous aerial vehicles

Unmanned air vehicles

USE: Autonomous aerial vehicles

AND

Remotely piloted aircraft

Unmanned airborne vehicles

USE: Autonomous aerial vehicles

Unmanned automobiles

USE: Autonomous automobiles

Unmanned autonomous cars

USE: Autonomous automobiles

Unmanned autonomous vehicles

USE: Autonomous vehicles

Unmanned surface vehicles

USE: Autonomous vehicles

Unmanned systems

USE: Autonomous systems

Unmanned underwater vehicles

USE: Autonomous underwater

vehicles

Unmanned vehicles

USE: Autonomous vehicles

Unpiloted aerial vehicles

USE: Autonomous aerial vehicles

Unpiloted air vehicles

USE: Autonomous aerial vehicles

Unsolicited e-mail

UF: Junk e-mail
Junk email
Spam
Spamming

mail

Unsolicited electronic e-

Unsolicited electronic email

Unsolicited email

BT: Electronic mail

RT: Computer crime

Cyberbullying

Office automation

Privacy-invasive software

Unsolicited electronic e-mail

USE: Unsolicited e-mail

Unsolicited electronic email

USE: Unsolicited e-mail

Unsolicited email

USE: Unsolicited e-mail

Unsupervised learning

BT: Learning systems

RT: Autoencoders

Deep learning

Formal concept analysis

Generative adversarial

networks

Perception evolution

networks

Reinforcement learning

Self-supervised learning

Semisupervised learning

Supervised learning

Temporal difference

learning

Text summarization

Weak supervision

NT: Competitive learning

Upcycle

BT: Recycling

Uplink

BT: Satellite communications

Upper bound

BT: Boundary conditions

RT: Lower bound

Uranium

BT: Chemical elements

Urban agriculture

UF: Urban farming

BT: Agriculture

RT: Urban areas



January 2026 IEEE Thesaurus

Urban air mobility

UF: UAM
BT: Aircraft navigation
RT: Air traffic control
Autonomous aerial vehicles
Helicopters
Urban areas
Vertiports

Urban areas

UF: Cities and towns
City
Metropolitan areas
Urban environments
BT: Geography
RT: Public infrastructure
Public transportation
Urban agriculture
Urban air mobility
Urban air mobility
Vertiports
NT: Smart cities
Urban planning

Urban environments

USE: Urban areas

Urban farming

USE: Urban agriculture

Urban heat islands

USE: Thermal pollution

Urban modeling

USE: Urban planning

Urban planning

UF: City planning
Urban modeling
BT: Urban areas
RT: Public infrastructure
NT: Urban policy

Urban policy

BT: Urban planning

Urban pollution

BT: Pollution

Urinary calculosis

USE: Kidney stones

URL

USE: Uniform resource locators

URLLC

USE: Ultra reliable low latency communication

Urogenital system

BT: Anatomy
NT: Bladder
Kidney
Reproductive system

Urology

UF: Genitourinary surgery
BT: Medical specialties

US activities

USE: IEEE United States activities

US Department of Agriculture

BT: US Government

US Department of Commerce

BT: US Government
NT: NIST
NTIA
National Oceanic and Atmospheric Administration

US Department of Defense

UF: DoD
BT: US Government

US Department of Energy

UF: DoE
BT: US Government

US Department of Health and Human Services

BT: US Government agencies
NT: National Institutes of Health

US Department of Homeland Security

BT: US Government
RT: Biological weapons
Chemical weapons
Cyberattack
Nuclear weapons
Terrorism
Weapons of mass

destruction

US Department of Transportation

UF: DOT
BT: US Government



January 2026 IEEE Thesaurus

US Government

BT: Government
 NT: US Department of
 Agriculture
 US Department of
 Commerce
 US Department of Defense
 US Department of Energy
 US Department of
 Homeland Security
 US Department of
 Transportation
 US Government agencies
 US local government

US Government agencies

BT: US Government
 RT: Patents
 NT: FAA
 FCC
 FDA
 NASA
 US Department of Health
 and Human Services

US local government

BT: US Government

Usability

BT: Software design

USB

USE: Universal Serial Bus

User centered design

UF: User-centered design
 User-centred design
 BT: Design methodology
 RT: Technology acceptance
 model

User computer interfaces

USE: User interfaces

User experience

BT: Ergonomics
 RT: Affective computing
 Human computer
 interaction
 Human factors
 Metaverse
 Mobile computing
 Quality of experience
 Technology acceptance
 model

User interfaces
 NT: Cyberbullying

User friendliness

USE: Human computer
 interaction

User generated content

USE: User-generated content

User interfaces

UF: Man-machine interfaces
 User computer interfaces
 User-computer interfaces
 Systems, man, and
 cybernetics
 BT: Adaptive learning
 Affordances
 Ambient intelligence
 Browsers
 Computer interfaces
 Computer peripherals
 Displays
 Gaze tracking
 Metaverse
 User experience
 Web design
 NT: Audio user interfaces
 Brain-computer interfaces
 Data visualization
 Emotion recognition
 Exoskeletons
 Graphical user interfaces
 Human computer
 interaction
 Human vehicle systems
 Human-robot interaction
 Intent recognition
 Smart cards

User preference

BT: Behavioral sciences
 RT: Decision making

User-centered design

USE: User centered design

User-centred design

USE: User centered design

User-computer interfaces

USE: User interfaces

User-created content

USE: User-generated content



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User-generated content

media
 UF: Consumer-generated
 User generated content
 User-created content
 BT: Data acquisition

USV

USE: Underactuated surface
 vessels

Uterus

BT: Reproductive system
 NT: Cervix

Utility programs

BT: System software

Utility theory

BT: Mathematics
 RT: Supply and demand

UV sources

USE: Ultraviolet sources

UWB antennas

USE: Ultra wideband antennas

UWB communication

USE: Ultra wideband
 communication

UWB radar

USE: Ultra wideband radar

UWB technology

USE: Ultra wideband technology

V2G

USE: Vehicle-to-grid

V2I

USE: Vehicle-to-infrastructure

V2V

USE: Vehicular ad hoc networks

V2X

USE: Vehicle-to-everything

Vacancy defects

BT: Crystallography

Vaccines

BT: Medical services
 RT: Immune response

Vacuum arc remelting

BT: Melt processing

Vacuum arcs

BT: Vacuum breakdown
 RT: Electron emission
 Thermionic emission
 Vacuum systems

Vacuum breakdown

BT: Dielectric breakdown
 RT: Electron emission
 Vacuum systems
 NT: Vacuum arcs

Vacuum electronics

BT: Vacuum technology

Vacuum energy

USE: Elementary particle vacuum

Vacuum plating

USE: Vapor deposition

Vacuum systems

BT: Vacuum technology
 RT: Bellows
 Casimir effect
 Leak detection
 Vacuum arcs
 Vacuum breakdown
 Gettering
 NT: Gettering

Vacuum technology

BT: Electron devices
 RT: Field emitter arrays
 Gettering
 Space charge
 NT: Photomultipliers
 Vacuum electronics
 Vacuum systems

Vacuum tubes

USE: Electron tubes

VAD

USE: Voice activity detection

Valuation

USE: Cost accounting



January 2026 IEEE Thesaurus

Value chain emissions

BT: Corporate social responsibility

Valves

BT: Fluid flow
Hydraulic equipment
Machine components
RT: Fluid flow control
Manifolds
NT: Microvalves

Vanadium

BT: Chemical elements

Vanes

USE: Blades

VANET

USE: Vehicular ad hoc networks

Vanpools

USE: Shared transport

Vapor deposition

UF: Vacuum plating
Vapour deposition
BT: Materials processing
Surface treatment
RT: Coatings
Electrochemical deposition
MOCVD
Pulsed laser deposition
Substrates
Thin films

Vapor trails

USE: Contrails

Vapour deposition

USE: Vapor deposition

Vapour trails

USE: Contrails

VAR

USE: Reactive power

Varactors

BT: Capacitors
Semiconductor diodes

Variable frequency drives

USE: Variable speed drives

Variable optical attenuators

USE: Optical attenuators

Variable selection

USE: Input variables

Variable speed drives

UF: Variable frequency drives
BT: Drives
RT: Magnetic gears
Motor drives
NT: Pitch control (audio)

Variable structure systems

BT: Adaptive systems

Varistors

BT: Resistors
RT: Arresters
Semiconductor devices

Vascular system

USE: Circulatory system

VCO

USE: Voltage-controlled oscillators

VCR

USE: Video recording

VCSEL

USE: Vertical cavity surface emitting lasers

Vector optimization

USE: Pareto optimization

Vector processors

BT: Microprocessors

Vector quantisation

USE: Vector quantization

Vector quantization

UF: Vector quantisation
BT: Quantization (signal)
RT: Codes
Encoding
Image coding
MPEG 4 Standard
Speech coding
Transform coding
Video coding



January 2026 IEEE Thesaurus

Vectors

BT: Linear algebra
RT: Eigenvalues and
eigenfunctions
Signal processing

Vegetable oils

BT: Oils
RT: Food products

Vegetation

BT: Biology
RT: Enhanced vegetation index
Forestry
Vegetation mapping
NT: Crops
Forests
Marine vegetation
Normalized difference
vegetation index

Vegetation mapping

BT: Geoscience and remote
sensing
RT: Agriculture
Forestry
Geophysical measurement
techniques
Remote sensing
Terrain mapping
Vegetation

Vehicle crash testing

BT: Automotive engineering
Product safety engineering
RT: Collision avoidance
Vehicles

Vehicle detection

BT: Automotive engineering
NT: License plate recognition

Vehicle driving

BT: Automotive engineering
NT: Autonomous driving
Driver behavior

Vehicle dynamics

BT: Automotive engineering
RT: Hardware-in-the-loop
simulation
Vehicles
NT: Rollover

Vehicle license plate recognition

USE: License plate recognition

Vehicle plate detection

USE: License plate recognition

Vehicle routing

BT: Traffic control
RT: Intelligent vehicles
Path planning

Vehicle safety

UF: Automobile safety
Automotive safety
BT: Automotive engineering
Safety
RT: Pedestrians
Vehicle-to-everything
NT: Advanced driver assistance
systems
Lane departure warning
systems
Lane detection

Vehicle to vehicle communication

USE: Vehicular ad hoc networks

Vehicle-to-everything

UF: V2X
BT: Communication systems
Intelligent vehicles
RT: Advanced driver assistance
systems
On board unit
Road safety
Vehicle safety
Vehicular ad hoc networks
NT: Vehicle-to-infrastructure

Vehicle-to-grid

UF: V2G
BT: Electric vehicles
Smart grids
RT: Battery powered vehicles
Demand side management
Distributed power
generation
Fuel cell vehicles
Hybrid electric vehicles
Load management
Propulsion
Solar powered vehicles



January 2026 IEEE Thesaurus

Vehicle-to-infrastructure

UF: V2I
BT: Vehicle-to-everything

Vehicle-to-vehicle

USE: Vehicular ad hoc networks

Vehicles

BT: Transportation
Vehicular and wireless
technologies
RT: Mobile robots
Tires
Vehicle crash testing
Vehicle dynamics
Vehicular ad hoc networks
NT: Bicycles
Connected vehicles
Electric vehicles
Hydrogen powered vehicles
Hypersonic vehicles
Intelligent vehicles
Internet of Vehicles
Land vehicles
Military vehicles
Remotely guided vehicles
Space vehicles

Vehicular ad hoc networks

UF: Ad hoc vehicle networks
Ad hoc vehicular networks
V2V
VANET
Vehicle to vehicle
communication
BT: Vehicle-to-vehicle
Ad hoc networks
RT: Dedicated short range
communication
Internet of Vehicles
Mobile communication
Vehicle-to-everything
Vehicles
Vehicular automation
NT: Road side unit

Vehicular and wireless technologies

UF: Vehicular technologies
NT: Automotive engineering
Land mobile radio
equipment
Navigation
Propulsion
Vehicles
Wireless sensor networks

Vehicular automation

BT: Automation
RT: Autonomous vehicles
Intelligent vehicles
Mechatronics
Mobile robots
Multi-agent systems
Vehicular ad hoc networks

Vehicular technologies

USE: Vehicular and wireless
technologies

Veins

BT: Blood vessels

Velocity control

UF: Rotational measurement
Rotational speed
Speed control
BT: Mechanical variables
control
RT: Aerospace control
Angular velocity
Cruise control
Motion control
Motor drives
Servosystems
NT: Angular velocity control

Velocity measurement

UF: Speed measurement
BT: Mechanical variables
measurement
RT: Angular velocity
Doppler measurement
Flowmeters
Motion measurement
Slow light
Tracking
NT: Terminal velocity

Vented

USE: Ventilators

Ventilation

BT: Cooling
RT: Air conditioning
HVAC
Temperature control
Vents
NT: Fans



January 2026 IEEE Thesaurus

Ventilators

UF: Vented
BT: Biomedical equipment
RT: Intubation
Respiratory system

Ventricle system

BT: Brain

Vents

BT: Mechanical products
RT: Air conditioning
Buildings
Ducts
Space heating
Ventilation
Windows

Venture capital

BT: Economics
Financial management
RT: Business continuity
Enterprise resource

planning

Entrepreneurship
Research and development

management

Risk analysis

Venus

BT: Planets

Vermin control

USE: Pest control

Veroboard

USE: Stripboard circuit

Vertical cavity surface emitting lasers

UF: VCSEL
Vertical cavity surface-

emitting lasers

Vertical-cavity surface-

emitting lasers

BT: Surface emitting lasers
RT: Distributed Bragg reflectors
P-i-n diodes

Vertical cavity surface-emitting lasers

USE: Vertical cavity surface

emitting lasers

Vertical recording

USE: Perpendicular magnetic

recording

Vertical takeoff and landing aircraft

UF: VTOL aircraft
BT: Aircraft
RT: Vertiports
NT: Helicopters

Vertical-cavity surface-emitting lasers

USE: Vertical cavity surface
emitting lasers

Vertiports

BT: Airports
RT: Air traffic control
Aircraft navigation
Urban air mobility
Urban areas
Vertical takeoff and landing
aircraft

Very high speed integrated circuits

UF: VHSIC
BT: Integrated circuits

Very large scale integration

UF: VLSI
Very-large-scale-integration
BT: Circuits
Integrated circuits
Large scale integration
RT: Damascene integration
Nanotechnology
Parameter extraction
NT: Neuromorphics

Very long instruction word

USE: VLIW

Very-high-frequency circuits

USE: VHF circuits

Very-high-frequency devices

USE: VHF devices

Very-large-scale-integration

USE: Very large scale integration

Very-long-instruction-word

USE: VLIW

Vesicles

BT: Cells (biology)

VHDL

UF: VHSIC Hardware
Description Language



January 2026 IEEE Thesaurus

BT: Hardware design
languages
RT: Electronic design
automation and methodology
Field programmable gate
arrays
Integrated circuits
Parallel programming

VHF circuits

UF: Very-high-frequency circuits
BT: Circuits
RT: Analog circuits
Helical antennas
VHF devices

VHF devices

UF: Very-high-frequency
devices
BT: Communications
technology
RT: VHF circuits

VHSIC

USE: Very high speed integrated
circuits

VHSIC Hardware Description Language

USE: VHDL

Vibrating bodies

USE: Vibrations

Vibration control

BT: Mechanical variables
control
RT: Damping
Isolation technology
Shock absorbers
Vibration measurement
Vibrations

Vibration measurement

BT: Mechanical variables
measurement
RT: Modal analysis
Vibration control
Vibrations
Vibrometers

Vibrational signal processing

USE: Signal processing

Vibrations

UF: Mechanical vibrations

Tuning forks
Vibrating bodies
BT: Mechanical factors
RT: Acoustic noise
Acoustics
Damping
Dynamics
Elastodynamics
Nanogenerators
Oscillators
Resonance
Seismocardiography
Vibration control
Vibration measurement

Vibrometers

BT: Meters
RT: Vibration measurement

Video analytics

USE: Visual analytics

Video annotation

USE: Image annotation

Video chats

USE: Web conferencing

Video codecs

BT: Codecs
Communication equipment
Video equipment
RT: Decoding
Image coding
MPEG 4 Standard
MPEG standards
Video coding

Video coding

UF: Advanced video coding
Videocoding
BT: Video signal processing
RT: Image coding
MPEG 4 Standard
MPEG standards
Rate distortion theory
Streaming media
Transcoding
Vector quantization
Video codecs
NT: DVD
High efficiency video coding



January 2026 IEEE Thesaurus

Video compression

BT: Video signal processing
RT: Data compression

Video conferencing

UF: Videoconferencing
BT: Communication systems
RT: Meetings

Video conversion

USE: Video restoration

Video description

BT: Assistive technologies
Audio systems
Videos

Video equipment

UF: Camcorders
BT: Communication equipment
RT: Consumer electronics
TV equipment
Video recording
NT: Optical projectors
Video codecs
Videos

Video games

UF: Computer games
Video-games
Videogames
BT: Videos
NT: Role playing games

Video on demand

BT: Streaming media
RT: Broadband communication
Digital multimedia

broadcasting

Video recording

UF: VCR
VTR
BT: Recording
RT: DVD
Image storage
Mobile video
Video equipment
NT: High definition video
Video restoration
Videos
Webcams

Video restoration

UF: Video conversion
BT: Video recording
RT: Computer vision
Deblurring
Image restoration
Video sequences

Video reviews

BT: IEEE indexing

Video sequence

USE: Video sequences

Video sequences

UF: Video sequence
BT: Computer graphics
RT: Image databases
Image processing
Multimedia computing
Video restoration

Video signal processing

BT: Multidimensional signal processing
RT: Authentication
Fall detection
Firewire
Gaze tracking
IEEE 1394 Standard
Image annotation
Image recognition
MPEG 4 Standard
MPEG standards
Motion detection
Object tracking
Streaming media
Time-frequency analysis
NT: Motion artifacts
Text detection
Video coding
Video compression

Video surveillance

BT: Surveillance
RT: Motion detection

Video tracking

BT: Image motion analysis
Tracking

Video-games

USE: Video games



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<i>Videocoding</i>					Smart assistants
USE:	Video coding		BT:	Computer applications	Conversational artificial
Videoconferences				intelligence	
BT:	Collaborative tools		RT:	Digital humans	Smart devices
<i>Videoconferencing</i>			NT:	Personal voice assistants	
USE:	Video conferencing				
<i>Videogames</i>			Virtual colonoscopy		
USE:	Video games		BT:	Colonoscopy	
Videophone systems			<i>Virtual currency</i>		
UF:	Picture phones		USE:	Online banking	
	Picturephones				
BT:	Communication systems		Virtual enterprises		
RT:	Image communication		BT:	Computer applications	Data processing
	Telephony			Economics	Operations research
	Visual communication		RT:	Electronic commerce	Internet
Videos				Research and development	Virtual manufacturing
UF:	Multimedia products			Virtual reality	
BT:	Video equipment				
	Video recording				
NT:	Deepfakes		Virtual environments		
	Text to video		BT:	Virtual reality	
	Video description		RT:	Internet of Things	Management information
	Video games				
Videotex			base		
UF:	Viewdata		NT:	Metaverse	
BT:	Communication systems			Virtual links	Virtualization
	Information services				
RT:	Data communication		<i>Virtual factories</i>		
	Teletext		USE:	Virtual manufacturing	
<i>Viewdata</i>			Virtual groups		
USE:	Videotex		UF:	Virtual teams	
Virology			BT:	Collaboration	
BT:	Microbiology				
RT:	Epidemics		Virtual LAN		
	Viruses (medical)		UF:	VLAN	Virtual local area network
Virtual artifact			BT:	Local area networks	Network function
BT:	Artificial intelligence			virtualization	Software defined
	Brain		networking		
	Digital systems		RT:	Virtual private networks	
	Virtual reality				
Virtual assistants			<i>Virtual learning</i>		
UF:	Alexa		USE:	Electronic learning	
	Bixby				
	Cortana				
	Siri				



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Virtual links

BT: Communication networks
Routing protocols
Telecommunication control
Virtual environments
RT: Ethernet
Routing

Cultural differences
Digital art
Digital photography
Digital preservation
Digital recording
Digital representation
Virtual reality

Virtual local area network

USE: Virtual LAN

Virtual office

USE: Remote working

Virtual machine monitors

UF: Hypervisors
VMMs
BT: Computers and information
processing
RT: Platform virtualization

Virtual power plants

BT: Distributed power
generation
RT: Cloud computing
Power engineering
computing
Power system control
Power system management

Virtual machines

UF: VM
BT: Virtualization
NT: Process virtual machines
System virtual machines

Virtual private networks

UF: VPN
BT: Computer networks
RT: Data security
Internet
Local area networks
Virtual LAN
Wide area networks
NT: Extranets

Virtual machining

BT: Machining
Virtual manufacturing
RT: Network function
virtualization
Software defined
networking
Virtual prototyping
Virtual reality

Virtual prototyping

BT: Design methodology
RT: Product development
Prototypes
Rapid prototyping
Research and development
Virtual machining
Virtual manufacturing
Virtual reality

Virtual manufacturing

UF: Digital factories
Virtual factories
BT: Computer applications
RT: CAD/CAM
Computer integrated
manufacturing
Concurrent engineering
Research and development
Virtual enterprises
Virtual prototyping
Virtual reality
NT: Virtual machining

Virtual reality

BT: Computer graphics
Graphics
RT: 3D audio
Affordances
Cyberspace
Digital humans
Digital representation
Digital transformation
Digital twins
Immersive audio
Metaverse
Mixed reality
Solid modeling
Spatial computing

Virtual meetings

USE: Web conferencing

Virtual museums

BT: Digital images
Museums
RT: Augmented reality



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	Virtual enterprises		Zika virus
	Virtual machining		
	Virtual manufacturing	<i>Viruses (microorganisms)</i>	
	Virtual museums	USE:	Microorganisms
	Virtual prototyping		
	Virtualization	<i>Visa gold</i>	
NT:	Augmented reality	USE:	Credit cards
	Augmented virtuality		
	Avatars	Viscera	
	Cybersickness	BT:	Body regions
	Extended reality		
	Immersive experience	Viscosity	
	Immersive learning	BT:	Fluids
	Photorealistic images		Measurement
	Virtual artifact		Resistance
	Virtual environments	RT:	Navier-Stokes equations
			Rheology
<i>Virtual seminars</i>			
USE:	Web conferencing	Visible light communication	
<i>Virtual sensing</i>		UF:	VLC
USE:	Soft sensors	BT:	Data communication
			Optical fiber communication
<i>Virtual sensors</i>		RT:	Light emitting diodes
USE:	Soft sensors		Light fidelity
			Lighting
<i>Virtual teams</i>			
USE:	Virtual groups	<i>Vision (biological)</i>	
		USE:	Visual systems
<i>virtual twins</i>			
USE:	Digital twins	<i>Vision Based Robot Control</i>	
		USE:	Visual servoing
Virtualization			
BT:	Virtual environments	<i>Vision captioning</i>	
RT:	Network function	USE:	Closed captioning
virtualization			
	Software defined	Vision defects	
networking		UF:	Amblyopia
	Virtual reality		Color blindness
NT:	Virtual machines		Myopia
		BT:	Visual systems
		RT:	Eye diseases
			Visual impairment
Virulence			
BT:	Microbiology	<i>Vision impairment</i>	
		USE:	Visual impairment
<i>Viruses (computer)</i>			
USE:	Computer viruses	Vision language model	
		BT:	Neural networks
Viruses (medical)		RT:	Computer vision
BT:	Microorganisms		Data visualization
RT:	COVID-19		Natural language
	Measles		
	Virology	processing	
NT:	Bacteriophages		Text recognition
	Coronaviruses		
	Influenza		



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Vision sensors

BT: Sensors
RT: Image processing

Vision systems (nonbiological)

USE: Machine vision

Vision transformers

UF: ViTs
BT: Computer vision
Transformers
RT: Convolutional neural

networks

Visual

USE: Visualization

Visual analytics

UF: Video analytics
BT: Visualization
RT: Information representation

Visual BASIC

BT: Computer languages
RT: Software engineering
Software tools
System software

Visual captioning

USE: Closed captioning

Visual communication

BT: Communication systems
RT: Image communication
Image resolution
TV
Videophone systems

Visual databases

UF: Moving object databases
BT: Databases
NT: Point cloud compression

Visual effects

BT: Visualization
RT: Animation
Computer graphics

Visual impairment

UF: Vision impairment
BT: Medical conditions
RT: Blindness
Cataracts
Eye diseases
Vision defects

NT: Macular degeneration

Visual information retrieval

BT: Information retrieval
RT: Cross modal retrieval
Image matching

Visual odometry

BT: Computer vision
Robots

Visual perception

BT: Visual systems

Visual place recognition

UF: VPR
BT: Image matching
RT: Image recognition
Location awareness

Visual prostheses

USE: Visual prosthesis

Visual prosthesis

UF: Electronic visual prosthesis
Visual prostheses
BT: Prosthetics
RT: Blindness

Visual servoing

UF: Vision Based Robot Control
BT: Motion control
Robot vision systems

Visual systems

UF: Vision (biological)
BT: Sense organs
RT: Head
Machine vision
Saliency detection
NT: Vision defects
Visual perception

Visualisation

USE: Visualization

Visualization

UF: Visual
Visualisation
BT: Computer graphics
Graphics
RT: Animation
Confusion matrices
Design tools
Educational technology



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	Image forensics		Vocabulary	
	YOLO		BT:	Information retrieval
NT:	Curve fitting			Lexicon
	Surface reconstruction		RT:	Ranking (statistics)
	Visual analytics			
	Visual effects		Vocational training	
Viterbi algorithm			UF:	NVQ
BT:	Algorithms			National vocational
RT:	Dynamic programming		qualification	
	Information theory		BT:	Training
	Mathematics		RT:	Industrial training
	Multiaccess communication			Multiskilling
	Probability		Vocoders	
	Speaker recognition		BT:	Communication equipment
	Stochastic processes			Telephone equipment
	Turbo codes		RT:	Speech codecs
				Speech coding
Vitrification			<i>Voice activity</i>	
BT:	Chemical technology		USE:	Voice activity detection
RT:	Radioactive waste disposal			
<i>ViTs</i>			Voice activity detection	
USE:	Vision transformers		UF:	Speech activity detection
				Speech detection
Vivaldi antennas				VAD
UF:	Vivaldi-antennas			Voice activity
BT:	Broadband antennas		BT:	Speech processing
			RT:	Personal voice assistants
<i>Vivaldi-antennas</i>				Speech coding
USE:	Vivaldi antennas			Speech recognition
				Speech synthesis
<i>VLAN</i>			Voice mail	
USE:	Virtual LAN		BT:	Message systems
<i>VLC</i>			RT:	Electronic mail
USE:	Visible light communication			Office automation
VLIW			<i>Voice over Internet protocol</i>	
UF:	Very long instruction word		USE:	Internet telephony
	Very-long-instruction-word			
BT:	Central Processing Unit		<i>Voice over IP</i>	
			USE:	Internet telephony
<i>VLPR</i>			<i>Voice print</i>	
USE:	License plate recognition		USE:	Spectrogram
<i>VLSI</i>			<i>Voice recognition</i>	
USE:	Very large scale integration		USE:	Speech recognition
<i>VM</i>			<i>Voice response systems</i>	
USE:	Virtual machines		USE:	Speech synthesis
<i>VMMs</i>			<i>Voice tract</i>	
USE:	Virtual machine monitors		USE:	Larynx



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Voice-over-Internet protocol

USE: Internet telephony

Voicegram

USE: Spectrogram

Voiceprint

USE: Spectrogram

VOIP

USE: Internet telephony

Volatile organic compounds

BT: Organic compounds

Volcanic activity

BT: Volcanoes

Volcanic ash

BT: Volcanoes

RT: Ash
Lava

Volcano

USE: Volcanoes

Volcanoes

UF: Volcano
BT: Geoscience
NT: Lava
Volcanic activity
Volcanic ash

Voltage

BT: Electric variables
RT: Automatic voltage control
Capacitance-voltage

characteristics

Electrophysiology
Phase frequency detectors
Voltage control
Voltage measurement
NT: Breakdown voltage
Dynamic voltage scaling
Threshold voltage
Voltage fluctuations

Voltage breakdown

USE: Dielectric breakdown

Voltage control

UF: Voltage mode control
Voltage regulation
Voltage-mode control
BT: Electric variables control

RT: Electric current control

Limiting
Motor drives
On load tap changers
Phase frequency detectors
Power distribution control
Power factor correction
Reactive power control
Regulators
Three-phase electric power
Voltage
Voltage multipliers
Automatic voltage control

NT:

Voltage controlled oscillators

USE: Voltage-controlled
oscillators

Voltage fluctuations

BT: Voltage
RT: Power systems

Voltage measurement

BT: Electric variables
measurement
RT: Automatic voltage control
Potentiometers
Voltage
Voltage transformers
Voltmeters
NT: Low voltage
Medium voltage

Voltage mode control

USE: Voltage control

Voltage multipliers

BT: Circuits
RT: AC-DC power converters
Charge pumps
Particle accelerators
Rectifiers
Transformers
Voltage control
NT: Capacitors
Diodes

Voltage regulation

USE: Voltage control

Voltage sags

USE: Power quality

Voltage source inverters

BT: Inverters



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Voltage transformers

UF: Potential transformers
BT: Instrument transformers
RT: Voltage measurement

Voltage-controlled oscillators

UF: VCO
Voltage controlled

oscillators

BT: Oscillators
RT: Ring oscillators

Voltage-mode control

USE: Voltage control

Voltage-source converters

UF: Modular multi-level
converters
VSC
Voltage-source convertors
BT: Converters
Power conversion
RT: AC-DC power converters
HVDC transmission
Power electronics
Pulse width modulation
converters

Voltage-source convertors

USE: Voltage-source converters

Voltmeters

BT: Meters
RT: Voltage measurement

Volume estimation

USE: Volume measurement

Volume measurement

UF: Volume estimation
BT: Mechanical variables
measurement
RT: Size measurement

Volume relaxation

BT: Mechanical factors

Volunteer computing

USE: Computer applications AND
Distributed processing

Vortices, optical

USE: Optical vortices

Voting

BT: Government
NT: Electronic voting

VPN

USE: Virtual private networks

VPR

USE: Visual place recognition

VPU

USE: Graphics processing units

VSC

USE: Voltage-source converters

VTOL aircraft

USE: Vertical takeoff and landing
aircraft

VTR

USE: Video recording

W3C

UF: World Wide Web
Consortium
BT: Standards organizations

W3C Standards

BT: Standards publications

Wafer bonding

BT: Bonding processes
Semiconductor device
manufacture

Wafer level packaging

USE: Wafer scale integration

Wafer scale integration

UF: Wafer level packaging
Wafer-level packaging
BT: Circuits
Integrated circuits
Large scale integration

Wafer-level packaging

USE: Wafer scale integration

Walking

USE: Legged locomotion

WAMS

USE: Wide area measurements



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<i>WAN</i>	USE:	Wide area networks	NT:	Waste reduction Sewage treatment Sludge treatment Waste handling equipment Wastewater treatment
<i>WANs</i>	USE:	Wide area networks		
<i>WAP</i>			Waste handling equipment	
AND	USE:	Wireless access points	BT:	Waste handling
		Wireless application	RT:	Materials handling
protocol			equipment	Remote handling
			equipment	
Warehousing			Waste heat	
	BT:	Material storage	BT:	Energy conversion
	RT:	Production facilities	RT:	Boilers
		Stacking		Cogeneration
		Storage automation		Energy conservation
<i>Warning systems</i>				Industrial waste
	USE:	Alarm systems		Thermal pollution
Warranties			<i>Waste incineration</i>	
	UF:	Product warranties	USE:	Incineration
		Product warranty		
	BT:	Product liability	Waste management	
Washing machines			BT:	Environmental
	BT:	Electric machines	management	
		Electrical products	RT:	Anaerobic digestion
		Home appliances		Biodegradation
		Home automation		Circular economy
				Composting
<i>Wasm</i>				Effluents
	USE:	Webassembly		Pollution control
<i>Waste compaction</i>				Production management
	USE:	Waste reduction		Radioactive waste
Waste disposal				Sanitary engineering
	BT:	Waste management	NT:	Slag
	RT:	Effluents		Waste materials
		Pollution		Wastewater treatment
		Radioactive waste		Waste disposal
		Sanitary engineering		Waste handling
		Slag		Waste recovery
	NT:	Incineration	Waste materials	Waste reduction
		Landfills	UF:	Refuse
		Radioactive waste disposal		Solid waste
<i>Waste electrical and electronic equipment</i>			BT:	Materials
	USE:	Electronic waste	RT:	Fuels
Waste handling				Radioactive pollution
	BT:	Waste management		Sanitary engineering
	RT:	Radioactive waste disposal		Waste management
				Waste recovery
				Water pollution
			NT:	Effluents



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	Electronic waste		Hydrodynamics
	Food waste		Hydrologic measurements
	Industrial waste		Hydrology
	Plastic waste		Lakes
	Post-consumer waste		Oceans
	Radioactive waste		Reservoirs
	Slurries		Rivers
	Wastewater		Steam engines
Waste recovery			Wastewater
BT:	Waste management		Water heating
RT:	Waste materials		Water pollution
	Waste reduction		Water resources
			Water splitting
			Water storage
Waste reduction			Wetlands
UF:	Waste compaction	NT:	Coastlines
BT:	Waste management		Freshwater
RT:	Design for disassembly		Groundwater
	Waste handling		Normalized difference
	Waste recovery	water index	
NT:	Compaction		Streams
Wastewater			Water cycle
BT:	Waste materials		Water quality
RT:	Industrial waste	Water conservation	
	Sanitary engineering	UF:	Water recycling
	Sludge treatment	BT:	Environmental
	Wastewater treatment	management	
	Water	RT:	Wastewater treatment
	Water pollution		Water resources
	Water pumps		Water storage
		NT:	Desalination
Wastewater treatment		Water cycle	
UF:	Dissolved air flotation	BT:	Water
BT:	Waste handling	NT:	Evaporation
RT:	Rubber products		Precipitation
	Sanitary engineering		Transpiration
	Sludge treatment		
	Waste management	<i>Water energy</i>	
	Wastewater	USE:	Hydroelectric power
	Water conservation	generation	
	Water pollution	Water heating	
NT:	Blackwater	BT:	Heating systems
	Greywater	RT:	District heating
	Ozonation		Water
Watches		Water jet cutting	
BT:	Clocks	UF:	Abrasive water jet cutting
RT:	Consumer products	BT:	Cutting tools
	Time measurement		
Water		Water monitoring	
UF:	H2O	BT:	Monitoring
BT:	Liquids		Water resources
RT:	Evaporation		



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Water pollution

BT: Oceanic engineering and
marine technology

RT: Pollution
Effluents
Industrial pollution
Lakes
Oils
Pesticides
Rivers
Sanitary engineering
Sewage treatment
Thermal pollution
Waste materials
Wastewater
Wastewater treatment
Water
Water quality
Water resources
NT: Marine pollution

Water power

USE: Hydroelectric power
generation

Water pumps

BT: Pumps
RT: Automotive components
Cooling
Hydraulic equipment
Irrigation
Photovoltaic systems
Wastewater

Water quality

BT: Water
RT: Water pollution
NT: Turbidity

Water recycling

USE: Water conservation

Water resources

BT: Environmental
management
RT: Lakes
Remote sensing
Rivers
Water
Water conservation
Water pollution
Watersheds
NT: Aquifers
Desalination
Eutrophication

Reservoirs
Stormwater
Water monitoring

Water splitting

BT: Chemical processes
RT: Chemical reactions
Hydrogen
Photochemistry
Water

Water storage

UF: Water-storage
BT: Material storage
RT: Aquifers
Crops
Lakes
Land use planning
Water
Water conservation
NT: Dams
Reservoirs

Water trees

USE: Trees - insulation

Water-storage

USE: Water storage

Watermark

USE: Watermarking

Watermarking

UF: Audio watermarking
Digital watermarking
Image watermarking
Watermark
BT: Security
RT: Copyright protection
Embossing
Internet of Things

Watersheds

BT: Hydrology
Land surface
RT: Rivers
Water resources

Watt hour meters

USE: Watthour meters

Watt-hour meters

USE: Watthour meters



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Watthour meters

UF: Watt hour meters
Watt-hour meters
BT: Wattmeters
RT: Energy measurement

Wattmeters

BT: Meters
RT: Power measurement
NT: Watthour meters

WAVE

USE: Wireless Access in
Vehicular Environments

Wave diffraction

USE: Diffraction

Wave energy conversion

UF: Wave energy converters
BT: Energy conversion
Hydroelectric power

generation

Wave energy converters

USE: Wave energy conversion

Wave equations

USE: Propagation

Wave functions

BT: Waves
RT: Conformal mapping
Elementary particle

exchange interactions

NT: Functional analysis
Wavelet analysis
Wavelet domain

Wave power

BT: Energy resources
Ocean waves
RT: Renewable energy sources

Wave propagation

USE: Propagation

Wave scattering

USE: Scattering

Waveform analysis

USE: Signal analysis

Waveform generators

USE: Signal generators

Waveguide components

BT: Electromagnetic
waveguides
RT: Circulators
Conformal mapping
Gap waveguide
Helical antennas
Waveguide theory
NT: Optical waveguides
Power combiners
Power dividers

Waveguide discontinuities

UF: Irises
Waveguide obstacles
BT: Transmission line
discontinuities
RT: Electromagnetic
waveguides
Loaded waveguides
Waveguide theory
NT: Reflection coefficient
Waveguide transitions

Waveguide junctions

BT: Junctions

Waveguide lasers

BT: Electromagnetic
waveguides
RT: Lasers
NT: Substrate integrated
waveguides

Waveguide obstacles

USE: Waveguide discontinuities

Waveguide theory

UF: Guided electromagnetic
wave propagation
BT: Electromagnetic
waveguides
RT: Antennas
Conformal mapping
Mathematics
Mode matching methods
Waveguide components
Waveguide discontinuities
Waveguide transitions

Waveguide transitions

BT: Waveguide discontinuities
RT: Waveguide theory



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Wavelength assignment

BT: Optical fiber networks

Wavelength conversion

BT: Optical fibers
RT: Wavelength converters

Wavelength converters

UF: Wavelength convertors
BT: Converters
RT: Wavelength conversion

Wavelength convertors

USE: Wavelength converters

Wavelength division multiplexed

USE: Wavelength division multiplexing

Wavelength division multiplexing

UF: WDM
Wavelength division multiplexed
Wavelength-division multiplexing
BT: Multiplexing
RT: Bragg gratings
Multicast communication
NT: WDM networks

Wavelength measurement

BT: Measurement
RT: Acoustic measurements
Electromagnetic measurements
Frequency measurement
Hyperspectral sensors
Optical variables
measurement

Wavelength routing

BT: Routing

Wavelength-division multiplexing

USE: Wavelength division multiplexing

Wavelet analysis

BT: Wave functions
RT: Wavelet transforms
NT: Multiresolution analysis

Wavelet coefficients

BT: Wavelet transforms

Wavelet domain

BT: Wave functions

Wavelet neural networks

USE: Neural networks

Wavelet packets

BT: Wavelet transforms

Wavelet transforms

BT: Transforms
RT: Harmonic analysis
Signal analysis
Signal processing
Signal representation
Wavelet analysis
NT: Biorthogonal modulation
Continuous wavelet transforms
Discrete wavelet transforms
Wavelet coefficients
Wavelet packets

Waves

BT: Physics
RT: Acoustic propagation
Acoustic scattering
Electromagnetic propagation
Electromagnetic radiation
Electromagnetic scattering
Time-domain analysis
NT: Atmospheric waves
Berry phase
Doppler effect
Electrodynamics
Magnetostatic waves
Matter waves
Plasma waves
Propagation
Reflectivity
Seismic waves
Shock waves
Solitons
Surface acoustic waves
Wave functions

WDM

USE: Wavelength division multiplexing

WDM networks

BT: Wavelength division multiplexing



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Weak supervision

UF: Weakly supervised learning
BT: Learning systems
RT: Machine learning
Semisupervised learning
Supervised learning
Unsupervised learning

Weakly supervised learning

USE: Weak supervision

Weapons

UF: Bomb
Munitions
Ordinance
BT: Military equipment
RT: Defense industry
Electronic countermeasures
Military systems
Terrorism
NT: Biological weapons
Chemical weapons
Guns
Missiles
Nuclear weapons
Projectiles
Weapons of mass

destruction

Weapons of mass destruction

UF: WMD
BT: Weapons
RT: Biological weapons
Chemical weapons
National security
Terrorism
US Department of

Homeland Security

Wearable antennas

BT: Antennas
Wearable devices
RT: Body area networks
Textile antennas
Wearable sensors

Wearable computers

USE: Wearable devices

Wearable computing

USE: Wearable devices

Wearable devices

UF: Body borne computers
Wearable computers

Wearable computing
Wearable electronics
Wearables

BT: Computers
Pervasive computing
RT: Fall detection
Internet of Medical Things
Smart textiles
Soft electronics
Textile electrodes
NT: Smart glasses
Wearable Health Monitoring

Systems

Wearable antennas

Wearable electronics

USE: Wearable devices

Wearable Health Monitoring Systems

UF: Apple watch
Fitbit
Smartwatch
WHMS
BT: Smart healthcare
Wearable devices
RT: Electronic healthcare
Smart devices

Wearable robots

UF: Hardsuit
Powered armor
Powered exoskeleton
BT: Robots
RT: Assistive technologies
Biomechanics
Human-robot interaction
Medical robotics
Military equipment
Military robotics
Military systems
Mobile robots
Orthotics
Prosthetics
Service robots
Soft robotics
Wearable sensors
NT: Supernumery robotic limbs

Wearable sensors

BT: Sensors
RT: Biosensors
Body sensor networks
Image sensors
Infrared detectors
Magnetic sensors



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	Nanosensors		<i>Web browsers</i>	
	Optical sensors		USE:	Browsers
	Pressure sensors			
	Sensor arrays		<i>Web cams</i>	
	Sensor fusion		USE:	Webcams
	Tactile sensors			
	Wearable antennas		Web conferencing	
	Wearable robots		UF:	Online conferencing
	Wireless sensor networks			Video chats
<i>Wearables</i>				Virtual meetings
USE:	Wearable devices			Virtual seminars
				Web meetings
Weather				Web seminars
BT:	Meteorology			Webcasts
NT:	Geomagnetic storms		BT:	Webinars
				Internet
Weather forecasting				Online services
UF:	Weather prediction		RT:	Web services
BT:	Meteorology			Teleconferencing
NT:	Wind forecasting			
			Web design	
Weather modification			UF:	Web site design
BT:	Meteorology		BT:	Web sites
RT:	Climate variability		RT:	Authoring systems
				Content management
<i>Weather prediction</i>				Software design
USE:	Weather forecasting		NT:	User interfaces
				Web page design
				Web pages
Weathering			<i>Web filters</i>	
BT:	Geological processes		USE:	Information filters
RT:	Biodegradation			
	Erosion		<i>Web meetings</i>	
Weaving			USE:	Web conferencing
BT:	Textile technology			
RT:	Cotton		Web mining	
	Fabrics		BT:	Data mining
	Textile fibers			
	Textile industry		<i>Web ontology language</i>	
	Textiles		USE:	OWL
Web 2.0			Web page design	
BT:	Internet		BT:	Web design
<i>Web 3.0</i>			Web pages	
USE:	Semantic Web		BT:	Web design
Web and internet services			<i>Web real-time communications</i>	
UF:	Baidu		USE:	WebRTC
	Internet services			
BT:	Web services		<i>Web robot</i>	
<i>Web assembly</i>			USE:	Bot (Internet)
USE:	Webassembly			



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Web search

BT: Search methods
RT: Metasearch
NT: Crawlers

World Wide Web
NT: Uniform resource locators
Web design

Web seminars

USE: Web conferencing

Web television

USE: Web TV

Web servers

BT: Servers
Web services

Web TV

UF: Web television
BT: Broadcasting
TV
Web services

Web services

BT: Internet
Middleware
RT: Asynchronous
communication
Cloud computing
Dark Web
Service computing
Webcams
NT: Business Process

Webassembly

UF: Wasm
Web assembly
BT: Binary codes
RT: Browsers
C++ languages
Java

Execution Language

Mashups
Message services
Service-oriented
architecture
Simple object access
protocol
Web TV
Web and internet services
Web conferencing
Web servers
WebRTC

Webcams

UF: Web cams
BT: Cameras
Video recording
RT: Web services

Webcasts

USE: Web conferencing

Webinars

USE: Web conferencing

WebRTC

UF: Web real-time
communications
BT: Application programming
interfaces
Real-time systems
Web services

Web services business process execution language

USE: Business Process
Execution Language

Web site design

USE: Web design

WeChat

USE: Message services AND
Social networking (online)

Web sites

UF: Reddit
BT: Computer applications
Information retrieval
RT: Computer networks
Content management
Electronic commerce
Extranets
Internet
Portals
Session hijacking
Social networking (online)

WEEE

USE: Electronic waste

Weibo

USE: Blogs

Weibull distribution

BT: Statistical distributions
RT: Failure analysis
Probability



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Reliability engineering
Statistics

Lakes
Rivers
Water

Weibull fading channels

BT: Fading channels

Weight control

BT: Mechanical variables
control

Weight measurement

BT: Mechanical variables
measurement

Weighted linear combination

USE: Weighted sum model

Weighted sum model

UF: SAW
Simple additive weighting
WLC
WSM
Weighted linear
combination
BT: Decision theory

Welding

BT: Fabrication
Joining processes
RT: Bonding processes
Brazing
Fasteners
Manufacturing
Materials processing
NT: Spot welding

Well logging

BT: Geophysics
Petroleum industry
RT: Oil drilling
Seismology

Wet electrodes

BT: Electrodes
RT: Biomedical electrodes

Wet etching

BT: Etching

Wetlands

BT: Ecosystems
Geoscience
RT: Aquatic ecosystems
Estuaries
Hydrology

Whale optimization algorithms

BT: Algorithms
RT: Biomimetics

Whales

BT: Marine animals

WhatsApp

USE: Freeware AND
Internet telephony

Wheelchairs

BT: Assistive technologies

Wheels

BT: Mechanical products
RT: Automobile manufacture
Automotive components
Automotive engineering
Axles
Flanges
Machine components
Manufacturing
Production
Steering systems
Structural plates
Tires

Whispering gallery modes

UF: Whispering-gallery modes
BT: Optics
RT: Microcavities

Whispering-gallery modes

USE: Whispering gallery modes

White blood cells

BT: Blood
RT: Leukemia
NT: Eosinophils
Lymphocytes
Mast cells

White box

USE: Glass box

White matter

BT: Central nervous system
RT: Action potentials
Axons
Brain



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	Learning systems		WANs
White noise		BT:	Communication systems
BT:	Noise		Computer networks
RT:	AWGN channels	RT:	Electronic learning
	Music		Frame relay
	Random number		IEEE 802.3 Standard
generation			Internetworking
NT:	AWGN		LAN interconnection
			LoRa
			Multiprocessor
White spaces		interconnection	
BT:	Radio spectrum		Open systems
management			Protocols
			Token networks
<i>White-box</i>			Virtual private networks
USE:	Glass box	NT:	Low-power wide area
		networks	
<i>Whitebox</i>			
USE:	Glass box		
		Wide band gap semiconductors	
<i>WHMS</i>		BT:	Semiconductor materials
USE:	Wearable Health Monitoring	RT:	Gallium alloys
Systems			
		<i>Wide dynamic range</i>	
Whole body imaging		USE:	High dynamic range
BT:	Biomedical image		
processing		<i>Wide-area measurement systems</i>	
		USE:	Wide area measurements
Whole-body PET			
BT:	Positron emission	<i>Wide-area measurements</i>	
tomography		USE:	Wide area measurements
<i>wi-fi</i>		Wideband	
USE:	Wireless fidelity	BT:	Bandwidth
			Communication systems
<i>Wi-Fi 6</i>		RT:	Narrowband
USE:	IEEE 802.11ax Standard		
		<i>Wideband amplifiers</i>	
<i>Wi-Max</i>		USE:	Broadband amplifiers
USE:	WiMAX		
		<i>Wideband antennas</i>	
<i>Wide area measurement systems</i>		USE:	Broadband antennas
USE:	Wide area measurements		
		Wiener filters	
Wide area measurements		BT:	Noise reduction
UF:	WAMS		
	Wide area measurement	<i>wifi</i>	
systems		USE:	Wireless fidelity
	Wide-area measurement		
systems		<i>Wiggler magnets</i>	
	Wide-area measurements	USE:	Undulators
BT:	Measurement		
		<i>WiGig</i>	
Wide area networks		USE:	IEEE 802.11 Standard
UF:	WAN		



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Wild fires

USE: Wildfires

Wildfires

UF: Wild fires
BT: Fires

Wildlife

BT: Animals

WiMax

USE: WiMAX

Wimax

USE: WiMAX

WiMAX

UF: Wi-Max
WiMax
Wimax
Worldwide Interoperability

for Microwave Access

BT: Wireless communication
RT: IEEE 802.16 Standard

Winches

BT: Materials handling
equipment
RT: Cables
Lifting equipment

Wind

BT: Meteorology
RT: Sea surface
Wind energy
Wind power generation
NT: Wind forecasting
Wind speed
Wind stress

Wind energy

UF: Wind-energy
BT: Energy resources
RT: Turbines
Wind
Wind forecasting
Wind power generation
Wind turbines
NT: Wind energy conversion

Wind energy conversion

BT: Energy conversion
Wind energy
Wind power generation

Wind energy generation

BT: Power generation
RT: Wind forecasting
Wind turbines
NT: Wind energy integration

Wind energy integration

UF: Wind integration
Wind power grid integration
BT: Power systems
Wind energy generation
RT: Power grids

Wind farm

USE: Wind farms

Wind farms

UF: Wind farm
BT: Energy resources

Wind forecasting

BT: Weather forecasting
Wind
RT: Wind energy
Wind energy generation
Wind turbines

Wind integration

USE: Wind energy integration

Wind power

USE: Wind power generation

Wind power generation

UF: Wind power
BT: Power generation
RT: Renewable Portfolio

Standard

Turbogenerators
Wind
Wind energy
NT: Wind energy conversion

Wind power grid integration

USE: Wind energy integration

Wind speed

BT: Wind
RT: Wind stress

Wind stress

BT: Stress
Wind
RT: Wind speed



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Wind tunnels

BT: Aerospace testing
Test facilities
RT: Aerodynamics
Aerospace simulation

Wind turbines

BT: Turbines
RT: Doubly fed induction
generators
Wind energy
Wind energy generation
Wind forecasting

Wind-energy

USE: Wind energy

Windings

UF: Transformer windings
BT: Electromagnetic fields
RT: AC machines
Coils
Electric machines
Magnetic circuits
Power transformers
Rotating machines
Transformers
NT: Machine windings

Windows

BT: Building materials
Manufactured products
RT: Glass products
Vents

Windpipe

USE: Trachea

Windscreen wipers

USE: Automotive components

Windscreens

USE: Automotive components

Windshield wipers

USE: Automotive components

Windshields

USE: Automotive components

Windup

BT: Feedback control

Wine industry

BT: Industry applications

NT: Wineries

Wineries

BT: Wine industry

Wire

BT: Materials
RT: Communication cables
Conductors
Wiring

Wire drawing

BT: Wires
RT: Manufacturing
Production

Wireless Access in Vehicular Environments

UF: WAVE
BT: Wireless networks
RT: IEEE 802.11p Standard
Intelligent vehicles

Wireless access networks

USE: Wireless networks

Wireless access points

UF: WAP
BT: Computer networks
Hardware
Mobile computing
Wireless communication
RT: IEEE 802.11 Standard
Routing protocols
Wireless LAN
Wireless fidelity

Wireless ad hoc network

USE: Mobile ad hoc networks

Wireless application protocol

UF: WAP
BT: Protocols
Wireless communication

Wireless cellular systems

BT: Wireless networks

Wireless charging

USE: Inductive charging

Wireless communication

UF: Wireless systems
BT: Communication systems
RT: Bluetooth
Dynamic spectrum access



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	IEEE 802.11 Standard		Wireless Metropolitan Area
	IEEE 802.11p Standard	Networks	
	IEEE 802.22 Standard		Wireless local area
	Inductive charging	networks	
	Light fidelity	BT:	Local area networks
	Location awareness	RT:	Ad hoc networks
	Long Term Evolution		Bluetooth
	Machine-to-machine		Butler matrices
communications			IEEE 802.11 Standard
	Mobile applications		IEEE 802.11e Standard
	Paging systems		IEEE 802.11g Standard
	Regional area networks		IEEE 802.11n Standard
	Wireless LAN		IEEE 802.15 Standard
	Wireless fidelity		LAN interconnection
NT:	Cognitive radio		Personal area networks
	Cooperative communication		Radio communication
communication	Dedicated short range		Time-varying channels
			Wireless access points
	GSM		Wireless communication
	MIMO		Wireless sensor networks
	MISO	NT:	Light fidelity
	Open wireless architecture		Wireless fidelity
	Point-to-multipoint		
communications		<i>Wireless local area networks</i>	
	Roaming	USE:	Wireless LAN
	SISO		
	Smart devices	Wireless mesh networks	
	Spatial diversity	BT:	Communication systems
	WRAN	RT:	Wireless sensor networks
	WiMAX		
	Wireless access points	<i>Wireless Metropolitan Area Networks</i>	
	Wireless application	USE:	Wireless LAN
protocol			
	Wireless networks	Wireless networks	
		UF:	Over-the-air
<i>Wireless digital devices</i>			Wireless access networks
USE:	Personal digital devices	BT:	Wireless communication
		RT:	Acoustic communication
<i>Wireless energy transmission</i>		(telecommunication)	
USE:	Wireless power transfer		IEEE 802.11p Standard
			IEEE 802.22 Standard
Wireless fidelity			Nanocommunication
UF:	wi-fi		WRAN
	wifi		Wireless power
BT:	Wireless LAN	transmission	
RT:	IEEE 802.11 Standard	NT:	AODV
	Light fidelity		Self-organizing networks
	Radio frequency		Wireless Access in
	Wireless access points	Vehicular Environments	
	Wireless communication		Wireless cellular systems
Wireless LAN		Wireless personal area networks	
UF:	Radio LAN	UF:	WPAN
	WLAN	BT:	Personal area networks



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Wireless power transfer

UF: Wireless energy
transmission
BT: Wireless power
transmission
NT: Simultaneous wireless
information and power transfer

Conductors
Layout
Metallization
Printed circuits
Wire

Wireless power transmission

BT: Power transmission
RT: Conductors
Wireless networks
NT: Wireless power transfer

Wireless regional area networks

USE: WRAN

Wireless security

USE: Communication system
security

Wireless sensor networks

UF: Underwater sensor
networks
BT: Communication systems
Vehicular and wireless
technologies
RT: Ad hoc networks
Cyber-physical systems
Edge computing
Internet of Things
Machine-to-machine
communications
Microsensors
Nanocommunication
Sensors
Wearable sensors
Wireless LAN
Wireless mesh networks
NT: Body sensor networks
Event detection

Wireless systems

USE: Wireless communication

Wires

BT: Structural shapes
RT: Nanowires
Springs
NT: Wire drawing

Wiring

BT: Electric variables
RT: Building services
Cables

Wishcycling

BT: Recycling

WLAN

USE: Wireless LAN

WLC

USE: Weighted sum model

WMD

USE: Weapons of mass
destruction

Wolfram

USE: Tungsten

Women's issues

USE: Gender issues

Wood alcohol

USE: Methanol

Wood industry

BT: Industries
RT: Forestry
Pulp and paper industry
Pulp manufacturing

Wood naphtha

USE: Methanol

Wood poles

USE: Poles and towers

Wood spirits

USE: Methanol

Wool

BT: Agricultural products
Textiles
RT: Clothing
Fabrics
Natural fibers
Textile fibers
Yarn

Word cloud

USE: Tag clouds



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Word processing

USE: Text processing

Work-place

USE: Employment

Workability

BT: Mechanical factors

Workflow management software

UF: Workflow management system

BT: Office automation

Workflow management system

USE: Workflow management software

Working conditions

USE: Employee welfare

Working environment noise

UF: Environmental noise

BT: Acoustic noise

RT: Ergonomics
Hazards
Occupational health
Occupational safety

Workplace

USE: Employment

Workshops

USE: Conferences

Workstation clusters

USE: Cluster computing

Workstations

BT: Microcomputers
Cluster computing
Computer displays
Computer graphics
Peer-to-peer computing

World Wide Web

UF: WWW
BT: Computer applications
RT: Cyberspace
InterPlanetary File System
Internet
Web sites
NT: Bot (Internet)
Mashups

World Wide Web Consortium

USE: W3C

Worldwide Interoperability for Microwave Access

USE: WiMAX

Worm gears

USE: Gears

Worms

USE: Grippers

Worms (computer)

USE: Computer worms

Wounds

BT: Injuries

Woven fabric composites

USE: Fabrics

WP-BPEL

USE: Business Process Execution Language

WPAN

USE: Wireless personal area networks

WRAN

UF: Wireless regional area networks
BT: Regional area networks
Wireless communication
RT: IEEE 802.22 Standard
Wireless networks

Wrapping

BT: Packaging
RT: Packaging machines

Wrist

BT: Arm
RT: Carpal tunnel syndrome

Writing

UF: Business writing
Engineering writing
Report writing
Technical reports
Technical writing
BT: Professional communication
RT: Manuals
Proposals



January 2026 IEEE Thesaurus

NT:	Translation Abstracts Bibliographies Biographies Braille Dictionaries Documentation Grammar Readability metrics Resumes Reviews Thesauri		X-ray applications X-ray detectors X-ray imaging
		X-ray detectors	BT: Ionizing radiation sensors RT: Crystallography Electromagnetic radiation Gamma-ray detectors Radiation detectors X-ray detection X-ray imaging X-rays
<i>Written character recognition</i>	USE: Handwriting recognition	X-ray diffraction	BT: Electromagnetic diffraction
<i>Written characters</i>	USE: Handwriting recognition	X-ray imaging	BT: X-ray applications RT: Gamma-ray detectors Phantoms Radiography X-ray detection X-ray detectors NT: Plasma x-ray sources
<i>Written-character recognition</i>	USE: Handwriting recognition		
<i>WSM</i>	USE: Weighted sum model		
<i>WWW</i>	USE: World Wide Web	X-ray lasers	BT: Lasers X-ray applications RT: Plasma x-ray sources X-rays
<i>WWW robot</i>	USE: Bot (Internet)		
<i>X reality</i>	USE: Extended reality	X-ray lithography	BT: Lithography
X-ray applications	BT: X-rays RT: Collimators Phantoms X-ray detection X-ray telescopes NT: X-ray imaging X-ray lasers	X-ray scattering	UF: XRD BT: X-rays
X-ray astronomy	BT: Astronomy X-rays RT: X-ray telescopes	X-ray telescopes	BT: Telescopes X-rays RT: Aerospace electronics X-ray applications X-ray astronomy
X-ray detection	BT: X-rays RT: Diagnostic radiography Diffraction Electromagnetic radiation Phantoms Radiography	X-ray tomography	BT: X-rays
		X-rays	BT: Medical services RT: Collimators Electromagnetic radiation Synchrotron radiation Undulators



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	NT:	X-ray detectors X-ray lasers X-ray applications X-ray astronomy X-ray detection X-ray scattering X-ray telescopes X-ray tomography			
X3D	BT:	Computer graphics Three-dimensional displays		Yarn	BT: Textile fibers RT: Wool
	RT:	ISO Standards		<i>YBa2Cu3O7</i>	USE: Yttrium barium copper
<i>XAI</i>	USE:	Explainable AI		oxide	
Xenon	BT:	Gases		<i>YBCO</i>	USE: Yttrium barium copper
<i>Xerography</i>	USE:	Electrophotography		oxide	
XML	UF:	Automatic Test Markup Language		<i>Yield estimate</i>	USE: Yield estimation
		Extensible Markup Language		Yield estimation	UF: Yield estimate BT: Estimation RT: Circuit analysis Crops Microprocessor chips
	BT:	Markup languages		YOLO	UF: YOLOv3 YOLOv5 You only look once BT: Object detection RT: Real-time systems Visualization
<i>Xplainable AI</i>	USE:	Explainable AI		<i>YOLOv3</i>	USE: YOLO
<i>XPM</i>	USE:	Cross-phase modulation		<i>YOLOv5</i>	USE: YOLO
<i>XR</i>	USE:	Extended reality		<i>You only look once</i>	USE: YOLO
<i>XRD</i>	USE:	X-ray scattering		<i>Young modulus</i>	USE: Young's modulus
<i>XSS</i>	USE:	Cross-site scripting		Young's modulus	UF: Tensile modulus Young modulus BT: Solids
<i>Y-Ba-Cu-O</i>	USE:	Yttrium barium copper oxide		Ytterbium	BT: Chemical elements
<i>Yachts</i>	USE:	Boats		Yttrium	BT: Chemical elements Metals NT: Yttrium compounds
Yagi-Uda antennas	BT:	Antennas			



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Yttrium barium copper oxide

UF: Y-Ba-Cu-O
YBCO
YBa₂Cu₃O₇
BT: High-temperature
superconductors
RT: Yttrium compounds
Barium compounds

Yttrium compounds

BT: Yttrium
RT: Alloying
NT: Yttrium barium copper
oxide

ZCS

USE: Zero current switching

Zero correlation zone

BT: Codes
Multiaccess communication
Sequential analysis

Zero current switching

UF: ZCS
Zero-current switching
BT: Switching circuits
RT: Inverters
Switching converters

Zero knowledge proof

BT: Cryptography
Protocols

Zero shot learning

UF: ZSL
Zero-shot learning
BT: Deep learning

Zero touch networks

USE: Autonomous networks

Zero Trust

BT: Security
RT: Access control
Authentication
Permission

Zero voltage switching

UF: ZVS
Zero-voltage switching
BT: Switching circuits
RT: Inverters
Switching converters

Zero waste

BT: Net zero

Zero-current switching

USE: Zero current switching

Zero-shot learning

USE: Zero shot learning

Zero-touch networks

USE: Autonomous networks

Zero-voltage switching

USE: Zero voltage switching

Zeros

USE: Poles and zeros

Zigbee

BT: Radio communication
RT: Automation
Biomedical equipment
Bluetooth
IEEE 802.15 Standard
Personal area networks
Personal communication

networks

Smoke detectors

Zika virus

BT: Viruses (medical)

Zinc

UF: Zn
BT: Metals
NT: Zinc compounds

Zinc compounds

BT: Zinc
NT: Zinc oxide

Zinc oxide

UF: ZnO
BT: Zinc compounds
NT: Indium gallium zinc oxide

ZINDO

USE: Computational modeling

Zip fasteners

USE: Fasteners

Zirconium

BT: Chemical elements
NT: Zirconium compounds



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Zirconium compounds

BT: Zirconium

Zn

USE: Zinc

ZnO

USE: Zinc oxide

Zoology

BT: Biology

NT: Animals

Entomology

Ornithology

Zooplankton

BT: Plankton

ZSL

USE: Zero shot learning

ZVS

USE: Zero voltage switching

